

Dynamical Systems Foundations

A Student Textbook Series

Topological and ergodic foundations, cocycles, arithmetic, hyperbolicity,
small divisors, thermodynamic formalism, parameter methods, and
renormalization

Reader edition — Foundation volumes F01-F16

Rigorous proofs are combined with a dedicated pedagogical exposition layer.

Preface to the reader

This Foundation series is meant to be *studied*, not merely consulted. A rigorous student textbook must do more than state and prove theorems: it must explain why the objects are introduced, how the results fit together, what typical examples look like when one actually computes them, and which hypotheses are doing real work. This edition therefore surrounds the proofs with a dedicated exposition layer without weakening or abbreviating the mathematics.

The sixteen volumes form a connected path. The first four move from invariant measures to ergodic averages, subadditivity, and Lyapunov exponents. Volumes F05–F09 develop the arithmetic and linear equations that create small-divisor phenomena and cocycle dynamics. Volumes F10–F12 explain uniform and nonuniform hyperbolicity and the iterative schemes used to solve nonlinear conjugacy problems. Volumes F13–F16 collect potential theory, thermodynamic formalism, parameter-exclusion methods, and abstract renormalization—shared tools needed later in KAM theory, spectral theory, billiards, and complex dynamics.

How to use the series

For a first reading, begin each chapter with its motivation, learning goals, and conceptual roadmap. Read the definitions next, but return to the guiding example whenever the notation becomes abstract. For a major theorem, first read the statement and the opening paragraph of the proof; identify the mechanism before following estimates line by line. The worked examples are intentionally computational: write them out by hand. The boundary examples are equally important because they show what the hypotheses prevent. End each chapter by answering the checkpoint questions before attempting the exercises.

Conventions

The books assume the standard graduate prerequisites in real and complex analysis, measure theory, functional analysis, differential topology, and basic probability. A symbol is nevertheless reintroduced when its role changes, and notation is recalled whenever a long dependency chain makes that useful.

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Part I

Foundation

Topological Dynamics and Invariant Measures

Why this chapter matters

A trajectory is deterministic, but many of the questions we ask about it are statistical. If a point moves as $x, Tx, T^2x, \dots$, what does it mean for the orbit to spend a definite fraction of its time in different regions of the state space? The answer begins with invariant probability measures. They are the probability laws that do not change when the dynamics advances by one step.

This chapter develops the compact topological setting in which invariant measures are unavoidable: time averages of point masses have weak-* limit points, and every such limit is invariant. Ergodic measures then play the role of statistically indecomposable states. Unique ergodicity is stronger still: it forces the same long-time average for every initial condition, uniformly for continuous observables. These ideas are the measure-theoretic language used throughout the rest of the series.

Prerequisites. Compact metric spaces, continuous maps, Borel probability measures, weak-* convergence, and the Riesz representation theorem.

Learning goals

- Construct invariant measures from empirical orbit measures.
- Interpret ergodicity as extremality and understand ergodic decomposition.
- Use unique ergodicity to turn pointwise orbit averages into uniform statements.
- Distinguish existence of an invariant measure from uniqueness and from recurrence.

Conceptual roadmap

The chapter follows the chain

orbit averages $\longrightarrow$ invariant measures
 $\longrightarrow$ ergodic components
 $\longrightarrow$ unique ergodicity.

Krylov-Bogolyubov gives existence. Compact convex geometry organizes all invariant measures, and ergodic measures appear as the extreme points. Empiri-

cal measures connect this geometry back to single trajectories. The final results show how uniqueness upgrades subsequential information to uniform convergence.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

1.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F01.1. Dynamical system

A *discrete-time dynamical system* is a pair (X, T) , where X is the state space and $T : X \rightarrow X$ is the time-one map. In continuous time we use a family $(\varphi_t)_{t \in \mathbb{R}}$ satisfying $\varphi_0 = \text{id}$ and $\varphi_{s+t} = \varphi_s \circ \varphi_t$. The category—topological, measurable, or smooth—is always specified.

Definition F01.2. Invariant probability measure

Let $T : X \rightarrow X$ be measurable. A Borel probability measure μ is *invariant* if $\mu(T^{-1}A) = \mu(A)$ for every Borel set A . Equivalently, $T_*\mu = \mu$.

Definition F01.3. Ergodic measure

An invariant probability μ is *ergodic* if every measurable set A satisfying $T^{-1}A = A$ modulo μ has $\mu(A) \in \{0, 1\}$.

Definition F01.4. Unique ergodicity

A continuous map T of a compact space X is *uniquely ergodic* if it has exactly one invariant Borel probability measure.

1.2 Main results

Theorem F01.1. Krylov-Bogolyubov existence theorem

Statement. Let X be a nonempty compact metric space and let $T : X \rightarrow X$ be continuous. Then there exists a Borel probability measure μ on X such that $T_*\mu = \mu$.

Proof. Fix any point $x \in X$. The idea is to average the point masses along its forward orbit and then take a limit.

Step 1: construct empirical probabilities. For $N \geq 1$ define

$$\mu_N := \frac{1}{N} \sum_{j=0}^{N-1} \delta_{T^j x}.$$

Each μ_N is a Borel probability measure. Because X is compact metric, $C(X)$ is separable and the set $\mathcal{P}(X)$ of Borel probability measures is weak-* compact and metrizable: view probabilities as positive norm-one functionals on $C(X)$, apply Banach-Alaoglu, and use separability to obtain metrizability on bounded sets. In particular, weak-* compactness is sequential compactness here. Therefore there are integers $N_\ell \rightarrow \infty$ and $\mu \in \mathcal{P}(X)$ such that

$$\mu_{N_\ell} \Longrightarrow \mu. \quad (1)$$

This is the only compactness step in the proof.

Step 2: show that the empirical measures become invariant. Let $f \in C(X)$. Since T is continuous, $f \circ T \in C(X)$. A finite telescoping sum gives

$$\begin{aligned} \int f \circ T d\mu_N - \int f d\mu_N &= \frac{1}{N} \sum_{j=0}^{N-1} (f(T^{j+1}x) - f(T^j x)) \\ &= \frac{f(T^N x) - f(x)}{N}. \end{aligned} \quad (2)$$

Because X is compact, f is bounded, so

$$\left| \frac{f(T^N x) - f(x)}{N} \right| \leq \frac{2\|f\|_\infty}{N} \rightarrow 0. \quad (3)$$

Step 3: pass invariance to the subsequential limit. Weak convergence in (1) means convergence of integrals against every continuous function. Applying it to both f and $f \circ T$ in (2), and using (3), yields

$$\int f \circ T d\mu = \int f d\mu \quad \text{for every } f \in C(X). \quad (4)$$

By definition of pushforward, $\int f d(T_*\mu) = \int f \circ T d\mu$. Hence (4) says that $T_*\mu$ and μ have the same integrals against all continuous functions. Uniqueness in the Riesz representation theorem gives

$$T_*\mu = \mu.$$

Thus μ is a T -invariant Borel probability measure on X , which is exactly the required conclusion. $\square$

Proposition F01.2. Invariant measures form a compact convex set

Statement. Let X be a nonempty compact metric space and let $T : X \rightarrow X$ be continuous. Then

$$\mathcal{M}_T(X) := \{\mu \in \mathcal{P}(X) : T_*\mu = \mu\}$$

is a nonempty compact convex subset of $\mathcal{P}(X)$ in the weak-* topology.

Proof. Let $\mathcal{P}(X)$ denote the set of Borel probability measures on the compact metric space X , and let

$$\mathcal{M}_T(X) := \{\mu \in \mathcal{P}(X) : T_*\mu = \mu\}$$

be the subset of T -invariant probabilities. We prove nonemptiness, convexity, and weak-* compactness separately.

Step 1: $\mathcal{M}_T(X)$ is nonempty. This is exactly the conclusion of the Krylov–Bogolyubov theorem (*Krylov–Bogolyubov existence theorem*), which applies because $T : X \rightarrow X$ is continuous and X is compact metric.

Step 2: $\mathcal{M}_T(X)$ is convex. Take $\mu_0, \mu_1 \in \mathcal{M}_T(X)$ and $0 \leq t \leq 1$, and put $\mu_t = (1-t)\mu_0 + t\mu_1$. Then μ_t is again a Borel probability measure. For every $f \in C(X)$,

$$\int_X f \circ T d\mu_t = (1-t) \int_X f \circ T d\mu_0 + t \int_X f \circ T d\mu_1 = (1-t) \int_X f d\mu_0 + t \int_X f d\mu_1 = \int_X f d\mu_t.$$

By the Riesz representation theorem, equality against every continuous f is equivalent to $T_*\mu_t = \mu_t$. Hence μ_t is invariant.

Step 3: $\mathcal{P}(X)$ is weak- compact.* The Riesz representation theorem identifies every $\mu \in \mathcal{P}(X)$ with the positive linear functional

$$L_\mu : C(X) \rightarrow \mathbb{R}, \quad L_\mu(f) = \int_X f d\mu,$$

of norm 1 satisfying $L_\mu(1) = 1$. Thus $\mathcal{P}(X)$ is contained in the closed unit ball of $C(X)^*$. The properties

$$L(1) = 1, \quad L(f) \geq 0 \quad \text{whenever } f \geq 0$$

are weak-* closed, because each condition is expressed by evaluation at a fixed element of $C(X)$. Therefore $\mathcal{P}(X)$ is a weak-* closed subset of the weak-* compact unit ball supplied by Banach–Alaoglu, and hence is weak-* compact. Since X is compact metric, $C(X)$ is separable, so on this bounded set the weak-* topology is metrizable; in particular the compactness may equivalently be used sequentially later.

Step 4: invariance is weak- closed.* For each $f \in C(X)$ define

$$\Phi_f(\mu) := \int_X (f \circ T - f) d\mu.$$

Because T and f are continuous, $f \circ T - f \in C(X)$, so Φ_f is weak-* continuous. A probabil-

ity measure μ is T -invariant exactly when $\Phi_f(\mu) = 0$ for every $f \in C(X)$. Consequently

$$\mathcal{M}_T(X) = \mathcal{P}(X) \cap \bigcap_{f \in C(X)} \Phi_f^{-1}(\{0\})$$

is weak-* closed in $\mathcal{P}(X)$. Since $\mathcal{P}(X)$ is weak-* compact, $\mathcal{M}_T(X)$ is weak-* compact as well.

Together with Steps 1 and 2, this proves that the invariant Borel probability measures form a nonempty weak-* compact convex set. $\square$

Theorem F01.3. Ergodic measures are extreme points

Statement. Let X be a compact metric space, let $T : X \rightarrow X$ be continuous, and let $\mu \in \mathcal{M}_T(X)$. Then μ is ergodic if and only if μ is an extreme point of the compact convex set $\mathcal{M}_T(X)$.

Proof. Write $\mathcal{M}_T(X)$ for the compact convex set of invariant Borel probability measures from *Invariant measures form a compact convex set*. We prove both implications.

If μ is not ergodic, then μ is not extreme. By nonergodicity there is a measurable set A such that $T^{-1}A = A$ modulo μ and

$$0 < \mu(A) < 1.$$

Define normalized restrictions

$$\mu_A(B) := \frac{\mu(B \cap A)}{\mu(A)}, \quad \mu_{A^c}(B) := \frac{\mu(B \cap A^c)}{\mu(A^c)}.$$

They are probability measures. We check invariance for μ_A ; the other case is identical. Since $T^{-1}A = A$ modulo μ ,

$$\begin{aligned} \mu_A(T^{-1}B) &= \frac{\mu(T^{-1}B \cap A)}{\mu(A)} = \frac{\mu(T^{-1}B \cap T^{-1}A)}{\mu(A)} \\ &= \frac{\mu(T^{-1}(B \cap A))}{\mu(A)} = \frac{\mu(B \cap A)}{\mu(A)} = \mu_A(B). \end{aligned}$$

Thus both normalized restrictions lie in $\mathcal{M}_T(X)$, they are distinct because one gives A mass 1 and the other mass 0, and

$$\mu = \mu(A)\mu_A + \mu(A^c)\mu_{A^c}$$

is a nontrivial convex decomposition. Hence μ is not extreme.

If μ is ergodic, then μ is extreme. Assume

$$\mu = t\nu_1 + (1-t)\nu_2, \quad 0 < t < 1, \quad \nu_1, \nu_2 \in \mathcal{M}_T(X). \quad (1)$$

From (1), for every measurable B , $t\nu_1(B) \leq \mu(B)$, so $\nu_1 \ll \mu$ and its Radon-Nikodym density

$$h := \frac{d\nu_1}{d\mu}$$

satisfies $0 \leq h \leq t^{-1}$ almost everywhere. In particular $h \in L^2(\mu)$.

Let $U : L^2(\mu) \rightarrow L^2(\mu)$ be the Koopman operator $Ug = g \circ T$. Since μ is T -invariant, U is an isometry. The invariance of ν_1 says that for every bounded measurable g ,

$$\int_X g \circ T h d\mu = \int_X g h d\mu. \quad (2)$$

On the other hand, invariance of μ applied to the integrable function gh gives

$$\int_X g \circ T h \circ T d\mu = \int_X (gh) \circ T d\mu = \int_X gh d\mu. \quad (3)$$

Subtracting (3) from (2),

$$\int_X (g \circ T)(h - h \circ T) d\mu = 0 \quad (4)$$

for every bounded measurable g . By L^2 density, (4) holds for every $g \in L^2(\mu)$. Thus $h - Uh$ is orthogonal to the range of U . In particular it is orthogonal to Uh , because Uh belongs to that range. Therefore

$$\|h\|_2^2 = \|Uh + (h - Uh)\|_2^2 = \|Uh\|_2^2 + \|h - Uh\|_2^2.$$

But U is an isometry, so $\|Uh\|_2 = \|h\|_2$. The last identity forces $\|h - Uh\|_2 = 0$, hence

$$h = h \circ T \quad \mu\text{-a.e.} \quad (5)$$

It remains to use ergodicity. If a real measurable function h obeys (5), then for every $c \in \mathbb{R}$ its superlevel set $\{h > c\}$ is invariant modulo μ . Ergodicity therefore gives $\mu\{h > c\} \in \{0, 1\}$ for every c . This forces h to be constant almost everywhere: otherwise there would exist numbers $a < b$ for which both $\{h \leq a\}$ and $\{h \geq b\}$ had positive measure, and choosing $c \in (a, b)$ would produce an invariant superlevel set of measure strictly between 0 and 1.

Since ν_1 is a probability,

$$1 = \nu_1(X) = \int_X h d\mu,$$

so the constant value of h is 1. Hence $\nu_1 = \mu$. Substituting into (1) then also gives $\nu_2 = \mu$. Thus every convex decomposition of μ inside $\mathcal{M}_T(X)$ is trivial, so μ is an extreme point.

The two implications prove the claimed equivalence between ergodicity and extremality. $\square$

Theorem F01.4. Ergodic decomposition

Statement. Let X be a nonempty compact metrizable space, $T : X \rightarrow X$ continuous, and let $\mathcal{M}_T(X)$ be the compact convex set of T -invariant Borel probability measures. Its extreme-point set is $\mathcal{M}_T^{\text{erg}}(X)$, the ergodic invariant probabilities, and $\mathcal{M}_T(X)$ is a metrizable Choquet simplex. Consequently, for every $\mu \in \mathcal{M}_T(X)$

there exists a unique Borel probability κ_μ on $\mathcal{M}_T^{\text{erg}}(X)$ such that, for every $f \in C(X)$,

$$\int_X f d\mu = \int_{\mathcal{M}_T^{\text{erg}}(X)} \left(\int_X f d\nu \right) d\kappa_\mu(\nu).$$

Here uniqueness means uniqueness among Borel probabilities carried by the ergodic-measure set.

Proof. Put

$$K := \mathcal{M}_T(X).$$

By *Invariant measures form a compact convex set*, K is a nonempty compact metrizable convex subset of the locally convex space of finite signed Borel measures on X , equipped with the weak-* topology. By *Ergodic measures are extreme points*,

$$\text{ex } K = \mathcal{M}_T^{\text{erg}}(X). \quad (1)$$

Thus existence of some extreme-point representation follows already from the metrizable Choquet representation theorem. The point requiring care is the *uniqueness* asserted in the present theorem. We verify the simplex property before invoking the uniqueness part of Choquet–Meyer.

Step 1: invariant densities stay invariant, even when T is not invertible. Let ρ and η be finite positive T -invariant Borel measures and assume $\rho \ll \eta$. Write

$$h = \frac{d\rho}{d\eta}.$$

We claim

$$h \circ T = h \quad \eta\text{-almost everywhere.} \quad (2)$$

The argument is the same Hilbert-space point that was needed in *Ergodic measures are extreme points*, but we record it here because it is the load-bearing simplex mechanism.

If h is bounded, then $h \in L^2(\eta)$. Let $Uq = q \circ T$ be the Koopman isometry on $L^2(\eta)$. Invariance of ρ gives, for bounded q ,

$$\int q \circ T h d\eta = \int q h d\eta, \quad (3)$$

while invariance of η gives

$$\int q \circ T h \circ T d\eta = \int q h d\eta. \quad (4)$$

Hence $h - Uh$ is orthogonal to the range of U , in particular to Uh . Since U is an isometry,

$$\|h\|_2^2 = \|Uh + (h - Uh)\|_2^2 = \|Uh\|_2^2 + \|h - Uh\|_2^2$$

forces $h = Uh$.

For a general nonnegative $h \in L^1(\eta)$, apply the bounded argument to the invariant

finite measures with densities

$$h_m := \min(h, m).$$

To justify this without circularity, first apply the bounded argument to $h/(1+h)$ relative to the invariant measure $\eta + \rho$: both $d\eta/d(\eta + \rho) = 1/(1+h)$ and $d\rho/d(\eta + \rho) = h/(1+h)$ are bounded invariant densities. Their ratio on the set where the first density is positive is h , so (2) follows. Thus invariant absolute-continuous finite measures have invariant Radon-Nikodym densities with respect to an invariant dominating measure.

Step 2: the cone of finite invariant measures is a lattice. Let ρ_1, ρ_2 be finite positive invariant measures and put

$$\eta := \rho_1 + \rho_2.$$

Then η is invariant and

$$h_i := \frac{d\rho_i}{d\eta}, \quad 0 \leq h_i \leq 1,$$

are invariant by Step 1. Define

$$\rho_1 \wedge \rho_2 := \min(h_1, h_2) \eta, \quad \rho_1 \vee \rho_2 := \max(h_1, h_2) \eta. \quad (5)$$

Because pointwise min and max of invariant functions are invariant, both measures in (5) are invariant. They are plainly the greatest lower bound and least upper bound in the usual order on finite measures: if $\theta \leq \rho_1, \rho_2$ and $k = d\theta/d\eta$, then $k \leq h_1, h_2$, hence $k \leq \min(h_1, h_2)$; the upper-bound statement is similar. Therefore the positive cone

$$C_T := \{\text{finite positive } T\text{-invariant Borel measures}\}$$

is a lattice cone.

Every nonzero element $\rho \in C_T$ has the unique normalization

$$\rho = \rho(X) \hat{\rho}, \quad \hat{\rho} \in K, \quad (6)$$

so K is a compact convex base of this lattice cone.

Step 3: apply the Choquet-Meyer simplex theorem. We use the standard Choquet-Meyer characterization from graduate convex analysis: a compact convex base of a lattice cone is a simplex; if the base is metrizable, every point has a unique representing Borel probability carried by its extreme points. Steps 1-2 verify exactly the hypothesis that is special to invariant measures, while *Invariant measures form a compact convex set* supplies compact metrizability. Hence K is a metrizable Choquet simplex.

Therefore, for every $\mu \in K$, there is a unique Borel probability κ_μ carried by $\text{ex } K$ such that μ is its barycenter. In the weak-* realization of K , the continuous affine functionals include

$$L_f(\nu) := \int_X f d\nu, \quad f \in C(X).$$

Thus the barycenter identity is precisely

$$\int_X f d\mu = \int_{\text{ex } K} \left(\int_X f d\nu \right) d\kappa_\mu(\nu) \quad (f \in C(X)). \quad (7)$$

Finally (1) identifies $\text{ex } K$ with $\mathcal{M}_T^{\text{erg}}(X)$, so (7) is the asserted ergodic decomposition. The uniqueness supplied by the simplex theorem is exactly uniqueness among Borel probabilities carried by the ergodic measures. $\square$

Theorem F01.5. Empirical-measure subsequential invariance

Statement. Let X be a nonempty compact metric space and let $T : X \rightarrow X$ be continuous. For every $x \in X$, define the empirical measures

$$\mu_n := \frac{1}{n} \sum_{j=0}^{n-1} \delta_{T^j x}.$$

Every weak-* subsequential limit of $(\mu_n)_{n \geq 1}$ is a T -invariant Borel probability measure.

Proof. Fix $x \in X$ and define the empirical measures

$$\mu_N := \frac{1}{N} \sum_{j=0}^{N-1} \delta_{T^j x}, \quad N \geq 1.$$

Suppose that a subsequence μ_{N_k} converges weak-* to a Borel probability measure μ . We must prove $T_*\mu = \mu$.

Let $f \in C(X)$. Since X is compact, f is bounded. Directly from the definition of μ_N ,

$$\begin{aligned} \int_X f \circ T d\mu_N - \int_X f d\mu_N &= \frac{1}{N} \sum_{j=0}^{N-1} (f(T^{j+1}x) - f(T^j x)) \\ &= \frac{f(T^N x) - f(x)}{N}. \end{aligned}$$

Every term in the middle sum cancels except the first negative term and the last positive term. Hence

$$\left| \int_X f \circ T d\mu_N - \int_X f d\mu_N \right| \leq \frac{2\|f\|_\infty}{N} \rightarrow 0. \quad (1)$$

Now pass to the convergent subsequence. Because T is continuous, $f \circ T \in C(X)$, so weak-* convergence gives

$$\int_X f d\mu_{N_k} \rightarrow \int_X f d\mu, \quad \int_X f \circ T d\mu_{N_k} \rightarrow \int_X f \circ T d\mu.$$

Combining these limits with (1) yields

$$\int_X f \circ T d\mu = \int_X f d\mu \quad \text{for every } f \in C(X).$$

By the Riesz representation theorem this is equivalent to $T_*\mu = \mu$. Thus every weak-* subsequential limit of the empirical measures is invariant. $\square$

Theorem F01.6. Uniform ergodic theorem for unique ergodicity

Statement. Let X be a compact metric space and let $T : X \rightarrow X$ be uniquely ergodic, with unique invariant Borel probability μ . Then for every $f \in C(X)$,

$$\sup_{x \in X} \left| \frac{1}{N} \sum_{j=0}^{N-1} f(T^j x) - \int_X f d\mu \right| \rightarrow 0 \quad (N \rightarrow \infty).$$

Proof. Let μ be the unique T -invariant Borel probability measure on X , and fix $f \in C(X)$. Write

$$A_N f(x) := \frac{1}{N} \sum_{j=0}^{N-1} f(T^j x), \quad c := \int_X f d\mu.$$

We prove that $A_N f \rightarrow c$ uniformly on X .

Suppose, to the contrary, that uniform convergence fails. Then there are $\varepsilon > 0$, integers $N_k \rightarrow \infty$, and points $x_k \in X$ such that

$$|A_{N_k} f(x_k) - c| \geq \varepsilon \quad \text{for every } k. \quad (1)$$

Associate to each selected orbit segment the empirical measure

$$\eta_k := \frac{1}{N_k} \sum_{j=0}^{N_k-1} \delta_{T^j x_k}.$$

The space $\mathcal{P}(X)$ of Borel probability measures on compact metric X is weak-* compact and metrizable. Hence, after passing to a subsequence, we may assume

$$\eta_k \xrightarrow{*} \eta$$

for some Borel probability measure η .

Empirical-measure subsequential invariance applies verbatim even though the base point varies with k : for every $g \in C(X)$,

$$\int g \circ T d\eta_k - \int g d\eta_k = \frac{g(T^{N_k} x_k) - g(x_k)}{N_k},$$

whose absolute value is at most $2\|g\|_\infty/N_k$. Passing to the limit shows that η is T -invariant. Unique ergodicity therefore forces

$$\eta = \mu. \quad (2)$$

Finally,

$$A_{N_k} f(x_k) = \int_X f d\eta_k \rightarrow \int_X f d\eta = \int_X f d\mu = c,$$

where the convergence follows from weak-* convergence and continuity of f . This contradicts the fixed discrepancy (1).

Thus for every $\varepsilon > 0$ there is N_0 such that

$$\sup_{x \in X} |A_N f(x) - c| < \varepsilon \quad (N \geq N_0),$$

which is exactly uniform convergence of the Birkhoff averages to the space average. $\square$

Corollary F01.7. Minimal rotations are uniquely ergodic

Statement. Let $\alpha \in \mathbb{R}^d$ satisfy

$$\langle k, \alpha \rangle \notin \mathbb{Z} \quad \text{for every } 0 \neq k \in \mathbb{Z}^d.$$

Then the translation $R_\alpha : x \mapsto x + \alpha$ on $\mathbb{T}^d$ is uniquely ergodic, and its unique invariant Borel probability is Haar measure. In dimension $d = 1$, this is the irrational-rotation case.

Proof. Write

$$\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d, \quad R_\omega(x) = x + \omega \pmod{\mathbb{Z}^d}.$$

For a torus translation, the relevant irrationality hypothesis is

$$\langle k, \omega \rangle \notin \mathbb{Z} \quad \text{for every } k \in \mathbb{Z}^d \setminus \{0\}. \quad (1)$$

Equivalently, no nonconstant character of the torus is fixed by the translation. We prove that this forces every invariant probability to be normalized Haar measure.

Step 1: invariance determines the Fourier coefficients. Let μ be an R_ω -invariant Borel probability measure. For $k \in \mathbb{Z}^d$ define

$$\hat{\mu}(k) := \int_{\mathbb{T}^d} e^{-2\pi i \langle k, x \rangle} d\mu(x).$$

Because μ is invariant, integration against μ is unchanged when we compose the integrand with R_ω . Hence

$$\begin{aligned} \hat{\mu}(k) &= \int_{\mathbb{T}^d} e^{-2\pi i \langle k, R_\omega x \rangle} d\mu(x) \\ &= e^{-2\pi i \langle k, \omega \rangle} \hat{\mu}(k). \end{aligned}$$

Thus

$$(1 - e^{-2\pi i \langle k, \omega \rangle}) \hat{\mu}(k) = 0.$$

For $k \neq 0$, condition (1) makes the factor in parentheses nonzero, so

$$\hat{\mu}(k) = 0 \quad (k \neq 0),$$

whereas $\hat{\mu}(0) = \mu(\mathbb{T}^d) = 1$.

Step 2: compare with an invariant measure that we already know. Let m be normalized Haar measure on $\mathbb{T}^d$. By the defining translation-invariance of Haar measure, m

is R_ω -invariant for every ω . Its Fourier coefficients are

$$\widehat{m}(0) = 1, \quad \widehat{m}(k) = 0 \quad (k \neq 0).$$

Indeed, if $k \neq 0$, choose $y \in \mathbb{T}^d$ with $e^{-2\pi i \langle k, y \rangle} \neq 1$. Translation invariance gives

$$\widehat{m}(k) = \int e^{-2\pi i \langle k, x+y \rangle} dm(x) = e^{-2\pi i \langle k, y \rangle} \widehat{m}(k),$$

which forces $\widehat{m}(k) = 0$.

Therefore μ and m have the same integral against every character, and hence against every trigonometric polynomial. Trigonometric polynomials are uniformly dense in $C(\mathbb{T}^d)$ (for example by the Stone-Weierstrass theorem). If $f \in C(\mathbb{T}^d)$ and $p_j \rightarrow f$ uniformly, then

$$\left| \int (f - p_j) d\mu \right| \leq \|f - p_j\|_\infty, \quad \left| \int (f - p_j) dm \right| \leq \|f - p_j\|_\infty.$$

Passing to the limit shows

$$\int f d\mu = \int f dm \quad \text{for every } f \in C(\mathbb{T}^d).$$

Uniqueness in the Riesz representation theorem therefore gives $\mu = m$. Since m is invariant, there exists exactly one invariant Borel probability measure: R_ω is uniquely ergodic.

For the circle, $d = 1$ and $\omega = \alpha$. If $\alpha \notin \mathbb{Q}$, then $k\alpha \notin \mathbb{Z}$ for every nonzero $k \in \mathbb{Z}$, so the same argument proves unique ergodicity of the irrational rotation R_α . $\square$

Proposition F01.8. Support structure of invariant measures

Statement. Let $T : X \rightarrow X$ be continuous on a compact metric space and let μ be a T -invariant Borel probability. Then $\text{supp } \mu$ is closed and forward invariant: $T(\text{supp } \mu) \subseteq \text{supp } \mu$. If T is a homeomorphism, then $T(\text{supp } \mu) = \text{supp } \mu$.

Scope note

Recurrence is a separate assertion proved in the Poincaré recurrence theorem in F02.

Proof. Recall the neighborhood characterization of the support:

$$\text{supp } \mu = \{x \in X : \mu(U) > 0 \text{ for every open neighborhood } U \ni x\}. \quad (1)$$

We prove the clauses of the statement separately.

Closedness. If $x \notin \text{supp } \mu$, then by (1) there is an open neighborhood $U_x \ni x$ with $\mu(U_x) = 0$. Every point of U_x also has the zero-measure neighborhood U_x , so $U_x \subseteq X \setminus \text{supp } \mu$. Hence the complement of the support is open, and $\text{supp } \mu$ is closed.

Forward invariance. Let $x \in \text{supp } \mu$, and let U be an arbitrary open neighborhood of

Tx . Since T is continuous, $T^{-1}U$ is an open neighborhood of x . By (1),

$$\mu(T^{-1}U) > 0.$$

The T -invariance of μ says $\mu(T^{-1}U) = \mu(U)$, so $\mu(U) > 0$. This holds for every neighborhood U of Tx ; therefore $Tx \in \text{supp } \mu$. We have proved

$$T(\text{supp } \mu) \subseteq \text{supp } \mu. \quad (2)$$

The homeomorphism case. Suppose now that T is a homeomorphism. We first check rather than assume that T^{-1} preserves μ . If A is Borel, then $T(A)$ is Borel and invariance under T gives

$$\mu(T(A)) = \mu(T^{-1}(T(A))) = \mu(A).$$

Thus

$$\mu((T^{-1})^{-1}A) = \mu(T(A)) = \mu(A),$$

so μ is invariant under T^{-1} . Applying the already-proved forward-invariance argument to T^{-1} yields

$$T^{-1}(\text{supp } \mu) \subseteq \text{supp } \mu. \quad (3)$$

Applying T to (3) gives $\text{supp } \mu \subseteq T(\text{supp } \mu)$; combining this with (2) proves

$$T(\text{supp } \mu) = \text{supp } \mu, \quad T^{-1}(\text{supp } \mu) = \text{supp } \mu.$$

This proves exactly the support assertions. Recurrence is not being used in this argument: it is proved separately in *Poincaré recurrence theorem*. $\square$

1.3 Worked examples

Worked example: Topological Dynamics and Invariant Measures: a model computation

Let $\alpha = \sqrt{2} - 1$ and let $R_\alpha(x) = x + \alpha \pmod{1}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. We verify directly why Lebesgue measure is the statistical law seen by every orbit.

For the Fourier character $e_k(x) = e^{2\pi i k x}$, $k \in \mathbb{Z}$,

$$\frac{1}{N} \sum_{j=0}^{N-1} e_k(R_\alpha^j x) = e^{2\pi i k x} \frac{1 - e^{2\pi i k N \alpha}}{N(1 - e^{2\pi i k \alpha})} \quad (k \neq 0).$$

Because α is irrational, $e^{2\pi i k \alpha} \neq 1$ for $k \neq 0$; hence the numerator is bounded by 2 while the denominator grows like N . Therefore this average tends to 0. For $k = 0$ the average is 1. Thus the orbit average agrees with the Lebesgue integral for every trigonometric polynomial. Uniform approximation of continuous functions

by trigonometric polynomials then gives

$$\frac{1}{N} \sum_{j=0}^{N-1} f(x + j\alpha) \longrightarrow \int_{\mathbb{T}} f(t) dt$$

uniformly in x .

There are two lessons. First, the calculation identifies the invariant measure rather than merely asserting its existence. Second, irrationality enters exactly where the denominator $1 - e^{2\pi i k \alpha}$ must not vanish. This same denominator will reappear in F08 as the simplest small divisor.

Boundary case: what fails at the edge of the hypotheses

Take instead $\alpha = p/q$ in lowest terms. Every orbit of R_α is a periodic cycle of length q . For each starting point x ,

$$\mu_x = \frac{1}{q} \sum_{j=0}^{q-1} \delta_{x+jp/q}$$

is invariant. If $x - y \notin q^{-1}\mathbb{Z}/\mathbb{Z}$, then $\mu_x \neq \mu_y$. Thus invariant measures still exist, but uniqueness fails. The Fourier denominator also vanishes for $k = q$, exactly matching the periodic obstruction.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

1.4 Chapter summary

What to remember

- Compactness converts empirical orbit measures into invariant measures.
- Ergodic measures are the indecomposable extreme points of the invariant-measure simplex.
- Unique ergodicity forces uniform convergence of continuous time averages.
- Existence, ergodicity, uniqueness, and recurrence are different properties and should never be conflated.

Looking ahead

F02 fixes one invariant probability measure and asks what happens for almost every point with respect to that measure. Birkhoff's theorem will turn the statis-

tical intuition of this chapter into an almost-everywhere convergence theorem.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

1.5 Exercises and problems

Exercise 1.1

Difficulty. Core.

Related results. Krylov-Bogolyubov existence theorem; Invariant measures form a compact convex set; Ergodic measures are extreme points.

Problem. Work through the worked example 'irrational rotation R_α on the circle with $\alpha = \sqrt{2} - 1$ '. For Krylov-Bogolyubov existence theorem, Invariant measures form a compact convex set, and Ergodic measures are extreme points, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 1.2

Difficulty. Core.

Related results. Ergodic decomposition; Empirical-measure subsequential invariance; Uniform ergodic theorem for unique ergodicity.

Problem. Analyze the boundary example 'a rational rotation and a non-uniquely-ergodic compact system'. For Ergodic decomposition, Empirical-measure subsequential invariance, and Uniform ergodic theorem for unique ergodicity, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 1.3

Difficulty. Advanced.

Related results. Ergodic measures are extreme points.

Problem. For Ergodic measures are extreme points, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 1.4

Difficulty. Advanced.

Related results. Ergodic decomposition.

Problem. Stress-test Ergodic decomposition. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 1.5

Difficulty. Advanced.

Related results. Krylov–Bogolyubov existence theorem; Invariant measures form a compact convex set.

Problem. Synthesis problem. Use Krylov–Bogolyubov existence theorem and Invariant measures form a compact convex set to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 1.6

Difficulty. Core.

Related results. Minimal rotations are uniquely ergodic.

Problem. Work directly with Minimal rotations are uniquely ergodic. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 1.7

Difficulty. Core.

Related results. Support structure of invariant measures.

Problem. Work directly with Support structure of invariant measures. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Measure-Preserving Ergodic Theory

Why this chapter matters

Once an invariant measure has been chosen, the central question changes. We no longer ask whether a statistical law exists; we ask whether a typical orbit reveals it. Poincaré recurrence says that positive-measure sets are revisited. Birkhoff says much more: time averages converge almost everywhere, and in the ergodic case the limit is the space average.

The chapter also separates several notions that are often blurred on a first encounter. Ergodicity is not mixing; mixing is not independence; recurrence alone gives no numerical average. The examples show these distinctions concretely.

Prerequisites. F01, measure theory, conditional expectation, and basic Hilbert-space orthogonality.

Learning goals

- Use recurrence and maximal inequalities to understand why pointwise ergodic averages converge.
- Interpret the Birkhoff limit as conditional expectation onto the invariant sigma-algebra.
- Distinguish ergodicity, weak mixing, mixing, and independence.
- Apply Borel-Cantelli and ratio ergodic ideas in simple models.

Conceptual roadmap

The logical spine is

recurrence $\rightarrow$ maximal inequality $\rightarrow$ Birkhoff convergence.

The mean ergodic theorem gives an L^2 companion. Ergodicity criteria identify when the limiting conditional expectation is constant. Mixing results then quantify decay of memory, while Borel-Cantelli controls repeated visits to changing targets.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

2.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F02.1. Measure-preserving transformation

A measurable map $T : (X, \mathcal{A}, \mu) \rightarrow (X, \mathcal{A}, \mu)$ is *measure preserving* if $\mu(T^{-1}A) = \mu(A)$ for every $A \in \mathcal{A}$.

Definition F02.2. Conditional expectation

For $f \in L^1(\mu)$ and a sub- σ -algebra $\mathcal{I} \subset \mathcal{A}$, the conditional expectation $\mathbb{E}(f \mid \mathcal{I})$ is the $\mathcal{I}$ -measurable integrable function characterized by $\int_A \mathbb{E}(f \mid \mathcal{I}) d\mu = \int_A f d\mu$ for every $A \in \mathcal{I}$.

Definition F02.3. Mixing and weak mixing

A measure-preserving map T is *mixing* if $\mu(T^{-n}A \cap B) \rightarrow \mu(A)\mu(B)$ for all measurable A, B . It is *weakly mixing* if the Cesaro averages of the absolute correlation deviations tend to zero.

Definition F02.4. Generic point

For a continuous map T of compact metric X and an invariant probability μ , a point x is *generic for μ* if, for every $f \in C(X)$, $N^{-1} \sum_{j=0}^{N-1} f(T^j x) \rightarrow \int_X f d\mu$.

2.2 Main results

Theorem F02.1. Poincaré recurrence theorem

Statement. Let $(X, \mathcal{A}, \mu)$ be a probability space and let $T : X \rightarrow X$ be measurable and measure preserving. For every $A \in \mathcal{A}$, for μ -almost every $x \in A$ there are infinitely many integers $n \geq 1$ such that $T^n x \in A$.

Proof. Let A be measurable. If $\mu(A) = 0$, the assertion “almost every point of A ” is vacuous, so assume $\mu(A) > 0$.

Step 1: points that never return form a null set. Define

$$B := A \cap \bigcap_{n \geq 1} T^{-n}(X \setminus A).$$

Thus $x \in B$ precisely when $x \in A$ and $T^n x \notin A$ for every $n \geq 1$.

We claim that the sets $T^{-n}B$, $n \geq 0$, are pairwise disjoint. If $0 \leq m < n$ and $x \in$

$T^{-m}B \cap T^{-n}B$, then $T^m x \in B \subseteq A$ and also

$$T^{n-m}(T^m x) = T^n x \in B \subseteq A.$$

But $n - m \geq 1$, contradicting the defining property of B : a point of B has no positive iterate in A . Hence the inverse images are disjoint.

Since T preserves μ ,

$$\mu(T^{-n}B) = \mu(B) \quad (n \geq 0).$$

Therefore, for every $N \geq 1$,

$$1 = \mu(X) \geq \mu\left(\bigcup_{n=0}^{N-1} T^{-n}B\right) = \sum_{n=0}^{N-1} \mu(T^{-n}B) = N\mu(B).$$

Letting $N \rightarrow \infty$ forces $\mu(B) = 0$.

Step 2: finitely many returns reduce to the never-return case. Let

$$F := \{x \in A : x \text{ returns to } A \text{ only finitely many times}\}.$$

For $x \in F$, the set $\{n \geq 0 : T^n x \in A\}$ is nonempty (it contains 0) and finite, so it has a largest element; call it $r(x)$. Then $T^{r(x)}x \in A$, and by maximality of $r(x)$ no positive iterate of $T^{r(x)}x$ lies in A . Thus $T^{r(x)}x \in B$, or equivalently

$$x \in T^{-r(x)}B.$$

Consequently

$$F \subseteq \bigcup_{r=0}^{\infty} T^{-r}B.$$

Every set on the right has measure $\mu(B) = 0$, so countable subadditivity gives $\mu(F) = 0$.

Hence, outside a null subset of A , every point of A returns to A infinitely often. Notice that no inverse map was used; the proof is valid for noninvertible probability-preserving transformations as stated. $\square$

Theorem F02.2. Maximal ergodic theorem

Statement. Let T preserve a probability measure μ and let $f \in L^1(\mu)$. Put $S_n f = \sum_{j=0}^{n-1} f \circ T^j$ and, for $\alpha \in \mathbb{R}$, $E_\alpha = \{x : \sup_{n \geq 1} S_n f(x) > \alpha\}$. Then $\alpha \mu(E_\alpha) \leq \int_{E_\alpha} f d\mu$. Equivalently, for every $g \in L^1$, if $E = \{\sup_{n \geq 1} S_n g > 0\}$, then $\int_E g d\mu \geq 0$.

Proof. We first prove the equivalent positive-part formulation, because it contains the essential argument.

Let $g \in L^1(\mu)$ and define

$$S_0 g := 0, \quad S_n g := \sum_{j=0}^{n-1} g \circ T^j \quad (n \geq 1).$$

For a fixed $N \geq 1$, set

$$F_N := \max_{0 \leq n \leq N} S_n g, \quad E_N := \{x : F_N(x) > 0\}.$$

Notice that $F_N \geq 0$ everywhere and $F_N = 0$ on $X \setminus E_N$.

Step 1: the key pointwise identity on E_N . For $n \geq 1$,

$$S_n g(x) = g(x) + S_{n-1} g(Tx).$$

Therefore

$$\max_{1 \leq n \leq N} S_n g(x) = g(x) + \max_{0 \leq m \leq N-1} S_m g(Tx) = g(x) + F_{N-1}(Tx).$$

Since F_N is the maximum of this quantity and $S_0 g = 0$, whenever $x \in E_N$ the positive maximum cannot be the 0 term. Hence

$$F_N(x) = g(x) + F_{N-1}(Tx), \quad x \in E_N. \quad (1)$$

Equivalently,

$$g = F_N - F_{N-1} \circ T \quad \text{on } E_N. \quad (2)$$

Step 2: integrate the identity. Using $F_N = 0$ off E_N , nonnegativity of F_{N-1} , and preservation of μ ,

$$\begin{aligned} \int_{E_N} g \, d\mu &= \int_{E_N} F_N \, d\mu - \int_{E_N} F_{N-1} \circ T \, d\mu \\ &= \int_X F_N \, d\mu - \int_{E_N} F_{N-1} \circ T \, d\mu \\ &\geq \int_X F_N \, d\mu - \int_X F_{N-1} \circ T \, d\mu \\ &= \int_X F_N \, d\mu - \int_X F_{N-1} \, d\mu \\ &\geq 0, \end{aligned} \quad (3)$$

because $F_{N-1} \leq F_N$ pointwise.

The sets E_N increase with N , and

$$E := \bigcup_{N \geq 1} E_N = \left\{ x : \sup_{n \geq 1} S_n g(x) > 0 \right\}.$$

Since $1_{E_N} g \rightarrow 1_E g$ pointwise and $|1_{E_N} g| \leq |g| \in L^1(\mu)$, dominated convergence applied to (3) gives

$$\int_E g \, d\mu \geq 0. \quad (4)$$

This proves the Hopf maximal inequality.

Step 3: derive the stated parameter form. Fix $f \in L^1(\mu)$ and $\alpha \in \mathbb{R}$, and apply (4) to $g = f - \alpha$. Since

$$S_n(f - \alpha) = S_n f - n\alpha,$$

we have

$$\left\{ \sup_{n \geq 1} S_n(f - \alpha) > 0 \right\} = \left\{ \sup_{n \geq 1} \frac{S_n f}{n} > \alpha \right\} = E_\alpha.$$

Thus

$$0 \leq \int_{E_\alpha} (f - \alpha) d\mu = \int_{E_\alpha} f d\mu - \alpha \mu(E_\alpha),$$

which is exactly

$$\alpha \mu(E_\alpha) \leq \int_{E_\alpha} f d\mu.$$

□

Theorem F02.3. von Neumann mean ergodic theorem

Statement. Let U be a unitary Koopman operator on $L^2(\mu)$ and let P denote the orthogonal projection onto the fixed subspace $\text{Fix}(U) = \ker(I - U)$. Then for every $f \in L^2(\mu)$,

$$\frac{1}{N} \sum_{j=0}^{N-1} U^j f \longrightarrow Pf$$

in $L^2(\mu)$ as $N \rightarrow \infty$.

Proof. Let $H = L^2(\mu)$ and let $U : H \rightarrow H$ be the unitary Koopman operator appearing in the statement. Define the fixed subspace

$$M := \ker(I - U) = \{h \in H : Uh = h\}.$$

We will show that the Cesàro averages converge to the orthogonal projection onto M .

Step 1: identify the orthogonal complement of the fixed space. For any bounded operator B on a Hilbert space,

$$\text{Ran}(B)^\perp = \ker(B^*),$$

because y is orthogonal to every Bx exactly when $\langle x, B^*y \rangle = 0$ for every x . Taking $B = I - U$ gives

$$\text{Ran}(I - U)^\perp = \ker(I - U^*).$$

Since U is unitary, $U^* = U^{-1}$, and

$$U^{-1}h = h \iff Uh = h.$$

Hence $\ker(I - U^*) = M$. Taking orthogonal complements and using $S^{\perp\perp} = \overline{S}$ yields the orthogonal decomposition

$$H = M \oplus \overline{\text{Ran}(I - U)}. \quad (1)$$

Step 2: compute the averages on each summand. For $N \geq 1$ set

$$A_N := \frac{1}{N} \sum_{j=0}^{N-1} U^j.$$

If $h \in M$, then $U^j h = h$ for every j , so

$$A_N h = h. \quad (2)$$

If $h = (I - U)g \in \text{Ran}(I - U)$, then the sum telescopes:

$$\begin{aligned} A_N h &= \frac{1}{N} \sum_{j=0}^{N-1} U^j (I - U)g \\ &= \frac{1}{N} \sum_{j=0}^{N-1} (U^j g - U^{j+1} g) \\ &= \frac{g - U^N g}{N}. \end{aligned}$$

Unitary operators preserve norm, so

$$\|A_N h\|_2 \leq \frac{\|g\|_2 + \|U^N g\|_2}{N} = \frac{2\|g\|_2}{N} \rightarrow 0. \quad (3)$$

Step 3: pass from coboundaries to their closure. Because $\|U^j\| = 1$,

$$\|A_N\| \leq \frac{1}{N} \sum_{j=0}^{N-1} \|U^j\| = 1. \quad (4)$$

Let $h \in \overline{\text{Ran}(I - U)}$ and choose $h_m \in \text{Ran}(I - U)$ with $h_m \rightarrow h$. Then

$$\|A_N h\|_2 \leq \|A_N(h - h_m)\|_2 + \|A_N h_m\|_2 \leq \|h - h_m\|_2 + \|A_N h_m\|_2.$$

For fixed m , the second term tends to zero by (3); then letting $m \rightarrow \infty$ proves $A_N h \rightarrow 0$.

Finally write $h = h_0 + h_1$ according to (1), with $h_0 \in M$ and $h_1 \in M^\perp$. Equations (2)-(4) give

$$A_N h \rightarrow h_0 = P_M h \quad \text{in } L^2(\mu),$$

where P_M is the orthogonal projection onto the invariant subspace. This is exactly the stated mean ergodic theorem. $\square$

Theorem F02.4. Birkhoff pointwise ergodic theorem

Statement. Let $(X, \mathcal{A}, \mu, T)$ be a probability-preserving system and let

$$\mathcal{I} := \{A \in \mathcal{A} : \mu(T^{-1}A \triangle A) = 0\}$$

be the invariant σ -algebra. For every $f \in L^1(\mu)$,

$$\frac{1}{N} \sum_{j=0}^{N-1} f \circ T^j \rightarrow \mathbb{E}(f \mid \mathcal{I})$$

μ -almost everywhere as $N \rightarrow \infty$. If T is ergodic, the limit is the constant $\int_X f d\mu$.

Proof. For $f \in L^1(\mu)$ write

$$A_N f := \frac{1}{N} \sum_{j=0}^{N-1} f \circ T^j.$$

Let

$$\mathcal{I} := \{A \in \mathcal{A} : T^{-1}A = A \text{ modulo } \mu\}$$

be the invariant σ -algebra. We prove in four stages that $A_N f$ converges almost everywhere and that its limit is $\mathbb{E}(f \mid \mathcal{I})$.

Step 1: a dense class with explicit convergence. Let U denote the Koopman isometry $Ug = g \circ T$ on $L^2(\mu)$. von Neumann mean ergodic theorem gives the Hilbert-space decomposition

$$L^2(\mu) = \text{Fix}(U) \oplus \overline{\text{Ran}(I - U)}. \quad (1)$$

Define

$$\mathcal{D} := \{h + (v - v \circ T) : h \in \text{Fix}(U) \cap L^\infty, v \in L^\infty\}.$$

We claim that $\mathcal{D}$ is dense in $L^1(\mu)$. Indeed, (1) first shows that invariant L^2 functions plus L^2 coboundaries are dense in L^2 . Truncating an invariant L^2 function preserves invariance, while truncating $v \in L^2$ approximates v in L^2 and therefore approximates $v - v \circ T$ because U is an isometry. Hence $\mathcal{D}$ is dense in L^2 . Since $\mu(X) = 1$, L^2 convergence implies L^1 convergence, and every L^1 function can first be truncated to a bounded function, so $\mathcal{D}$ is dense in L^1 as claimed.

If $d = h + (v - v \circ T) \in \mathcal{D}$, then the coboundary telescopes:

$$A_N d = h + \frac{1}{N} \sum_{j=0}^{N-1} (v \circ T^j - v \circ T^{j+1}) = h + \frac{v - v \circ T^N}{N}.$$

Because v is bounded,

$$\left\| \frac{v - v \circ T^N}{N} \right\|_\infty \leq \frac{2\|v\|_\infty}{N} \rightarrow 0. \quad (2)$$

Thus $A_N d \rightarrow h$ uniformly, hence pointwise, for every $d \in \mathcal{D}$.

Step 2: maximal control of perturbations. For any $r \in L^1$ and $\lambda > 0$, Maximal ergodic theorem applied to r and $-r$ gives

$$\mu \left\{ \sup_{N \geq 1} |A_N r| > \lambda \right\} \leq \frac{2\|r\|_1}{\lambda}. \quad (3)$$

Indeed, on $E_+ = \{\sup_N A_N r > \lambda\}$ the maximal inequality gives $\lambda \mu(E_+) \leq \int_{E_+} r \leq \|r\|_1$, and the same estimate for $-r$ controls the negative deviations.

For a real-valued f , define its pointwise oscillation

$$\Omega_f := \limsup_{N \rightarrow \infty} A_N f - \liminf_{N \rightarrow \infty} A_N f.$$

If $d \in \mathcal{D}$, then $A_N d$ converges, and therefore

$$\Omega_f \leq 2 \sup_{N \geq 1} |A_N(f - d)|.$$

Consequently, by (3),

$$\mu\{\Omega_f > 2\lambda\} \leq \frac{2\|f - d\|_1}{\lambda}. \quad (4)$$

Since $\mathcal{D}$ is L^1 -dense, the right-hand side can be made arbitrarily small. Hence $\mu\{\Omega_f > 2\lambda\} = 0$ for every $\lambda > 0$, and thus $\Omega_f = 0$ almost everywhere.

We must also rule out a common infinite value of the limsup and liminf. Applying (3) directly to $r = f$ gives

$$\mu\left\{\sup_N |A_N f| > M\right\} \leq \frac{2\|f\|_1}{M}.$$

Letting $M \rightarrow \infty$ shows that $\sup_N |A_N f| < \infty$ almost everywhere. Therefore $A_N f$ converges almost everywhere to a finite measurable function; denote it by f^* .

For complex-valued f , apply the argument to the real and imaginary parts separately.

Step 3: identify the limit on the dense class. The fixed space of U is precisely $L^2(\mathcal{I})$: an invariant function has invariant level sets, and conversely $\mathcal{I}$ -simple functions are invariant and are dense in $L^2(\mathcal{I})$. Hence for $h \in \text{Fix}(U)$,

$$\mathbb{E}(h \mid \mathcal{I}) = h.$$

If $v \in L^\infty$ and $A \in \mathcal{I}$, then $1_A \circ T = 1_A$ almost everywhere, so

$$\int_A v \circ T \, d\mu = \int_X (1_A v) \circ T \, d\mu = \int_A v \, d\mu.$$

Thus $\mathbb{E}(v - v \circ T \mid \mathcal{I}) = 0$. Therefore for every $d = h + (v - v \circ T) \in \mathcal{D}$,

$$\lim_{N \rightarrow \infty} A_N d = h = \mathbb{E}(d \mid \mathcal{I}). \quad (5)$$

Step 4: identify the limit for arbitrary $f \in L^1$. Fix $\lambda > 0$ and choose $d \in \mathcal{D}$. Using the pointwise convergence of both averages and (5),

$$|f^* - \mathbb{E}(f \mid \mathcal{I})| \leq \sup_N |A_N(f - d)| + |\mathbb{E}(f - d \mid \mathcal{I})| \quad \text{a.e.}$$

Hence

$$\begin{aligned} \mu\{|f^* - \mathbb{E}(f \mid \mathcal{I})| > 2\lambda\} &\leq \mu\left\{\sup_N |A_N(f - d)| > \lambda\right\} \\ &\quad + \mu\{|\mathbb{E}(f - d \mid \mathcal{I})| > \lambda\} \\ &\leq \frac{2\|f - d\|_1}{\lambda} + \frac{\|\mathbb{E}(f - d \mid \mathcal{I})\|_1}{\lambda} \\ &\leq \frac{3\|f - d\|_1}{\lambda}, \end{aligned} \quad (6)$$

where the last step uses the L^1 contraction property of conditional expectation. Let d approach f in L^1 . The right-hand side of (6) tends to zero, so

$$f^* = \mathbb{E}(f \mid \mathcal{I}) \quad \mu\text{-a.e.}$$

This proves the pointwise ergodic theorem.

If T is ergodic, every $\mathcal{I}$ -measurable integrable function is constant almost everywhere. The defining integral identity for conditional expectation then gives

$$\mathbb{E}(f \mid \mathcal{I}) = \int_X f d\mu,$$

so in the ergodic case the Birkhoff averages converge almost everywhere to the space average. $\square$

Theorem F02.5. Generic points for ergodic measures

Statement. Let X be compact metrizable, $T : X \rightarrow X$ continuous, and μ an ergodic T -invariant Borel probability. Then for μ -almost every x and every $f \in C(X)$,

$$\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \longrightarrow \int_X f d\mu.$$

Equivalently, $N^{-1} \sum_{n=0}^{N-1} \delta_{T^n x}$ converges weak-* to μ for μ -almost every x .

Proof. Because X is compact metrizable, the Banach space $C(X)$ with the supremum norm is separable. Choose a countable dense set

$$\{f_m : m \geq 1\} \subset C(X).$$

For each fixed m , *Birkhoff pointwise ergodic theorem* applied to the ergodic invariant probability μ gives a full-measure set Y_m such that

$$\frac{1}{N} \sum_{n=0}^{N-1} f_m(T^n x) \longrightarrow \int_X f_m d\mu \quad (x \in Y_m). \quad (1)$$

Let

$$Y := \bigcap_{m=1}^{\infty} Y_m.$$

Since this is a countable intersection of full-measure sets, $\mu(Y) = 1$. We now show that every $x \in Y$ is generic for μ .

Fix $x \in Y$, $f \in C(X)$, and $\varepsilon > 0$. Choose m so that

$$\|f - f_m\|_{\infty} < \varepsilon.$$

For every N ,

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} (f - f_m)(T^n x) \right| \leq \varepsilon,$$

and also

$$\left| \int_X (f - f_m) d\mu \right| \leq \varepsilon.$$

Therefore

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) - \int_X f d\mu \right| \leq 2\varepsilon + \left| \frac{1}{N} \sum_{n=0}^{N-1} f_m(T^n x) - \int_X f_m d\mu \right|.$$

By (1), the last term tends to 0. Hence the limsup of the left-hand side is at most 2ε . Since $\varepsilon > 0$ was arbitrary,

$$\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \longrightarrow \int_X f d\mu$$

for every $f \in C(X)$. Thus every $x \in Y$ is generic and generic points have full μ -measure.

Finally, define the empirical measures

$$\eta_N^x := \frac{1}{N} \sum_{n=0}^{N-1} \delta_{T^n x}.$$

For $f \in C(X)$,

$$\int_X f d\eta_N^x = \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x).$$

Hence the just-proved convergence for every continuous f is precisely

$$\eta_N^x \xrightarrow{*} \mu.$$

This proves the equivalent empirical-measure formulation as well. □

Proposition F02.6. Ergodicity criteria

Statement. Let T preserve a probability measure μ . The following are equivalent: (i) every T -invariant measurable set has measure 0 or 1; (ii) every T -invariant function in $L^2(\mu)$ is constant almost everywhere; (iii) for every $f, g \in L^2(\mu)$,

$$\frac{1}{N} \sum_{n=0}^{N-1} \int f(g \circ T^n) d\mu \longrightarrow \left(\int f d\mu \right) \left(\int g d\mu \right).$$

Proof. Let $U : L^2(\mu) \rightarrow L^2(\mu)$ be the Koopman isometry $Ug = g \circ T$. We prove

$$(i) \iff (ii) \iff (iii).$$

$(i) \Rightarrow (ii)$. Assume T is ergodic, and let $g \in L^2(\mu)$ satisfy $g \circ T = g$ almost everywhere. It is enough to treat real-valued g , since the real and imaginary parts of an invariant complex function are invariant.

For every $c \in \mathbb{R}$, the superlevel set

$$A_c := \{x : g(x) > c\}$$

is invariant modulo μ , because $g \circ T = g$ almost everywhere. Ergodicity gives $\mu(A_c) \in \{0, 1\}$ for every c . If g were not constant almost everywhere, there would be $a < b$ such that both $\{g \leq a\}$ and $\{g \geq b\}$ had positive measure. For any $c \in (a, b)$ this would imply $0 < \mu(A_c) < 1$, a contradiction. Hence g is constant almost everywhere.

(ii) $\Rightarrow$ (i). If A is invariant modulo μ , then $1_A \circ T = 1_A$ almost everywhere. By (ii), 1_A is constant almost everywhere. Because an indicator only takes the values 0 and 1, that constant is 0 or 1, so $\mu(A) \in \{0, 1\}$. Thus T is ergodic.

(ii) $\Rightarrow$ (iii). For $g \in L^2$, define

$$A_N g := \frac{1}{N} \sum_{n=0}^{N-1} U^n g.$$

von Neumann mean ergodic theorem says that $A_N g$ converges in L^2 to the orthogonal projection Pg of g onto $\text{Fix}(U)$. Under (ii), this fixed space is exactly the one-dimensional space of constants. The orthogonal projection onto constants is

$$Pg = \left(\int_X g \, d\mu \right) 1,$$

because $\mu(X) = 1$. Therefore, for $f, g \in L^2$,

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} \int_X f(g \circ T^n) \, d\mu &= \int_X f A_N g \, d\mu \\ &\longrightarrow \left(\int_X f \, d\mu \right) \left(\int_X g \, d\mu \right), \end{aligned}$$

where the last passage follows from Cauchy-Schwarz and L^2 convergence.

(iii) $\Rightarrow$ (ii). Let $g \in L^2$ be invariant. Then $g \circ T^n = g$ almost everywhere for every n , so taking $f = \bar{g}$ in (iii) gives

$$\int_X |g|^2 \, d\mu = \left| \int_X g \, d\mu \right|^2.$$

But

$$\int_X \left| g - \int_X g \, d\mu \right|^2 \, d\mu = \int_X |g|^2 \, d\mu - \left| \int_X g \, d\mu \right|^2 = 0.$$

Thus g is constant almost everywhere, proving (ii).

All three conditions are therefore equivalent. □

Theorem F02.7. Ergodic theorem for measure-preserving flows

Statement. Let $(X, \mathcal{A}, \mu)$ be a probability space and let $(\phi_t)_{t \in \mathbb{R}}$ be a jointly measurable measure-preserving flow. Write

$$\mathcal{I}_\phi := \{A \in \mathcal{A} : \mu(\phi_t^{-1} A \Delta A) = 0 \text{ for every } t \in \mathbb{R}\}.$$

For every $f \in L^1(\mu)$,

$$A_T f(x) := \frac{1}{T} \int_0^T f(\phi_t x) dt \longrightarrow \mathbb{E}(f \mid \mathcal{I}_\phi)(x)$$

for μ -almost every x as $T \rightarrow \infty$. If the flow is ergodic, then the limit is the constant $\int_X f d\mu$.

Proof. Let

$$T := \phi_1$$

and define the one-unit averaged observable

$$F(x) := \int_0^1 f(\phi_t x) dt. \quad (1)$$

Joint measurability of $(t, x) \mapsto \phi_t x$ and Fubini's theorem show that F is measurable. Since every ϕ_t preserves μ ,

$$\begin{aligned} \|F\|_{L^1} &\leq \int_X \int_0^1 |f(\phi_t x)| dt d\mu(x) \\ &= \int_0^1 \int_X |f(\phi_t x)| d\mu(x) dt = \|f\|_{L^1}. \end{aligned} \quad (2)$$

Thus $F \in L^1(\mu)$.

Step 1: integer-time averages. For every integer $N \geq 1$, the flow property gives

$$\begin{aligned} \int_0^N f(\phi_t x) dt &= \sum_{n=0}^{N-1} \int_n^{n+1} f(\phi_t x) dt \\ &= \sum_{n=0}^{N-1} \int_0^1 f(\phi_s(\phi_n x)) ds = \sum_{n=0}^{N-1} F(T^n x). \end{aligned} \quad (3)$$

By *Birkhoff pointwise ergodic theorem* applied to the measure-preserving transformation $T = \phi_1$,

$$\frac{1}{N} \int_0^N f(\phi_t x) dt = \frac{1}{N} \sum_{n=0}^{N-1} F(T^n x) \longrightarrow L(x) := \mathbb{E}(F \mid \mathcal{I}_1)(x) \quad (4)$$

for almost every x , where

$$\mathcal{I}_1 := \{A \in \mathcal{A} : \mu(T^{-1}A \triangle A) = 0\}.$$

The remaining work is to identify L with the conditional expectation onto the smaller sigma-algebra invariant under the entire flow.

Step 2: the time-one Birkhoff limit is invariant under every flow time. Fix $0 \leq s \leq 1$, and put

$$H_s(x) := \int_0^s f(\phi_t x) dt \in L^1(\mu). \quad (5)$$

A direct change of variables in time gives the coboundary identity

$$\begin{aligned}
 F(\phi_s x) - F(x) &= \int_s^{1+s} f(\phi_t x) dt - \int_0^1 f(\phi_t x) dt \\
 &= \int_1^{1+s} f(\phi_t x) dt - \int_0^s f(\phi_t x) dt \\
 &= H_s(Tx) - H_s(x).
 \end{aligned} \tag{6}$$

Conditional expectation onto $\mathcal{I}_1$ annihilates a T -coboundary: for $A \in \mathcal{I}_1$,

$$\int_A H_s \circ T d\mu = \int_A H_s d\mu,$$

so

$$\mathbb{E}(H_s \circ T - H_s \mid \mathcal{I}_1) = 0. \tag{7}$$

Because ϕ_s commutes with T , it maps $\mathcal{I}_1$ to itself modulo null sets. Conditional expectation therefore commutes with composition by ϕ_s :

$$\mathbb{E}(F \circ \phi_s \mid \mathcal{I}_1) = \mathbb{E}(F \mid \mathcal{I}_1) \circ \phi_s = L \circ \phi_s. \tag{8}$$

For completeness, (8) follows by testing both sides on $A \in \mathcal{I}_1$ and changing variables under the invertible measure-preserving map ϕ_s .

Taking conditional expectation of (6) and using (7)–(8) yields

$$L \circ \phi_s = L \quad \text{a.e.} \tag{9}$$

for every fixed $s \in [0, 1]$. Since L is already T -invariant, the flow law extends (9) to every $s \in \mathbb{R}$. Hence L is measurable with respect to the completed full-flow invariant sigma-algebra $\mathcal{I}_\phi$.

Step 3: identify the conditional expectation. If $A \in \mathcal{I}_\phi$, then in particular $A \in \mathcal{I}_1$. Using (1), Fubini, and invariance of A under every ϕ_t ,

$$\begin{aligned}
 \int_A L d\mu &= \int_A F d\mu \\
 &= \int_0^1 \int_A f(\phi_t x) d\mu(x) dt \\
 &= \int_0^1 \int_A f(x) d\mu(x) dt = \int_A f d\mu.
 \end{aligned} \tag{10}$$

Together with $\mathcal{I}_\phi$ -measurability, (10) is exactly the characterizing property of conditional expectation. Therefore

$$L = \mathbb{E}(f \mid \mathcal{I}_\phi) \quad \text{a.e.} \tag{11}$$

Step 4: pass from integer to arbitrary time. Define the nonnegative one-unit envelope

$$G(x) := \int_0^1 |f(\phi_t x)| dt. \tag{12}$$

As in (2), $G \in L^1$. We need the elementary fact

$$\frac{G(T^N x)}{N} \longrightarrow 0 \quad \text{for a.e. } x. \quad (13)$$

Indeed, for every $\varepsilon > 0$, measure preservation gives

$$\sum_{N=1}^{\infty} \mu\{G(T^N x) > \varepsilon N\} = \sum_{N=1}^{\infty} \mu\{G > \varepsilon N\}.$$

Because the tail function $t \mapsto \mu\{G > t\}$ is decreasing,

$$\varepsilon \sum_{N=1}^{\infty} \mu\{G > \varepsilon N\} \leq \int_0^{\infty} \mu\{G > t\} dt = \int_X G d\mu < \infty. \quad (14)$$

The first Borel-Cantelli argument now proves (13), first for each rational $\varepsilon > 0$ and hence altogether.

Write an arbitrary $T_* > 1$ as

$$T_* = N + r, \quad N = \lfloor T_* \rfloor, \quad 0 \leq r < 1.$$

Then

$$\frac{1}{T_*} \int_0^{T_*} f(\phi_t x) dt = \frac{N}{T_*} \left(\frac{1}{N} \int_0^N f(\phi_t x) dt \right) + \frac{1}{T_*} \int_0^r f(\phi_t(T^N x)) dt. \quad (15)$$

The absolute value of the last term is at most

$$\frac{G(T^N x)}{T_*} \leq \frac{G(T^N x)}{N} \longrightarrow 0 \quad (16)$$

for almost every x , while $N/T_* \rightarrow 1$. Equations (4), (11), (15), and (16) prove

$$\frac{1}{T_*} \int_0^{T_*} f(\phi_t x) dt \longrightarrow \mathbb{E}(f \mid \mathcal{I}_\phi)(x)$$

for almost every x .

If the flow is ergodic, $\mathcal{I}_\phi$ is trivial modulo null sets, so the conditional expectation is the constant $\int_X f d\mu$. $\square$

Proposition F02.8. Mixing hierarchy

Statement. Mixing implies weak mixing, and weak mixing implies ergodicity. Neither converse holds in general: irrational circle rotations are ergodic but not weakly mixing, while there exist weakly mixing transformations that are not strongly mixing.

Proof. We use the convention of *the definition of mixing*. Thus strong mixing means

$$\mu(T^{-n}A \cap B) - \mu(A)\mu(B) \longrightarrow 0 \quad (n \rightarrow \infty) \quad (1)$$

for every measurable A, B , while weak mixing means

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(T^{-n}A \cap B) - \mu(A)\mu(B)| \longrightarrow 0. \quad (2)$$

Step 1: strong mixing implies weak mixing. Fix A, B and put

$$a_n := |\mu(T^{-n}A \cap B) - \mu(A)\mu(B)|.$$

Under strong mixing, $a_n \rightarrow 0$. Every convergent sequence and its Cesaro means have the same limit, so

$$\frac{1}{N} \sum_{n=0}^{N-1} a_n \longrightarrow 0.$$

This is exactly (2).

Step 2: weak mixing implies ergodicity. Let A be invariant modulo null sets. Then $\mu(T^{-n}A \cap A) = \mu(A)$ for every n , and therefore the weak-mixing condition with $B = A$ gives

$$0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\mu(A) - \mu(A)^2| = \mu(A)(1 - \mu(A)).$$

Hence $\mu(A) \in \{0, 1\}$. Thus T is ergodic. This is consistent with, and is a direct set-theoretic proof of, the implication encoded in *Ergodicity criteria*.

Step 3: ergodic does not imply weakly mixing. Let

$$R_\alpha(x) = x + \alpha \pmod{1}, \quad \alpha \notin \mathbb{Q},$$

on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with Haar probability m . *Minimal rotations are uniquely ergodic* gives unique ergodicity, hence ergodicity. Take $f(x) = e^{2\pi i x}$. Then $\int f \, dm = 0$ and

$$f \circ R_\alpha^n = e^{2\pi i n \alpha} f,$$

so

$$\left| \int_{\mathbb{T}} \bar{f} (f \circ R_\alpha^n) \, dm \right| = 1 \quad \text{for every } n. \quad (3)$$

Weak mixing in the set formulation (2) is equivalent, by linearity and L^2 approximation of functions by simple functions, to

$$\frac{1}{N} \sum_{n=0}^{N-1} \left| \int f (g \circ T^n) \, d\mu - \int f \, d\mu \int g \, d\mu \right| \rightarrow 0 \quad (4)$$

for all $f, g \in L^2(\mu)$. Equation (3) contradicts (4), so the irrational rotation is not weakly mixing.

Step 4: a concrete weakly mixing transformation that is not strongly mixing. We give the classical Chacon construction and include the two properties needed here.

Start with one interval (the stage-0 tower) of width $w_0 = 2/3$ and height $h_0 = 1$.

At stage n , cut every level of the tower into three pieces of equal width, producing three subcolumns. Put one spacer level above the middle subcolumn and stack, from bottom to top, the left subcolumn, the middle subcolumn together with its spacer, and the right subcolumn. Then

$$h_{n+1} = 3h_n + 1, \quad w_{n+1} = \frac{w_n}{3}. \quad (5)$$

The transformation maps each level linearly and measure-preservingly to the level immediately above it. These partial maps are compatible from one stage to the next and therefore define, outside a null set, an invertible Lebesgue-measure-preserving transformation T_C on the union of the towers. Since

$$h_n w_n \nearrow w_0 + \sum_{j \geq 1} w_j = \frac{2}{3} \sum_{j \geq 0} 3^{-j} = 1, \quad (6)$$

the union has total measure one. Realizing the construction by nested interval cuts, the tower partitions generate the Borel sigma-algebra.

We first record ergodicity. Let E be T_C -invariant. On every pair of levels in the same stage- n tower, invariance and the level-to-level bijections show that the relative density of E is the same. Call it a_n . Because the stage- n tower has measure tending to one, $a_n \rightarrow \mu(E)$. On the other hand the rank-one tower partitions generate, so there are unions U_n of stage- n levels with $\mu(E \Delta U_n) \rightarrow 0$. Inside the stage- n tower, every included level contributes relative error $1 - a_n$ and every omitted level contributes relative error a_n . Consequently

$$\mu(E \Delta U_n) \geq \min(a_n, 1 - a_n) \mu(\text{stage-}n \text{ tower}) + o(1). \quad (7)$$

Letting $n \rightarrow \infty$ gives $\min(\mu(E), 1 - \mu(E)) = 0$. Thus T_C is ergodic.

We next show that T_C has no nonconstant L^2 eigenfunction. Suppose $f \circ T_C = \lambda f$ with $|\lambda| = 1$ and $f \neq 0$. Ergodicity makes $|f|$ constant almost everywhere, so normalize $|f| = 1$. Let $\mathcal{P}_n$ be the stage- n tower partition and put $f_n = \mathbb{E}(f \mid \mathcal{P}_n)$. Since the tower partitions generate, $f_n \rightarrow f$ in L^2 . Choose a stage- n level L_n on which the conditional constant c_n satisfies $|c_n| \rightarrow 1$ and on which the L^2 error on each of its three equal subpieces tends to zero after normalization by the subpiece measure. Such levels exist because the global L^2 error tends to zero and the towers exhaust the space.

In the next tower the left third of L_n is carried by $T_C^{h_n}$ onto the middle third of the same old level, while the middle third is carried by $T_C^{h_n+1}$ onto its right third; the extra 1 is exactly the spacer. Since f is close to the same number c_n on all three thirds, the eigenfunction identity yields

$$|\lambda^{h_n} - 1| \rightarrow 0, \quad |\lambda^{h_n+1} - 1| \rightarrow 0. \quad (8)$$

Dividing the second relation by the first on the unit circle gives $\lambda = 1$. Hence every eigenfunction has eigenvalue 1, and ergodicity then makes it constant.

For completeness, recall why this implies weak mixing. On $H = L_0^2(\mu)$ let U be the Koopman unitary. By the unitary spectral theorem, an atom of a spectral measure is the spectral projection onto an eigenspace. Thus absence of nonconstant eigenfunc-

tions means every spectral measure on H is nonatomic. Wiener's elementary Fourier lemma then gives, for $f, g \in H$,

$$\frac{1}{N} \sum_{n=0}^{N-1} |\langle U^n f, g \rangle|^2 \rightarrow 0. \quad (9)$$

(For $f = g$ this is Wiener's lemma; the general case follows by polarization.) Cauchy-Schwarz turns (9) into the Cesaro-absolute correlation limit (4). Hence T_C is weakly mixing.

It remains to see that T_C is not strongly mixing. Fix a level A of some stage m with $\mu(A) < 1/3$; for example one may take the base of the stage-1 tower. For every $n \geq m$, A is a union of stage- n levels. When the stage- n tower is cut into three copies, $T_C^{h_n}$ carries the left third of each such level onto its middle third. Therefore

$$\mu(T_C^{-h_n} A \cap A) \geq \frac{1}{3} \mu(A) \quad (n \geq m). \quad (10)$$

Because $0 < \mu(A) < 1/3$,

$$\frac{1}{3} \mu(A) > \mu(A)^2. \quad (11)$$

Thus the correlations along the subsequence h_n cannot converge to $\mu(A)^2$, contradicting strong mixing.

The two implications and both failures of the converses have now been proved. $\square$

Theorem F02.9. Hopf ratio ergodic theorem (selected form)

Statement. Let T preserve an ergodic probability μ . If $f, g \in L^1(\mu)$, $g > 0$ almost everywhere, and $\int g d\mu > 0$, then for μ -almost every x ,

$$\frac{\sum_{j=0}^{n-1} f(T^j x)}{\sum_{j=0}^{n-1} g(T^j x)} \rightarrow \frac{\int f d\mu}{\int g d\mu}.$$

Proof. For $h \in L^1(\mu)$ write

$$S_n h(x) := \sum_{j=0}^{n-1} h(T^j x).$$

By *Birkhoff pointwise ergodic theorem* and ergodicity, there is a full-measure set X_f on which

$$\frac{S_n f(x)}{n} \rightarrow \int_X f d\mu, \quad (1)$$

and a full-measure set X_g on which

$$\frac{S_n g(x)}{n} \rightarrow \int_X g d\mu. \quad (2)$$

We also need to know that the denominator appearing in the ratio is not silently zero. Let $N := \{x : g(x) \leq 0\}$. The hypothesis $g > 0$ almost everywhere says $\mu(N) = 0$. Since T

preserves μ ,

$$\mu\left(\bigcup_{j \geq 0} T^{-j}N\right) = 0. \quad (3)$$

Thus on the full-measure set

$$X_+ := X \setminus \bigcup_{j \geq 0} T^{-j}N$$

we have $g(T^j x) > 0$ for every $j \geq 0$, and hence $S_n g(x) > 0$ for every $n \geq 1$.

Now fix

$$x \in X_f \cap X_g \cap X_+.$$

Set

$$a := \int_X f \, d\mu, \quad b := \int_X g \, d\mu > 0.$$

Equations (1)–(2) say

$$\frac{S_n f(x)}{n} \rightarrow a, \quad \frac{S_n g(x)}{n} \rightarrow b.$$

Since $b > 0$, ordinary division of convergent real sequences yields

$$\frac{S_n f(x)}{S_n g(x)} = \frac{S_n f(x)/n}{S_n g(x)/n} \rightarrow \frac{a}{b} = \frac{\int f \, d\mu}{\int g \, d\mu}.$$

This holds on a full-measure set, proving the selected ratio theorem. $\square$

Theorem F02.10. Borel-Cantelli lemmas

Statement. Let $(E_n)_{n \geq 1}$ be events in a probability space. If $\sum_n \mathbb{P}(E_n) < \infty$, then $\mathbb{P}(\limsup_n E_n) = 0$. If the events are independent and $\sum_n \mathbb{P}(E_n) = \infty$, then $\mathbb{P}(\limsup_n E_n) = 1$.

Proof. Recall that

$$\limsup_{n \rightarrow \infty} E_n = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n.$$

A point belongs to this set exactly when it lies in infinitely many of the events E_n .

First Borel-Cantelli lemma. Assume $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$. For every N , countable sub-additivity gives

$$\mathbb{P}\left(\bigcup_{n \geq N} E_n\right) \leq \sum_{n \geq N} \mathbb{P}(E_n). \quad (1)$$

The tail unions decrease as N increases. Since we are in a probability space, continuity

from above applies, and therefore

$$\begin{aligned}\mathbb{P}(\limsup E_n) &= \lim_{N \rightarrow \infty} \mathbb{P}\left(\bigcup_{n \geq N} E_n\right) \\ &\leq \lim_{N \rightarrow \infty} \sum_{n \geq N} \mathbb{P}(E_n) = 0.\end{aligned}$$

Thus only finitely many E_n occur almost surely.

Second Borel-Cantelli lemma. Now assume that the events E_n are independent and $\sum_n \mathbb{P}(E_n) = \infty$. For fixed N and $M \geq N$, the complements are also independent, so

$$\begin{aligned}\mathbb{P}\left(\bigcap_{n=N}^M E_n^c\right) &= \prod_{n=N}^M (1 - \mathbb{P}(E_n)) \\ &\leq \exp\left(-\sum_{n=N}^M \mathbb{P}(E_n)\right),\end{aligned}\tag{2}$$

using $1 - t \leq e^{-t}$ for $0 \leq t \leq 1$. Because every tail of a divergent series of nonnegative terms also diverges, the right-hand side of (2) tends to 0 as $M \rightarrow \infty$. The finite intersections in (2) decrease with M , hence continuity from above gives

$$\mathbb{P}\left(\bigcap_{n=N}^{\infty} E_n^c\right) = 0 \quad \text{for every } N.\tag{3}$$

Finally,

$$(\limsup E_n)^c = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_n^c,$$

which is the event that from some index onward none of the E_n occurs. By (3) it is a countable union of null sets, so it is null. Therefore

$$\mathbb{P}(\limsup E_n) = 1.$$

The independence assumption entered only in the product identity (2); no divergent-series conclusion under weaker dependence has been used. $\square$

2.3 Worked examples

Worked example: Measure-Preserving Ergodic Theory: a model computation

Let $X = \{0, 1\}^{\mathbb{Z}}$ with the Bernoulli measure $\mu = (\frac{1}{2}, \frac{1}{2})^{\mathbb{Z}}$, and let T be the left shift. Take the cylinder observable

$$f(x) = \mathbf{1}_{\{x_0=1\}}.$$

Then

$$\frac{1}{N} \sum_{j=0}^{N-1} f(T^j x) = \frac{1}{N} \# \{0 \leq j < N : x_j = 1\},$$

the frequency of ones in the first N coordinates. Birkhoff's theorem gives convergence for almost every sequence. Because the Bernoulli shift is ergodic, the limit is

$$\int f d\mu = \frac{1}{2}.$$

Mixing can be seen directly for cylinder sets. Suppose A depends only on coordinates in $[-r, r]$ and B only on coordinates in $[-s, s]$. For $n > r + s$, the coordinate blocks defining $T^{-n}A$ and B are disjoint. Product independence gives

$$\mu(T^{-n}A \cap B) = \mu(A)\mu(B).$$

Thus, for cylinders, the mixing limit is eventually an exact equality. Approximation by cylinders extends mixing to all measurable sets.

Boundary case: what fails at the edge of the hypotheses

Let R_α be an irrational circle rotation and take $f(x) = e^{2\pi i x}$. Lebesgue measure is invariant and the system is ergodic, but

$$\int f(R_\alpha^n x) \overline{f(x)} dx = e^{2\pi i n \alpha}.$$

Its modulus is always 1, so the correlation does not tend to 0. Irrational rotation is therefore ergodic but not mixing. This example is the quickest antidote to the common misconception that ergodicity means rapid loss of memory.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

2.4 Chapter summary

What to remember

- Poincaré recurrence is qualitative; Birkhoff provides quantitative long-time averages.
- Birkhoff limits are conditional expectations onto invariant information.
- Ergodicity makes invariant information trivial, but does not imply mixing.
- Mixing and Borel-Cantelli are stronger tools for repeated-event statistics.

Looking ahead

F03 replaces additive sums by subadditive quantities such as $\log \|A_n(x)\|$. Kingman's theorem is the bridge from ordinary ergodic averages to growth rates of matrix products.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

2.5 Exercises and problems

Exercise 2.1

Difficulty. Core.

Related results. Poincaré recurrence theorem; Maximal ergodic theorem; Birkhoff pointwise ergodic theorem.

Problem. Work through the worked example 'the two-symbol Bernoulli shift with a cylinder observable'. For Poincaré recurrence theorem, Maximal ergodic theorem, and Birkhoff pointwise ergodic theorem, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 2.2

Difficulty. Core.

Related results. von Neumann mean ergodic theorem; Ergodicity criteria; Ergodic theorem for measure-preserving flows.

Problem. Analyze the boundary example 'an irrational rotation: ergodic but not mixing'. For von Neumann mean ergodic theorem, Ergodicity criteria, and Ergodic theorem for measure-preserving flows, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 2.3

Difficulty. Advanced.

Related results. Birkhoff pointwise ergodic theorem.

Problem. For Birkhoff pointwise ergodic theorem, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 2.4

Difficulty. Advanced.

Related results. von Neumann mean ergodic theorem.

Problem. Stress-test von Neumann mean ergodic theorem. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 2.5

Difficulty. Advanced.

Related results. Poincaré recurrence theorem; Maximal ergodic theorem.

Problem. Synthesis problem. Use Poincaré recurrence theorem and Maximal ergodic theorem to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 2.6

Difficulty. Core.

Related results. Mixing hierarchy.

Problem. Work directly with Mixing hierarchy. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 2.7**Difficulty.** Core.**Related results.** Hopf ratio ergodic theorem (selected form).**Problem.** Work directly with Hopf ratio ergodic theorem (selected form). Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.**Exercise 2.8****Difficulty.** Core.**Related results.** Generic points for ergodic measures.**Problem.** Work directly with Generic points for ergodic measures. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.**Exercise 2.9****Difficulty.** Core.**Related results.** Borel-Cantelli lemmas.**Problem.** Work directly with Borel-Cantelli lemmas. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Subadditive Ergodic Theory

Why this chapter matters

Many dynamical quantities are not additive. Matrix norms satisfy $\|AB\| \leq \|A\|\|B\|$, so logarithms of cocycle norms are subadditive rather than additive. Distances, partition functions, first-passage times, and optimization costs behave similarly. Kingman's theorem is the ergodic theorem adapted to this inequality. The conceptual payoff is large: once a dynamical quantity can be organized into a subadditive process, an asymptotic rate often follows almost automatically.

Prerequisites. F02, integrability, matrix norms, and elementary subadditive sequences.

Learning goals

- Recognize deterministic and random subadditive processes.
- Understand how Fekete's lemma becomes Kingman's ergodic theorem.
- Derive matrix-product growth rates from subadditivity.
- See exactly what ergodicity adds: constancy of an already existing invariant limit.

Conceptual roadmap

Fekete's lemma handles numerical sequences. Kingman adds a moving base point and measure. Applying Kingman to $f_n(x) = \log \|A_n(x)\|$ produces the top Lyapunov exponent. F04 will refine one number into an entire invariant spectrum and splitting.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

3.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F03.1. Subadditive process

A sequence of measurable extended-real functions $(f_n)_{n \geq 1}$ is a *subadditive process over T* if $f_{m+n}(x) \leq f_m(x) + f_n(T^m x)$ almost everywhere, together with the integrability assumptions stated in the theorem being used.

Definition F03.2. Kingman limit

The *Kingman limit* is the almost-everywhere T -invariant extended-real function $f_\infty(x) = \lim_{n \rightarrow \infty} f_n(x)/n$ furnished by Kingman's subadditive ergodic theorem.

Definition F03.3. Stationary process

A process is *stationary* if its joint law is invariant under time shift. In deterministic applications in this book, stationarity is encoded by placing the process over a measure-preserving base transformation.

Definition F03.4. Top Lyapunov exponent

For a measurable matrix cocycle A over T , the *top Lyapunov exponent* at x is $\lambda_1(x) = \lim_{n \rightarrow \infty} n^{-1} \log \|A_n(x)\|$ when the limit exists; the value $-\infty$ is allowed in noninvertible settings.

3.2 Main results

Lemma F03.1. Fekete subadditive lemma

Statement. Let $(a_n)_{n \geq 1}$ be a real sequence satisfying $a_{m+n} \leq a_m + a_n$ for all $m, n \geq 1$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n} \in [-\infty, \infty).$$

Proof. Assume first, as in the main statement, that every a_n is real and that

$$a_{m+n} \leq a_m + a_n \quad (m, n \geq 1). \tag{1}$$

Set

$$L := \inf_{k \geq 1} \frac{a_k}{k} \in [-\infty, \infty).$$

The value $L = -\infty$ is allowed even though every a_k is finite. Since $L \leq a_n/n$ for every n ,

$$L \leq \liminf_{n \rightarrow \infty} \frac{a_n}{n}. \quad (2)$$

We now prove the reverse inequality for the limsup.

Fix one block length $k \geq 1$. For each n use the division algorithm

$$n = qk + r, \quad q \geq 0, \quad 0 \leq r < k.$$

Repeated application of (1) gives $a_{qk} \leq qa_k$. If $r > 0$, one more application yields

$$a_n = a_{qk+r} \leq qa_k + a_r.$$

If $r = 0$, simply use $a_n = a_{qk} \leq qa_k$. Because only the finitely many remainders $0, 1, \dots, k-1$ can occur, the constant

$$C_k := \max\{0, a_1, \dots, a_{k-1}\}$$

(with $C_1 = 0$) gives the uniform estimate

$$a_n \leq qa_k + C_k. \quad (3)$$

Divide by n :

$$\frac{a_n}{n} \leq \frac{qk}{n} \frac{a_k}{k} + \frac{C_k}{n}.$$

For fixed k , as $n \rightarrow \infty$ we have $qk/n = 1 - r/n \rightarrow 1$ and $C_k/n \rightarrow 0$. Therefore

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \frac{a_k}{k}. \quad (4)$$

Since (4) holds for every k ,

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \inf_{k \geq 1} \frac{a_k}{k} = L. \quad (5)$$

Combining (2) and (5) shows that the limit exists in the extended real line and equals L :

$$\boxed{\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}.}$$

In particular, if the infimum is $-\infty$, then (5) says the limsup is $-\infty$, so the normalized sequence tends to $-\infty$.

If one uses the optional extended-real convention in which some term a_m itself may equal $-\infty$ (and no $+\infty$ values are present), then (1) implies $a_{m+r} = -\infty$ for every $r \geq 1$. Hence all sufficiently large terms are $-\infty$, and both sides of the boxed identity are $-\infty$. This is the advertised extended-real variant. $\square$

Theorem F03.2. Kingman subadditive ergodic theorem

Statement. Let $(X, \mathcal{A}, \mu, T)$ be a probability-preserving system and let $f_n : X \rightarrow \mathbb{R} \cup \{-\infty\}$ be measurable, with no $+\infty$ values. Assume $f_1^+ \in L^1(\mu)$ and $f_{m+n}(x) \leq f_m(x) + f_n(T^m x)$ a.e. for all $m, n \geq 1$, using the convention $-\infty + a = -\infty$. Then f_n/n

converges a.e. to a T -invariant $f : X \rightarrow \mathbb{R} \cup \{-\infty\}$; $f^+ \in L^1(\mu)$, and the extended integral satisfies $\int f d\mu = \lim_{n \rightarrow \infty} n^{-1} \int f_n d\mu = \inf_{n \geq 1} n^{-1} \int f_n d\mu \in [-\infty, \infty)$.

Proof. We first prove the theorem when every f_n is finite-valued almost everywhere. The possible value $-\infty$ will be restored in Step 5 by a second, outer truncation argument.

Write

$$a_n := \int_X f_n d\mu \in [-\infty, \infty).$$

This extended integral is well defined because the positive part of every f_n is integrable. Indeed, repeated subadditivity gives

$$f_n(x) \leq \sum_{j=0}^{n-1} f_1(T^j x), \quad (1)$$

so

$$f_n^+(x) \leq \sum_{j=0}^{n-1} f_1^+(T^j x), \quad \int f_n^+ d\mu \leq n \int f_1^+ d\mu < \infty. \quad (2)$$

Integrating the subadditivity inequality and using invariance of μ gives

$$a_{m+n} \leq a_m + a_n. \quad (3)$$

Hence *Fekete subadditive lemma*, with the usual extended-real interpretation, yields

$$L := \lim_{n \rightarrow \infty} \frac{1}{n} \int f_n d\mu = \inf_{n \geq 1} \frac{1}{n} \int f_n d\mu \in [-\infty, \infty). \quad (4)$$

We must prove that f_n/n converges almost everywhere and that its limit has integral L .

Define

$$\underline{f}(x) := \liminf_{n \rightarrow \infty} \frac{f_n(x)}{n}, \quad \bar{f}(x) := \limsup_{n \rightarrow \infty} \frac{f_n(x)}{n}. \quad (5)$$

They take values in $[-\infty, \infty]$ a priori, although the argument below will rule out $+\infty$ almost everywhere.

Step 1: the liminf and limsup are T -invariant almost everywhere. Subadditivity with $m = 1$ gives

$$f_n(x) \leq f_1(x) + f_{n-1}(Tx).$$

Dividing by n and taking liminf or limsup gives

$$\underline{f}(x) \leq \underline{f}(Tx), \quad \bar{f}(x) \leq \bar{f}(Tx) \quad (6)$$

whenever $f_1(x)$ is finite; this is true almost everywhere because f_1 is real valued. For each rational c , (6) implies

$$\{\underline{f} \geq c\} \subseteq T^{-1}\{\underline{f} \geq c\}.$$

The two sets have the same measure because T preserves μ , so their set difference is

null. Intersecting over countably many rational c shows

$$\underline{f} \circ T = \underline{f} \quad \text{a.e.} \quad (7)$$

The same argument applies to $\bar{f}$.

Step 2: identify the integral of the liminf. We first assume that there is a constant $C \geq 0$ such that

$$f_n(x) \geq -Cn \quad \text{for a.e. } x \text{ and every } n \geq 1. \quad (8)$$

Then $f_n/n + C \geq 0$, so Fatou's lemma and (4) give

$$\int \underline{f} d\mu \leq \liminf_{n \rightarrow \infty} \frac{1}{n} \int f_n d\mu = L. \quad (9)$$

In particular $\underline{f}$ is integrable because it is bounded below by $-C$ and its positive part has finite integral by (9).

Fix $\varepsilon > 0$. For $k \geq 1$ set

$$E_k := \left\{ x : \text{for some } 1 \leq j \leq k, \frac{f_j(x)}{j} < \underline{f}(x) + \varepsilon \right\}. \quad (10)$$

By the definition of $\liminf$, $E_k \uparrow X$ up to a null set. Define

$$\psi_k(x) := \begin{cases} \underline{f}(x) + \varepsilon, & x \in E_k, \\ f_1(x), & x \notin E_k. \end{cases} \quad (11)$$

Under (8), $f_1 \geq -C$, while $f_1^+ \in L^1$; hence $f_1 \in L^1$ and $\psi_k \in L^1$.

We now prove the stopping-block estimate that is the heart of the argument. Fix a point x for which (7) holds along its entire forward orbit and for which all subadditivity inequalities hold along that orbit. Starting at time 0, whenever the orbit is outside E_k we advance one step and charge the cost f_1 ; whenever it first enters E_k , choose a block length $j \in \{1, \dots, k\}$ satisfying

$$f_j(T^r x) < j(\underline{f}(T^r x) + \varepsilon) = j(\underline{f}(x) + \varepsilon), \quad (12)$$

and advance by that whole block. Continue until the next chosen block would extend past time n . Subadditivity then decomposes $f_n(x)$ into the costs of all completed one-step and selected blocks, plus a final unfinished tail of length at most k .

Every completed one-step segment outside E_k has cost $f_1(T^i x) = \psi_k(T^i x)$. Every selected block beginning at time r has, by (12), cost at most

$$\sum_{i=r}^{r+j-1} (\underline{f}(T^i x) + \varepsilon) \leq \sum_{i=r}^{r+j-1} \psi_k(T^i x). \quad (13)$$

Only the final tail, containing at most k indices, is not covered by this first sum. Bound-

ing its one-step costs by $\psi_k \vee f_1$ gives, for $n \geq k$,

$$f_n(x) \leq \sum_{i=0}^{n-k-1} \psi_k(T^i x) + \sum_{i=n-k}^{n-1} (\psi_k \vee f_1)(T^i x). \quad (14)$$

This is the precise form in which the stopping construction will be used.

Integrating (14), using T -invariance of μ , and dividing by n gives

$$\frac{1}{n} \int f_n d\mu \leq \frac{n-k}{n} \int \psi_k d\mu + \frac{k}{n} \int (\psi_k \vee f_1) d\mu.$$

Letting $n \rightarrow \infty$ and using (4),

$$L \leq \int \psi_k d\mu. \quad (15)$$

Because $E_k \uparrow X$, the functions ψ_k converge almost everywhere to $\underline{f} + \varepsilon$. Moreover, under (8), both $\underline{f}$ and f_1 are integrable and

$$|\psi_k| \leq |\underline{f}| + \varepsilon + |f_1|,$$

so dominated convergence in (15) gives

$$L \leq \int \underline{f} d\mu + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ and combining with (9), we obtain

$$\int \underline{f} d\mu = L \quad \text{under the lower bound (8).} \quad (16)$$

We now remove (8). For $C > 0$ define

$$f_n^{(C)} := \max\{f_n, -Cn\}. \quad (17)$$

The elementary inequality

$$\max\{a+b, c+d\} \leq \max\{a, c\} + \max\{b, d\}$$

shows that $(f_n^{(C)})$ is again subadditive. It satisfies the linear lower bound $f_n^{(C)} \geq -Cn$, and

$$\underline{f}^{(C)} := \liminf_n \frac{f_n^{(C)}}{n} = \max\{\underline{f}, -C\}. \quad (18)$$

Applying (16) to the truncated sequence gives

$$\int (\underline{f} \vee (-C)) d\mu = \inf_{n \geq 1} \frac{1}{n} \int (f_n \vee (-Cn)) d\mu. \quad (19)$$

For fixed n , the integrands on the right decrease pointwise to f_n as $C \rightarrow \infty$ and have the same integrable positive part as f_n ; therefore their extended integrals decrease to

$\int f_n$. Taking the infimum first in n and then in C gives

$$\inf_{C>0} \int (f \vee (-C)) d\mu = \inf_{n \geq 1} \frac{1}{n} \int f_n d\mu = L. \quad (20)$$

The left side is, by the definition of the extended integral from below, $\int \underline{f} d\mu$. Thus

$$\boxed{\int \underline{f} d\mu = L.} \quad (21)$$

In particular $\underline{f}^+ \in L^1$: for example $\underline{f} \vee (-1)$ is integrable by (19), and its positive part is exactly $\underline{f}^+$.

Step 3: an orbit-tail lemma. We shall need the following elementary consequence of Borel-Cantelli. If $g \in L^1(X)$, then

$$\frac{g(T^n x)}{n} \longrightarrow 0 \quad \text{for a.e. } x. \quad (22)$$

Indeed, for $\varepsilon > 0$, invariance gives

$$\sum_{n \geq 1} \mu\{|g \circ T^n| \geq \varepsilon n\} = \sum_{n \geq 1} \mu\{|g| \geq \varepsilon n\}.$$

The tail-sum estimate

$$\sum_{n \geq 1} \mu\{|g| \geq \varepsilon n\} \leq \varepsilon^{-1} \int |g| d\mu + 1 < \infty \quad (23)$$

shows, by *Borel-Cantelli lemmas*, that the event $|g(T^n x)| \geq \varepsilon n$ occurs only finitely often. Taking a countable sequence $\varepsilon \downarrow 0$ proves (22).

Fix $k \geq 1$. We next show

$$\limsup_{n \rightarrow \infty} \frac{f_{kn}(x)}{n} = k \bar{f}(x) \quad \text{for a.e. } x. \quad (24)$$

One inequality is immediate because the subsequence kn cannot have a larger normalized limsup than the full sequence:

$$\limsup_n \frac{f_{kn}}{kn} \leq \bar{f}. \quad (25)$$

For the converse, write an arbitrary large integer as $N = km + r$ with $0 \leq r < k$. Repeated subadditivity gives

$$f_N \leq f_{km} + g \circ T^{km}, \quad (26)$$

where we may take

$$g := \max\{0, f_1^+, \dots, f_{k-1}^+\} \in L^1;$$

when $r = 0$ the remainder term is simply omitted. By (22), $g(T^{km}x)/N \rightarrow 0$, while

$km/N \rightarrow 1$. Taking limsup in (26) yields

$$\bar{f} \leq \limsup_m \frac{f_{km}}{km}. \quad (27)$$

Together with (25), this proves (24).

Step 4: the opposite integral inequality for the limsup. Again begin under the lower bound (8). Fix $k \geq 1$ and define, for the measure-preserving map T^k , the additive sequence

$$F_n := - \sum_{j=0}^{n-1} f_k \circ T^{jk}. \quad (28)$$

Since $f_k \geq -Ck$, we have $F_1^+ = (-f_k)^+ \leq Ck$, so Step 2 applies to this additive sequence. If

$$\underline{F} := \liminf_n \frac{F_n}{n},$$

then (21), applied with T^k , gives

$$\int \underline{F} d\mu = \inf_n \frac{1}{n} \int F_n d\mu = - \int f_k d\mu. \quad (29)$$

Subadditivity implies

$$f_{kn} \leq \sum_{j=0}^{n-1} f_k \circ T^{jk} = -F_n,$$

so by (24)

$$-\underline{F} = \limsup_n \frac{1}{n} \sum_{j=0}^{n-1} f_k \circ T^{jk} \geq \limsup_n \frac{f_{kn}}{n} = k\bar{f}. \quad (30)$$

Integrating and using (29),

$$\int \bar{f} d\mu \leq \frac{1}{k} \int f_k d\mu. \quad (31)$$

Since this holds for every k ,

$$\int \bar{f} d\mu \leq L. \quad (32)$$

But always $\underline{f} \leq \bar{f}$, while (16) says $\int \underline{f} = L$ under (8). Hence

$$\underline{f} = \bar{f} \quad \text{a.e.} \quad (33)$$

for every subadditive sequence satisfying a linear lower bound.

Finally apply (33) to each truncated sequence (17). By (18) and its limsup analogue,

$$\underline{f} \vee (-C) = \bar{f} \vee (-C) \quad \text{a.e.} \quad (34)$$

for every positive integer C . Outside the union of the corresponding null sets, let $C \rightarrow \infty$ in (34). We obtain

$$\underline{f} = \bar{f} =: f \quad \text{a.e.} \quad (35)$$

Thus $f_n/n \rightarrow f$ almost everywhere. Step 1 gives $f \circ T = f$ almost everywhere, Step 2 gives $f^+ \in L^1$, and (21) together with (4) gives

$$\boxed{\int f \, d\mu = \lim_{n \rightarrow \infty} \frac{1}{n} \int f_n \, d\mu = \inf_{n \geq 1} \frac{1}{n} \int f_n \, d\mu.} \quad (36)$$

All equalities are understood in $[-\infty, \infty)$, so the case $L = -\infty$ is included for the finite-valued process.

Step 5: allow the original process to take the value $-\infty$. Now let the f_n have values in $\mathbb{R} \cup \{-\infty\}$ as in the statement. For every integer $C \geq 1$ define

$$f_n^{[C]} := f_n \vee (-Cn). \quad (37)$$

Each $f_n^{[C]}$ is finite valued, $(f_1^{[C]})^+ \leq f_1^+$, and the same max inequality used in (17) shows that $(f_n^{[C]})$ is subadditive. Therefore the finite-valued case just proved gives an invariant limit g_C satisfying

$$\frac{f_n^{[C]}}{n} \rightarrow g_C \quad \text{a.e.} \quad (38)$$

and

$$\int g_C \, d\mu = \inf_{n \geq 1} \frac{1}{n} \int f_n^{[C]} \, d\mu. \quad (39)$$

Pointwise,

$$\frac{f_n^{[C]}}{n} = \max \left\{ \frac{f_n}{n}, -C \right\}.$$

Hence

$$g_C = \max \left\{ \liminf_n \frac{f_n}{n}, -C \right\} = \max \left\{ \limsup_n \frac{f_n}{n}, -C \right\} \quad (40)$$

outside one null set (after intersecting over countably many C). Since (40) holds for every integer $C \geq 1$, letting $C \rightarrow \infty$ forces the liminf and limsup of f_n/n to coincide. Denote their common value by $f \in \mathbb{R} \cup \{-\infty\}$. Then

$$g_C = f \vee (-C), \quad (41)$$

so $g_C \downarrow f$ and, because each g_C is invariant, f is invariant as well. Also $g_C^+ = f^+$ for every C , and the finite-valued theorem implies $g_C^+ \in L^1$; thus $f^+ \in L^1$.

Finally take the infimum over C in (39). The left side decreases to the extended integral $\int f \, d\mu$. On the right, infima commute and for fixed n the functions $f_n^{[C]} \downarrow f_n$ with unchanged positive part, so

$$\begin{aligned} \int f \, d\mu &= \inf_C \inf_n \frac{1}{n} \int f_n^{[C]} \, d\mu \\ &= \inf_n \inf_C \frac{1}{n} \int f_n^{[C]} \, d\mu \\ &= \inf_n \frac{1}{n} \int f_n \, d\mu. \end{aligned} \quad (42)$$

The scalar sequence of extended integrals is subadditive exactly as in (3), so *Fekete*

subadditive lemma gives the corresponding limit in n . This proves (36) at the full extended-real scope stated in the theorem. $\square$

Corollary F03.3. Ergodic Kingman specialization

Statement. Under the hypotheses of Kingman's theorem, if T is ergodic then the invariant limit is constant almost everywhere and equals

$$\inf_{n \geq 1} \frac{1}{n} \int f_n d\mu,$$

with the value $-\infty$ allowed.

Proof. Let

$$g(x) := \lim_{n \rightarrow \infty} \frac{f_n(x)}{n}$$

be the Kingman limit supplied by *Kingman subadditive ergodic theorem*. That theorem gives

$$g \circ T = g \quad \text{a.e.} \quad (1)$$

and $g : X \rightarrow \mathbb{R} \cup \{-\infty\}$ is measurable.

We first spell out why ergodicity forces such an invariant extended-real function to be constant. For every rational q , the set

$$E_q := \{x : g(x) < q\}$$

is invariant modulo null sets by (1). Since T is ergodic,

$$\mu(E_q) \in \{0, 1\}. \quad (2)$$

The family $(E_q)_{q \in \mathbb{Q}}$ is increasing in q .

If $\mu(E_q) = 1$ for every rational q , then

$$\mu\left(\bigcap_{m \geq 1} E_{-m}\right) = 1,$$

so $g = -\infty$ almost everywhere.

Otherwise define

$$c := \inf\{q \in \mathbb{Q} : \mu(E_q) = 1\} \in \mathbb{R}. \quad (3)$$

For every rational $q < c$, (2) and the definition of c give $\mu(E_q) = 0$, hence $g \geq q$ a.e. Letting $q \uparrow c$ through rationals yields $g \geq c$ a.e. Conversely, for every rational $q > c$ we have $\mu(E_q) = 1$, so $g < q$ a.e.; letting $q \downarrow c$ gives $g \leq c$ a.e. Therefore

$$g = c \quad \text{a.e.} \quad (4)$$

Thus in all cases the Kingman limit is almost surely constant, with value $c \in \mathbb{R} \cup \{-\infty\}$.

Finally *Kingman subadditive ergodic theorem* gives

$$\int g d\mu = \inf_{n \geq 1} \frac{1}{n} \int f_n d\mu. \quad (5)$$

Because μ is a probability measure and $g = c$ a.e., the left side is exactly c in the extended-real sense. Hence

$$c = \inf_{n \geq 1} \frac{1}{n} \int f_n d\mu,$$

with $c = -\infty$ allowed. □

Corollary F03.4. Top Lyapunov exponent from Kingman

Statement. Let A be a measurable matrix cocycle with $\log^+ \|A\| \in L^1$. Then $\phi_n(x) := \log \|A_n(x)\|$ is subadditive, and $n^{-1}\phi_n(x)$ converges almost everywhere to an invariant function $\lambda_{\max}(x) \in [-\infty, \infty)$. If the base is ergodic, this limit is constant almost everywhere.

Proof. Let $A : X \rightarrow M_d(\mathbb{R})$ be the measurable cocycle generator and define

$$A_n(x) := A(T^{n-1}x) \cdots A(Tx)A(x), \quad n \geq 1. \quad (1)$$

We use the convention

$$\log 0 = -\infty. \quad (2)$$

Thus

$$\phi_n(x) := \log \|A_n(x)\|$$

is measurable with values in $\mathbb{R} \cup \{-\infty\}$.

The cocycle identity is

$$A_{m+n}(x) = A_n(T^m x)A_m(x). \quad (3)$$

By submultiplicativity of the operator norm,

$$\|A_{m+n}(x)\| \leq \|A_n(T^m x)\| \|A_m(x)\|. \quad (4)$$

Taking logarithms, with the convention (2), gives

$$\phi_{m+n}(x) \leq \phi_m(x) + \phi_n(T^m x). \quad (5)$$

Hence (ϕ_n) is a subadditive process in exactly the sense required by *Kingman subadditive ergodic theorem*.

Its one-step positive part is

$$\phi_1^+(x) = \log^+ \|A(x)\|, \quad (6)$$

which is integrable by hypothesis. Therefore *Kingman subadditive ergodic theorem* applies and produces a measurable T -invariant function

$$\lambda_{\max} : X \rightarrow \mathbb{R} \cup \{-\infty\}$$

such that

$$\lambda_{\max}(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|A_n(x)\| \quad (7)$$

for almost every x .

This is precisely the top Lyapunov exponent defined in *the definition of top Lyapunov exponent*. Notice that (7) still makes sense when some $A_n(x)$ vanishes: then the relevant logarithm is $-\infty$, and the extended-real form of Kingman's theorem covers that case.

If T is ergodic, *Ergodic Kingman specialization* applied to the subadditive process (ϕ_n) says that the invariant limit in (7) is constant almost everywhere. This proves the final assertion. $\square$

Theorem F03.5. Furstenberg-Kesten-type matrix product growth

Statement. Let $(X, \mathcal{A}, \mu, T)$ be an ergodic probability-preserving system and let $A : X \rightarrow M_d(\mathbb{R})$ be measurable with $\log^+ \|A\| \in L^1(\mu)$. Put $Y_n(x) = A(T^{n-1}x) \cdots A(Tx)A(x)$ and $\log 0 = -\infty$. Then there is a constant $\gamma \in [-\infty, \infty)$ such that $n^{-1} \log \|Y_n(x)\| \rightarrow \gamma$ for a.e. x , and $\gamma = \lim_{n \rightarrow \infty} n^{-1} \int \log \|Y_n\| d\mu = \inf_{n \geq 1} n^{-1} \int \log \|Y_n\| d\mu$.

Proof. Define

$$Y_n(x) = A(T^{n-1}x) \cdots A(Tx)A(x)$$

and

$$\phi_n(x) := \log \|Y_n(x)\|, \quad \log 0 := -\infty. \quad (1)$$

This is precisely the cocycle process treated in *Top Lyapunov exponent from Kingman*. Hence

$$\frac{1}{n} \log \|Y_n(x)\| \longrightarrow \lambda_{\max}(x) \quad (2)$$

for almost every x , where $\lambda_{\max}$ is T -invariant. Since the base system is ergodic, *Top Lyapunov exponent from Kingman* also gives a constant $\gamma \in [-\infty, \infty)$ such that

$$\lambda_{\max}(x) = \gamma \quad \text{for a.e. } x. \quad (3)$$

This proves the almost-sure growth-rate assertion.

For completeness, we identify the constant without appealing to any additional random-matrix theorem. The cocycle identity

$$Y_{m+n}(x) = Y_n(T^m x) Y_m(x)$$

and submultiplicativity of the norm show that (ϕ_n) is subadditive, while

$$\phi_1^+ = \log^+ \|A\| \in L^1(\mu).$$

Therefore the integral conclusion of *Kingman subadditive ergodic theorem* gives

$$\int \lambda_{\max} d\mu = \lim_{n \rightarrow \infty} \frac{1}{n} \int \log \|Y_n\| d\mu = \inf_{n \geq 1} \frac{1}{n} \int \log \|Y_n\| d\mu. \quad (4)$$

Because $\mu(X) = 1$ and $\lambda_{\max} = \gamma$ almost everywhere, the left side of (4) is γ in the extended-real sense. Thus

$$\gamma = \lim_{n \rightarrow \infty} \frac{1}{n} \int \log \|Y_n\| d\mu = \inf_{n \geq 1} \frac{1}{n} \int \log \|Y_n\| d\mu.$$

This is the stated Furstenberg-Kesten-type growth formula. $\square$

Proposition F03.6. Subadditive bounds with error terms

Statement. Let $(a_n)_{n \geq 1}$ be real and suppose that for some $C \geq 0$, $a_{m+n} \leq a_m + a_n + C$ for all $m, n \geq 1$. Then $b_n := a_n + C$ is subadditive. Hence a_n/n converges and $\lim_{n \rightarrow \infty} a_n/n = \inf_{q \geq 1} (a_q + C)/q$. More quantitatively, if $n = kq + r$ with $k \geq 0$ and $0 \leq r < q$, then $b_n \leq kb_q + b_r$ when $r > 0$, while $b_{kq} \leq kb_q$; in particular, for fixed q , $\limsup_{n \rightarrow \infty} a_n/n \leq (a_q + C)/q$.

Proof. The key observation is that a uniform additive error can be absorbed once and for all. Define

$$b_n := a_n + C.$$

Then, for all $m, n \geq 1$,

$$b_{m+n} = a_{m+n} + C \leq a_m + a_n + 2C = b_m + b_n. \quad (1)$$

Thus (b_n) is an ordinary subadditive sequence.

By *Fekete subadditive lemma*,

$$\lim_{n \rightarrow \infty} \frac{b_n}{n} = \inf_{q \geq 1} \frac{b_q}{q} = \inf_{q \geq 1} \frac{a_q + C}{q}. \quad (2)$$

Since $C/n \rightarrow 0$,

$$\frac{a_n}{n} = \frac{b_n}{n} - \frac{C}{n}$$

has the same limit. Hence

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{q \geq 1} \frac{a_q + C}{q}. \quad (3)$$

We now record the quantitative block estimate that is repeatedly useful later. Fix $q \geq 1$ and write

$$n = kq + r, \quad k \geq 0, \quad 0 \leq r < q.$$

If $r = 0$, repeated use of (1) gives

$$b_{kq} \leq kb_q. \quad (4)$$

If $r > 0$, first apply (4) and then one further subadditive step:

$$b_{kq+r} \leq b_{kq} + b_r \leq kb_q + b_r. \quad (5)$$

For fixed q , the finitely many remainder values $b_1, \dots, b_{q-1}$ are bounded above by some

number M_q . Dividing (5) by $n = kq + r$ gives, for $r > 0$,

$$\frac{a_n}{n} = \frac{b_n - C}{n} \leq \frac{kb_q + M_q - C}{kq + r}.$$

As $n \rightarrow \infty$ with q fixed, necessarily $k \rightarrow \infty$, and therefore

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \frac{b_q}{q} = \frac{a_q + C}{q}. \quad (6)$$

The case $r = 0$ gives the same bound directly from (4). Because q was arbitrary, (6) is consistent with and recovers the infimum formula (3). This proves both the block decomposition and the limiting-slope statement. $\square$

3.3 Worked examples

Worked example: Subadditive Ergodic Theory: a model computation

Let (X, T, μ) be ergodic and let

$$A(x) = \begin{pmatrix} e^{c+g(x)-g(Tx)} & 0 \\ 0 & e^{-c-g(x)+g(Tx)} \end{pmatrix}, \quad c > 0,$$

where g is bounded. The product telescopes:

$$A_n(x) = \begin{pmatrix} e^{nc+g(x)-g(T^n x)} & 0 \\ 0 & e^{-nc-g(x)+g(T^n x)} \end{pmatrix}.$$

Hence

$$\log \|A_n(x)\| = nc + g(x) - g(T^n x)$$

for all sufficiently large n , uniformly up to a bounded error. Dividing by n ,

$$\frac{1}{n} \log \|A_n(x)\| \longrightarrow c.$$

The coboundary term $g - g \circ T$ changes finite-time growth but not the asymptotic exponent. This elementary telescoping model is the linear-cocycle analogue of the fact that adding a discrete derivative does not change a long-time average.

Boundary case: what fails at the edge of the hypotheses

Let $X = X_1 \sqcup X_2$ with both pieces invariant and of positive measure. Put $A(x) = e$ on X_1 and $A(x) = e^2$ on X_2 . Then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log |A_n(x)| = \begin{cases} 1, & x \in X_1, \\ 2, & x \in X_2. \end{cases}$$

Kingman's limit exists and is invariant, but it is not constant because the base

is not ergodic. Ergodicity is therefore a constancy hypothesis, not an existence hypothesis.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

3.4 Chapter summary

What to remember

- Subadditivity is the natural algebraic structure behind norm growth.
- Kingman produces an invariant asymptotic rate under integrability.
- Ergodicity turns the invariant rate into an almost-sure constant.
- Matrix cocycles are the principal dynamical application developed next.

Looking ahead

F04 uses exterior powers and singular values to extract all Lyapunov exponents, not just the top one, and then builds the Oseledets filtration and splitting.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

3.5 Exercises and problems

Exercise 3.1

Difficulty. Core.

Related results. Fekete subadditive lemma; Kingman subadditive ergodic theorem; Ergodic Kingman specialization.

Problem. Work through the worked example 'a diagonal matrix cocycle $A(x) = \text{diag}(e^{f(x)}, e^{-f(x)})'$ '. For Fekete subadditive lemma, Kingman subadditive ergodic theorem, and Ergodic Kingman specialization, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypoth-

esis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 3.2

Difficulty. Core.

Related results. Top Lyapunov exponent from Kingman; Subadditive bounds with error terms; Furstenberg-Kesten-type matrix product growth.

Problem. Analyze the boundary example 'a subadditive sequence with limit -infinity and a nonergodic base'. For Top Lyapunov exponent from Kingman, Subadditive bounds with error terms, and Furstenberg-Kesten-type matrix product growth, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 3.3

Difficulty. Advanced.

Related results. Ergodic Kingman specialization.

Problem. For Ergodic Kingman specialization, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 3.4

Difficulty. Advanced.

Related results. Top Lyapunov exponent from Kingman.

Problem. Stress-test Top Lyapunov exponent from Kingman. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 3.5

Difficulty. Advanced.

Related results. Fekete subadditive lemma; Kingman subadditive ergodic theorem.

Problem. Synthesis problem. Use Fekete subadditive lemma and Kingman subadditive ergodic theorem to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Linear Cocycles and Oseledets Theory

Why this chapter matters

A single exponential growth rate cannot describe a multidimensional linear system. One direction may expand while another contracts, and those directions may vary measurably with the base point. Oseledets' theorem is the nonlinear-dynamical analogue of diagonalizing a matrix at exponential scale.

The theorem is both geometric and quantitative: it produces Lyapunov exponents and invariant subspaces. Exterior powers are the key bookkeeping device because sums of the largest exponents become top exponents of new cocycles, bringing the problem back within the reach of Kingman.

Prerequisites. F03, finite-dimensional linear algebra, singular values, exterior algebra, and measurable dynamics.

Learning goals

- Compute singular-value growth and understand exterior powers as exponent sums.
- Distinguish the filtration and invertible splitting forms of Oseledets' theorem.
- Use exponential gaps to force invariant subspaces.
- Apply determinant and symplectic constraints to Lyapunov spectra.

Conceptual roadmap

Start with singular values. Passing to $\Lambda^k A$ converts the product of the first k singular values into one operator norm. Kingman therefore produces asymptotic sums $\lambda_1 + \cdots + \lambda_k$. Differences recover the individual exponents. Tempered estimates then turn asymptotic growth information into invariant filtrations and, for invertible systems, invariant splittings.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

4.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F04.1. Linear cocycle

A rank- d linear cocycle over $T : X \rightarrow X$ is a measurable map $A : X \rightarrow M_d(\mathbb{R})$ (or $GL(d, \mathbb{R})$ in the invertible case) with iterates $A_n(x) = A(T^{n-1}x) \cdots A(Tx)A(x)$.

Definition F04.2. Singular values

For a linear map A on a Euclidean d -space, the singular values $\sigma_1(A) \geq \cdots \geq \sigma_d(A) \geq 0$ are the square roots of the eigenvalues of A^*A , counted with multiplicity.

Definition F04.3. Exterior power

The k -th exterior power $\Lambda^k A$ is the induced map on $\Lambda^k \mathbb{R}^d$ determined by $(\Lambda^k A)(v_1 \wedge \cdots \wedge v_k) = Av_1 \wedge \cdots \wedge Av_k$.

Definition F04.4. Oseledets splitting

At an Oseledets-regular point x , an *Oseledets splitting* is an equivariant direct sum $\mathbb{R}^d = E_1(x) \oplus \cdots \oplus E_r(x)$ such that every nonzero $v \in E_i(x)$ has exponential growth rate $\lambda_i(x)$ under the cocycle (and, in the two-sided theorem, the corresponding backward rate).

4.2 Main results

Proposition F04.1. Singular values and exterior powers

Statement. Let A act on a d -dimensional Euclidean space with singular values $\sigma_1(A) \geq \cdots \geq \sigma_d(A) \geq 0$. Then

$$\|\Lambda^k A\| = \sigma_1(A) \cdots \sigma_k(A).$$

For decomposable k -vectors, this is the maximal k -dimensional volume expansion.

Proof. Let A act on a d -dimensional Euclidean space, and fix $1 \leq k \leq d$. Choose a singular-value decomposition

$$A = U\Sigma V^*, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_d), \quad \sigma_1 \geq \cdots \geq \sigma_d \geq 0,$$

where U and V are orthogonal.

Step 1: reduce to the diagonal singular-value matrix. Exterior powers are functorial, so

$$\Lambda^k A = (\Lambda^k U)(\Lambda^k \Sigma)(\Lambda^k V^*). \quad (1)$$

Equip $\Lambda^k \mathbb{R}^d$ with the Euclidean inner product for which

$$e_{i_1} \wedge \cdots \wedge e_{i_k}, \quad 1 \leq i_1 < \cdots < i_k \leq d,$$

is an orthonormal basis whenever $(e_1, \dots, e_d)$ is orthonormal in $\mathbb{R}^d$. Orthogonal maps preserve this induced inner product; therefore $\Lambda^k U$ and $\Lambda^k V^*$ are isometries. From (1),

$$\|\Lambda^k A\| = \|\Lambda^k \Sigma\|. \quad (2)$$

Step 2: read the operator norm from the wedge basis. For each increasing k -tuple,

$$(\Lambda^k \Sigma)(e_{i_1} \wedge \cdots \wedge e_{i_k}) = (\sigma_{i_1} \cdots \sigma_{i_k}) e_{i_1} \wedge \cdots \wedge e_{i_k}.$$

Thus $\Lambda^k \Sigma$ is diagonal in an orthonormal basis. Its operator norm is therefore the largest absolute diagonal entry. Since the singular values are nonnegative and decreasing,

$$\|\Lambda^k A\| = \max_{i_1 < \cdots < i_k} \sigma_{i_1} \cdots \sigma_{i_k} = \sigma_1 \cdots \sigma_k. \quad (3)$$

Step 3: interpret the same number as maximal k -volume expansion. Let $\xi = v_1 \wedge \cdots \wedge v_k \neq 0$; equivalently, the vectors $v_1, \dots, v_k$ are linearly independent. The induced norm satisfies

$$\|\xi\|^2 = \det(\langle v_i, v_j \rangle)_{i,j=1}^k,$$

so $\|\xi\|$ is exactly the Euclidean k -volume of the parallelotope spanned by the v_i . Also

$$\Lambda^k A(\xi) = Av_1 \wedge \cdots \wedge Av_k.$$

Hence, by the definition of operator norm and (3),

$$\frac{\text{Vol}_k(Av_1, \dots, Av_k)}{\text{Vol}_k(v_1, \dots, v_k)} = \frac{\|\Lambda^k A \xi\|}{\|\xi\|} \leq \sigma_1 \cdots \sigma_k.$$

Choose $v_1, \dots, v_k$ to be orthonormal right singular vectors corresponding to $\sigma_1, \dots, \sigma_k$. Then equality holds. Therefore $\sigma_1 \cdots \sigma_k$ is precisely the maximal k -dimensional volume expansion, as claimed. $\square$

Lemma F04.2. Exterior-power cocycle construction

Statement. For every k ,

$$\Lambda^k(A_n(x)) = (\Lambda^k A)_n(x).$$

Moreover $\log^+ \|\Lambda^k A(x)\| \leq k \log^+ \|A(x)\|$, so the Kingman integrability hypothesis passes to every exterior power.

Proof. Fix $1 \leq k \leq d$. Recall that for a linear map $B : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the induced map on the

k th exterior power is defined on decomposable vectors by

$$(\Lambda^k B)(v_1 \wedge \cdots \wedge v_k) := Bv_1 \wedge \cdots \wedge Bv_k. \quad (1)$$

From this definition one immediately obtains the functorial identity

$$\Lambda^k(BC) = (\Lambda^k B)(\Lambda^k C). \quad (2)$$

Indeed, both sides send $v_1 \wedge \cdots \wedge v_k$ to $BCv_1 \wedge \cdots \wedge BCv_k$, and decomposable k -vectors span $\Lambda^k \mathbb{R}^d$.

For the cocycle product

$$A_n(x) = A(T^{n-1}x) \cdots A(x),$$

repeated use of (2) therefore gives

$$\begin{aligned} \Lambda^k(A_n(x)) &= \Lambda^k(A(T^{n-1}x)) \cdots \Lambda^k(A(x)) \\ &= (\Lambda^k A)_n(x). \end{aligned} \quad (3)$$

This proves the cocycle identity claimed in the first part of the statement.

For the integrability estimate, let

$$\sigma_1(A(x)) \geq \cdots \geq \sigma_d(A(x)) \geq 0$$

be the singular values. *Singular values and exterior powers* gives

$$\|\Lambda^k A(x)\| = \sigma_1(A(x)) \cdots \sigma_k(A(x)). \quad (4)$$

Since every singular value is at most the largest one,

$$\sigma_j(A(x)) \leq \sigma_1(A(x)) = \|A(x)\|,$$

so

$$\|\Lambda^k A(x)\| \leq \|A(x)\|^k. \quad (5)$$

If $A(x) = 0$, both sides of (5) are zero and the logarithmic inequality below is interpreted with $\log 0 = -\infty$. Otherwise taking logarithms in (5) gives

$$\log \|\Lambda^k A(x)\| \leq k \log \|A(x)\|.$$

Taking positive parts yields the pointwise estimate

$$\boxed{\log^+ \|\Lambda^k A(x)\| \leq k \log^+ \|A(x)\|}. \quad (6)$$

Consequently, if $\log^+ \|A\| \in L^1(\mu)$, then $\log^+ \|\Lambda^k A\| \in L^1(\mu)$. Thus every exterior-power cocycle satisfies the same one-step positive-log integrability condition needed for the subadditive Kingman argument. $\square$

Theorem F04.3. Top Lyapunov exponent for each exterior power

Statement. For each $1 \leq k \leq d$, if $\log^+ \|A\| \in L^1$, then

$$\frac{1}{n} \log \|\Lambda^k A_n(x)\| \longrightarrow \Gamma_k(x)$$

almost everywhere, where Γ_k is invariant and may be $-\infty$. In the ergodic case Γ_k is constant almost everywhere.

Proof. Fix $k \in \{1, \dots, d\}$ and set

$$B(x) := \Lambda^k A(x) \quad \text{on } \Lambda^k \mathbb{R}^d. \quad (1)$$

By *Exterior-power cocycle construction*, the n -step product of B is exactly

$$B_n(x) = \Lambda^k(A_n(x)), \quad (2)$$

and

$$\log^+ \|B(x)\| \leq k \log^+ \|A(x)\|. \quad (3)$$

Thus the assumed integrability of $\log^+ \|A\|$ implies $\log^+ \|B\| \in L^1(\mu)$.

Define

$$g_n^{(k)}(x) := \log \|B_n(x)\| = \log \|\Lambda^k A_n(x)\|, \quad \log 0 = -\infty. \quad (4)$$

The cocycle identity

$$B_{m+n}(x) = B_n(T^m x) B_m(x)$$

and submultiplicativity of the operator norm give

$$g_{m+n}^{(k)}(x) \leq g_m^{(k)}(x) + g_n^{(k)}(T^m x). \quad (5)$$

Moreover $(g_1^{(k)})^+ = \log^+ \|B\| \in L^1$. Therefore *Kingman subadditive ergodic theorem* applies. It yields a measurable T -invariant function

$$\Gamma_k : X \rightarrow \mathbb{R} \cup \{-\infty\}$$

such that

$$\Gamma_k(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|\Lambda^k A_n(x)\| \quad (6)$$

for almost every x .

If T is ergodic, *Ergodic Kingman specialization* applied to the subadditive process $(g_n^{(k)})$ shows that Γ_k is constant almost everywhere. Since k was arbitrary, the conclusion holds for every exterior power $1 \leq k \leq d$. $\square$

Theorem F04.4. Oseledets multiplicative ergodic theorem: filtration form

Statement. Let $(X, \mathcal{A}, \mu, T)$ be a probability-preserving system and let $A : X \rightarrow M_d(\mathbb{R})$ be measurable with $\log^+ \|A\| \in L^1(\mu)$. Put $A_n(x) = A(T^{n-1}x) \cdots A(x)$ and use

$\log 0 = -\infty$. Then for μ -a.e. x there are distinct exponents

$$\lambda_1(x) > \cdots > \lambda_{k(x)}(x) \geq -\infty$$

and a measurable filtration

$$\mathbb{R}^d = V_1(x) \supsetneq \cdots \supsetneq V_{k(x)+1}(x) = \{0\}$$

such that $A(x)V_i(x) \subseteq V_i(Tx)$ and, for every $v \in V_i(x) \setminus V_{i+1}(x)$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|A_n(x)v\| = \lambda_i(x).$$

The exponent list and the multiplicities $\dim V_i - \dim V_{i+1}$ are T -invariant measurable data; in particular they are constant almost everywhere when T is ergodic.

Proof. Write

$$A_n(x) := A(T^{n-1}x) \cdots A(Tx)A(x), \quad A_0(x) = I, \quad (1)$$

and use the convention $\log 0 = -\infty$. The proof has two parts. We first isolate the deterministic matrix statement that converts exterior-power growth into a vectorwise filtration. We then apply that statement on a full-measure set of orbits.

Step 1: exterior powers determine all singular-value rates. For $1 \leq q \leq d$, *Top Lyapunov exponent for each exterior power* gives a measurable T -invariant function $\Gamma_q : X \rightarrow \mathbb{R} \cup \{-\infty\}$ such that, on a common full-measure set,

$$\Gamma_q(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|\Lambda^q A_n(x)\|. \quad (2)$$

Let

$$0 \leq s_{1,n}(x) \leq \cdots \leq s_{d,n}(x) \quad (3)$$

be the singular values of $A_n(x)$ in nondecreasing order. By *Singular values and exterior powers*,

$$\|\Lambda^q A_n(x)\| = s_{d-q+1,n}(x) \cdots s_{d,n}(x). \quad (4)$$

Consequently every individual limit

$$\chi_j(x) := \lim_{n \rightarrow \infty} \frac{1}{n} \log s_{j,n}(x) \in \mathbb{R} \cup \{-\infty\}, \quad 1 \leq j \leq d, \quad (5)$$

exists. Here is the small extended-real point that is sometimes suppressed. Starting from the largest singular value, $n^{-1} \log s_{d,n} = n^{-1} \log \|A_n\|$ converges. As long as the already recovered sum of the larger rates is finite, the next rate is obtained by subtracting two consecutive limits in (4). If at some stage the new exterior-power limit equals $-\infty$, then the corresponding singular value has rate $-\infty$; by the ordering in (3), all smaller singular values also have rate $-\infty$. Thus no undefined expression $(-\infty) - (-\infty)$ is needed.

The ordered d -tuple

$$\chi_1(x) \leq \cdots \leq \chi_d(x) \quad (6)$$

is measurable. It is also T -invariant almost everywhere because it is recovered, with multiplicity, from the invariant family $(\Gamma_q)_{q=1}^d$.

Step 2: the one-step matrices are subexponential along almost every orbit. Set

$$h(x) := \log^+ \|A(x)\| \in L^1(\mu).$$

For every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mu\{x : h(x) > \varepsilon n\} < \infty. \quad (7)$$

Indeed, the layer-cake formula gives $\int_0^\infty \mu\{h > t\} dt = \int h d\mu < \infty$, and comparison of the monotone tail function with its integral yields (7). *Borel-Cantelli lemmas* therefore implies that for almost every x and every rational $\varepsilon > 0$, $h(T^n x) \leq \varepsilon n$ for all sufficiently large n . Hence, on a T -invariant full-measure set,

$$\boxed{\lim_{n \rightarrow \infty} \frac{1}{n} \log^+ \|A(T^n x)\| = 0.} \quad (8)$$

Step 3: deterministic Raghunathan-Ruelle singular-flag lemma. We now prove the finite-dimensional statement needed below.

Claim. Let $(B_n)_{n \geq 1}$ be real $d \times d$ matrices and $P_n = B_n \cdots B_1$. Assume

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log^+ \|B_{n+1}\| = 0 \quad (9)$$

and assume that $n^{-1} \log \|\Lambda^q P_n\|$ converges in $\mathbb{R} \cup \{-\infty\}$ for every $1 \leq q \leq d$. Then there are distinct numbers

$$\alpha_1 < \cdots < \alpha_s, \quad \alpha_1 \in \mathbb{R} \cup \{-\infty\}, \quad \alpha_2, \dots, \alpha_s \in \mathbb{R}, \quad (10)$$

and a flag

$$\{0\} = W_0 \subsetneq W_1 \subsetneq \cdots \subsetneq W_s = \mathbb{R}^d \quad (11)$$

such that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|P_n v\| = \alpha_r \quad (v \in W_r \setminus W_{r-1}). \quad (12)$$

Moreover the flag is obtained as a limit of right-singular subspaces of P_n .

Proof of the claim. Let $s_{1,n} \leq \cdots \leq s_{d,n}$ be the singular values of P_n . Exactly as in Step 1, the exterior-power hypotheses imply the existence of

$$\beta_j := \lim_{n \rightarrow \infty} \frac{1}{n} \log s_{j,n} \in \mathbb{R} \cup \{-\infty\}. \quad (13)$$

Group equal values among the β_j as in (10), and let m_r be the total multiplicity of $\alpha_1, \dots, \alpha_r$. Choose an orthogonal singular-value decomposition

$$P_n = L_n D_n R_n^*, \quad (14)$$

where $D_n = \text{diag}(s_{1,n}, \dots, s_{d,n})$ and the columns of R_n are right singular vectors. Write

$E_{r,n}$ for the span of the columns whose singular-value rate is α_r , and

$$W_{r,n} := E_{1,n} \oplus \cdots \oplus E_{r,n}. \quad (15)$$

The dimensions of these spaces are independent of n by construction: the blocks are defined by the fixed index intervals determined by the limiting rates, not by accidental finite-time equalities.

We next prove that the singular blocks at different large times become almost orthogonal unless they correspond to the same rate. Fix $\delta > 0$ smaller than one quarter of every positive gap $\alpha_{r+1} - \alpha_r$ between finite adjacent rates. If $\alpha_1 = -\infty$, temporarily replace it in the estimates by an arbitrary finite number $-M < \alpha_2 - 1$; at the end M will be allowed to tend to infinity.

By (9) and (13), after increasing $N = N(\delta)$, for all $n \geq N$ we have

$$\|B_{n+1}\| \leq e^{\delta n}, \quad e^{n(\alpha_r - \delta)} \leq s_{j,n} \leq e^{n(\alpha_r + \delta)} \quad (16)$$

whenever j belongs to the r -th finite-rate block. In the $-\infty$ block, the upper estimate in (16) is replaced by $s_{j,n} \leq e^{-Mn}$.

Let $Q_n := R_{n+1}^* R_n$, the orthogonal change-of-basis matrix from the right singular basis at time n to the one at time $n+1$. Suppose first that $r < r'$ and $u \in E_{r,n}$ is a unit vector. If z denotes the orthogonal projection of u onto $E_{r',n+1}$, then the images under P_{n+1} of the components of u in the distinct right-singular blocks at time $n+1$ are orthogonal. Hence

$$s_{r',\min}(n+1) \|z\| \leq \|P_{n+1}u\| \leq \|B_{n+1}\| \|P_n u\|, \quad (17)$$

where $s_{r',\min}(n+1)$ is the smallest singular value in the r' -th block. Using (16) gives, after harmlessly enlarging the coefficient of δ ,

$$\|\text{proj}_{E_{r',n+1}} u\| \leq \exp\{-n(\alpha_{r'} - \alpha_r - 4\delta)\}. \quad (18)$$

The same estimate holds with $E_{r,n}$ replaced by $W_{r,n}$, up to a fixed factor depending only on d .

For later use we need the analogous estimate in the opposite triangular direction. We record the elementary linear-algebra argument rather than hide it. Let $Q = (q_{ij})$ be an orthogonal matrix written in the individual singular bases. Let $b(i)$ be the limiting-rate block containing the i -th coordinate and attach to that coordinate the block weight $a_i = e^{n(\alpha_{b(i)} - 4\delta b(i))}$. Thus all coordinates in one limiting-rate block carry the same weight. Estimate (18) says, after a uniform constant is inserted, that

$$|q_{ij}| \leq C a_i / a_j \quad (\beta_i < \beta_j). \quad (19)$$

Because $Q^{-1} = Q^*$, an entry of Q^{-1} is a cofactor of Q divided by $\det Q$, whose absolute value is 1. Expand that cofactor into permutation monomials. In every monomial contributing to the (i, j) entry with $\beta_i < \beta_j$, the permutation path that begins at i and ends at j must cross the rate blocks from the block of i to that of j . Multiplying (19) along those crossings telescopes the weights and contributes at most $C_d a_i / a_j$; all

factors that remain inside a block are at most 1 because Q is orthogonal. There are at most $(d-1)!$ monomials. Thus

$$|(Q^{-1})_{ij}| \leq C'_d a_i / a_j. \quad (20)$$

Since $(Q^{-1})_{ij} = q_{ji}$, (20) supplies the missing lower-triangular angle estimate. This is precisely the compact-matrix inverse estimate used by Raghunathan; in the present orthogonal setting no determinant lower bound other than $|\det Q| = 1$ is required.

Iterating the one-step change of singular basis now gives a uniform estimate between time n and every later time. We spell out the induction because it is the point at which a merely qualitative “the singular spaces converge” argument would be insufficient. Let

$$Q(n, k) := R_{n+k}^* R_n = Q_{n+k-1} \cdots Q_n. \quad (21)$$

For two distinct rate blocks r, r' , expand the (r', r) block of this product by inserting the intermediate block projections after every factor. Thus each term is represented by a block path

$$r = i_0, i_1, \dots, i_\ell = r' \quad (22)$$

and by strictly increasing crossing times $n \leq m_1 < \cdots < m_\ell < n + k$; between two crossing times the path remains in one block. Diagonal block factors have norm at most 1, while every genuine crossing $i_{a-1} \neq i_a$ is bounded by (18)–(20):

$$C \exp\{-m_a(|\alpha_{i_a} - \alpha_{i_{a-1}}| - 4\delta)\}. \quad (23)$$

This path expansion is finite for fixed k , so it is legitimate term by term. We now bound it uniformly in k .

Let

$$g := \min_{1 \leq j < s} (\alpha_{j+1} - \alpha_j) > 0 \quad (24)$$

when there is more than one finite rate block, and choose $4\delta < g/4s$. For a path (22), put

$$L(i_0, \dots, i_\ell) := \sum_{a=1}^{\ell} |\alpha_{i_a} - \alpha_{i_{a-1}}|.$$

The triangle inequality gives $L \geq |\alpha_{r'} - \alpha_r|$. More is true when the path backtracks: after deleting consecutive excursions that reverse the order of the rate blocks, one obtains a monotone path with at most $s-1$ crossings, while every deleted extra crossing contributes at least g to L . Hence

$$L \geq |\alpha_{r'} - \alpha_r| + g \max\{0, \ell - (s-1)\}. \quad (25)$$

Because every $m_a \geq n$, the product of the exponential factors in (23) is at most

$$C^\ell e^{-n(|\alpha_{r'} - \alpha_r| - 4s\delta)} e^{-n(g-4\delta) \max\{0, \ell - (s-1)\}}. \quad (26)$$

For fixed ℓ , summing over the increasing crossing times only decreases the bound by geometric factors: writing $m_a = n + j_a$ contributes the convergent sum of $e^{-(g-4\delta)(j_1 + \cdots + j_\ell)}$ over $0 \leq j_1 < \cdots < j_\ell$. There are at most $s^{\ell-1}$ choices of the intermediate block indices. The final factor in (26) therefore makes the sum over all ℓ converge once n is sufficiently

large; the finitely many short paths $\ell \leq s - 1$ contribute only a fixed constant. We conclude that, after renaming $4s\delta$ as δ , there is $C_\delta < \infty$ such that

$$\|\text{proj}_{E_{r',n+k}}|_{E_{r,n}}\| \leq C_\delta e^{-n(|\alpha_{r'} - \alpha_r| - \delta)} \quad (27)$$

for every $k \geq 0$, every sufficiently large n , and every $r \neq r'$. This is the Raghunathan-Ruelle cross-time angle estimate. In the adjacent-block case its mechanism is simply

$$\sum_{j=0}^{\infty} e^{-(n+j)(\alpha_{r+1} - \alpha_r - 4\delta)} = O\left(e^{-n(\alpha_{r+1} - \alpha_r - 4\delta)}\right), \quad (28)$$

and (25)–(26) show explicitly why arbitrarily many backtracking crossings do not destroy the estimate.

Estimate (27) implies that, for each r , the sequence of subspaces $E_{r,n}$ is Cauchy in the Grassmannian. Indeed, the squared norm of the projection of a unit vector in $E_{r,n}$ onto the orthogonal complement of $E_{r,n+k}$ is the sum of the squared projections onto the finitely many blocks $E_{r',n+k}$, $r' \neq r$, and hence tends to zero exponentially in n , uniformly in k . Let

$$E_r := \lim_{n \rightarrow \infty} E_{r,n}, \quad W_r := E_1 \oplus \cdots \oplus E_r. \quad (29)$$

The limiting blocks remain mutually orthogonal and have the prescribed multiplicities.

We finally compute the growth of a nonzero $v \in E_r$. Let $v_{j,n} = \text{proj}_{E_{j,n}} v$. Sending $k \rightarrow \infty$ in (27) gives, for $j \neq r$ and every sufficiently small $\delta > 0$,

$$\|v_{j,n}\| \leq C_{\delta,v} e^{-n(|\alpha_j - \alpha_r| - \delta)}, \quad (30)$$

while $v_{r,n} \rightarrow v$, so $\|v_{r,n}\| \geq \|v\|/2$ for large n . Because the images under P_n of the right-singular blocks are orthogonal,

$$\|P_n v\|^2 = \sum_{j=1}^s \|P_n v_{j,n}\|^2. \quad (31)$$

If α_r is finite, (13), (30), and (31) yield

$$e^{n(\alpha_r - \delta)} \frac{\|v\|}{2} \leq \|P_n v\| \leq C'_{\delta,v} e^{n(\alpha_r + 3\delta)} \quad (32)$$

for large n . The terms $j < r$ are already slower than α_r , and for $j > r$ the faster singular-value growth is cancelled by the exponential angle factor in (30). Letting $\delta \downarrow 0$ proves

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|P_n v\| = \alpha_r. \quad (33)$$

We now pass from the orthogonal limiting block E_r to the whole filtration layer. Write $v = v_r + w$ with $0 \neq v_r \in E_r$ and $w \in W_{r-1}$. By the result just proved, $\|P_n v_r\| = e^{n(\alpha_r + o(1))}$. Inducting upward on r , every vector in W_{r-1} has exponent at most $\alpha_{r-1} < \alpha_r$, so $\|P_n w\| =$

$e^{n(\alpha_{r-1}+o(1))}$ (or is eventually zero). Therefore

$$\|P_n v_r\| - \|P_n w\| \leq \|P_n v\| \leq \|P_n v_r\| + \|P_n w\|,$$

and division by the dominant quantity $\|P_n v_r\|$ shows that the two bounds have the same exponential rate α_r . Thus (12) holds for every $v \in W_r \setminus W_{r-1}$; no cancellation between distinct limiting blocks is being assumed away.

If $\alpha_1 = -\infty$, repeat the upper-bound argument for the first block with $-M$ in place of α_1 . Since $M > 0$ is arbitrary, every nonzero vector in E_1 has exponent $-\infty$. All higher blocks are handled by the finite-rate argument. This proves (12) and the claim. $\triangleleft$

Step 4: apply the deterministic lemma orbit by orbit. Intersect the full-measure sets from Steps 1 and 2 and make the intersection forward T -invariant by deleting the null union of its inverse iterates. Fix x in the resulting set and put

$$B_{n+1} = A(T^n x), \quad P_n = A_n(x).$$

Hypotheses (9) and the exterior-power hypotheses of the claim are exactly (8) and (2). We therefore obtain distinct increasing exponents

$$\alpha_1(x) < \cdots < \alpha_{s(x)}(x) \tag{34}$$

and an increasing slow flag

$$\{0\} = W_0(x) \subsetneq W_1(x) \subsetneq \cdots \subsetneq W_{s(x)}(x) = \mathbb{R}^d \tag{35}$$

with vectorwise growth $\alpha_r(x)$ on $W_r(x) \setminus W_{r-1}(x)$.

To match the statement's decreasing convention, define

$$k(x) := s(x), \quad \lambda_i(x) := \alpha_{s(x)-i+1}(x), \quad V_i(x) := W_{s(x)-i+1}(x), \tag{36}$$

for $1 \leq i \leq k(x)$, and $V_{k(x)+1}(x) = \{0\}$. Then

$$\mathbb{R}^d = V_1(x) \supsetneq \cdots \supsetneq V_{k(x)+1}(x) = \{0\}, \tag{37}$$

and every $v \in V_i(x) \setminus V_{i+1}(x)$ satisfies

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|A_n(x)v\| = \lambda_i(x). \tag{38}$$

Step 5: measurability of the flag. The deterministic construction above can be chosen measurably in x . One convenient way to see this without making an arbitrary measurable choice of singular bases is to use the positive semidefinite matrices

$$H_n(x) := (A_n(x)^* A_n(x))^{1/(2n)}. \tag{39}$$

The proof of the claim shows that their eigenvalues converge to $e^{\chi_1(x)}, \dots, e^{\chi_d(x)}$ (with $e^{-\infty} = 0$) and that every spectral block separated by a limiting gap has a convergent spectral projection. Hence $H_n(x)$ converges in operator norm to a positive semidefinite

matrix $H(x)$ whose cumulative eigenspaces are precisely the $W_r(x)$.

Each H_n is a measurable matrix-valued function because matrix multiplication, transpose, and the continuous functional calculus on positive semidefinite matrices are Borel operations. Therefore $H = \lim H_n$ is measurable. On the measurable set where a fixed multiplicity pattern occurs, the spectral projection onto a cluster separated by a gap is a continuous function of H (equivalently, it is obtained from the finite-dimensional spectral calculus). Thus $x \mapsto W_r(x)$, and hence $x \mapsto V_i(x)$, is a measurable Grassmannian-valued map. The exponents and multiplicities are measurable for the same reason, or directly from the measurable limits in (2).

Step 6: equivariance of the filtration. Let $v \in V_i(x)$. If $A(x)v = 0$, then certainly $A(x)v \in V_i(Tx)$. Otherwise the cocycle identity gives

$$A_n(Tx)A(x)v = A_{n+1}(x)v. \quad (40)$$

If v lies in the j -th layer with $j \geq i$, its exponent is $\lambda_j(x) \leq \lambda_i(x)$. From (40),

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|A_n(Tx)A(x)v\| = \lambda_j(x). \quad (41)$$

The individual singular-value rates, hence the list of layer exponents, are T -invariant. Therefore (38) places $A(x)v$ in the slow subspace $V_i(Tx)$. Since $V_i(x)$ is spanned by its successive layers,

$$\boxed{A(x)V_i(x) \subseteq V_i(Tx)}. \quad (42)$$

Step 7: invariance and the ergodic case. As observed in Step 1, the ordered singular-value rates are recovered from the T -invariant exterior-power limits Γ_q . Hence the distinct exponents λ_i , their multiplicities

$$m_i(x) := \dim V_i(x) - \dim V_{i+1}(x), \quad (43)$$

and $k(x)$ are T -invariant measurable data. If T is ergodic, every such finite-valued invariant multiplicity function and every invariant extended-real exponent function is constant almost everywhere. This proves all assertions of the filtration form of the multiplicative ergodic theorem. $\square$

Corollary F04.5. Lyapunov spectrum from exterior powers

Statement. Let $\chi_1(x) \geq \dots \geq \chi_d(x)$ be the Lyapunov exponents, repeated with multiplicity, and let $\Gamma_k(x) = \lim_{n \rightarrow \infty} n^{-1} \log \|\Lambda^k A_n(x)\|$. Whenever the sum is finite,

$$\Gamma_k(x) = \chi_1(x) + \dots + \chi_k(x).$$

On the finite-exponent range, $\chi_k = \Gamma_k - \Gamma_{k-1}$; the remaining directions in the terminal rank filtration carry exponent $-\infty$.

Proof. Fix a point x in the full-measure regular set of *Oseledets multiplicative ergodic theorem: filtration form*. Write the Lyapunov exponents with multiplicity in nonin-

creasing order as

$$\chi_1(x) \geq \chi_2(x) \geq \cdots \geq \chi_d(x), \quad \chi_j(x) \in \mathbb{R} \cup \{-\infty\}. \quad (1)$$

Let

$$0 \leq s_{d,n}(x) \leq \cdots \leq s_{1,n}(x) \quad (2)$$

be the singular values of $A_n(x)$, now written in nonincreasing order to match (1). The deterministic singular-flag step in the proof of *Oseledets multiplicative ergodic theorem: filtration form* proves, for every j ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log s_{j,n}(x) = \chi_j(x). \quad (3)$$

This is not an additional MET hypothesis: it is the mechanism by which the filtration and its layer exponents were constructed.

By *Singular values and exterior powers*,

$$\|\Lambda^k A_n(x)\| = s_{1,n}(x) s_{2,n}(x) \cdots s_{k,n}(x). \quad (4)$$

If $\chi_1(x), \dots, \chi_k(x)$ are finite, taking logarithms, dividing by n , and using (3) gives

$$\Gamma_k(x) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \|\Lambda^k A_n(x)\| = \chi_1(x) + \cdots + \chi_k(x). \quad (5)$$

The left-hand limit is the one already supplied by *Top Lyapunov exponent for each exterior power*. With $\Gamma_0 := 0$, whenever Γ_{k-1} is finite we therefore recover

$$\boxed{\chi_k = \Gamma_k - \Gamma_{k-1}}. \quad (6)$$

It remains only to make the $-\infty$ boundary precise. Suppose $\chi_j = -\infty$ for the first time at index j . Since the exponents are ordered nonincreasingly,

$$\chi_j = \chi_{j+1} = \cdots = \chi_d = -\infty. \quad (7)$$

Conversely, if $\Gamma_j = -\infty$ while Γ_{j-1} is finite, then (4) shows

$$\frac{1}{n} \log s_{j,n} = \frac{1}{n} \log \|\Lambda^j A_n\| - \frac{1}{n} \log \|\Lambda^{j-1} A_n\| \longrightarrow -\infty,$$

so $\chi_j = -\infty$, and again all slower exponents are $-\infty$ by ordering. Thus the terminal $-\infty$ part of the spectrum is exactly the terminal rank filtration described in the statement. $\square$

Theorem F04.6. Oseledets theorem: invertible splitting form

Statement. Let $(X, \mathcal{A}, \mu, T)$ be probability-preserving with T invertible and bimeasurable. Let $A : X \rightarrow GL(d, \mathbb{R})$ be measurable and assume $\log^+ \|A\| + \log^+ \|A^{-1}\| \in L^1(\mu)$. For $n > 0$ define $A_n(x) = A(T^{n-1}x) \cdots A(x)$, $A_{-n}(x) = A_n(T^{-n}x)^{-1}$, and $A_0 = I$. Then for μ -a.e. x there are distinct finite exponents $\lambda_1(x) > \cdots > \lambda_{k(x)}(x)$ and a

measurable equivariant splitting

$$\mathbb{R}^d = E_1(x) \oplus \cdots \oplus E_{k(x)}(x), \quad A(x)E_i(x) = E_i(Tx),$$

such that every $0 \neq v \in E_i(x)$ satisfies

$$\lim_{n \rightarrow +\infty} \frac{1}{n} \log \|A_n(x)v\| = \lim_{n \rightarrow -\infty} \frac{1}{n} \log \|A_n(x)v\| = \lambda_i(x).$$

The exponent and multiplicity data are T -invariant and hence constant almost everywhere in the ergodic case.

Proof. For $n \geq 1$ put

$$A_n(x) = A(T^{n-1}x) \cdots A(x), \quad A_{-n}(x) := A_n(T^{-n}x)^{-1}, \quad A_0(x) = I. \quad (1)$$

Then $A_{m+n}(x) = A_m(T^n x)A_n(x)$ for all $m, n \in \mathbb{Z}$.

Step 1: forward and backward filtrations, and finiteness of the exponents. Apply *Oseledets multiplicative ergodic theorem: filtration form* to A over T . On a full-measure invariant set we obtain distinct forward exponents

$$\lambda_1(x) > \cdots > \lambda_{k(x)}(x) \quad (2)$$

and a decreasing filtration

$$\mathbb{R}^d = V_1^+(x) \supsetneq \cdots \supsetneq V_{k(x)+1}^+(x) = \{0\}. \quad (3)$$

All the exponents in (2) are finite under the present two-sided integrability hypothesis. Indeed, for every nonzero v ,

$$\|A_n(x)v\| \geq \|A_n(x)^{-1}\|^{-1}\|v\|, \quad (4)$$

and

$$\log^+ \|A_n(x)^{-1}\| \leq \sum_{j=0}^{n-1} \log^+ \|A(T^j x)^{-1}\|. \quad (5)$$

Birkhoff pointwise ergodic theorem applied to the integrable function $\log^+ \|A^{-1}\|$ makes the right side of (5), divided by n , converge to a finite conditional expectation almost everywhere. Thus no nonzero vector can have forward exponent $-\infty$.

For backward time let $S := T^{-1}$ and define the one-step inverse cocycle

$$B(x) := A(T^{-1}x)^{-1}. \quad (6)$$

Its n -step product over S is exactly

$$B_n(x) = A_{-n}(x). \quad (7)$$

Because T preserves μ , $\log^+ \|B\| \in L^1(\mu)$; the same argument using $B^{-1}(x) = A(T^{-1}x)$

shows that the backward Lyapunov exponents are also finite. *Oseledets multiplicative ergodic theorem: filtration form* applied to (S, B) therefore gives a backward filtration and a finite backward spectrum.

Step 2: the backward spectrum is the negative reverse of the forward spectrum. We prove this point rather than calling it “obvious from invertibility.” Write the forward exponents with multiplicity as

$$\chi_1(x) \geq \cdots \geq \chi_d(x), \quad (8)$$

and the positive- n exponents of $B_n = A_{-n}$ as

$$\beta_1(x) \geq \cdots \geq \beta_d(x). \quad (9)$$

Let Γ_q^+ and Γ_q^- be the top exterior-power growth functions for A over T and B over S , respectively, with $\Gamma_0^+ = 0$. By *Lyapunov spectrum from exterior powers*,

$$\Gamma_q^+ = \sum_{j=1}^q \chi_j, \quad \Gamma_q^- = \sum_{j=1}^q \beta_j. \quad (10)$$

For every invertible $d \times d$ matrix M , the singular values give the exact identity

$$\|\Lambda^q M^{-1}\| = \frac{\|\Lambda^{d-q} M\|}{|\det M|} = \frac{\|\Lambda^{d-q} M\|}{\|\Lambda^d M\|}. \quad (11)$$

Using $B_n(x) = A_n(T^{-n}x)^{-1}$, (11) becomes

$$\log \|\Lambda^q B_n(x)\| = \log \|\Lambda^{d-q} A_n(T^{-n}x)\| - \log \|\Lambda^d A_n(T^{-n}x)\|. \quad (12)$$

The two-sided integrability assumptions make every term in (12) integrable: the positive parts are bounded by sums of $\log^+ \|A\|$, while the negative parts are bounded using $\|A_n^{-1}\|$ and sums of $\log^+ \|A^{-1}\|$.

Let E be any T -invariant measurable set of positive measure. It is also S -invariant. Integrate (12) over E and change variables $y = T^{-n}x$; invariance of E and μ gives

$$\frac{1}{n} \int_E \log \|\Lambda^q B_n\| d\mu = \frac{1}{n} \int_E \log \|\Lambda^{d-q} A_n\| d\mu - \frac{1}{n} \int_E \log \|\Lambda^d A_n\| d\mu. \quad (13)$$

Apply the integral conclusion of *Kingman subadditive ergodic theorem* to each exterior-power subadditive process, restricted to the normalized probability measure on E . Letting $n \rightarrow \infty$ in (13) yields

$$\int_E \Gamma_q^- d\mu = \int_E (\Gamma_{d-q}^+ - \Gamma_d^+) d\mu. \quad (14)$$

Both sides of the desired pointwise identity are finite T -invariant measurable functions. Since (14) holds on every invariant E , it follows that

$$\boxed{\Gamma_q^- = \Gamma_{d-q}^+ - \Gamma_d^+ \quad \text{a.e.}} \quad (15)$$

Indeed, otherwise one of the invariant sets where the difference is positive or negative by at least $1/m$ would have positive measure and contradict (14).

Combining (10) and (15),

$$\sum_{j=1}^q \beta_j = - \sum_{j=d-q+1}^d \chi_j. \quad (16)$$

Successive differences now give

$$\boxed{\beta_j = -\chi_{d-j+1} \quad (1 \leq j \leq d).} \quad (17)$$

Thus the distinct backward exponents are precisely $-\lambda_k > \dots > -\lambda_1$, with the same multiplicities in reverse order.

Step 3: relabel the backward filtration as a forward-fast flag. Because only finitely many multiplicity patterns can occur, we may work on one invariant measurable stratum on which k and the multiplicities

$$m_i := \dim V_i^+ - \dim V_{i+1}^+ \quad (18)$$

are fixed; the strata together still have full measure.

Relabel the filtration supplied by the backward MET so that

$$\{0\} = U_0(x) \subsetneq U_1(x) \subsetneq \dots \subsetneq U_k(x) = \mathbb{R}^d, \quad (19)$$

where

$$\dim U_i = m_1 + \dots + m_i \quad (20)$$

and every

$$0 \neq v \in U_i(x) \setminus U_{i-1}(x)$$

satisfies

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|A_{-n}(x)v\| = -\lambda_i(x). \quad (21)$$

Since $A(x)$ is invertible and the dimensions of all filtration spaces are invariant, the inclusions from the forward MET are equalities. Thus

$$A(x)V_i^+(x) = V_i^+(Tx), \quad A(x)U_i(x) = U_i(Tx). \quad (22)$$

The second equality is the backward-filtration equivariance rewritten in forward time.

Step 4: transversality of the forward and backward flags. We claim that for every $1 \leq i < k$,

$$\boxed{V_{i+1}^+(x) \cap U_i(x) = \{0\} \quad \text{for a.e. } x.} \quad (23)$$

This is the recurrence step in Ruelle's proof.

Let $I_i(x) := V_{i+1}^+(x) \cap U_i(x)$ and suppose the bad set $S := \{x : I_i(x) \neq \{0\}\}$ has positive measure. By (22), S is invariant. The rank $x \mapsto \dim I_i(x)$ is measurable and invariant. Since it takes only the finitely many values $1, \dots, d$, after replacing S by one of its positive-measure invariant rank strata we may, and do, assume that $\dim I_i(x) = r \geq 1$ is constant on S . Thus I_i is a measurable rank- r subbundle on S ; in particular measurable

orthonormal frames, and hence operator norms of restrictions to I_i , are available there. Since $\lambda_i - \lambda_{i+1} > 0$ on S and the gap is invariant, for some rational $\delta > 0$ the invariant subset

$$S_\delta := \{x \in S : \lambda_i(x) - \lambda_{i+1}(x) > 4\delta\} \quad (24)$$

has positive measure.

For each $x \in S_\delta$, every nonzero vector in $I_i(x)$ has forward exponent at most $\lambda_{i+1}(x)$ and backward positive- n exponent at most $-\lambda_i(x)$. These upper bounds are uniform on the unit sphere of $I_i(x)$ after a sufficiently large time. To justify this often-suppressed point, choose a basis of the finite-dimensional space $I_i(x)$. The growth estimate holds for each basis vector; equivalence of the coefficient norm with the ambient norm then gives the same exponential upper bound, with an extra fixed multiplicative constant, for every unit vector. That constant is absorbed into $e^{\delta n}$ for large n .

Consequently S_δ is the increasing union of measurable sets S_N on which, for every $n \geq N$ and every $u \in I_i(x)$,

$$\|A_n(x)u\| \leq e^{n(\lambda_{i+1}(x)+\delta)}\|u\|, \quad (25)$$

and

$$\|A_{-n}(x)u\| \leq e^{n(-\lambda_i(x)+\delta)}\|u\|. \quad (26)$$

Measurability of S_N follows by taking operator norms of the restrictions to the measurable fixed-rank bundle I_i (equivalently, in a measurable orthonormal frame chosen on the rank stratum).

Choose N so large that $2\mu(S_\delta \setminus S_N) < \mu(S_\delta)$. Because S_δ is invariant, for every $n \geq N$,

$$\mu(S_N \cap T^{-n}S_N) \geq \mu(S_\delta) - 2\mu(S_\delta \setminus S_N) > 0. \quad (27)$$

Fix such an n and a point x in this intersection. Then both x and $T^n x$ belong to S_N . For $0 \neq u \in I_i(x)$ put $w = A_n(x)u \in I_i(T^n x)$, using (22). Apply (26) at $T^n x$:

$$\|u\| = \|A_{-n}(T^n x)w\| \leq e^{n(-\lambda_i(x)+\delta)}\|w\|,$$

where invariance of λ_i was used. Hence

$$\|A_n(x)u\| = \|w\| \geq e^{n(\lambda_i(x)-\delta)}\|u\|. \quad (28)$$

But (25) at x gives the opposite upper bound

$$\|A_n(x)u\| \leq e^{n(\lambda_{i+1}(x)+\delta)}\|u\|. \quad (29)$$

Equations (28)–(29) imply $\lambda_i(x) - \lambda_{i+1}(x) \leq 2\delta$, contradicting (24). Thus (23) holds.

The dimensions now force the complementary relation

$$\boxed{V_{i+1}^+(x) \oplus U_i(x) = \mathbb{R}^d.} \quad (30)$$

Indeed,

$$\dim V_{i+1}^+ = m_{i+1} + \cdots + m_k = d - (m_1 + \cdots + m_i) = d - \dim U_i.$$

Step 5: construct the Oseledets splitting. Define

$$E_i(x) := V_i^+(x) \cap U_i(x), \quad 1 \leq i \leq k. \quad (31)$$

For $i = k$ this means $E_k = V_k^+$ because $U_k = \mathbb{R}^d$. From (30), every $v \in V_i^+$ decomposes uniquely as $v = v_{i+1} + u_i$ with $v_{i+1} \in V_{i+1}^+$ and $u_i \in U_i$. Since both v and v_{i+1} lie in V_i^+ , the vector u_i belongs to E_i . Therefore

$$V_i^+ = V_{i+1}^+ \oplus E_i. \quad (32)$$

Iterating (32) from $V_1^+ = \mathbb{R}^d$ gives the direct sum

$$\boxed{\mathbb{R}^d = E_1(x) \oplus \cdots \oplus E_k(x),} \quad (33)$$

and $\dim E_i = m_i$.

The maps $x \mapsto E_i(x)$ are measurable. On each fixed-multiplicity stratum, V_i^+ and U_i are measurable Grassmannian-valued maps, and (30) gives a measurable transverse direct-sum chart; intersection is continuous on the open set of pairs having the fixed transverse dimensions. Equation (22) and invertibility imply

$$\boxed{A(x)E_i(x) = E_i(Tx).} \quad (34)$$

Step 6: the two time-direction growth rates. If $0 \neq v \in E_i(x)$, then $v \in V_i^+(x)$. For $i < k$, relation (23) with index i gives $V_{i+1}^+(x) \cap U_i(x) = \{0\}$, and since $v \in U_i(x)$ we obtain $v \notin V_{i+1}^+(x)$. For $i = k$ the same conclusion is automatic because $V_{k+1}^+(x) = \{0\}$. Hence the forward MET gives

$$\lim_{n \rightarrow +\infty} \frac{1}{n} \log \|A_n(x)v\| = \lambda_i(x). \quad (35)$$

Likewise $v \in U_i(x)$. If $i > 1$, relation (23) with index $i-1$ says $V_i^+(x) \cap U_{i-1}(x) = \{0\}$; since $v \in V_i^+(x)$ this implies $v \notin U_{i-1}(x)$. For $i = 1$ the conclusion is automatic from $U_0(x) = \{0\}$. By (21),

$$\lim_{n \rightarrow +\infty} \frac{1}{n} \log \|A_{-n}(x)v\| = -\lambda_i(x). \quad (36)$$

Writing $m = -n \rightarrow -\infty$ turns (36) into

$$\boxed{\lim_{m \rightarrow -\infty} \frac{1}{m} \log \|A_m(x)v\| = \lambda_i(x).} \quad (37)$$

Together, (35) and (37) are exactly the asserted two-sided Lyapunov law.

Finally the functions λ_i and the multiplicities m_i are the same T -invariant data already supplied by *Oseledets multiplicative ergodic theorem: filtration form*. They are therefore constant almost everywhere when T is ergodic. This completes the proof. $\square$

Lemma F04.7. Tempering-kernel lemma

Statement. Let $(X, \mathcal{A}, \mu, T)$ be probability-preserving with T invertible and bimeasurable. Let $r : X \rightarrow [1, \infty)$ be measurable. If $\log r \in L^1(\mu)$, then for almost every

x ,

$$\lim_{|n| \rightarrow \infty} \frac{1}{|n|} \log r(T^n x) = 0.$$

A second sufficient criterion, which does not require $\log r \in L^1$, is the following: if $r : X \rightarrow (0, \infty)$ is finite measurable and $h := \log r \circ T - \log r \in L^1(\mu)$, then the same two-sided orbit-sublinear conclusion holds. More generally, whenever this orbit-sublinear conclusion holds, then for every $\varepsilon > 0$

$$r_\varepsilon(x) := \sup_{n \in \mathbb{Z}} r(T^n x) e^{-\varepsilon|n|}$$

is finite almost everywhere, measurable, satisfies $r_\varepsilon \geq r$, and obeys

$$e^{-\varepsilon} r_\varepsilon(x) \leq r_\varepsilon(Tx) \leq e^\varepsilon r_\varepsilon(x), \quad r_\varepsilon(T^n x) \leq r_\varepsilon(x) e^{\varepsilon|n|} \quad (n \in \mathbb{Z}).$$

Moreover r_ε is itself tempered in the exact sense $|n|^{-1} \log r_\varepsilon(T^n x) \rightarrow 0$. Thus it is a tempered majorant that is also ε -slowly varying. In particular, if $s : X \rightarrow (0, \infty)$ and $|\log s| \in L^1$, applying the lemma to $r = \max\{s, s^{-1}\}$ gives one tempered slowly varying majorant controlling both s and s^{-1} .

Proof. We separate the argument into the orbit-tail statement and the construction of the slowly varying envelope.

Step 1: an L^1 logarithm is sublinear along almost every orbit. Assume first that $r \geq 1$ and $\log r \in L^1(\mu)$. Fix $\eta > 0$ and put $g := \log r \geq 0$. For positive time, invariance of μ gives

$$\sum_{n=1}^{\infty} \mu\{x : g(T^n x) > \eta n\} = \sum_{n=1}^{\infty} \mu\{x : g(x) > \eta n\}. \quad (1)$$

The series on the right is finite. Indeed, for every $t \geq 0$,

$$\sum_{n=1}^{\infty} \mathbf{1}_{\{t > \eta n\}} \leq \frac{t}{\eta},$$

so Tonelli's theorem yields

$$\sum_{n=1}^{\infty} \mu\{g > \eta n\} \leq \frac{1}{\eta} \int_X g d\mu < \infty. \quad (2)$$

By *Borel-Cantelli lemmas* (the first Borel-Cantelli lemma), for almost every x there is $N_+(x, \eta)$ such that

$$g(T^n x) \leq \eta n \quad (n \geq N_+(x, \eta)). \quad (3)$$

Because T is invertible and measure preserving, exactly the same calculation with T^{-1} gives, for almost every x , an $N_-(x, \eta)$ such that

$$g(T^{-n} x) \leq \eta n \quad (n \geq N_-(x, \eta)). \quad (4)$$

Intersect the full-measure sets obtained for the countable sequence $\eta = 1/m$, $m \geq 1$.

Since $g \geq 0$, (3)-(4) then imply

$$\boxed{\lim_{|n| \rightarrow \infty} \frac{g(T^n x)}{|n|} = 0} \quad (5)$$

for every x in the resulting full-measure invariant set. This proves the first assertion.

Notice that Birkhoff averages are not needed here: the quantity in (5) is a *single orbit value divided by time*, not an average. The summability in (2) is the precise mechanism that gives the claim.

Step 1b: an integrable one-step logarithmic ratio also implies temperedness.

The preceding sufficient condition $\log r \in L^1$ is convenient, but later regularity functions need not themselves have integrable logarithm. Let now $r : X \rightarrow (0, \infty)$ be finite and measurable and assume instead that

$$h(x) := \log r(Tx) - \log r(x) \in L^1(\mu). \quad (5a)$$

Birkhoff pointwise ergodic theorem gives

$$\frac{1}{n} \sum_{j=0}^{n-1} h(T^j x) \longrightarrow \mathbb{E}(h \mid \mathcal{I})(x) \quad \text{a.e.} \quad (5b)$$

The left side telescopes:

$$\frac{1}{n} \sum_{j=0}^{n-1} h(T^j x) = \frac{\log r(T^n x) - \log r(x)}{n}. \quad (5c)$$

We claim that the invariant limit in (5b) is zero. This needs a short argument because $\log r$ itself was not assumed integrable. Put $f := \log r$ and $X_M := \{|f| \leq M\}$. Since f is finite almost everywhere, $\bigcup_{M \geq 1} X_M$ has full measure. By Poincaré recurrence (*Poincaré recurrence theorem*), almost every point of every X_M returns to X_M infinitely often. Along such a return sequence $n_\ell \rightarrow \infty$,

$$\left| \frac{f(T^{n_\ell} x) - f(x)}{n_\ell} \right| \leq \frac{2M}{n_\ell} \longrightarrow 0. \quad (5d)$$

But the full sequence in (5c) has the limit given by (5b); hence that limit must be zero at almost every recurrent point of X_M . Taking the union over M proves

$$\frac{\log r(T^n x)}{n} \longrightarrow 0 \quad (n \rightarrow +\infty). \quad (5e)$$

Applying the same argument to T^{-1} (whose one-step coboundary is $-h \circ T^{-1} \in L^1$) gives the negative-time half. Therefore

$$\boxed{\lim_{|n| \rightarrow \infty} \frac{\log r(T^n x)}{|n|} = 0.} \quad (5f)$$

Thus an integrable logarithmic one-step ratio is a second, independent route to temperedness.

Step 2: the tempering envelope is finite and measurable. Now assume only the orbit-sublinear property (5) for the given measurable $r : X \rightarrow [1, \infty)$. Fix $\varepsilon > 0$ and define

$$r_\varepsilon(x) := \sup_{n \in \mathbb{Z}} r(T^n x) e^{-\varepsilon|n|}. \quad (6)$$

The supremum is over a countable family of measurable functions, hence r_ε is measurable. The term $n = 0$ shows immediately that $r_\varepsilon(x) \geq r(x)$.

For an x satisfying (5), choose $N = N(x, \varepsilon)$ so large that

$$\log r(T^n x) \leq \frac{\varepsilon}{2}|n| \quad (|n| \geq N). \quad (7)$$

Therefore, for $|n| \geq N$,

$$r(T^n x) e^{-\varepsilon|n|} \leq e^{-\varepsilon|n|/2} \leq 1. \quad (8)$$

Only the finitely many terms with $|n| < N$ can be larger than 1, so the supremum in (6) is finite. Thus $r_\varepsilon(x) < \infty$ almost everywhere.

Step 3: one-step slow variation. Shift the index in (6):

$$r_\varepsilon(Tx) = \sup_{n \in \mathbb{Z}} r(T^{n+1}x) e^{-\varepsilon|n|} = \sup_{m \in \mathbb{Z}} r(T^m x) e^{-\varepsilon|m-1|}. \quad (9)$$

The elementary inequality

$$|m| - 1 \leq |m - 1| \leq |m| + 1 \quad (10)$$

gives

$$e^{-\varepsilon} e^{-\varepsilon|m|} \leq e^{-\varepsilon|m-1|} \leq e^\varepsilon e^{-\varepsilon|m|}. \quad (11)$$

Multiplying by $r(T^m x)$ and taking suprema over m yields

$$\boxed{e^{-\varepsilon} r_\varepsilon(x) \leq r_\varepsilon(Tx) \leq e^\varepsilon r_\varepsilon(x)}. \quad (12)$$

Iterating (12) forward and backward gives, for every $n \in \mathbb{Z}$,

$$\boxed{r_\varepsilon(T^n x) \leq r_\varepsilon(x) e^{\varepsilon|n|}}. \quad (13)$$

This is the required ε -slow variation. We also verify the standard, stronger terminology: r_ε itself is tempered, meaning that its logarithm is sublinear along the orbit. Fix $0 < \delta < \varepsilon$. From (5), for the present x there is M such that

$$\log r(T^m x) \leq \delta|m| \quad (|m| \geq M). \quad (14)$$

For such m and every integer k , the triangle inequality gives

$$\log r(T^m x) - \varepsilon|m - k| \leq \delta|m| - \varepsilon|m - k| \leq \delta|k| - (\varepsilon - \delta)|m - k| \leq \delta|k|. \quad (15)$$

For the finitely many indices $|m| < M$, put $C := \max_{|m| < M} \log r(T^m x)$. Their contribution to the logarithm of $r_\varepsilon(T^k x)$ is at most

$$C - \varepsilon \min_{|m| < M} |m - k|, \quad (16)$$

which tends to $-\infty$ linearly as $|k| \rightarrow \infty$. Hence, for all sufficiently large $|k|$,

$$0 \leq \frac{1}{|k|} \log r_\varepsilon(T^k x) \leq \delta. \quad (17)$$

Since $\delta > 0$ was arbitrary,

$$\lim_{|k| \rightarrow \infty} \frac{1}{|k|} \log r_\varepsilon(T^k x) = 0. \quad (18)$$

Thus “tempered” and “ ε -slowly varying” are both justified, but they are logically distinct properties.

Step 4: simultaneous control of a function and its reciprocal. Let $s : X \rightarrow (0, \infty)$ satisfy $|\log s| \in L^1$. Set

$$r(x) := \max\{s(x), s(x)^{-1}\}. \quad (19)$$

Then $r \geq 1$ and

$$\log r = |\log s| \in L^1. \quad (20)$$

Applying Steps 1-3 to this r gives a finite measurable r_ε satisfying (12)-(13) and

$$s(x) \leq r(x) \leq r_\varepsilon(x), \quad s(x)^{-1} \leq r(x) \leq r_\varepsilon(x). \quad (21)$$

Thus one and the same tempered majorant controls both s and s^{-1} , as claimed. $\square$

Proposition F04.8. Determinant and symplectic constraints on spectra

Statement. If $A : X \rightarrow SL(d, \mathbb{R})$ is invertible and $\log^+ \|A\| + \log^+ \|A^{-1}\| \in L^1$, then the multiplicity-weighted Lyapunov exponents satisfy

$$\sum_{j=1}^d \chi_j(x) = 0$$

almost everywhere. If instead $A : X \rightarrow Sp(2d, \mathbb{R})$ satisfies the same two-sided logarithmic integrability hypothesis, then, after ordering with multiplicity,

$$\chi_j(x) = -\chi_{2d+1-j}(x)$$

almost everywhere.

Proof. We treat the determinant-one and symplectic assertions separately.

Step 1: determinant one forces zero total Lyapunov sum. Assume $A(x) \in SL(d, \mathbb{R})$ almost everywhere. Then every product $A_n(x)$ also belongs to $SL(d, \mathbb{R})$, so

$$\det A_n(x) = 1 \quad (n \geq 1). \quad (1)$$

The one-dimensional space $\Lambda^d \mathbb{R}^d$ carries the scalar map $\Lambda^d A_n$, whose operator norm is $|\det A_n|$. Hence

$$\frac{1}{n} \log \|\Lambda^d A_n(x)\| = \frac{1}{n} \log |\det A_n(x)| = 0. \quad (2)$$

By *Lyapunov spectrum from exterior powers* the limit on the left equals the sum of all Lyapunov exponents repeated with multiplicity. Therefore

$$\boxed{\chi_1(x) + \cdots + \chi_d(x) = 0} \quad (3)$$

for almost every x .

The two-sided logarithmic integrability assumed in the statement ensures that all χ_j are finite, so (3) is an ordinary real equality, not an extended-real convention.

Step 2: singular values of a symplectic matrix are reciprocal in pairs. Let $M \in Sp(2d, \mathbb{R})$ and let

$$J = \begin{pmatrix} 0 & I_d \\ -I_d & 0 \end{pmatrix}.$$

The symplectic identity is

$$M^T J M = J. \quad (4)$$

Since $J^{-1} = -J$ and J is orthogonal, (4) gives

$$M^{-1} = J^{-1} M^T J = -J M^T J. \quad (5)$$

Left and right multiplication by orthogonal matrices does not change singular values, and transposition does not change them either. Thus M^{-1} and M have the same singular values as unordered multisets.

On the other hand, if

$$\sigma_1(M) \geq \cdots \geq \sigma_{2d}(M) > 0,$$

then the singular values of the inverse, written in decreasing order, are

$$\sigma_1(M^{-1}) = \sigma_{2d}(M)^{-1}, \quad \dots, \quad \sigma_{2d}(M^{-1}) = \sigma_1(M)^{-1}. \quad (6)$$

Because the ordered singular-value lists of M^{-1} and M are equal, (6) implies term by term

$$\boxed{\sigma_j(M) \sigma_{2d+1-j}(M) = 1 \quad (1 \leq j \leq 2d).} \quad (7)$$

This proves the finite-time reciprocal-pairing identity directly from the symplectic equation.

Step 3: pass the finite-time identity to Lyapunov exponents. If $A(x) \in Sp(2d, \mathbb{R})$ almost everywhere, then every cocycle product $A_n(x)$ is symplectic. Apply (7) with $M = A_n(x)$:

$$\log \sigma_j(A_n(x)) = -\log \sigma_{2d+1-j}(A_n(x)). \quad (8)$$

To pass to the limit using only the stated exterior-power conclusions, note that *Singular values and exterior powers* gives, for $1 \leq j \leq 2d$,

$$\log \sigma_j(A_n) = \log \|\Lambda^j A_n\| - \log \|\Lambda^{j-1} A_n\|. \quad (9)$$

Under the present two-sided integrability all Lyapunov exponents are finite, so *Lya-*

punov spectrum from exterior powers applies to both exterior-power terms and yields

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sigma_j(A_n(x)) = \Gamma_j(x) - \Gamma_{j-1}(x) = \chi_j(x). \quad (10)$$

Divide (8) by n and let $n \rightarrow \infty$. Using (10) for both indices gives

$$\boxed{\chi_j(x) = -\chi_{2d+1-j}(x) \quad (1 \leq j \leq 2d)} \quad (11)$$

for almost every x , which is the desired symplectic spectral symmetry.

In particular, if $2d$ is even there is no unpaired central index; zero exponents, when present, occur in a self-opposite block of even total multiplicity unless additional symplectic structure further decomposes it. $\square$

Lemma F04.9. Exponential gap forces invariant splitting

Statement. Let T preserve a probability measure and let $0 \rightarrow E \rightarrow V \xrightarrow{p} F \rightarrow 0$ be a measurable short exact sequence of finite-rank linear cocycles whose one-step maps are fiberwise isomorphisms. Equip the bundles with measurable Euclidean norms and assume $\log^+ \|T_V(x)\| \in L^1$. Suppose that for almost every x there are numbers $\lambda_E > \lambda_F$, with $\lambda_E \in \mathbb{R}$ and $\lambda_F \in \mathbb{R} \cup \{-\infty\}$, such that $N^{-1} \log \|T_E^N(x)e\| \rightarrow \lambda_E$ and $N^{-1} \log \|T_F^N(x)f\| \rightarrow \lambda_F$, uniformly over the corresponding unit spheres in the fixed fiber. Then there exists a measurable lift $\sigma_x : F_x \rightarrow V_x$ with $p_x \sigma_x = \text{id}_{F_x}$ such that $V_x = E_x \oplus \sigma_x(F_x)$ and $T_V(x)\sigma_x(F_x) = \sigma_{Tx}(F_{Tx})$ for almost every x . Thus the extension splits measurably and invariantly.

Proof. We write the proof in enough detail to make clear where the exponent gap is used and why no form of the multiplicative ergodic theorem is being used implicitly. All statements below are restricted to one common T -invariant full-measure set on which the hypotheses and the orbitwise estimates that we derive hold.

Step 1: choose a measurable reference complement and write the cocycle in blocks. Give V its measurable Euclidean metric. The orthogonal complement $E_x^\perp$ is a measurable subbundle and the quotient map $p_x : E_x^\perp \rightarrow F_x$ is an isomorphism. Its inverse gives a measurable reference lift

$$\sigma_{0,x} : F_x \longrightarrow V_x, \quad p_x \circ \sigma_{0,x} = \text{id}_{F_x},$$

and hence

$$V_x = E_x \oplus \sigma_{0,x}(F_x). \quad (1)$$

We transport the quotient norm to F_x through this identification. Relative to (1), the one-step cocycle has the upper-triangular form

$$T_V(x) = \begin{pmatrix} T_E(x) & U(x) \\ 0 & T_F(x) \end{pmatrix}, \quad (2)$$

where

$$U(x) : F_x \longrightarrow E_{Tx}.$$

Because the coordinate projections in the orthogonal splitting have norm one,

$$\|U(x)\| \leq \|T_V(x)\|. \quad (3)$$

Thus $g(x) := \log^+ \|U(x)\|$ belongs to L^1 .

Step 2: the off-diagonal block has no exponential one-step spikes along a typical orbit. We claim that for almost every x ,

$$\frac{1}{n} \log^+ \|U(T^n x)\| \longrightarrow 0. \quad (4)$$

Fix $\varepsilon > 0$. Since $g \geq 0$ is integrable,

$$\sum_{n=1}^{\infty} \mu\{g > \varepsilon n\} \leq \frac{1}{\varepsilon} \int g \, d\mu + 1 < \infty. \quad (5)$$

Indeed, the sum of the tail probabilities is bounded, up to the harmless first term, by the layer-cake integral of g/ε . The first Borel–Cantelli lemma, *Borel–Cantelli lemmas*, therefore says that for almost every x only finitely many n satisfy $g(T^n x) > \varepsilon n$. Since ε is arbitrary, first over a countable sequence $\varepsilon = 1/m$ and then by monotonicity, (4) follows.

This elementary estimate is the only place where the ambient integrability hypothesis is used in the proof.

Step 3: translate the uniform exponent hypotheses into operator estimates. For $n \geq 0$ write

$$T_E^n(x) : E_x \rightarrow E_{T^n x}, \quad T_F^n(x) : F_x \rightarrow F_{T^n x}$$

for the iterates, with the zeroth iterate equal to the identity.

First suppose that λ_F is finite. Choose $\varepsilon > 0$ so small that

$$\lambda_E - \lambda_F - 3\varepsilon > 0. \quad (6)$$

Uniform convergence on the unit sphere of E_x gives, for all sufficiently large n ,

$$\inf_{\|e\|=1} \|T_E^n(x)e\| \geq e^{(\lambda_E - \varepsilon)n}.$$

Since $T_E^n(x)$ is invertible,

$$\|(T_E^n(x))^{-1}\| \leq e^{-(\lambda_E - \varepsilon)n}. \quad (7)$$

Likewise, uniform convergence on the unit sphere of F_x gives

$$\|T_F^n(x)\| \leq e^{(\lambda_F + \varepsilon)n} \quad (8)$$

for all sufficiently large n . Finally, (4) gives

$$\|U(T^n x)\| \leq e^{\varepsilon n} \quad (9)$$

for all sufficiently large n .

If $\lambda_F = -\infty$, choose instead a finite number $\beta < \lambda_E$ and then $\varepsilon > 0$ with

$$\lambda_E - \beta - 2\varepsilon > 0. \quad (10)$$

Uniform convergence to $-\infty$ means precisely that, for all sufficiently large n ,

$$\|T_F^n(x)\| \leq e^{\beta n}. \quad (11)$$

The estimates (7) and (9) remain valid. Thus both cases give an exponential margin between the inverse growth on E and the forward growth on F .

Step 4: solve the invariant-graph equation by a convergent series. Any other lift of F_x into V_x has the form

$$\sigma_x(f) = \sigma_{0,x}(f) + \tau_x(f), \quad \tau_x : F_x \rightarrow E_x. \quad (12)$$

The graph of τ is invariant exactly when

$$T_E(x)\tau_x + U(x) = \tau_{T_x}T_F(x). \quad (13)$$

Motivated by solving (13) backwards along the fast E direction, define

$$\tau_x := - \sum_{n=0}^{\infty} (T_E^{n+1}(x))^{-1} U(T^n x) T_F^n(x). \quad (14)$$

This is the graph-series formula used in Filip's proof.

We now prove that (14) converges in operator norm. In the finite λ_F case, (7)-(9) imply that all sufficiently large summands have norm at most

$$\exp(-(\lambda_E - \varepsilon)(n+1) + \varepsilon n + (\lambda_F + \varepsilon)n) \leq C_x e^{-\delta n}, \quad (15)$$

where

$$\delta := \lambda_E - \lambda_F - 3\varepsilon > 0.$$

In the $\lambda_F = -\infty$ case, (7), (9), and (11) give the same conclusion with

$$\delta := \lambda_E - \beta - 2\varepsilon > 0. \quad (16)$$

The finitely many initial terms are harmless. Hence the series converges absolutely in the finite-dimensional operator space $\text{Hom}(F_x, E_x)$.

Each partial sum in (14) is measurable in x on the full-measure set where it is defined, and the pointwise operator-norm limit of measurable finite-dimensional matrix-valued maps is measurable. Thus $x \mapsto \tau_x$ is measurable.

Step 5: verify the graph equation rather than treating the series formally. For $n \geq 1$ the cocycle identities give

$$T_E^{n+1}(x) = T_E^n(Tx)T_E(x), \quad T_F^n(x) = T_F^{n-1}(Tx)T_F(x). \quad (17)$$

Therefore

$$T_E(x)(T_E^{n+1}(x))^{-1} = (T_E^n(Tx))^{-1}. \quad (18)$$

Because the series is absolutely convergent, we may multiply it by the fixed one-step maps and reindex term by term. Using (18),

$$\begin{aligned} T_E(x)\tau_x + U(x) &= - \sum_{n=1}^{\infty} (T_E^n(Tx))^{-1} U(T^n x) T_F^n(x) \\ &= - \sum_{m=0}^{\infty} (T_E^{m+1}(Tx))^{-1} U(T^{m+1}x) T_F^m(Tx) T_F(x) \\ &= \tau_{Tx} T_F(x). \end{aligned}$$

Thus (13) holds almost everywhere.

Step 6: obtain the invariant direct-sum splitting. Define σ_x by (12). Since $\tau_x(f) \in E_x$ and $p_x \sigma_{0,x}(f) = f$,

$$p_x \sigma_x(f) = f. \quad (19)$$

Consequently σ_x is injective and

$$V_x = E_x \oplus \sigma_x(F_x). \quad (20)$$

Finally, (13) gives for every $f \in F_x$

$$\begin{aligned} T_V(x)\sigma_x(f) &= T_E(x)\tau_x(f) + U(x)f + \sigma_{0,Tx}(T_F(x)f) \\ &= \tau_{Tx}(T_F(x)f) + \sigma_{0,Tx}(T_F(x)f) \\ &= \sigma_{Tx}(T_F(x)f). \end{aligned}$$

Hence

$$T_V(x)\sigma_x(F_x) = \sigma_{Tx}(F_{Tx}),$$

because $T_F(x)$ is onto. The complement is therefore measurable and invariant, which is exactly the claimed splitting. $\square$

4.3 Worked examples

Worked example: Linear Cocycles and Oseledets Theory: a model computation

Consider the constant cocycle

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

Then

$$A^n = \begin{pmatrix} 3^n & 0 \\ 0 & 2^{-n} \end{pmatrix}.$$

The singular values are $\sigma_1(A^n) = 3^n$ and $\sigma_2(A^n) = 2^{-n}$, so

$$\lambda_1 = \log 3, \quad \lambda_2 = -\log 2.$$

The Oseledets spaces are the coordinate axes: $E_1 = \mathbb{R}e_1$ and $E_2 = \mathbb{R}e_2$. On the second exterior power, $\Lambda^2 A$ is multiplication by $\det A = 3/2$, and indeed

$$\lambda_1 + \lambda_2 = \log(3/2).$$

This one calculation contains the general exterior-power principle in its simplest form.

Boundary case: what fails at the edge of the hypotheses

Take

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = 2I + N, \quad N^2 = 0.$$

Then

$$A^n = 2^n \left(I + \frac{n}{2} N \right).$$

The polynomial factor n disappears after dividing $\log \|A^n\|$ by n , so both Lyapunov exponents equal $\log 2$. With no exponential gap, asymptotic growth alone does not force a preferred two-way splitting. This is why gap hypotheses appear in invariant-splitting criteria.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

4.4 Chapter summary

What to remember

- Lyapunov exponents measure exponential growth along invariant directions.
- Exterior powers turn sums of exponents into top-exponent problems.
- An exponential gap is what converts rate information into a distinguished splitting.
- Temperedness controls the nonuniform constants that later enter Pesin theory.

Looking ahead

F05 changes direction from growth to arithmetic. The same Fourier and resonance estimates developed there will later control the small divisors that appear in cohomological equations and KAM theory.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

4.5 Exercises and problems

Exercise 4.1

Difficulty. Core.

Related results. Singular values and exterior powers; Exterior-power cocycle construction; Top Lyapunov exponent for each exterior power.

Problem. Work through the worked example 'a constant diagonal $GL(2, \mathbb{R})$ cocycle with two exponents'. For Singular values and exterior powers, Exterior-power cocycle construction, and Top Lyapunov exponent for each exterior power, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 4.2

Difficulty. Core.

Related results. Oseledets multiplicative ergodic theorem: filtration form; Oseledets theorem: invertible splitting form; Lyapunov spectrum from exterior powers.

Problem. Analyze the boundary example 'a triangular cocycle with coincident exponents where invariant splitting is not forced'. For Oseledets multiplicative ergodic theorem: filtration form, Oseledets theorem: invertible splitting form, and Lyapunov spectrum from exterior powers, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement

statement.

Exercise 4.3

Difficulty. Advanced.

Related results. Top Lyapunov exponent for each exterior power.

Problem. For Top Lyapunov exponent for each exterior power, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 4.4

Difficulty. Advanced.

Related results. Oseledets multiplicative ergodic theorem: filtration form.

Problem. Stress-test Oseledets multiplicative ergodic theorem: filtration form. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 4.5

Difficulty. Advanced.

Related results. Singular values and exterior powers; Exterior-power cocycle construction.

Problem. Synthesis problem. Use Singular values and exterior powers and Exterior-power cocycle construction to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 4.6

Difficulty. Core.

Related results. Temperedness lemma.

Problem. Work directly with Temperedness lemma. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the exam-

ple, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 4.7

Difficulty. Core.

Related results. Determinant and symplectic constraints on spectra.

Problem. Work directly with Determinant and symplectic constraints on spectra. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 4.8

Difficulty. Core.

Related results. Exponential gap forces invariant splitting.

Problem. Work directly with Exponential gap forces invariant splitting. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Arithmetic of Frequencies

Why this chapter matters

Rotations and quasiperiodic systems are simple geometrically but subtle arithmetically. The question is not merely whether a frequency is irrational; it is how well it can be approximated by rationals. That approximation rate governs the size of Fourier denominators and therefore the stability of conjugacy equations. This chapter builds the arithmetic vocabulary used throughout KAM theory, co-cycle reducibility, circle dynamics, and spectral theory: continued fractions in one dimension, Diophantine inequalities in several dimensions, resonant slabs, and the Brjuno condition.

Prerequisites. Elementary number theory, Euclidean algorithm, Lebesgue measure, and basic Fourier notation.

Learning goals

- Compute continued-fraction convergents and use best-approximation properties.
- Translate Diophantine inequalities into lower bounds for Fourier denominators.
- Estimate the measure of resonant frequency sets.
- Distinguish Diophantine, Liouville, and Brjuno arithmetic.

Conceptual roadmap

Continued fractions give exact one-dimensional approximation data. Dirichlet shows that good rational approximations always exist; Diophantine conditions say they are not *too* good. Resonant-slab estimates quantify how much parameter space violates a chosen inequality. Borel–Cantelli then turns summable estimates into almost-everywhere arithmetic conclusions.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

5.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F05.1. Continued fraction

For irrational α , write $\alpha = [a_0; a_1, a_2, \dots]$. Its convergents p_n/q_n are defined by $p_n = a_n p_{n-1} + p_{n-2}$ and $q_n = a_n q_{n-1} + q_{n-2}$, with the standard initial conditions.

Definition F05.2. Diophantine frequency

A vector $\omega \in \mathbb{R}^d$ is (γ, τ) -*Diophantine* if $\text{dist}(\langle k, \omega \rangle, \mathbb{Z}) \geq \gamma |k|^{-\tau}$ for every $0 \neq k \in \mathbb{Z}^d$, using the lattice norm fixed in the surrounding theorem.

Definition F05.3. Resonance module

At a specified tolerance, a *resonance module* for ω is the $\mathbb{Z}$ -span of integer vectors k for which $\langle k, \omega \rangle$ is exactly resonant or lies within the stated quantitative resonance threshold. Its rank is the rank of that $\mathbb{Z}$ -module.

Definition F05.4. Brjuno condition

An irrational number α with continued-fraction denominators q_n is *Brjuno* if $\sum_{n \geq 0} \log q_{n+1}/q_n < \infty$.

5.2 Main results

Theorem F05.1. Dirichlet simultaneous approximation

Statement. Let $\omega \in \mathbb{R}^d$ and $Q \geq 1$ be an integer. Then there exist $1 \leq q \leq Q^d$ and $p \in \mathbb{Z}^d$ such that $\|q\omega - p\|_\infty < Q^{-1}$. Dually, for every $\theta \in \mathbb{R}^d$ and integer $Q \geq 1$ there exist $0 \neq k \in \mathbb{Z}^d$ with $\|k\|_\infty \leq Q$ and $m \in \mathbb{Z}$ such that $|\langle k, \theta \rangle - m| < Q^{-d}$.

Proof. We prove the two assertions separately. Both are direct applications of the pigeonhole principle, but it is useful to keep track of the counting and of the strict inequalities because those are exactly what determines the powers of Q in the conclusion.

Simultaneous approximation. For a vector $y \in \mathbb{R}^d$, write $\{y\} \in [0, 1)^d$ for the vector obtained by taking the fractional part of each coordinate. Consider the $Q^d + 1$ points

$$\{0\omega\}, \{1\omega\}, \dots, \{Q^d\omega\} \quad \text{in } [0, 1)^d.$$

Partition the half-open cube $[0, 1)^d$ into the Q^d half-open cubes

$$\prod_{j=1}^d \left[\frac{r_j}{Q}, \frac{r_j + 1}{Q} \right), \quad r_j \in \{0, 1, \dots, Q - 1\}.$$

Since there are more points than cubes, two of the fractional vectors, say $\{n_1\omega\}$ and $\{n_2\omega\}$ with $0 \leq n_1 < n_2 \leq Q^d$, lie in the same small cube. Set

$$q := n_2 - n_1.$$

Then

$$1 \leq q \leq Q^d.$$

For each $i = 1, 2$ choose $p_i \in \mathbb{Z}^d$ so that $n_i\omega = p_i + \{n_i\omega\}$. Subtracting gives

$$q\omega - (p_2 - p_1) = \{n_2\omega\} - \{n_1\omega\}.$$

Because the two fractional vectors lie in the same half-open cube of side Q^{-1} , every coordinate of the vector on the right has absolute value strictly smaller than Q^{-1} . With $p := p_2 - p_1 \in \mathbb{Z}^d$ we obtain

$$\|q\omega - p\|_\infty < Q^{-1},$$

which is the simultaneous assertion.

Dual approximation. Let

$$\mathcal{N} := \{0, 1, \dots, Q\}^d.$$

There are $(Q + 1)^d$ vectors in $\mathcal{N}$, hence strictly more than Q^d . To every $n \in \mathcal{N}$ associate the scalar fractional part

$$x_n := \{\langle n, \theta \rangle\} \in [0, 1).$$

Partition $[0, 1)$ into the Q^d half-open intervals

$$\left[\frac{r}{Q^d}, \frac{r + 1}{Q^d} \right), \quad r = 0, 1, \dots, Q^d - 1.$$

Two distinct vectors $n, n' \in \mathcal{N}$ have $x_n, x_{n'}$ in the same interval. Put

$$k := n - n'.$$

Then $k \neq 0$ and, coordinate by coordinate, $|k_j| \leq Q$, so $\|k\|_\infty \leq Q$. Write

$$\langle n, \theta \rangle = m_n + x_n, \quad \langle n', \theta \rangle = m_{n'} + x_{n'}$$

with $m_n, m_{n'} \in \mathbb{Z}$. Subtraction yields

$$\langle k, \theta \rangle - (m_n - m_{n'}) = x_n - x_{n'}.$$

The two fractional parts lie in the same interval of length Q^{-d} , so

$$|x_n - x_{n'}| < Q^{-d}.$$

Taking $m := m_n - m_{n'}$ proves

$$|\langle k, \theta \rangle - m| < Q^{-d}.$$

This is the required dual statement. $\square$

Theorem F05.2. Continued-fraction convergents

Statement. Let $\alpha = [a_0; a_1, a_2, \dots]$ be irrational and set $p_{-2} = 0, p_{-1} = 1, q_{-2} = 1, q_{-1} = 0, p_n = a_n p_{n-1} + p_{n-2}$ and $q_n = a_n q_{n-1} + q_{n-2}$. Then $p_n/q_n = [a_0; \dots, a_n]$, $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$, and with the complete quotient $\alpha_{n+1} = [a_{n+1}; a_{n+2}, \dots]$, $\alpha = (p_n \alpha_{n+1} + p_{n-1}) / (q_n \alpha_{n+1} + q_{n-1})$. Consequently $\alpha - p_n/q_n = (-1)^n / [q_n (q_n \alpha_{n+1} + q_{n-1})]$ and $1/[q_n (q_n + q_{n+1})] < |\alpha - p_n/q_n| < 1/(q_n q_{n+1})$.

Proof. We divide the argument into the algebraic identities and the approximation estimate.

Step 1: the recursion produces the finite convergents. For a real parameter $t > 0$, define

$$[a_0; \dots, a_n, t] := a_0 + \frac{1}{a_1 + \frac{1}{\ddots + \frac{1}{a_n + 1/t}}}.$$

A convenient induction gives the more general identity

$$[a_0; \dots, a_n, t] = \frac{p_n t + p_{n-1}}{q_n t + q_{n-1}}. \quad (1)$$

For $n = 0$, the right-hand side equals

$$\frac{a_0 t + 1}{t} = a_0 + \frac{1}{t},$$

so (1) holds. If it holds at level $n - 1$, then replacing the terminal parameter there by $a_n + 1/t$ gives

$$\frac{p_{n-1}(a_n + 1/t) + p_{n-2}}{q_{n-1}(a_n + 1/t) + q_{n-2}} = \frac{(a_n p_{n-1} + p_{n-2})t + p_{n-1}}{(a_n q_{n-1} + q_{n-2})t + q_{n-1}},$$

which is exactly (1) at level n by the defining recursion.

Letting $t \rightarrow \infty$ in (1) yields

$$[a_0; \dots, a_n] = \frac{p_n}{q_n}. \quad (2)$$

(The denominator q_n is positive for $n \geq 0$.)

Step 2: determinant identity. Set

$$D_n := p_n q_{n-1} - p_{n-1} q_n.$$

Using the recursion once,

$$\begin{aligned} D_n &= (a_n p_{n-1} + p_{n-2}) q_{n-1} - p_{n-1} (a_n q_{n-1} + q_{n-2}) \\ &= p_{n-2} q_{n-1} - p_{n-1} q_{n-2} = -D_{n-1}. \end{aligned}$$

At $n = 0$,

$$D_0 = p_0q_{-1} - p_{-1}q_0 = -1.$$

Therefore

$$p_nq_{n-1} - p_{n-1}q_n = (-1)^{n-1} \quad (3)$$

for every $n \geq 0$. In particular the numerator and denominator of each convergent are coprime, since any common divisor would divide the left-hand side of (3).

Step 3: insert the infinite tail. Let

$$\alpha_{n+1} := [a_{n+1}; a_{n+2}, \dots].$$

The continued fraction for α is obtained from the finite prefix by placing α_{n+1} after a_n . Thus (1) gives the complete-quotient identity

$$\alpha = \frac{p_n\alpha_{n+1} + p_{n-1}}{q_n\alpha_{n+1} + q_{n-1}}. \quad (4)$$

Subtracting p_n/q_n and using (3),

$$\begin{aligned} \alpha - \frac{p_n}{q_n} &= \frac{q_n(p_n\alpha_{n+1} + p_{n-1}) - p_n(q_n\alpha_{n+1} + q_{n-1})}{q_n(q_n\alpha_{n+1} + q_{n-1})} \\ &= \frac{q_np_{n-1} - p_nq_{n-1}}{q_n(q_n\alpha_{n+1} + q_{n-1})} \\ &= \frac{(-1)^n}{q_n(q_n\alpha_{n+1} + q_{n-1})}. \end{aligned} \quad (5)$$

This is the claimed exact error formula.

Step 4: compare the exact denominator with adjacent convergents. Because

$$\alpha_{n+1} = a_{n+1} + \frac{1}{\alpha_{n+2}}, \quad \alpha_{n+2} > 1,$$

we have

$$q_n\alpha_{n+1} + q_{n-1} = q_{n+1} + \frac{q_n}{\alpha_{n+2}}. \quad (6)$$

Since $0 < 1/\alpha_{n+2} < 1$, (6) implies the strict inequalities

$$q_{n+1} < q_n\alpha_{n+1} + q_{n-1} < q_{n+1} + q_n. \quad (7)$$

All quantities are positive, so taking reciprocals and multiplying by $1/q_n$ in (5) gives

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| \alpha - \frac{p_n}{q_n} \right| < \frac{1}{q_nq_{n+1}}.$$

This proves every assertion in the statement. □

Theorem F05.3. Best approximation property

Statement. Let p_n/q_n be the n th continued-fraction convergent of an irrational α . Then

$$\frac{1}{q_n + q_{n+1}} < |q_n \alpha - p_n| < \frac{1}{q_{n+1}}.$$

Moreover, if $0 < q < q_{n+1}$ and $p \in \mathbb{Z}$, then

$$|q\alpha - p| \geq |q_n \alpha - p_n|$$

unless $(p, q) = \pm(p_n, q_n)$.

Proof. Put

$$\delta_j := q_j \alpha - p_j.$$

By *Continued-fraction convergents*,

$$\delta_j = \frac{(-1)^j}{q_j \alpha_{j+1} + q_{j-1}}, \quad (1)$$

so consecutive errors have opposite signs. The same theorem also gives immediately

$$\frac{1}{q_n + q_{n+1}} < |\delta_n| < \frac{1}{q_{n+1}}. \quad (2)$$

Thus only the best-approximation assertion remains.

The recurrence for p_j, q_j gives

$$\delta_{n+1} = a_{n+1} \delta_n + \delta_{n-1}. \quad (3)$$

Multiplying all three errors by -1 if necessary, we may reason with the orientation

$$\delta_n > 0, \quad \delta_{n-1} < 0.$$

Write

$$r := \frac{|\delta_{n-1}|}{|\delta_n|} > 0.$$

Since δ_{n+1} has the sign opposite to δ_n , (3) implies

$$a_{n+1} |\delta_n| - |\delta_{n-1}| < 0,$$

and hence

$$r > a_{n+1}. \quad (4)$$

Now let $p \in \mathbb{Z}$ and $0 < q < q_{n+1}$. The determinant identity in *Continued-fraction convergents* says

$$\det \begin{pmatrix} q_n & q_{n-1} \\ p_n & p_{n-1} \end{pmatrix} = \pm 1.$$

Therefore the two integer vectors (q_n, p_n) and (q_{n-1}, p_{n-1}) form a basis of $\mathbb{Z}^2$. There are

unique integers A, B such that

$$(q, p) = A(q_n, p_n) + B(q_{n-1}, p_{n-1}). \quad (5)$$

Taking the linear form $(s, t) \mapsto s\alpha - t$ in (5) gives

$$q\alpha - p = A\delta_n + B\delta_{n-1} = (A - Br)\delta_n. \quad (6)$$

Suppose first, for contradiction, that

$$|q\alpha - p| < |\delta_n|.$$

By (6),

$$|A - Br| < 1. \quad (7)$$

There are three possibilities.

If $B = 0$, then (7) gives $|A| < 1$, hence $A = 0$; (5) would give $q = 0$, contrary to the hypothesis.

If $B > 0$, then (7) and (4) imply

$$A > Br - 1 > Ba_{n+1} - 1.$$

Because A and Ba_{n+1} are integers, $A \geq Ba_{n+1}$. Consequently

$$q = Aq_n + Bq_{n-1} \geq B(a_{n+1}q_n + q_{n-1}) = Bq_{n+1} \geq q_{n+1},$$

again a contradiction.

Finally, if $B < 0$, write $B = -C$ with $C > 0$. Then (7) reads $|A + Cr| < 1$. Using (4),

$$A < 1 - Cr < 1 - Ca_{n+1},$$

so integrality gives $A \leq -Ca_{n+1}$. Hence

$$q = Aq_n - Cq_{n-1} \leq -C(a_{n+1}q_n + q_{n-1}) = -Cq_{n+1} < 0,$$

contradicting $q > 0$.

We have proved

$$|q\alpha - p| \geq |q_n\alpha - p_n| \quad (0 < q < q_{n+1}). \quad (8)$$

For completeness, let us also identify the equality case. If equality holds in (8), then $|A - Br| = 1$. The same estimates as above show that $B > 0$ would force $q \geq q_{n+1}$ and $B < 0$ would force $q < 0$. Thus $B = 0$, and $|A| = 1$. Since $q > 0$, equation (5) forces $A = 1$ and therefore

$$(q, p) = (q_n, p_n).$$

If one allows denominators of either sign, the simultaneous negative pair $(-q_n, -p_n)$ is the only additional equality case. This is precisely the “up to sign” qualification in the statement.

Combining (8) with (2) proves the theorem. $\square$

Proposition F05.4. Diophantine classes and continued fractions

Statement. Let $\alpha = [a_0; a_1, \dots]$ be irrational with convergent denominators q_n . Then $\sup_n a_n < \infty$ iff $\sup_n q_{n+1}/q_n < \infty$ iff there is $\gamma > 0$ with $\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-1}$ for all $q \geq 1$. More generally, for every $\tau \geq 1$, such a bound $\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-\tau}$ holds for some $\gamma > 0$ iff $q_{n+1} = O(q_n^\tau)$. Finally, with $w(\alpha) = \sup\{w : \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-w} \text{ infinitely often}\}$, one has $w(\alpha) = \limsup_n (\log q_{n+1})/(\log q_n)$; hence α is Liouville iff this limsup is infinite.

Proof. Let

$$\alpha = [a_0; a_1, a_2, \dots] \in \mathbb{R} \setminus \mathbb{Q}, \quad \frac{p_n}{q_n}$$

be its convergents, with $q_n > 0$. Write

$$\|t\|_{\mathbb{R}/\mathbb{Z}} := \text{dist}(t, \mathbb{Z}).$$

Best approximation property gives the two facts used throughout:

$$\frac{1}{q_n + q_{n+1}} < \|q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{q_{n+1}}, \quad (1)$$

and, whenever $0 < q < q_{n+1}$,

$$\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \|q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \quad (2)$$

except for the harmless equality case at the convergent itself.

Step 1: bounded type. The continued-fraction recurrence is

$$q_{n+1} = a_{n+1}q_n + q_{n-1}, \quad 0 < q_{n-1} \leq q_n. \quad (3)$$

Hence

$$a_{n+1} \leq \frac{q_{n+1}}{q_n} \leq a_{n+1} + 1. \quad (4)$$

Thus the partial quotients (a_n) are bounded if and only if $\sup_n q_{n+1}/q_n < \infty$.

Assume first that

$$q_{n+1} \leq Mq_n \quad (5)$$

for every n , after increasing M to absorb finitely many initial indices. Given $q \geq 1$, choose n with $q_n \leq q < q_{n+1}$. From (1)–(2),

$$\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{1}{q_n + q_{n+1}} \geq \frac{1}{(1+M)q_n} \geq \frac{1}{(1+M)q}. \quad (6)$$

So α is badly approximable, equivalently it is $(\gamma, 1)$ -Diophantine in dimension one with $\gamma = (1+M)^{-1}$.

Conversely, suppose that for some $\gamma > 0$,

$$\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-1} \quad (q \geq 1). \quad (7)$$

Taking $q = q_n$ and using the upper bound in (1),

$$\frac{\gamma}{q_n} \leq \|q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{q_{n+1}},$$

so

$$q_{n+1} < \gamma^{-1} q_n. \quad (8)$$

Together with (4), this proves the classical equivalence

$$\sup_n a_n < \infty \iff \sup_n \frac{q_{n+1}}{q_n} < \infty \iff \inf_{q \geq 1} q \|q \alpha\|_{\mathbb{R}/\mathbb{Z}} > 0. \quad (9)$$

Step 2: general one-dimensional Diophantine exponents. Fix $\tau \geq 1$. Suppose

$$\|q \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-\tau} \quad (q \geq 1). \quad (10)$$

At a convergent denominator, (1) gives

$$\gamma q_n^{-\tau} < q_{n+1}^{-1},$$

hence

$$q_{n+1} < \gamma^{-1} q_n^\tau. \quad (11)$$

Conversely assume that

$$q_{n+1} \leq C q_n^\tau \quad (12)$$

for all n . If $q_n \leq q < q_{n+1}$, (1)-(2) imply

$$\|q \alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n+1}} \quad (5.1)$$

$$\geq \frac{1}{q_n + C q_n^\tau} \geq \frac{1}{(1+C) q_n^\tau} \geq \frac{1}{(1+C) q^\tau}. \quad (13)$$

Thus

$$\alpha \text{ is } (\gamma, \tau)\text{-Diophantine for some } \gamma > 0 \iff q_{n+1} = O(q_n^\tau). \quad (14)$$

The constants are quantitative: (10) gives $C = \gamma^{-1}$, while (12) gives $\gamma = (1+C)^{-1}$.

Step 3: Liouville growth. Define the approximation exponent in the distance normalization by

$$w(\alpha) := \sup \left\{ w > 0 : \|q \alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-w} \text{ for infinitely many } q \in \mathbb{N} \right\}. \quad (15)$$

We claim

$$w(\alpha) = \limsup_{n \rightarrow \infty} \frac{\log q_{n+1}}{\log q_n}. \quad (16)$$

The lower bound follows immediately from (1): along $q = q_n$,

$$\log \|q_n \alpha\|^{-1} > \log q_{n+1}. \quad (17)$$

For the reverse bound, if $q_n \leq q < q_{n+1}$, then by (1)–(2) and $q_{n+1} > q_n$,

$$\|q\alpha\|^{-1} \leq \|q_n\alpha\|^{-1} < q_n + q_{n+1} < 2q_{n+1}. \quad (18)$$

Since $\log q \geq \log q_n$, (18) yields

$$\frac{\log \|q\alpha\|^{-1}}{\log q} \leq \frac{\log 2 + \log q_{n+1}}{\log q_n}. \quad (19)$$

Taking limsups proves (16).

If Liouville means that for every $N > 0$ there are infinitely many rationals p/q with $|\alpha - p/q| < q^{-N}$, then this is equivalent to $w(\alpha) = \infty$ (the two exponents differ by one because $|\alpha - p/q| = |q\alpha - p|/q$). Hence (16) gives the denominator-growth criterion

$$\boxed{\alpha \text{ is Liouville} \iff \limsup_{n \rightarrow \infty} \frac{\log q_{n+1}}{\log q_n} = \infty.} \quad (20)$$

This completes all three claimed correspondences. $\square$

Theorem F05.5. Kronecker density criterion for torus translations

Statement. For $\omega \in \mathbb{R}^d$, the translation $R_\omega : x \mapsto x + \omega$ on $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is minimal (equivalently every orbit is dense) if and only if $\langle k, \omega \rangle \notin \mathbb{Z}$ for every $0 \neq k \in \mathbb{Z}^d$, equivalently $1, \omega_1, \dots, \omega_d$ are linearly independent over $\mathbb{Q}$.

Proof. Write $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ and let m denote normalized Haar (equivalently normalized Lebesgue) measure on the torus. For $k \in \mathbb{Z}^d$ let

$$\chi_k(x) := e^{2\pi i \langle k, x \rangle}$$

be the corresponding Fourier character.

A resonance prevents density. Suppose that some $0 \neq k \in \mathbb{Z}^d$ satisfies $\langle k, \omega \rangle \in \mathbb{Z}$. Then for every $x \in \mathbb{T}^d$ and $n \in \mathbb{Z}$,

$$\chi_k(x + n\omega) = \chi_k(x) e^{2\pi i n \langle k, \omega \rangle} = \chi_k(x).$$

Thus the whole orbit of x is contained in the level set $\chi_k^{-1}(\{\chi_k(x)\})$. Because $k \neq 0$, χ_k is not constant on $\mathbb{T}^d$, so this level set is a proper closed subset. The orbit cannot be dense. Hence minimality implies the absence of all nonzero integer resonances.

Absence of resonances forces density. Assume now that

$$\langle k, \omega \rangle \notin \mathbb{Z} \quad \text{for every } 0 \neq k \in \mathbb{Z}^d. \quad (1)$$

Fix $x \in \mathbb{T}^d$. For $k \neq 0$,

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} \chi_k(x + n\omega) &= \chi_k(x) \frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i n \langle k, \omega \rangle} \\ &= \chi_k(x) \frac{1 - e^{2\pi i N \langle k, \omega \rangle}}{N(1 - e^{2\pi i \langle k, \omega \rangle})}. \end{aligned} \quad (2)$$

The denominator in (2) is nonzero by (1), while the numerator is bounded by 2. Hence the average tends to 0, uniformly in x . For $k = 0$ the average is identically 1. Therefore, for every trigonometric polynomial P ,

$$\frac{1}{N} \sum_{n=0}^{N-1} P(x + n\omega) \longrightarrow \int_{\mathbb{T}^d} P \, dm, \quad (3)$$

uniformly in x .

Trigonometric polynomials are uniformly dense in $C(\mathbb{T}^d)$ by the Stone-Weierstrass theorem. If $\|f - P\|_\infty < \varepsilon$, then

$$\left| \frac{1}{N} \sum_{n=0}^{N-1} f(x + n\omega) - \int f \, dm \right| \leq 2\varepsilon + \left| \frac{1}{N} \sum_{n=0}^{N-1} P(x + n\omega) - \int P \, dm \right|.$$

Using (3) and then letting $\varepsilon \downarrow 0$ shows that

$$\frac{1}{N} \sum_{n=0}^{N-1} f(x + n\omega) \longrightarrow \int f \, dm \quad (4)$$

for every continuous f and every x .

Suppose the orbit of our fixed x were not dense. Its closure K would be a proper closed subset of the compact torus, so the complement contains a nonempty open set U . Choose a nonnegative continuous function f , not identically zero, whose support is contained in U . Then $\int f \, dm > 0$, because every nonempty open subset of the torus has positive Haar measure. But f vanishes on the whole orbit of x , so the left-hand side of (4) is 0 for every N , a contradiction. Hence the orbit of every x is dense, which is exactly minimality.

Finally, condition (1) is equivalent to rational independence of $1, \omega_1, \dots, \omega_d$. Indeed, a rational relation among these numbers can be cleared of denominators to give integers $m, k_1, \dots, k_d$, not all zero, with

$$k_1\omega_1 + \dots + k_d\omega_d = m.$$

The coefficients k_j cannot all vanish unless $m = 0$, so this is exactly a nonzero integer resonance. The converse is the same implication read backwards. $\square$

Theorem F05.6. Metric abundance of Diophantine frequencies

Statement. For $\tau > d$, define

$$\text{DC}_d(\gamma, \tau) := \left\{ \omega \in [0, 1]^d : \|\langle k, \omega \rangle\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma |k|^{-\tau} \text{ for every } 0 \neq k \in \mathbb{Z}^d \right\}.$$

There is $C_{d,\tau} < \infty$ such that

$$\text{Leb}([0, 1]^d \setminus \text{DC}_d(\gamma, \tau)) \leq C_{d,\tau} \gamma, \quad 0 < \gamma \leq 1.$$

Consequently $\bigcup_{\gamma>0} \text{DC}_d(\gamma, \tau)$ has full Lebesgue measure.

Proof. For $k \in \mathbb{Z}^d \setminus \{0\}$ and $\varepsilon > 0$, define the resonance strip

$$R(k, \varepsilon) := \left\{ \omega \in [0, 1]^d : \|\langle k, \omega \rangle\|_{\mathbb{R}/\mathbb{Z}} < \varepsilon \right\}.$$

The proof has two ingredients: a uniform estimate for one resonance strip and a summation over all lattice vectors.

Step 1: one resonance strip has measure at most 2ε . Choose an index j for which $k_j \neq 0$. Fix all coordinates of ω except ω_j . With the other coordinates frozen,

$$\langle k, \omega \rangle = k_j \omega_j + c$$

for a constant c . The map

$$[0, 1] \ni t \longmapsto k_j t + c \pmod{1} \in \mathbb{R}/\mathbb{Z}$$

is a $|k_j|$ -fold covering and preserves normalized Lebesgue measure. If $0 < \varepsilon \leq 1/2$, the arc $\{y : \|y\|_{\mathbb{R}/\mathbb{Z}} < \varepsilon\}$ has circle measure 2ε . Thus every one-dimensional slice of $R(k, \varepsilon)$ in the j -th coordinate has length 2ε . Fubini's theorem gives

$$\text{Leb } R(k, \varepsilon) = 2\varepsilon \quad (0 < \varepsilon \leq 1/2).$$

For $\varepsilon > 1/2$ the trivial bound by 1 suffices. Hence in all cases

$$\text{Leb } R(k, \varepsilon) \leq \min(1, 2\varepsilon) \leq 2\varepsilon. \tag{1}$$

Step 2: sum the resonance strips. By the definition of $\text{DC}_d(\gamma, \tau)$,

$$[0, 1]^d \setminus \text{DC}_d(\gamma, \tau) = \bigcup_{k \in \mathbb{Z}^d \setminus \{0\}} R(k, \gamma |k|^{-\tau}).$$

Therefore (1) and countable subadditivity give

$$\text{Leb}([0, 1]^d \setminus \text{DC}_d(\gamma, \tau)) \leq 2\gamma \sum_{k \in \mathbb{Z}^d \setminus \{0\}} |k|^{-\tau}. \tag{2}$$

For any fixed norm on $\mathbb{R}^d$, the number of lattice points in the shell $m \leq |k| < m+1$ is

$O_d(m^{d-1})$. Consequently

$$\sum_{k \neq 0} |k|^{-\tau} \leq C_d \sum_{m=1}^{\infty} m^{d-1-\tau} < \infty \quad \text{when } \tau > d.$$

Set

$$C_{d,\tau} := 2 \sum_{k \neq 0} |k|^{-\tau} < \infty.$$

Then (2) becomes the advertised quantitative estimate

$$\text{Leb}([0, 1]^d \setminus DC_d(\gamma, \tau)) \leq C_{d,\tau} \gamma, \quad 0 < \gamma \leq 1. \quad (3)$$

Step 3: pass from quantitative abundance to full measure. Take $\gamma_m = 1/m$. As m increases, the lower bound in the Diophantine condition weakens, so the sets $DC_d(1/m, \tau)$ increase. By (3),

$$\text{Leb}([0, 1]^d \setminus DC_d(1/m, \tau)) \leq \frac{C_{d,\tau}}{m} \rightarrow 0.$$

Continuity from below for the increasing good sets, or equivalently continuity from above for their decreasing complements, gives

$$\text{Leb}\left(\bigcup_{m \geq 1} DC_d(1/m, \tau)\right) = 1.$$

Since this countable union is contained in $\bigcup_{\gamma > 0} DC_d(\gamma, \tau)$, the latter set also has full Lebesgue measure. No independence or Borel-Cantelli input is needed for this estimate. $\square$

Lemma F05.7. Resonant slab measure estimate

Statement. Let $\Omega \subset [-R, R]^d$ be measurable, $k \in \mathbb{Z}^d \setminus \{0\}$, $m \in \mathbb{Z}$, and $\varepsilon > 0$. With $|k|_\infty = \max_j |k_j|$, $\text{Leb}\{\omega \in \Omega : |\langle k, \omega \rangle - m| < \varepsilon\} \leq 2(2R)^{d-1} \varepsilon / |k|_\infty$. More generally, for any finite family of resonances $(k_\alpha, m_\alpha, \varepsilon_\alpha)$, the measure of their union in Ω is at most $2(2R)^{d-1} \sum_\alpha \varepsilon_\alpha / |k_\alpha|_\infty$.

Proof. Choose an index $j \in \{1, \dots, d\}$ such that

$$|k_j| = |k|_\infty > 0.$$

Write a point of $\mathbb{R}^d$ as $\omega = (\omega_j, \omega')$, where ω' denotes the remaining $d-1$ coordinates. For fixed ω' , the resonance inequality is

$$\left| k_j \omega_j + \sum_{\ell \neq j} k_\ell \omega_\ell - m \right| < \varepsilon. \quad (1)$$

As an inequality in the single real variable ω_j , (1) describes an open interval of length

exactly

$$\frac{2\varepsilon}{|k_j|} = \frac{2\varepsilon}{|k|_\infty}, \quad (2)$$

possibly truncated when we intersect with $[-R, R]$.

Because $\Omega \subset [-R, R]^d$, every fiber in the remaining coordinates lies inside $[-R, R]^{d-1}$. Fubini's theorem and (2) give

$$\begin{aligned} \text{Leb}\{\omega \in \Omega : |\langle k, \omega \rangle - m| < \varepsilon\} \\ \leq \int_{[-R, R]^{d-1}} \frac{2\varepsilon}{|k|_\infty} d\omega' = 2(2R)^{d-1} \frac{\varepsilon}{|k|_\infty}. \end{aligned} \quad (3)$$

This proves the single-slab estimate.

Now let $\{(k_\alpha, m_\alpha, \varepsilon_\alpha)\}_{\alpha \in A}$ be a finite family and denote the corresponding resonant subsets of Ω by S_α . Subadditivity of Lebesgue measure and (3) yield

$$\text{Leb}\left(\bigcup_{\alpha \in A} S_\alpha\right) \leq \sum_{\alpha \in A} \text{Leb}(S_\alpha) \leq 2(2R)^{d-1} \sum_{\alpha \in A} \frac{\varepsilon_\alpha}{|k_\alpha|_\infty}.$$

No disjointness or independence of the resonant slabs is required. $\square$

Corollary F05.8. Borel-Cantelli Diophantine consequence

Statement. Let $\Omega \subset [-R, R]^d$ be measurable, and let $(k_n, m_n, \varepsilon_n)_{n \geq 1}$ satisfy $k_n \in \mathbb{Z}^d \setminus \{0\}$, $m_n \in \mathbb{Z}$, and $\varepsilon_n > 0$. If

$$\sum_{n=1}^{\infty} \frac{\varepsilon_n}{|k_n|_\infty} < \infty,$$

then for Lebesgue-almost every $\omega \in \Omega$ there exists $N(\omega)$ such that

$$|\langle k_n, \omega \rangle - m_n| \geq \varepsilon_n \quad \text{for every } n \geq N(\omega).$$

Equivalently, the set of parameters lying in infinitely many of the prescribed resonant slabs has Lebesgue measure zero.

Proof. For each $n \geq 1$, define the resonant slab inside the parameter domain by

$$E_n := \{\omega \in \Omega : |\langle k_n, \omega \rangle - m_n| < \varepsilon_n\}. \quad (1)$$

The set E_n is measurable because the map $\omega \mapsto \langle k_n, \omega \rangle - m_n$ is continuous and Ω is measurable.

By the resonant-slab estimate *Resonant slab measure estimate*,

$$\text{Leb}(E_n) \leq 2(2R)^{d-1} \frac{\varepsilon_n}{|k_n|_\infty}. \quad (2)$$

Consequently,

$$\sum_{n=1}^{\infty} \text{Leb}(E_n) \leq 2(2R)^{d-1} \sum_{n=1}^{\infty} \frac{\varepsilon_n}{|k_n|_\infty} < \infty. \quad (3)$$

Thus the convergence half of the Borel–Cantelli lemma, *Borel–Cantelli lemmas*, applies to the measure space Ω equipped with Lebesgue measure. It yields

$$\text{Leb}\left(\limsup_{n \rightarrow \infty} E_n\right) = 0, \quad \limsup_{n \rightarrow \infty} E_n = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n. \quad (4)$$

The set on the right of (4) is precisely the set of parameters that belong to infinitely many resonant slabs.

Therefore, for Lebesgue-almost every $\omega \in \Omega$, the point ω belongs to only finitely many E_n . For such an ω there is an integer $N(\omega)$ such that $\omega \notin E_n$ for every $n \geq N(\omega)$. Returning to the definition (1), this means

$$|\langle k_n, \omega \rangle - m_n| \geq \varepsilon_n \quad (n \geq N(\omega)),$$

which is exactly the claimed eventual nonresonance statement.

Notice that no independence of the slabs is required: only the convergence half of Borel–Cantelli is used. Also, the theorem makes no claim that the exceptional finite set of resonances can be chosen uniformly in ω ; the threshold $N(\omega)$ is allowed to depend on the parameter, as the conclusion states. $\square$

Theorem F05.9. Brjuno arithmetic package

Statement. Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ have continued-fraction convergents p_n/q_n , and put

$$B(\alpha) := \sum_{n \geq 0} \frac{\log q_{n+1}}{q_n}, \quad \Psi_\alpha(Q) := \max_{1 \leq q \leq Q} \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}^{-1} \quad (Q \geq 1).$$

Then the following are equivalent:

$$B(\alpha) < \infty, \quad \sum_{\ell \geq 0} \frac{\log \Psi_\alpha(2^\ell)}{2^\ell} < \infty, \quad \int_1^\infty \frac{\log \Psi_\alpha(Q)}{Q^2} dQ < \infty.$$

Every Diophantine irrational is Brjuno, while non-Brjuno irrationals exist.

Scope note

Only the arithmetic equivalences and examples are asserted here; holomorphic linearization consequences belong to the Complex Dynamics series.

Proof. Let

$$\delta(q) := \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \Psi_\alpha(Q) := \max_{1 \leq q \leq Q} \delta(q)^{-1}. \quad (1)$$

Changing finitely many initial continued-fraction terms cannot affect convergence of any series below, so whenever convenient we begin at an index for which $q_{n+1} > q_n \geq 2$.

Step 1: denominator intervals are the exact small-divisor scales. *Best approximation property* says that if

$$q_n \leq Q < q_{n+1}, \quad (2)$$

then among all integers $1 \leq q \leq Q$ the minimum of $\delta(q)$ is attained at $q = q_n$. Hence

$$\Psi_\alpha(Q) = \delta(q_n)^{-1} \quad (q_n \leq Q < q_{n+1}). \quad (3)$$

The same theorem gives

$$\frac{1}{q_n + q_{n+1}} < \delta(q_n) < \frac{1}{q_{n+1}},$$

and therefore

$$q_{n+1} < \Psi_\alpha(Q) < q_n + q_{n+1} < 2q_{n+1} \quad (q_n \leq Q < q_{n+1}). \quad (4)$$

Thus, up to an additive constant $\log 2$, the logarithmic small divisor on the whole n th denominator interval is exactly $\log q_{n+1}$.

We shall repeatedly use the elementary fact

$$\sum_{n \geq 0} \frac{1 + \log q_n}{q_n} < \infty. \quad (5)$$

Indeed, the denominator recurrence from *Continued-fraction convergents* gives $q_{n+2} \geq q_{n+1} + q_n$, hence $q_n \geq c\varphi^n$ for a fixed $c > 0$ and the golden ratio $\varphi > 1$. Since $x \mapsto (1 + \log x)/x$ is decreasing for all sufficiently large x , (5) follows by comparison with $C(1 + n)\varphi^{-n}$.

Step 2: the continued-fraction Brjuno sum is equivalent to the dyadic small-divisor sum. Put

$$D(\alpha) := \sum_{\ell \geq 0} \frac{\log \Psi_\alpha(2^\ell)}{2^\ell}. \quad (6)$$

For each sufficiently large n , let

$$I_n := \{\ell \in \mathbb{N} : q_n \leq 2^\ell < q_{n+1}\}. \quad (7)$$

The nonempty sets I_n partition all sufficiently large dyadic indices. If $I_n \neq \emptyset$, let $\ell_n = \min I_n$. Then

$$q_n \leq 2^{\ell_n} < 2q_n,$$

so

$$\frac{1}{2q_n} < 2^{-\ell_n} \leq \frac{1}{q_n}, \quad \sum_{\ell \in I_n} 2^{-\ell} \leq 2^{1-\ell_n} \leq \frac{2}{q_n}. \quad (8)$$

Using (4),

$$\frac{\log q_{n+1}}{2q_n} \leq \sum_{\ell \in I_n} \frac{\log \Psi_\alpha(2^\ell)}{2^\ell} \leq \frac{2 \log(2q_{n+1})}{q_n}. \quad (9)$$

Consequently the part of $D(\alpha)$ indexed by nonempty I_n converges if and only if

$$\sum_{I_n \neq \emptyset} \frac{\log q_{n+1}}{q_n} < \infty. \quad (10)$$

It remains to see that omitted denominator intervals never matter. If I_n is empty, there is no power of two in $[q_n, q_{n+1})$. The first power of two not smaller than q_n is $< 2q_n$,

so emptiness forces

$$q_{n+1} < 2q_n. \quad (11)$$

Hence, by (5),

$$\sum_{I_n=\emptyset} \frac{\log q_{n+1}}{q_n} \leq \sum_n \frac{\log 2 + \log q_n}{q_n} < \infty. \quad (12)$$

Combining (9)-(12),

$$\boxed{B(\alpha) < \infty \iff D(\alpha) < \infty.} \quad (13)$$

This is the scale-grouping step in which continued fractions enter essentially.

Step 3: the dyadic series is equivalent to the Bruno-Russmann integral. The function $Q \mapsto \Psi_\alpha(Q)$ is nondecreasing. Therefore, for every $\ell \geq 0$ and $2^\ell \leq Q \leq 2^{\ell+1}$,

$$\log \Psi_\alpha(2^\ell) \leq \log \Psi_\alpha(Q) \leq \log \Psi_\alpha(2^{\ell+1}). \quad (14)$$

Since

$$\int_{2^\ell}^{2^{\ell+1}} \frac{dQ}{Q^2} = 2^{-\ell-1}, \quad (15)$$

we obtain

$$\frac{\log \Psi_\alpha(2^\ell)}{2^{\ell+1}} \leq \int_{2^\ell}^{2^{\ell+1}} \frac{\log \Psi_\alpha(Q)}{Q^2} dQ \leq \frac{\log \Psi_\alpha(2^{\ell+1})}{2^{\ell+1}}. \quad (16)$$

Summing (16) over ℓ shows

$$\boxed{D(\alpha) < \infty \iff \int_1^\infty \frac{\log \Psi_\alpha(Q)}{Q^2} dQ < \infty.} \quad (17)$$

Together, (13) and (17) prove all three announced arithmetic equivalences.

Step 4: every Diophantine irrational is Brjuno. Suppose that α is Diophantine: for some $\gamma > 0$ and $\tau \geq 1$,

$$\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-\tau} \quad (q \geq 1). \quad (18)$$

By the continued-fraction characterization in *Diophantine classes and continued fractions*, equivalently

$$q_{n+1} \leq C q_n^\tau \quad (19)$$

for some $C > 0$. Hence

$$\frac{\log q_{n+1}}{q_n} \leq \frac{\log C + \tau \log q_n}{q_n}. \quad (20)$$

The series of the right-hand side converges by (5), so $B(\alpha) < \infty$.

Step 5: non-Brjuno irrationals exist. We give an explicit continued-fraction construction to make clear that the condition is not automatic. Start with any finite irrational continued-fraction prefix and then choose the partial quotients recursively so large that

$$a_{n+1} \geq e^{q_n}. \quad (21)$$

The denominator recurrence gives

$$q_{n+1} = a_{n+1}q_n + q_{n-1} \geq e^{q_n} q_n, \quad (22)$$

so

$$\frac{\log q_{n+1}}{q_n} \geq 1 + \frac{\log q_n}{q_n} \geq 1. \quad (23)$$

Thus the Brjuno series diverges. The infinite continued fraction defines an irrational number, so non-Brjuno irrationals indeed exist.

The proof is entirely arithmetic. In particular, we have not asserted that the Brjuno condition is sufficient or necessary for any holomorphic, smooth, Hamiltonian, or co-cycle linearization problem; those conclusions require additional dynamical/analytic theorems and are treated in the appropriate specialist volumes. $\square$

5.3 Worked examples

Worked example: Arithmetic of Frequencies: a model computation

Let

$$\alpha = \frac{\sqrt{5} - 1}{2} = [0; 1, 1, 1, \dots].$$

Its convergent denominators are Fibonacci numbers:

$$q_0 = 1, \quad q_1 = 1, \quad q_2 = 2, \quad q_3 = 3, \quad q_4 = 5, \dots,$$

and $q_{n+1} = q_n + q_{n-1}$. The convergents are

$$1, \quad \frac{1}{2}, \quad \frac{2}{3}, \quad \frac{3}{5}, \quad \frac{5}{8}, \dots$$

(up to the initial indexing convention). Since all partial quotients are bounded by 1, α is badly approximable: there is $c > 0$ such that

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c}{q^2}$$

for every rational p/q . Equivalently,

$$\text{dist}(k\alpha, \mathbb{Z}) \geq \frac{c}{|k|} \quad (k \neq 0).$$

Thus the golden-ratio frequency gives the cleanest model of a one-dimensional Diophantine frequency.

Boundary case: what fails at the edge of the hypotheses

Define the Liouville number

$$L = \sum_{m=1}^{\infty} 10^{-m!}.$$

Let p_m/q_m be the truncation after the m -th term, so $q_m = 10^{m!}$. The tail satisfies

$$0 < \left| L - \frac{p_m}{q_m} \right| < 2 \cdot 10^{-(m+1)!} = 2q_m^{-(m+1)}.$$

For every power $q^{-\tau}$, infinitely many approximants eventually beat it. Consequently no fixed Diophantine inequality can hold. In Fourier problems, this produces exceptionally small denominators along a sparse sequence of modes.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

5.4 Chapter summary

What to remember

- Continued fractions encode optimal one-dimensional rational approximations.
- Diophantine conditions are quantitative nonresonance conditions.
- Resonant sets are thin enough to estimate geometrically in parameter space.
- Liouville behavior explains why small divisors can overwhelm regularity.

Looking ahead

F06 applies this arithmetic to circle maps and rotation numbers. F08 will later convert the same lower bounds into norm estimates for cohomological equations.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

5.5 Exercises and problems

Exercise 5.1

Difficulty. Core.

Related results. Continued-fraction convergents; Best approximation property; Diophantine classes and continued fractions.

Problem. Work through the worked example ‘continued fractions of the golden ratio and its convergents’. For Continued-fraction convergents, Best approximation property, and Diophantine classes and continued fractions, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 5.2

Difficulty. Core.

Related results. Kronecker density criterion for torus translations; Metric abundance of Diophantine frequencies; Resonant slab measure estimate.

Problem. Analyze the boundary example ‘a Liouville number exhibiting exceptionally small divisors’. For Kronecker density criterion for torus translations, Metric abundance of Diophantine frequencies, and Resonant slab measure estimate, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 5.3

Difficulty. Advanced.

Related results. Diophantine classes and continued fractions.

Problem. For Diophantine classes and continued fractions, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 5.4**Difficulty.** Advanced.**Related results.** Kronecker density criterion for torus translations.**Problem.** Stress-test Kronecker density criterion for torus translations. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.**Solution sketch.** Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.**Exercise 5.5****Difficulty.** Advanced.**Related results.** Continued-fraction convergents; Best approximation property.**Problem.** Synthesis problem. Use Continued-fraction convergents and Best approximation property to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.**Solution sketch.** State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.**Exercise 5.6****Difficulty.** Lab.**Related results.** None.**Problem.** Computational laboratory: carry out the reproducible experiment 'continued fractions of the golden ratio and its convergents'. Use the computation only as numerical evidence or visualization, not as proof. Include a refinement or convergence test and identify one numerical artifact that could mislead a reader.**Solution sketch.** State parameters and truncation/time horizon, compare with an analytically controlled identity or exactly solvable case, repeat after refinement, and explain what the numerical experiment can and cannot establish.**Exercise 5.7****Difficulty.** Core.**Related results.** Borel-Cantelli Diophantine consequence.**Problem.** Work directly with Borel-Cantelli Diophantine consequence. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 5.8

Difficulty. Core.

Related results. Brjuno arithmetic package.

Problem. Use the definition of Brjuno arithmetic package in three settings: one example satisfying it, one near-miss failing exactly one clause, and one example in which the definition reduces to a familiar special case. Prove each classification directly from the definition.

Solution sketch. Write the definition first, test its clauses one at a time, and make the failed clause explicit in the near-miss example. Avoid invoking deeper theorems when the definition alone suffices.

Exercise 5.9

Difficulty. Core.

Related results. Dirichlet simultaneous approximation.

Problem. Work directly with Dirichlet simultaneous approximation. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Circle and Torus Dynamics

Why this chapter matters

Circle dynamics is the first setting where topology, arithmetic, and smoothness interact in a sharp way. Every orientation-preserving circle homeomorphism has a rotation number, but the rotation number alone does not determine the geometry. Under sufficient regularity, irrational rotation number forces conjugacy-like behavior; below the Denjoy threshold, wandering intervals can appear. The chapter is therefore a model for a recurring theme: arithmetic describes the desired normal form, while regularity decides whether the dynamics can actually be straightened to it.

Prerequisites. F01, F05, orientation-preserving circle homeomorphisms, and elementary real analysis.

Learning goals

- Construct lifts and compute rotation numbers.
- Use periodic-orbit and semiconjugacy criteria.
- Understand the role of distortion estimates in Denjoy's theorem.
- See why irrational rotation number does not by itself imply smooth or even topological conjugacy.

Conceptual roadmap

Lifts turn a circle map into an order-preserving map of the line. Subadditivity/order arguments give the rotation number. Rational rotation number corresponds to periodic behavior. Irrational rotation number gives a semiconjugacy to a rigid rotation; Denjoy distortion is what prevents the semiconjugacy from collapsing intervals, thereby upgrading it to conjugacy under the stated regularity.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

6.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F06.1. Circle lift

If $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ is an orientation-preserving circle homeomorphism, a *lift* is an increasing homeomorphism $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $F(x + 1) = F(x) + 1$ and projecting to f .

Definition F06.2. Rotation number

For a lift F of f , the rotation number is $\rho(f) = \lim_{n \rightarrow \infty} (F^n(x) - x)/n \pmod{\mathbb{Z}}$. The limit is independent of x ; replacing F by another lift changes the real value by an integer.

Definition F06.3. Semiconjugacy

A *semiconjugacy* from (X, T) to (Y, S) is a continuous surjection $h : X \rightarrow Y$ satisfying $h \circ T = S \circ h$. If h is a homeomorphism, it is a conjugacy.

Definition F06.4. Denjoy distortion

Denjoy distortion denotes quantitative control of $\log(f^n)'$ on closest-return intervals, obtained from bounded variation of $\log f'$ or the regularity hypothesis stated in the theorem.

6.2 Main results

Theorem F06.1. Existence of rotation number

Statement. For an orientation-preserving circle homeomorphism with lift $F : \mathbb{R} \rightarrow \mathbb{R}$, the limit

$$\tau(F) := \lim_{n \rightarrow \infty} \frac{F^n(x) - x}{n}$$

exists and is independent of x . Its class modulo $\mathbb{Z}$ is the rotation number of the circle homeomorphism.

Proof. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a lift of an orientation-preserving circle homeomorphism. Thus F is strictly increasing and

$$F(x + 1) = F(x) + 1. \quad (1)$$

By induction, (1) implies

$$F^n(x + 1) = F^n(x) + 1 \quad (n \geq 1). \quad (2)$$

Define the n -step displacement

$$D_n(x) := F^n(x) - x.$$

Equation (2) shows that D_n is continuous and 1-periodic, hence it has finite maximum and minimum on $[0, 1]$.

Step 1: the displacement varies by at most one with the base point. Suppose $x \leq y \leq x + 1$. Monotonicity of F^n and (2) give

$$F^n(x) \leq F^n(y) \leq F^n(x + 1) = F^n(x) + 1.$$

Subtracting y and comparing with $D_n(x) = F^n(x) - x$ yields

$$-1 \leq D_n(y) - D_n(x) \leq 1. \quad (3)$$

For arbitrary x, y , use the 1-periodicity of D_n to replace y by an integer translate lying in $[x, x + 1]$. Thus (3) implies

$$\text{osc}(D_n) := \sup_x D_n(x) - \inf_x D_n(x) \leq 1. \quad (4)$$

Set

$$a_n := \sup_x D_n(x), \quad b_n := \inf_x D_n(x).$$

Then (4) says

$$0 \leq a_n - b_n \leq 1. \quad (5)$$

Step 2: the extremal displacements are sub- and superadditive. For all $m, n \geq 1$,

$$\begin{aligned} D_{m+n}(x) &= F^{m+n}(x) - x \\ &= (F^m(F^n x) - F^n x) + (F^n x - x) \\ &= D_m(F^n x) + D_n(x). \end{aligned} \quad (6)$$

Taking suprema in (6) gives

$$a_{m+n} \leq a_m + a_n, \quad (7)$$

and taking infima gives

$$b_{m+n} \geq b_m + b_n. \quad (8)$$

Therefore (a_n) is subadditive and $(-b_n)$ is subadditive. *Fekete subadditive lemma* gives the limits

$$\alpha := \lim_{n \rightarrow \infty} \frac{a_n}{n}, \quad \beta := \lim_{n \rightarrow \infty} \frac{b_n}{n}. \quad (9)$$

By (5),

$$0 \leq \frac{a_n - b_n}{n} \leq \frac{1}{n} \rightarrow 0,$$

so the limits in (9) agree. Write their common value as $\rho(F)$.

Step 3: every base point has the same asymptotic displacement. For every x ,

$$b_n \leq D_n(x) \leq a_n.$$

After division by n , both outer terms converge to $\rho(F)$, so the squeeze theorem yields

$$\lim_{n \rightarrow \infty} \frac{F^n(x) - x}{n} = \rho(F) \quad \text{for every } x \in \mathbb{R}. \quad (10)$$

Thus the real limit is independent of the base point for a fixed lift.

Step 4: changing the lift changes the real number by an integer. Every other lift has the form $G = F + m$ for some $m \in \mathbb{Z}$. Equation (1) implies $F(x + m) = F(x) + m$ for every integer m , and induction then gives

$$G^n(x) = F^n(x) + nm.$$

Consequently

$$\rho(G) = \rho(F) + m.$$

Hence the class

$$\rho(f) := \rho(F) \pmod{\mathbb{Z}}$$

is independent of the chosen lift. Together with (10), this proves existence and base-point independence of the rotation number exactly as stated. $\square$

Theorem F06.2. Periodic orbit criterion

Statement. Let $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ be an orientation-preserving circle homeomorphism, let F be a lift, and let $\rho(F)$ be its translation number. If $p/q \in \mathbb{Q}$ is in lowest terms with $q \geq 1$, then $\rho(F) = p/q$ if and only if there exists $x \in \mathbb{R}$ such that $F^q(x) = x + p$. Equivalently, f has a periodic orbit; for p/q reduced, the least period of the corresponding orbit is exactly q .

Proof. Assume first that there exists $x \in \mathbb{R}$ with

$$F^q(x) = x + p. \quad (1)$$

Because F commutes with integer translations, iterating (1) gives

$$F^{kq}(x) = x + kp \quad (k \geq 1).$$

Hence, using *Existence of rotation number*,

$$\rho(F) = \lim_{k \rightarrow \infty} \frac{F^{kq}(x) - x}{kq} = \frac{p}{q}. \quad (2)$$

Conversely, suppose

$$\rho(F) = \frac{p}{q}. \quad (3)$$

Define

$$G(x) := F^q(x) - p.$$

Then G is again an increasing degree-one lift, and by the iterate and integer-shift rules for translation numbers,

$$\rho(G) = q\rho(F) - p = 0. \quad (4)$$

We claim that G has a fixed point. If not, the continuous 1-periodic function

$$h(x) := G(x) - x$$

never vanishes. Since $\mathbb{R}/\mathbb{Z}$ is connected, h has one strict sign everywhere.

Suppose first that $h > 0$. Compactness of $[0, 1]$ gives

$$\delta := \min_{x \in [0, 1]} h(x) > 0,$$

so $G(y) \geq y + \delta$ for every $y \in \mathbb{R}$. Inductively,

$$G^n(x) \geq x + n\delta.$$

Therefore

$$\rho(G) = \lim_{n \rightarrow \infty} \frac{G^n(x) - x}{n} \geq \delta > 0,$$

contradicting (4). If $h < 0$, the same argument applied to $-h$ gives a number $\delta > 0$ with $G^n(x) \leq x - n\delta$, hence $\rho(G) \leq -\delta < 0$, again contradicting (4). Thus some x satisfies $G(x) = x$, which is exactly

$$F^q(x) = x + p. \quad (5)$$

Projecting (5) to the circle shows that $f^q([x]) = [x]$, so f has a periodic orbit.

It remains to verify the claimed least period when p/q is reduced. Let r be the least positive period of $[x]$. Then for some $s \in \mathbb{Z}$,

$$F^r(x) = x + s.$$

The already proved forward implication yields

$$\rho(F) = \frac{s}{r}.$$

Together with $\rho(F) = p/q$ and $\gcd(p, q) = 1$, this implies $q \mid r$. On the other hand, (5) gives $f^q([x]) = [x]$, so the least period r divides q . Hence $r = q$. $\square$

Theorem F06.3. Poincaré semiconjugacy theorem

Statement. Let $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ be orientation preserving, let F be a lift with irrational translation number $\rho \in \mathbb{R} \setminus \mathbb{Q}$. Then there exists a continuous nondecreasing map $H : \mathbb{R} \rightarrow \mathbb{R}$ with $H(x+1) = H(x) + 1$ and $H(F(x)) = H(x) + \rho$. Hence H descends to a continuous monotone surjective degree-one map h satisfying $h \circ f = R_\rho \circ h$.

Proof. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a lift of f and let $\rho \in \mathbb{R} \setminus \mathbb{Q}$ be its translation number from *Existence of rotation number*. Thus

$$F(x+1) = F(x) + 1, \quad \lim_{m \rightarrow \infty} \frac{F^m(x) - x}{m} = \rho. \quad (1)$$

We construct a lift of the desired semiconjugacy.

Step 1: bounded deviations from the rigid rotation. Fix an integer $n \neq 0$ and set

$$D_n(x) := F^n(x) - x.$$

Since F^n is again an increasing degree-one lift, if $x \leq y \leq x + 1$ then

$$F^n(x) \leq F^n(y) \leq F^n(x) + 1.$$

It follows exactly as in the rotation-number proof that

$$\sup_x D_n(x) - \inf_x D_n(x) \leq 1. \quad (2)$$

The translation number of the lift F^n is $n\rho$: indeed, for $m \rightarrow \infty$,

$$\frac{(F^n)^m(x) - x}{m} = n \frac{F^{nm}(x) - x}{nm} \rightarrow n\rho. \quad (3)$$

Consequently $n\rho$ lies between $\inf D_n$ and $\sup D_n$. To see this without invoking any extra theorem, if for example $\inf D_n > n\rho$, then $F^n(y) \geq y + c$ for some $c > n\rho$ and every y ; iterating gives $(F^n)^m(x) \geq x + mc$, contradicting (3). The case $\sup D_n < n\rho$ is analogous. Combining this observation with (2),

$$|F^n(x) - x - n\rho| \leq 1 \quad (x \in \mathbb{R}, n \in \mathbb{Z}). \quad (4)$$

For $n = 0$ the estimate is trivial. Notice that the argument covers negative n because every F^n is still an increasing degree-one lift.

Step 2: construct a monotone degree-one solution of the lift equation. Define

$$H(x) := \sup_{n \in \mathbb{Z}} (F^n(x) - n\rho). \quad (5)$$

Estimate (4) shows

$$x - 1 \leq H(x) \leq x + 1, \quad (6)$$

so the supremum is finite. Every function $x \mapsto F^n(x) - n\rho$ is increasing; hence H is nondecreasing. Also

$$H(x + 1) = \sup_n (F^n(x + 1) - n\rho) \quad (6.1)$$

$$= \sup_n (F^n(x) + 1 - n\rho) = H(x) + 1, \quad (7)$$

and reindexing the supremum gives

$$H(F(x)) = \sup_{n \in \mathbb{Z}} (F^{n+1}(x) - n\rho) \quad (6.2)$$

$$= \rho + \sup_{m \in \mathbb{Z}} (F^m(x) - m\rho) = H(x) + \rho. \quad (8)$$

Thus all algebraic requirements for a lift of a semiconjugacy are already satisfied. It remains to prove that H is continuous.

Step 3: rule out jumps. A monotone real function has finite one-sided limits. Write

$$J(x) := H(x+) - H(x-) \geq 0$$

for its jump at x . Because F is an increasing homeomorphism, $t \rightarrow x^\pm$ implies $F(t) \rightarrow F(x)^\pm$. Taking one-sided limits in (8) therefore gives

$$J(F(x)) = J(x). \quad (9)$$

Suppose $J(x_0) = a > 0$. Then every point $F^m(x_0)$ has the same jump a . These points are pairwise distinct modulo $\mathbb{Z}$. Indeed, if $F^q(x_0) = x_0 + p$ for some $q \neq 0$ and $p \in \mathbb{Z}$, then iterating would give $F^{q\ell}(x_0) = x_0 + \ell p$, and (1) would imply $\rho = p/q \in \mathbb{Q}$, contrary to hypothesis.

Choose arbitrarily many distinct representatives of this orbit in a single interval of length one. For a nondecreasing map satisfying $H(x+1) = H(x) + 1$, the sum of the jumps at any finite set of points in one period is at most its total variation over that period, namely 1. Hence $Na \leq 1$ for every N , impossible. Therefore $J(x) = 0$ for every x . A monotone function with no jumps is continuous, so H is continuous.

Step 4: descend to the circle. Equation (7) makes

$$h : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}, \quad h([x]) := [H(x)],$$

well defined. It is continuous and monotone. Since $H(x+1) = H(x) + 1$, the intermediate value theorem shows that $H([x, x+1])$ contains an interval of length one, so h is onto and has degree one. Finally (8) descends to

$$h \circ f = R_\rho \circ h.$$

Thus f is semiconjugate to the irrational rigid rotation of angle $\rho \pmod{1}$. $\square$

Proposition F06.4. Continuity and monotonicity of rotation number

Statement. Let $F, G : \mathbb{R} \rightarrow \mathbb{R}$ be lifts of orientation-preserving circle homeomorphisms. If $F(x) \leq G(x)$ for every x , then $\rho(F) \leq \rho(G)$. If $F_n \rightarrow F$ uniformly, then $\rho(F_n) \rightarrow \rho(F)$. Hence the translation number is monotone in the lift order and continuous in the C^0 topology.

Proof. We first prove monotonicity. Assume

$$F(x) \leq G(x) \quad \text{for every } x \in \mathbb{R}. \quad (1)$$

Both maps are increasing. If $F^n(x) \leq G^n(x)$, then

$$F^{n+1}(x) = F(F^n(x)) \leq F(G^n(x)) \leq G(G^n(x)) = G^{n+1}(x).$$

By induction,

$$F^n(x) \leq G^n(x) \quad (n \geq 1).$$

Subtracting x , dividing by n , and taking the limits from *Existence of rotation number*

gives

$$\rho(F) \leq \rho(G). \quad (2)$$

We next prove continuity. Fix $\varepsilon > 0$. Choose rational numbers

$$\frac{p}{q} < \rho(F) < \frac{p'}{q'} \quad (3)$$

with

$$\rho(F) - \frac{p}{q} < \varepsilon, \quad \frac{p'}{q'} - \rho(F) < \varepsilon. \quad (4)$$

The rationals in (3) are strictly different from $\rho(F)$. By *Periodic orbit criterion*, the continuous 1-periodic functions

$$x \mapsto F^q(x) - x - p, \quad x \mapsto F^{q'}(x) - x - p'$$

have no zeros. Their signs are determined by (3). Indeed, if the first were everywhere negative, then the lift $F^q - p$ would have negative translation number, whereas

$$\rho(F^q - p) = q\rho(F) - p > 0.$$

Thus it is everywhere positive. Similarly the second is everywhere negative. Compactness of $[0, 1]$ therefore gives numbers $\eta_-, \eta_+ > 0$ such that

$$F^q(x) - x - p \geq \eta_- \quad \text{and} \quad F^{q'}(x) - x - p' \leq -\eta_+ \quad (5)$$

for all $x \in \mathbb{R}$.

We need one elementary continuity fact about finite iterates. If $G_n \rightarrow F$ uniformly, then for each fixed $r \geq 1$,

$$G_n^r \rightarrow F^r \quad \text{uniformly on } \mathbb{R}. \quad (6)$$

To see this, note that a degree-one lift has the form $F(x) = x + \psi(x)$ with ψ continuous and 1-periodic; hence F is uniformly continuous on $\mathbb{R}$. Starting from the case $r = 1$, write

$$\begin{aligned} |G_n^{r+1}(x) - F^{r+1}(x)| &\leq |G_n(G_n^r(x)) - F(G_n^r(x))| \\ &\quad + |F(G_n^r(x)) - F(F^r(x))|. \end{aligned}$$

The first term is at most $\|G_n - F\|_\infty$, and the second tends uniformly to zero by the uniform continuity of F and the induction hypothesis. This proves (6).

Apply (6) with $r = q$ and $r = q'$. For every lift G sufficiently close to F in the uniform norm, the strict inequalities (5) persist in the weaker form

$$G^q(x) - x - p > 0, \quad G^{q'}(x) - x - p' < 0 \quad (7)$$

for every x . The same uniform-drift argument used in *Periodic orbit criterion* now gives

$$q\rho(G) - p > 0, \quad q'\rho(G) - p' < 0.$$

Hence

$$\frac{p}{q} < \rho(G) < \frac{p'}{q'}. \quad (8)$$

Combining (4) and (8),

$$|\rho(G) - \rho(F)| < \varepsilon.$$

Thus ρ is continuous for the uniform topology on normalized lifts. Together with (2), this proves both assertions. $\square$

Lemma F06.5. Denjoy distortion inequality

Statement. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving C^1 diffeomorphism with irrational rotation number α , let p_n/q_n be the continued-fraction convergents of α , and assume $\log Df$ has bounded variation $V = \text{Var}(\log Df)$. Then for every n and every $x \in \mathbb{T}$, $|\log Df^{q_n}(x)| \leq V$, equivalently $e^{-V} \leq Df^{q_n}(x) \leq e^V$; the same bounds hold for Df^{-q_n} . In particular, the hypothesis is automatic for C^2 circle diffeomorphisms.

Proof. Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let

$$\alpha = \rho(f) \in \mathbb{R} \setminus \mathbb{Q},$$

and let p_n/q_n be the continued-fraction convergents fixed in F05. Put

$$\varphi := \log Df, \quad V := \text{Var}_{\mathbb{T}}(\varphi) < \infty. \quad (1)$$

Because f is an orientation-preserving C^1 diffeomorphism, Df is continuous and strictly positive, so φ is a well-defined continuous function.

We first prove the bounded-variation orbit-sum estimate that drives Denjoy's inequality. This is done here because invoking Denjoy's theorem or unique ergodicity of f at this point would be circular.

Step 1: choose an invariant measure and identify its factor measure. By *Krylov-Bogolyubov existence theorem*, f admits an invariant Borel probability μ . By the *Poincaré semiconjugacy theorem*, there is a continuous monotone degree-one surjection

$$h : \mathbb{T} \longrightarrow \mathbb{T}, \quad h \circ f = R_\alpha \circ h. \quad (2)$$

Therefore $h_*\mu$ is R_α -invariant. The irrational rotation is uniquely ergodic by *Minimal rotations are uniquely ergodic*, so

$$h_*\mu = m, \quad (3)$$

where m is normalized Haar (Lebesgue) measure on $\mathbb{T}$. Notice that we have not asserted unique ergodicity of f .

Step 2: build the equal-mass return-time partition. Fix n and abbreviate

$$p = p_n, \quad q = q_n, \quad \delta = q\alpha - p. \quad (4)$$

Continued-fraction convergents gives

$$|\alpha - p/q| < \frac{1}{qq_{n+1}}, \quad \text{hence} \quad |\delta| < \frac{1}{q_{n+1}} \leq \frac{1}{q}. \quad (5)$$

Fix $x \in \mathbb{T}$ and put $y = h(x)$. Since p and q are coprime,

$$y, y + \frac{p}{q}, \dots, y + \frac{(q-1)p}{q}$$

is exactly the translate by y of the q -point rational grid $\{0, 1/q, \dots, (q-1)/q\}$, in a permuted order.

If $\delta > 0$, let I_j be the half-open grid interval of length $1/q$ starting at $y + jp/q$ and going in the positive direction. If $\delta < 0$, let I_j be the half-open grid interval of length $1/q$ ending at $y + jp/q$. The I_j , $0 \leq j < q$, form a Borel partition of the circle. Moreover

$$h(f^j x) = y + j\alpha = y + \frac{jp}{q} + \frac{j\delta}{q} \pmod{1}. \quad (6)$$

For $0 \leq j < q$, (5) gives

$$\left| \frac{j\delta}{q} \right| < \frac{1}{q},$$

with the sign chosen in the definition of I_j . Hence

$$h(f^j x) \in I_j. \quad (7)$$

(The case $q = 1$ is included and is immediate.)

Set

$$J_j := h^{-1}(I_j). \quad (8)$$

Because h is monotone, the J_j are cyclically ordered arcs, up to fibers over their endpoints. By (3), every endpoint fiber has μ -measure zero and

$$\mu(J_j) = m(I_j) = \frac{1}{q}. \quad (9)$$

Thus, after assigning the null boundary fibers to one adjacent arc, the J_j form a measurable partition, $f^j x \in J_j$, and each has the same μ -mass $1/q$.

Step 3: the Denjoy-Koksma estimate at q_n . Let $\psi : \mathbb{T} \rightarrow \mathbb{R}$ be any function of bounded variation. Using (9),

$$\begin{aligned} \sum_{j=0}^{q-1} \psi(f^j x) - q \int_{\mathbb{T}} \psi d\mu &= \sum_{j=0}^{q-1} \left(\psi(f^j x) - q \int_{J_j} \psi(t) d\mu(t) \right) \\ &= q \sum_{j=0}^{q-1} \int_{J_j} (\psi(f^j x) - \psi(t)) d\mu(t). \end{aligned} \quad (10)$$

For $t \in J_j$, both t and $f^j x$ lie in the same cyclic interval J_j , so

$$|\psi(f^j x) - \psi(t)| \leq \text{Var}_{J_j}(\psi).$$

The interiors of the J_j are disjoint and occur in cyclic order; hence variation is super-additive over their concatenation:

$$\sum_{j=0}^{q-1} \text{Var}_{J_j}(\psi) \leq \text{Var}_{\mathbb{T}}(\psi). \quad (11)$$

Taking absolute values in (10), then using (9) and (11), gives the sharp return-time estimate

$$\left| \sum_{j=0}^{q_n-1} \psi(f^j x) - q_n \int \psi d\mu \right| \leq \text{Var}_{\mathbb{T}}(\psi) \quad (12)$$

for every x and every n .

Step 4: apply (12) to $\log Df$. The chain rule gives

$$\log Df^{q_n}(x) = \sum_{j=0}^{q_n-1} \varphi(f^j x). \quad (13)$$

Set

$$c := \int_{\mathbb{T}} \varphi d\mu. \quad (14)$$

Applying (12) with $\psi = \varphi$ and using (1),

$$|\log Df^{q_n}(x) - q_n c| \leq V \quad (x \in \mathbb{T}). \quad (15)$$

Equivalently,

$$e^{q_n c - V} \leq Df^{q_n}(x) \leq e^{q_n c + V}. \quad (16)$$

Step 5: the invariant mean c is zero. Let F be a degree-one lift of f . Then $F^{q_n}(x+1) = F^{q_n}(x) + 1$, so the fundamental theorem of calculus yields

$$\int_0^1 Df^{q_n}(x) dx = F^{q_n}(1) - F^{q_n}(0) = 1. \quad (17)$$

If $q_n c > V$, the lower bound in (16) would be strictly larger than 1 at every point, contradicting (17). If $q_n c < -V$, the upper bound in (16) would be strictly smaller than 1 everywhere, again contradicting (17). Therefore

$$|q_n c| \leq V \quad \text{for every } n. \quad (18)$$

Since $q_n \rightarrow \infty$, (18) forces

$$c = 0. \quad (19)$$

Substituting (19) into (15) proves

$$\boxed{|\log Df^{q_n}(x)| \leq V} \quad (n \geq 0, x \in \mathbb{T}), \quad (20)$$

or, equivalently,

$$\boxed{e^{-V} \leq Df^{q_n}(x) \leq e^V.} \quad (21)$$

Step 6: inverse iterates and the C^2 specialization. For every x ,

$$Df^{-q_n}(x) = \frac{1}{Df^{q_n}(f^{-q_n}x)}. \quad (22)$$

Applying (21) at $f^{-q_n}x$ and taking reciprocals gives the same bounds for Df^{-q_n} .

Finally, if f is C^2 , then $Df > 0$ and

$$(\log Df)' = \frac{D^2f}{Df}$$

is continuous on the compact circle. Hence $\log Df$ is C^1 and therefore has finite total variation. Thus the bounded-variation hypothesis is automatic in the C^2 case. $\square$

Theorem F06.6. Denjoy theorem

Statement. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ be an orientation-preserving C^1 diffeomorphism with irrational rotation number α . If $\log Df$ has bounded variation (equivalently, since $Df > 0$ is continuous on the compact circle, if Df has bounded variation), then f has no wandering interval and the Poincaré semiconjugacy is a homeomorphism. Hence f is topologically conjugate to the rigid rotation R_α . In particular the hypothesis holds for every orientation-preserving C^2 circle diffeomorphism.

Proof. Let

$$V := \text{Var}(\log Df) < \infty. \quad (1)$$

Because the rotation number α is irrational, *Poincaré semiconjugacy theorem* gives a continuous monotone degree-one surjection

$$h : \mathbb{T} \longrightarrow \mathbb{T}, \quad h \circ f = R_\alpha \circ h. \quad (2)$$

We prove that h is injective. Since a continuous bijection from the compact circle to the Hausdorff circle is a homeomorphism, this will prove the claimed conjugacy.

Step 1: a nontrivial fiber of h produces a wandering interval. Choose lifts $F, H : \mathbb{R} \rightarrow \mathbb{R}$ so that H is nondecreasing, $H(x+1) = H(x) + 1$, and

$$H(F(x)) = H(x) + \alpha. \quad (3)$$

Suppose that h is not injective. Then, after choosing representatives in an interval of length one, there are $x < y < x+1$ with $H(x) = H(y)$. Monotonicity forces

$$H \equiv H(x) \quad \text{on } [x, y]. \quad (4)$$

Let $I \subset \mathbb{T}$ be the nonempty open arc obtained by projecting (x, y) . For every $k \in \mathbb{Z}$, (2)

implies that h is constant on $f^k(I)$, with value

$$R_\alpha^k(h(I)) = h(I) + k\alpha \pmod{1}. \quad (5)$$

If $k \neq \ell$, irrationality of α gives $R_\alpha^k(h(I)) \neq R_\alpha^\ell(h(I))$. Distinct fibers of a function are disjoint, so

$$f^k(I) \cap f^\ell(I) = \emptyset \quad (k \neq \ell). \quad (6)$$

Thus I is a wandering interval. Notice that this conclusion used only monotonicity of the semiconjugacy and irrationality; no differentiability has entered yet.

Step 2: Denjoy distortion forbids the wandering interval. Let p_n/q_n be the continued-fraction convergents of α . By *Denjoy distortion inequality*,

$$e^{-V} \leq Df^{q_n}(z) \leq e^V \quad \text{for every } z \in \mathbb{T} \text{ and every } n. \quad (7)$$

Choose a lift of the arc I to a real interval and use the corresponding lift of f^{q_n} on it. Since the map is orientation preserving, the fundamental theorem of calculus gives

$$|f^{q_n}(I)| = \int_I Df^{q_n}(z) dz \geq e^{-V} |I|. \quad (8)$$

The denominators q_n are strictly increasing after at most finitely many initial indices. Hence (6) says that the infinitely many intervals $f^{q_n}(I)$ are pairwise disjoint. Their total length is therefore at most the circumference of the circle, while (8) gives

$$1 \geq \sum_{n \geq n_0} |f^{q_n}(I)| \geq \sum_{n \geq n_0} e^{-V} |I| = +\infty, \quad (9)$$

a contradiction. Thus h has no nontrivial fiber and is injective.

Consequently h is a homeomorphism, and (2) becomes

$$h \circ f \circ h^{-1} = R_\alpha. \quad (10)$$

In particular no wandering interval exists.

Finally, the two common bounded-variation normalizations are equivalent here. Since f is an orientation-preserving C^1 diffeomorphism, Df is continuous and $0 < m \leq Df \leq M < \infty$. The functions $\log$ on $[m, M]$ and $\exp$ on $[\log m, \log M]$ are Lipschitz, so composition with either one preserves bounded variation up to a finite multiplicative constant. Hence

$$Df \in BV \iff \log Df \in BV. \quad (11)$$

If $f \in C^2$, then $Df \in C^1$ on the compact circle and therefore has bounded variation, proving the final assertion. $\square$

Corollary F06.7. Unique ergodicity of irrational circle diffeomorphisms under Denjoy hypotheses

Statement. Let $f : \mathbb{T} \rightarrow \mathbb{T}$ satisfy the hypotheses of Theorem F06.6, let α be its irrational rotation number, and let h be the Denjoy conjugacy $h \circ f = R_\alpha \circ h$. Then

f is uniquely ergodic. Its unique invariant Borel probability is

$$\mu_f = (h^{-1})_* m,$$

where m is normalized Lebesgue (Haar) measure on $\mathbb{T}$. Consequently, for every $\varphi \in C(\mathbb{T})$,

$$\frac{1}{N} \sum_{j=0}^{N-1} \varphi(f^j x) \longrightarrow \int_{\mathbb{T}} \varphi d\mu_f$$

uniformly in x .

Proof. By Denjoy theorem there is a circle homeomorphism h such that

$$h \circ f = R_\alpha \circ h, \tag{1}$$

where $\alpha \notin \mathbb{Q}$. Let m denote normalized Haar measure on $\mathbb{T}$, and define

$$\mu_f := (h^{-1})_* m. \tag{2}$$

For every Borel set $E \subset \mathbb{T}$, using (1) and invariance of m under R_α ,

$$\begin{aligned} (f_* \mu_f)(E) &= \mu_f(f^{-1}E) = m(h(f^{-1}E)) \\ &= m(R_\alpha^{-1}(h(E))) = m(h(E)) = \mu_f(E). \end{aligned}$$

Thus μ_f is f -invariant.

Now let ν be any f -invariant Borel probability. Push it forward by h : $\eta := h_* \nu$. From (1),

$$(R_\alpha)_* \eta = (R_\alpha)_* h_* \nu = h_* f_* \nu = h_* \nu = \eta.$$

Hence η is invariant under the irrational rotation. By *Minimal rotations are uniquely ergodic*, R_α is uniquely ergodic, so $\eta = m$. Since h is invertible,

$$\nu = (h^{-1})_* \eta = (h^{-1})_* m = \mu_f.$$

Therefore μ_f is the unique f -invariant Borel probability.

Finally, f is continuous on the compact circle and is uniquely ergodic. *Uniform ergodic theorem for unique ergodicity* therefore applies: for every $\varphi \in C(\mathbb{T})$,

$$\frac{1}{N} \sum_{j=0}^{N-1} \varphi(f^j x) \longrightarrow \int_{\mathbb{T}} \varphi d\mu_f$$

uniformly in x . This is the asserted final clause. $\square$

Proposition F06.8. Fiber/torus rotation conventions

Statement. For a lift $F : \mathbb{R} \rightarrow \mathbb{R}$ of an orientation-preserving circle homeomorphism, replacing F by $F + m$ changes its translation number by m , so the circle rotation number is a class in $\mathbb{R}/\mathbb{Z}$. For a torus homeomorphism f isotopic to the

identity with lift $\tilde{f} : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the displacement cocycle is $\varphi(x) = \tilde{f}(\tilde{x}) - \tilde{x}$. The rotation vector of an f -invariant probability μ is $\int \varphi d\mu$ modulo $\mathbb{Z}^d$, independent of the lift up to the corresponding integer vector.

Proof. We treat the circle and torus statements separately.

Circle lifts. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a lift of an orientation-preserving circle homeomorphism f . Any other lift G of the same map satisfies

$$G(x) - F(x) \in \mathbb{Z}$$

for every x , because $F(x)$ and $G(x)$ project to the same point of $\mathbb{R}/\mathbb{Z}$. The difference is a continuous integer-valued function of x , hence is constant. Thus

$$G = F + m \quad \text{for some } m \in \mathbb{Z}. \quad (1)$$

Since every degree-one lift commutes with integer translations, $F(x + m) = F(x) + m$. From (1), induction gives

$$G^n(x) = F^n(x) + nm. \quad (2)$$

Therefore

$$\frac{G^n(x) - x}{n} = \frac{F^n(x) - x}{n} + m.$$

Taking the limit supplied by *Existence of rotation number*,

$$\rho(G) = \rho(F) + m. \quad (3)$$

Hence the class $\rho(F) \bmod \mathbb{Z}$ is independent of the chosen lift and is exactly the circle rotation number.

Torus displacement. Now let $f : \mathbb{T}^d \rightarrow \mathbb{T}^d$ be isotopic to the identity and let $\tilde{f} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a lift satisfying

$$\tilde{f}(x + n) = \tilde{f}(x) + n \quad (n \in \mathbb{Z}^d). \quad (4)$$

For $x \in \mathbb{T}^d$, choose a lift $\tilde{x} \in \mathbb{R}^d$ and set

$$\phi(x) := \tilde{f}(\tilde{x}) - \tilde{x}. \quad (5)$$

This is independent of the representative $\tilde{x}$: replacing $\tilde{x}$ by $\tilde{x} + n$ and using (4) leaves the difference in (5) unchanged. Thus $\phi : \mathbb{T}^d \rightarrow \mathbb{R}^d$ is a well-defined continuous displacement cocycle.

Let μ be an f -invariant Borel probability measure. Its rotation vector relative to the chosen lift is

$$\text{rot}_{\tilde{f}}(\mu) := \int_{\mathbb{T}^d} \phi d\mu \in \mathbb{R}^d. \quad (6)$$

If $\hat{f}$ is another lift, then, exactly as in the circle case, $\hat{f} - \tilde{f}$ is a continuous $\mathbb{Z}^d$ -valued function on connected $\mathbb{R}^d$, hence a constant vector $\ell \in \mathbb{Z}^d$. The corresponding displacement is therefore

$$\hat{\phi} = \phi + \ell,$$

and so

$$\text{rot}_{\tilde{f}}(\mu) = \text{rot}_{\tilde{f}}(\mu) + \ell. \quad (7)$$

Consequently the class of (6) in $\mathbb{R}^d/\mathbb{Z}^d$ is independent of the chosen lift, with exactly the advertised integer normalization ambiguity. $\square$

6.3 Worked examples

Worked example: Circle and Torus Dynamics: a model computation

Start with $R_\alpha(x) = x + \alpha \pmod{1}$, whose lift is $F(x) = x + \alpha$. Then

$$F^n(x) - x = n\alpha,$$

so $\rho(R_\alpha) = \alpha \pmod{1}$.

Now choose

$$H(x) = x + \frac{\varepsilon}{2\pi} \sin(2\pi x), \quad |\varepsilon| < 1.$$

Since $H'(x) = 1 + \varepsilon \cos(2\pi x) > 0$, H is an increasing lift of a circle diffeomorphism h . Define

$$f = h^{-1} \circ R_\alpha \circ h.$$

A lift is $\tilde{F} = H^{-1} \circ (F \circ H)$, and conjugacy gives

$$\tilde{F}^n = H^{-1} \circ (x \mapsto x + n\alpha) \circ H.$$

Because $H(x) - x$ is bounded and periodic,

$$\frac{\tilde{F}^n(x) - x}{n} \longrightarrow \alpha.$$

Thus rotation number is invariant under orientation-preserving conjugacy, while the geometry of individual orbits has been nonlinearly deformed.

Boundary case: what fails at the edge of the hypotheses

Denjoy's construction begins with an irrational rotation orbit and replaces each orbit point by a small interval, choosing lengths with finite total sum. The resulting monotone factor map collapses those intervals back to the rotation orbit. One can arrange a C^1 circle diffeomorphism whose rotation number is irrational but for which the inserted intervals wander and never return to themselves. The map is semiconjugate, not conjugate, to the rigid rotation. The missing hypothesis is precisely the bounded-variation/distortion control on $\log f'$ used by Denjoy's theorem.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

6.4 Chapter summary**What to remember**

- Rotation number is a topological invariant extracted from a lift.
- Rational rotation number detects periodic behavior; irrational rotation number leads to a rotation factor.
- Distortion control prevents wandering intervals and upgrades semiconjugacy to conjugacy.
- Arithmetic and smoothness play distinct roles and should be tracked separately.

Looking ahead

F07 develops the analytic and tame norms needed to quantify regularity. Those estimates become the technical language for F08 and later KAM iterations.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

6.5 Exercises and problems**Exercise 6.1**

Difficulty. Core.

Related results. Existence of rotation number; Periodic orbit criterion; Poincaré semiconjugacy theorem.

Problem. Work through the worked example 'an irrational rigid rotation and a smooth conjugate of it'. For Existence of rotation number, Periodic orbit criterion, and Poincaré semiconjugacy theorem, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypoth-

esis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 6.2

Difficulty. Core.

Related results. Continuity and monotonicity of rotation number; Denjoy distortion inequality; Denjoy theorem.

Problem. Analyze the boundary example 'a Denjoy C^1 example with wandering intervals when BV regularity fails'. For Continuity and monotonicity of rotation number, Denjoy distortion inequality, and Denjoy theorem, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 6.3

Difficulty. Advanced.

Related results. Poincaré semiconjugacy theorem.

Problem. For Poincaré semiconjugacy theorem, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 6.4

Difficulty. Advanced.

Related results. Continuity and monotonicity of rotation number.

Problem. Stress-test Continuity and monotonicity of rotation number. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 6.5

Difficulty. Advanced.

Related results. Existence of rotation number; Periodic orbit criterion.

Problem. Synthesis problem. Use Existence of rotation number and Periodic orbit

criterion to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 6.6

Difficulty. Lab.

Related results. None.

Problem. Computational laboratory: carry out the reproducible experiment ‘an irrational rigid rotation and a smooth conjugate of it’. Use the computation only as numerical evidence or visualization, not as proof. Include a refinement or convergence test and identify one numerical artifact that could mislead a reader.

Solution sketch. State parameters and truncation/time horizon, compare with an analytically controlled identity or exactly solvable case, repeat after refinement, and explain what the numerical experiment can and cannot establish.

Exercise 6.7

Difficulty. Core.

Related results. Unique ergodicity of irrational circle diffeomorphisms under Denjoy hypotheses.

Problem. Work directly with Unique ergodicity of irrational circle diffeomorphisms under Denjoy hypotheses. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 6.8

Difficulty. Core.

Related results. Fiber/torus rotation conventions.

Problem. Work directly with Fiber/torus rotation conventions. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Quantitative Function Spaces and Tame Calculus

Why this chapter matters

A qualitative statement such as “the solution is analytic” is too weak for an iteration. KAM and Nash-Moser arguments spend regularity at every step, so one must know exactly how much is lost under differentiation, multiplication, composition, inversion, and smoothing.

This chapter is a toolbox of quantitative estimates. Its purpose is not to create new dynamics but to make later dynamical arguments honest: every strip width, derivative loss, and tame constant must have a place to live.

Prerequisites. Fourier series, several-variable calculus, Sobolev spaces, and complex analysis on strips.

Learning goals

- Estimate Fourier coefficients of analytic and Gevrey functions.
- Use Cauchy estimates to convert lost strip width into derivative bounds.
- Control products, compositions, and near-identity inverses.
- Understand why smoothing restores derivatives in Nash-Moser schemes.

Conceptual roadmap

Analytic strip norms and Fourier decay are two views of the same regularity. Cauchy estimates explain derivative loss. Product and composition bounds control nonlinear operations. Smoothing and interpolation then provide a way to trade high-frequency information for regularity, leading to tame estimates on scales of spaces.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

7.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F07.1. Analytic strip norm

For $h > 0$, $\mathbb{T}_h^d = \{z \in \mathbb{C}^d / \mathbb{Z}^d : |\operatorname{Im} z_j| < h \text{ for all } j\}$. Unless another analytic norm is declared, $\|f\|_h = \sup_{z \in \mathbb{T}_h^d} |f(z)|$.

Definition F07.2. Smoothing operator

On a Banach scale, a family (S_N) is a *smoothing family* if it satisfies direct estimates such as $\|S_N u\|_r \leq C N^{r-s} \|u\|_s$ and approximation estimates $\|(I - S_N)u\|_s \leq C N^{s-r} \|u\|_r$ in the stated index ranges.

Definition F07.3. Tame estimate

A *tame estimate* on a graded scale is an inequality in which the highest norm of a nonlinear expression occurs at most linearly, multiplied by controlled lower norms, with constants uniform on a prescribed bounded set.

Definition F07.4. Gevrey class

A function is *Gevrey of order* $s \geq 1$ if its derivatives satisfy $\|\partial^\alpha f\| \leq C R^{|\alpha|} (\alpha!)^s$ for all multi-indices α . In the local real-variable convention, $s = 1$ corresponds to analyticity.

7.2 Main results

Proposition F07.1. Gevrey Fourier characterization

Statement. Fix $\sigma \geq 1$ and the Fourier convention $f(x) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k) e^{2\pi i \langle k, x \rangle}$. If $\|\partial^\alpha f\|_\infty \leq M R^{|\alpha|} (\alpha!)^\sigma$ for every multi-index α , then $|\widehat{f}(k)| \leq 2M \exp[-(\log 2/2)(\pi|k|_\infty/R)^{1/\sigma}]$ for every k . Conversely, if $|\widehat{f}(k)| \leq A e^{-a|k|_1^{1/\sigma}}$, then $f \in G^\sigma(\mathbb{T}^d)$ and $\|\partial^\alpha f\|_\infty \leq A S_{a,\sigma,d} [2\pi d^\sigma (2\sigma/a)^\sigma]^{|\alpha|} (\alpha!)^\sigma$, where $S_{a,\sigma,d} = \sum_k e^{-(a/2)|k|_1^{1/\sigma}} < \infty$.

Proof. Fix $\sigma \geq 1$ and use

$$f(x) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k) e^{2\pi i \langle k, x \rangle}.$$

We prove the two implications separately.

Step 1: Gevrey derivative bounds imply stretched-exponential Fourier decay. Assume

$$\|\partial^\alpha f\|_{L^\infty} \leq MR^{|\alpha|}(\alpha!)^\sigma \quad (\alpha \in \mathbb{N}^d). \quad (1)$$

For $k \neq 0$, choose j with $|k_j| = |k|_\infty =: K$. Integrating the Fourier coefficient by parts m times in x_j gives, for every integer $m \geq 0$,

$$|\widehat{f}(k)| \leq \frac{\|\partial_j^m f\|_{L^\infty}}{(2\pi K)^m} \leq M \left(\frac{R(m!)^\sigma}{(2\pi K)^m} \right). \quad (2)$$

Using $m! \leq m^m$,

$$|\widehat{f}(k)| \leq M \left(\frac{Rm^\sigma}{2\pi K} \right)^m. \quad (3)$$

Put

$$A := \left(\frac{\pi K}{R} \right)^{1/\sigma}.$$

If $A \geq 2$, choose $m = \lfloor A \rfloor$. Then $m \leq A$, so the factor in parentheses in (3) is at most $1/2$; moreover $m \geq A/2$. Hence

$$|\widehat{f}(k)| \leq M2^{-m} \leq M \exp \left(-\frac{\log 2}{2} \left(\frac{\pi K}{R} \right)^{1/\sigma} \right). \quad (4)$$

If $A < 2$, the trivial estimate $|\widehat{f}(k)| \leq \|f\|_\infty \leq M$ and $\exp[-(\log 2)A/2] > 1/2$ give the same bound with an extra factor 2. Thus for every $k \in \mathbb{Z}^d$,

$$|\widehat{f}(k)| \leq 2M \exp \left[-\frac{\log 2}{2} \left(\frac{\pi |k|_\infty}{R} \right)^{1/\sigma} \right]. \quad (5)$$

(The $k = 0$ coefficient is covered by $|\widehat{f}(0)| \leq M$.)

Step 2: stretched-exponential Fourier decay implies Gevrey derivative bounds. Conversely suppose that, for some $A, a > 0$,

$$|\widehat{f}(k)| \leq Ae^{-a|k|_1^{1/\sigma}} \quad (k \in \mathbb{Z}^d). \quad (6)$$

The right-hand side is summable, and remains summable after multiplication by any power of $|k|_1$. Consequently the Fourier series and all of its formal derivatives converge normally on the real torus; in particular $f \in C^\infty$ and termwise differentiation is legitimate.

Let α be a multi-index and write $m = |\alpha|$. Since $|k^\alpha| \leq |k|_1^m$,

$$\|\partial^\alpha f\|_\infty \leq A(2\pi)^m \sum_{k \in \mathbb{Z}^d} |k|_1^m e^{-a|k|_1^{1/\sigma}}. \quad (7)$$

Split the exponential into two equal factors. For $t \geq 0$, set $y = t^{1/\sigma}$. Elementary calculus

gives

$$\sup_{t \geq 0} t^m e^{-(a/2)t^{1/\sigma}} = \sup_{y \geq 0} y^{\sigma m} e^{-(a/2)y} = \left(\frac{2\sigma m}{ae} \right)^{\sigma m}. \quad (8)$$

The elementary Stirling lower bound $m! \geq (m/e)^m$ therefore yields

$$\sup_{t \geq 0} t^m e^{-(a/2)t^{1/\sigma}} \leq \left(\frac{2\sigma}{a} \right)^{\sigma m} (m!)^\sigma. \quad (9)$$

Define the finite constant

$$S_{a,\sigma,d} := \sum_{k \in \mathbb{Z}^d} e^{-(a/2)|k|_1^{1/\sigma}} < \infty. \quad (10)$$

Using (9) for one half of the exponential in (7) and (10) for the other,

$$\|\partial^\alpha f\|_\infty \leq AS_{a,\sigma,d} (2\pi)^m \left(\frac{2\sigma}{a} \right)^{\sigma m} (m!)^\sigma. \quad (11)$$

Finally

$$\frac{m!}{\alpha!} = \binom{m}{\alpha_1, \dots, \alpha_d} \leq d^m, \quad (12)$$

so $(m!)^\sigma \leq d^{\sigma m} (\alpha!)^\sigma$. Hence

$$\boxed{\|\partial^\alpha f\|_\infty \leq AS_{a,\sigma,d} \left[2\pi d^\sigma \left(\frac{2\sigma}{a} \right)^\sigma \right]^{|\alpha|} (\alpha!)^\sigma.} \quad (13)$$

This is precisely the Gevrey- σ derivative bound of *the definition of Gevrey class*. Together, (5) and (13) give the claimed quantitative equivalence. $\square$

Theorem F07.2. Cauchy estimates on complex strips

Statement. Let $0 < h' < h$, let f be a bounded holomorphic scalar-, vector-, or matrix-valued function on $\mathbb{T}_h^d$, and let $\beta \in \mathbb{N}^d$ be a multi-index. With the strip sup norm of Definition F07.1, $\|\partial^\beta f\|_{h'} \leq \beta! (h - h')^{-|\beta|} \|f\|_h$ (componentwise for vector/matrix values).

Proof. Let

$$\beta = (\beta_1, \dots, \beta_d) \in \mathbb{N}^d, \quad |\beta| := \beta_1 + \dots + \beta_d, \quad \beta! := \beta_1! \cdots \beta_d!.$$

Fix a point $z = (z_1, \dots, z_d) \in \mathbb{T}_{h'}^d$ and choose

$$0 < \rho < h - h'.$$

Because $|\operatorname{Im} z_j| < h'$ for every j , changing any coordinate z_j by a complex number of modulus at most ρ keeps that coordinate inside the strip $|\operatorname{Im} w_j| < h$. Hence the closed coordinate polydisc

$$\{w : |w_j - z_j| \leq \rho \text{ for all } j\}$$

is contained in $\mathbb{T}_h^d$ after passing to the periodic quotient.

For a scalar-valued f , apply the one-variable Cauchy integral formula β_j times in the j th coordinate, successively for $j = 1, \dots, d$. This gives

$$\partial^\beta f(z) = \frac{\beta!}{(2\pi i)^d} \int_{|\zeta_1 - z_1|=\rho} \cdots \int_{|\zeta_d - z_d|=\rho} \frac{f(\zeta_1, \dots, \zeta_d)}{\prod_{j=1}^d (\zeta_j - z_j)^{\beta_j+1}} d\zeta_d \cdots d\zeta_1. \quad (1)$$

Taking absolute values in (1), each contour contributes a factor $2\pi\rho$, while the denominator contributes ρ^{β_j+1} in the j th variable. Therefore

$$|\partial^\beta f(z)| \leq \beta! \rho^{-|\beta|} \|f\|_h. \quad (2)$$

The bound is independent of $z \in \mathbb{T}_{h'}^d$, so

$$\|\partial^\beta f\|_{h'} \leq \beta! \rho^{-|\beta|} \|f\|_h.$$

Since this holds for every $\rho < h - h'$, letting $\rho \uparrow h - h'$ yields

$$\|\partial^\beta f\|_{h'} \leq \beta! (h - h')^{-|\beta|} \|f\|_h.$$

For vector-valued functions the argument is applied componentwise. For matrix-valued functions it is applied entrywise (or, equivalently in finite dimension, after composing with norm-one linear functionals); the same stated componentwise estimate follows. This proves the theorem. $\square$

Proposition F07.3. Fourier decay of analytic periodic functions

Statement. With the Fourier convention

$$f(x) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k) e^{2\pi i \langle k, x \rangle},$$

if f is bounded and holomorphic on $\mathbb{T}_h^d$, then

$$|\widehat{f}(k)| \leq \|f\|_h e^{-2\pi h |k|_1} \quad (k \in \mathbb{Z}^d).$$

Conversely, if coefficients (a_k) satisfy $\sum_k |a_k| e^{2\pi h' |k|_1} < \infty$ for every $0 < h' < h$, then $\sum_k a_k e^{2\pi i \langle k, z \rangle}$ converges normally on every $\mathbb{T}_{h'}^d$, defines a holomorphic periodic function there, and

$$\|f\|_{h'} \leq \sum_k |a_k| e^{2\pi h' |k|_1}.$$

Proof. We use the Fourier convention from the statement,

$$\widehat{f}(k) = \int_{[0,1]^d} f(x) e^{-2\pi i \langle k, x \rangle} dx.$$

Forward implication: analyticity gives exponential decay. Fix $k \in \mathbb{Z}^d$ and $0 < s < h$. Define a real vector $y = (y_1, \dots, y_d)$ by

$$y_j := -s \operatorname{sgn}(k_j),$$

where $\text{sgn}(0) = 0$. Periodicity and holomorphy allow the integration contour in each coordinate to be shifted from x_j to $x_j + iy_j$. Performing these shifts successively gives

$$\widehat{f}(k) = \int_{[0,1]^d} f(x + iy) e^{-2\pi i \langle k, x+iy \rangle} dx. \quad (1)$$

Now

$$\left| e^{-2\pi i \langle k, x+iy \rangle} \right| = e^{2\pi \langle k, y \rangle} = e^{-2\pi s |k|_1}.$$

Since $x + iy \in \mathbb{T}_h^d$, (1) yields

$$|\widehat{f}(k)| \leq \|f\|_h e^{-2\pi s |k|_1}.$$

This is true for every $s < h$. Letting $s \uparrow h$ gives

$$\boxed{|\widehat{f}(k)| \leq \|f\|_h e^{-2\pi h |k|_1}}. \quad (2)$$

Converse implication: weighted summability reconstructs an analytic function. Let $(a_k)_{k \in \mathbb{Z}^d}$ satisfy

$$\sum_{k \in \mathbb{Z}^d} |a_k| e^{2\pi h' |k|_1} < \infty \quad \text{for every } 0 < h' < h. \quad (3)$$

Fix one such h' . If $z = x + iy$ with $|y_j| < h'$, then

$$\left| e^{2\pi i \langle k, z \rangle} \right| = e^{-2\pi \langle k, y \rangle} \leq e^{2\pi h' |k|_1}. \quad (4)$$

Therefore (3) and the Weierstrass M -test show that

$$\sum_k a_k e^{2\pi i \langle k, z \rangle} \quad (5)$$

converges absolutely and uniformly on $\mathbb{T}_{h'}^d$, in the sup-norm sense (equivalently on every compact fundamental-domain closure strictly inside that strip), and

$$\left\| \sum_k a_k e^{2\pi i \langle k, z \rangle} \right\|_{h'} \leq \sum_k |a_k| e^{2\pi h' |k|_1}. \quad (6)$$

Each finite partial sum is entire and $\mathbb{Z}^d$ -periodic. To justify holomorphy of the limit at an arbitrary point of $\mathbb{T}_{h'}^d$, choose h'' with

$$h' < h'' < h.$$

The same argument gives uniform convergence on the larger strip $\mathbb{T}_{h''}^d$, hence locally uniform convergence in a neighborhood of every point of $\mathbb{T}_{h'}^d$. By the Weierstrass theorem, the limit is holomorphic there. Since $h' < h$ was arbitrary, (5) defines a holomorphic periodic function on the whole open strip $\mathbb{T}_h^d$, and (6) gives the claimed bound on every smaller strip. $\square$

Proposition F07.4. Analytic algebra/product estimates

Statement. Let f, g be bounded holomorphic scalar functions on $\mathbb{T}_h^d$. Then $\|fg\|_h \leq \|f\|_h \|g\|_h$. For matrix-valued A, B and a fixed submultiplicative matrix norm, $\|AB\|_h \leq \|A\|_h \|B\|_h$ and $\|[A, B]\|_h \leq 2\|A\|_h \|B\|_h$. The same estimates hold on every smaller strip without change of constant.

Proof. The point of this proposition is that no derivative or strip loss occurs: the sup norm is already an algebra norm.

For scalar-valued f and g , at every $z \in \mathbb{T}_h^d$ we have

$$|f(z)g(z)| \leq |f(z)| |g(z)| \leq \|f\|_h \|g\|_h.$$

Taking the supremum over z gives

$$\|fg\|_h \leq \|f\|_h \|g\|_h. \quad (1)$$

The product is holomorphic because products of holomorphic functions are holomorphic.

Now let A and B be matrix-valued, and let $|\cdot|$ be the fixed submultiplicative matrix norm, so

$$|MN| \leq |M| |N|$$

for all matrices of the relevant size. Pointwise,

$$|A(z)B(z)| \leq |A(z)| |B(z)|,$$

and therefore

$$\|AB\|_h \leq \|A\|_h \|B\|_h. \quad (2)$$

For the commutator $[A, B] = AB - BA$, the triangle inequality and (2) give

$$\begin{aligned} \|[A, B]\|_h &\leq \|AB\|_h + \|BA\|_h \\ &\leq \|A\|_h \|B\|_h + \|B\|_h \|A\|_h \\ &= 2\|A\|_h \|B\|_h. \end{aligned} \quad (3)$$

Finally, if $0 < h' < h$, restricting a function from $\mathbb{T}_h^d$ to $\mathbb{T}_{h'}^d$ can only decrease its sup norm. Thus the same arguments give (1)–(3) on every smaller strip with the same constants. $\square$

Proposition F07.5. Composition estimates

Statement. Let $0 < h' < h$, $\delta = h - h'$, and let $f : \mathbb{T}_h^d \rightarrow \mathbb{C}^m$ be bounded holomorphic. If $u : \mathbb{T}_{h'}^d \rightarrow \mathbb{C}^d$ is holomorphic and periodic with $\eta := \|u\|_{h'} < \delta$, then $\Phi = \text{Id} + u$ maps $\mathbb{T}_{h'}^d$ into $\mathbb{T}_h^d$ and $\|f \circ \Phi\|_{h'} \leq \|f\|_h$, while $\|f \circ \Phi - f\|_{h'} \leq d\eta(\delta - \eta)^{-1} \|f\|_h$. For integers $r \geq 1$, if $f \in C^r(U, \mathbb{R}^m)$ and $\Phi = \text{Id} + u \in C^r(V, U)$, then $\|f \circ \Phi\|_{C^r(V)} \leq C_{r,d,m} \|f\|_{C^r(U)} (1 + \|u\|_{C^r(V)})^r$.

Proof. We begin with the analytic estimate. Put

$$\delta := h - h' > 0, \quad \eta := \|u\|_{h'} < \delta, \quad \Phi(z) := z + u(z).$$

For $z \in \mathbb{T}_{h'}^d$ and each coordinate j ,

$$|\operatorname{Im} \Phi_j(z)| \leq |\operatorname{Im} z_j| + |u_j(z)| < h' + \eta < h.$$

Hence

$$\Phi(\mathbb{T}_{h'}^d) \subset \mathbb{T}_{h'+\eta}^d \subset \mathbb{T}_h^d. \quad (1)$$

The first bound follows immediately:

$$\|f \circ \Phi\|_{h'} \leq \|f\|_h. \quad (2)$$

For the difference estimate, fix $z \in \mathbb{T}_{h'}^d$ and consider the segment

$$z_t := z + tu(z), \quad 0 \leq t \leq 1.$$

By the same estimate as in (1), every z_t lies in $\mathbb{T}_{h'+\eta}^d$. The fundamental theorem of calculus gives

$$f(z + u(z)) - f(z) = \int_0^1 Df(z_t) u(z) dt. \quad (3)$$

The distance from $\mathbb{T}_{h'+\eta}^d$ to the boundary of $\mathbb{T}_h^d$ in each coordinate is

$$h - (h' + \eta) = \delta - \eta.$$

Applying *Cauchy estimates on complex strips* to each first partial derivative,

$$\|\partial_j f\|_{h'+\eta} \leq \frac{\|f\|_h}{\delta - \eta} \quad (1 \leq j \leq d). \quad (4)$$

Using the coordinate sup norm for vectors,

$$|Df(z_t)u(z)| \leq \sum_{j=1}^d |\partial_j f(z_t)| |u_j(z)| \leq d \frac{\|f\|_h}{\delta - \eta} \eta.$$

Taking the supremum in (3) proves

$$\boxed{\|f \circ \Phi - f\|_{h'} \leq d \frac{\eta}{\delta - \eta} \|f\|_h.} \quad (5)$$

We now prove the C^r estimate. Let $f \in C^r(U, \mathbb{R}^m)$ and $\Phi = \operatorname{Id} + u \in C^r(V, U)$. We use the norm

$$\|g\|_{C^r} := \max_{0 \leq j \leq r} \sup \|D^j g\|,$$

with any fixed equivalent operator norms on the finite-dimensional multilinear spaces. For $1 \leq k \leq r$, the multivariable chain rule (Faà di Bruno formula) expresses $D^k(f \circ \Phi)$

as a finite sum of terms of the form

$$(D^\ell f)(\Phi(x)) [D^{j_1} \Phi(x), \dots, D^{j_\ell} \Phi(x)], \quad (6)$$

where

$$1 \leq \ell \leq k, \quad j_i \geq 1, \quad j_1 + \dots + j_\ell = k.$$

The number and combinatorial coefficients of these terms depend only on r, d, m . Since

$$D\Phi = I + Du, \quad D^j \Phi = D^j u \quad (j \geq 2),$$

every derivative factor in (6) is bounded by

$$1 + \|u\|_{C^r(V)}.$$

Also

$$\|(D^\ell f) \circ \Phi\|_{C^0(V)} \leq \|f\|_{C^r(U)}.$$

Therefore each term in (6) is bounded by

$$\|f\|_{C^r(U)} (1 + \|u\|_{C^r(V)})^\ell \leq \|f\|_{C^r(U)} (1 + \|u\|_{C^r(V)})^r.$$

Summing the finitely many terms and including $k = 0$ gives a constant $C_{r,d,m}$ such that

$$\|f \circ \Phi\|_{C^r(V)} \leq C_{r,d,m} \|f\|_{C^r(U)} (1 + \|u\|_{C^r(V)})^r. \quad (7)$$

Equations (2), (5), and (7) are exactly the three advertised composition bounds. $\square$

Lemma F07.6. Quantitative inverse-function estimate

Statement. Let $0 < h' < h$ and let $u : \mathbb{T}_h^d \rightarrow \mathbb{C}^d$ be bounded holomorphic and periodic. Put $\Phi = \text{Id} + u$, $\eta = \|u\|_h$, and $\kappa = \|Du\|_h$ in the operator norm induced by the coordinate sup norm. If $\kappa < 1$ and $h' + \eta < h$, then every $y \in \mathbb{T}_{h'}^d$ has a unique preimage $x \in \mathbb{T}_h^d$ with $\|x - y\|_\infty \leq \eta$. Thus the inverse branch on $\mathbb{T}_{h'}^d$ has the form $\Phi^{-1} = \text{Id} + v$ with $\|v\|_{h'} \leq \eta$ and $\|Dv\|_{h'} \leq \kappa/(1 - \kappa)$.

Proof. Work first with the periodic lifts on $\mathbb{C}^d$. Fix $y \in \mathbb{T}_{h'}^d$, and choose a representative, still denoted y , with $|\text{Im } y_j| < h'$ for all j . Consider the closed sup-norm ball

$$B_y := \{x \in \mathbb{C}^d : \|x - y\|_\infty \leq \eta\}.$$

Because $h' + \eta < h$, every point of B_y lies in the strip $\mathbb{T}_h^d$ at the level of the lift.

Define

$$\mathcal{C}_y(x) := y - u(x).$$

For $x \in B_y$,

$$\|\mathcal{C}_y(x) - y\|_\infty = \|u(x)\|_\infty \leq \eta,$$

so $\mathcal{C}_y$ maps B_y into itself. Moreover, for $x_1, x_2 \in B_y$, the straight segment joining them

remains in the convex strip, and the mean-value formula gives

$$\|u(x_1) - u(x_2)\|_\infty \leq \kappa \|x_1 - x_2\|_\infty.$$

Hence

$$\|\mathcal{C}_y(x_1) - \mathcal{C}_y(x_2)\|_\infty \leq \kappa \|x_1 - x_2\|_\infty. \quad (1)$$

Since $\kappa < 1$, the contraction mapping theorem produces a unique $x \in B_y$ satisfying

$$x = \mathcal{C}_y(x) = y - u(x),$$

or equivalently

$$\Phi(x) = x + u(x) = y. \quad (2)$$

The construction gives immediately

$$\|x - y\|_\infty \leq \eta. \quad (3)$$

The solution is compatible with integer translations. If $n \in \mathbb{Z}^d$, periodicity of u shows that $x + n$ is the unique solution associated with $y + n$. Thus the solution descends to the torus strip and defines an inverse branch

$$\Phi^{-1}(y) = y + v(y), \quad v(y) := x - y.$$

By (3),

$$\|v\|_{h'} \leq \eta. \quad (4)$$

We next justify regularity and the derivative bound. At every solution x , the derivative

$$D\Phi(x) = I + Du(x)$$

is invertible because $\|Du(x)\| \leq \kappa < 1$; indeed the Neumann series gives

$$\|(I + Du(x))^{-1}\| \leq \frac{1}{1 - \kappa}. \quad (5)$$

The holomorphic implicit-function theorem therefore makes the unique solution $x = \Phi^{-1}(y)$ holomorphic in y . Differentiating $\Phi(\Phi^{-1}(y)) = y$ yields

$$D\Phi^{-1}(y) = (I + Du(x))^{-1}.$$

Since $\Phi^{-1} = I + v$,

$$Dv(y) = (I + Du(x))^{-1} - I = -(I + Du(x))^{-1} Du(x).$$

Using (5),

$$\boxed{\|Dv\|_{h'} \leq \frac{\kappa}{1 - \kappa}}. \quad (6)$$

Together, (2), (4), and (6) give the asserted inverse and its explicit norm/domain-loss bounds. $\square$

Theorem F07.7. Smoothing operator estimates

Statement. For $s \leq r$ and $N \geq 1$, let S_N be Fourier truncation on $\mathbb{T}^d$ to frequencies $|k| \leq N$. Then

$$\|S_N u\|_{H^r} \leq (1 + N^2)^{(r-s)/2} \|u\|_{H^s}.$$

For $r \leq s$,

$$\|(I - S_N)u\|_{H^r} \leq (1 + N^2)^{(r-s)/2} \|u\|_{H^s}.$$

The same family satisfies $S_N S_M = S_{\min\{N, M\}}$ and commutes with derivatives.

Proof. Write the Fourier series of u as

$$u(x) = \sum_{k \in \mathbb{Z}^d} \widehat{u}(k) e^{2\pi i \langle k, x \rangle}$$

and use the Sobolev norm

$$\|u\|_{H^s}^2 := \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^s |\widehat{u}(k)|^2.$$

For $N \geq 1$, let

$$S_N u := \sum_{|k| \leq N} \widehat{u}(k) e^{2\pi i \langle k, x \rangle}.$$

The two estimates come from comparing the same Fourier weights on two different frequency regions.

Low-frequency estimate. Assume $s \leq r$. For $|k| \leq N$,

$$(1 + |k|^2)^r = (1 + |k|^2)^{r-s} (1 + |k|^2)^s \leq (1 + N^2)^{r-s} (1 + |k|^2)^s.$$

Therefore

$$\begin{aligned} \|S_N u\|_{H^r}^2 &= \sum_{|k| \leq N} (1 + |k|^2)^r |\widehat{u}(k)|^2 \\ &\leq (1 + N^2)^{r-s} \sum_{|k| \leq N} (1 + |k|^2)^s |\widehat{u}(k)|^2 \\ &\leq (1 + N^2)^{r-s} \|u\|_{H^s}^2. \end{aligned}$$

Taking square roots yields

$$\|S_N u\|_{H^r} \leq (1 + N^2)^{(r-s)/2} \|u\|_{H^s}. \quad (1)$$

High-frequency tail estimate. Assume now $r \leq s$. Since $r - s \leq 0$, the function $t \mapsto t^{r-s}$ decreases on $(0, \infty)$. Hence, on $|k| > N$,

$$(1 + |k|^2)^{r-s} \leq (1 + N^2)^{r-s}.$$

It follows that

$$\begin{aligned} \|(I - S_N)u\|_{H^r}^2 &= \sum_{|k| > N} (1 + |k|^2)^r |\widehat{u}(k)|^2 \\ &\leq (1 + N^2)^{r-s} \sum_{|k| > N} (1 + |k|^2)^s |\widehat{u}(k)|^2 \\ &\leq (1 + N^2)^{r-s} \|u\|_{H^s}^2. \end{aligned}$$

Thus

$$\|(I - S_N)u\|_{H^r} \leq (1 + N^2)^{(r-s)/2} \|u\|_{H^s}. \quad (2)$$

Notice that the exponent in (2) is nonpositive, so the tail decreases with N whenever $s > r$.

Algebraic properties. On the k -th Fourier coefficient, S_N is multiplication by $\mathbf{1}_{\{|k| \leq N\}}$. Therefore

$$S_N S_M = S_{\min(N, M)}.$$

A derivative ∂^α multiplies the same Fourier coefficient by $(2\pi i k)^\alpha$, which commutes with the cutoff multiplier. Consequently

$$\partial^\alpha S_N u = S_N \partial^\alpha u$$

whenever the indicated derivative belongs to the relevant distribution or Sobolev class. Equations (1)–(2) and these coefficientwise identities prove every clause of the statement. $\square$

Theorem F07.8. Interpolation inequalities

Statement. (i) Let $s_0 \leq s \leq s_1$ be real and write $s = (1 - \theta)s_0 + \theta s_1$ with $0 \leq \theta \leq 1$. For every $u \in H^{s_1}(\mathbb{T}^d)$,

$$\|u\|_{H^s} \leq \|u\|_{H^{s_0}}^{1-\theta} \|u\|_{H^{s_1}}^\theta.$$

(ii) If $0 < \alpha < \beta \leq 1$ and v is bounded and β -Hölder on $\mathbb{T}^d$, then

$$[v]_{C^\alpha} \leq (2\|v\|_{L^\infty})^{1-\alpha/\beta} [v]_{C^\beta}^{\alpha/\beta}.$$

Consequently, for $u \in C^{m, \beta}$ and every multi-index γ with $|\gamma| = m$, the same estimate applied to $v = \partial^\gamma u$ interpolates the $C^{m, \alpha}$ seminorm between the C^m sup norm and the $C^{m, \beta}$ seminorm.

Proof. Sobolev interpolation. Use the Fourier norm

$$\|u\|_{H^r}^2 = \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^r |\widehat{u}(k)|^2.$$

Let

$$s = (1 - \theta)s_0 + \theta s_1, \quad 0 \leq \theta \leq 1.$$

For each Fourier mode,

$$(1 + |k|^2)^s |\widehat{u}(k)|^2 = \left((1 + |k|^2)^{s_0} |\widehat{u}(k)|^2 \right)^{1-\theta} \times \left((1 + |k|^2)^{s_1} |\widehat{u}(k)|^2 \right)^\theta.$$

If $0 < \theta < 1$, Hölder's inequality for sums with exponents $(1 - \theta)^{-1}$ and θ^{-1} gives

$$\|u\|_{H^s}^2 \leq (\|u\|_{H^{s_0}}^2)^{1-\theta} (\|u\|_{H^{s_1}}^2)^\theta. \quad (2)$$

Taking square roots proves

$$\|u\|_{H^s} \leq \|u\|_{H^{s_0}}^{1-\theta} \|u\|_{H^{s_1}}^\theta. \quad (3)$$

For $\theta = 0$ or 1 , (3) is simply equality at an endpoint, so all cases are covered.

Hölder interpolation. Let

$$A := 2\|v\|_{L^\infty}, \quad B := [v]_{C^\beta}.$$

If $A = 0$ or $B = 0$, then v is constant and the desired estimate is immediate. Assume $A, B > 0$ and put

$$r_0 := \left(\frac{A}{B} \right)^{1/\beta}. \quad (4)$$

For distinct $x, y \in \mathbb{T}^d$, write $r = d(x, y)$. The same increment has the two elementary bounds

$$|v(x) - v(y)| \leq A, \quad |v(x) - v(y)| \leq Br^\beta. \quad (5)$$

Therefore

$$\frac{|v(x) - v(y)|}{r^\alpha} \leq \min\{Ar^{-\alpha}, Br^{\beta-\alpha}\}. \quad (6)$$

The first function in the minimum decreases with r , while the second increases. They agree at $r = r_0$. Hence their minimum is bounded for all $r > 0$ by their common value there:

$$\begin{aligned} Ar_0^{-\alpha} &= A \left(\frac{A}{B} \right)^{-\alpha/\beta} \\ &= A^{1-\alpha/\beta} B^{\alpha/\beta}. \end{aligned} \quad (7)$$

Taking the supremum over $x \neq y$ in (6) gives

$$[v]_{C^\alpha} \leq (2\|v\|_{L^\infty})^{1-\alpha/\beta} [v]_{C^\beta}^{\alpha/\beta}. \quad (8)$$

This argument does not require the distance r_0 to be realized on the torus: if r_0 lies outside the available range of distances, the minimum in (6) stays on only one side of its crossing point and is even smaller than the right-hand side of (7).

Finally, if $u \in C^{m,\beta}$ and $|\gamma| = m$, apply (8) to $v = \partial^\gamma u$. This gives

$$[\partial^\gamma u]_{C^\alpha} \leq (2\|\partial^\gamma u\|_\infty)^{1-\alpha/\beta} [\partial^\gamma u]_{C^\beta}^{\alpha/\beta},$$

which is the announced $C^{m,\alpha}$ interpolation consequence. $\square$

Proposition F07.9. Sobolev tame product estimate

Statement. Let $s_0 > d/2$ and $s \geq s_0$ be real. There is a constant $C = C(s, s_0, d)$ such that, for all scalar-valued $u, v \in H^s(\mathbb{T}^d)$,

$$\|uv\|_{H^s} \leq C(\|u\|_{H^s}\|v\|_{H^{s_0}} + \|u\|_{H^{s_0}}\|v\|_{H^s}).$$

The same conclusion holds componentwise for vector- and matrix-valued functions, with the constant enlarged only by the fixed finite target dimension.

Scope note

This proposition is only a product estimate. Sobolev composition and diffeomorphism inversion require separate hypotheses and results.

Proof. We use the Fourier convention

$$u(x) = \sum_{k \in \mathbb{Z}^d} \widehat{u}(k) e^{2\pi i \langle k, x \rangle}, \quad \langle k \rangle = (1 + |k|^2)^{1/2},$$

so that

$$\|u\|_{H^s}^2 = \sum_{k \in \mathbb{Z}^d} \langle k \rangle^{2s} |\widehat{u}(k)|^2. \quad (1)$$

The proof has two elementary ingredients.

Step 1: the low Sobolev norm controls the Fourier ℓ^1 norm. Because $s_0 > d/2$,

$$C_{s_0, d}^{(1)} := \left(\sum_{k \in \mathbb{Z}^d} \langle k \rangle^{-2s_0} \right)^{1/2} < \infty. \quad (2)$$

Hence Cauchy-Schwarz gives

$$\sum_k |\widehat{u}(k)| = \sum_k \langle k \rangle^{-s_0} \langle k \rangle^{s_0} |\widehat{u}(k)| \leq C_{s_0, d}^{(1)} \|u\|_{H^{s_0}}. \quad (3)$$

Thus the Fourier coefficients of every H^{s_0} function are absolutely summable.

Step 2: put the high weight on only one factor. For $k, \ell \in \mathbb{Z}^d$, the triangle inequality implies

$$\langle k \rangle \leq \langle \ell \rangle + \langle k - \ell \rangle.$$

Since $s \geq 0$, there is a constant $C_s^{(2)}$ such that

$$\langle k \rangle^s \leq C_s^{(2)} (\langle \ell \rangle^s + \langle k - \ell \rangle^s). \quad (4)$$

The product has Fourier coefficients

$$\widehat{uv}(k) = \sum_{\ell \in \mathbb{Z}^d} \widehat{u}(\ell) \widehat{v}(k - \ell). \quad (5)$$

Introduce the nonnegative sequences

$$a_s(\ell) = \langle \ell \rangle^s |\widehat{u}(\ell)|, \quad a_0(\ell) = |\widehat{u}(\ell)|,$$

$$b_s(m) = \langle m \rangle^s |\widehat{v}(m)|, \quad b_0(m) = |\widehat{v}(m)|.$$

Multiplying (5) by $\langle k \rangle^s$ and using (4),

$$\langle k \rangle^s |\widehat{uv}(k)| \leq C_s^{(2)} [(a_s * b_0)(k) + (a_0 * b_s)(k)]. \quad (6)$$

Taking the ℓ^2 norm and applying Young's convolution inequality $\ell^2 * \ell^1 \rightarrow \ell^2$,

$$\begin{aligned} \|uv\|_{H^s} &\leq C_s^{(2)} (\|a_s * b_0\|_{\ell^2} + \|a_0 * b_s\|_{\ell^2}) \\ &\leq C_s^{(2)} (\|a_s\|_{\ell^2} \|b_0\|_{\ell^1} + \|a_0\|_{\ell^1} \|b_s\|_{\ell^2}). \end{aligned} \quad (7)$$

By (1),

$$\|a_s\|_{\ell^2} = \|u\|_{H^s}, \quad \|b_s\|_{\ell^2} = \|v\|_{H^s},$$

and by (3),

$$\|a_0\|_{\ell^1} \leq C_{s_0, d}^{(1)} \|u\|_{H^{s_0}}, \quad \|b_0\|_{\ell^1} \leq C_{s_0, d}^{(1)} \|v\|_{H^{s_0}}.$$

Substituting these estimates into (7) gives

$$\boxed{\|uv\|_{H^s} \leq C(s, s_0, d) (\|u\|_{H^s} \|v\|_{H^{s_0}} + \|u\|_{H^{s_0}} \|v\|_{H^s}).} \quad (8)$$

For vector- or matrix-valued functions, apply (8) to each scalar component of the bilinear product and sum the finitely many component bounds. Only a target-dimension constant is added.

The proof deliberately stops here. A Sobolev composition or inverse-map estimate requires additional hypotheses on integer/fractional regularity, Jacobian control, and the precise diffeomorphism neighborhood. Those claims are therefore not smuggled into (8); the already precise analytic and C^r statements remain in *Composition estimates* and *Quantitative inverse-function estimate*. $\square$

7.3 Worked examples

Worked example: Quantitative Function Spaces and Tame Calculus: a model computation

Let

$$f(z) = \cos(2\pi z) + \frac{1}{2} \cos(6\pi z).$$

Its only nonzero Fourier coefficients occur at $k = \pm 1, \pm 3$, with sizes $1/2, 1/2, 1/4, 1/4$.

On the strip $|\Im z| < h$,

$$|e^{2\pi i k z}| \leq e^{2\pi |k| h},$$

so

$$\|f\|_h \leq \cosh(2\pi h) + \frac{1}{2} \cosh(6\pi h).$$

If $0 < \delta < h$, Cauchy's estimate gives

$$\|f'\|_{h-\delta} \leq \delta^{-1} \|f\|_h$$

(up to the convention-dependent Cauchy constant). Directly,

$$f'(z) = -2\pi \sin(2\pi z) - 3\pi \sin(6\pi z),$$

which exhibits the same principle: high Fourier modes cost more under differentiation. If a Fourier cutoff S_N keeps only $|k| \leq N$, then for this particular f , $S_1 f = \cos(2\pi z)$ and the discarded tail is exactly $\frac{1}{2} \cos(6\pi z)$.

Boundary case: what fails at the edge of the hypotheses

Set

$$f_N(x) = N^{-r} \sin(2\pi N x).$$

Its derivatives satisfy $\|f_N^{(r)}\|_\infty \asymp 1$, but

$$\|f_N^{(r+1)}\|_\infty \asymp N.$$

Thus a sequence can remain bounded in C^r while one more derivative diverges. Any inverse operator that loses a derivative cannot be iterated indefinitely on a single fixed C^r space; one needs either extra regularity or a smoothing mechanism.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

7.4 Chapter summary

What to remember

- Analyticity is quantified by strip width and exponential Fourier decay.
- Differentiation consumes regularity; Cauchy estimates make the cost explicit.
- Nonlinear operations require product/composition/inverse estimates, not just pointwise algebra.
- Smoothing and tame estimates are the mechanism that makes derivative-losing iterations possible.

Looking ahead

F08 puts these estimates to work on cohomological and homological equations, where the analytic loss competes directly with arithmetic small divisors.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

7.5 Exercises and problems

Exercise 7.1

Difficulty. Core.

Related results. Cauchy estimates on complex strips; Fourier decay of analytic periodic functions; Analytic algebra/product estimates.

Problem. Work through the worked example 'a trigonometric polynomial on a complex strip and its Fourier tail'. For Cauchy estimates on complex strips, Fourier decay of analytic periodic functions, and Analytic algebra/product estimates, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 7.2

Difficulty. Core.

Related results. Composition estimates; Quantitative inverse-function estimate; Smoothing operator estimates.

Problem. Analyze the boundary example 'a sequence showing derivative loss without smoothing'. For Composition estimates, Quantitative inverse-function estimate, and Smoothing operator estimates, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 7.3**Difficulty.** Advanced.**Related results.** Analytic algebra/product estimates.**Problem.** For Analytic algebra/product estimates, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.**Solution sketch.** Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.**Exercise 7.4****Difficulty.** Advanced.**Related results.** Composition estimates.**Problem.** Stress-test Composition estimates. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.**Solution sketch.** Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.**Exercise 7.5****Difficulty.** Advanced.**Related results.** Cauchy estimates on complex strips; Fourier decay of analytic periodic functions.**Problem.** Synthesis problem. Use Cauchy estimates on complex strips and Fourier decay of analytic periodic functions to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.**Solution sketch.** State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.**Exercise 7.6****Difficulty.** Core.**Related results.** Interpolation inequalities.**Problem.** Work directly with Interpolation inequalities. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 7.7

Difficulty. Core.

Related results. Sobolev tame product estimate.

Problem. Work directly with Sobolev tame product estimate. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 7.8

Difficulty. Core.

Related results. Gevrey Fourier characterization.

Problem. Work directly with Gevrey Fourier characterization. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Cohomological Equations and Small Divisors

Why this chapter matters

The equation

$$u(x + \omega) - u(x) = f(x)$$

is the linear heart of many conjugacy problems. Fourier transformation makes it look trivial—divide each Fourier coefficient by $e^{2\pi i \langle k, \omega \rangle} - 1$. The difficulty is that these denominators can be extremely small or exactly zero.

This chapter makes the obstruction and the loss of regularity visible. It is the technical bridge from arithmetic to nonlinear KAM and reducibility arguments.

Prerequisites. F05 and F07, Fourier series on tori, and basic linear algebra.

Learning goals

- Solve scalar cohomological equations Fourier mode by Fourier mode.
- Identify zero-average and resonance obstructions before attempting inversion.
- Derive analytic and Sobolev loss estimates from Diophantine bounds.
- Extend the scalar picture to parameter-dependent and matrix homological equations.

Conceptual roadmap

First isolate the invariant obstruction (the average) and exact resonances. On nonresonant modes, divide by the Fourier denominator. Diophantine arithmetic supplies a polynomial lower bound for that denominator, while analytic Fourier decay pays for the polynomial loss after shrinking the strip. Resonant/nonresonant decompositions are the finite-scale version used inside iterative schemes.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

8.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F08.1. Cohomological equation

A *cohomological equation* over T has the form $u \circ T - u = f$, or $u \circ T - u = f - c$ after removing an invariant obstruction. Solvability is always stated in a specified function space.

Definition F08.2. Small divisor

A *small divisor* is a Fourier denominator whose modulus can become arbitrarily small along integer modes; for translation by ω , the basic example is $e^{2\pi i \langle k, \omega \rangle} - 1$.

Definition F08.3. Homological equation

A *homological equation* is the linear equation for a near-identity conjugacy or normal-form correction, typically $\mathcal{L}u = f$, where $\mathcal{L}$ is generated by the unperturbed dynamics.

Definition F08.4. Resonant mode

A Fourier mode $k \in \mathbb{Z}^d$ is *resonant at a specified scale* if its homological divisor vanishes or lies below the quantitative threshold chosen for that scale.

8.2 Main results

Theorem F08.1. Analytic scalar cohomological equation

Statement. Let f be bounded, holomorphic and $\mathbb{Z}^d$ -periodic on $\mathbb{T}_h^d$, with $\hat{f}(0) = 0$. Assume $\text{dist}(\langle k, \omega \rangle, \mathbb{Z}) \geq \gamma |k|_1^{-\tau}$ for every $0 \neq k \in \mathbb{Z}^d$. For $0 < \delta \leq \min\{h, 1\}$ there is a unique zero-average solution of $u(x + \omega) - u(x) = f(x)$ holomorphic on $\mathbb{T}_{h-\delta}^d$, and $\|u\|_{h-\delta} \leq C_{d,\tau} \gamma^{-1} \delta^{-(\tau+d)} \|f\|_h$, where, with $p = \tau + d - 1$, $C_{d,\tau} = 2^{d+p-2} d^{d-1} (d-1)!^{-1} ((2\pi)^{-1} + \Gamma(p+1)(2\pi)^{-(p+1)})$.

Proof. Write the Fourier series

$$f(x) = \sum_{k \in \mathbb{Z}^d} \hat{f}(k) e^{2\pi i \langle k, x \rangle}, \quad \hat{f}(0) = 0. \quad (1)$$

If a zero-average solution exists, its nonzero Fourier coefficients must satisfy

$$(e^{2\pi i \langle k, \omega \rangle} - 1) \hat{u}(k) = \hat{f}(k), \quad k \neq 0. \quad (2)$$

Thus the only issue is to show that the formal coefficients

$$\widehat{u}(0) := 0, \quad \widehat{u}(k) := \frac{\widehat{f}(k)}{e^{2\pi i \langle k, \omega \rangle} - 1} \quad (k \neq 0) \quad (3)$$

define a holomorphic function on the smaller strip with the claimed norm bound.

Step 1: lower-bound the small divisor. For $t \in \mathbb{R}$, set $r = \text{dist}(t, \mathbb{Z}) \in [0, 1/2]$. Then

$$|e^{2\pi i t} - 1| = 2|\sin(\pi r)|.$$

Concavity of $\sin$ on $[0, \pi/2]$ gives $\sin(\pi r) \geq 2r$, hence

$$|e^{2\pi i t} - 1| \geq 4 \text{dist}(t, \mathbb{Z}). \quad (4)$$

Applying the Diophantine hypothesis to $t = \langle k, \omega \rangle$ yields

$$|e^{2\pi i \langle k, \omega \rangle} - 1| \geq 4\gamma |k|_1^{-\tau}, \quad k \neq 0. \quad (5)$$

In particular no nonzero divisor vanishes.

Step 2: combine divisor control with analytic Fourier decay. Fourier decay of analytic periodic functions gives

$$|\widehat{f}(k)| \leq \|f\|_h e^{-2\pi h |k|_1}. \quad (6)$$

From (3), (5), and (6),

$$|\widehat{u}(k)| \leq \frac{\|f\|_h}{4\gamma} |k|_1^\tau e^{-2\pi h |k|_1}. \quad (7)$$

For $z = x + iy \in \mathbb{T}_{h-\delta}^d$,

$$|e^{2\pi i \langle k, z \rangle}| \leq e^{2\pi(h-\delta)|k|_1}.$$

Therefore the Fourier series (3) is absolutely dominated by

$$\frac{\|f\|_h}{4\gamma} \sum_{k \neq 0} |k|_1^\tau e^{-2\pi \delta |k|_1}. \quad (8)$$

It remains to estimate this lattice sum.

Step 3: an explicit lattice-shell estimate. Let

$$N_d(n) := \#\{k \in \mathbb{Z}^d : |k|_1 = n\}, \quad n \geq 1.$$

Choose the signs of the coordinates and then a weak composition of n into d nonnegative parts. This overcounts vectors with zero coordinates, but gives the convenient bound

$$N_d(n) \leq 2^d \binom{n+d-1}{d-1} \leq \frac{2^d d^{d-1}}{(d-1)!} n^{d-1}. \quad (9)$$

Put

$$p := \tau + d - 1, \quad a := 2\pi\delta.$$

Then (9) implies

$$\sum_{k \neq 0} |k|_1^\tau e^{-2\pi\delta|k|_1} \leq \frac{2^d d^{d-1}}{(d-1)!} \sum_{n \geq 1} n^p e^{-an}. \quad (10)$$

For $x \in [n-1, n]$ we have $(x+1)^p \geq n^p$ and $e^{-ax} \geq e^{-an}$, so

$$\sum_{n \geq 1} n^p e^{-an} \leq \int_0^\infty (x+1)^p e^{-ax} dx. \quad (11)$$

Since $(x+1)^p \leq 2^p(1+x^p)$,

$$\int_0^\infty (x+1)^p e^{-ax} dx \leq 2^p \left(a^{-1} + \Gamma(p+1) a^{-(p+1)} \right). \quad (12)$$

Because $0 < \delta \leq 1$,

$$a^{-1} = (2\pi)^{-1} \delta^{-1} \leq (2\pi)^{-1} \delta^{-(p+1)}. \quad (13)$$

Combining (10)–(13) gives

$$\sum_{k \neq 0} |k|_1^\tau e^{-2\pi\delta|k|_1} \leq K_{d,\tau} \delta^{-(\tau+d)}, \quad (14)$$

where

$$K_{d,\tau} = \frac{2^{d+p} d^{d-1}}{(d-1)!} \left(\frac{1}{2\pi} + \frac{\Gamma(p+1)}{(2\pi)^{p+1}} \right). \quad (15)$$

Thus (8) yields

$$\|u\|_{h-\delta} \leq C_{d,\tau} \gamma^{-1} \delta^{-(\tau+d)} \|f\|_h, \quad (16)$$

with $C_{d,\tau} = K_{d,\tau}/4$, exactly the constant displayed in the statement.

The same majorant proves uniform convergence on every closed smaller strip; therefore the Fourier series defines a holomorphic periodic function on $\mathbb{T}_{h-\delta}^d$. Equation (2) then holds term by term and gives $u(x+\omega) - u(x) = f(x)$.

Finally, if u_1, u_2 are two zero-average holomorphic solutions, their difference v satisfies

$$(e^{2\pi i \langle k, \omega \rangle} - 1) \hat{v}(k) = 0$$

for every k . By (5), all nonzero Fourier coefficients vanish, and the zero-average normalization gives $\hat{v}(0) = 0$. Hence $v = 0$, proving uniqueness. $\square$

Theorem F08.2. Finite-regularity cohomological equation

Statement. Let $s \in \mathbb{R}$ and $\tau \geq 0$. Assume $f \in H^s(\mathbb{T}^d)$ has $\hat{f}(0) = 0$ and ω satisfies $\text{dist}(\langle k, \omega \rangle, \mathbb{Z}) \geq \gamma |k|_1^{-\tau}$ for every $0 \neq k \in \mathbb{Z}^d$. Then there is a unique zero-average $u \in H^{s-\tau}(\mathbb{T}^d)$ satisfying $u(x+\omega) - u(x) = f(x)$ in the distribution sense, and $\|u\|_{H^{s-\tau}} \leq d^{\tau/2} (4\gamma)^{-1} \|f\|_{H^s}$, where $\|v\|_{H^r}^2 = \sum_k (1 + |k|_2^2)^r |\hat{v}(k)|^2$.

Proof. We use the Fourier convention

$$f(x) = \sum_{k \in \mathbb{Z}^d} \hat{f}(k) e^{2\pi i \langle k, x \rangle}$$

and, for every real r , the Sobolev norm

$$\|v\|_{H^r}^2 := \sum_{k \in \mathbb{Z}^d} (1 + |k|_2^2)^r |\widehat{v}(k)|^2. \quad (1)$$

The zero-average condition is $\widehat{f}(0) = 0$.

Step 1: solve every nonzero Fourier mode. If $u(x + \omega) - u(x) = f(x)$, then for $k \neq 0$

$$(e^{2\pi i \langle k, \omega \rangle} - 1) \widehat{u}(k) = \widehat{f}(k). \quad (2)$$

Set $\widehat{u}(0) = 0$ and, for $k \neq 0$, define

$$\widehat{u}(k) := \frac{\widehat{f}(k)}{e^{2\pi i \langle k, \omega \rangle} - 1}. \quad (3)$$

For $t \in \mathbb{R}$ we have

$$|e^{2\pi i t} - 1| \geq 4 \operatorname{dist}(t, \mathbb{Z}), \quad (4)$$

as proved in *Analytic scalar cohomological equation*. The present proof does not need that theorem as a black box: (4) follows immediately by writing the left side as $2|\sin(\pi r)|$, where $r = \operatorname{dist}(t, \mathbb{Z}) \in [0, 1/2]$, and using $\sin(\pi r) \geq 2r$. Hence the (γ, τ) -Diophantine assumption, written with the ℓ^1 lattice norm, gives

$$|e^{2\pi i \langle k, \omega \rangle} - 1| \geq 4\gamma |k|_1^{-\tau}, \quad k \neq 0. \quad (5)$$

In particular the coefficient formula (3) is well defined and

$$|\widehat{u}(k)| \leq \frac{1}{4\gamma} |k|_1^\tau |\widehat{f}(k)|. \quad (6)$$

Step 2: estimate the Sobolev norm. Because $|k|_1 \leq \sqrt{d} |k|_2$, for $k \neq 0$,

$$|k|_1^{2\tau} \leq d^\tau |k|_2^{2\tau} \leq d^\tau (1 + |k|_2^2)^\tau. \quad (7)$$

Using (6)–(7) in (1),

$$\|u\|_{H^{s-\tau}}^2 = \sum_{k \neq 0} (1 + |k|_2^2)^{s-\tau} |\widehat{u}(k)|^2 \quad (8.1)$$

$$\leq \frac{1}{16\gamma^2} \sum_{k \neq 0} (1 + |k|_2^2)^{s-\tau} |k|_1^{2\tau} |\widehat{f}(k)|^2 \quad (8.2)$$

$$\leq \frac{d^\tau}{16\gamma^2} \sum_{k \neq 0} (1 + |k|_2^2)^s |\widehat{f}(k)|^2 \quad (8.3)$$

$$\leq \frac{d^\tau}{16\gamma^2} \|f\|_{H^s}^2. \quad (8)$$

Therefore

$$\|u\|_{H^{s-\tau}} \leq \frac{d^{\tau/2}}{4\gamma} \|f\|_{H^s}. \quad (9)$$

This also proves that the coefficient sequence (3) actually belongs to $H^{s-\tau}$, so it defines a Sobolev distribution/function of precisely the regularity claimed in the statement. Multiplying its Fourier coefficients by the bounded factors $e^{2\pi i \langle k, \omega \rangle} - 1$ recovers the Fourier coefficients of f ; hence (2) holds in $H^{s-\tau}$, equivalently in the sense of distributions. Since its right-hand side is the given H^s function, this is the desired cohomological equation.

Step 3: uniqueness under the zero-average normalization. Suppose $u_1, u_2 \in H^{s-\tau}$ are zero-average solutions and put $v = u_1 - u_2$. Then

$$(e^{2\pi i \langle k, \omega \rangle} - 1)\widehat{v}(k) = 0$$

for every k . By (5) the multiplier is nonzero for every $k \neq 0$, so $\widehat{v}(k) = 0$ there; the zero-average condition also gives $\widehat{v}(0) = 0$. Thus $v = 0$ in $H^{s-\tau}$.

Notice that no lower bound on $s - \tau$ was used: the Fourier definition of $H^r(\mathbb{T}^d)$ makes the argument valid for every real s (and hence also when the solution has negative Sobolev index). $\square$

Proposition F08.3. Obstruction by averages and resonances

Statement. Let $R_\omega(x) = x + \omega$ on $\mathbb{T}^d$. If $u, f \in L^1$ satisfy $u \circ R_\omega - u = f$, then $\int f = 0$ and, more generally, $\widehat{f}(k) = 0$ for every $k \in \mathbb{Z}^d$ with $\langle k, \omega \rangle \in \mathbb{Z}$. If R_ω is ergodic, any two finite measurable solutions of the same cohomological equation differ almost everywhere by a constant.

Proof. Let $R_\omega(x) = x + \omega$ on $\mathbb{T}^d$, let Haar probability measure be denoted by m , and suppose

$$u \circ R_\omega - u = f \tag{1}$$

with $u, f \in L^1(m)$.

Step 1: the average obstruction. Translation invariance of Haar measure gives

$$\int_{\mathbb{T}^d} u \circ R_\omega dm = \int_{\mathbb{T}^d} u dm.$$

Integrating (1),

$$\boxed{\int_{\mathbb{T}^d} f dm = 0.} \tag{2}$$

Thus zero average is necessary for every integrable solution, regardless of any Diophantine assumption.

Step 2: exact resonances give mode-by-mode obstructions. Because $u, f \in L^1$, their Fourier coefficients are defined. Taking the k -th Fourier coefficient of (1) gives

$$(e^{2\pi i \langle k, \omega \rangle} - 1)\widehat{u}(k) = \widehat{f}(k). \tag{3}$$

Hence every resonant vector

$$k \in \mathbb{Z}^d, \quad \langle k, \omega \rangle \in \mathbb{Z}, \tag{4}$$

must satisfy

$$\widehat{f}(k) = 0. \quad (5)$$

The average obstruction (2) is exactly the special case $k = 0$. For an ergodic torus translation, there are no nonzero resonant integer vectors, so the only Fourier obstruction of this exact type is the mean.

Step 3: uniqueness modulo constants in the ergodic case. Now assume R_ω is ergodic and let u_1, u_2 be two finite measurable solutions of the same equation (integrability is not needed for this uniqueness clause). Their difference

$$v := u_1 - u_2$$

satisfies

$$v \circ R_\omega = v \quad \text{a.e.} \quad (6)$$

For every rational a , the sublevel set

$$E_a := \{x : v(x) < a\}$$

is invariant modulo null sets. Ergodicity therefore gives $m(E_a) \in \{0, 1\}$ for every $a \in \mathbb{Q}$. Since v is finite almost everywhere, there is a cut value

$$c := \inf\{a \in \mathbb{Q} : m(E_a) = 1\} \in \mathbb{R}$$

(with the evident equivalent essential-infimum/essential-supremum interpretation). For every $a < c$, $m(E_a) = 0$, while for every $a > c$, $m(E_a) = 1$. Taking rational sequences $a \uparrow c$ and $a \downarrow c$ shows

$$v = c \quad \text{a.e.} \quad (7)$$

Thus measurable solutions are unique up to an additive constant. $\square$

Lemma F08.4. Resonant/nonresonant Fourier decomposition

Statement. Let f be bounded and holomorphic on $\mathbb{T}_h^d$, with Fourier convention $f(x) = \sum_k \widehat{f}(k) e^{2\pi i \langle k, x \rangle}$. Fix $\omega \in \mathbb{R}^d$, an integer cutoff $N \geq 1$, and a threshold $\kappa > 0$. For $0 < |k|_\infty \leq N$ set $D_k(\omega) = e^{2\pi i \langle k, \omega \rangle} - 1$, and split these modes into $\mathcal{N} = \{k : |D_k(\omega)| \geq \kappa\}$ and $\mathcal{R} = \{k : |D_k(\omega)| < \kappa\}$. Then

$$f = \widehat{f}(0) + f_{\mathcal{N}} + f_{\mathcal{R}} + f_{>N}.$$

On the nonresonant part,

$$u_{\mathcal{N}}(x) := \sum_{k \in \mathcal{N}} \frac{\widehat{f}(k)}{D_k(\omega)} e^{2\pi i \langle k, x \rangle}$$

solves $u_{\mathcal{N}}(x + \omega) - u_{\mathcal{N}}(x) = f_{\mathcal{N}}(x)$. For every $0 < \delta < h$, with

$$A_d(\delta) := \sum_{k \in \mathbb{Z}^d} e^{-2\pi\delta|k|_1} = \left(\frac{1 + e^{-2\pi\delta}}{1 - e^{-2\pi\delta}} \right)^d,$$

one has

$$\|u_{\mathcal{N}}\|_{h-\delta} \leq \kappa^{-1} A_d(\delta) \|f\|_h, \quad \|f_{>N}\|_{h-\delta} \leq e^{-\pi\delta N} A_d(\delta/2) \|f\|_h.$$

Scope note

No quantitative smallness of the low resonant block follows without an additional arithmetic or parameter-exclusion hypothesis.

Proof. For $0 < |k|_{\infty} \leq N$, write

$$D_k(\omega) = e^{2\pi i \langle k, \omega \rangle} - 1$$

and define

$$\mathcal{N} := \{k : 0 < |k|_{\infty} \leq N, |D_k(\omega)| \geq \kappa\}, \quad \mathcal{R} := \{k : 0 < |k|_{\infty} \leq N, |D_k(\omega)| < \kappa\}. \quad (1)$$

These sets are disjoint and exhaust all nonzero modes below the cutoff. Consequently the Fourier series splits exactly as

$$f = \widehat{f}(0) + f_{\mathcal{N}} + f_{\mathcal{R}} + f_{>N}, \quad (2)$$

where the four terms use respectively the zero mode, the modes in $\mathcal{N}$, the modes in $\mathcal{R}$, and the modes with $|k|_{\infty} > N$.

On $\mathcal{N}$ the divisor never vanishes, so define

$$u_{\mathcal{N}}(x) := \sum_{k \in \mathcal{N}} \frac{\widehat{f}(k)}{D_k(\omega)} e^{2\pi i \langle k, x \rangle}. \quad (3)$$

The sum is finite. Translating one Fourier mode by ω multiplies it by $e^{2\pi i \langle k, \omega \rangle}$; therefore

$$\begin{aligned} u_{\mathcal{N}}(x + \omega) - u_{\mathcal{N}}(x) &= \sum_{k \in \mathcal{N}} \frac{\widehat{f}(k)}{D_k(\omega)} (e^{2\pi i \langle k, \omega \rangle} - 1) e^{2\pi i \langle k, x \rangle} \\ &= \sum_{k \in \mathcal{N}} \widehat{f}(k) e^{2\pi i \langle k, x \rangle} = f_{\mathcal{N}}(x). \end{aligned} \quad (4)$$

Thus the small-divisor inversion is exact on the nonresonant modes.

We next prove the quantitative bounds. By *Fourier decay of analytic periodic functions*,

$$|\widehat{f}(k)| \leq \|f\|_h e^{-2\pi h |k|_1}. \quad (5)$$

If $z \in \mathbb{T}_{h-\delta}^d$, then

$$|e^{2\pi i \langle k, z \rangle}| \leq e^{2\pi(h-\delta)|k|_1}. \quad (6)$$

Since $|D_k(\omega)| \geq \kappa$ for $k \in \mathcal{N}$, (3)–(6) give

$$\begin{aligned} |u_{\mathcal{N}}(z)| &\leq \kappa^{-1} \|f\|_h \sum_{k \in \mathcal{N}} e^{-2\pi\delta|k|_1} \\ &\leq \kappa^{-1} \|f\|_h \sum_{k \in \mathbb{Z}^d} e^{-2\pi\delta|k|_1}. \end{aligned} \quad (7)$$

The last lattice sum factorizes coordinatewise. With $q = e^{-2\pi\delta}$,

$$\sum_{n \in \mathbb{Z}} q^{|n|} = 1 + 2 \sum_{n \geq 1} q^n = \frac{1+q}{1-q},$$

so

$$A_d(\delta) := \sum_{k \in \mathbb{Z}^d} e^{-2\pi\delta|k|_1} = \left(\frac{1+e^{-2\pi\delta}}{1-e^{-2\pi\delta}} \right)^d. \quad (8)$$

Taking the supremum in (7) proves

$$\|u_{\mathcal{N}}\|_{h-\delta} \leq \kappa^{-1} A_d(\delta) \|f\|_h. \quad (9)$$

For the tail, (5)–(6) similarly give

$$\|f_{>N}\|_{h-\delta} \leq \|f\|_h \sum_{|k|_{\infty} > N} e^{-2\pi\delta|k|_1}. \quad (10)$$

If $|k|_{\infty} > N$, then $|k|_1 > N$, and hence

$$e^{-2\pi\delta|k|_1} = e^{-\pi\delta|k|_1} e^{-\pi\delta|k|_1} \leq e^{-\pi\delta N} e^{-\pi\delta|k|_1}. \quad (11)$$

Using (8) with $\delta/2$ in (10)–(11) yields

$$\|f_{>N}\|_{h-\delta} \leq e^{-\pi\delta N} A_d(\delta/2) \|f\|_h. \quad (12)$$

Nothing in the argument bounds $f_{\mathcal{R}}$ by a small factor: those are exactly the modes on which the divisor is below the prescribed threshold. Their treatment requires additional arithmetic information, a parameter exclusion, or a normal-form decision, and is deliberately not hidden inside this lemma. $\square$

Theorem F08.5. Parameter-dependent cohomological equation

Statement. Let (Λ, d_{Λ}) be a metric parameter space. For each $\lambda \in \Lambda$, let $\omega_{\lambda} \in \mathbb{R}^d$ and let f_{λ} be bounded and holomorphic on $\mathbb{T}_h^d$ with $\hat{f}_{\lambda}(0) = 0$. Assume the uniform Diophantine bound

$$\text{dist}(\langle k, \omega_{\lambda} \rangle, \mathbb{Z}) \geq \gamma |k|_1^{-\tau} \quad (k \neq 0, \lambda \in \Lambda),$$

and set

$$F_0 := \sup_{\lambda} \|f_{\lambda}\|_h, \quad F_{\text{Lip}} := \sup_{\lambda \neq \mu} \frac{\|f_{\lambda} - f_{\mu}\|_h}{d_{\Lambda}(\lambda, \mu)}, \quad \Omega_{\text{Lip}} := \sup_{\lambda \neq \mu} \frac{|\omega_{\lambda} - \omega_{\mu}|_{\infty}}{d_{\Lambda}(\lambda, \mu)},$$

all finite. For $a \geq 0$ and $\delta > 0$ write $\mathcal{S}_{d,a}(\delta) = \sum_{k \neq 0} |k|_1^a e^{-2\pi\delta|k|_1}$. Then for every $0 < \delta < \min\{h, 1\}$ there is a unique zero-average family u_{λ} , holomorphic on $\mathbb{T}_{h-\delta}^d$, satisfying $u_{\lambda}(x + \omega_{\lambda}) - u_{\lambda}(x) = f_{\lambda}(x)$ and

$$\sup_{\lambda} \|u_{\lambda}\|_{h-\delta} \leq \frac{F_0}{4\gamma} \mathcal{S}_{d,\tau}(\delta),$$

$$\sup_{\lambda \neq \mu} \frac{\|u_{\lambda} - u_{\mu}\|_{h-\delta}}{d_{\Lambda}(\lambda, \mu)} \leq \frac{F_{\text{Lip}}}{4\gamma} \mathcal{S}_{d,\tau}(\delta) + \frac{\pi F_0 \Omega_{\text{Lip}}}{8\gamma^2} \mathcal{S}_{d,2\tau+1}(\delta).$$

Scope note

For a Cantor parameter set the conclusion gives only zeroth-order Lipschitz/Whitney control; higher parameter jets require additional hypotheses.

Proof. For $k \in \mathbb{Z}^d$ and $\lambda \in \Lambda$, write

$$D_k(\lambda) := e^{2\pi i \langle k, \omega_{\lambda} \rangle} - 1. \quad (1)$$

The uniform Diophantine hypothesis and the elementary inequality used in *Analytic scalar cohomological equation* give, for every $k \neq 0$,

$$|D_k(\lambda)| \geq 4 \operatorname{dist}(\langle k, \omega_{\lambda} \rangle, \mathbb{Z}) \geq 4\gamma |k|_1^{-\tau}. \quad (2)$$

In particular, no nonzero Fourier divisor vanishes for any parameter.

Step 1: solve at each parameter. By *Analytic scalar cohomological equation*, for every λ the unique zero-average solution is specified mode by mode by

$$\widehat{u}_{\lambda}(0) = 0, \quad \widehat{u}_{\lambda}(k) = \frac{\widehat{f}_{\lambda}(k)}{D_k(\lambda)}, \quad k \neq 0. \quad (3)$$

Fourier decay of analytic periodic functions gives analytic Fourier decay

$$|\widehat{f}_{\lambda}(k)| \leq \|f_{\lambda}\|_h e^{-2\pi h|k|_1} \leq F_0 e^{-2\pi h|k|_1}. \quad (4)$$

Combining (2)–(4), and then evaluating the Fourier series on the strip $\mathbb{T}_{h-\delta}^d$, yields

$$\|u_{\lambda}\|_{h-\delta} \leq \frac{F_0}{4\gamma} \sum_{k \neq 0} |k|_1^{\tau} e^{-2\pi\delta|k|_1} = \frac{F_0}{4\gamma} \mathcal{S}_{d,\tau}(\delta). \quad (5)$$

The lattice sum is finite for every $\delta > 0$, so the series converges normally on the smaller strip. Taking the supremum over λ proves the first displayed estimate in the statement.

Step 2: estimate the variation of one Fourier coefficient. Fix distinct $\lambda, \mu \in \Lambda$ and $k \neq 0$. From (3),

$$\widehat{u}_\lambda(k) - \widehat{u}_\mu(k) = \frac{\widehat{f}_\lambda(k) - \widehat{f}_\mu(k)}{D_k(\lambda)} + \widehat{f}_\mu(k) \left(\frac{1}{D_k(\lambda)} - \frac{1}{D_k(\mu)} \right). \quad (6)$$

The first term is controlled by one small divisor. Indeed, Fourier decay applied to $f_\lambda - f_\mu$ gives

$$|\widehat{f}_\lambda(k) - \widehat{f}_\mu(k)| \leq F_{\text{Lip}} d_\Lambda(\lambda, \mu) e^{-2\pi h|k|_1}, \quad (7)$$

and hence by (2)

$$\left| \frac{\widehat{f}_\lambda(k) - \widehat{f}_\mu(k)}{D_k(\lambda)} \right| \leq \frac{F_{\text{Lip}}}{4\gamma} d_\Lambda(\lambda, \mu) |k|_1^\tau e^{-2\pi h|k|_1}. \quad (8)$$

The second term contains the moving divisor. The function $t \mapsto e^{2\pi i t}$ is 2π -Lipschitz, so

$$|D_k(\lambda) - D_k(\mu)| \leq 2\pi |\langle k, \omega_\lambda - \omega_\mu \rangle| \leq 2\pi |k|_1 \Omega_{\text{Lip}} d_\Lambda(\lambda, \mu). \quad (9)$$

Using (2) twice,

$$\left| \frac{1}{D_k(\lambda)} - \frac{1}{D_k(\mu)} \right| = \frac{|D_k(\lambda) - D_k(\mu)|}{|D_k(\lambda) D_k(\mu)|} \leq \frac{\pi}{8\gamma^2} |k|_1^{2\tau+1} \Omega_{\text{Lip}} d_\Lambda(\lambda, \mu). \quad (10)$$

Together with $|\widehat{f}_\mu(k)| \leq F_0 e^{-2\pi h|k|_1}$, equations (6), (8), and (10) imply

$$\frac{|\widehat{u}_\lambda(k) - \widehat{u}_\mu(k)|}{d_\Lambda(\lambda, \mu)} \leq e^{-2\pi h|k|_1} \left[\frac{F_{\text{Lip}}}{4\gamma} |k|_1^\tau + \frac{\pi F_0 \Omega_{\text{Lip}}}{8\gamma^2} |k|_1^{2\tau+1} \right]. \quad (11)$$

Step 3: sum on the smaller analytic strip. For $z \in \mathbb{T}_{h-\delta}^d$,

$$|e^{2\pi i \langle k, z \rangle}| \leq e^{2\pi(h-\delta)|k|_1}.$$

Multiply (11) by this factor and sum over $k \neq 0$. Normal convergence gives

$$\frac{\|u_\lambda - u_\mu\|_{h-\delta}}{d_\Lambda(\lambda, \mu)} \leq \frac{F_{\text{Lip}}}{4\gamma} \mathcal{S}_{d,\tau}(\delta) + \frac{\pi F_0 \Omega_{\text{Lip}}}{8\gamma^2} \mathcal{S}_{d,2\tau+1}(\delta). \quad (12)$$

Taking the supremum over distinct parameters proves the second displayed estimate.

The same Fourier formula (3) gives uniqueness at each parameter, exactly as in *Analytic scalar cohomological equation*: the difference of two zero-average solutions has every nonzero Fourier coefficient annihilated by a nonzero divisor and has zero constant coefficient. Hence the family $\lambda \mapsto u_\lambda$ is the unique zero-average family with the asserted Lipschitz control. When Λ is a closed Cantor parameter set, (12) is precisely the zeroth-order Lipschitz/Whitney control used at this stage of KAM iteration; no higher parameter jet is claimed here. $\square$

Proposition F08.6. Matrix homological equation near a constant

Statement. Let $D = \text{diag}(\lambda_1, \dots, \lambda_m) \in GL(m, \mathbb{C})$, $\omega \in \mathbb{R}^d$, and let $G : \mathbb{T}_h^d \rightarrow M_m(\mathbb{C})$ be

bounded and holomorphic. Fix $N \geq 1$ and $\kappa > 0$, and for $|k|_\infty \leq N$ put

$$\Delta_{ij}(k) := e^{2\pi i \langle k, \omega \rangle} \lambda_j - \lambda_i.$$

Split the low matrix/Fourier triples into $\mathcal{N} = \{(i, j, k) : |\Delta_{ij}(k)| \geq \kappa\}$ and $\mathcal{R} = \{(i, j, k) : |\Delta_{ij}(k)| < \kappa\}$, and write $G = G_{\mathcal{N}} + G_{\mathcal{R}} + G_{>N}$. Define X by $\widehat{X}_{ij}(k) = \widehat{G}_{ij}(k)/\Delta_{ij}(k)$ on $\mathcal{N}$ and zero otherwise. Then

$$X(x + \omega)D - DX(x) = G_{\mathcal{N}}(x).$$

For $0 < \delta < h$, with $\|H\|_{h, \max} = \max_{i,j} \|H_{ij}\|_h$ and A_d as in Lemma F08.4,

$$\|X\|_{h-\delta, \max} \leq \kappa^{-1} A_d(\delta) \|G\|_{h, \max}, \quad \|G_{>N}\|_{h-\delta, \max} \leq e^{-\pi \delta N} A_d(\delta/2) \|G\|_{h, \max}.$$

Moreover the near-identity conjugacy has the first-order expansion

$$(I - \varepsilon X(x + \omega))(D + \varepsilon G(x))(I + \varepsilon X(x)) = D + \varepsilon(G_{\mathcal{R}} + G_{>N}) + O(\varepsilon^2).$$

Scope note

No smallness of the low resonant block is assumed or concluded here, and the nondiagonalizable-constant case requires a separate reduction.

Proof. Write

$$G(x) = \sum_{k \in \mathbb{Z}^d} \widehat{G}(k) e^{2\pi i \langle k, x \rangle}, \quad X(x) = \sum_{k \in \mathbb{Z}^d} \widehat{X}(k) e^{2\pi i \langle k, x \rangle}. \quad (1)$$

Since $D = \text{diag}(\lambda_1, \dots, \lambda_m)$, the (i, j) entry of the k th Fourier coefficient of the homological operator is

$$(X(\cdot + \omega)D - DX)_{ij}(k) = (e^{2\pi i \langle k, \omega \rangle} \lambda_j - \lambda_i) \widehat{X}_{ij}(k) = \Delta_{ij}(k) \widehat{X}_{ij}(k). \quad (2)$$

Thus the matrix equation has become a family of scalar algebraic equations, one for each triple (i, j, k) .

Step 1: resonant/nonresonant decomposition. For $|k|_\infty \leq N$, define

$$\mathcal{N} = \{(i, j, k) : |\Delta_{ij}(k)| \geq \kappa\}, \quad \mathcal{R} = \{(i, j, k) : |\Delta_{ij}(k)| < \kappa\}. \quad (3)$$

Let $G_{\mathcal{N}}$ contain exactly the Fourier entries indexed by $\mathcal{N}$, let $G_{\mathcal{R}}$ contain the complementary entries with $|k|_\infty \leq N$, and let $G_{>N}$ contain the modes with $|k|_\infty > N$. Then, entry by entry,

$$G = G_{\mathcal{N}} + G_{\mathcal{R}} + G_{>N}. \quad (4)$$

This is the matrix analogue of *Resonant/nonresonant Fourier decomposition*; unlike the scalar case, the resonance decision depends on both the Fourier mode and the ordered pair of eigenvalues.

Define X by

$$\hat{X}_{ij}(k) = \begin{cases} \hat{G}_{ij}(k)/\Delta_{ij}(k), & (i, j, k) \in \mathcal{N}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Because only finitely many modes $|k|_\infty \leq N$ occur, X is a trigonometric polynomial and hence is entire in the complexified torus variables. Substituting (5) into (2) gives exactly

$$X(x + \omega)D - DX(x) = G_{\mathcal{N}}(x). \quad (6)$$

No equation is imposed on a resonant triple; those entries are deliberately left in the normal form.

Step 2: analytic estimate for the correction. For a scalar bounded holomorphic function g on $\mathbb{T}_h^d$, *Resonant/nonresonant Fourier decomposition* uses the Fourier bound

$$|\hat{g}(k)| \leq \|g\|_h e^{-2\pi h|k|_1}. \quad (7)$$

Apply this entrywise to G . On every nonresonant triple, $|\Delta_{ij}(k)|^{-1} \leq \kappa^{-1}$, so for $z \in \mathbb{T}_{h-\delta}^d$,

$$\begin{aligned} |X_{ij}(z)| &\leq \kappa^{-1} \|G\|_{h,\max} \sum_{k \in \mathbb{Z}^d} e^{-2\pi \delta |k|_1} \\ &= \kappa^{-1} A_d(\delta) \|G\|_{h,\max}. \end{aligned} \quad (8)$$

Taking the maximum over i, j proves

$$\|X\|_{h-\delta,\max} \leq \kappa^{-1} A_d(\delta) \|G\|_{h,\max}. \quad (9)$$

The high-frequency estimate is likewise entrywise the tail estimate from *Resonant/nonresonant Fourier decomposition*:

$$\|G_{>N}\|_{h-\delta,\max} \leq e^{-\pi \delta N} A_d(\delta/2) \|G\|_{h,\max}. \quad (10)$$

There is intentionally no analogous smallness assertion for $G_{\mathcal{R}}$: a low divisor has been declared resonant precisely because this linear step does not safely invert it.

Step 3: relation with a near-identity cocycle conjugacy. Consider the formal change of coordinates $I + \varepsilon X(x)$. Its inverse has the first-order expansion

$$(I + \varepsilon X(x + \omega))^{-1} = I - \varepsilon X(x + \omega) + O(\varepsilon^2). \quad (11)$$

Therefore

$$\begin{aligned} (I - \varepsilon X(x + \omega))(D + \varepsilon G(x))(I + \varepsilon X(x)) \\ = D + \varepsilon(G(x) + DX(x) - X(x + \omega)D) + O(\varepsilon^2). \end{aligned} \quad (12)$$

By the homological equation (6),

$$DX(x) - X(x + \omega)D = -G_{\mathcal{N}}(x). \quad (13)$$

Using (4) in (12), the first-order perturbation becomes

$$G - G_N = G_{\mathcal{R}} + G_{>N}.$$

Hence

$$(I - \varepsilon X(x + \omega))(D + \varepsilon G(x))(I + \varepsilon X(x)) = D + \varepsilon(G_{\mathcal{R}} + G_{>N}) + O(\varepsilon^2). \quad (14)$$

This identity explains the terminology: the homological correction removes all low nonresonant matrix Fourier entries at first order, whereas the low resonant block is retained and must be incorporated into the next normal form or treated by a separate resonance step. $\square$

Corollary F08.7. Small-divisor estimates for circle linearization

Statement. Let $F_\varepsilon(x) = x + \alpha + \varepsilon v(x) + O(\varepsilon^2)$ and seek $H_\varepsilon(x) = x + \varepsilon u(x) + O(\varepsilon^2)$, $\beta_\varepsilon = \alpha + \varepsilon c + O(\varepsilon^2)$ with $H_\varepsilon \circ F_\varepsilon = H_\varepsilon + \beta_\varepsilon$. Then the first-order equation is $u(x + \alpha) - u(x) = c - v(x)$ and necessarily $c = \widehat{v}(0)$. If α is (γ, τ) -Diophantine, the unique zero-average correction satisfies the analytic estimate from Theorem F08.1 and, for $v \in H^s$, the Sobolev estimate $\|u\|_{H^{s-\tau}} \leq (4\gamma)^{-1} \|v - \widehat{v}(0)\|_{H^s}$.

Scope note

This corollary concerns the linearized conjugacy equation; nonlinear linearization requires an additional iteration theorem.

Proof. Work with lifts to $\mathbb{R}$. Consider a differentiable one-parameter family through the rigid rotation,

$$F_\varepsilon(x) = x + \alpha + \varepsilon v(x) + O(\varepsilon^2), \quad (1)$$

and seek a conjugating lift and corrected rotation number of the form

$$H_\varepsilon(x) = x + \varepsilon u(x) + O(\varepsilon^2), \quad \beta_\varepsilon = \alpha + \varepsilon c + O(\varepsilon^2). \quad (2)$$

The lift conjugacy equation is

$$H_\varepsilon(F_\varepsilon(x)) = H_\varepsilon(x) + \beta_\varepsilon. \quad (3)$$

We identify its first-order part before invoking any small-divisor theorem.

By (1)–(2),

$$\begin{aligned} H_\varepsilon(F_\varepsilon(x)) &= F_\varepsilon(x) + \varepsilon u(F_\varepsilon(x)) + O(\varepsilon^2) \\ &= x + \alpha + \varepsilon v(x) + \varepsilon u(x + \alpha) + O(\varepsilon^2), \end{aligned}$$

where replacing $u(F_\varepsilon(x))$ by $u(x + \alpha)$ changes the expression only at order ε^2 . The right side of (3) is

$$x + \alpha + \varepsilon(u(x) + c) + O(\varepsilon^2).$$

Equating coefficients of ε gives

$$u(x + \alpha) - u(x) = c - v(x). \quad (4)$$

Averaging (4) over the circle forces

$$c = \int_{\mathbb{T}} v(x) dx = \widehat{v}(0). \quad (5)$$

Thus the shape correction solves

$$u(x + \alpha) - u(x) = -(v(x) - \widehat{v}(0)), \quad (6)$$

whose right-hand side has zero average.

Assume now that

$$\text{dist}(k\alpha, \mathbb{Z}) \geq \gamma|k|^{-\tau} \quad (k \in \mathbb{Z} \setminus \{0\}). \quad (7)$$

If v is bounded and holomorphic on the strip $\mathbb{T}_h$, then *Analytic scalar cohomological equation* applied to $f = -(v - \widehat{v}(0))$ gives, for $0 < \delta \leq \min\{h, 1\}$, a unique zero-average analytic solution on $\mathbb{T}_{h-\delta}$ and

$$\|u\|_{h-\delta} \leq C_{1,\tau} \gamma^{-1} \delta^{-(\tau+1)} \|v - \widehat{v}(0)\|_h. \quad (8)$$

Since $|\widehat{v}(0)| \leq \|v\|_h$,

$$\|v - \widehat{v}(0)\|_h \leq 2\|v\|_h,$$

so also

$$\|u\|_{h-\delta} \leq 2C_{1,\tau} \gamma^{-1} \delta^{-(\tau+1)} \|v\|_h. \quad (9)$$

Likewise, for any real s , if $v \in H^s(\mathbb{T})$, then *Finite-regularity cohomological equation* gives a unique zero-average $u \in H^{s-\tau}$ with

$$\|u\|_{H^{s-\tau}} \leq \frac{1}{4\gamma} \|v - \widehat{v}(0)\|_{H^s}. \quad (10)$$

Here the dimension factor in *Finite-regularity cohomological equation* equals one.

Equations (5), (8), and (10) are precisely the small-divisor estimates for the linearized circle conjugacy problem. They do not assert that the full nonlinear conjugacy (3) exists; that requires the later Newton/KAM or Nash–Moser machinery. $\square$

8.3 Worked examples

Worked example: Cohomological Equations and Small Divisors: a model computation

Fix an irrational α and an integer $m \neq 0$. Solve

$$u(x + \alpha) - u(x) = \cos(2\pi mx).$$

Write

$$\cos(2\pi mx) = \frac{1}{2}e^{2\pi imx} + \frac{1}{2}e^{-2\pi imx}.$$

Seeking a zero-average solution with the same two modes gives

$$\widehat{u}(m) = \frac{1}{2(e^{2\pi i m \alpha} - 1)}, \quad \widehat{u}(-m) = \frac{1}{2(e^{-2\pi i m \alpha} - 1)}.$$

Therefore

$$u(x) = \frac{e^{2\pi i m x}}{2(e^{2\pi i m \alpha} - 1)} + \frac{e^{-2\pi i m x}}{2(e^{-2\pi i m \alpha} - 1)}.$$

The solution is real because the two coefficients are complex conjugates. Moreover,

$$|e^{2\pi i m \alpha} - 1| = 2|\sin(\pi m \alpha)| \asymp \text{dist}(m\alpha, \mathbb{Z}).$$

Thus the size of u is exactly controlled by the arithmetic quality of α . This is the small-divisor mechanism with no nonlinear complications hiding it.

Boundary case: what fails at the edge of the hypotheses

Let $\alpha = p/q$ and choose $m = q$. Then $e^{2\pi i m \alpha} = 1$, so the Fourier denominator vanishes. The obstruction can also be seen without Fourier series: summing $u(x + \alpha) - u(x) = \cos(2\pi q x)$ along one period gives

$$0 = \sum_{j=0}^{q-1} \cos(2\pi q(x + jp/q)) = q \cos(2\pi q x),$$

which is false for generic x . Exact resonance is therefore a genuine solvability obstruction, not a defect of the estimate.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

8.4 Chapter summary

What to remember

- Cohomological equations reduce conjugacy problems to linear inversion along the dynamics.
- Averages and resonant modes are compatibility conditions, not technical nuisances.
- Diophantine inequalities control the inverse but force regularity loss.
- Finite-scale resonance decompositions are the basic unit of KAM and cocycle schemes.

Looking ahead

F09 replaces scalar functions by matrix cocycles and projective dynamics. The homological equations of this chapter then become conjugacy equations for linear systems.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

8.5 Exercises and problems

Exercise 8.1

Difficulty. Core.

Related results. Analytic scalar cohomological equation; Finite-regularity cohomological equation; Obstruction by averages and resonances.

Problem. Work through the worked example ' $u(x+\alpha)-u(x)=\cos(2\pi m x)$ solved explicitly'. For Analytic scalar cohomological equation, Finite-regularity cohomological equation, and Obstruction by averages and resonances, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 8.2

Difficulty. Core.

Related results. Resonant/nonresonant Fourier decomposition; Parameter-dependent cohomological equation; Matrix homological equation near a constant.

Problem. Analyze the boundary example 'a resonant/rational frequency where the denominator vanishes'. For Resonant/nonresonant Fourier decomposition, Parameter-dependent cohomological equation, and Matrix homological equation near a constant, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 8.3**Difficulty.** Advanced.**Related results.** Obstruction by averages and resonances.**Problem.** For Obstruction by averages and resonances, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.**Solution sketch.** Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.**Exercise 8.4****Difficulty.** Advanced.**Related results.** Resonant/nonresonant Fourier decomposition.**Problem.** Stress-test Resonant/nonresonant Fourier decomposition. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.**Solution sketch.** Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.**Exercise 8.5****Difficulty.** Advanced.**Related results.** Analytic scalar cohomological equation; Finite-regularity cohomological equation.**Problem.** Synthesis problem. Use Analytic scalar cohomological equation and Finite-regularity cohomological equation to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.**Solution sketch.** State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.**Exercise 8.6****Difficulty.** Core.**Related results.** Small-divisor estimates for circle linearization.**Problem.** Work directly with Small-divisor estimates for circle linearization. Re-state its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include

the calculations rather than only a verbal discussion.

Matrix Cocycles and Projective Dynamics

Why this chapter matters

A matrix cocycle is a linear dynamical system driven by another dynamical system. Its norm growth belongs to Oseledec's theory, but many finer questions are geometric: does it preserve stable and unstable directions? Can it be conjugated to a simpler cocycle? What does its action look like after projecting directions to projective space?

In dimension two, projective dynamics turns linear algebra into circle dynamics. This viewpoint is central both to hyperbolicity and to later spectral theory.

Prerequisites. F04, F06, F08, projective space, and basic symplectic linear algebra.

Learning goals

- Compute how cocycle conjugacies change matrix iterates.
- Understand the projective action of $SL(2, \mathbb{R})$ and fibered rotation number.
- Recognize uniform hyperbolicity through invariant directions, exponential dichotomies, or cone fields.
- Use determinant and symplectic structure to constrain spectra.

Conceptual roadmap

Conjugacy calculus comes first because every normal-form argument uses it. Projectivization then converts invariant vector bundles into invariant points or graphs in a fiber circle. Uniform hyperbolicity is characterized in several equivalent languages: invariant stable/unstable directions, exponential dichotomy, cone invariance, and dominated splitting.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

9.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F09.1. Cocycle conjugacy

Two G -valued cocycles (T, A) and (T, A') are *conjugate* by $B : X \rightarrow G$ if $A'(x) = B(Tx)A(x)B(x)^{-1}$. Regularity and boundedness of B are part of the equivalence notion in use.

Definition F09.2. Projective action

For $A \in GL(d, \mathbb{R})$, the induced projective action is $[v] \mapsto [Av]$ on $\mathbb{P}^{d-1}(\mathbb{R})$. For $SL(2, \mathbb{R})$ this is a projective-circle (Möbius) action.

Definition F09.3. Fibered rotation number

For a homotopically admissible $SL(2, \mathbb{R})$ cocycle over an irrational rotation, the *fibered rotation number* is the asymptotic average displacement of a lift of its projective skew product, taken modulo the normalization lattice.

Definition F09.4. Uniform hyperbolicity

A continuous cocycle is *uniformly hyperbolic* if there is a continuous invariant splitting $E_x^s \oplus E_x^u$ and constants $C > 0$, $\lambda > 1$ such that $\|A_n(x)v^s\| \leq C\lambda^{-n}\|v^s\|$ and $\|A_{-n}(x)v^u\| \leq C\lambda^{-n}\|v^u\|$ for all $n \geq 0$.

9.2 Main results

Proposition F09.1. Cocycle conjugacy calculus

Statement. Let $T : X \rightarrow X$, let $A, A' : X \rightarrow GL(d, \mathbb{R})$, and let $B : X \rightarrow GL(d, \mathbb{R})$ satisfy

$$A'(x) = B(Tx)A(x)B(x)^{-1}.$$

Then for every $n \geq 1$,

$$A'_n(x) = B(T^n x)A_n(x)B(x)^{-1}.$$

If $\|B\|$ and $\|B^{-1}\|$ are uniformly bounded by M , then for every singular-value index j ,

$$M^{-2}\sigma_j(A_n(x)) \leq \sigma_j(A'_n(x)) \leq M^2\sigma_j(A_n(x)),$$

so the two cocycles have the same Lyapunov exponents wherever the corresponding normalized singular-value limits exist. If, in addition, X is compact, T is

a homeomorphism, and A, A', B are continuous, then continuous invariant splittings are carried to continuous invariant splittings by $E'(x) = B(x)E(x)$. Uniform-hyperbolicity estimates are preserved up to the factor M^2 ; more generally, any domination estimate

$$\|A_n(x)|_{E(x)}\| \|(A_n(x)|_{F(x)})^{-1}\| \leq C\tau^n, \quad 0 < \tau < 1,$$

is transferred to the conjugated splitting with constant M^4C and the same τ .

Proof. For $n = 1$ the asserted identity is the definition of conjugacy. Assume

$$A'_n(x) = B(T^n x)A_n(x)B(x)^{-1}.$$

Then

$$\begin{aligned} A'_{n+1}(x) &= A'(T^n x)A'_n(x) \\ &= B(T^{n+1}x)A(T^n x)B(T^n x)^{-1}B(T^n x)A_n(x)B(x)^{-1} \\ &= B(T^{n+1}x)A_{n+1}(x)B(x)^{-1}. \end{aligned}$$

Induction gives

$$A'_n(x) = B(T^n x)A_n(x)B(x)^{-1} \tag{1}$$

for every $n \geq 1$.

Assume now

$$\sup_x \max\{\|B(x)\|, \|B(x)^{-1}\|\} \leq M. \tag{2}$$

For invertible matrices P, Q, C , the min-max characterization of singular values gives

$$\sigma_j(PCQ) \leq \|P\| \|Q\| \sigma_j(C). \tag{3}$$

Applying (3) to (1) yields

$$\sigma_j(A'_n(x)) \leq M^2 \sigma_j(A_n(x)). \tag{4}$$

Solving (1) for A_n ,

$$A_n(x) = B(T^n x)^{-1}A'_n(x)B(x),$$

and applying (3) again gives

$$\sigma_j(A_n(x)) \leq M^2 \sigma_j(A'_n(x)). \tag{5}$$

Combining (4)-(5),

$$M^{-2} \sigma_j(A_n(x)) \leq \sigma_j(A'_n(x)) \leq M^2 \sigma_j(A_n(x)). \tag{6}$$

Thus

$$\left| \frac{1}{n} \log \sigma_j(A'_n(x)) - \frac{1}{n} \log \sigma_j(A_n(x)) \right| \leq \frac{2 \log M}{n}. \tag{7}$$

Whenever either normalized singular-value limit exists, (7) shows that the other exists and is equal to it. Hence the Lyapunov spectra agree wherever these limits define them.

For the geometric assertions assume the additional compactness and continuity

hypotheses. If $E(x)$ is a continuous A -invariant subbundle, put

$$E'(x) := B(x)E(x).$$

Continuity follows from continuity of B and E , and

$$A'(x)E'(x) = B(Tx)A(x)E(x) = B(Tx)E(Tx) = E'(Tx), \quad (8)$$

so the transformed bundle is invariant. Direct sums are preserved because $B(x)$ is invertible.

If $v' = B(x)v$ with $v \in E^s(x)$, then by (1)-(2)

$$\|A'_n(x)v'\| = \|B(T^n x)A_n(x)v\| \leq M C \lambda^{-n} \|v\| \leq M^2 C \lambda^{-n} \|v'\|. \quad (9)$$

The same argument applied to negative iterates on the unstable bundle gives the corresponding estimate there. Thus uniform hyperbolicity is preserved; applying the same reasoning to the inverse conjugacy proves the converse.

Finally suppose $E \oplus F$ satisfies the domination inequality in the statement. From (1), restricted to the transformed bundles,

$$\|A'_n|_{E'}\| \leq M^2 \|A_n|_E\|, \quad \|(A'_n|_{F'})^{-1}\| \leq M^2 \|(A_n|_F)^{-1}\|.$$

Multiplication gives

$$\|A'_n|_{E'}\| \|(A'_n|_{F'})^{-1}\| \leq M^4 C \tau^n,$$

which is the claimed transferred domination estimate. $\square$

Proposition F09.2. Projective action of $SL(2, \mathbb{R})$

Statement. Under the identification $\mathbb{RP}^1 \cong \widehat{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$ given by $z \leftrightarrow [(z, 1)]$ and $\infty \leftrightarrow [(1, 0)]$, a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{R})$$

acts by the Möbius map $z \mapsto (az + b)/(cz + d)$ with the usual conventions at ∞ . For a cocycle $A(x)$ over T , a projective line field $z(x)$ is invariant if and only if $z(Tx) = A(x) \cdot z(x)$. In particular, for

$$A_E(x) = \begin{pmatrix} E - v(x) & -1 \\ 1 & 0 \end{pmatrix},$$

the affine-chart equation is $z(Tx) = E - v(x) - 1/z(x)$ whenever both sides are finite, with its projective extension covering $z = 0$ and $z = \infty$.

Proof. Recall that a point of $\mathbb{RP}^1$ is a one-dimensional linear subspace of $\mathbb{R}^2$. Every line other than the vertical line $\mathbb{R}(1, 0)$ has a unique representative of the form $(z, 1)$ with $z \in \mathbb{R}$. Thus

$$z \longleftrightarrow [(z, 1)], \quad \infty \longleftrightarrow [(1, 0)] \quad (1)$$

identifies $\mathbb{RP}^1$ with the extended real line $\widehat{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{R}).$$

For a finite coordinate z ,

$$A \begin{pmatrix} z \\ 1 \end{pmatrix} = \begin{pmatrix} az + b \\ cz + d \end{pmatrix}. \quad (2)$$

If $cz + d \neq 0$, rescaling the vector in (2) so that its second coordinate is 1 gives

$$A \cdot z = \frac{az + b}{cz + d}. \quad (3)$$

If $cz + d = 0$, then the image vector has second coordinate zero. It is nonzero because A is invertible, so the image line is ∞ . Likewise, from

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix}$$

we obtain

$$A \cdot \infty = \begin{cases} a/c, & c \neq 0, \\ \infty, & c = 0. \end{cases} \quad (4)$$

Equations (3)–(4) are exactly the usual Möbius extension to $\widehat{\mathbb{R}}$. Notice that invertibility is what prevents the numerator and denominator in (3) from vanishing simultaneously.

Now let $A : X \rightarrow SL(2, \mathbb{R})$ be a cocycle over $T : X \rightarrow X$. A line field $\ell(x) \in \mathbb{RP}^1$ is invariant precisely when

$$A(x)\ell(x) = \ell(Tx) \quad (5)$$

for every x in the domain under consideration. Writing $\ell(x) = [(z(x), 1)]$ whenever the affine chart is finite, (5) and (3) become

$$z(Tx) = \frac{a(x)z(x) + b(x)}{c(x)z(x) + d(x)}. \quad (6)$$

Conversely, if a map $z(x) \in \widehat{\mathbb{R}}$ satisfies (6) with the projective conventions (3)–(4), then the associated line field satisfies (5). Thus the line-field and Möbius formulations are equivalent, not merely one-way implications.

For the Schrödinger-form matrix

$$A_E(x) = \begin{pmatrix} E - v(x) & -1 \\ 1 & 0 \end{pmatrix},$$

substitution into (3) gives, whenever $z \neq 0$ and the image is finite,

$$A_E(x) \cdot z = \frac{(E - v(x))z - 1}{z} = E - v(x) - \frac{1}{z}. \quad (7)$$

Therefore an invariant projective line field satisfies the Riccati coordinate equation

$$\boxed{z(Tx) = E - v(x) - \frac{1}{z(x)}}. \quad (8)$$

If $z(x) = 0$, the denominator in (7) vanishes and the projective image is ∞ ; if $z(x) = \infty$, equation (4) gives the corresponding finite value $E - v(x)$. Thus the apparent singularities of the affine Riccati formula are merely chart changes on $\mathbb{RP}^1$.

No spectral assertion has been made: (8) is only the projective-coordinate form of the cocycle action, exactly as claimed. $\square$

Theorem F09.3. Fibered rotation number

Statement. Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and $A \in C^0(\mathbb{T}, SL(2, \mathbb{R}))$ be homotopic to the identity. Choose a continuous degree-one lift of the induced projective skew product,

$$\tilde{F}_A(x, y) = (x + \alpha, f_x(y)), \quad f_x(y + 1) = f_x(y) + 1.$$

Then there is a real number $\tilde{\rho}(\alpha, A)$ such that

$$\frac{\pi_2(\tilde{F}_A^n(x, y)) - y}{n} \rightarrow \tilde{\rho}(\alpha, A)$$

uniformly in $(x, y) \in \mathbb{T} \times \mathbb{R}$. A different lift changes $\tilde{\rho}$ by an integer, so $\rho(\alpha, A) := \tilde{\rho}(\alpha, A) \bmod \mathbb{Z}$ is well defined and is continuous in A for the C^0 topology. If $B : \mathbb{T} \rightarrow SL(2, \mathbb{R})$ is continuous and homotopic to the identity and $A'(x) = B(x + \alpha)A(x)B(x)^{-1}$, then $\rho(\alpha, A') = \rho(\alpha, A)$ in $\mathbb{R}/\mathbb{Z}$.

Scope note

The conjugacy invariance stated here is only for conjugacies homotopic to the identity.

Proof. Let $R_\alpha(x) = x + \alpha$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. By *Projective action of $SL(2, R)$* , the projective action of each $A(x)$ is an orientation-preserving circle homeomorphism. Because A is homotopic to the identity, the projective skew product admits a continuous lift

$$\tilde{F}_A(x, y) = (x + \alpha, f_x(y)), \quad f_x(y + 1) = f_x(y) + 1, \quad (1)$$

where every $f_x : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing and the map $(x, y) \mapsto f_x(y)$ is continuous.

For $n \geq 1$, write

$$f_x^{(n)} := f_{x+(n-1)\alpha} \circ \cdots \circ f_x, \quad \Phi_n(x, y) := f_x^{(n)}(y) - y. \quad (2)$$

Thus Φ_n is the fiber displacement of $\tilde{F}_A^n$. We shall prove that Φ_n/n converges uniformly to a constant.

Step 1: a uniform one-unit fiber oscillation bound. Every $f_x^{(n)}$ is again an increasing degree-one lift. Hence, if $y < z < y + 1$, then

$$0 < f_x^{(n)}(z) - f_x^{(n)}(y) < 1.$$

Consequently

$$\left| (f_x^{(n)}(z) - z) - (f_x^{(n)}(y) - y) \right| < 1. \quad (3)$$

Since $\Phi_n(x, y+1) = \Phi_n(x, y)$, define the continuous functions

$$a_n(x) := \max_{0 \leq y \leq 1} \Phi_n(x, y), \quad b_n(x) := \min_{0 \leq y \leq 1} \Phi_n(x, y). \quad (4)$$

Equation (3) gives

$$0 \leq a_n(x) - b_n(x) \leq 1 \quad (x \in \mathbb{T}, n \geq 1). \quad (5)$$

In particular,

$$b_n(x) \leq \Phi_n(x, y) \leq a_n(x) \quad (x \in \mathbb{T}, y \in \mathbb{R}). \quad (6)$$

Step 2: subadditive and superadditive envelopes. The cocycle identity is

$$\Phi_{m+n}(x, y) = \Phi_n(x, y) + \Phi_m(x + n\alpha, f_x^{(n)}(y)). \quad (7)$$

Taking the maximum or minimum in the fiber gives

$$\begin{aligned} a_{m+n}(x) &\leq a_n(x) + a_m(x + n\alpha), \\ b_{m+n}(x) &\geq b_n(x) + b_m(x + n\alpha). \end{aligned} \quad (8)$$

Let $m_{\mathbb{T}}$ denote Haar probability measure and put

$$A_n := \int_{\mathbb{T}} a_n dm_{\mathbb{T}}, \quad B_n := \int_{\mathbb{T}} b_n dm_{\mathbb{T}}. \quad (9)$$

Since $m_{\mathbb{T}}$ is invariant under R_{α} , (8) implies

$$A_{m+n} \leq A_m + A_n, \quad B_{m+n} \geq B_m + B_n. \quad (10)$$

Fekete's lemma therefore yields

$$\rho_+ := \lim_{n \rightarrow \infty} \frac{A_n}{n} = \inf_{n \geq 1} \frac{A_n}{n}, \quad \rho_- := \lim_{n \rightarrow \infty} \frac{B_n}{n} = \sup_{n \geq 1} \frac{B_n}{n}. \quad (11)$$

But (5) gives $0 \leq A_n - B_n \leq 1$, so

$$\rho_+ = \rho_- =: \tilde{\rho}(\alpha, A). \quad (12)$$

This is the candidate real rotation number for the chosen lift.

Step 3: unique ergodicity upgrades mean bounds to uniform convergence. Fix an integer $q \geq 1$. Write $n = kq + r$ with $0 \leq r < q$. Iterating (8) in q -blocks gives

$$a_n(x) \leq \sum_{j=0}^{k-1} a_q(x + jq\alpha) + \max_{0 \leq s < q} \|a_s\|_{\infty}, \quad (13)$$

and similarly

$$b_n(x) \geq \sum_{j=0}^{k-1} b_q(x + jq\alpha) - \max_{0 \leq s < q} \|b_s\|_{\infty}. \quad (14)$$

Here we take $a_0 = b_0 = 0$.

Because α is irrational, $q\alpha$ is irrational. Thus $R_{q\alpha}$ is uniquely ergodic by *Minimal rotations are uniquely ergodic*. Applying the uniform ergodic theorem *Uniform ergodic theorem for unique ergodicity* to the continuous functions a_q and b_q yields, uniformly in x ,

$$\frac{1}{k} \sum_{j=0}^{k-1} a_q(x + jq\alpha) \longrightarrow A_q, \quad \frac{1}{k} \sum_{j=0}^{k-1} b_q(x + jq\alpha) \longrightarrow B_q. \quad (15)$$

Divide (13)-(14) by $n = kq + r$ and let $n \rightarrow \infty$. Uniformly in x ,

$$\limsup_{n \rightarrow \infty} \frac{a_n(x)}{n} \leq \frac{A_q}{q}, \quad \liminf_{n \rightarrow \infty} \frac{b_n(x)}{n} \geq \frac{B_q}{q}. \quad (16)$$

Now choose q so that $A_q/q < \tilde{\rho} + \varepsilon$ and $B_q/q > \tilde{\rho} - \varepsilon$, which is possible by (11)-(12). Together with (6), (16) shows that for all sufficiently large n , uniformly in (x, y) ,

$$\tilde{\rho} - 2\varepsilon < \frac{\Phi_n(x, y)}{n} < \tilde{\rho} + 2\varepsilon. \quad (17)$$

Since $\varepsilon > 0$ is arbitrary,

$$\boxed{\frac{f_x^{(n)}(y) - y}{n} \longrightarrow \tilde{\rho}(\alpha, A) \text{ uniformly in } (x, y).} \quad (18)$$

In particular the limit exists and is independent of the initial base point and projective fiber coordinate.

Step 4: independence of the chosen lift modulo $\mathbb{Z}$. Let $\tilde{F}'_A$ be another continuous degree-one lift of the same projective skew product. The two fiber coordinates differ by a continuous integer-valued function on the connected space $\mathbb{T} \times \mathbb{R}$; hence there is a single $p \in \mathbb{Z}$ such that

$$\tilde{F}'_A(x, y) = \tilde{F}_A(x, y) + (0, p). \quad (19)$$

Because integer translations commute with every degree-one lift, iteration gives an additional displacement np . Therefore

$$\tilde{\rho}'(\alpha, A) = \tilde{\rho}(\alpha, A) + p. \quad (20)$$

Thus

$$\rho(\alpha, A) := \tilde{\rho}(\alpha, A) \pmod{\mathbb{Z}} \quad (21)$$

is intrinsic.

Step 5: continuity in the C^0 topology. Fix A_0 and a lift $\tilde{F}_{A_0}$. For A sufficiently C^0 -close to A_0 , covering-space lifting gives a unique lift $\tilde{F}_A$ whose fiber map is uniformly close to that of $\tilde{F}_{A_0}$. In this local choice the lift depends continuously on A in the uniform topology. Hence, for every fixed N , finite composition shows that

$$A \longmapsto a_N(\cdot; A), \quad A \longmapsto b_N(\cdot; A) \quad (22)$$

are continuous in the uniform norm, and so are their integrals $A_N(A)$ and $B_N(A)$.

For every such A , (11)–(12) imply the finite-iterate enclosure

$$\frac{B_N(A)}{N} \leq \tilde{\rho}(\alpha, A) \leq \frac{A_N(A)}{N}, \quad 0 \leq \frac{A_N(A) - B_N(A)}{N} \leq \frac{1}{N}. \quad (23)$$

Given $\varepsilon > 0$, first choose N with $1/N < \varepsilon$ and then choose A close enough to A_0 that both endpoints in (23) differ from the corresponding endpoints for A_0 by less than ε . The two intervals have length at most ε , so

$$|\tilde{\rho}(\alpha, A) - \tilde{\rho}(\alpha, A_0)| < 3\varepsilon. \quad (24)$$

Thus the locally chosen real representative is continuous; passing to $\mathbb{R}/\mathbb{Z}$ proves continuity of $\rho(\alpha, A)$ globally.

Step 6: invariance under degree-zero conjugacy. Let $B : \mathbb{T} \rightarrow SL(2, \mathbb{R})$ be continuous and homotopic to the identity, and set

$$A'(x) = B(x + \alpha)A(x)B(x)^{-1}. \quad (25)$$

The induced projective coordinate change admits a degree-one lift

$$\tilde{H}_B(x, y) = (x, h_x(y)), \quad h_x(y + 1) = h_x(y) + 1. \quad (26)$$

The displacement $h_x(y) - y$ is continuous and 1-periodic in y ; therefore

$$M := \sup_{x, y} |h_x(y) - y| < \infty. \quad (27)$$

Choose the lift of the conjugated skew product to be

$$\tilde{F}_{A'} := \tilde{H}_B \circ \tilde{F}_A \circ \tilde{H}_B^{-1}. \quad (28)$$

Then

$$\tilde{F}_{A'}^n(\tilde{H}_B(x, y)) = \tilde{H}_B(\tilde{F}_A^n(x, y)). \quad (29)$$

If $\tilde{F}_A^n(x, y) = (x + n\alpha, y_n)$, comparison of fiber displacements in (29) gives

$$|(h_{x+n\alpha}(y_n) - h_x(y)) - (y_n - y)| \leq 2M. \quad (30)$$

Divide by n and use (18). The two real rotation numbers for these compatible lifts are equal. An arbitrary lift changes either one only by an integer, so

$$\boxed{\rho(\alpha, A') = \rho(\alpha, A) \text{ in } \mathbb{R}/\mathbb{Z}.} \quad (31)$$

The proof deliberately stops here: conjugacies of nonzero degree acquire the separate degree correction fixed by the volume convention and are not part of this theorem. $\square$

Theorem F09.4. Uniform hyperbolicity via invariant directions

Statement. Let $T : X \rightarrow X$ be a homeomorphism of a compact metric space and $A : X \rightarrow SL(2, \mathbb{R})$ continuous. The cocycle is uniformly hyperbolic if and only if

there are continuous invariant line fields $E^s(x) \neq E^u(x)$ and constants $C \geq 1$, $\lambda > 1$ such that for every $n \geq 0$, $v^s \in E^s(x)$ and $v^u \in E^u(x)$,

$$\|A_n(x)v^s\| \leq C\lambda^{-n}\|v^s\|, \quad \|A_n(x)v^u\| \geq C^{-1}\lambda^n\|v^u\|.$$

Equivalently, the unstable estimate may be written as backward contraction $\|A_{-n}(x)v^u\| \leq C\lambda^{-n}\|v^u\|$. The two invariant line fields are uniformly transverse and uniquely determined by these contraction/expansion properties.

Proof. For $n \geq 1$ let

$$A_n(x) = A(T^{n-1}x) \cdots A(x), \quad A_{-n}(x) = A_n(T^{-n}x)^{-1}. \quad (1)$$

We prove carefully that the two common formulations of uniform hyperbolicity are equivalent and then establish the asserted uniform transversality and uniqueness.

Step 1: backward unstable contraction implies forward unstable expansion. Assume the cocycle is uniformly hyperbolic in the sense of the definition of uniform hyperbolicity. Thus there are continuous invariant lines E^s, E^u and constants $C \geq 1$, $\lambda > 1$ such that

$$\|A_n(x)v^s\| \leq C\lambda^{-n}\|v^s\|, \quad v^s \in E^s(x), \quad (2)$$

and

$$\|A_{-n}(x)v^u\| \leq C\lambda^{-n}\|v^u\|, \quad v^u \in E^u(x). \quad (3)$$

Fix $v^u \in E^u(x)$ and put

$$w = A_n(x)v^u \in E^u(T^n x), \quad (4)$$

where invariance of E^u was used. By the cocycle identities, $A_{-n}(T^n x)w = v^u$. Apply (3) at the base point $T^n x$:

$$\|v^u\| = \|A_{-n}(T^n x)w\| \leq C\lambda^{-n}\|w\|. \quad (5)$$

Therefore

$$\|A_n(x)v^u\| \geq C^{-1}\lambda^n\|v^u\|. \quad (6)$$

Together with (2), this is the forward contraction/expansion formulation in the statement.

Step 2: forward unstable expansion implies backward unstable contraction. Conversely, suppose continuous invariant transverse lines E^s, E^u satisfy (2) and (6). Let $w \in E^u(x)$ and set

$$v = A_{-n}(x)w \in E^u(T^{-n}x). \quad (7)$$

Then $A_n(T^{-n}x)v = w$. Applying (6) at $T^{-n}x$ gives

$$\|w\| \geq C^{-1}\lambda^n\|v\|. \quad (8)$$

Equivalently,

$$\|A_{-n}(x)w\| = \|v\| \leq C\lambda^{-n}\|w\|. \quad (9)$$

Thus (2) and (9) are exactly the estimates in the definition of uniform hyperbolicity.

Hence the cocycle is uniformly hyperbolic. This proves the equivalence.

Step 3: transversality is uniform on a compact base. For each x , the two lines are distinct. Let

$$\vartheta(x) = \angle(E^s(x), E^u(x)) \in (0, \pi/2] \quad (10)$$

be the acute angle between them. The line fields are continuous, so ϑ is continuous. Since X is compact and $\vartheta(x) > 0$ at every point,

$$\vartheta_* := \min_{x \in X} \vartheta(x) > 0. \quad (11)$$

Thus the splitting is not merely pointwise transverse; its angle is bounded uniformly away from zero.

Equivalently, if $P^s(x)$ and $P^u(x)$ are the projections associated with

$$\mathbb{R}^2 = E^s(x) \oplus E^u(x), \quad (12)$$

then continuity and compactness give a constant $M < \infty$ such that

$$\|P^s(x)\|, \|P^u(x)\| \leq M \quad (x \in X). \quad (13)$$

We use this quantitative form in the uniqueness argument.

Step 4: uniqueness of the stable line. Suppose $\ell(x)$ is another line at a fixed base point x whose nonzero vectors satisfy a forward exponential contraction estimate

$$\|A_n(x)v\| \leq C_0 \lambda_0^{-n} \|v\| \quad (n \geq 0) \quad (14)$$

for some $C_0 \geq 1$, $\lambda_0 > 1$. We show $\ell(x) = E^s(x)$.

Take $0 \neq v \in \ell(x)$ and decompose it using (12):

$$v = v^s + v^u, \quad v^s \in E^s(x), \quad v^u \in E^u(x). \quad (15)$$

If $v^u \neq 0$, invariance of the splitting and (13) imply

$$\|A_n(x)v\| \geq M^{-1} \|A_n(x)v^u\|. \quad (16)$$

Using the unstable expansion (6),

$$\|A_n(x)v\| \geq M^{-1} C^{-1} \lambda^n \|v^u\|. \quad (17)$$

The right-hand side grows exponentially, contradicting (14). Hence $v^u = 0$, so $v \in E^s(x)$ and therefore $\ell(x) = E^s(x)$.

The same argument with time reversed shows that $E^u(x)$ is the unique line whose vectors are exponentially contracted under backward iteration. Since this holds at every x , the invariant stable and unstable line fields are uniquely determined.

All claimed properties now follow. $\square$

Theorem F09.5. Exponential dichotomy equivalence

Statement. Let $T : X \rightarrow X$ be a homeomorphism of a compact metric space and $A : X \rightarrow SL(2, \mathbb{R})$ continuous. The cocycle is uniformly hyperbolic iff there is a continuous rank-one projection $P(x)$ with $A(x)P(x) = P(Tx)A(x)$ and constants $K \geq 1$, $a > 0$ such that $\|A_n(x)P(x)\| \leq Ke^{-an}$ and $\|A_{-n}(x)(I - P(x))\| \leq Ke^{-an}$ for every $n \geq 0$, where $A_{-n}(x) = A_n(T^{-n}x)^{-1}$. In this equivalence $E^s(x) = \text{im } P(x)$ and $E^u(x) = \ker P(x)$.

Proof. Let $T : X \rightarrow X$ be a homeomorphism of a compact metric space and $A : X \rightarrow SL(2, \mathbb{R})$ continuous. For $n \geq 1$ write

$$A_n(x) := A(T^{n-1}x) \cdots A(x),$$

and define the negative-time cocycle by

$$A_{-n}(x) := A_n(T^{-n}x)^{-1}. \quad (1)$$

Thus $A_{-n}(x)$ maps the fiber over x to the fiber over $T^{-n}x$.

We call a continuous family of rank-one projections $P(x)$ an *exponential dichotomy* if

$$A(x)P(x) = P(Tx)A(x) \quad (2)$$

and there are $K \geq 1$ and $a > 0$ such that for every $n \geq 0$,

$$\|A_n(x)P(x)\| \leq Ke^{-an}, \quad (3)$$

$$\|A_{-n}(x)(I - P(x))\| \leq Ke^{-an}. \quad (4)$$

We prove that this is equivalent to uniform hyperbolicity.

Uniform hyperbolicity implies an exponential dichotomy. Assume first that the cocycle is uniformly hyperbolic. Thus there is a continuous invariant splitting

$$\mathbb{R}^2 = E^s(x) \oplus E^u(x) \quad (5)$$

and constants $C \geq 1$, $\lambda > 1$ such that

$$\|A_n(x)v^s\| \leq C\lambda^{-n}\|v^s\|, \quad (6)$$

for $v^s \in E^s(x)$, and

$$\|A_{-n}(x)v^u\| \leq C\lambda^{-n}\|v^u\|, \quad (7)$$

for $v^u \in E^u(x)$.

Let $P(x)$ be the projection onto $E^s(x)$ along $E^u(x)$. Continuity and transversality of the two line fields imply continuity of P ; compactness of X gives

$$M := \sup_x \max\{\|P(x)\|, \|I - P(x)\|\} < \infty. \quad (8)$$

Invariance of (5) means $A(x)E^{s/u}(x) = E^{s/u}(Tx)$, and therefore the two linear maps

$A(x)P(x)$ and $P(Tx)A(x)$ agree on both summands of (5). Hence (2) holds.

For arbitrary $v \in \mathbb{R}^2$, $P(x)v \in E^s(x)$, so (6) and (8) give

$$\|A_n(x)P(x)v\| \leq C\lambda^{-n}\|P(x)v\| \leq CM\lambda^{-n}\|v\|.$$

Thus (3) holds with $K = CM$ and $a = \log \lambda$. Equation (7) gives (4) in exactly the same way. Hence the cocycle has an exponential dichotomy.

An exponential dichotomy implies uniform hyperbolicity. Conversely suppose a continuous rank-one projection $P(x)$ satisfies (2)–(4). Define

$$E^s(x) := \text{im } P(x), \quad E^u(x) := \ker P(x) = \text{im}(I - P(x)). \quad (9)$$

Since $P(x)$ is a projection,

$$\mathbb{R}^2 = E^s(x) \oplus E^u(x). \quad (10)$$

Continuity of a constant-rank projection implies continuity of its image and kernel in the Grassmannian; in dimension two this can also be seen directly by applying $P(x)$ and $I - P(x)$ to a local vector not annihilated at a chosen base point.

From (2), if $v \in E^s(x)$ then

$$P(Tx)A(x)v = A(x)P(x)v = A(x)v,$$

so $A(x)v \in E^s(Tx)$. If $v \in E^u(x)$, then $P(x)v = 0$ and (2) gives $P(Tx)A(x)v = 0$, so $A(x)v \in E^u(Tx)$. Since $A(x)$ is invertible and both spaces are one-dimensional, these inclusions are equalities. Thus the splitting is invariant.

For $v^s \in E^s(x)$ we have $P(x)v^s = v^s$, hence (3) gives

$$\|A_n(x)v^s\| \leq Ke^{-an}\|v^s\|. \quad (11)$$

For $v^u \in E^u(x)$, $(I - P(x))v^u = v^u$, so (4) gives

$$\|A_{-n}(x)v^u\| \leq Ke^{-an}\|v^u\|. \quad (12)$$

Taking $C = K$ and $\lambda = e^a > 1$, (11)–(12) are exactly the stable and inverse-unstable estimates of the definition of uniform hyperbolicity. Therefore the cocycle is uniformly hyperbolic. $\square$

Proposition F09.6. Cone criterion for cocycles

Statement. Let $T : X \rightarrow X$ be a homeomorphism of a compact metric space and $A : X \rightarrow SL(2, \mathbb{R})$ continuous. Suppose there are an integer $N \geq 1$ and a continuous family $\mathcal{C}_x \subset \mathbb{RP}^1$ of closed proper projective intervals with nonempty interior such that

$$A_N(x)\mathcal{C}_x \subset \text{int}\mathcal{C}_{T^N x} \quad (x \in X).$$

Then the cocycle is uniformly hyperbolic. More precisely, the forward projective iterates of the cones determine a unique continuous invariant unstable line E^u , the inverse iterates of the complementary cones determine a unique continuous

invariant stable line E^s , the two lines are uniformly transverse, and there are $C \geq 1$, $\lambda > 1$ with the uniform estimates of Theorem F09.4. Conversely, every uniformly hyperbolic $SL(2, \mathbb{R})$ cocycle admits such a strictly invariant cone field for some N .

Proof. Write

$$S = T^N, \quad B(x) = A_N(x). \quad (1)$$

Thus B is a continuous $SL(2, \mathbb{R})$ cocycle over the homeomorphism S . We first construct the invariant lines for the N -step cocycle and then recover invariance under the original one-step cocycle.

Step 1: an elementary projective-interval contraction fact. If $I \subset \mathbb{RP}^1$ is a closed proper interval with nonempty interior, its interior carries the Hilbert metric. In an affine coordinate in which $I = [a, b]$ this is

$$d_I(s, t) = \left| \log \frac{(t-a)(b-s)}{(s-a)(b-t)} \right|, \quad a < s, t < b. \quad (2)$$

The formula is a cross-ratio, so projective transformations preserve it: if M is projective and $M(I) = J$, then

$$d_J(Ms, Mt) = d_I(s, t). \quad (3)$$

We shall also use the following elementary consequence. If $J \Subset \text{int } I$, then the inclusion $(J, d_J) \hookrightarrow (I, d_I)$ is Lipschitz with some constant $q(I, J) < 1$. Indeed, after sending I to $[-1, 1]$ projectively, the two endpoints of J lie in a compact subinterval $[-1 + \delta, 1 - \delta]$. Differentiating the one-dimensional formulas for d_I and d_J shows that the ratio of the corresponding infinitesimal metrics is bounded by a number strictly smaller than one; integrating along the interval gives the claim. The bound depends continuously on the four endpoints as long as J stays strictly inside I .

For our family put

$$J_x = B(x)\mathcal{C}_x. \quad (4)$$

By hypothesis $J_x \Subset \text{int } \mathcal{C}_{Sx}$. The cone field, B , and the endpoints vary continuously, and X is compact. Hence the preceding contraction constants can be chosen uniformly: there exists

$$0 < q < 1 \quad (5)$$

such that for all x and all projective points $\xi, \eta \in \text{int } \mathcal{C}_x$,

$$d_{\mathcal{C}_{Sx}}(B(x)\xi, B(x)\eta) \leq q d_{\mathcal{C}_x}(\xi, \eta) \quad (6)$$

whenever both points are first restricted to the compact subinterval that appears after one forward step. Equivalently, after the first iterate, each additional B -iterate contracts projective Hilbert diameter by at least the factor q .

Step 2: nested forward images select a unique unstable line. For $m \geq 1$ define the compact projective interval

$$K_m(x) = B_m(S^{-m}x)\mathcal{C}_{S^{-m}x} \subset \text{int } \mathcal{C}_x. \quad (7)$$

Strict invariance gives

$$K_{m+1}(x) \subset K_m(x). \quad (8)$$

Moreover the Hilbert diameter of $K_m(x)$, measured in $\mathcal{C}_x$, tends to zero uniformly in x : the first image has uniformly bounded Hilbert diameter, and (6) contributes a factor q^{m-1} . Therefore

$$\bigcap_{m \geq 1} K_m(x) \quad (9)$$

consists of a single projective point. Denote the corresponding line by $E^u(x)$.

The convergence of the intervals $K_m(x)$ to a point is uniform in x , and each finite-stage interval depends continuously on x . Hence $x \mapsto E^u(x)$ is continuous. From (7), shifting m by one shows

$$B(x)E^u(x) = E^u(Sx). \quad (10)$$

Thus E^u is the unique B -invariant line lying inside the forward cone field.

Step 3: the complementary cones select a stable line. Let

$$\mathcal{D}_x = \overline{\mathbb{RP}^1 \setminus \text{int } \mathcal{C}_x}. \quad (11)$$

This is again a closed proper projective interval. Since a projective $SL(2, \mathbb{R})$ map is a homeomorphism of $\mathbb{RP}^1$, the strict inclusion in the hypothesis is equivalent to

$$B(x)^{-1}\mathcal{D}_{Sx} \subset \text{int } \mathcal{D}_x. \quad (12)$$

Apply Steps 1-2 to the inverse cocycle over S^{-1} . We obtain a unique continuous line $E^s(x) \subset \text{int } \mathcal{D}_x$ satisfying

$$B(x)E^s(x) = E^s(Sx). \quad (13)$$

Because the first forward cone images and first backward complementary-cone images stay a uniform positive distance from the relevant boundaries, $E^u(x)$ and $E^s(x)$ are separated by a positive projective distance uniformly in x . In particular,

$$\inf_{x \in X} \angle(E^s(x), E^u(x)) > 0. \quad (14)$$

Step 4: projective contraction gives vector expansion in determinant one. We record the elementary identity that converts projective contraction into norm growth. Let $M \in SL(2, \mathbb{R})$, let v be a unit vector, and choose a unit vector w with $\det(v, w) = 1$. The curve of lines $[v + tw]$ has unit angular speed at $t = 0$. Since $\det(Mv, Mw) = 1$, differentiating the angle of $[M(v + tw)]$ gives

$$|D(\mathbb{P}M)_{[v]}| = \frac{1}{\|Mv\|^2}. \quad (15)$$

On the compact subintervals containing $E^u(x)$, Hilbert distance and ordinary angular distance are uniformly bi-Lipschitz. Iterating (6) therefore gives constants $C_0 \geq 1$ and $0 < q < 1$ such that

$$|D(\mathbb{P}B_m(x))_{E^u(x)}| \leq C_0 q^m. \quad (16)$$

Using (15), for a unit $v^u \in E^u(x)$,

$$\|B_m(x)v^u\| \geq C_0^{-1/2} q^{-m/2}. \quad (17)$$

The same argument for the inverse cocycle on E^s yields backward expansion on E^s . Reversing time, exactly as in *Uniform hyperbolicity via invariant directions*, gives forward contraction:

$$\|B_m(x)v^s\| \leq C_1 q^{m/2} \|v^s\|, \quad v^s \in E^s(x), \quad (18)$$

for a uniform C_1 .

Thus the N -step cocycle has a continuous uniformly transverse splitting with exponential contraction and expansion.

Step 5: the splitting is invariant under one-step iteration. At this stage invariance has only been proved for $B = A_N$. We now use the uniqueness part of *Uniform hyperbolicity via invariant directions* to recover invariance under A .

Fix x and let $0 \neq v \in E^u(x)$. For every $m \geq 0$ the cocycle identity

$$B_m(Tx)A(x) = A(T^{mN}x)B_m(x) \quad (19)$$

holds. The norms of $A(y)$ and $A(y)^{-1}$ are uniformly bounded on the compact base. Hence (17) implies that vectors on the line $A(x)E^u(x)$ expand exponentially under the B -cocycle based at Tx . By uniqueness of the exponentially expanding line for B ,

$$A(x)E^u(x) = E^u(Tx). \quad (20)$$

The same argument, with time reversed, gives

$$A(x)E^s(x) = E^s(Tx). \quad (21)$$

Finally write an arbitrary $n \geq 0$ as $n = mN + r$ with $0 \leq r < N$. The finitely many remainder products $A_r(y)$ and their inverses are uniformly bounded. Combining those bounds with (17)–(18) produces constants $C \geq 1$ and $\lambda > 1$ such that

$$\|A_n(x)v^s\| \leq C\lambda^{-n}\|v^s\|, \quad \|A_n(x)v^u\| \geq C^{-1}\lambda^n\|v^u\|. \quad (22)$$

Uniform hyperbolicity via invariant directions now identifies this splitting as uniform hyperbolicity.

Step 6: converse—uniform hyperbolicity produces strict cones. Assume conversely that the cocycle is uniformly hyperbolic. By *Uniform hyperbolicity via invariant directions* there are continuous uniformly transverse invariant lines E^s, E^u and constants C, λ . Use the splitting coordinates $v = v^u + v^s$. For a fixed small $\alpha > 0$ let

$$\mathcal{C}_x = \{[v^u + v^s] : \|v^s\| \leq \alpha\|v^u\|\}. \quad (23)$$

Uniform transversality makes this a continuous proper projective interval field. For $v = v^u + v^s$ in this cone,

$$\frac{\|A_N(x)v^s\|}{\|A_N(x)v^u\|} \leq C^2\lambda^{-2N} \frac{\|v^s\|}{\|v^u\|}. \quad (24)$$

Choose N so large that $C^2\lambda^{-2N} < 1$. Then (24) sends the closed cone strictly into the cone with a smaller slope, hence

$$A_N(x)\mathcal{C}_x \subset \text{int}\mathcal{C}_{T^N x} \quad (25)$$

uniformly in x . This proves the converse and completes the cone criterion. $\square$

Theorem F09.7. Dominated splitting criterion in dimension two

Statement. Let $T : X \rightarrow X$ be a homeomorphism of a compact metric space and let $A : X \rightarrow GL(2, \mathbb{R})$ be continuous. Write $A_n(x) = A(T^{n-1}x) \cdots A(x)$ and let $\sigma_1(B) \geq \sigma_2(B) > 0$ denote the singular values of $B \in GL(2, \mathbb{R})$. The following are equivalent:

1. there are continuous A -invariant line fields E^u, E^s with $\mathbb{R}^2 = E^u(x) \oplus E^s(x)$ and constants $C \geq 1, 0 < \rho < 1$ such that

$$\frac{\|A_n(x)|_{E^s(x)}\|}{m(A_n(x)|_{E^u(x)})} \leq C\rho^n \quad (x \in X, n \geq 0);$$

2. there are constants $C_0 \geq 1, 0 < \theta < 1$ such that

$$\frac{\sigma_2(A_n(x))}{\sigma_1(A_n(x))} \leq C_0\theta^n \quad (x \in X, n \geq 1).$$

When these conditions hold, E^u, E^s are uniquely determined. They are also equivalent to the existence, for some iterate $N \geq 1$, of a continuous proper projective cone field $\mathcal{C}_x \subset \mathbb{RP}^1$ with nonempty interior such that $A_N(x)\mathcal{C}_x \subset \text{int}\mathcal{C}_{T^N x}$; the cone may be chosen around E^u .

Proof. Set

$$r_n(x) := \frac{\sigma_2(A_n(x))}{\sigma_1(A_n(x))}. \quad (1)$$

Because A and A^{-1} are continuous on the compact space X , there is $M \geq 1$ such that

$$\|A(x)\| \leq M, \quad \|A(x)^{-1}\| \leq M \quad (x \in X). \quad (2)$$

We prove the two implications separately. The direction from a singular-value gap to an invariant splitting is the substantial one.

Step 1: domination implies an exponential singular-value gap. Assume there are continuous invariant transverse lines E^u, E^s and constants $C \geq 1, 0 < \rho < 1$ such that

$$q_n(x) := \frac{\|A_n(x)|_{E^s(x)}\|}{m(A_n(x)|_{E^u(x)})} \leq C\rho^n. \quad (3)$$

Both spaces are one-dimensional, so the norm and conorm of the restriction to one of the lines are the same number.

Choose continuous unit vectors locally in E^u, E^s . Equivalently, and without making any global orientability assumption, equip each fibre with the continuous inner prod-

uct for which $E^u(x)$ and $E^s(x)$ are orthogonal and unit vectors in the original Euclidean metric have uniformly comparable length. Compactness and uniform transversality imply that the identity between this adapted norm and the Euclidean norm has condition number bounded by one constant $K \geq 1$, uniformly in x .

In the adapted norms the map $A_n(x)$ is diagonal with respect to the invariant splittings at x and $T^n x$. Hence, once $q_n(x) < 1$, its two singular values in the adapted norms are exactly the expansion factors on E^u and E^s , so their ratio is $q_n(x)$. Changing the norm at the source and target by uniformly K -conditioned linear identifications changes each singular value by at most a fixed multiplicative factor. Therefore there is $K_1 \geq 1$ such that, for all sufficiently large n ,

$$r_n(x) \leq K_1 q_n(x) \leq K_1 C \rho^n. \quad (4)$$

There are only finitely many remaining values of n ; enlarging the constant if necessary gives

$$r_n(x) \leq C_0 \rho^n \quad (x \in X, n \geq 1). \quad (5)$$

This is condition (2).

Step 2: finite-time slow singular directions converge uniformly. Assume now that

$$r_n(x) \leq C_0 \theta^n, \quad 0 < \theta < 1. \quad (6)$$

Choose n_0 so large that $C_0 \theta^n < 1$ for $n \geq n_0$. For such n the two singular values of $A_n(x)$ are distinct. Let $S_n(x) \in \mathbb{RP}^1$ be the most-contracted right singular line and let $U_n(x) = S_n(x)^\perp$ be the most-expanded right singular line. Because the singular value is simple, $S_n(x)$ depends continuously on x .

We estimate the angle between two consecutive slow singular directions. Let v be a unit vector in $S_n(x)$ and decompose it orthogonally relative to $S_{n+1}(x) \oplus U_{n+1}(x)$ as $v = s + w$. The images of these two singular lines under $A_{n+1}(x)$ are orthogonal, hence

$$\|A_{n+1}(x)v\| \geq \sigma_1(A_{n+1}(x)) \|w\|. \quad (7)$$

On the other hand,

$$\|A_{n+1}(x)v\| = \|A(T^n x)A_n(x)v\| \leq M \sigma_2(A_n(x)). \quad (8)$$

Also

$$\sigma_1(A_{n+1}(x)) \geq m(A(T^n x)) \sigma_1(A_n(x)) \geq M^{-1} \sigma_1(A_n(x)). \quad (9)$$

Since $\|w\|$ is the sine of the acute angle between $S_n(x)$ and $S_{n+1}(x)$, (7)–(9) give

$$\sin \angle(S_{n+1}(x), S_n(x)) \leq M^2 r_n(x) \leq M^2 C_0 \theta^n. \quad (10)$$

For small angles, the angle is bounded by a fixed multiple of its sine. Thus the series of successive angular distances is uniformly summable. The sequence S_n therefore converges uniformly to a continuous line field, which we denote by

$$E^s(x) := \lim_{n \rightarrow \infty} S_n(x). \quad (11)$$

Moreover there is $C_s > 0$ such that

$$\angle(E^s(x), S_n(x)) \leq C_s \theta^n \quad (n \geq n_0, x \in X). \quad (12)$$

Step 3: the limiting slow line is invariant. We compare $A(x)S_{n+1}(x)$ with $S_n(Tx)$. Let v be a unit vector in $S_{n+1}(x)$ and put

$$z := \frac{A(x)v}{\|A(x)v\|}. \quad (13)$$

Decompose $z = s + w$ with $s \in S_n(Tx)$ and $w \in U_n(Tx)$. Since the singular images of $S_n(Tx)$ and $U_n(Tx)$ under $A_n(Tx)$ are orthogonal,

$$\sigma_1(A_n(Tx))\|w\| \leq \|A_n(Tx)z\| = \frac{\sigma_2(A_{n+1}(x))}{\|A(x)v\|}. \quad (14)$$

Now $\|A(x)v\| \geq M^{-1}$, while the elementary singular-value inequality $m(BC) \leq \|C\|m(B)$ gives

$$\sigma_2(A_{n+1}(x)) = m(A_n(Tx)A(x)) \leq M\sigma_2(A_n(Tx)). \quad (15)$$

Consequently

$$\sin \angle(A(x)S_{n+1}(x), S_n(Tx)) \leq M^2 r_n(Tx) \leq M^2 C_0 \theta^n. \quad (16)$$

Letting $n \rightarrow \infty$ and using (11) yields

$$A(x)E^s(x) = E^s(Tx). \quad (17)$$

Step 4: construct the fast invariant line by reversing time. Consider the inverse cocycle over T^{-1} ,

$$B(x) := A(T^{-1}x)^{-1}. \quad (18)$$

Its n -step product is

$$B_n(x) = A_n(T^{-n}x)^{-1}. \quad (19)$$

For a 2×2 invertible matrix L , the singular values of L^{-1} are $\sigma_2(L)^{-1} \geq \sigma_1(L)^{-1}$. Therefore

$$\frac{\sigma_2(B_n(x))}{\sigma_1(B_n(x))} = r_n(T^{-n}x) \leq C_0 \theta^n. \quad (20)$$

Applying Steps 2-3 to (T^{-1}, B) gives a continuous B -invariant slow line. Call it $E^u(x)$. Its B -invariance is equivalent to

$$A(x)E^u(x) = E^u(Tx). \quad (21)$$

At this stage E^u and E^s are continuous invariant line fields, but we have not yet assumed or proved that they are everywhere distinct. That is precisely the point of the next two steps.

Step 5: at every Oseledets-regular point the two lines are the fast and slow Lyapunov lines. Fix an arbitrary T -invariant Borel probability measure μ . The integrability hypotheses of the two-sided multiplicative ergodic theorem *Oseledets theorem: invert-*

ible splitting form are automatic here because A and A^{-1} are bounded on X . Thus at μ -almost every point x there is an Oseledets splitting.

We first identify $E^s(x)$. From (12), a unit vector $v \in E^s(x)$ differs from a unit vector in $S_n(x)$ by $O(\theta^n)$. Therefore, for a uniform constant C_1 ,

$$\|A_n(x)v\| \leq \sigma_2(A_n(x)) + C_1\theta^n\sigma_1(A_n(x)) \leq C_2\theta^n\sigma_1(A_n(x)). \quad (22)$$

At an Oseledets-regular point, the exponential growth rate of $\sigma_1(A_n(x)) = \|A_n(x)\|$ equals the largest Lyapunov exponent. Indeed, a fixed basis adapted to the finitely many Oseledets subspaces gives the upper bound, while any nonzero vector in the top Oseledets space gives the matching lower bound. Taking logarithms in (22) shows that every nonzero vector in $E^s(x)$ has exponent at most

$$\lambda_1(x) + \log \theta < \lambda_1(x). \quad (23)$$

Hence the spectrum has two distinct exponents and $E^s(x)$ is the slow Oseledets line. Applying the same argument to the inverse cocycle shows that $E^u(x)$ is the fast Oseledets line. In particular, at every regular point,

$$E^u(x) \neq E^s(x) \quad (24)$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{\|A_n(x)|_{E^s(x)}\|}{\|A_n(x)|_{E^u(x)}\|} = \lambda_2(x) - \lambda_1(x) \leq \log \theta. \quad (25)$$

The last inequality follows also directly from (6) by taking logarithmic singular-value limits at the regular point.

Step 6: upgrade the almost-everywhere Lyapunov gap to a uniform finite-time domination inequality. Because the two line fields are continuous and invariant, the function

$$\psi(x) := \log \|A(x)|_{E^s(x)}\| - \log \|A(x)|_{E^u(x)}\| \quad (26)$$

is continuous even at a hypothetical point where the two lines coincide. For every $n \geq 1$,

$$\sum_{j=0}^{n-1} \psi(T^j x) = \log \frac{\|A_n(x)|_{E^s(x)}\|}{\|A_n(x)|_{E^u(x)}\|}. \quad (27)$$

By (25), for every invariant probability measure μ , the left side divided by n converges for μ -almost every x to a value at most $\log \theta$. Since ψ is bounded and μ is invariant,

$$\int_X \psi d\mu = \int_X \frac{1}{n} \sum_{j=0}^{n-1} \psi \circ T^j d\mu \leq \log \theta \quad (28)$$

after passing to the limit by dominated convergence.

Choose κ with $\theta < \kappa < 1$. We claim that there is an $N \geq 1$ such that

$$\frac{1}{N} \sum_{j=0}^{N-1} \psi(T^j x) < \log \kappa \quad \text{for every } x \in X. \quad (29)$$

If not, for every arbitrarily large n one could choose x_n for which the left side is at least $\log \kappa$. The empirical measures

$$\nu_n := \frac{1}{n} \sum_{j=0}^{n-1} \delta_{T^j x_n} \quad (30)$$

have a weak-* convergent subsequence by compactness of the space of probability measures on X . As in the Krylov–Bogolyubov argument *Krylov–Bogolyubov existence theorem*, every such limit ν is T -invariant. Passing to the limit in $\int \psi d\nu_n$ would give

$$\int \psi d\nu \geq \log \kappa, \quad (31)$$

contradicting (28). Thus (29) holds.

Exponentiating (29) gives

$$\frac{\|A_N(x)|_{E^s(x)}\|}{\|A_N(x)|_{E^u(x)}\|} < \kappa < 1 \quad (x \in X). \quad (32)$$

This inequality has two consequences at once. First, $E^u(x)$ and $E^s(x)$ cannot coincide at any point, because if they did then their invariance would make the numerator and denominator in (32) equal. Therefore

$$\mathbb{R}^2 = E^u(x) \oplus E^s(x) \quad (33)$$

for every x . Second, iterating (32) gives

$$\frac{\|A_{kN}(x)|_{E^s(x)}\|}{\|A_{kN}(x)|_{E^u(x)}\|} \leq \kappa^k. \quad (34)$$

The finitely many remainders $0 \leq r < N$ are uniformly bounded by compactness. Writing $n = kN + r$ therefore yields constants $C \geq 1$ and $\rho = \kappa^{1/N} < 1$ such that

$$\frac{\|A_n(x)|_{E^s(x)}\|}{m(A_n(x)|_{E^u(x)})} \leq C\rho^n. \quad (35)$$

Thus condition (1) holds.

Step 7: uniqueness of the dominated splitting. Suppose $\tilde{E}^u \oplus \tilde{E}^s$ were a second dominated splitting. Apply Step 1 to it and then the construction of Steps 2–3. The limit in (11) is determined solely by the matrices $A_n(x)$, so its slow line must equal both E^s and $\tilde{E}^s$. Applying the inverse-cocycle construction gives $E^u = \tilde{E}^u$. Hence the dominated splitting is unique.

Step 8: equivalence with a strict invariant cone field. Assume first that the dominated splitting exists. Increase N if necessary so that the ratio in (32) is at most some $\kappa < 1$. In the splitting coordinates $v = v^u + v^s$, define a projective cone around $E^u(x)$ by

$$\mathcal{C}_x(\eta) := \{[v^u + v^s] : \|v^s\| \leq \eta\|v^u\|\} \quad (36)$$

for a fixed $\eta > 0$. Invariance of the lines and (32) imply that the slope $\|v^s\|/\|v^u\|$ is multiplied by at most κ . Hence

$$A_N(x)\mathcal{C}_x(\eta) \subset \text{int } \mathcal{C}_{T^N x}(\eta). \quad (37)$$

Conversely, suppose a continuous proper cone field satisfies (37) for some N . Multiplying each $A(x)$ by the positive scalar $|\det A(x)|^{-1/2}$ does not change either its projective action or any ratio of singular values, and it does not change a domination ratio because the same scalar multiplies both one-dimensional restrictions. We may therefore assume $|\det A(x)| = 1$.

The compact strict inclusion in (37) gives a uniform contraction in the Hilbert metric of the projective intervals. Nested forward cone images then shrink to a unique continuous invariant line E^u ; applying the same argument to inverse images of the complementary intervals gives a continuous invariant line E^s . The two limit lines lie in disjoint compact subintervals of projective space and are therefore uniformly transverse. This is exactly the projective-contraction argument used in *Cone criterion for cocycles*; its metric part does not use the sign of the determinant.

For completeness, the vector estimate follows from the elementary projective derivative identity

$$|D(\mathbb{P}L)_{[v]}| = \frac{|\det L|}{\|Lv\|^2} \quad (\|v\| = 1) \quad (38)$$

when $|\det L| = 1$. Uniform projective contraction along E^u therefore gives uniform expansion there for a sufficiently large iterate, while inverse projective contraction gives uniform contraction along E^s . Their quotient is < 1 uniformly for that iterate, hence (35) follows by the same block-and-remainder argument as above. Restoring the discarded scalar factors leaves the quotient unchanged. Thus the strict cone condition is equivalent to domination.

All assertions in the theorem are proved. $\square$

Proposition F09.8. Symplectic cocycle spectral pairing

Statement. Under the hypotheses of Theorem F04.6, assume in addition that $A(x) \in Sp(2d, \mathbb{R})$ almost everywhere. On the full-measure Oseledets set write $E_\lambda(x)$ for the Lyapunov subspace of exponent λ , and let ω be the standard symplectic form. If $v \in E_\lambda(x)$ and $w \in E_\mu(x)$ with $\lambda + \mu \neq 0$, then $\omega(v, w) = 0$. Consequently the pairing $E_\lambda(x) \times E_{-\lambda}(x) \rightarrow \mathbb{R}$ is nondegenerate, $\dim E_\lambda = \dim E_{-\lambda}$, and the zero-exponent space $E_0(x)$ (when present) is symplectic and hence even-dimensional. The stable and unstable sums are isotropic; if there are no zero exponents, both have dimension d and are Lagrangian.

Proof. Fix a point x in the full-measure Oseledets set from *Oseledets theorem: invertible splitting form*. Because every $A_n(x)$ is symplectic,

$$\omega(A_n(x)v, A_n(x)w) = \omega(v, w) \quad (1)$$

for all integers n and all $v, w \in \mathbb{R}^{2d}$. Choose nonzero $v \in E_\lambda(x)$ and $w \in E_\mu(x)$.

Suppose first that $\lambda + \mu < 0$. Pick $\varepsilon > 0$ with $\lambda + \mu + 2\varepsilon < 0$. By the defining Oseledets limits, for all sufficiently large positive n ,

$$\|A_n(x)v\| \leq e^{n(\lambda+\varepsilon)}, \quad \|A_n(x)w\| \leq e^{n(\mu+\varepsilon)}. \quad (2)$$

(For the two fixed nonzero vectors, any fixed multiplicative constants are absorbed once n is large.) Since $|\omega(u, z)| \leq C_\omega \|u\| \|z\|$, (1)–(2) imply

$$|\omega(v, w)| \leq C e^{n(\lambda+\mu+2\varepsilon)} \rightarrow 0.$$

Hence $\omega(v, w) = 0$.

If $\lambda + \mu > 0$, apply the same argument to negative iterates. The two-sided conclusion of *Oseledets theorem: invertible splitting form* gives

$$\lim_{n \rightarrow +\infty} \frac{1}{n} \log \|A_{-n}(x)v\| = -\lambda, \quad \lim_{n \rightarrow +\infty} \frac{1}{n} \log \|A_{-n}(x)w\| = -\mu.$$

Now $-\lambda - \mu < 0$, so (1) with n replaced by $-n$ again forces $\omega(v, w) = 0$. We have proved

$$\boxed{\lambda + \mu \neq 0 \implies \omega(E_\lambda(x), E_\mu(x)) = 0.} \quad (3)$$

Fix an exponent λ . Suppose $0 \neq v \in E_\lambda(x)$ were symplectically orthogonal to all of $E_{-\lambda}(x)$. By (3), it is automatically orthogonal to every $E_\mu(x)$ with $\mu \neq -\lambda$ as well. The Oseledets spaces sum to all of $\mathbb{R}^{2d}$, so v would be orthogonal to the whole ambient space, contradicting nondegeneracy of ω . Thus the bilinear pairing

$$E_\lambda(x) \times E_{-\lambda}(x) \longrightarrow \mathbb{R}, \quad (v, w) \longmapsto \omega(v, w), \quad (4)$$

is nondegenerate in its first variable. Interchanging λ and $-\lambda$ shows nondegeneracy in the second variable. Therefore

$$\dim E_\lambda(x) = \dim E_{-\lambda}(x). \quad (5)$$

This also recovers, at the bundle level, the multiplicity pairing from *Determinant and symplectic constraints on spectra*.

For $\lambda = 0$, equation (3) says that $E_0(x)$ is orthogonal to every nonzero-exponent space. If a vector in $E_0(x)$ were also orthogonal to all of $E_0(x)$, it would again be orthogonal to the whole ambient space. Hence $\omega|_{E_0(x)}$ is nondegenerate: $E_0(x)$ is a symplectic subspace and therefore has even dimension.

Let

$$E^s(x) := \bigoplus_{\lambda < 0} E_\lambda(x), \quad E^u(x) := \bigoplus_{\lambda > 0} E_\lambda(x).$$

If $v, w \in E^s(x)$, every pair of exponent components has negative sum, so (3) gives $\omega(v, w) = 0$. Thus $E^s(x)$ is isotropic; the same is true of $E^u(x)$. By (5), $\dim E^s = \dim E^u$. If there is no zero exponent, their direct sum is all of $\mathbb{R}^{2d}$, hence both dimensions equal d . An isotropic subspace of a $2d$ -dimensional symplectic vector space with dimension d is Lagrangian. Therefore both stable and unstable Oseledets sums are Lagrangian in the zero-free case. $\square$

9.3 Worked examples

Worked example: Matrix Cocycles and Projective Dynamics: a model computation

Let

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}, \quad \lambda > 1.$$

The projective line can be parametrized by slopes $t = y/x$, together with the vertical direction. For $x \neq 0$,

$$t \mapsto \frac{\lambda^{-1}y}{\lambda x} = \lambda^{-2}t.$$

Thus the horizontal direction $[e_1]$ is attracting in forward projective time, while the vertical direction $[e_2]$ is repelling projectively. In the original vector dynamics the roles are expressed as

$$\|A^n e_2\| = \lambda^{-n}, \quad \|A^{-n} e_1\| = \lambda^{-n}.$$

Hence $E^s = \mathbb{R}e_2$, $E^u = \mathbb{R}e_1$, and the cocycle is uniformly hyperbolic with $C = 1$. A cone such as $|y| \leq c|x|$ is mapped strictly inside itself because its slope is multiplied by $\lambda^{-2} < 1$.

Boundary case: what fails at the edge of the hypotheses

For the parabolic matrix

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad P^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

there is only one projective fixed direction and growth is polynomial, not exponential. Stable and unstable directions coalesce, no invariant cone is strictly contracted in projective angle, and no exponential dichotomy exists. This is the boundary between hyperbolic and neutral projective behavior.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

9.4 Chapter summary

What to remember

- Conjugacy changes coordinates without changing the underlying cocycle dynamics.
- Projective dynamics records the evolution of directions rather than lengths.
- Uniform hyperbolicity has several equivalent geometric and operator-theoretic formulations.
- Parabolic behavior is a canonical mechanism by which hyperbolicity degenerates.

Looking ahead

F10 moves from linear cocycles to nonlinear diffeomorphisms. The same invariant-splitting idea becomes the stable manifold theorem, shadowing, structural stability, and normal hyperbolicity.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

9.5 Exercises and problems

Exercise 9.1

Difficulty. Core.

Related results. Cocycle conjugacy calculus; Projective action of $SL(2, \mathbb{R})$; Fibered rotation number.

Problem. Work through the worked example 'a constant hyperbolic $SL(2, \mathbb{R})$ matrix and its projective fixed directions'. For Cocycle conjugacy calculus, Projective action of $SL(2, \mathbb{R})$, and Fibered rotation number, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 9.2

Difficulty. Core.

Related results. Uniform hyperbolicity via invariant directions; Exponential dichotomy equivalence; Cone criterion for cocycles.

Problem. Analyze the boundary example 'a parabolic matrix where stable/unstable directions coalesce'. For Uniform hyperbolicity via invariant directions, Exponential dichotomy equivalence, and Cone criterion for cocycles, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 9.3

Difficulty. Advanced.

Related results. Fibered rotation number.

Problem. For Fibered rotation number, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 9.4

Difficulty. Advanced.

Related results. Uniform hyperbolicity via invariant directions.

Problem. Stress-test Uniform hyperbolicity via invariant directions. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 9.5

Difficulty. Advanced.

Related results. Cocycle conjugacy calculus; Projective action of $SL(2, \mathbb{R})$.

Problem. Synthesis problem. Use Cocycle conjugacy calculus and Projective action of $SL(2, \mathbb{R})$ to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each

result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 9.6

Difficulty. Lab.

Related results. None.

Problem. Computational laboratory: carry out the reproducible experiment ‘a constant hyperbolic $SL(2, \mathbb{R})$ matrix and its projective fixed directions’. Use the computation only as numerical evidence or visualization, not as proof. Include a refinement or convergence test and identify one numerical artifact that could mislead a reader.

Solution sketch. State parameters and truncation/time horizon, compare with an analytically controlled identity or exactly solvable case, repeat after refinement, and explain what the numerical experiment can and cannot establish.

Exercise 9.7

Difficulty. Core.

Related results. Dominated splitting criterion in dimension two.

Problem. Work directly with Dominated splitting criterion in dimension two. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 9.8

Difficulty. Core.

Related results. Symplectic cocycle spectral pairing.

Problem. Work directly with Symplectic cocycle spectral pairing. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Uniform Hyperbolicity and Dominated Splittings

Why this chapter matters

Hyperbolicity is the setting in which nonlinear dynamics behaves most like its linearization. Stable and unstable directions persist, nearby pseudo-orbits are shadowed by true orbits, and the qualitative dynamics survives small perturbations. These properties are not separate miracles: they are consequences of uniform exponential separation.

This chapter develops the geometric toolkit that later reappears in billiards, partially hyperbolic dynamics, and renormalization.

Prerequisites. F09, smooth manifolds, the inverse/contraction mapping theorems, and differential calculus.

Learning goals

- Construct local stable and unstable manifolds from a hyperbolic splitting.
- Use local product structure and shadowing to organize nearby orbits.
- Understand structural stability and the inclination lemma geometrically.
- Distinguish full hyperbolicity, domination, and normal hyperbolicity.

Conceptual roadmap

The local stable-manifold theorem is the basic nonlinear graph-transform result. Uniformity over a compact hyperbolic set gives stable laminations and local product structure. Shadowing converts approximate orbits into true ones, which underlies structural stability. Dominated and normally hyperbolic splittings retain only the relative rates needed for robust invariant geometry.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

10.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F10.1. Hyperbolic set

A compact invariant set Λ of a C^1 diffeomorphism f is *hyperbolic* if $T_\Lambda M = E^s \oplus E^u$ continuously and Df -invariantly, with uniform exponential contraction on E^s and uniform backward contraction on E^u .

Definition F10.2. Local stable manifold

The local stable manifold $W_{\text{loc}}^s(x)$ is the embedded disk through x , tangent to E_x^s , consisting locally of points whose forward iterates remain near the orbit of x and converge to it exponentially.

Definition F10.3. Dominated splitting

A Df -invariant splitting $E \oplus F$ is *dominated* if there exist $N \geq 1$ and $0 < \theta < 1$ such that $\|Df^N|_{E_x}\| \|(Df^N|_{F_x})^{-1}\| \leq \theta$ for every x in the invariant set.

Definition F10.4. Normal hyperbolicity

Let N be a compact invariant C^r submanifold. It is *r -normally hyperbolic* when there is a continuous invariant splitting $T_N M = E^s \oplus TN \oplus E^u$ for which contraction/expansion in the normal bundles dominates the corresponding tangential rates to the order required by r . The precise one-step inequalities are those stated in Theorem F10.10.

10.2 Main results

Lemma F10.1. Adapted norm for a hyperbolic splitting

Statement. Let $L = L_s \oplus L_u$ be a hyperbolic linear map. For every $a < 1$ strictly larger than both the stable spectral radius and the spectral radius of L_u^{-1} , there is an equivalent norm for which

$$\|L_s\| \leq a, \quad \|L_u^{-1}\| \leq a.$$

Proof. We first prove the elementary norm-construction lemma that will be used on both halves of the hyperbolic splitting.

Let L be a linear operator on a finite-dimensional normed space and assume

$$r(L) < a. \tag{1}$$

Choose b with $r(L) < b < a$. By the spectral-radius formula,

$$\lim_{n \rightarrow \infty} \|L^n\|^{1/n} = r(L),$$

so $\|L^n\| \leq b^n$ for all sufficiently large n . Enlarging a constant to absorb the finitely many initial powers, there is $C \geq 1$ such that

$$\|L^n\| \leq Cb^n \quad (n \geq 0). \quad (2)$$

Define

$$\|v\|_a := \sup_{n \geq 0} a^{-n} \|L^n v\|. \quad (3)$$

The supremum is finite because (2) gives

$$a^{-n} \|L^n v\| \leq C(b/a)^n \|v\| \leq C\|v\|.$$

Homogeneity and the triangle inequality in (3) follow term by term from the original norm, and the $n = 0$ term gives

$$\|v\| \leq \|v\|_a.$$

Together with the preceding upper bound,

$$\|v\| \leq \|v\|_a \leq C\|v\|, \quad (4)$$

so (3) defines an equivalent norm.

Now shift the index in the supremum:

$$\begin{aligned} \|Lv\|_a &= \sup_{n \geq 0} a^{-n} \|L^{n+1}v\| \\ &= a \sup_{m \geq 1} a^{-m} \|L^m v\| \\ &\leq a\|v\|_a. \end{aligned}$$

Thus the operator norm of L in the adapted norm is at most a .

Apply this construction to the stable restriction L_s and to L_u^{-1} on the unstable subspace. The hypothesis of the result is precisely

$$\max\{r(L_s), r(L_u^{-1})\} < a < 1,$$

so the construction produces norms $\|\cdot\|_{s,a}$ and $\|\cdot\|_{u,a}$ satisfying

$$\|L_s v_s\|_{s,a} \leq a\|v_s\|_{s,a}, \quad \|L_u^{-1} v_u\|_{u,a} \leq a\|v_u\|_{u,a}. \quad (5)$$

On the invariant direct sum $E^s \oplus E^u$, define

$$\|v_s + v_u\|_* := \max\{\|v_s\|_{s,a}, \|v_u\|_{u,a}\}. \quad (6)$$

Because the splitting is a direct sum in finite dimension and both component norms are equivalent to their original restrictions, (6) is an equivalent norm on the whole

space. Its restrictions satisfy, by (5),

$$\|L_s\|_* \leq a, \quad \|L_u^{-1}\|_* \leq a.$$

Since $a < 1$, both the stable action and the inverse unstable action are strict contractions in this adapted norm, which is exactly the claimed conclusion. $\square$

Lemma F10.2. Parameter-dependent contraction theorem

Statement. Let P, Y be Banach spaces, $U \subset P$ open, $r \geq 1$, and $\Phi : U \times Y \rightarrow Y$ be C^r . Assume a uniform contraction bound $\|\Phi(\xi, y_1) - \Phi(\xi, y_2)\| \leq q\|y_1 - y_2\|$ with $0 \leq q < 1$. Then every $\Phi(\xi, \cdot)$ has a unique fixed point $y_*(\xi)$, the map $y_* : U \rightarrow Y$ is C^r , and $Dy_*(\xi) = (I - D_y\Phi(\xi, y_*(\xi)))^{-1}D_\xi\Phi(\xi, y_*(\xi))$, with $\|(I - D_y\Phi)^{-1}\| \leq (1 - q)^{-1}$.

Proof. Let P and Y be Banach spaces, let $U \subset P$ be open, and let

$$\Phi : U \times Y \longrightarrow Y$$

be C^r , where $r \geq 1$. Assume that there is a single number $0 \leq q < 1$ such that

$$\|\Phi(\xi, y_1) - \Phi(\xi, y_2)\|_Y \leq q\|y_1 - y_2\|_Y \quad (1)$$

for all $\xi \in U$ and $y_1, y_2 \in Y$.

Step 1: existence and uniqueness for each parameter. Fix $\xi \in U$. Since Y is complete, (1) says that $y \mapsto \Phi(\xi, y)$ is a contraction of Y into itself. Banach's fixed-point theorem therefore supplies a unique point $y_*(\xi) \in Y$ such that

$$y_*(\xi) = \Phi(\xi, y_*(\xi)). \quad (2)$$

Thus a fixed-point map $y_* : U \rightarrow Y$ is well defined.

Step 2: the fixed point varies continuously. For $\xi, \eta \in U$, use (2) twice and insert $\Phi(\xi, y_*(\eta))$:

$$\|y_*(\xi) - y_*(\eta)\| \leq \|\Phi(\xi, y_*(\xi)) - \Phi(\xi, y_*(\eta))\| \quad (10.1)$$

$$+ \|\Phi(\xi, y_*(\eta)) - \Phi(\eta, y_*(\eta))\| \quad (10.2)$$

$$\leq q\|y_*(\xi) - y_*(\eta)\| + \|\Phi(\xi, y_*(\eta)) - \Phi(\eta, y_*(\eta))\|. \quad (10.3)$$

Hence

$$\|y_*(\xi) - y_*(\eta)\| \leq \frac{1}{1 - q} \|\Phi(\xi, y_*(\eta)) - \Phi(\eta, y_*(\eta))\|. \quad (3)$$

For fixed η , the last factor tends to zero as $\xi \rightarrow \eta$ because Φ is continuous. Thus y_* is continuous.

Step 3: invertibility of the derivative in the fixed-point variable. Because Φ is differentiable, (1) implies

$$\|D_y\Phi(\xi, y)\|_{\mathcal{L}(Y, Y)} \leq q \quad (4)$$

for every (ξ, y) : apply (1) to y and $y + tv$, divide by $|t|$, and let $t \rightarrow 0$. Therefore

$$I - D_y \Phi(\xi, y)$$

is invertible, with the norm-convergent Neumann series

$$(I - D_y \Phi)^{-1} = \sum_{n=0}^{\infty} (D_y \Phi)^n, \quad \|(I - D_y \Phi)^{-1}\| \leq \frac{1}{1 - q}. \quad (5)$$

Step 4: C^r dependence on the parameter. Define

$$F : U \times Y \rightarrow Y, \quad F(\xi, y) := y - \Phi(\xi, y).$$

Then F is C^r , equation (2) is precisely $F(\xi, y_*(\xi)) = 0$, and

$$D_y F(\xi, y_*(\xi)) = I - D_y \Phi(\xi, y_*(\xi))$$

is invertible by (5). The Banach-space implicit-function theorem thus shows, near every $\xi_0 \in U$, that the zero set of F is the graph of a C^r map of the parameter. That local graph must equal y_* , because for each parameter the fixed point is unique. These local representations agree on overlaps, so $y_* : U \rightarrow Y$ is globally C^r .

Differentiating (2) gives the useful first-derivative identity

$$Dy_*(\xi) = (I - D_y \Phi(\xi, y_*(\xi)))^{-1} D_\xi \Phi(\xi, y_*(\xi)), \quad (6)$$

and therefore

$$\|Dy_*(\xi)\| \leq \frac{1}{1 - q} \|D_\xi \Phi(\xi, y_*(\xi))\|. \quad (7)$$

Higher C^r regularity is exactly what the implicit-function theorem provides after the invertibility check (5). $\square$

Theorem F10.3. Local stable manifold theorem for a hyperbolic fixed point

Statement. Let $F : M \rightarrow M$ be a C^r diffeomorphism, $r \geq 1$, and let p be a hyperbolic fixed point. Write $T_p M = E^s \oplus E^u$. Then there are a chart centered at p , a number $b \in (0, 1)$, a neighborhood U of p , and a unique germ of a C^r embedded disk $W_{\text{loc}}^s(p)$ through p such that $T_p W_{\text{loc}}^s(p) = E^s$, $F(W_{\text{loc}}^s(p)) \cap U \subset W_{\text{loc}}^s(p)$, and points on the disk converge to p at rate $O(b^n)$. More precisely, after shrinking U , the disk equals the b -exponentially convergent local stable set

$$\{x \in U : F^n(x) \in U \ \forall n \geq 0, \sup_{n \geq 0} b^{-n} d(F^n(x), p) < \infty\}.$$

In adapted local coordinates it is the graph of a C^r map $h : E^s \supset B_\delta(0) \rightarrow E^u$ with $h(0) = 0$ and $Dh(0) = 0$, and there is $C \geq 1$ such that $d(F^n(x), p) \leq Cb^n d(x, p)$ on a smaller stable disk. The local unstable statement follows by applying the theorem to F^{-1} .

Proof. We prove the stable statement. The unstable statement then follows by apply-

ing what we prove to F^{-1} , whose stable space at p is E^u .

Step 1: local coordinates and an adapted norm. Choose a smooth chart sending p to 0. In these coordinates write

$$F(z) = Lz + R(z), \quad L = DF(0), \quad R(0) = 0, \quad DR(0) = 0. \quad (1)$$

Hyperbolicity gives the L -invariant splitting

$$\mathbb{R}^m = E^s \oplus E^u, \quad L = L_s \oplus L_u, \quad (2)$$

where the spectrum of L_s lies strictly inside and the spectrum of L_u strictly outside the unit circle. By *Adapted norm for a hyperbolic splitting*, after replacing the norm by an equivalent one we may choose numbers

$$0 < a < b < 1 \quad (3)$$

so that

$$\|L_s\| \leq a, \quad \|L_u^{-1}\| \leq a. \quad (4)$$

We use the maximum norm associated with the splitting in (2).

Because $DR(0) = 0$, the nonlinear remainder is as Lipschitz-small as we wish on a sufficiently small ball. It is convenient to remove domain issues in the forthcoming infinite series. Choose a smooth cutoff which is 1 on a small ball about 0 and 0 outside a slightly larger ball, and replace R by a compactly supported C^r map $\tilde{R}$ which agrees with R near 0 and satisfies

$$\tilde{R}(0) = 0, \quad \text{Lip}(\tilde{R}) \leq \eta. \quad (5)$$

Here $\eta > 0$ can be chosen arbitrarily small: on a ball of radius ρ , $\sup \|DR\| \rightarrow 0$ as $\rho \rightarrow 0$, while $\sup_{\|z\| \leq \rho} \|R(z)\|/\rho \rightarrow 0$; these two facts control respectively the derivative of R and the derivative of the cutoff. We choose η so that

$$q := \eta \max \left\{ \frac{1}{b-a}, \frac{a}{1-ab} \right\} < 1. \quad (6)$$

Set

$$\tilde{F}(z) := Lz + \tilde{R}(z). \quad (7)$$

Near 0, $\tilde{F} = F$.

Step 2: the weighted sequence space and the Lyapunov-Perron map. Let $\mathcal{Y}_b$ be the Banach space of sequences $z = (z_n)_{n \geq 0}$, $z_n = (x_n, y_n) \in E^s \oplus E^u$, for which

$$\|z\|_b := \sup_{n \geq 0} b^{-n} \|z_n\| < \infty. \quad (8)$$

Thus membership in $\mathcal{Y}_b$ means exponential decay at least as fast as b^n .

For a stable parameter $\xi \in E^s$, define $\mathcal{T}_\xi : \mathcal{Y}_b \rightarrow \mathcal{Y}_b$ by

$$(\mathcal{T}_\xi z)_n^s = L_s^n \xi + \sum_{j=0}^{n-1} L_s^{n-1-j} \tilde{R}_s(z_j), \quad (9)$$

and

$$(\mathcal{T}_\xi z)_n^u = - \sum_{j=n}^{\infty} L_u^{-(j-n+1)} \tilde{R}_u(z_j). \quad (10)$$

The empty sum in (9) is 0. The series in (10) converges absolutely because $\|L_u^{-k}\| \leq a^k$ and $\|z_j\| \leq b^j \|z\|_b$.

We now verify the contraction estimate rather than hiding it in the phrase “standard Lyapunov-Perron argument.” From (4)–(5),

$$\begin{aligned} b^{-n} \|(\mathcal{T}_\xi z - \mathcal{T}_\xi w)_n^s\| &\leq \eta \|z - w\|_b b^{-n} \sum_{j=0}^{n-1} a^{n-1-j} b^j \\ &\leq \frac{\eta}{b-a} \|z - w\|_b, \end{aligned} \quad (11)$$

while

$$\begin{aligned} b^{-n} \|(\mathcal{T}_\xi z - \mathcal{T}_\xi w)_n^u\| &\leq \eta \|z - w\|_b b^{-n} \sum_{j=n}^{\infty} a^{j-n+1} b^j \\ &= \frac{\eta a}{1-ab} \|z - w\|_b. \end{aligned} \quad (12)$$

Hence, by (6),

$$\|\mathcal{T}_\xi z - \mathcal{T}_\xi w\|_b \leq q \|z - w\|_b. \quad (13)$$

Also

$$\|\mathcal{T}_\xi 0\|_b = \sup_{n \geq 0} b^{-n} \|L_s^n \xi\| \leq \|\xi\|, \quad (14)$$

so Banach’s fixed-point theorem gives a unique sequence $z(\xi) = (z_n(\xi))_{n \geq 0} \in \mathcal{Y}_b$ with

$$\mathcal{T}_\xi z(\xi) = z(\xi), \quad \|z(\xi)\|_b \leq \frac{1}{1-q} \|\xi\|. \quad (15)$$

Step 3: a fixed sequence is an actual orbit. Write $z_n = (x_n, y_n)$. Subtracting the formula (9) at successive indices gives

$$x_{n+1} = L_s x_n + \tilde{R}_s(z_n). \quad (16)$$

Similarly, multiplying (10) by L_u and separating its first term gives

$$y_{n+1} = L_u y_n + \tilde{R}_u(z_n). \quad (17)$$

Thus

$$z_{n+1} = \tilde{F}(z_n) \quad (n \geq 0). \quad (18)$$

At $n = 0$, (9) says $x_0 = \xi$. Define

$$h(\xi) := y_0(\xi). \quad (19)$$

Then

$$z_0(\xi) = (\xi, h(\xi)), \quad (20)$$

and (15), (18) show that the orbit of this point converges to 0 at the uniform rate b^n .

Step 4: C^r dependence and tangency. We record why smoothness does not disappear inside the sequence-space construction. Because $\tilde{R}$ is C^r with bounded derivatives, the substitution map

$$z = (z_n) \mapsto (\tilde{R}(z_n))_{n \geq 0}$$

is C^r on $\mathcal{Y}_b$: its k th derivative is obtained coordinatewise, and the factor b^{-n} in (8) is compensated by the k input factors b^n ; for $k \geq 1$, $b^{(k-1)n} \leq 1$. The summation operators in (9)–(10) are bounded linear maps by (11)–(12). Hence $(\xi, z) \mapsto \mathcal{T}_\xi z$ is C^r and has the uniform contraction constant $q < 1$. *Parameter-dependent contraction theorem* therefore implies that

$$\xi \mapsto z(\xi) \in \mathcal{Y}_b \quad (21)$$

is C^r . Evaluation at $n = 0$ is continuous linear, so h is C^r .

At $\xi = 0$, uniqueness gives $z(0) = 0$, hence $h(0) = 0$. Differentiate the fixed-point equations at $\xi = 0$. Since $D\tilde{R}(0) = DR(0) = 0$, the derivative in a direction $v \in E^s$ satisfies

$$(Dz(0)v)_n^s = L_s^n v, \quad (Dz(0)v)_n^u = 0. \quad (22)$$

In particular

$$Dh(0)v = 0. \quad (23)$$

Thus the graph of h is tangent to E^s at the origin.

Step 5: return from the cutoff map to the original diffeomorphism. Let $B_\rho(0)$ be a ball on which $\tilde{F} = F$. By (15), if $\|\xi\| < (1 - q)\rho$, then every term of the fixed sequence lies in $B_\rho(0)$. For such ξ , (18) is therefore an orbit of the original map F , not merely of the cutoff map. After reducing the parameter radius once more, the graph

$$W_{\text{loc}}^s(0) := \{(\xi, h(\xi)) : \|\xi\| < \delta\} \quad (24)$$

is a C^r embedded disk.

Its forward invariance is encoded by the sequence itself. Indeed, the tail $(z_{n+1}(\xi))_{n \geq 0}$ is again a b -decaying orbit. If $\xi_1 = P_s z_1(\xi)$, uniqueness of the fixed sequence with stable initial coordinate ξ_1 gives

$$z_{n+1}(\xi) = z_n(\xi_1). \quad (25)$$

Consequently, whenever both sides remain in the chosen local chart, $F(\xi, h(\xi)) = (\xi_1, h(\xi_1))$.

Moreover, (15) gives

$$\|F^n(\xi, h(\xi))\| = \|z_n(\xi)\| \leq \frac{b^n}{1 - q} \|\xi\| \leq \frac{b^n}{1 - q} \|(\xi, h(\xi))\|. \quad (26)$$

Equivalence of the adapted coordinate norm with the Riemannian metric turns (26) into

$$d(F^n(x), p) \leq C b^n d(x, p) \quad (27)$$

on a smaller stable disk, for a uniform constant $C \geq 1$.

Step 6: characterization by exponential convergence. It remains to prove the converse, which is the step that identifies the graph with the local stable set rather than

merely constructing an invariant disk. Let z_0 be sufficiently close to 0, suppose all iterates $z_n = F^n(z_0)$ stay in the cutoff-equality neighborhood, and assume

$$\sup_{n \geq 0} b^{-n} \|z_n\| < \infty. \quad (28)$$

Then $z = (z_n)$ belongs to $\mathcal{Y}_b$. Writing the stable recurrence forward gives exactly (9) with $\xi = P_s z_0$.

For the unstable coordinate, iterate the recurrence $y_{j+1} = L_u y_j + \tilde{R}_u(z_j)$ from n to N and solve backward:

$$y_n = L_u^{-(N+1-n)} y_{N+1} - \sum_{j=n}^N L_u^{-(j-n+1)} \tilde{R}_u(z_j). \quad (29)$$

If $M = \|z\|_b$, then the first term is bounded by

$$\|L_u^{-(N+1-n)} y_{N+1}\| \leq a^{N+1-n} M b^{N+1}, \quad (30)$$

which tends to 0 as $N \rightarrow \infty$ because $ab < 1$. Letting $N \rightarrow \infty$ in (29) yields exactly (10). Hence z is a fixed point of $\mathcal{T}_{P_s z_0}$. By uniqueness of that fixed point,

$$z_0 = (P_s z_0, h(P_s z_0)) \in W_{\text{loc}}^s(0). \quad (31)$$

This proves equality with the b -exponentially convergent local stable set in the statement.

Finally, if another local C^r stable disk through 0 has the same stated invariance and exponential-convergence property, every sufficiently small point on it satisfies (28), so (31) places it on the graph (24). Since both are embedded disks of dimension $\dim E^s$ through 0, their germs agree. Thus the local stable disk is unique in the stated sense. Returning through the coordinate chart proves the theorem at p . $\square$

Theorem F10.4. Stable manifold theorem for hyperbolic sets

Statement. Let $f : M \rightarrow M$ be a C^r diffeomorphism, $r \geq 1$, and let Λ be a compact hyperbolic invariant set with splitting $T_\Lambda M = E^s \oplus E^u$. There exist $\varepsilon > 0$, $C \geq 1$, and $0 < b < 1$ such that for every $x \in \Lambda$ there are C^r embedded disks $W_\varepsilon^s(x)$ and $W_\varepsilon^u(x)$ through x , tangent respectively to E_x^s and E_x^u , depending continuously on x in local C^1 graph coordinates, with

$$f(W_\varepsilon^s(x)) \cap B(fx, \varepsilon) \subset W_\varepsilon^s(fx), \quad f^{-1}(W_\varepsilon^u(x)) \cap B(f^{-1}x, \varepsilon) \subset W_\varepsilon^u(f^{-1}x),$$

and

$$d(f^n y, f^n x) \leq C b^n d(y, x) \quad (y \in W_\varepsilon^s(x), n \geq 0),$$

with the time-reversed estimate on $W_\varepsilon^u(x)$. After shrinking ε , $W_\varepsilon^s(x)$ is exactly the set of points whose forward orbit stays ε -close to the orbit of x for all $n \geq 0$; analogously for $W_\varepsilon^u(x)$ in negative time. All constants are uniform in $x \in \Lambda$.

Proof. Let the hyperbolicity constants in some background Riemannian norm be $C_0 \geq 1$

and $0 < \lambda_0 < 1$:

$$\|Df_x^n v^s\| \leq C_0 \lambda_0^n \|v^s\|, \quad \|Df_x^{-n} v^u\| \leq C_0 \lambda_0^n \|v^u\|. \quad (1)$$

We give the stable construction. The unstable one is the same argument for f^{-1} .

Step 1: a uniform adapted norm along the hyperbolic set. Choose

$$\lambda_0 < a < 1. \quad (2)$$

For $v^s \in E_x^s$ and $v^u \in E_x^u$ define

$$|v^s|_{s,x} = \sum_{n=0}^{\infty} a^{-n} \|Df_x^n v^s\|, \quad |v^u|_{u,x} = \sum_{n=0}^{\infty} a^{-n} \|Df_x^{-n} v^u\|. \quad (3)$$

The series converge uniformly on unit vectors by (1), because $\lambda_0/a < 1$. They therefore define continuous norms on the two bundles, uniformly equivalent to the Riemannian norm. Put

$$|v^s + v^u|_x = \max\{|v^s|_{s,x}, |v^u|_{u,x}\}. \quad (4)$$

A direct index shift in (3) gives

$$|Df_x v^s|_{fx} \leq a |v^s|_x, \quad |Df_x^{-1} v^u|_{f^{-1}x} \leq a |v^u|_x. \quad (5)$$

Thus the one-step linear dynamics has the same contraction constant a at every point of Λ .

Step 2: uniform nonlinear charts along an orbit. Fix $x \in \Lambda$ and write $x_n = f^n x$. Using the exponential map, or any uniformly chosen finite atlas around the compact set Λ , define for z small in $T_{x_n} M$

$$F_n(z) = \exp_{x_{n+1}}^{-1}(f(\exp_{x_n} z)). \quad (6)$$

Then

$$F_n(0) = 0, \quad DF_n(0) = L_n := Df_{x_n}, \quad (7)$$

and, writing

$$F_n(z) = L_n z + R_n(z), \quad (8)$$

we have $R_n(0) = 0$ and $DR_n(0) = 0$.

Compactness of Λ and uniform continuity of Df imply the following uniform statement: for every $\eta > 0$ there is a radius $\rho > 0$, independent of x and n , such that on the ρ -ball

$$|R_n(z) - R_n(z')|_{x_{n+1}} \leq \eta |z - z'|_{x_n}. \quad (9)$$

The same charts and estimates work for f^{-1} after decreasing ρ . Let P_n^s, P_n^u denote the splitting projections at x_n . Since the adapted norm is the max norm relative to the splitting, these projections have norm one.

Choose a number

$$a < b < 1. \quad (10)$$

We shall take η so small that both the weighted and unweighted Lyapunov-Perron operators below are contractions.

Step 3: the nonautonomous Lyapunov-Perron equations. For $0 \leq i < j$ write

$$L_{i,j}^s = L_{j-1}|_{E_{x_{j-1}}^s} \cdots L_i|_{E_{x_i}^s}, \quad (11)$$

and let $L_{i,j}^u$ denote the analogous unstable product. From (5),

$$\|L_{i,j}^s\| \leq a^{j-i}, \quad \|(L_{i,j}^u)^{-1}\| \leq a^{j-i}. \quad (12)$$

Fix a small stable initial vector $\xi \in E_x^s$. On sequences $z = (z_n)_{n \geq 0}$, $z_n \in T_{x_n}M$, use the weighted norm

$$\|z\|_b = \sup_{n \geq 0} b^{-n} |z_n|_{x_n}. \quad (13)$$

The orbit equation $z_{n+1} = F_n(z_n)$ together with the requirement of forward decay is equivalent to

$$\begin{aligned} (\Phi_{x,\xi} z)_n^s &= L_{0,n}^s \xi + \sum_{j=0}^{n-1} L_{j+1,n}^s P_{j+1}^s R_j(z_j), \\ (\Phi_{x,\xi} z)_n^u &= - \sum_{j=n}^{\infty} (L_{n,j+1}^u)^{-1} P_{j+1}^u R_j(z_j). \end{aligned} \quad (14)$$

Here the empty sum is zero. The first equation is forward variation of constants. The second is obtained by solving the unstable recurrence backwards from the condition that the unstable component cannot grow at positive time.

Let z, z' lie in a small weighted ball on which (9) holds. Using (12),

$$\begin{aligned} b^{-n} |(\Phi z - \Phi z')_n^s| &\leq \eta \|z - z'\|_b b^{-n} \sum_{j=0}^{n-1} a^{n-1-j} b^j \leq \frac{\eta}{b-a} \|z - z'\|_b, \\ b^{-n} |(\Phi z - \Phi z')_n^u| &\leq \eta \|z - z'\|_b b^{-n} \sum_{j=n}^{\infty} a^{j+1-n} b^j = \frac{a\eta}{1-ab} \|z - z'\|_b. \end{aligned} \quad (15)$$

Hence

$$\|\Phi z - \Phi z'\|_b \leq Q_b \eta \|z - z'\|_b, \quad Q_b := \frac{1}{b-a} + \frac{a}{1-ab}. \quad (16)$$

Choose η so that $Q_b \eta < 1/2$. For $|\xi|$ sufficiently small, $\Phi_{x,\xi}$ also maps a fixed small weighted ball into itself. Banach's fixed-point theorem gives a unique fixed sequence

$$z(x, \xi) = (z_n(x, \xi))_{n \geq 0} \quad (17)$$

and the same estimate gives

$$\|z(x, \xi)\|_b \leq C_1 |\xi|_x \quad (18)$$

with C_1 independent of x .

Define

$$h_x(\xi) = P_0^u z_0(x, \xi). \quad (19)$$

Because the stable equation at $n = 0$ gives $P_0^s z_0 = \xi$, the set

$$\mathcal{W}_x^s = \{\xi + h_x(\xi) : |\xi| < \delta\} \subset T_x M \quad (20)$$

is a graph over E_x^s . The fixed-point equations imply that its orbit in the moving charts is exactly the sequence (17). In particular, (18) yields uniform exponential convergence at rate b .

Step 4: smoothness, tangency, and continuous dependence on the base point. The maps $\Phi_{x,\xi}$ are C^r in the finite-dimensional variable ξ and in the sequence variable on the small ball, with contraction constant strictly below one uniformly in x . In local trivializations of $E^s \oplus E^u$, their dependence on $x \in \Lambda$ is continuous in the C^1 operator topology. *Parameter-dependent contraction theorem*, applied in each such trivialization, shows that the fixed point is C^r in ξ and continuous in x in local C^1 graph coordinates. Uniqueness makes the local constructions agree on overlapping trivializations.

At $\xi = 0$ the fixed point is the zero sequence. Differentiating (14) at $(\xi, z) = (0, 0)$, where $DR_n(0) = 0$, gives zero unstable derivative at time zero. Thus

$$h_x(0) = 0, \quad Dh_x(0) = 0, \quad (21)$$

so the graph is tangent to E_x^s .

Returning by $\exp_x$, and decreasing δ uniformly if necessary, defines a C^r embedded disk

$$W_\varepsilon^s(x) = \exp_x(\mathcal{W}_x^s) \quad (22)$$

for a uniform $\varepsilon > 0$.

Step 5: local invariance and the uniform contraction estimate. If $y = \exp_x z_0(x, \xi)$, the fixed sequence satisfies $z_{n+1} = F_n(z_n)$. Therefore its tail $(z_{n+1})_{n \geq 0}$ is the unique fixed sequence associated with the base point fx and stable coordinate $P_1^s z_1$. Consequently, whenever the image remains in the chosen local chart,

$$f(W_\varepsilon^s(x)) \cap B(fx, \varepsilon) \subset W_\varepsilon^s(fx). \quad (23)$$

By (18) and uniform bi-Lipschitz equivalence of the charts and adapted norms, there is $C \geq 1$, independent of x , such that

$$d(f^n y, f^n x) \leq C b^n d(y, x) \quad (24)$$

for y on a slightly smaller local stable disk and every $n \geq 0$.

Step 6: exact characterization by staying close. It remains to justify a point that is often compressed in statements of the stable-manifold theorem: a point whose forward orbit merely stays in the small moving neighborhoods must already lie on the graph above.

Choose η even smaller, if necessary, so that

$$Q_\infty \eta < 1, \quad Q_\infty := \frac{1}{1-a} + \frac{a}{1-a}. \quad (25)$$

The same formulas (14), now on the Banach space of bounded sequences with norm $\|z\|_\infty = \sup_n |z_n|$, define a contraction with constant $Q_\infty \eta$.

Suppose y is so close to x that

$$d(f^n y, f^n x) < \varepsilon \quad (n \geq 0). \quad (26)$$

In moving coordinates let z_n represent $f^n y$. The sequence is bounded. Forward variation of constants gives the stable equation in (14). For the unstable coordinate, solve the recurrence from time n to a terminal time $m > n$:

$$P_n^u z_n = (L_{n,m}^u)^{-1} P_m^u z_m - \sum_{j=n}^{m-1} (L_{n,j+1}^u)^{-1} P_{j+1}^u R_j(z_j). \quad (27)$$

By (12) and boundedness of z_m , the first term tends to zero as $m \rightarrow \infty$. Hence the bounded orbit satisfies the full equations (14). For its stable initial coordinate $\xi = P_0^s z_0$, it is therefore a fixed point of the unweighted Lyapunov-Perron contraction. The weighted fixed point constructed in Step 3 is also bounded and is a fixed point of the same operator. Uniqueness in the bounded-sequence space forces the two to coincide. Thus $y \in W_\varepsilon^s(x)$, and (24) follows automatically.

This proves the asserted local-stable-set characterization.

Step 7: unstable disks. Apply Steps 1-6 to f^{-1} and the unstable bundle E^u . The constants may be replaced by the worse of the stable and unstable constants, still uniform on the compact set. We obtain $W_\varepsilon^u(x)$, tangent to E_x^u , with the stated negative-time invariance, exponential estimate, continuous dependence, and local-orbit characterization.

All conclusions follow. $\square$

Theorem F10.5. Local product structure

Statement. Let Λ be a compact hyperbolic set of a C^1 diffeomorphism f . For sufficiently small local-manifold radius $\varepsilon > 0$ there exist $\delta > 0$ and $K \geq 1$ such that whenever $x, y \in \Lambda$ and $d(x, y) < \delta$, the disks $W_\varepsilon^u(x)$ and $W_\varepsilon^s(y)$ meet in a unique point

$$[x, y] := W_\varepsilon^u(x) \cap W_\varepsilon^s(y),$$

the intersection is uniformly transverse, $d([x, y], x) + d([x, y], y) \leq Kd(x, y)$, and $(x, y) \mapsto [x, y]$ is continuous. For a general hyperbolic set the bracket is an ambient point of M and need not belong to Λ . If Λ is locally maximal, then $[x, y] \in \Lambda$ for all sufficiently close $x, y \in \Lambda$, so Λ has local product structure as an invariant set.

Proof. Fix the local stable and unstable plaques supplied by *Stable manifold theorem for hyperbolic sets*. Because the splitting

$$T_x M = E_x^u \oplus E_x^s, \quad x \in \Lambda, \quad (1)$$

is continuous and Λ is compact, the angle between the two bundles is bounded below by a positive number. We use this uniform transversality to put all nearby plaques into graph form with uniform constants.

Step 1: uniform splitting charts and small graph slopes. Choose $0 < \kappa < 1/4$. By com-

pactness, the continuous C^1 dependence of the plaques in *Stable manifold theorem for hyperbolic sets*, and their tangency to E^u and E^s , there is a radius $r > 0$ with the following property. For every $x \in \Lambda$ we may choose a chart

$$\chi_x : B_r(0) \subset E_x^u \oplus E_x^s \longrightarrow M, \quad \chi_x(0) = x, \quad (2)$$

whose differential and inverse differential are uniformly bounded, and in which

$$\chi_x^{-1}(W_\varepsilon^u(x)) = \{(u, \beta_x(u)) : |u| < r\}, \quad (3)$$

with

$$\beta_x(0) = 0, \quad \|D\beta_x\| \leq \kappa. \quad (4)$$

If $y \in \Lambda$ is sufficiently close to x , the stable plaque through y , restricted to the same chart, is a graph over the E_x^s direction:

$$\chi_x^{-1}(W_\varepsilon^s(y)) = \{(\alpha_{x,y}(s), s) : s \in V_{x,y}\}, \quad (5)$$

where $V_{x,y}$ contains a fixed small ball around the stable coordinate s_y of $\chi_x^{-1}(y) = (u_y, s_y)$ and

$$\|D\alpha_{x,y}\| \leq \kappa. \quad (6)$$

For y sufficiently close to x , the same domain also contains $s = 0$. Since the stable graph passes through y , we have $\alpha_{x,y}(s_y) = u_y$. Uniform bi-Lipschitz control of the charts, together with (6), therefore gives

$$|\alpha_{x,y}(0)| + |s_y| \leq |u_y| + \kappa|s_y| + |s_y| \leq C_0 d(x, y). \quad (7)$$

All constants are uniform in x .

The graph descriptions (3)–(6) are the precise local form of the geometric fact that the plaques are C^1 and remain uniformly transverse.

Step 2: solve the intersection equation. A point belongs to both graphs exactly when

$$s = \beta_x(u), \quad u = \alpha_{x,y}(s). \quad (8)$$

Eliminating s gives the fixed-point equation

$$u = \Gamma_{x,y}(u) := \alpha_{x,y}(\beta_x(u)). \quad (9)$$

By (4) and (6),

$$\|D\Gamma_{x,y}\| \leq \kappa^2 < 1. \quad (10)$$

If $d(x, y)$ is small enough, (7) ensures that $\Gamma_{x,y}$ maps a fixed small ball in E_x^u into itself. Hence Banach's fixed-point theorem gives a unique solution $u_* = u_*(x, y)$. Put

$$s_* := \beta_x(u_*), \quad [x, y] := \chi_x(u_*, s_*). \quad (11)$$

Then

$$[x, y] \in W_\varepsilon^u(x) \cap W_\varepsilon^s(y). \quad (12)$$

The same contraction argument proves uniqueness inside the chosen local plaques.

Step 3: quantitative location of the bracket. From (9),

$$|u_*| \leq |\Gamma_{x,y}(0)| + \kappa^2 |u_*|. \quad (13)$$

Thus

$$|u_*| \leq \frac{1}{1 - \kappa^2} |\alpha_{x,y}(0)| \leq C_1 d(x, y). \quad (14)$$

By (4), $|s_*| \leq \kappa |u_*|$. The chart is uniformly bi-Lipschitz, so

$$d([x, y], x) \leq C_2 d(x, y). \quad (15)$$

Applying the same estimate with the stable graph centered at y , or simply using the triangle inequality together with (15), gives

$$d([x, y], x) + d([x, y], y) \leq K d(x, y) \quad (16)$$

for a uniform K .

Step 4: uniform transversality at the intersection. At (u_*, s_*) the tangent space of the unstable graph consists of vectors

$$(\dot{u}, D\beta_x(u_*)\dot{u}), \quad (17)$$

while the tangent space of the stable graph consists of

$$(D\alpha_{x,y}(s_*)\dot{s}, \dot{s}). \quad (18)$$

If a nonzero vector belonged to both tangent spaces, then

$$\dot{u} = D\alpha_{x,y}(s_*)D\beta_x(u_*)\dot{u}. \quad (19)$$

The operator on the right has norm at most $\kappa^2 < 1$, so (19) forces $\dot{u} = 0$, a contradiction. The same estimate gives a uniform lower bound for the angle of intersection. Thus the bracket intersection is uniformly transverse.

Step 5: continuity of the bracket map. The plaque family is continuous in local C^1 graph coordinates by *Stable manifold theorem for hyperbolic sets*. Hence $(x, y, u) \mapsto \Gamma_{x,y}(u)$ is continuous, with a contraction constant bounded by the same number $\kappa^2 < 1$. *Parameter-dependent contraction theorem*, in its continuous-parameter form, shows that $u_*(x, y)$ depends continuously on (x, y) . Equations (11) then imply that

$$(x, y) \longmapsto [x, y] \quad (20)$$

is continuous wherever $d(x, y) < \delta$.

Step 6: why the bracket need not lie in an arbitrary hyperbolic set. Up to this point we used only hyperbolicity of Λ . Therefore the point $[x, y]$ is an ambient point of M . Nothing in the preceding argument forces it to belong to Λ : an arbitrary invariant subset of a larger hyperbolic set may omit such a bracket point. This is why local

maximality must be separated from the ambient intersection theorem.

Assume now that Λ is locally maximal. Thus there is an open neighborhood U of Λ such that

$$\Lambda = \bigcap_{n \in \mathbb{Z}} f^n(U). \quad (21)$$

Shrink the plaque radius and δ so that the fixed neighborhood in which local stable/unstable orbits stay close to Λ is contained in U .

Let $z = [x, y]$. Since $z \in W_\varepsilon^s(y)$,

$$d(f^n z, f^n y) \longrightarrow 0 \quad (n \rightarrow +\infty), \quad (22)$$

and the entire local forward orbit remains in the chosen neighborhood of the orbit of $y \in \Lambda$. Hence $f^n z \in U$ for all $n \geq 0$. Similarly, $z \in W_\varepsilon^u(x)$ implies $f^n z \in U$ for every $n \leq 0$. Therefore

$$f^n z \in U \quad \text{for every } n \in \mathbb{Z}, \quad (23)$$

and (21) gives $z \in \Lambda$.

Thus locally maximal hyperbolic sets have local product structure as invariant sets, while every compact hyperbolic set has the ambient bracket construction proved above. $\square$

Proposition F10.6. Markov rectangles for basic hyperbolic sets: interface

Statement. Let Λ be a compact hyperbolic set of a C^1 diffeomorphism and assume that Λ has local product structure, so sufficiently close $y, z \in \Lambda$ have a bracket $[y, z] = W_{\text{loc}}^u(y) \cap W_{\text{loc}}^s(z)$ which belongs to Λ . For every sufficiently small $\delta > 0$ and every $x \in \Lambda$ there is a relative neighborhood $R_x \subset \Lambda$ of x , of diameter $< \delta$, such that $y, z \in R_x$ implies $[y, z] \in R_x$. In local product coordinates R_x is the image of a product of a stable slice and an unstable slice. Compactness gives a finite cover of Λ by such rectangles.

Scope note

This proposition gives local rectangles and a finite rectangle cover, not the Markov refinement/transition property of a Markov partition.

Proof. Fix $x \in \Lambda$. By the local product hypothesis and the continuity of the bracket map from *Local product structure*, there is a relative neighborhood $V_x \subset \Lambda$ of x in which every bracket used below is defined and belongs to Λ .

Step 1: stable and unstable slices through x . Set

$$S_x = V_x \cap W_{\text{loc}}^s(x), \quad U_x = V_x \cap W_{\text{loc}}^u(x). \quad (1)$$

For $y \in V_x$ define

$$s_x(y) := [y, x] \in S_x, \quad u_x(y) := [x, y] \in U_x. \quad (2)$$

Indeed, by the bracket convention,

$$[y, x] \in W_{\text{loc}}^u(y) \cap W_{\text{loc}}^s(x), \quad [x, y] \in W_{\text{loc}}^u(x) \cap W_{\text{loc}}^s(y). \quad (3)$$

Both maps in (2) are continuous because the bracket map is continuous.

Step 2: the bracket gives product coordinates. For $s \in S_x$ and $u \in U_x$ sufficiently close to x , define

$$\Phi_x(s, u) = [s, u]. \quad (4)$$

The point $[s, u]$ lies in Λ by local product structure. We claim that, after shrinking V_x , the map

$$\Psi_x : V_x \longrightarrow S_x \times U_x, \quad \Psi_x(y) = (s_x(y), u_x(y)), \quad (5)$$

is inverse to Φ_x .

Take $y \in V_x$ and put

$$s = [y, x], \quad u = [x, y]. \quad (6)$$

Because $s \in W_{\text{loc}}^u(y)$, the local unstable plaque through s is the same local unstable leaf germ as the one through y . Likewise, $u \in W_{\text{loc}}^s(y)$, so the local stable plaque through u is the same stable leaf germ as that through y . Therefore

$$[s, u] = W_{\text{loc}}^u(s) \cap W_{\text{loc}}^s(u) = W_{\text{loc}}^u(y) \cap W_{\text{loc}}^s(y) = y. \quad (7)$$

Conversely, if $s \in S_x$ and $u \in U_x$ and $y = [s, u]$, then $y \in W_{\text{loc}}^u(s)$ and $s \in W_{\text{loc}}^s(x)$. Hence $[y, x] = s$ by uniqueness of the local bracket. Similarly $[x, y] = u$. Thus

$$\Psi_x(\Phi_x(s, u)) = (s, u). \quad (8)$$

So Φ_x and Ψ_x are mutually inverse continuous maps on suitably small neighborhoods. In particular, a sufficiently small relative neighborhood of x in Λ has genuine product coordinates.

Step 3: choose a smaller product box. Let $S'_x \subset S_x$ and $U'_x \subset U_x$ be sufficiently small relative neighborhoods of x in the two slices, and define

$$R_x = \Phi_x(S'_x \times U'_x). \quad (9)$$

Because Φ_x is a homeomorphism onto its image, R_x is a relative neighborhood of x in Λ . By continuity of Φ_x , we may choose the two factors so small that

$$\text{diam } R_x < \delta. \quad (10)$$

It remains to verify the rectangle property. Write

$$y = [s_1, u_1], \quad z = [s_2, u_2], \quad (11)$$

with $s_i \in S'_x$ and $u_i \in U'_x$. The point y lies on the unstable plaque through s_1 , so its local unstable germ is that of s_1 ; the point z lies on the stable plaque through u_2 , so its local stable germ is that of u_2 . Hence

$$[y, z] = [s_1, u_2]. \quad (12)$$

Since $(s_1, u_2) \in S'_x \times U'_x$, (9) gives

$$[y, z] \in R_x. \quad (13)$$

Thus R_x is a rectangle in the precise local-product sense.

Step 4: finite cover. The sets R_x , $x \in \Lambda$, form a relative open cover of the compact set Λ . Therefore finitely many suffice:

$$\Lambda = R_{x_1} \cup \cdots \cup R_{x_m}. \quad (14)$$

Each has diameter $< \delta$ and is closed under the local bracket whenever the two bracket inputs lie in that same rectangle.

Nothing in this argument gives the additional forward/backward boundary compatibility required of a Markov partition. That refinement is a separate theorem, as stated in the scope boundary. $\square$

Theorem F10.7. Shadowing lemma

Statement. Let $f : M \rightarrow M$ be a C^1 diffeomorphism and Λ a compact hyperbolic invariant set. There exist $\delta_0 > 0$ and $K \geq 1$ such that every bi-infinite sequence $(x_n)_{n \in \mathbb{Z}} \subset \Lambda$ with $d(fx_n, x_{n+1}) \leq \delta \leq \delta_0$ is shadowed by a unique orbit $(f^n y)_{n \in \mathbb{Z}}$ in a fixed sufficiently small neighborhood of the pseudo-orbit, and

$$\sup_{n \in \mathbb{Z}} d(f^n y, x_n) \leq K\delta.$$

Uniqueness means uniqueness among orbits contained in that fixed local shadowing neighborhood. If Λ is locally maximal and the neighborhood is chosen inside an isolating neighborhood, then the shadowing point y belongs to Λ .

Proof. We prove a uniform local shadowing theorem directly from the hyperbolic splitting. The advantage of the sequence-space proof is that existence and uniqueness are obtained from the same contraction.

Step 1: adapted norms and the pseudo-orbit charts. As in the proof of *Stable manifold theorem for hyperbolic sets*, choose a continuous adapted norm on $T_\Lambda M = E^s \oplus E^u$ and a number $0 < a < 1$ such that

$$|Df_x v^s|_{f_x} \leq a|v^s|_x, \quad |Df_x^{-1} v^u|_{f^{-1}x} \leq a|v^u|_x. \quad (1)$$

All adapted norms are uniformly equivalent to the background Riemannian norm.

Let $(x_n)_{n \in \mathbb{Z}} \subset \Lambda$ satisfy

$$d(fx_n, x_{n+1}) \leq \delta. \quad (2)$$

For δ small and v in a fixed small ball of $T_{x_n} M$, define

$$F_n(v) = \exp_{x_{n+1}}^{-1}(f(\exp_{x_n} v)). \quad (3)$$

Unlike the moving charts along a true orbit, now

$$e_{n+1} := F_n(0) \quad (4)$$

need not vanish. Uniform smoothness of the charts gives

$$|e_{n+1}|_{x_{n+1}} \leq C_0 \delta. \quad (5)$$

Because $f x_n$ and x_{n+1} both lie in Λ and are $O(\delta)$ apart, continuity of the hyperbolic splitting implies that, after transport to the chart at x_{n+1} , $DF_n(0)$ is close to a map preserving the splitting $E_{x_n}^s \oplus E_{x_n}^u$ to $E_{x_{n+1}}^s \oplus E_{x_{n+1}}^u$.

Let P_n^s, P_n^u be the splitting projections and define the block-diagonal part

$$H_n = P_{n+1}^s DF_n(0) P_n^s + P_{n+1}^u DF_n(0) P_n^u. \quad (6)$$

Choose a number $a < \alpha < 1$. By taking δ small, the stable block of H_n and the inverse of its unstable block satisfy uniformly

$$\|H_n^s\| \leq \alpha, \quad \|(H_n^u)^{-1}\| \leq \alpha. \quad (7)$$

The off-diagonal part of $DF_n(0)$ tends uniformly to zero with δ . Consequently we can write

$$F_n(v) = H_n v + e_{n+1} + R_n(v), \quad (8)$$

where $R_n(0) = 0$ and, after also shrinking the chart radius r ,

$$|R_n(v) - R_n(w)|_{x_{n+1}} \leq \eta |v - w|_{x_n} \quad (9)$$

for every $|v|, |w| \leq r$, with $\eta > 0$ as small as we wish, uniformly in n and in the pseudo-orbit.

Step 2: invert the hyperbolic difference operator on bounded sequences. Let

$$\mathcal{X} = \left\{ v = (v_n)_{n \in \mathbb{Z}} : v_n \in T_{x_n} M, \quad \|v\|_\infty := \sup_n |v_n|_{x_n} < \infty \right\}. \quad (10)$$

For a bounded forcing sequence $w = (w_n)$, consider

$$v_{n+1} = H_n v_n + w_{n+1}. \quad (11)$$

Write, for $i < j$,

$$H_{i,j}^s = H_{j-1}^s \cdots H_i^s, \quad H_{i,j}^u = H_{j-1}^u \cdots H_i^u. \quad (12)$$

The unique bounded solution of (11) is

$$\begin{aligned} v_n^s &= \sum_{j=-\infty}^{n-1} H_{j+1,n}^s w_{j+1}^s, \\ v_n^u &= - \sum_{j=n}^{\infty} (H_{n,j+1}^u)^{-1} w_{j+1}^u. \end{aligned} \quad (13)$$

Indeed, the series converge absolutely by (7), direct substitution gives (11), and any bounded homogeneous solution must have zero stable component when iterated backwards and zero unstable component when iterated forwards. Thus it is zero.

Denote this Green operator by G . From (7),

$$\|Gw\|_\infty \leq Q\|w\|_\infty, \quad Q := \frac{1+\alpha}{1-\alpha}. \quad (14)$$

The displayed constant is slightly wasteful but uniform and convenient.

Step 3: solve the nonlinear orbit equation. An exact orbit shadowing the pseudo-orbit has chart coordinates $v = (v_n)$ satisfying

$$v_{n+1} = F_n(v_n). \quad (15)$$

By (8), this is equivalent to

$$v = G(e + R(v)), \quad (16)$$

where $e = (e_n)$ and $(R(v))_{n+1} = R_n(v_n)$.

Choose the uniform chart radius and δ_0 so that (9) holds with

$$Q\eta \leq \frac{1}{2} \quad (17)$$

and so that

$$2QC_0\delta_0 < r. \quad (18)$$

On the closed ball $\|v\|_\infty \leq r$, the map

$$\Psi(v) := G(e + R(v)) \quad (19)$$

has Lipschitz constant at most $Q\eta \leq 1/2$. Moreover, because $R(0) = 0$,

$$\|\Psi(0)\|_\infty \leq QC_0\delta. \quad (20)$$

Therefore, on the smaller closed ball of radius $2QC_0\delta$, Ψ maps the ball into itself and is a contraction. Banach's theorem gives a unique fixed point v and

$$\|v\|_\infty \leq 2QC_0\delta. \quad (21)$$

Define

$$y = \exp_{x_0}(v_0). \quad (22)$$

Equation (15) and induction show

$$f^n y = \exp_{x_n}(v_n) \quad (n \in \mathbb{Z}). \quad (23)$$

Uniform bi-Lipschitz control of the exponential charts and (21) give a constant K such that

$$\sup_{n \in \mathbb{Z}} d(f^n y, x_n) \leq K\delta. \quad (24)$$

This is the required shadowing estimate.

Step 4: local uniqueness. Suppose another full orbit $(f^n z)$ remains in the same fixed chart neighborhood of the pseudo-orbit. Let u_n be its chart coordinates. Then $u = (u_n)$

is bounded and satisfies the same exact orbit equations (15), hence the same fixed-point equation (16). If the local neighborhood is chosen inside the contraction ball, uniqueness of the fixed point gives

$$u = v, \quad (25)$$

and therefore $z = y$. Thus no separate expansivity theorem is needed for the local uniqueness asserted here; it is built into the hyperbolic sequence-space inverse.

Step 5: the locally maximal case. If Λ is locally maximal, choose an isolating neighborhood U with

$$\Lambda = \bigcap_{n \in \mathbb{Z}} f^n(U). \quad (26)$$

Because Λ is compact and U is open, shrink the fixed shadowing neighborhood so that it lies inside U around every point of Λ . Then (24) implies

$$f^n y \in U \quad (n \in \mathbb{Z}). \quad (27)$$

Equation (26) yields $y \in \Lambda$.

The proof is complete. $\square$

Theorem F10.8. Structural stability of Anosov diffeomorphisms

Statement. Let M be compact and let $f : M \rightarrow M$ be a C^1 Anosov diffeomorphism. For every $\eta > 0$ sufficiently small there is a C^1 neighborhood $\mathcal{U}$ of f such that every $g \in \mathcal{U}$ is Anosov and there is a homeomorphism $h_g : M \rightarrow M$ satisfying

$$h_g \circ g = f \circ h_g, \quad \sup_{x \in M} d(h_g(x), x) < \eta.$$

The conjugacy is unique among maps sufficiently C^0 -close to the identity that satisfy the same conjugacy equation. Equivalently, $h_g^{-1} \circ f = g \circ h_g^{-1}$.

Proof. We divide the proof into three logically separate ingredients: robustness of hyperbolicity, expansivity, and shadowing.

Step 1: C^1 -nearby diffeomorphisms remain Anosov. Since f is Anosov, $TM = E_f^s \oplus E_f^u$ continuously and invariantly. As in *Stable manifold theorem for hyperbolic sets*, choose a continuous adapted norm and $0 < a < 1$ such that

$$|Df_x v^s|_{f_x} \leq a |v^s|_x, \quad |Df_x^{-1} v^u|_{f^{-1}x} \leq a |v^u|_x. \quad (1)$$

Fix $0 < c < 1$. Define the unstable and stable cones

$$\begin{aligned} \mathcal{C}_x^u &= \{v = v^s + v^u : |v^s|_x \leq c |v^u|_x\}, \\ \mathcal{C}_x^s &= \{v = v^s + v^u : |v^u|_x \leq c |v^s|_x\}. \end{aligned} \quad (2)$$

For f , (1) gives strict cone invariance. For example, if $v \in \mathcal{C}_x^u$, then

$$\frac{|Df_x v^s|}{|Df_x v^u|} \leq a^2 \frac{|v^s|}{|v^u|} \leq a^2 c < c, \quad (3)$$

and the unstable component expands by at least a^{-1} .

Now let g be C^1 -close to f . Then gx is uniformly close to fx , the fixed bundles E_f^s, E_f^u at those nearby target points are close, and Dg_x is uniformly close to Df_x . Therefore, after making the C^1 neighborhood of f sufficiently small, there are numbers

$$0 < c_1 < c, \quad \mu > 1, \quad (4)$$

such that

$$Dg_x(\mathcal{C}_x^u) \subset \{w : |w^s|_{gx} \leq c_1 |w^u|_{gx}\} \subset \text{int } \mathcal{C}_{gx}^u, \quad (5)$$

and

$$|Dg_x v|_{gx} \geq \mu |v|_x \quad (v \in \mathcal{C}_x^u), \quad (6)$$

while the analogous statements hold for Dg^{-1} on the stable cones. Here the decomposition at gx is still taken with respect to the fixed f -bundles at that point; no g -invariant bundle has yet been assumed.

We now spell out the subspace construction rather than treating the intersection of cones as automatically linear. A d_u -plane contained in $\mathcal{C}_x^u$ is the graph of a linear map

$$L : E_f^u(x) \longrightarrow E_f^s(x), \quad \|L\| \leq c. \quad (7)$$

Relative to the fixed f -splittings at x and gx , write

$$Dg_x = \begin{pmatrix} A_x & B_x \\ C_x & D_x \end{pmatrix}, \quad (8)$$

where $D_x : E_f^u(x) \rightarrow E_f^u(gx)$. For $g = f$ the off-diagonal blocks vanish, $\|A_x\| \leq a$, and $m(D_x) \geq a^{-1}$. Hence, for g C^1 -close to f , $D_x + C_x L$ is invertible on the graph ball $\|L\| \leq c$, and the image plane is the graph of

$$\mathcal{G}_{g,x}^u(L) = (A_x L + B_x)(C_x L + D_x)^{-1}. \quad (9)$$

At $g = f$ the Lipschitz constant of this graph transform is at most a^2 . By uniform continuity and compactness, after shrinking the C^1 neighborhood there is a single $0 < \theta < 1$ such that every $\mathcal{G}_{g,x}^u$ maps the closed graph ball $\|L\| \leq c$ into the smaller ball $\|L\| \leq c_1$ and is θ -Lipschitz.

Starting at $g^{-n}x$ with any graph L_{-n} of norm at most c and composing the n forward graph transforms gives a graph at x . Two such choices differ by at most $2c\theta^n$. Therefore the graphs converge, uniformly and independently of the initial choice, to a limit $L_g^u(x)$. Its graph defines $E_g^u(x)$. Uniform convergence of continuous finite graph transforms gives continuity in x , and shifting the construction by one step gives

$$Dg_x E_g^u(x) = E_g^u(gx). \quad (10)$$

Applying the same construction to Dg^{-1} on the stable graph ball gives a continuous invariant bundle E_g^s . The two bundles remain uniformly transverse because they stay in the disjoint cones (2). Estimates (6) and its stable inverse counterpart give uniform exponential expansion/contraction on these invariant bundles. Thus every sufficiently C^1 -close g is Anosov.

Moreover, the graph-transform contraction, cone margins, expansion rates, and the C^1 bounds used above can all be chosen uniformly for g in a sufficiently small neighborhood of f . Inspecting the Green-operator proof of *Shadowing lemma* then shows that its local shadowing radius and expansivity scale also admit uniform positive lower bounds on this neighborhood.

Step 2: expansivity of f and of nearby g . We claim that an Anosov diffeomorphism is expansive. Choose local stable and unstable plaques so small that the ambient bracket of *Local product structure* is unique. There exists $\alpha_f > 0$ such that if

$$d(f^n x, f^n y) < \alpha_f \quad \text{for every } n \in \mathbb{Z}, \quad (9)$$

then y lies in both the local stable and local unstable plaque through x : the nonnegative inequalities put y in $W_{\text{loc}}^s(x)$ by the staying-close characterization of *Stable manifold theorem for hyperbolic sets*, and the nonpositive ones put it in $W_{\text{loc}}^u(x)$. Their unique local intersection is x , so $y = x$.

The same argument applies to each nearby Anosov g . By the uniform cone constants from Step 1 and the uniform proof of the local stable-manifold theorem, after shrinking the neighborhood of f we may choose a common positive lower bound

$$\alpha_* > 0 \quad (10)$$

for the relevant local expansivity scales of all such g and of f .

Step 3: shadow each g -orbit by an f -orbit. Let $\eta > 0$ be so small that

$$4\eta < \alpha_*. \quad (11)$$

Apply *Shadowing lemma* to f with a shadowing neighborhood smaller than η . There is $\delta_f > 0$ such that every sufficiently small bi-infinite pseudo-orbit of f is uniquely η -shadowed.

If g is uniformly C^0 -close enough to f , then for every $x \in M$ the bi-infinite g -orbit

$$x_n = g^n x \quad (12)$$

is a δ_f -pseudo-orbit for f , because

$$d(fx_n, x_{n+1}) = d(f(g^n x), g(g^n x)) \leq \|f - g\|_{C^0} < \delta_f. \quad (13)$$

Define $h_g(x)$ to be the unique point whose f -orbit shadows this pseudo-orbit:

$$d(f^n h_g(x), g^n x) < \eta \quad (n \in \mathbb{Z}). \quad (14)$$

At $n = 0$,

$$d(h_g(x), x) < \eta, \quad (15)$$

so h_g is uniformly close to the identity.

Shift the pseudo-orbit in (12) by one index. Both $f(h_g(x))$ and $h_g(gx)$ shadow the

shifted sequence $(g^{n+1}x)_{n \in \mathbb{Z}}$. Uniqueness in the shadowing theorem gives

$$f \circ h_g = h_g \circ g. \quad (16)$$

Step 4: continuity of the shadowing map. Let $x_j \rightarrow x$. Compactness of M implies that any subsequence of $h_g(x_j)$ has a convergent subsubsequence, say to z . For each fixed integer n , continuity of f^n and g^n and (14) yield

$$d(f^n z, g^n x) \leq \eta. \quad (17)$$

Thus z shadows the g -orbit of x . By local uniqueness, $z = h_g(x)$. Every convergent subsequence has this same limit; compactness then implies

$$h_g(x_j) \rightarrow h_g(x). \quad (18)$$

Hence h_g is continuous.

Step 5: construct the inverse map. Because Step 1 shows that g is Anosov, *Shadowing lemma* also applies to g . Shrinking the C^1 neighborhood of f if necessary, the f -orbit of every y is a sufficiently small pseudo-orbit for g . Define $k_g(y)$ to be its unique η -shadowing point for g :

$$d(g^n k_g(y), f^n y) < \eta \quad (n \in \mathbb{Z}). \quad (19)$$

Exactly as above, k_g is continuous and

$$g \circ k_g = k_g \circ f, \quad d(k_g(y), y) < \eta. \quad (20)$$

For $x \in M$, combine (14) and (19):

$$\begin{aligned} d(g^n k_g(h_g(x)), g^n x) &\leq d(g^n k_g(h_g(x)), f^n h_g(x)) \\ &\quad + d(f^n h_g(x), g^n x) < 2\eta < \alpha_* \end{aligned} \quad (21)$$

for all $n \in \mathbb{Z}$. Expansivity of g gives

$$k_g(h_g(x)) = x. \quad (22)$$

Similarly, expansivity of f gives

$$h_g(k_g(y)) = y. \quad (23)$$

Thus h_g is a homeomorphism with inverse k_g , and (16) is the desired conjugacy equation.

Step 6: uniqueness among near-identity conjugacies. Suppose h_1, h_2 both satisfy

$$f \circ h_i = h_i \circ g \quad (24)$$

and both are C^0 -close enough to the identity that $\sup_x d(h_1(x), h_2(x)) < \alpha_f$. Then for every $n \in \mathbb{Z}$,

$$\begin{aligned} d(f^n h_1(x), f^n h_2(x)) &= d(h_1(g^n x), h_2(g^n x)) \\ &< \alpha_f. \end{aligned} \quad (25)$$

Expansivity of f gives $h_1(x) = h_2(x)$ for every x . Hence the near-identity conjugacy is unique.

This proves structural stability. $\square$

Theorem F10.9. Inclination / lambda lemma

Statement. Let $f : M \rightarrow M$ be a C^1 diffeomorphism, let p be a hyperbolic fixed point, and let D be a C^1 embedded disk of dimension $\dim E_p^u$ meeting $W^s(p)$ transversely at a point q . Let Δ be a compact disk whose closure is contained in the interior of $W_{\text{loc}}^u(p)$. Then for all sufficiently large n , $f^n(D)$ contains a C^1 subdisk D_n which, in a fixed chart straightening $W_{\text{loc}}^s(p)$ and $W_{\text{loc}}^u(p)$, is a graph over the unstable-coordinate disk corresponding to Δ , and these graphing functions converge to zero in C^1 . Equivalently, $D_n \rightarrow \Delta$ in C^1 .

Proof. Let

$$d_s = \dim E_p^s, \quad d_u = \dim E_p^u. \quad (1)$$

By *Local stable manifold theorem for a hyperbolic fixed point* applied to f and f^{-1} , the local stable and unstable manifolds are C^1 disks through p , tangent to E_p^s and E_p^u . Since they meet transversely, a C^1 coordinate change in a small neighborhood of p identifies

$$p = (0, 0), \quad W_{\text{loc}}^s(p) = \{(s, 0)\}, \quad W_{\text{loc}}^u(p) = \{(0, u)\}, \quad (2)$$

with $s \in \mathbb{R}^{d_s}$ and $u \in \mathbb{R}^{d_u}$. Write the local map as

$$F(s, u) = (F_s(s, u), F_u(s, u)). \quad (3)$$

The invariance of the coordinate axes gives

$$F_s(0, u) = 0, \quad F_u(s, 0) = 0 \quad (4)$$

whenever the indicated points stay in the local chart.

Step 1: a local graph-transform estimate near the unstable axis. Choose an adapted norm at p so that for the block derivative

$$DF(0, 0) = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \quad (5)$$

we have, for some $0 < a < 1$,

$$\|A\| \leq a, \quad \|B^{-1}\| \leq a. \quad (6)$$

Choose numbers

$$a^2 < \theta < 1, \quad 1 < \nu < a^{-1}. \quad (7)$$

By continuity of DF , after shrinking to a box

$$Q = B_R^s \times B_R^u \quad (8)$$

we obtain the following uniform graph-transform facts.

If

$$\Gamma_\psi = \{(\psi(u), u) : u \in U\} \subset Q \quad (9)$$

is a C^1 graph with sufficiently small $\|\psi\|_{C^1}$, then the portion of $F(\Gamma_\psi)$ that remains in Q is again a graph

$$\Gamma_{\mathcal{T}\psi} = \{((\mathcal{T}\psi)(u'), u') : u' \in U'\}. \quad (10)$$

There are constants $0 < \kappa < 1$ and a modulus of continuity $\omega(t) \rightarrow 0$ as $t \downarrow 0$ such that

$$\|\mathcal{T}\psi\|_{C^0} \leq \kappa\|\psi\|_{C^0}, \quad (11)$$

and

$$\|D(\mathcal{T}\psi)\|_{C^0} \leq \theta\|D\psi\|_{C^0} + \omega(\|\psi\|_{C^0}). \quad (12)$$

The unstable projection also expands domains: after shrinking Q , its restriction to every such small graph is uniformly co-Lipschitz by a factor $\nu > 1$. Consequently an unstable-coordinate ball of radius r centered at the graph base point has image containing the ball of radius νr centered at the next base point, provided the target ball stays in Q .

For completeness, the slope estimate comes from the exact derivative formula. If the tangent graph is represented by a linear map $L : \mathbb{R}^{d_u} \rightarrow \mathbb{R}^{d_s}$, then

$$L' = (F_{s,s}L + F_{s,u})(F_{u,s}L + F_{u,u})^{-1}. \quad (13)$$

At $(0, 0)$ and $L = 0$ this becomes

$$L' = ALB^{-1}, \quad (14)$$

whose norm is at most $a^2\|L\|$. Along the unstable axis, $F_s(0, u) = 0$, so the zero graph is invariant. The normal derivative along that axis remains strictly contracting in a sufficiently small box, which gives (11). Uniform continuity of DF then gives (12): the deviation of the derivative coefficients at $(\psi(u), u)$ from those at $(0, u)$ is bounded by a modulus depending on $\|\psi\|_{C^0}$. This is the point that allows a C^1 proof without assuming F is C^2 .

Finally, the unstable-coordinate derivative is uniformly close to the expanding block B . Shrinking Q once more makes the unstable-coordinate map on each small graph co-Lipschitz with factor ν . The ball-in-image claim follows from the inverse function theorem (equivalently, from the inverse branch, whose Lipschitz constant is at most ν^{-1}).

Step 2: a forward iterate of D enters the unstable graph cone. Let

$$q_m = f^m(q). \quad (14)$$

Because $q \in W^s(p)$,

$$q_m \longrightarrow p. \quad (15)$$

The transversality hypothesis

$$T_q D \oplus T_q W^s(p) = T_q M \quad (16)$$

means that the projection of $T_q D$ to the unstable quotient is an isomorphism. Therefore

a small disk $D^{(0)} \subset D$ around q is a C^1 graph over an unstable complement to $T_q W^s(p)$.

Apply Df_q^m to the tangent graph. Along the orbit $q_m \rightarrow p$, the splitting between the tangent of the stable manifold and an unstable complement becomes arbitrarily close to the hyperbolic splitting at p . Transversality is preserved by the derivative because $Df(TW^s) = TW^s$, so every advanced tangent disk is still a graph over an unstable complement. Evaluating the slope formula (13) along the stable orbit $q_m \rightarrow p$ gives a recurrence

$$\ell_{m+1} \leq \theta \ell_m + \varepsilon_m, \quad \varepsilon_m \rightarrow 0, \quad (15)$$

for the tangent slopes once the orbit is in Q . Iterating this inequality shows $\ell_m \rightarrow 0$: for a given tolerance, first make the tail ε_m small and then let the geometric factor θ^m kill the finite initial part. Hence there is m_0 such that a sufficiently small subdisk

$$\tilde{D}_{m_0} \subset f^{m_0}(D^{(0)}) \quad (17)$$

containing q_{m_0} is a graph $\Gamma_{\psi_{m_0}}$ in Q with $\|\psi_{m_0}\|_{C^1}$ inside the graph-transform neighborhood of Step 1.

This is exactly where transversality is used. If $T_q D$ contained a nonzero stable direction instead of projecting isomorphically to the unstable quotient, the forward tangent could fail to enter the unstable graph cone.

Step 3: slopes and heights converge to the unstable manifold. For $k \geq 0$, iterate the graph transform on the local component that stays in Q and write

$$\psi_k := \psi_{m_0+k}, \quad h_k := \|\psi_k\|_{C^0}, \quad \ell_k := \|D\psi_k\|_{C^0}. \quad (18)$$

Equation (11) gives

$$h_k \leq \kappa^k h_0 \rightarrow 0. \quad (19)$$

Equation (12) gives

$$\ell_{k+1} \leq \theta \ell_k + \omega(h_k). \quad (20)$$

Iterating,

$$\ell_k \leq \theta^k \ell_0 + \sum_{j=0}^{k-1} \theta^{k-1-j} \omega(h_j). \quad (21)$$

The sum tends to zero. Indeed, choose J so large that $\omega(h_j)$ is uniformly small for $j \geq J$; the finitely many terms $j < J$ are killed by the factors θ^{k-1-j} , while the tail is bounded by a geometric series times that small uniform bound. Therefore

$$\|\psi_k\|_{C^1} = h_k + \ell_k \rightarrow 0. \quad (22)$$

The graph centers are the points q_{m_0+k} , whose unstable coordinate is zero because they lie on the straightened stable axis and which converge to p . Thus both the graph position and the tangent plane converge to those of the local unstable manifold.

Step 4: the expanding domain eventually covers any fixed compact unstable disk. Let $U_\Delta \Subset B_R^u$ be the unstable-coordinate domain corresponding to the prescribed compact disk

$$\Delta = \{(0, u) : u \in U_\Delta\} \Subset W_{\text{loc}}^u(p). \quad (20)$$

At time m_0 the graph in (17) contains an unstable-coordinate ball $B_{r_0}(0)$ for some $r_0 > 0$; the center is 0 because q_{m_0} lies on the stable axis. Choose a radius $R_\Delta < R$ such that $U_\Delta \subset B_{R_\Delta}(0)$ and such that the slightly larger ball $B_{\nu R_\Delta}(0)$ still lies in B_R^u .

As long as the current radius is below R_Δ , the domain-expansion estimate increases it by at least the factor ν . Hence after finitely many iterates the local image graph contains $B_{R_\Delta}(0)$. Once this has happened, every later local image also contains $B_{R_\Delta}(0)$: its unstable projection contains the larger ball $B_{\nu R_\Delta}(0)$ before we restrict back to the fixed box. Thus for all sufficiently large k , the local component of $f^{m_0+k}(D)$ contains a graph over the whole domain U_Δ .

Restrict that component to the subdisk whose unstable projection is exactly U_Δ and call it D_{m_0+k} . Then

$$D_{m_0+k} = \{(\psi_k(u), u) : u \in U_\Delta\}. \quad (23)$$

By (22),

$$\|\psi_k\|_{C^1(U_\Delta)} \longrightarrow 0. \quad (24)$$

The zero graph over U_Δ is exactly Δ . Hence

$$D_n \longrightarrow \Delta \quad \text{in } C^1 \quad (25)$$

through the subdisks just constructed. This is the inclination lemma. $\square$

Theorem F10.10. Normally hyperbolic invariant manifold persistence

Statement. Let $f : M \rightarrow M$ be a C^r diffeomorphism, $r \geq 1$, and let $N \subset M$ be a compact C^r invariant submanifold that is r -normally hyperbolic in the sense of Definition F10.4. There are a tubular neighborhood U of N and a C^1 neighborhood $\mathcal{U}$ of f such that, for every C^r diffeomorphism $g \in \mathcal{U}$, there is a unique C^r g -invariant submanifold $N_g \subset U$ among submanifolds sufficiently C^1 -close to N that project diffeomorphically onto N along the fixed tube. That projection restricts to a C^r diffeomorphism $N_g \rightarrow N$, and $N_g \rightarrow N$ in C^1 as $g \rightarrow f$ in C^1 .

The manifold N_g is again r -normally hyperbolic after slightly weakening the strict constants, with stable and unstable bundles of the same dimensions as those of N . Through every $z \in N_g$ there are local strong stable and strong unstable C^r disks tangent to the continued bundles, locally invariant and exponentially contracting in forward and backward time respectively. These disks vary continuously with z in local C^1 graph coordinates.

Scope note

Higher transverse regularity of the strong foliations requires additional bunching hypotheses.

Proof. We prove the theorem in the graph-transform form. Besides giving persistence, this form makes visible the precise reason for the two order- r inequalities in the definition of normal hyperbolicity.

Step 1: strict constants, tubular coordinates, and room for perturbation. Along N we

have the Df -invariant splitting

$$T_N M = E^s \oplus TN \oplus E^u. \quad (1)$$

By the definition of normal hyperbolicity, after replacing the ambient norm along N by an equivalent continuous norm, there are constants

$$0 < \nu, \bar{\nu}, \gamma, \bar{\gamma} < 1, \quad \nu < \gamma^r, \quad \bar{\nu} < \bar{\gamma}^r, \quad (2)$$

such that, for every $x \in N$,

$$\|Df_x|_{E_x^s}\| \leq \nu, \quad m(Df_x|_{TN}) \geq \gamma, \quad \|Df_x|_{TN}\| \leq \bar{\gamma}^{-1}, \quad \|Df_{fx}^{-1}|_{E_{fx}^u}\| \leq \bar{\nu}. \quad (3)$$

Here $m(L) = \inf_{\|v\|=1} \|Lv\| = \|L^{-1}\|^{-1}$ for an invertible linear map. The inequalities in (2) are strict. Hence we may choose slightly weaker constants

$$\nu < \nu_0 < \gamma_0^r < \gamma^r, \quad \bar{\nu} < \bar{\nu}_0 < \bar{\gamma}_0^r < \bar{\gamma}^r \quad (4)$$

with $0 < \gamma_0 < \gamma$ and $0 < \bar{\gamma}_0 < \bar{\gamma}$. By compactness of N and continuity of Df , all cone and conorm inequalities corresponding to (3)–(4) hold on a fixed neighborhood of N and remain valid, with the same strict margins, for every C^r diffeomorphism g in a sufficiently small C^1 neighborhood of f .

Choose a C^r tubular identification of a disk bundle in

$$E := E^s \oplus E^u \longrightarrow N \quad (5)$$

with a neighborhood of N . We write points as (x, s, u) , where $s \in E_x^s$ and $u \in E_x^u$. In finitely many bundle charts the map f has the form

$$(x, s, u) \longmapsto (f_N(x) + a(x, s, u), A_x^s s + b(x, s, u), A_x^u u + c(x, s, u)), \quad (6)$$

where $f_N = f|_N$. At $(s, u) = (0, 0)$ the error terms and their first normal derivatives vanish after the three invariant linear blocks have been extracted. A C^1 -nearby g has the same form with a uniformly small C^1 defect at the zero section. All estimates below are made in this fixed finite bundle atlas, so constants are uniform.

There is one harmless boundary issue in a local graph transform. Choose nested tube radii

$$0 < \rho < 2\rho < 3\rho \quad (7)$$

so small that all the preceding estimates hold on the 3ρ tube. Multiply the nonlinear normal terms in (6), in the annulus between 2ρ and 3ρ , by a fixed smooth cutoff. This gives a globally defined bundle model on the relevant disk bundles, equal to the original map on the 2ρ tube and satisfying the same strict first-derivative inequalities after ρ is decreased if necessary. Every fixed graph constructed below has height $< \rho/4$ over the ρ core; consequently it lies where no cutoff was made. Thus the auxiliary extension changes neither the persistent manifold nor any local strong plaque. We use it only to keep every graph transform on a fixed complete function space.

Step 2: the center-stable graph by a backward transform. The base of the center-

stable graph is a fixed small stable disk bundle $D^{cs} = E^s(\rho) \rightarrow N$; its tangent bundle along the zero section is naturally identified with $TN \oplus E^s$. For $G = g^{-1}$ consider graphs

$$u = h^{cs}(z), \quad z = (x, s) \in D^{cs}, \quad (8)$$

in the closed class

$$\|h^{cs}\|_{C^0} \leq \rho/4, \quad \text{Lip}(h^{cs}) \leq \eta, \quad (9)$$

where $\eta > 0$ is chosen small. With the sup norm this class is complete: a uniform limit of maps with a common Lipschitz constant has the same Lipschitz constant. We first write the derivative estimates for C^1 graphs; they extend to the whole Lipschitz class by approximation and the pointwise Lipschitz estimates. Write

$$G(z, h(z)) = (X_h(z), Y_h(z)) \quad (10)$$

in the tangent splitting $(TN \oplus E^s) \oplus E^u$. Along the zero section for f , the E^u block of DG has norm at most $\bar{\nu}$, while the $TN \oplus E^s$ block has conorm at least $\bar{\gamma}$: on TN this is the reciprocal of the upper center bound in (3), and on E^s it is at least $\nu^{-1} > 1$. Hence, after the choices above,

$$m(DX_h) \geq \bar{\gamma}_0 - O(\eta) > 0, \quad \|D_u Y_h\| \leq \bar{\nu}_0, \quad (11)$$

and all off-diagonal first derivatives are as small as desired. Moreover X_h is C^1 -close to the corresponding base map of the unperturbed bundle model, which is a diffeomorphism on the cutoff-extended disk bundle. The uniform conorm bound in (11), compactness, and this C^1 closeness therefore make X_h a diffeomorphism onto its image; that image contains the fixed core after the radii are chosen with the margin in (7). Define the backward center-stable graph transform by

$$\mathcal{T}_g^{cs} h := Y_h \circ X_h^{-1}. \quad (12)$$

We record the two estimates that will also be used for smoothness. If the graph height and slope bounds in (9), the tube, and the C^1 neighborhood of f are small enough, then there is $q_0 < 1$ such that

$$\|\mathcal{T}_g^{cs} h_1 - \mathcal{T}_g^{cs} h_2\|_{C^0} \leq q_0 \|h_1 - h_2\|_{C^0}. \quad (13)$$

Indeed, compare the two preimages of the same target base point. The lower bound in (11) controls their base displacement by the vertical difference of the two input graphs; the E^u component of DG contracts that vertical difference by at most $\bar{\nu}_0$, and the remaining off-diagonal terms are arbitrarily small. Since $\bar{\nu}_0 < 1$, the resulting coefficient is < 1 .

For derivatives the reparametrization in (12) matters. The inverse base derivative has norm at most $\bar{\gamma}_0^{-1} + o(1)$. Consequently the coefficient multiplying the highest j -th derivative of an input graph in the j -jet transform is bounded by

$$q_j^{cs} \leq \bar{\nu}_0 \bar{\gamma}_0^{-j} + o(1) < 1, \quad 1 \leq j \leq r. \quad (14)$$

The final inequality follows from (4); the neighborhood and η are then shrunk once so

that it remains strict simultaneously for all $j \leq r$.

The derivative cone estimate shows that the class (9) is preserved by the graph transform. Banach's fixed-point theorem applied to (13) on this complete Lipschitz class therefore gives a unique small fixed graph h_g^{cs} , Lipschitz with the same small slope. Denote its graph by

$$W_g^{cs} := \{(z, h_g^{cs}(z)) : z \in D^{cs}\}. \quad (15)$$

By construction $G(W_g^{cs}) = W_g^{cs}$ on the smaller core, equivalently W_g^{cs} is locally g -invariant.

We shall use the following trapping characterization. After shrinking the core, a point whose full forward orbit remains in the 2ρ tube belongs to W_g^{cs} . To see this, measure its E^u displacement from the fixed graph over the same center-stable base coordinate (x, s) . Reversing (11)–(13), one forward step expands every nonzero such transverse displacement by a factor > 1 once the graph slope and off-diagonal terms are small. A nonzero displacement therefore cannot remain inside the fixed tube for all positive iterates. Thus the transverse displacement is zero. This is the local uniqueness property of the center-stable graph that we need later.

Step 3: the center-unstable graph by a forward transform. Now use as base a fixed small unstable disk bundle $D^{cu} = E^u(\rho) \rightarrow N$, whose tangent along the zero section is $TN \oplus E^u$, and apply the same construction directly to g , using graphs

$$s = h^{cu}(x, u). \quad (16)$$

Along the zero section, the stable block contracts by at most ν , while the $TN \oplus E^u$ block has conorm at least γ : the center conorm is bounded below by γ , and the unstable conorm is at least $\bar{\nu}^{-1} > 1$. Hence there is a forward graph transform $\mathcal{T}_g^{cu}$ with

$$\|\mathcal{T}_g^{cu} h_1 - \mathcal{T}_g^{cu} h_2\|_{C^0} \leq q'_0 \|h_1 - h_2\|_{C^0}, \quad q'_0 < 1, \quad (17)$$

and its j -jet coefficient satisfies

$$q_j^{cu} \leq \nu_0 \gamma_0^{-j} + o(1) < 1, \quad 1 \leq j \leq r. \quad (18)$$

Its unique small fixed graph is denoted W_g^{cu} . It is locally invariant, and a point whose full backward orbit remains in the tube belongs to W_g^{cu} .

Step 4: the C^r section argument. We prove, rather than merely cite, the regularity mechanism implicit in (14) and (18). We do it for W_g^{cs} ; the other graph is identical.

Start the graph transform from a C^r graph h_0 satisfying (9), for example the zero graph after the cutoff extension, and put

$$h_{n+1} = \mathcal{T}_g^{cs} h_n. \quad (19)$$

Equation (13) implies uniform convergence $h_n \rightarrow h_g^{cs}$. For each $j \geq 1$, differentiate the identity defining (12) j times. Faà di Bruno's formula has exactly one term containing $D^j h_n$ linearly. After the inverse base map has been differentiated, the resulting j -jet

recurrence has the form

$$D^j h_{n+1} = A_{n,j}(D^j h_n \circ \xi_n) + B_{n,j}, \quad (20)$$

where ξ_n is the inverse base reparametrization. The coefficient $A_{n,j}$ contains one normal derivative and j inverse base derivatives; hence

$$\|A_{n,j}\|_{C^0} \leq q_j < 1 \quad (21)$$

uniformly, with q_j supplied by (14). Every term in $B_{n,j}$ contains only derivatives $D^\ell h_n$ with $\ell < j$ and derivatives of g of order at most j . This is the only derivative bookkeeping needed for the C^r section theorem.

For $j = 1$, the differentiated graph transform is the usual fractional-linear map on tangent graphs (a graph transform on the Grassmannian), not an affine map. Its derivative with respect to the slope variable has norm at most $q_1 < 1$ by (14). The coefficients are continuous functions of the graph point and Dg . Since $h_n \rightarrow h_g^{cs}$ uniformly and Dg is uniformly continuous on the compact tube, the induced maps on the Banach space of continuous first-jet sections converge uniformly on the preserved slope ball to a limiting q_1 -contraction. That limiting map has a unique fixed section P_1 . The elementary nonautonomous contraction estimate

$$d_{n+1} \leq q_1 d_n + \varepsilon_n, \quad \varepsilon_n \rightarrow 0, \quad (22)$$

with $d_n = \|Dh_n - P_1\|_{C^0}$ gives $d_n \rightarrow 0$: taking limsup in (22) yields $\limsup d_n \leq q_1 \limsup d_n$. Therefore $Dh_n \rightarrow P_1$ uniformly. Uniform convergence of h_n and Dh_n implies that h_g^{cs} is C^1 and $Dh_g^{cs} = P_1$.

Assume inductively that the derivatives through order $j - 1$ converge uniformly and that h_g^{cs} is C^{j-1} . Then the coefficients and forcing term in (20) converge uniformly, because g is C^r and $B_{n,j}$ uses only the already controlled lower jets. For $j \geq 2$ the highest jet enters affinely, and the limiting j -jet map is a contraction in that highest-jet variable, with constant $q_j < 1$. The same argument proves uniform convergence of $D^j h_n$ to a continuous limit. The standard repeated fundamental-theorem-of-calculus criterion then says that the uniform limit h_g^{cs} is C^j with that limit as its j -th derivative. Induction through $j = r$ proves

$$W_g^{cs}, W_g^{cu} \text{ are } C^r. \quad (23)$$

Notice an important point: only the first derivative of g has to be small. Higher derivatives of g enter the forcing terms $B_{n,j}$, not the contraction coefficient of the highest jet. Thus a C^1 -small C^r perturbation is enough, exactly as stated.

The same contraction argument also gives continuous dependence of the fixed graphs in the C^1 topology on g . Indeed, if two graph transforms have the common contraction constant $q < 1$, their fixed graphs differ by at most $(1 - q)^{-1}$ times the sup norm of the difference of the transforms; applying the analogous estimate to the first-jet transforms gives C^1 dependence. No claim of C^r dependence on g in the C^1 perturbation topology is needed.

Step 5: intersect the two persistent graphs. For $g = f$, the center-stable and center-

unstable manifolds meet along N and their tangent spaces satisfy

$$(TN \oplus E^s) \cap (TN \oplus E^u) = TN, \quad (TN \oplus E^s) + (TN \oplus E^u) = T_N M. \quad (24)$$

By the C^1 continuity just proved, the two persistent manifolds remain uniformly transverse when g is close to f .

More explicitly, in each normal fiber their equations are

$$u = h_g^{cs}(x, s), \quad s = h_g^{cu}(x, u). \quad (25)$$

Their slopes can be made smaller than a fixed $\eta < 1/4$. For each $x \in N$ solve

$$(s, u) = (h_g^{cu}(x, u), h_g^{cs}(x, s)). \quad (26)$$

The right side is a contraction on a fixed small product ball, with Lipschitz constant at most η . Hence (26) has a unique solution $(s_g(x), u_g(x))$. Since the two graph functions are C^r and the derivative of the left side minus the right side is invertible, the ordinary implicit-function theorem gives C^r dependence on x . Thus

$$N_g := W_g^{cs} \cap W_g^{cu} = \{(x, s_g(x), u_g(x)) : x \in N\} \quad (27)$$

is a compact C^r submanifold. Projection along the fixed tube is a C^r diffeomorphism $N_g \rightarrow N$. Since both W_g^{cs} and W_g^{cu} are locally invariant and N_g lies in the inner core, $g(N_g) = N_g$.

When $g \rightarrow f$ in C^1 , the two fixed graph transforms and their first-jet transforms converge, so (26) and its derivative imply

$$N_g \longrightarrow N \quad \text{in } C^1. \quad (28)$$

Step 6: local uniqueness of the persistent manifold. Let $\tilde{N}$ be another compact C^r g -invariant submanifold contained in the inner tube, sufficiently C^1 -close to N , and projecting diffeomorphically to N . Every point of $\tilde{N}$ has its entire positive and negative orbit in the inner tube because $\tilde{N}$ is invariant and compact. The trapping characterizations from Steps 2 and 3 therefore give

$$\tilde{N} \subset W_g^{cs}, \quad \tilde{N} \subset W_g^{cu}. \quad (29)$$

Hence, by the fiberwise uniqueness in (26),

$$\tilde{N} \subset W_g^{cs} \cap W_g^{cu} = N_g. \quad (30)$$

Both manifolds have dimension $\dim N$ and both project bijectively onto N ; therefore $\tilde{N} = N_g$. This is exactly the local uniqueness asserted in the theorem. No global uniqueness among distant invariant manifolds is claimed.

Step 7: persistence of the invariant splitting and of the spectral gaps. Along N_g , use the tubular identification to compare TN_g with TN and the ambient tangent bundle with (1). The derivative Dg is uniformly close to Df in these coordinates. Apply the graph transform to linear maps from E^s to $TN \oplus E^u$ and, in reverse time, from E^u to

$E^s \oplus TN$. The same strict cone inequalities used in Steps 2-3 give unique continuous invariant graphs

$$E_g^s, \quad E_g^u, \quad (31)$$

with the same dimensions as E^s, E^u and with uniformly small graph norms. Together with TN_g they form

$$TN_g M = E_g^s \oplus TN_g \oplus E_g^u. \quad (32)$$

Because all inequalities in (2)-(3) were made strict before perturbing, continuity of norms and conorms permits constants

$$\nu_0 < \nu' < (\gamma')^r < 1, \quad \bar{\nu}_0 < \bar{\nu}' < (\bar{\gamma}')^r < 1 \quad (33)$$

with center bounds $m(Dg|_{TN_g}) \geq \gamma'$ and $\|Dg|_{TN_g}\| \leq (\bar{\gamma}')^{-1}$, and with

$$\|Dg|_{E_g^s}\| \leq \nu', \quad \|Dg^{-1}|_{E_g^u}\| \leq \bar{\nu}'. \quad (34)$$

Thus N_g is again r -normally hyperbolic, with the same stable and unstable dimensions and slightly weakened constants.

Step 8: local strong stable and unstable plaques. Fix $z \in N_g$. Along the orbit $g^n z$, the splitting

$$E_g^s \oplus (TN_g \oplus E_g^u) \quad (35)$$

has a uniform forward gap: the first bundle contracts by at most ν' , whereas the conorm of the complementary dynamics is at least γ' , and $\nu' < \gamma'$. Run the same graph transform as in Step 3, now with the E_g^s coordinate as the contracting fiber and the complementary coordinate as the base. The j -jet coefficient is bounded by

$$\nu'(\gamma')^{-j} < 1, \quad 1 \leq j \leq r. \quad (36)$$

The argument of Step 4 therefore yields a local C^r strong stable disk $W_{\text{loc}}^{ss}(z)$ tangent to $E_g^s(z)$. Uniform contraction of the normal coordinate gives constants $C \geq 1$ and $0 < \theta < 1$, independent of z , such that

$$d(g^n y, g^n z) \leq C\theta^n d(y, z), \quad y \in W_{\text{loc}}^{ss}(z), \quad n \geq 0. \quad (37)$$

The graph construction also gives local forward invariance.

Apply the same argument to g^{-1} and the splitting $E_g^u \oplus (TN_g \oplus E_g^s)$, using $\bar{\nu}' < (\bar{\gamma}')^r$, to obtain the local C^r strong unstable disk and the corresponding backward exponential estimate. The graph transforms have uniform contraction constants and depend continuously on z ; hence the plaques vary continuously with the base point in local C^1 graph coordinates. Their leaves are C^r by the jet argument above.

Nothing in this proof gives C^r transverse dependence of the strong foliation, and no such conclusion is stated. Additional bunching hypotheses are needed for stronger transverse regularity. All conclusions of the theorem have now been proved. $\square$

Theorem F10.11. Dominated splitting robustness

Statement. Let $f : M \rightarrow M$ be a C^1 diffeomorphism, let $\Lambda \subset M$ be compact and f -invariant, and suppose

$$T_\Lambda M = E \oplus F, \quad \|Df^N|_{E_x}\| \|(Df^N|_{F_x})^{-1}\| \leq \theta < 1$$

for some $N \geq 1$ and every $x \in \Lambda$. Then, after replacing N by a multiple, there are a neighborhood U of Λ , continuous extensions of E, F to U , and mutually transverse cone fields about those extensions which are strictly invariant by Df^N and Df^{-N} whenever the relevant orbit segment stays in U . Consequently there is a C^1 neighborhood $\mathcal{U}$ of f such that for every $g \in \mathcal{U}$ and every compact g -invariant set $K \subset U$, the tangent cocycle over K has a dominated splitting $T_K M = E_g \oplus F_g$ with the same dimensions, lying in the prescribed cones and uniformly close to the original extended bundles; this continuation is unique among invariant splittings lying in those cone neighborhoods. The bundles depend continuously on the base point and continuously on the cocycle in local graph coordinates. One may choose an equivalent continuous Riemannian metric in which $E \perp F$ and the domination inequality is strict with constant < 1 for the chosen iterate.

Scope note

The statement includes an adapted metric for the chosen iterate, not the stronger one-step adapted-metric theorem.

Proof. Write $d_E = \dim E$ and $d_F = \dim F$. The argument has two parts: first we construct a robust cone/graph-transform mechanism on a neighborhood of Λ ; then we apply it to C^1 -nearby diffeomorphisms.

Step 1: strengthen the domination and choose an iterate-adapted metric. By hypothesis,

$$\|Df^N|_{E_x}\| \|(Df^N|_{F_x})^{-1}\| \leq \theta < 1. \quad (1)$$

Iterating (1) gives

$$\|Df^{kN}|_{E_x}\| \|(Df^{kN}|_{F_x})^{-1}\| \leq \theta^k. \quad (2)$$

Choose k so large that the right side is much smaller than one, and put $L = kN$.

Choose any continuous inner products on E and F and declare the two subbundles orthogonal. This gives a continuous Riemannian metric on $T_\Lambda M$. Relative to the original metric the two norms are uniformly equivalent, so after increasing k once more we may arrange

$$\|Df^L|_{E_x}\|_* \|(Df^L|_{F_x})^{-1}\|_* \leq \eta \quad (x \in \Lambda) \quad (3)$$

with, say, $0 < \eta < 1/10$. Extending the metric to a neighborhood of Λ proves the adapted-metric assertion made in the statement: the bundles are orthogonal and the domination is visible with constant < 1 at the fixed iterate L . We do not claim here that L can be reduced to one.

Step 2: construct strict cones on Λ . For $a > 0$ define the fast cone

$$\mathcal{C}_a^F(x) := \{e + v : e \in E_x, v \in F_x, \|e\|_* \leq a\|v\|_*\}, \quad (4)$$

and the slow cone

$$\mathcal{C}_a^E(x) := \{e + v : e \in E_x, v \in F_x, \|v\|_* \leq a\|e\|_*\}. \quad (5)$$

If $e + v \in \mathcal{C}_a^F(x)$, invariance of E, F and (3) give

$$\frac{\|Df^L e\|_*}{\|Df^L v\|_*} \leq \|Df^L|_{E_x}\|_* \|(Df^L|_{F_x})^{-1}\|_* \frac{\|e\|_*}{\|v\|_*} \leq \eta a. \quad (6)$$

Thus

$$Df_x^L \mathcal{C}_a^F(x) \subset \mathcal{C}_{\eta a}^F(f^L x) \subset \text{int } \mathcal{C}_a^F(f^L x) \cup \{0\}. \quad (7)$$

Applying the same calculation to f^{-L} gives strict backward invariance of $\mathcal{C}_a^E$.

Step 3: extend the cone mechanism off Λ . We first justify the standard extension step rather than hiding it in notation. Let $P_E(x)$ be the orthogonal projection onto E_x along F_x . In finitely many tangent-bundle trivializations its matrix entries are continuous on the closed set Λ ; extend those entries to a neighborhood and patch the extensions by a partition of unity. The resulting endomorphism $Q_E(y)$ equals $P_E(y)$ on Λ and is uniformly close to a projection after the neighborhood is shrunk. Replace it by the self-adjoint endomorphism $(Q_E + Q_E^*)/2$; this still equals the orthogonal projection P_E on Λ . For y sufficiently near Λ , its spectrum lies in two disjoint small intervals around 0 and 1. The orthogonal spectral projection onto the eigenspaces belonging to the cluster near 1 is therefore a continuous genuine projection $\tilde{P}_E(y)$ of constant rank. Its image defines a continuous extension $\tilde{E}$, and its kernel defines a complementary extension $\tilde{F}$. Thus, after shrinking a neighborhood U of Λ , we obtain continuous complementary bundles $\tilde{E}, \tilde{F}$ agreeing with E, F on Λ . Shrink U again so that their angle is uniformly bounded from zero.

Define cones over U by the same inequalities (4)–(5), with tildes. Because (7) is a strict inclusion on the compact set Λ , continuity gives a number $0 < \eta_1 < 1$ and a possibly smaller U such that whenever

$$x, fx, \dots, f^L x \in U, \quad (8)$$

we have

$$Df_x^L \mathcal{C}_a^F(x) \subset \mathcal{C}_{\eta_1 a}^F(f^L x), \quad (9)$$

and, for the reversed orbit segment,

$$Df_x^{-L} \mathcal{C}_a^E(x) \subset \mathcal{C}_{\eta_1 a}^E(f^{-L} x). \quad (10)$$

This is the desired strict cone field on a neighborhood.

Step 4: strict cone invariance is open in the C^1 topology. The closure of the narrower cone $\mathcal{C}_{\eta_1 a}^F(y)$ is a positive projective distance from the boundary of $\mathcal{C}_a^F(y)$, uniformly for y in the compact closure of a smaller neighborhood of Λ . The same holds for the slow

cones. The map

$$g \mapsto Dg^L \quad (11)$$

is continuous from the C^1 topology to the uniform topology on derivatives along orbit segments of bounded length. Hence there are a C^1 neighborhood $\mathcal{U}$ of f and $0 < \eta_2 < 1$ such that, for every $g \in \mathcal{U}$ and every g -orbit segment of length L contained in U ,

$$Dg_x^L \mathcal{C}_a^F(x) \subset \mathcal{C}_{\eta_2 a}^F(g^L x), \quad (12)$$

and analogously

$$Dg_x^{-L} \mathcal{C}_a^E(x) \subset \mathcal{C}_{\eta_2 a}^E(g^{-L} x). \quad (13)$$

This is the precise sense in which the cone criterion is robust.

Step 5: the graph transform produces the perturbed fast bundle. Let $K \subset U$ be compact and g -invariant. We work over K . In the splitting

$$T_x M = \tilde{E}_x \oplus \tilde{F}_x, \quad (14)$$

a d_F -plane contained in the interior of $\mathcal{C}_a^F(x)$ is uniquely the graph of a linear map

$$P_x : \tilde{F}_x \rightarrow \tilde{E}_x, \quad \|P_x\| < a. \quad (15)$$

Write the block matrix of Dg_x^L as

$$Dg_x^L = \begin{pmatrix} A_x & B_x \\ C_x & D_x \end{pmatrix} : \tilde{E}_x \oplus \tilde{F}_x \longrightarrow \tilde{E}_{g^L x} \oplus \tilde{F}_{g^L x}. \quad (16)$$

The image of the graph of P is the graph of

$$\mathcal{G}_x(P) = (A_x P + B_x)(C_x P + D_x)^{-1}. \quad (17)$$

Strict cone invariance guarantees that $C_x P + D_x$ is invertible throughout $\|P\| \leq a$ and that $\|\mathcal{G}_x(P)\| \leq \eta_2 a$.

At $g = f$ and $x \in \Lambda$ the off-diagonal blocks vanish and

$$\mathcal{G}_x(P) = Df^L|_{E_x} P (Df^L|_{F_x})^{-1}, \quad (18)$$

whose Lipschitz constant is at most $\eta < 1/10$ by (3). Differentiating (17) with respect to P and using compactness shows that, after shrinking U and $\mathcal{U}$ once more, there is a uniform $q < 1$ such that

$$\|\mathcal{G}_x(P) - \mathcal{G}_x(Q)\| \leq q \|P - Q\| \quad (19)$$

for all $x \in K$ and $\|P\|, \|Q\| \leq a$.

Starting with any continuous graph section $P^{(0)}$ with $\|P^{(0)}\| \leq a$ and repeatedly applying the graph transform from the past,

$$P_x^{(n+1)} = \mathcal{G}_{g^{-L}x}(P_{g^{-L}x}^{(n)}), \quad (20)$$

produces a uniformly Cauchy sequence by (19). Its uniform limit $P_g^F(x)$ is continuous

and satisfies the invariance equation

$$P_g^F(g^L x) = \mathcal{G}_x(P_g^F(x)). \quad (21)$$

Define

$$F_g(x) := \text{graph } P_g^F(x). \quad (22)$$

It is a continuous Dg^L -invariant d_F -plane field contained in the fast cone. Uniqueness follows immediately from the contraction (19).

Although we constructed invariance first for g^L , it is actually Dg -invariant. Since the fixed graph in (21) lies in the narrower cone $\mathcal{C}_{\eta_2 a}^F$, continuity of the one-step derivative and invariance of F on Λ allow U and $\mathcal{U}$ to be chosen so that Dg maps this narrower cone into $\mathcal{C}_a^F$. Hence the field $x \mapsto Dg_{g^{-1}x} F_g(g^{-1}x)$ is another continuous Dg^L -invariant d_F -plane field in the same fast cone. By uniqueness of the fixed graph, it equals $F_g(x)$.

Step 6: construct the slow bundle and prove domination. Apply the same graph-transform argument to g^{-L} and the slow cones. This gives a continuous Dg -invariant d_E -plane field E_g contained in $\mathcal{C}_a^E$. Choosing a small makes the two cone families uniformly transverse, hence

$$T_K M = E_g \oplus F_g. \quad (23)$$

Because all block matrices in (16) are uniformly close to the block-diagonal matrices for $Df^L|_{E \oplus F}$ on Λ , and the invariant graphs have uniformly small norm, (3) persists with a slightly larger constant. Thus, after the neighborhoods were chosen sufficiently small,

$$\|Dg^L|_{E_g(x)}\| \|(Dg^L|_{F_g(x)})^{-1}\| \leq \eta_3 < 1 \quad (x \in K). \quad (24)$$

This is a dominated splitting in the sense of *the definition of dominated splitting*.

Step 7: continuity with respect to the cocycle and uniqueness. Formula (17) depends continuously on the derivative cocycle. The contraction constant $q < 1$ is uniform. If two graph transforms differ by at most ε in the sup norm, their fixed graph sections differ by at most $\varepsilon/(1 - q)$. Hence E_g and F_g depend continuously on g in local graph coordinates and converge uniformly to E, F as the derivative cocycle converges to $Df|_\Lambda$ (with base points identified inside the chosen bundle charts).

Finally, the continuation is unique inside the prescribed cone neighborhood. Indeed, any invariant d_F -plane field contained in the fast cone is a fixed section of the graph transform and therefore equals F_g by (19); the time-reversed statement gives E_g . This is exactly the uniqueness needed for robust continuation of the original splitting.

This proves robustness, the cone-field statement, the claimed continuity, and the iterate-adapted metric assertion. $\square$

10.3 Worked examples

Worked example: Uniform Hyperbolicity and Dominated Splittings: a model computation

Consider

$$F(x, y) = (\lambda x, \mu y), \quad 0 < |\lambda| < 1 < |\mu|.$$

The origin is hyperbolic. For $z = (x, y)$,

$$F^n z = (\lambda^n x, \mu^n y).$$

Therefore $F^n z \rightarrow 0$ as $n \rightarrow \infty$ exactly when $y = 0$. Hence

$$W^s(0) = \{(x, 0) : x \in \mathbb{R}\}, \quad W^u(0) = \{(0, y) : y \in \mathbb{R}\}.$$

The local product of $(x_1, 0) \in W^s$ and $(0, y_2) \in W^u$ is simply (x_1, y_2) : stable and unstable coordinates can be chosen independently. In this linear model the graph transform is visible by inspection; in the nonlinear theorem, the same contraction survives after the derivative is perturbed by sufficiently small nonlinear terms.

Boundary case: what fails at the edge of the hypotheses

Take

$$G(x, y) = (x, 2y).$$

The derivative at the origin has eigenvalues 1 and 2. The unit eigenvalue creates a center direction. Points $(x, 0)$ do not converge to the origin unless $x = 0$, yet they also do not move away. The hyperbolic splitting required by the stable-manifold theorem is absent, so the theorem cannot produce a stable disk whose tangent complement is uniformly expanding. Center-manifold or normally hyperbolic theory is needed instead.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

10.4 Chapter summary

What to remember

- Uniform exponential contraction/expansion makes nonlinear invariant geometry robust.
- Stable manifolds, local product structure, shadowing, and structural stability

are different faces of the same hyperbolic mechanism.

- Dominated splitting requires relative separation, not necessarily contraction to zero.
- Normal hyperbolicity compares normal rates with tangential rates along an invariant manifold.

Looking ahead

F11 removes uniformity. Oseledets exponents replace fixed contraction rates, and Pesin blocks recover enough local uniformity to rebuild stable/unstable manifolds almost everywhere.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

10.5 Exercises and problems

Exercise 10.1

Difficulty. Core.

Related results. Local stable manifold theorem for a hyperbolic fixed point; Stable manifold theorem for hyperbolic sets; Local product structure.

Problem. Work through the worked example 'the linear saddle $(x,y) \rightarrow (\lambda x, \mu y)$, $|\lambda| < 1 < |\mu|$ '. For Local stable manifold theorem for a hyperbolic fixed point, Stable manifold theorem for hyperbolic sets, and Local product structure, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 10.2

Difficulty. Core.

Related results. Shadowing lemma; Structural stability of Anosov diffeomorphisms; Inclination / λ lemma.

Problem. Analyze the boundary example 'a neutral fixed point where the stable-manifold theorem hypotheses fail'. For Shadowing lemma, Structural stability of Anosov diffeomorphisms, and Inclination / λ lemma, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no

longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 10.3

Difficulty. Advanced.

Related results. Local product structure.

Problem. For Local product structure, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 10.4

Difficulty. Advanced.

Related results. Shadowing lemma.

Problem. Stress-test Shadowing lemma. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 10.5

Difficulty. Advanced.

Related results. Local stable manifold theorem for a hyperbolic fixed point; Stable manifold theorem for hyperbolic sets.

Problem. Synthesis problem. Use Local stable manifold theorem for a hyperbolic fixed point and Stable manifold theorem for hyperbolic sets to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 10.6

Difficulty. Lab.

Related results. None.

Problem. Computational laboratory: carry out the reproducible experiment 'the linear saddle $(x,y) \rightarrow (\lambda x, \mu y)$, $|\lambda| < 1 < |\mu|$ '. Use the computation only

as numerical evidence or visualization, not as proof. Include a refinement or convergence test and identify one numerical artifact that could mislead a reader.

Solution sketch. State parameters and truncation/time horizon, compare with an analytically controlled identity or exactly solvable case, repeat after refinement, and explain what the numerical experiment can and cannot establish.

Exercise 10.7

Difficulty. Core.

Related results. Normally hyperbolic invariant manifold persistence.

Problem. Work directly with Normally hyperbolic invariant manifold persistence. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 10.8

Difficulty. Core.

Related results. Dominated splitting robustness.

Problem. Work directly with Dominated splitting robustness. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 10.9

Difficulty. Core.

Related results. Markov rectangles for basic hyperbolic sets: interface.

Problem. Work directly with Markov rectangles for basic hyperbolic sets: interface. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 10.10

Difficulty. Core.

Related results. Adapted norm for a hyperbolic splitting.

Problem. Work directly with Adapted norm for a hyperbolic splitting. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 10.11

Difficulty. Core.

Related results. Parameter-dependent contraction theorem.

Problem. Work directly with Parameter-dependent contraction theorem. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Nonuniform Hyperbolicity and Pesin Theory

Why this chapter matters

Many important systems are hyperbolic only along typical orbits and only at asymptotic scales. The angles between stable and unstable directions may become small, and contraction constants may vary wildly from point to point. Pesin theory explains how to recover usable local geometry from Oseledets' nonuniform data.

The central idea is uniformization: restrict to Pesin blocks on which exponents, angles, and chart sizes have controlled bounds. Uniform hyperbolic arguments can then be run block by block, with measure theory used to assemble the conclusions.

Prerequisites. F04, F10, smooth ergodic theory, conditional measures, and differential geometry.

Learning goals

- Build Lyapunov norms and Pesin blocks from Oseledets data.
- Understand local stable/unstable manifolds in the nonuniform setting.
- Interpret absolute continuity and SRB unstable conditional measures.
- Connect positive Lyapunov exponents to entropy through Ruelle and Pesin formulae.

Conceptual roadmap

Oseledets gives rates but poor constants. Lyapunov norms absorb the nonuniformity, and Pesin blocks select regions where the resulting constants are uniform. Graph-transform arguments then produce local invariant manifolds. Absolute continuity makes those manifolds compatible with ambient volume. The entropy results and Hopf argument show how this geometry controls statistical behavior.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

11.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F11.1. Pesin block

A *Pesin block* is a measurable, often compact, subset of Oseledets-regular points on which Lyapunov exponents, angles between invariant subspaces, tempered constants, and admissible chart radii obey fixed uniform bounds.

Definition F11.2. Lyapunov chart

A *Lyapunov chart* is a local coordinate chart adapted to the Oseledets splitting and Lyapunov norm, chosen so that the derivative is nearly block diagonal with prescribed exponential rates and the nonlinear remainder is uniformly small on the chart scale.

Definition F11.3. Absolute continuity

A measurable foliation or lamination is *absolutely continuous* if holonomy between appropriate transversals sends sets of zero induced Lebesgue measure to sets of zero measure; quantitative Jacobian bounds are stated when needed.

Definition F11.4. SRB unstable-conditional property

An invariant probability μ has the *SRB unstable-conditional property* if its conditional measures on local unstable manifolds are absolutely continuous with respect to the induced Riemannian leaf volumes. Physicality is a separate property unless an equivalence theorem is invoked.

Definition F11.5. Partition subordinate to unstable manifolds

A measurable partition ξ is *subordinate to unstable manifolds* if, for μ -almost every x , the atom $\xi(x)$ is contained in $W^u(x)$, contains an open neighborhood of x in that unstable leaf, and is measurable in the Rokhlin sense needed to define conditional measures.

11.2 Main results

Lemma F11.1. Lyapunov norm and Pesin regularity functions

Statement. Let $(X, \mathcal{A}, \mu, T)$ be probability-preserving with T invertible and bimeasurable, and let $A : X \rightarrow GL(d, \mathbb{R})$ be measurable with $\log^+ \|A\| + \log^+ \|A^{-1}\| \in L^1(\mu)$. On a full-measure invariant subset of the Oseledets set supplied by Theorem F04.6, write $\mathbb{R}^d = E_1(x) \oplus \cdots \oplus E_k(x)$ with finite exponents $\lambda_1(x) > \cdots > \lambda_k(x)$. For every $\varepsilon > 0$ there is a measurable inner product with norm $\|\cdot\|_{x,\varepsilon}$, making the $E_i(x)$ mutually orthogonal, such that for $v \in E_i(x)$,

$$e^{\lambda_i(x)-\varepsilon}\|v\|_{x,\varepsilon} \leq \|A(x)v\|_{Tx,\varepsilon} \leq e^{\lambda_i(x)+\varepsilon}\|v\|_{x,\varepsilon}.$$

Moreover

$$\|v\| \leq \|v\|_{x,\varepsilon} \leq K_\varepsilon(x)\|v\|$$

for a finite measurable function $K_\varepsilon \geq 1$ which is tempered and ε -slowly varying:

$$e^{-\varepsilon}K_\varepsilon(x) \leq K_\varepsilon(Tx) \leq e^\varepsilon K_\varepsilon(x), \quad K_\varepsilon(T^n x) \leq K_\varepsilon(x)e^{\varepsilon|n|}.$$

In particular $|n|^{-1} \log K_\varepsilon(T^n x) \rightarrow 0$ almost everywhere.

Proof. Let $\mathcal{R}_0$ be the full-measure invariant Oseledets set from *Oseledets theorem: invertible splitting form*. Intersect $\mathcal{R}_0$ with the invariant full-measure sets on which the Birkhoff limits used in Step 5 and the tempering conclusion of *Temperedness lemma* hold, and denote the resulting invariant full-measure set by $\mathcal{R}$. All constructions below take place on $\mathcal{R}$. Fix $\varepsilon > 0$ and put

$$\eta := \frac{\varepsilon}{2}. \tag{1}$$

We first build a Lyapunov norm with error η and then use the remaining margin to obtain the desired slowly varying comparison constant.

Step 1: the weighted inner product on one Oseledets space. Fix $x \in \mathcal{R}$ and an exponent space $E_i(x)$. For $u, v \in E_i(x)$ define

$$\langle u, v \rangle_{x,\eta,i} := \sum_{n \in \mathbb{Z}} \langle A_n(x)u, A_n(x)v \rangle e^{-2\lambda_i(x)n-2\eta|n|}. \tag{2}$$

Here A_n is the integer-time cocycle from *Oseledets theorem: invertible splitting form*.

We first justify convergence. If $0 \neq u \in E_i(x)$, the two-sided Oseledets law says

$$\lim_{n \rightarrow +\infty} \frac{1}{n} \log \|A_n(x)u\| = \lambda_i(x), \quad \lim_{m \rightarrow +\infty} \frac{1}{m} \log \|A_{-m}(x)u\| = -\lambda_i(x). \tag{3}$$

Choose $0 < \delta < \eta$. For all sufficiently large n, m (depending on x, u, δ),

$$\|A_n(x)u\| \leq e^{(\lambda_i+\delta)n}, \quad \|A_{-m}(x)u\| \leq e^{(-\lambda_i+\delta)m}, \tag{4}$$

after absorbing the fixed norm of u and the finitely many initial iterates into a harmless

constant. Consequently the positive and negative tails of

$$\sum_{n \in \mathbb{Z}} \|A_n(x)u\|^2 e^{-2\lambda_i n - 2\eta|n|} \quad (5)$$

are bounded by geometric series with ratio $e^{-2(\eta-\delta)} < 1$. Thus (5) converges. By Cauchy-Schwarz, the bilinear series (2) converges absolutely for every $u, v \in E_i(x)$.

Every summand in (2) is positive semidefinite, and the $n = 0$ summand is the background inner product. Hence (2) is a positive-definite inner product and

$$\|u\|_{x,\eta,i} \geq \|u\| \quad (u \in E_i(x)). \quad (6)$$

Step 2: assemble the exponent spaces and normalize the lower bound. On an invariant measurable stratum on which the number $k = k(x)$ of distinct exponents and their multiplicities are fixed, let $P_i(x)$ denote the measurable projection onto $E_i(x)$ along the direct sum of the other Oseledets spaces. For $v, w \in \mathbb{R}^d$ define

$$\langle v, w \rangle_{x,\varepsilon} := k \sum_{i=1}^k \langle P_i(x)v, P_i(x)w \rangle_{x,\eta,i}. \quad (7)$$

The notation uses ε for the final norm even though the summation kernel uses the smaller number $\eta = \varepsilon/2$. By construction the spaces $E_i(x)$ are mutually orthogonal for (7).

The factor k is included for a specific reason. Writing $v = v_1 + \cdots + v_k$ with $v_i = P_i(x)v$, the $n = 0$ terms in (2) give

$$\|v\|_{x,\varepsilon}^2 \geq k \sum_{i=1}^k \|v_i\|^2 \geq \left(\sum_{i=1}^k \|v_i\| \right)^2 \geq \|v\|^2. \quad (8)$$

Thus the reader-facing lower comparison

$$\boxed{\|v\| \leq \|v\|_{x,\varepsilon}} \quad (9)$$

is literally valid even though the Oseledets spaces need not be orthogonal in the background Euclidean metric.

There are only finitely many multiplicity patterns. The Oseledets splitting, its projections, and the exponent functions are measurable on each invariant stratum. Every matrix coefficient in (7) is a countable sum of measurable functions. Therefore the inner product (7), and hence its norm, depends measurably on x ; the stratumwise definitions glue measurably on $\mathcal{R}$.

Step 3: the one-step Lyapunov inequalities. Let $v \in E_i(x)$. Since $A(x)E_i(x) = E_i(Tx)$

and $\lambda_i(Tx) = \lambda_i(x)$, the cocycle identity gives

$$\begin{aligned} \|A(x)v\|_{Tx,\eta,i}^2 &= \sum_{n \in \mathbb{Z}} \|A_n(Tx)A(x)v\|^2 e^{-2\lambda_i n - 2\eta|n|} \\ &= \sum_{m \in \mathbb{Z}} \|A_m(x)v\|^2 e^{-2\lambda_i(m-1) - 2\eta|m-1|}. \end{aligned} \quad (10)$$

The elementary estimate

$$|m| - 1 \leq |m - 1| \leq |m| + 1 \quad (11)$$

shows that the weight in the last line of (10) lies between $e^{2(\lambda_i - \eta)}$ and $e^{2(\lambda_i + \eta)}$ times the corresponding weight in (2). Hence

$$e^{\lambda_i - \eta} \|v\|_{x,\eta,i} \leq \|A(x)v\|_{Tx,\eta,i} \leq e^{\lambda_i + \eta} \|v\|_{x,\eta,i}. \quad (12)$$

Because $k(Tx) = k(x)$ and (7) only multiplies the component norm by the same factor $\sqrt{k}$, (12) is exactly the same inequality for the final norm:

$$e^{\lambda_i - \eta} \|v\|_{x,\varepsilon} \leq \|A(x)v\|_{Tx,\varepsilon} \leq e^{\lambda_i + \eta} \|v\|_{x,\varepsilon}. \quad (13)$$

Since $\eta = \varepsilon/2 < \varepsilon$, (13) implies the claimed bounds with $\lambda_i \pm \varepsilon$.

Step 4: the raw norm-equivalence constant is finite and measurable. For $x \in \mathcal{R}$ put

$$C_\eta(x) := \sup_{v \neq 0} \frac{\|v\|_{x,\varepsilon}}{\|v\|}. \quad (14)$$

The numerator is a finite norm on the finite-dimensional space $\mathbb{R}^d$, so $1 \leq C_\eta(x) < \infty$ by (9) and finite-dimensional norm equivalence. Measurability follows, for example, by taking the supremum in (14) over a fixed countable dense subset of the background unit sphere. Thus

$$\|v\| \leq \|v\|_{x,\varepsilon} \leq C_\eta(x) \|v\|. \quad (15)$$

The remaining issue is not finiteness but the orbit variation of C_η .

Step 5: the logarithmic one-step ratio of C_η is integrable. Define

$$L(x) := \max_{1 \leq i \leq k(x)} |\lambda_i(x)|. \quad (16)$$

We first record that $L \in L^1(\mu)$. By *Birkhoff pointwise ergodic theorem* applied to the two nonnegative integrable functions

$$g_+(x) := \log^+ \|A(x)\|, \quad g_-(x) := \log^+ \|A(x)^{-1}\|, \quad (17)$$

the Birkhoff averages converge almost everywhere to integrable invariant functions G_+ and G_- . For $0 \neq v \in E_i(x)$,

$$\|A_n(x)v\| \leq \|v\| \exp \left(\sum_{j=0}^{n-1} g_+(T^j x) \right), \quad (18)$$

while

$$\|A_n(x)v\| \geq \frac{\|v\|}{\|A_n(x)^{-1}\|} \geq \|v\| \exp\left(-\sum_{j=0}^{n-1} g_-(T^j x)\right). \quad (19)$$

Taking normalized logarithms and using the Oseledets limit gives

$$-G_-(x) \leq \lambda_i(x) \leq G_+(x). \quad (20)$$

Therefore

$$L(x) \leq G_+(x) + G_-(x) \in L^1(\mu). \quad (21)$$

Because the exponent spaces are orthogonal in the adapted norm, (13) gives, for arbitrary $v = \sum_i v_i$,

$$e^{-(L(x)+\eta)}\|v\|_{x,\varepsilon} \leq \|A(x)v\|_{Tx,\varepsilon} \leq e^{L(x)+\eta}\|v\|_{x,\varepsilon}. \quad (22)$$

Let $w \in \mathbb{R}^d$ have background norm 1 and write $v = A(x)^{-1}w$. The upper inequality in (22) and (15) give

$$\|w\|_{Tx,\varepsilon} \leq e^{L+\eta}C_\eta(x)\|A(x)^{-1}w\| \leq e^{L+\eta}C_\eta(x)\|A(x)^{-1}\|.$$

Taking the supremum over such w yields

$$\log C_\eta(Tx) - \log C_\eta(x) \leq L(x) + \eta + \log^+ \|A(x)^{-1}\|. \quad (23)$$

Conversely, the lower inequality in (22), rewritten as $\|v\|_{x,\varepsilon} \leq e^{L+\eta}\|A(x)v\|_{Tx,\varepsilon}$, gives for background-unit v

$$\|v\|_{x,\varepsilon} \leq e^{L+\eta}C_\eta(Tx)\|A(x)v\| \leq e^{L+\eta}C_\eta(Tx)\|A(x)\|.$$

Hence

$$\log C_\eta(x) - \log C_\eta(Tx) \leq L(x) + \eta + \log^+ \|A(x)\|. \quad (24)$$

Combining (23)-(24) with (21),

$$\boxed{|\log C_\eta(Tx) - \log C_\eta(x)| \in L^1(\mu)}. \quad (25)$$

Step 6: temper the raw constant. The integrable-coboundary clause of *Temperedness lemma* applied to $r = C_\eta$ and (25) shows that

$$\lim_{|n| \rightarrow \infty} \frac{1}{|n|} \log C_\eta(T^n x) = 0 \quad (26)$$

for almost every x . Apply the envelope part of the same lemma with parameter ε and define

$$K_\varepsilon(x) := \sup_{n \in \mathbb{Z}} C_\eta(T^n x) e^{-\varepsilon|n|}. \quad (27)$$

Then K_ε is finite and measurable almost everywhere, $K_\varepsilon \geq C_\eta \geq 1$, and

$$e^{-\varepsilon} K_\varepsilon(x) \leq K_\varepsilon(Tx) \leq e^\varepsilon K_\varepsilon(x), \quad (28)$$

$$K_\varepsilon(T^n x) \leq K_\varepsilon(x) e^{\varepsilon|n|} \quad (n \in \mathbb{Z}), \quad (29)$$

and

$$\lim_{|n| \rightarrow \infty} \frac{1}{|n|} \log K_\varepsilon(T^n x) = 0. \quad (30)$$

Finally, (15) and $C_\eta \leq K_\varepsilon$ give

$$\|v\| \leq \|v\|_{x,\varepsilon} \leq K_\varepsilon(x) \|v\|. \quad (31)$$

Equations (13), (28)–(31) prove every asserted property of the Lyapunov norm and its Pesin regularity function. $\square$

Proposition F11.2. Pesin blocks and uniformization

Statement. Let $f \in C^{1+\alpha}(M)$, $\alpha > 0$, be a diffeomorphism of a compact Riemannian manifold preserving a probability measure μ , and let $\mathcal{R}_{\text{hyp}}$ be the Oseledets-regular set on which all Lyapunov exponents are nonzero. For every integer $m \geq 1$ put $\varepsilon_m = \alpha/[100(1+\alpha)m]$ and let K_m be the Pesin regularity function supplied by Lemma F11.1 for the derivative cocycle and error ε_m . Then the measurable sets

$$\Lambda_{m,\ell} := \left\{ x \in \mathcal{R}_{\text{hyp}} : \min_i |\lambda_i(x)| \geq m^{-1}, K_m(x) \leq \ell \right\}, \quad \ell \in \mathbb{N},$$

cover $\mathcal{R}_{\text{hyp}}$. On each $\Lambda_{m,\ell}$ the Lyapunov splitting has a uniform angle bound, the stable and unstable derivative rates in Lyapunov norm are uniformly separated from one, and $K_m(f^n x) \leq \ell e^{\varepsilon_m|n|}$. There is also a block-dependent radius $\rho_{m,\ell} > 0$ such that Lyapunov charts of radius at least $\rho_{m,\ell}$ have uniformly small C^1 nonlinear remainder and the prescribed stable/unstable linear rates. Finally, for every m, ℓ and $\delta > 0$ there is a compact subset $\Lambda_{m,\ell,\delta} \subset \Lambda_{m,\ell}$ with $\mu(\Lambda_{m,\ell} \setminus \Lambda_{m,\ell,\delta}) < \delta$ on which the exponent data, splitting, Lyapunov coordinate changes and chart-radius function are continuous. Thus the hyperbolic regular set is exhausted modulo μ by compact Pesin blocks carrying uniform quantitative constants.

Proof. Let $\mathcal{R}_{\text{hyp}}$ denote the full set of Oseledets-regular points at which every Lyapunov exponent of Df is nonzero. The set need not have full μ -measure unless μ is hyperbolic; the proposition concerns precisely this hyperbolic regular part.

For $x \in \mathcal{R}_{\text{hyp}}$ write

$$T_x M = E_1(x) \oplus \cdots \oplus E_{k(x)}(x), \quad \lambda_1(x) > \cdots > \lambda_{k(x)}(x), \quad (1)$$

and define

$$\chi(x) := \min_{1 \leq i \leq k(x)} |\lambda_i(x)| > 0. \quad (2)$$

For $m \geq 1$ put

$$\varepsilon_m := \frac{\alpha}{100(1+\alpha)m}. \quad (3)$$

This deliberately conservative choice leaves room not only for the Lyapunov-norm error but also for the later H"older chart-rescaling loss ε_m/α ; in particular $\varepsilon_m + \varepsilon_m/\alpha < 1/(50m)$. Apply *Lyapunov norm and Pesin regularity functions* to the derivative cocycle with error ε_m . After intersecting the countably many full-measure regular sets involved, we may suppose simultaneously for every m that there are Lyapunov norms $\|\cdot\|_{x,m}$ and finite measurable functions $K_m(x) \geq 1$ satisfying

$$\|v\| \leq \|v\|_{x,m} \leq K_m(x)\|v\|, \quad (4)$$

$$e^{-\varepsilon_m} K_m(x) \leq K_m(fx) \leq e^{\varepsilon_m} K_m(x), \quad K_m(f^n x) \leq K_m(x) e^{\varepsilon_m |n|}, \quad (5)$$

and, for $v \in E_i(x)$,

$$e^{\lambda_i(x) - \varepsilon_m} \|v\|_{x,m} \leq \|Df_x v\|_{fx,m} \leq e^{\lambda_i(x) + \varepsilon_m} \|v\|_{x,m}. \quad (6)$$

Distinct $E_i(x)$ are orthogonal in $\|\cdot\|_{x,m}$.

Step 1: measurable blocks exhaust the hyperbolic regular set. For integers $m, \ell \geq 1$ define

$$\Lambda_{m,\ell} := \{x \in \mathcal{R}_{\text{hyp}} : \chi(x) \geq m^{-1}, K_m(x) \leq \ell\}. \quad (7)$$

The exponent functions and K_m are measurable, so every $\Lambda_{m,\ell}$ is measurable. If $x \in \mathcal{R}_{\text{hyp}}$, then $\chi(x) > 0$; choose m so large that $m^{-1} \leq \chi(x)$. For this m the number $K_m(x)$ is finite, so some integer ℓ satisfies $K_m(x) \leq \ell$. Hence

$$\mathcal{R}_{\text{hyp}} = \bigcup_{m=1}^{\infty} \bigcup_{\ell=1}^{\infty} \Lambda_{m,\ell} \quad (8)$$

up to the null set discarded when the countably many conclusions of *Lyapunov norm and Pesin regularity functions* were made simultaneous.

The blocks are not expected to be invariant. What replaces invariance is the controlled deterioration supplied by (5): if $x \in \Lambda_{m,\ell}$, then

$$K_m(f^n x) \leq \ell e^{\varepsilon_m |n|} \quad (n \in \mathbb{Z}). \quad (9)$$

This is the standard "tempered loss along the orbit" built into Pesin blocks.

Step 2: uniform stable and unstable rates. Let

$$E^s(x) := \bigoplus_{\lambda_i(x) < 0} E_i(x), \quad E^u(x) := \bigoplus_{\lambda_i(x) > 0} E_i(x). \quad (10)$$

Because the exponent spaces are orthogonal in the Lyapunov norm, (6) gives for $v^s \in E^s(x)$

$$\|Df_x v^s\|_{fx,m} \leq e^{-m^{-1} + \varepsilon_m} \|v^s\|_{x,m} \leq e^{-9/(10m)} \|v^s\|_{x,m}, \quad (11)$$

and for $v^u \in E^u(x)$

$$\|Df_x v^u\|_{fx,m} \geq e^{m^{-1} - \varepsilon_m} \|v^u\|_{x,m} \geq e^{9/(10m)} \|v^u\|_{x,m}. \quad (12)$$

Equivalently,

$$\|Df_x^{-1}w^u\|_{x,m} \leq e^{-9/(10m)}\|w^u\|_{fx,m} \quad (w^u \in E^u(fx)). \quad (13)$$

Thus each fixed m gives a uniform hyperbolicity margin in the adapted metric.

Step 3: a quantitative angle bound in the background metric. Fix $x \in \Lambda_{m,\ell}$. Let $u \in E^s(x)$ and $v \in E^u(x)$ be background-unit vectors, and replace v by $-v$ if necessary so that the background angle $\theta = \angle(u, v)$ lies in $[0, \pi/2]$. Orthogonality in the Lyapunov norm and (4) imply

$$\ell^2\|u - v\|^2 \geq \|u - v\|_{x,m}^2 = \|u\|_{x,m}^2 + \|v\|_{x,m}^2 \geq 2. \quad (14)$$

Since $\|u - v\| = 2 \sin(\theta/2)$,

$$\boxed{\theta \geq 2 \arcsin\left(\frac{1}{\sqrt{2}\ell}\right) > 0.} \quad (15)$$

Exactly the same argument applies to any two complementary sums of distinct Oseledets spaces. Thus the relevant splitting angles are uniformly bounded away from zero on $\Lambda_{m,\ell}$.

Step 4: uniform Lyapunov-chart radii. We now use the $C^{1+\alpha}$ hypothesis. Compactness of M gives a finite family of normal coordinate charts of a common radius $r_0 > 0$ on which the coordinate representations of f and f^{-1} have uniformly α -Hölder first derivative. Consequently there is a constant $H \geq 1$, depending only on f and the chosen background charts, such that whenever z is in such a chart,

$$\|D\hat{f}_x(z) - D\hat{f}_x(0)\| \leq H|z|^\alpha. \quad (16)$$

Here $\hat{f}_x$ denotes the background-coordinate representation centered at x and fx ; shrinking r_0 if necessary absorbs the uniformly bounded derivatives of the exponential charts into H .

Choose a linear map

$$C_x^{(m)} : \mathbb{R}^d \longrightarrow T_x M \quad (17)$$

that is an isometry from the Euclidean norm to the Lyapunov norm $\|\cdot\|_{x,m}$ and sends the coordinate blocks to the Oseledets spaces. From (4),

$$\|C_x^{(m)}\| \leq 1, \quad \|(C_x^{(m)})^{-1}\| \leq K_m(x). \quad (18)$$

In the corresponding Lyapunov coordinates the local map is

$$F_x := (C_{fx}^{(m)})^{-1} \circ \hat{f}_x \circ C_x^{(m)}. \quad (19)$$

Its derivative at the origin preserves the coordinate Oseledets blocks and, by (11)–(13), has the stated uniform stable/unstable rates. Moreover (16) and (18) give, for $|z|$ within the common background chart,

$$\|DF_x(z) - DF_x(0)\| \leq HK_m(fx)|z|^\alpha. \quad (20)$$

For $x \in \Lambda_{m,\ell}$, (5) implies $K_m(fx) \leq e^{\varepsilon_m} \ell$. Fix, for definiteness,

$$\delta_m := \frac{1 - e^{-1/(5m)}}{20} > 0 \quad (21)$$

and set

$$\rho_{m,\ell} := \min \left\{ r_0, \left(\frac{\delta_m}{H e^{\varepsilon_m} \ell} \right)^{1/\alpha} \right\} > 0. \quad (22)$$

Then every $x \in \Lambda_{m,\ell}$ has a Lyapunov chart of radius at least $\rho_{m,\ell}$ on which

$$\|DF_x(z) - DF_x(0)\| \leq \delta_m. \quad (23)$$

The particular numerical choice (21) is not important; what matters is that it depends only on the hyperbolicity margin m^{-1} and leaves a strictly positive gap after the nonlinear perturbation. Equations (11)–(13), (15), (22), and (23) are the promised uniform block constants.

A pointwise admissible radius can similarly be chosen as

$$\rho_m(x) := \min \left\{ r_0, \left(\frac{\delta_m}{H K_m(fx)} \right)^{1/\alpha} \right\}. \quad (24)$$

It is measurable. Since K_m is slowly varying, its logarithmic orbit variation is controlled; after applying the elementary tempering-envelope construction if one wants a symmetric one-step bound, ρ_m may be chosen tempered as well. On $\Lambda_{m,\ell}$, (24) is bounded below by (22).

Step 5: compact blocks with continuous data. The Grassmannians of subspaces of TM are compact metric bundles, and the Oseledets subspaces, exponent functions, K_m , and the radius function ρ_m are measurable after restricting to each of the finitely many multiplicity/signature strata. On each such stratum choose measurable orthonormal frames for the Oseledets subbundles (for example by measurable Gram-Schmidt applied to a countable family of background sections); this makes the coordinate changes $C_x^{(m)}$ measurable as well. Fix m, ℓ and $\delta > 0$. Apply Lusin's theorem on each stratum and then take a finite union after discarding total measure less than $\delta/2$. By inner regularity of μ on the compact metric space M , shrink once more to a compact set

$$\Lambda_{m,\ell,\delta} \subset \Lambda_{m,\ell} \quad (25)$$

with

$$\mu(\Lambda_{m,\ell} \setminus \Lambda_{m,\ell,\delta}) < \delta \quad (26)$$

on which all the listed measurable data are continuous.

The quantitative inequalities proved in Steps 2–4 remain valid on every such compact subset with the same constants, since those inequalities already held on the larger measurable block. For each fixed pair (m, ℓ) choose a sequence of such compact subsets with discarded measure tending to zero. The union of that sequence covers $\Lambda_{m,\ell}$ modulo μ . Enumerating the countable family of pairs (m, ℓ) and taking the union of all these compact approximants therefore covers $\mathcal{R}_{\text{hyp}}$ modulo μ by (8). No nesting in m

is asserted or needed. This proves the stated uniformization. $\square$

Theorem F11.3. Pesin local stable/unstable manifold theorem

Statement. Let $f \in C^{1+\alpha}(M)$, $\alpha > 0$, be a diffeomorphism of a compact Riemannian manifold preserving a probability measure μ . There is a full-measure invariant Pesin-regular subset of the Oseledets set with the following property. If x belongs to this subset and has no zero Lyapunov exponent, put

$$E^s(x) = \bigoplus_{\lambda_i(x) < 0} E_i(x), \quad E^u(x) = \bigoplus_{\lambda_i(x) > 0} E_i(x).$$

If $E^s(x) \neq \{0\}$, set $\chi_s(x) = \min_{\lambda_i < 0} |\lambda_i(x)|$. For every $0 < \vartheta_s < \chi_s(x)$ there are an embedded local C^1 disk $W_{\text{loc}}^s(x)$ tangent at x to $E^s(x)$, locally forward invariant, and a constant $C_{x,s} \geq 1$ such that

$$d(f^n y, f^n x) \leq C_{x,s} e^{-(\chi_s(x) - \vartheta_s)n} d(y, x) \quad (y \in W_{\text{loc}}^s(x), n \geq 0).$$

If $E^u(x) \neq \{0\}$, set $\chi_u(x) = \min_{\lambda_i > 0} \lambda_i(x)$. For every $0 < \vartheta_u < \chi_u(x)$ there are an embedded local C^1 disk $W_{\text{loc}}^u(x)$ tangent at x to $E^u(x)$, locally backward invariant, and a constant $C_{x,u} \geq 1$ such that

$$d(f^{-n} y, f^{-n} x) \leq C_{x,u} e^{-(\chi_u(x) - \vartheta_u)n} d(y, x) \quad (y \in W_{\text{loc}}^u(x), n \geq 0).$$

Thus the total stable/unstable contraction rates may be chosen arbitrarily close to the weakest negative/positive Lyapunov exponents. On every Pesin block from Proposition F11.2 the disk radii, splitting angles, graph-Lipschitz bounds, chart constants and some exponential contraction/expansion constants may be chosen uniformly. For prescribed rate losses ϑ_s, ϑ_u , the same uniform conclusion holds after choosing/refining the Pesin block with Lyapunov-error parameter sufficiently small relative to the requested losses and α .

Scope note

The displayed rates control the total stable and unstable bundles; sharper rates attached to individual faster Lyapunov subspaces require a refined flag/manifold statement.

Proof. We prove the stable assertion. The unstable assertion follows by applying the same construction to f^{-1} .

Let x lie in the invariant full-measure set on which *Lyapunov norm and Pesin regularity functions* and its countably many rational-error versions hold. Assume first that $E^s(x) \neq \{0\}$ and $E^u(x) \neq \{0\}$. Fix a stable exponent tolerance ϑ_s with $0 < \vartheta_s < \chi_s(x)$. Choose $\delta > 0$ so small that

$$2\delta + \frac{3\delta}{\alpha} < \vartheta_s, \quad \delta + \frac{\delta}{\alpha} < \frac{1}{4} \min\{\chi_s(x), \chi_u(x)\}. \quad (1)$$

Put

$$\tau := \frac{\delta}{\alpha}. \quad (2)$$

Apply *Lyapunov norm and Pesin regularity functions* with error δ . Along the orbit $x_n = f^n x$ we obtain Lyapunov norms $|\cdot|_n$ and a δ -slowly varying comparison function $K_n := K_\delta(x_n)$ such that

$$\|v\| \leq |v|_n \leq K_n \|v\|, \quad e^{-\delta} K_n \leq K_{n+1} \leq e^\delta K_n, \quad (4)$$

and, on the stable and unstable sums,

$$|Df_{x_n} v^s|_{n+1} \leq e^{-\chi_s(x)+\delta} |v^s|_n, \quad |Df_{x_n}^{-1} v^u|_n \leq e^{-\chi_u(x)+\delta} |v^u|_{n+1}. \quad (5)$$

The two sums are orthogonal in the Lyapunov inner product.

Step 1: slowly varying nonlinear chart radii. Because $f \in C^{1+\alpha}$ and M is compact, choose a common normal-chart radius $r_* > 0$ and $H \geq 1$ such that every background-coordinate map $\hat{f}_y$ centered at y and f_y satisfies

$$\|D\hat{f}_y(z) - D\hat{f}_y(0)\| \leq H\|z\|^\alpha \quad (6)$$

whenever $\|z\| \leq r_*$. We also shrink r_* once so that the source ball is mapped inside the target background normal chart; compactness makes this choice uniform in y .

For every regular y on the orbit choose a Lyapunov isometry $C_y : \mathbb{R}^d \rightarrow T_y M$ carrying the Euclidean orthogonal splitting $S_y \oplus U_y$ to $E^s(y) \oplus E^u(y)$. As in *Pesin blocks and uniformization*,

$$\|C_y\| \leq 1, \quad \|C_y^{-1}\| \leq K_\delta(y). \quad (7)$$

In these coordinates set

$$F_y := C_{f_y}^{-1} \circ \hat{f}_y \circ C_y. \quad (8)$$

Then $F_y(0) = 0$, $L_y := DF_y(0)$ preserves $S_y \oplus U_y$, and (5) is the Euclidean estimate for L_y .

We shall choose a nonlinear Lipschitz constant $\beta > 0$ later. Define

$$r(y) := \min \left\{ r_*, \left(\frac{\beta e^{-\tau}}{H K_\delta(fy)} \right)^{1/\alpha} \right\}. \quad (9)$$

The clipping by the constant r_* does not destroy multiplicative slow variation: from (4),

$$e^{-\tau} r(y) \leq r(fy) \leq e^\tau r(y). \quad (10)$$

Indeed the unclipped second term changes by at most the factor $e^{\pm\delta/\alpha} = e^{\pm\tau}$, and taking the minimum with the same positive constant preserves those two inequalities.

Write $r_n = r(x_n)$ and rescale every moving chart to unit size:

$$G_n(z) := r_{n+1}^{-1} F_{x_n}(r_n z), \quad |z| \leq 1. \quad (11)$$

Its linear part is

$$B_n := DG_n(0) = \frac{r_n}{r_{n+1}} L_{x_n}. \quad (12)$$

By (5) and (10),

$$a_s := e^{-\chi_s(x)+\delta+\tau} < 1, \quad a_u := e^{-\chi_u(x)+\delta+\tau} < 1, \quad (13)$$

and

$$\|B_n|_{S_n}\| \leq a_s, \quad \|(B_n|_{U_n})^{-1}\| \leq a_u. \quad (14)$$

Moreover, for $|z|, |z'| \leq 1$, (6)-(10) imply

$$\|DG_n(z) - DG_n(0)\| \leq \frac{r_n}{r_{n+1}} HK_{n+1} r_n^\alpha \leq \beta. \quad (15)$$

Thus, writing

$$G_n(z) = B_n z + R_n(z), \quad (16)$$

we have $R_n(0) = 0$, $DR_n(0) = 0$, and

$$|R_n(z) - R_n(z')| \leq \beta |z - z'| \quad (17)$$

on the unit ball, uniformly in n . The same Hölder estimate applied at two points gives, after decreasing the background chart radius if necessary,

$$\|DR_n(z) - DR_n(z')\| \leq \beta |z - z'|^\alpha, \quad (18)$$

so the sequence operator below is genuinely C^1 , not merely Lipschitz.

Choose now

$$a_s < b := e^{-\chi_s(x)+2\delta+2\tau} < 1. \quad (19)$$

The second inequality in (1) ensures $b < 1$. Let

$$Q_b := \frac{1}{b - a_s} + \frac{a_u}{1 - a_u b}. \quad (20)$$

Decrease β , and therefore the radii (9), so that

$$Q_b \beta < \frac{1}{4}. \quad (21)$$

Step 2: the nonautonomous Lyapunov-Perron operator. For $0 \leq i < j$ let $B_{i,j}^s$ and $B_{i,j}^u$ denote the stable and unstable products of the B_n . From (14),

$$\|B_{i,j}^s\| \leq a_s^{j-i}, \quad \|(B_{i,j}^u)^{-1}\| \leq a_u^{j-i}. \quad (22)$$

Let P_n^s, P_n^u be the Euclidean orthogonal projections for $S_n \oplus U_n$.

Fix a stable coordinate $\xi \in S_0$. On sequences $z = (z_n)_{n \geq 0}$ use the weighted norm

$$\|z\|_b := \sup_{n \geq 0} b^{-n} |z_n|. \quad (23)$$

For a fixed small number $R \in (0, 1/4)$, work on the closed ball $\mathcal{B}_R = \{z : \|z\|_b \leq R\}$. Every $z \in \mathcal{B}_R$ satisfies $|z_n| \leq R b^n < 1$, so all $R_n(z_n)$ in the formulas below are evaluated inside their domains.

The orbit equation $z_{n+1} = G_n(z_n)$ together with forward exponential decay is equiva-

lent to

$$\begin{aligned} (\Phi_\xi z)_n^s &= B_{0,n}^s \xi + \sum_{j=0}^{n-1} B_{j+1,n}^s P_{j+1}^s R_j(z_j), \\ (\Phi_\xi z)_n^u &= - \sum_{j=n}^{\infty} (B_{n,j+1}^u)^{-1} P_{j+1}^u R_j(z_j). \end{aligned} \quad (24)$$

The first line is forward variation of constants. The second solves the unstable recurrence backwards from the requirement that the unstable component not grow in positive time.

For $z, z' \in \mathcal{B}_R$, (17) and (22) give exactly

$$\begin{aligned} b^{-n} |(\Phi_\xi z - \Phi_\xi z')_n^s| &\leq \frac{\beta}{b - a_s} \|z - z'\|_b, \\ b^{-n} |(\Phi_\xi z - \Phi_\xi z')_n^u| &\leq \frac{a_u \beta}{1 - a_u b} \|z - z'\|_b. \end{aligned} \quad (25)$$

Hence

$$\|\Phi_\xi z - \Phi_\xi z'\|_b \leq Q_b \beta \|z - z'\|_b. \quad (26)$$

By (21) the contraction constant is less than $1/4$.

Also $\|\Phi_0(0)\|_b = 0$, and the parameter term satisfies

$$\sup_n b^{-n} |B_{0,n}^s \xi| \leq |\xi|. \quad (27)$$

Thus if $|\xi| \leq R/2$, then (25)–(27) imply $\Phi_\xi(\mathcal{B}_R) \subset \mathcal{B}_R$. Banach's theorem gives a unique fixed sequence

$$z(\xi) = (z_n(\xi))_{n \geq 0} \in \mathcal{B}_R. \quad (28)$$

Moreover, subtracting the fixed-point equations for ξ and 0 yields

$$\|z(\xi)\|_b \leq \frac{1}{1 - Q_b \beta} |\xi| =: C_0 |\xi|. \quad (29)$$

Step 3: the stable graph is C^1 and tangent to $E^s(x)$. At time zero the stable equation in (24) says $P_0^s z_0(\xi) = \xi$. Define

$$h_x(\xi) := P_0^u z_0(\xi), \quad |\xi| < R/2. \quad (30)$$

Because $2R < 1$, the same formulas define a C^1 map on the open sequence ball $\|z\|_b < 2R$ for $|\xi| < R$; the fixed points constructed above lie in the smaller closed ball $\mathcal{B}_R$. The derivative series converge uniformly by the same geometric estimates as (25), and (18) gives continuity of the derivative in the sequence variable. *Parameter-dependent contraction theorem* therefore implies that $z(\xi)$, and hence h_x , is C^1 . At $\xi = 0$ the fixed point is the zero sequence. Differentiating (24) there, using $DR_n(0) = 0$, shows that the derivative of the unstable component at time zero is zero:

$$h_x(0) = 0, \quad Dh_x(0) = 0. \quad (31)$$

Thus the graph is tangent to S_0 .

Return to the manifold by

$$\Psi_x(z) := \exp_x(C_x(r_0 z)), \quad r_0 := r(x), \quad (32)$$

and set, after reducing the parameter radius from $R/2$ to a smaller fixed $r_s > 0$ if necessary,

$$W_{\text{loc}}^s(x) := \{\Psi_x(\xi + h_x(\xi)) : \xi \in S_0, |\xi| < r_s\}. \quad (33)$$

For r_s sufficiently small, (32) is injective on the graph. Hence (33) is an embedded C^1 disk and

$$T_x W_{\text{loc}}^s(x) = E^s(x). \quad (34)$$

Step 4: local invariance. The fixed sequence satisfies the genuine orbit relation $z_{n+1} = G_n(z_n)$. Its tail $(z_{n+1})_{n \geq 0}$ satisfies the same Lyapunov-Perron equations based at $x_1 = fx$. By uniqueness, that tail is the stable fixed sequence corresponding to its stable coordinate $P_1^s z_1$.

We only need to make sure this new stable coordinate stays inside the chosen parameter disk. From (29), the fixed point satisfies $|z_0(\xi)| \leq C_0 |\xi|$. Therefore, using (14), (17), and $R_0(0) = 0$,

$$|P_1^s z_1| \leq a_s |\xi| + \beta |z_0(\xi)| \leq (a_s + \beta C_0) |\xi|. \quad (35)$$

Since $Q_b \beta < 1/4$ gives $C_0 \leq 4/3$, we may decrease β once more so that $a_s + \beta C_0 < 1$. Then any sufficiently small fixed parameter radius r_s is mapped strictly into itself. Consequently

$$\boxed{f(W_{\text{loc}}^s(x)) \subset W_{\text{loc}}^s(fx)} \quad (36)$$

whenever both sides are interpreted with these compatible Pesin charts.

Step 5: contraction in the manifold metric. Equation (29) gives

$$|z_n(\xi)| \leq C_0 b^n |\xi|. \quad (37)$$

Undoing the scale in (11), the Lyapunov-coordinate displacement at time n is $r_n z_n$. From (10),

$$r_n \leq r_0 e^{\tau n} \quad (n \geq 0). \quad (38)$$

Since $Dh_x(0) = 0$, reduce r_s once more so that $\|Dh_x\| \leq 1$ on the stable parameter ball. The exponential chart is bi-Lipschitz on the chosen small background radius, and (7) says that the smallest background singular value of C_x is at least $K_\delta(x)^{-1}$. Therefore there are finite constants $c_x, C'_x > 0$ such that

$$c_x r_0 |\xi| \leq d(\Psi_x(\xi + h_x(\xi)), x) \leq C'_x r_0 |\xi|. \quad (39)$$

At time n , $\|C_{x_n}\| \leq 1$, so (37)–(38) and the upper bi-Lipschitz chart bound give

$$d(f^n y, f^n x) \leq C''_x r_0 e^{\tau n} C_0 b^n |\xi|. \quad (40)$$

Combining (39)–(40) produces a finite constant $C_x \geq 1$ with

$$d(f^n y, f^n x) \leq C_x (be^\tau)^n d(y, x). \quad (41)$$

By (19),

$$be^\tau = e^{-\chi_s(x) + 2\delta + 3\tau}. \quad (42)$$

The first inequality in (1) gives $2\delta + 3\tau < \vartheta_s$, and hence

$$be^\tau \leq e^{-(\chi_s(x) - \vartheta_s)}. \quad (43)$$

Therefore

$$\boxed{d(f^n y, f^n x) \leq C_x e^{-(\chi_s(x) - \vartheta_s)n} d(y, x) \quad (n \geq 0).} \quad (44)$$

This is the claimed rate arbitrarily close to the weakest negative exponent.

Step 6: unstable manifold. Fix any $0 < \vartheta_u < \chi_u(x)$ and apply Steps 1–5 to f^{-1} , choosing its auxiliary δ from the analogue of (1) with ϑ_u . The positive Lyapunov exponents of f become the negative exponents of f^{-1} , *Lyapunov norm and Pesin regularity functions* applied to the derivative cocycle of f^{-1} supplies the corresponding slowly varying Lyapunov norms and regularity functions, and the construction gives an embedded C^1 disk $W_{\text{loc}}^u(x)$ tangent to $E^u(x)$ such that

$$f^{-1}(W_{\text{loc}}^u(x)) \subset W_{\text{loc}}^u(f^{-1}x) \quad (45)$$

and

$$\boxed{d(f^{-n}y, f^{-n}x) \leq C_x e^{-(\chi_u(x) - \vartheta_u)n} d(y, x) \quad (n \geq 0).} \quad (46)$$

If $E^s(x) = \{0\}$, take $W_{\text{loc}}^s(x) = \{x\}$ and omit the stable estimate; if $E^u(x) = \{0\}$, do the analogous thing for the unstable disk. If one side is nonzero and the other zero, the above construction simplifies by deleting the corresponding sums in (24).

Step 7: uniformity on Pesin blocks. Fix a block $\Lambda_{m,\ell}$ from *Pesin blocks and uniformization*. Its normalization $\varepsilon_m = \alpha/[100(1 + \alpha)m]$ was chosen precisely so that both the Lyapunov error and the Hölder radius-rescaling loss are small compared with the exponent margin $1/m$. Repeating Steps 1–4 with $\delta = \varepsilon_m$ therefore gives constants $a_s, a_u < 1$, a nonlinear bound β , a graph radius, graph-Lipschitz bound, and an exponential contraction factor depending only on m, ℓ, α and the global $C^{1+\alpha}$ bounds of f . *Pesin blocks and uniformization* supplies the uniform lower chart radius and angle bound. Hence the disk radii, angles, graph bounds, chart constants, and some stable/unstable exponential estimates are uniform on the entire block.

If one prescribes smaller rate losses $\vartheta_s, \vartheta_u > 0$, choose a Lyapunov error δ satisfying the corresponding analogues of (1) uniformly relative to the block exponent margin and refine the block by imposing an upper bound on the corresponding K_δ . The countable-block construction of *Pesin blocks and uniformization* applies verbatim to this error parameter. On that refinement the rate in (44) (and its unstable analogue) is uniform with the prescribed rate losses.

The theorem only claims uniform constants on the blocks; no continuity of the leaf family in the base point is needed here. Such continuity on suitably chosen compact

Pesin sets can be developed later if required for absolute continuity or holonomy arguments.

This proves every assertion. $\square$

Theorem F11.4. Absolute continuity of Pesin stable/unstable laminations

Statement. Let $f \in C^{1+\alpha}(M)$, $\alpha > 0$, be a diffeomorphism of a compact Riemannian manifold preserving a probability measure μ , and let Λ be a compact Pesin block contained in one fixed hyperbolic Oseledets-index stratum, so that $s = \dim E^s$ and $u = \dim E^u$ are constant on Λ . After shrinking to a sufficiently small block-dependent product neighborhood U , the local stable disks $\{W_{\text{loc}}^s(z) : z \in \Lambda \cap U\}$ form a lamination of their union. Let $T^1, T^2 \subset U$ be C^1 disks of dimension u whose tangent spaces are uniformly transverse to the stable cones and such that every stable plaque in the chosen foliation box meets each T^i exactly once. On the trace

$$A^1 := T^1 \cap \bigcup_{z \in \Lambda \cap U} W_{\text{loc}}^s(z),$$

the stable holonomy $h^s : A^1 \rightarrow A^2 \subset T^2$, obtained by following local stable plaques, is absolutely continuous with respect to the induced Riemannian volumes m_{T^1}, m_{T^2} . There is a constant $C_\Lambda \geq 1$, depending only on the quantitative constants of the block and on fixed transversality/ C^1 bounds for the allowed transversals, such that its Radon–Nikodym Jacobian satisfies

$$C_\Lambda^{-1} \leq Jh^s(y) \leq C_\Lambda \quad \text{for } m_{T^1}\text{-a.e. } y \in A^1.$$

Equivalently, for every Borel set $B \subset A^1$,

$$C_\Lambda^{-1} m_{T^1}(B) \leq m_{T^2}(h^s B) \leq C_\Lambda m_{T^1}(B).$$

The analogous statement holds for unstable holonomy, with f^{-1} in place of f and transversals of dimension s .

Proof. We prove the stable assertion. Applying the same argument to f^{-1} proves absolute continuity of the unstable lamination.

Fix a compact Pesin block Λ contained in one fixed hyperbolic Oseledets-index stratum; write $s = \dim E^s$ and $r = \dim E^u$, which are therefore constant on Λ . We shall use only the quantitative conclusions already obtained in *Pesin blocks and uniformization* and *Pesin local stable/unstable manifold theorem*. After refining the block once, if necessary, there are constants

$$0 < a < 1, \quad \rho > 0, \quad K \geq 1, \quad \gamma_0 > 0 \tag{1}$$

with the following uniform properties. Every $z \in \Lambda$ has a Lyapunov chart of radius at least ρ ; in these charts the stable and unstable subspaces are orthogonal, the angle between their images in the background metric is at least γ_0 , and the local stable and unstable disks have uniform graph size and graph slope. By choosing the Lyapunov error in the block sufficiently small, the one-step graph transforms in the stable and

unstable cones have contraction factor at most a .

The constants below may change when the product neighborhood is shrunk, but all of them depend only on this fixed block, the global $C^{1+\alpha}$ norm of f , and the fixed transversality and C^1 bounds imposed on the two transversals.

Step 1: a local stable product box. Choose $x_0 \in \Lambda$. On a sufficiently small background coordinate ball U about x_0 , identify every tangent space with $\mathbb{R}^d$ by a smooth orthonormal trivialization. The continuous splitting on the compact block from *Pesin blocks and uniformization* then lies in fixed transverse cones

$$C^s \quad \text{and} \quad C^u \tag{2}$$

around $E^s(x_0)$ and $E^u(x_0)$.

The Lyapunov–Perron operator used in *Pesin local stable/unstable manifold theorem* has, on this compact block, a contraction constant bounded uniformly below 1. Its coefficients (the base point, derivative blocks, chart changes and nonlinear remainders) vary continuously on the compact Lusin block. *Parameter-dependent contraction theorem* therefore implies continuous dependence of its fixed point, and of the derivative of the fixed point, on the base point. Consequently, after shrinking U , the local stable disks meeting U are C^1 graphs with a common graph radius and slope bound and vary continuously in the C^1 topology. Distinct stable disks cannot cross: if two of them met, forward contraction and uniqueness in the local stable-manifold construction would identify their germs, and hence their plaques in U . Thus

$$\mathcal{F}_\Lambda^s(U) := \bigcup_{z \in \Lambda \cap U} W_{\text{loc}}^s(z) \cap U \tag{3}$$

is a lamination of its (not necessarily open) union.

A C^1 disk tangent to the unstable cone and sufficiently small meets each stable plaque in at most one point. By the product-box hypothesis in the statement, each plaque under consideration meets both T^1 and T^2 . Stable plaques therefore define a bijection

$$h^s : A^1 \longrightarrow A^2, \quad A^i = T^i \cap \mathcal{F}_\Lambda^s(U). \tag{4}$$

The C^1 dependence just proved implies that h^s and $(h^s)^{-1}$ are continuous on their trace sets. In particular, images of Borel subsets of A^1 are Borel in A^2 .

Step 2: the finite-time graph lemma. We record explicitly the estimate that will replace the phrase “bounded distortion.” There exist numbers

$$0 < \eta < 1, \quad q_0 > 0, \quad C_0 \geq 1 \tag{5}$$

such that the following holds. Let $w \in \Lambda \cap U$, let $y_i = T^i \cap W_{\text{loc}}^s(w)$, and write

$$w_n = f^n w, \quad y_{i,n} = f^n y_i. \tag{6}$$

For every $0 < q \leq q_0$ and every sufficiently large n , there are neighborhoods $P_n^i(w, q) \subset f^n(T^i)$ of $y_{i,n}$ which, in the Lyapunov chart at w_n , are graphs over the unstable space

on a ball of radius

$$q_n = q\eta^n, \quad (7)$$

and satisfy

$$\|D\psi_n^i\| \leq C_0\eta^n. \quad (8)$$

Moreover the plaques can be chosen compatibly in time:

$$f^{-1}P_k^i(w, q) \subset P_{k-1}^i(w, q), \quad 1 \leq k \leq n. \quad (9)$$

If z_k, z'_k belong to the same $P_k^i(w, q)$ and are the k th iterates of points in its pullback, then

$$d(z_k, z'_k) \leq C_0q\eta^k \quad (10)$$

and, after parallel transport to one tangent space,

$$d_{\text{Gr}}(T_{z_k}P_k^i, T_{z'_k}P_k^i) \leq C_0(\eta_1^k + q^\alpha\eta_1^k) \quad (11)$$

for some $0 < \eta_1 < 1$ depending only on the block.

Here is the graph-transform estimate behind this lemma. In a Lyapunov chart write the derivative at a point in block form relative to the stable/unstable splitting,

$$Df = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (12)$$

A tangent plane in the unstable cone is the graph of a linear map L from the unstable to the stable coordinate. Its image is the graph

$$\mathcal{G}(L) = (AL + B)(CL + D)^{-1}, \quad (13)$$

where the denominator is invertible on the fixed cone. The hyperbolic block bounds and the small nonlinear derivative imply, after shrinking the chart,

$$\|\mathcal{G}(L)\| \leq a\|L\| + Cd(\text{base point}, w)^\alpha, \quad (14)$$

and, for two base points and two graphs,

$$\|\mathcal{G}_p(L) - \mathcal{G}_{p'}(L')\| \leq a\|L - L'\| + Cd(p, p')^\alpha. \quad (15)$$

These inequalities follow directly by subtracting (13), using $X^{-1} - Y^{-1} = X^{-1}(Y - X)Y^{-1}$, and the uniform bounds on the inverse of $CL + D$. The $C^{1+\alpha}$ hypothesis gives the last term in (14)–(15). Iterating (14) gives (8) once the initial C^1 slope bound is fixed. Applying (15) along points at distance $O(q\eta^k)$ gives

$$\Delta_{k+1} \leq a\Delta_k + Cq^\alpha\eta^{\alpha k}. \quad (16)$$

Hence

$$\Delta_k \leq a^k\Delta_0 + Cq^\alpha \sum_{j=0}^{k-1} a^{k-1-j}\eta^{\alpha j} \leq C_0(1 + q^\alpha)\eta_1^k \quad (17)$$

for any fixed $\eta_1 \in (\max\{a, \eta^\alpha\}, 1)$. This proves the tangent estimate (11). It remains only to justify the domains in (9). On the fixed unstable cone the unstable coordinate of the linear part expands by a factor at least $\nu_u > 1$, while the nonlinear derivative is as small as required. Thus, after a final chart shrink, the forward image of an unstable graph disk of radius q_{k-1} contains an unstable graph disk of radius at least $c\nu_u q_{k-1}$, with $c > 0$ uniform. Choose η in (5) still smaller if necessary so that $\eta < c\nu_u$. Since $q_k = \eta q_{k-1}$, the graph disk of radius q_k can be chosen inside the forward image; this is (9). Its graph slope is uniformly bounded, so any two points in it are at distance at most a uniform multiple of q_k , which is (10). Thus every assertion of the finite-time graph lemma follows from explicit uniform graph-transform estimates.

Step 3: summable inverse-Jacobian distortion. On the compact set of pairs (p, V) for which $p \in M$ and V is an r -plane in the fixed unstable cone, define

$$\Phi(p, V) := \log \text{Jac}(Df_p^{-1}|_V). \quad (18)$$

Because f is a $C^{1+\alpha}$ diffeomorphism on a compact manifold, Df^{-1} and $(Df^{-1})^{-1}$ are uniformly bounded. The determinant on an r -plane is smooth in the matrix and in the Grassmannian variable away from singularity. Therefore there is C_1 such that, after parallel transport,

$$|\Phi(p, V) - \Phi(p', V')| \leq C_1(d(p, p')^\alpha + d_{\text{Gr}}(V, V')) \quad (19)$$

throughout these cones.

For $z \in f^{-n}P_n^i(w, q)$ put $z_k = f^k z$ and define

$$H_i(z, n) := \text{Jac}(Df_{z_n}^{-n}|_{T_{z_n}P_n^i(w, q)}). \quad (20)$$

Let y_i be the center point and $y_{i,k} = f^k y_i$. By the chain rule,

$$\log \frac{H_i(z, n)}{H_i(y_i, n)} = \sum_{k=1}^n [\Phi(z_k, T_{z_k}P_k^i) - \Phi(y_{i,k}, T_{y_{i,k}}P_k^i)]. \quad (21)$$

Using (10), (11), and (19),

$$\left| \log \frac{H_i(z, n)}{H_i(y_i, n)} \right| \leq C_2 \sum_{k=1}^n (q^\alpha \eta^{\alpha k} + \eta_1^k) \leq D_0 < \infty. \quad (22)$$

Thus

$$e^{-D_0} \leq \frac{H_i(z, n)}{H_i(y_i, n)} \leq e^{D_0} \quad (23)$$

uniformly in i, n, w and z .

The two centers $y_{1,k}$ and $y_{2,k}$ lie on the same local stable plaque. *Pesin local stable/unstable manifold theorem* gives, uniformly on the block,

$$d(y_{1,k}, y_{2,k}) \leq C_3 \sigma^k \quad (24)$$

for some $0 < \sigma < 1$. By (8), the tangent spaces of both forward transversals converge

to the same unstable cone center; applying (15) to the two graph sequences yields

$$d_{\text{Gr}}(T_{y_{1,k}} P_k^1, T_{y_{2,k}} P_k^2) \leq C_4 \sigma_1^k \quad (25)$$

for some $0 < \sigma_1 < 1$. Hence another application of (19) gives a summable series and therefore

$$e^{-D_1} \leq \frac{H_1(y_1, n)}{H_2(y_2, n)} \leq e^{D_1} \quad (26)$$

uniformly in n .

Equations (23) and (26) are the quantitative heart of absolute continuity. They say that pulling a small terminal unstable-like plaque back to the original transversal produces uniformly comparable densities on the two sides.

Step 4: comparable paired plaque volumes. Because both $P_n^1(w, q)$ and $P_n^2(w, q)$ are graphs over the same r -dimensional unstable ball of radius q_n , with slopes bounded by $C_0 \eta^n$, the area formula gives a constant $C_5 \geq 1$ such that

$$C_5^{-1} \leq \frac{m_{f^n T^1}(P_n^1(w, q))}{m_{f^n T^2}(P_n^2(w, q))} \leq C_5. \quad (27)$$

Indeed the graph volume density is $\sqrt{\det(I + (D\psi)^* D\psi)}$, which lies between 1 and a uniform constant on the fixed graph cone.

The change-of-variables formula and (20) give

$$m_{T^i}(f^{-n} P_n^i) = \int_{P_n^i} H_i(f^{-n} \zeta, n) dm_{f^n T^i}(\zeta). \quad (28)$$

Using (23), replace H_i in the integral by its value at the center up to the factor $e^{\pm D_0}$; then use (26) and (27). We obtain one constant $C_6 \geq 1$ such that

$$C_6^{-1} \leq \frac{m_{T^2}(f^{-n} P_n^2(w, q))}{m_{T^1}(f^{-n} P_n^1(w, q))} \leq C_6 \quad (29)$$

for every sufficiently large n , every allowed center w , and every $0 < q \leq q_0$.

Step 5: from paired plaques to holonomy measure bounds. We now pass from (29), which compares one pair of small plaques, to arbitrary Borel subsets of the trace. We include the covering step because this is where an informal proof often hides the actual absolute-continuity argument.

Let $B \subset A^1$ be Borel and first assume that B is relatively compact in the product box. Fix a sufficiently large n . At time n , the family of terminal plaques $P_n^1(w, q)$, with w ranging over the stable plaques through points of B and with $q > 0$ arbitrarily small, is a Vitali-type cover of $f^n B$. In the C^1 coordinates on $f^n T^1$ these terminal plaques are uniformly small graphs over unstable balls and hence have uniformly bounded eccentricity. The finite-dimensional Besicovitch covering theorem therefore supplies a countable subcover $\{P_{n,j}^1\}$ of all of $f^n B$, of multiplicity at most N_r , where N_r depends only on r and the fixed cone geometry. Put

$$Q_j^1 := f^{-n} P_{n,j}^1 \subset T^1 \quad (30a)$$

and let $Q_j^2 := f^{-n}P_{n,j}^2 \subset T^2$ be the pullback of the paired terminal plaque. Because $f^{-n}|_{f^n T^1}$ is a diffeomorphism, pulling back preserves the overlap multiplicity: the family $\{Q_j^1\}$ still has multiplicity at most N_r . For fixed n , continuity of f^{-n} also lets us choose the terminal radii so small that every Q_j^1 lies in any prescribed relative neighborhood B_δ of B . Notice that we do *not* claim the pullback plaques themselves have bounded eccentricity; that unnecessary assertion would generally be false.

Stable contraction implies that, after taking n large and the terminal radius q small, the paired plaques approximate holonomy in the following precise sense: if Q_j^1 is centered at y , then every point of $h^s(B \cap Q_j^1)$ lies in the pullback $\hat{Q}_j^2 = f^{-n}\hat{P}_{n,j}^2$ of a fixed small enlargement $\hat{P}_{n,j}^2$ of the paired terminal plaque $P_{n,j}^2$. The terminal enlargement factor tends to 1 as the terminal graph slopes and stable separation tend to zero. Bounded overlap of the first terminal cover and the small displacement of paired centers imply a bounded overlap constant N'_r for the enlarged second terminal cover; pulling back preserves this multiplicity. The terminal graph-volume estimate and the same distortion calculation as in (28)–(29) show $m_{T^2}(\hat{Q}_j^2) \leq C_7 m_{T^1}(Q_j^1)$ with a uniform constant. Hence, with a constant C_7 independent of B ,

$$\begin{aligned} m_{T^2}(h^s B) &\leq \sum_j m_{T^2}(\hat{Q}_j^2) \\ &\leq C_7 \sum_j m_{T^1}(Q_j^1) \\ &\leq C_7 N_r m_{T^1}(B_\delta), \end{aligned} \tag{30}$$

where B_δ is an arbitrarily small relative neighborhood of B in T^1 . Outer regularity and then $\delta \downarrow 0$ yield

$$m_{T^2}(h^s B) \leq C_8 m_{T^1}(B). \tag{31}$$

For completeness, the approximation assertion used in (30) follows directly from the product geometry. At time n , corresponding points on one stable plaque are $O(\sigma^n)$ apart by (24), while both terminal transversals have slope $O(\eta^n)$ by (8). Projection along the stable cone from one terminal transversal to the other therefore differs from the identity in the common unstable coordinate by $O(\sigma^n + \eta^n q_n)$. Thus the image of a q_n -plaque lies between $(1 - \epsilon_n)q_n$ and $(1 + \epsilon_n)q_n$ graph plaques with $\epsilon_n \rightarrow 0$. This is exactly the enlargement used above and also shows that its overlap multiplicity remains uniform.

Apply the same argument with T^1 and T^2 interchanged. Since the inverse holonomy is the stable holonomy in the opposite direction,

$$m_{T^1}(B) \leq C_8 m_{T^2}(h^s B). \tag{32}$$

Combining (31) and (32),

$$C_8^{-1} m_{T^1}(B) \leq m_{T^2}(h^s B) \leq C_8 m_{T^1}(B) \tag{33}$$

for every Borel $B \subset A^1$ (general B follows by exhausting the product box by relatively compact subboxes).

Equation (31) already proves that h^s sends m_{T^1} -null sets to m_{T^2} -null sets. Hence the

measure

$$\nu(B) := m_{T^2}(h^s B) \quad (34)$$

is absolutely continuous with respect to $m_{T^1}|_{A^1}$. By the Radon-Nikodym theorem there is a measurable density $Jh^s = d\nu/dm_{T^1}$. The two inequalities in (33), applied to all Borel sets, imply

$$C_8^{-1} \leq Jh^s \leq C_8 \quad m_{T^1}\text{-almost everywhere on } A^1. \quad (35)$$

This is precisely the asserted stable absolute continuity and two-sided Jacobian bound.

Step 6: unstable holonomy. The inverse map f^{-1} is again $C^{1+\alpha}$ on the compact manifold. Its stable Pesin disks are the unstable Pesin disks of f , and its compact Pesin blocks have the same type of uniform angle, chart, graph and distortion bounds. Repeating Steps 1-5 for f^{-1} proves the identical statement for unstable holonomy between s -dimensional transversals.

This proves the theorem. $\square$

Theorem F11.5. Ruelle inequality

Statement. Let f be a C^1 diffeomorphism of a compact Riemannian manifold and μ an f -invariant Borel probability. Then

$$h_\mu(f) \leq \int_M \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x) d\mu(x),$$

with the right-hand side interpreted through Oseledets' theorem.

Proof. Put $d = \dim M$ and

$$L := \max \left\{ 1, \sup_{x \in M} \|D_x f\| \right\}. \quad (1)$$

Since f is C^1 and M is compact, $L < \infty$. Apply Oseledets to the derivative cocycle. At every Oseledets-regular point let

$$S(x) := \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x), \quad (2)$$

and define $S = 0$ on the null exceptional set. The Lyapunov data are measurable and invariant modulo null sets, so after changing S on a null set we may and shall take S measurable and f -invariant. Moreover $0 \leq S(x) \leq d \log L$ almost everywhere.

We shall prove

$$h_\mu(f) \leq \int_M S d\mu. \quad (3)$$

The proof is organized into five elementary steps. The entropy facts needed below are proved in the argument, so no theorem from the later thermodynamic formalism volume is being used.

Step 1: the finite-time expanding-volume quantity. Let

$$\sigma_1(A) \geq \cdots \geq \sigma_d(A) > 0$$

be the singular values of an invertible linear map A , and set

$$P(A) := \prod_{j=1}^d \max\{1, \sigma_j(A)\}. \quad (4)$$

By *Singular values and exterior powers*,

$$\|\wedge^k A\| = \sigma_1(A) \cdots \sigma_k(A). \quad (5)$$

Hence

$$\log P(A) = \max_{0 \leq k \leq d} \log \|\wedge^k A\|, \quad (6)$$

where $\wedge^0 A$ has norm 1. Indeed, as k increases the partial sum $\sum_{j \leq k} \log \sigma_j(A)$ increases exactly while $\sigma_k(A) > 1$ and decreases after the singular values drop below 1.

Apply (6) to $A = D_x f^n$. *Lyapunov spectrum from exterior powers* gives, at every regular point,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|\wedge^k D_x f^n\| = \lambda_1(x) + \cdots + \lambda_k(x) \quad (7)$$

with the Lyapunov exponents repeated according to multiplicity. Since the maximum in (6) is over only $d + 1$ values, (7) implies

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log P(D_x f^n) = S(x). \quad (8)$$

Notice that this is valid without ergodicity: $S(x)$ is allowed to vary with x .

Fix a number $\eta > 0$. For $m \geq 1$ define

$$R_m = R_m(\eta) := \left\{ x \text{ regular} : P(D_x f^n) \leq e^{n(S(x)+\eta)} \text{ for every } n \geq m \right\}. \quad (9)$$

The sets R_m are measurable and increasing, and (8) gives

$$\mu(R_m) \longrightarrow 1. \quad (10)$$

Step 2: a geometric partition and the one-step packing estimate for f^m . Fix m . We first record a uniform local consequence of differentiability of the fixed map f^m .

Choose a radius $\rho_0 > 0$ below the injectivity radius of M . Compactness allows us to choose all exponential-coordinate maps on balls of radius ρ_0 with uniformly bounded metric and volume distortion. For $x \in M$ write

$$F_x(v) := \exp_{f^m x}^{-1}(f^m(\exp_x v)).$$

Uniform continuity of $D(f^m)$ on the compact manifold gives a number $r_m > 0$ such that, whenever $0 < r \leq r_m$ and $\|v\| \leq 3r$,

$$\|F_x(v) - D_x f^m v\| \leq r \quad (11)$$

uniformly in x . By decreasing r_m if necessary, all sets occurring in (11), together with their $3r$ -neighborhoods, remain inside the chosen normal-coordinate balls.

For such an r , take a maximal r -separated set $\Gamma = \{c_1, \dots, c_N\} \subset M$. Assign every point to a nearest center, breaking ties by the index of the center, and let $\beta = \{C_1, \dots, C_N\}$ be the resulting Borel Voronoi partition. Then

$$B(c_i, r/2) \subset C_i \subset B(c_i, r), \quad \text{diam } C_i \leq 2r. \quad (12)$$

The smaller balls $B(c_i, r/3)$ are pairwise disjoint. Uniform small-ball volume estimates on the compact Riemannian manifold give constants $c_0, C_0 > 0$, independent of m, r and the partition, such that

$$c_0 r^d \leq \text{vol } B(z, r/3), \quad \text{vol}(\exp_z E) \leq C_0 \text{vol}_{T_z M}(E) \quad (13)$$

for the small sets used below.

Let $T = f^m$ and let $D = T(C')$ be an atom of $T\beta$. Pick any $x \in C'$. Since $C' \subset B(x, 2r)$, (11) and the uniform metric comparison in normal coordinates show that the $2r$ -neighborhood of D is contained, after applying $\exp_{T_x}^{-1}$, in a set of the form

$$D_x f^m(B(0, Ar)) + B(0, Ar), \quad (14)$$

where $A \geq 1$ depends only on the fixed Riemannian-coordinate constants and not on m, r, x .

We need only a crude but uniform volume estimate for (14). In left singular coordinates for $D_x f^m$, the Minkowski sum in (14) lies in a rectangular box whose j -th half-side is at most $Ar(\sigma_j(D_x f^m) + 1)$. Consequently (13) gives

$$\text{vol}(D_{2r}) \leq C_1 r^d \prod_{j=1}^d (1 + \sigma_j(D_x f^m)) \leq C_1 2^d r^d P(D_x f^m), \quad (15)$$

where D_{2r} denotes the metric $2r$ -neighborhood of D .

If $C_i \cap D \neq \emptyset$, then c_i lies in the r -neighborhood of D , hence $B(c_i, r/3) \subset D_{2r}$. These $r/3$ -balls are disjoint. Combining (13) and (15), we obtain a constant $K \geq 1$, depending only on the Riemannian dimension/coordinate constants, such that

$$\#\{C \in \beta : C \cap D \neq \emptyset\} \leq KP(D_x f^m). \quad (16)$$

The crucial point is that K is independent of m, r, D , and x ; r_m may shrink with m , but the scale r^d cancels from the packing ratio.

Two consequences of (16) will be used. For arbitrary x , $\sigma_j(D_x f^m) \leq \|D_x f^m\| \leq L^m$, so

$$\#\{C : C \cap D \neq \emptyset\} \leq KL^{md}. \quad (17)$$

If, in addition, $x \in R_m$, then (9) gives the sharper estimate

$$\#\{C : C \cap D \neq \emptyset\} \leq Ke^{m(S(x)+\eta)}. \quad (18)$$

Step 3: the elementary entropy interface. We record three finite-partition facts and prove them here.

For a finite partition $\mathcal{P}$, write $H_\mu(\mathcal{P})$ for its Shannon entropy, $H_\mu(\mathcal{P} \mid \mathcal{Q})$ for conditional entropy, and $h_\mu(T, \mathcal{P})$ for the usual entropy rate.

First, for any finite partitions α, β ,

$$h_\mu(T, \alpha) \leq h_\mu(T, \beta) + H_\mu(\alpha \mid \beta). \quad (19)$$

Indeed, by the chain rule and monotonicity of conditional entropy,

$$H_\mu\left(\bigvee_{k=0}^{n-1} T^{-k}\alpha\right) \leq H_\mu\left(\bigvee_{k=0}^{n-1} T^{-k}\beta\right) + \sum_{k=0}^{n-1} H_\mu(T^{-k}\alpha \mid T^{-k}\beta).$$

Invariance makes every term in the sum equal to $H_\mu(\alpha \mid \beta)$. Divide by n and let $n \rightarrow \infty$.

Second, if ζ is a finite T -invariant partition modulo null sets, then

$$h_\mu(T, \beta) = h_\mu(T, \beta \vee \zeta) \leq H_\mu(\beta \mid T\beta \vee \zeta). \quad (20)$$

For the equality, the n -fold orbit joins of β and of $\beta \vee \zeta$ differ by at most the single partition ζ , so their entropies differ by at most $H_\mu(\zeta)$; division by n gives equality. For the inequality, since T is invertible,

$$h_\mu(T, \beta \vee \zeta) = \lim_{n \rightarrow \infty} H_\mu\left(\beta \vee \zeta \mid \bigvee_{k=1}^n T^k(\beta \vee \zeta)\right).$$

Conditional entropy decreases when the conditioning partition is refined, so this limit is at most $H_\mu(\beta \vee \zeta \mid T\beta \vee \zeta) = H_\mu(\beta \mid T\beta \vee \zeta)$.

Third,

$$h_\mu(f^m) = mh_\mu(f). \quad (21)$$

For completeness, if $\mathcal{P}$ is finite, the n -orbit join for f^m is coarser than the mn -orbit join for f , whence $h_\mu(f^m, \mathcal{P}) \leq mh_\mu(f)$ and therefore $h_\mu(f^m) \leq mh_\mu(f)$. Conversely, for a finite partition $\mathcal{A}$ set $\mathcal{Q} = \bigvee_{j=0}^{m-1} f^{-j}\mathcal{A}$. Then

$$\bigvee_{k=0}^{n-1} (f^m)^{-k}\mathcal{Q} = \bigvee_{j=0}^{mn-1} f^{-j}\mathcal{A},$$

so $h_\mu(f^m, \mathcal{Q}) = mh_\mu(f, \mathcal{A})$. Taking the supremum over $\mathcal{A}$ proves the reverse inequality in (21).

We also need the following approximation observation. Given an arbitrary finite partition $\alpha = \{A_1, \dots, A_q\}$ and $\gamma > 0$, the radius r in Step 2 may be chosen so small that the corresponding Voronoi partition β satisfies

$$H_\mu(\alpha \mid \beta) < \gamma. \quad (22)$$

To see this, choose compact sets $K_i \subset A_i$ with $\mu(M \setminus \bigcup_i K_i) < \theta$, where $\theta > 0$ will be chosen in a moment. The disjoint compact sets K_i have positive pairwise distance. If $2r$ is smaller than that distance, every atom of β meets at most one K_i . From the β -atom one can therefore predict the α -label correctly except on a set of measure at

most θ . Introducing the error indicator and applying the entropy chain rule gives the elementary finite-alphabet (Fano) estimate

$$H_\mu(\alpha \mid \beta) \leq H_b(\theta) + \theta \log(q - 1), \quad (23)$$

with the final term interpreted as 0 if $q = 1$. Since the right-hand side tends to 0 with θ , (22) follows. Thus the geometric partitions used in Step 2 can approximate the entropy rate of *any* prescribed finite partition; no generating-partition theorem is being assumed.

Step 4: condition on finite bins of the nonergodic Lyapunov data. Let

$$B := d \log L.$$

Fix $\delta > 0$. Partition the interval $[0, B]$ into finitely many half-open intervals of length δ (with the last interval truncated at B), and let $\zeta = \{Z_j\}$ be the corresponding measurable partition of M according to the value of $S(x)$. Null exceptional points may be put in any atom. Because $S \circ f = S$ almost everywhere, ζ is invariant modulo μ for both f and $T = f^m$.

For each bin Z_j let b_j be its upper endpoint, enlarged to the next multiple of δ if necessary. Thus

$$x \in Z_j \implies S(x) \leq b_j \leq S(x) + \delta, \quad (24)$$

and hence

$$\sum_j b_j \mu(Z_j) \leq \int_M S d\mu + \delta. \quad (25)$$

Apply (20) to the geometric partition β of Step 2. An atom of the conditioning partition $T\beta \vee \zeta$ has the form $E = D \cap Z_j$, where $D = T(C')$ for a unique atom $C' \in \beta$. The conditional distribution of β on E is supported on at most

$$N(D) := \#\{C \in \beta : C \cap D \neq \emptyset\}$$

atoms. A probability vector supported on at most N points has entropy at most $\log N$. Therefore

$$h_\mu(T, \beta) \leq \sum_{D,j} \mu(D \cap Z_j) \log N(D). \quad (26)$$

Call the pair (D, j) *good* if

$$C' \cap Z_j \cap R_m \neq \emptyset. \quad (27)$$

For a good pair choose x in this intersection. Then (18) and (24) imply

$$\log N(D) \leq \log K + m(b_j + \eta). \quad (28)$$

For a bad pair use the crude estimate (17):

$$\log N(D) \leq \log K + md \log L. \quad (29)$$

The total μ -mass of all bad conditioning atoms is at most $\mu(M \setminus R_m)$. Indeed, since T is

invertible and Z_j is T -invariant modulo null sets,

$$\mu(D \cap Z_j) = \mu(C' \cap Z_j), \quad (30)$$

and for a bad pair $C' \cap Z_j \subset M \setminus R_m$. The sets $C' \cap Z_j$ are disjoint as (C', j) varies.

Insert (28)–(30) into (26). Since the conditioning atoms have total mass one and the mass of good atoms in the j -th bin is at most $\mu(Z_j)$, we get

$$\begin{aligned} h_\mu(T, \beta) &\leq \log K + m\eta + m \sum_j b_j \mu(Z_j) + md \log L \mu(M \setminus R_m) \\ &\leq \log K + m \left(\int_M S d\mu + \eta + \delta \right) + md \log L \mu(M \setminus R_m). \end{aligned} \quad (31)$$

This is the required nonergodic replacement for the usual step in which the Lyapunov spectrum is declared constant by ergodicity.

Step 5: remove the geometric partition, the exceptional set, and the error parameters. Let α be an arbitrary finite measurable partition and let $\gamma > 0$. Choose $r \leq r_m$ so small that the geometric partition β satisfies (22). By (19) and (31),

$$\begin{aligned} h_\mu(T, \alpha) &\leq h_\mu(T, \beta) + \gamma \\ &\leq \log K + \gamma + m \left(\int_M S d\mu + \eta + \delta \right) + md \log L \mu(M \setminus R_m). \end{aligned} \quad (32)$$

The right-hand side is independent of α . Taking the supremum over all finite α gives the same estimate for $h_\mu(T) = h_\mu(f^m)$. Using (21) and dividing by m ,

$$h_\mu(f) \leq \int_M S d\mu + \eta + \delta + d \log L \mu(M \setminus R_m) + \frac{\log K + \gamma}{m}. \quad (33)$$

Now let $m \rightarrow \infty$. By (10) the exceptional-set term tends to zero, and the last term also tends to zero. Thus

$$h_\mu(f) \leq \int_M S d\mu + \eta + \delta.$$

Finally let $\eta \downarrow 0$ and $\delta \downarrow 0$. With the definition (2) of S , this is exactly

$$h_\mu(f) \leq \int_M \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x) d\mu(x).$$

No ergodicity of μ was used. □

Theorem F11.6. Pesin entropy formula

Statement. Let f be a $C^{1+\alpha}$ diffeomorphism of a compact Riemannian manifold and let μ be an f -invariant probability absolutely continuous with respect to Rie-

mannian volume. Then

$$h_\mu(f) = \int_M \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x) d\mu(x),$$

where the $\lambda_i(x)$ are the distinct Lyapunov exponents and $m_i(x)$ their multiplicities.

Proof.

Proof strategy

The upper bound is Ruelle's inequality. For the reverse inequality we first establish the countable Shannon–McMillan–Breiman/local-entropy input needed on nonuniform Pesin blocks. We then compare dynamical Bowen sets with unstable-volume contraction and the unstable Jacobian, and finally integrate the pointwise lower bound. Keeping these three layers separate prevents measure-theoretic entropy from being conflated with the geometric Jacobian estimate.

Put

$$\chi^+(x) := \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x). \quad (1)$$

Because f is a C^1 diffeomorphism of a compact manifold, all Lyapunov exponents are bounded in absolute value by a uniform constant depending only on $\|Df\|_\infty$ and $\|Df^{-1}\|_\infty$. Thus $\chi^+ \in L^\infty(\mu)$. The Ruelle inequality, *Ruelle inequality*, gives

$$h_\mu(f) \leq \int_M \chi^+ d\mu. \quad (2)$$

It remains to prove the reverse inequality. We give the complete lower-bound argument, including the measure-theoretic local-entropy input that is sometimes hidden behind the Shannon–McMillan–Breiman theorem.

Auxiliary lemma 1: countable Shannon–McMillan–Breiman. Let $(X, \mathcal{B}, \eta, T)$ be a probability-preserving system and let $\mathcal{P}$ be a finite or countable measurable partition with $H_\eta(\mathcal{P}) < \infty$. Write

$$\mathcal{P}_1^n := \bigvee_{j=1}^n T^{-j} \mathcal{P}, \quad \mathcal{P}^n := \bigvee_{j=0}^n T^{-j} \mathcal{P}. \quad (3)$$

For a partition $\mathcal{A}$ and a sigma-algebra $\mathcal{G}$, denote by $I_\eta(\mathcal{A} | \mathcal{G})$ the conditional information function. Then

$$\frac{1}{n+1} I_\eta(\mathcal{P}^n)(x) \longrightarrow h_{\mathcal{P}}(x) \quad \text{for } \eta\text{-a.e. } x, \quad (4)$$

where $h_{\mathcal{P}}$ is T -invariant and integrable, and

$$\int_X h_{\mathcal{P}} d\eta = h_\eta(T, \mathcal{P}). \quad (5)$$

No ergodicity is assumed.

We prove this statement. For $n \geq 1$ set

$$q_n := I_\eta(\mathcal{P} \mid \sigma(\mathcal{P}_1^n)), \quad q_0 := I_\eta(\mathcal{P}). \quad (6)$$

We first need an integrable domination for (q_n) . For an atom $A \in \mathcal{P}$ put

$$q_n^A := -\log \mathbb{E}_\eta(\mathbf{1}_A \mid \sigma(\mathcal{P}_1^n)). \quad (7)$$

Fix $L > 0$ and let B_n^A be the first-crossing set on which $q_n^A > L$ but $q_j^A \leq L$ for $1 \leq j < n$. The sets B_n^A are disjoint and belong to $\sigma(\mathcal{P}_1^n)$. Therefore

$$\begin{aligned} \eta(B_n^A \cap A) &= \int_{B_n^A} \mathbb{E}_\eta(\mathbf{1}_A \mid \sigma(\mathcal{P}_1^n)) d\eta \\ &= \int_{B_n^A} e^{-q_n^A} d\eta \leq e^{-L} \eta(B_n^A). \end{aligned} \quad (8)$$

Summing over n gives

$$\eta\{x \in A : \sup_{n \geq 1} q_n(x) > L\} \leq e^{-L}. \quad (9)$$

The left side is also at most $\eta(A)$. Hence, with $q^* := \sup_{n \geq 1} q_n$,

$$\begin{aligned} \int_A q^* d\eta &\leq \int_0^\infty \min\{\eta(A), e^{-L}\} dL \\ &= \eta(A)(1 - \log \eta(A)) \end{aligned} \quad (10)$$

for every positive-measure atom A . Summing over A yields

$$\boxed{\int_X q^* d\eta \leq H_\eta(\mathcal{P}) + 1 < \infty.} \quad (11)$$

For each fixed A , the conditional probabilities in (7) form a bounded martingale along the increasing sigma-algebras $\sigma(\mathcal{P}_1^n)$. Martingale convergence therefore gives a.e. convergence of q_n^A , hence of q_n , to an information function q . By (11), dominated convergence also gives

$$q_n \longrightarrow q \quad \text{in } L^1(\eta). \quad (12)$$

The information chain rule, applied from the terminal symbol backwards, gives the exact identity

$$I_\eta(\mathcal{P}^n) = q_n + q_{n-1} \circ T + \cdots + q_0 \circ T^n. \quad (13)$$

To compare this triangular sum with the Birkhoff average of q , put $r_k := |q_k - q|$. Then $r_k \rightarrow 0$ a.e. and in L^1 , and $r_k \leq q_0 + q^* + q =: R_0 \in L^1$ for every $k \geq 0$. For $N \geq 1$ let

$$R_N := \sup_{k \geq N} r_k. \quad (14)$$

Then $R_N \downarrow 0$ a.e. and in L^1 . For $n > N$,

$$\frac{1}{n+1} \sum_{i=0}^n r_{n-i} \circ T^i \leq \frac{1}{n+1} \sum_{i=0}^{n-N} R_N \circ T^i + \frac{1}{n+1} \sum_{i=n-N+1}^n R_0 \circ T^i. \quad (15)$$

By Birkhoff, the limsup of the first term is $\mathbb{E}_\eta(R_N | \mathcal{I}_T)$ a.e. For the second term we use the elementary fact that if $R \in L^1_+$, then $R(T^n x)/n \rightarrow 0$ a.e.: indeed $\sum_n \eta\{R > \varepsilon n\} < \infty$ follows by comparing that sum with $\varepsilon^{-1} \int R d\eta$, and Borel-Cantelli applies. The second term in (15), a sum of only N such shifted terms divided by $n+1$, therefore tends to zero a.e. Finally

$$\int \mathbb{E}_\eta(R_N | \mathcal{I}_T) d\eta = \int R_N d\eta \rightarrow 0,$$

and the conditional expectations decrease to zero. Thus

$$\frac{1}{n+1} \sum_{i=0}^n r_{n-i} \circ T^i \rightarrow 0 \quad \text{a.e.} \quad (16)$$

The same convergence holds in L^1 directly from (12) and Cesàro averaging of the norms. Combining (13), (16), and Birkhoff for q gives

$$\frac{1}{n+1} I_\eta(\mathcal{P}^n) \rightarrow \mathbb{E}_\eta(q | \mathcal{I}_T) =: h_{\mathcal{P}} \quad \text{a.e. and in } L^1. \quad (17)$$

Integrating and using $\int I_\eta(\mathcal{P}^n) d\eta = H_\eta(\mathcal{P}^n)$ proves (5), because $(n+1)^{-1} H_\eta(\mathcal{P}^n)$ converges by the definition of $h_\eta(T, \mathcal{P})$. This proves Auxiliary lemma 1.

Auxiliary lemma 2: a variable-radius entropy lower bound. Let (X, d) be a compact d -dimensional Riemannian manifold, let $T : X \rightarrow X$ be measurable and preserve a probability η , and let m be a probability measure with $\eta \ll m$. Let $\rho : X \rightarrow (0, 1)$ be measurable with $\log \rho \in L^1(\eta)$. Define

$$B_n(T, \rho, x) := \{y : d(T^j x, T^j y) \leq \rho(T^j x), 0 \leq j \leq n\} \quad (18)$$

and

$$h_m(T, \rho, x) := \limsup_{n \rightarrow \infty} -\frac{1}{n} \log m(B_n(T, \rho, x)). \quad (19)$$

Then

$$\boxed{h_\eta(T) \geq \int_X h_m(T, \rho, x) d\eta(x).} \quad (20)$$

First we construct a countable finite-entropy partition adapted to ρ . Compactness gives constants $C_0, r_0 > 0$ such that for every $0 < r < r_0$ there is a finite measurable partition $\mathcal{Q}_r$ whose atoms have diameter at most r and whose cardinality is at most $C_0 r^{-d}$. For example, take a maximal $r/3$ -separated set; the $r/3$ -balls cover, standard volume comparison bounds the number of centers by $C_0 r^{-d}$, and assigning each point to its first covering ball produces a measurable partition of diameter at most $2r/3 < r$. After replacing ρ by $\min(\rho, r_0/2)$, which changes neither the argument nor integrability,

put

$$U_k := \{e^{-(k+1)} < \rho \leq e^{-k}\}, \quad k \geq 0. \quad (21)$$

The integrability of $-\log \rho$ implies

$$\sum_{k \geq 0} k \eta(U_k) < \infty. \quad (22)$$

We also need

$$\sum_{k \geq 0} -\eta(U_k) \log \eta(U_k) < \infty. \quad (23)$$

To see this, use the elementary series lemma: if $0 \leq a_k \leq 1$ and $\sum k a_k < \infty$, then $\sum a_k \log(1/a_k) < \infty$. Indeed, on indices where $\log(1/a_k) \leq k$ the desired summand is bounded by $k a_k$; on the remaining indices $a_k \leq e^{-k}$, and $\sqrt{t} \log(1/t) \leq 2/e$ gives a summable $O(e^{-k/2})$ bound.

Intersect U_k with the atoms of $\mathcal{Q}_{e^{-(k+1)}}$ and discard null pieces. The resulting countable partition $\mathcal{P}$ satisfies

$$\text{diam } \mathcal{P}(x) \leq \rho(x) \quad \text{for } \eta\text{-a.e. } x. \quad (24)$$

Moreover, by the entropy chain rule,

$$H_\eta(\mathcal{P}) \leq H_\eta(\{U_k\}) + \sum_k \eta(U_k) \log |\mathcal{Q}_{e^{-(k+1)}}| < \infty \quad (25)$$

by (22), (23), and the polynomial covering bound.

Let $\mathcal{P}^n = \bigvee_{j=0}^n T^{-j} \mathcal{P}$. By (24),

$$\mathcal{P}^n(x) \subset B_n(T, \rho, x) \quad \text{for a.e. } x. \quad (26)$$

Apply Auxiliary lemma 1. Its limit $h_{\mathcal{P}}(x)$ satisfies $\int h_{\mathcal{P}} d\eta = h_\eta(T, \mathcal{P}) \leq h_\eta(T)$. Let $\mathcal{G}_\infty = \sigma(\bigcup_n \mathcal{P}^n)$. Since $\eta \ll m$, the restriction of η to $\mathcal{G}_\infty$ has a Radon-Nikodym derivative k with respect to the restriction of m . Conditional expectation with respect to $\sigma(\mathcal{P}^n)$ gives, on the atom containing x ,

$$\frac{\eta(\mathcal{P}^n(x))}{m(\mathcal{P}^n(x))} = \mathbb{E}_m(k \mid \sigma(\mathcal{P}^n))(x) \longrightarrow k(x) \quad (27)$$

for m -a.e. x . On an η -full set, $0 < k(x) < \infty$; hence

$$\frac{1}{n} \log \frac{\eta(\mathcal{P}^n(x))}{m(\mathcal{P}^n(x))} \longrightarrow 0. \quad (28)$$

Equations (17), (26), and (28) imply

$$h_m(T, \rho, x) \leq h_{\mathcal{P}}(x) \quad \text{for } \eta\text{-a.e. } x. \quad (29)$$

Integrating proves (20).

Auxiliary lemma 3: the corrected finite-dimensional graph transform. Let $E =$

$E^0 \oplus E^u$ and $E' = F^0 \oplus F^u$ be finite-dimensional normed spaces and suppose the norms of both projections associated with each splitting are at most Γ . Let $F : B_r(0) \subset E \rightarrow E'$ be a C^1 embedding, $F(0) = 0$, such that $D_0F(E^0) = F^0$, $D_0F(E^u) = F^u$, and

$$\|D_0Fv\| \geq \Lambda\|v\| \quad (v \in E^u), \quad \|D_0F|_{E^0}\| \leq B, \quad \Lambda > B > 1. \quad (30)$$

Fix $c > 0$. If

$$\sup_{z \in B_r(0)} \|D_zF - D_0F\| \leq \delta \quad (31)$$

and

$$\Gamma\delta(1+c) < \Lambda, \quad \Gamma\delta(1+c)^2 < c(\Lambda - B), \quad (32)$$

then the image of every (E^0, E^u) graph of Lipschitz dispersion at most c , contained in $B_r(0)$, is an (F^0, F^u) graph of dispersion at most c .

Indeed, in the two splittings write

$$F(u, v) = (Lu + p(u, v), Tv + q(u, v)). \quad (33)$$

The derivatives of p and q have norm at most $\Gamma\delta$. If $u = \psi(v)$ with $\text{Lip } \psi \leq c$, then the second component of the image is $\Phi(v) = Tv + q(\psi(v), v)$ and

$$\|\Phi(v) - \Phi(w)\| \geq [\Lambda - \Gamma\delta(1+c)]\|v - w\|. \quad (34)$$

Thus Φ is one-to-one and has a Lipschitz inverse on its open image (invariance of domain supplies openness). The first component has Lipschitz constant at most $Bc + \Gamma\delta(1+c)$ as a function of v . Therefore the new graph has slope at most

$$\frac{Bc + \Gamma\delta(1+c)}{\Lambda - \Gamma\delta(1+c)} \leq c \quad (35)$$

by the second inequality in (32). This proves the lemma. Notice that the same δ works after replacing (Λ, B) by (Λ^q, B^q) for any $q \geq 1$, because both restrictions in (32) become weaker. This is the uniformity point required by the published correction to the original proof.

Step 1: reduce to fixed unstable dimension. On the Oseledets set let

$$E^u(x) := \bigoplus_{\lambda_i(x) > 0} E_i(x), \quad E^0(x) := \bigoplus_{\lambda_i(x) \leq 0} E_i(x). \quad (36)$$

For $0 \leq j \leq d := \dim M$ set

$$\Sigma_j := \{x : \dim E^u(x) = j\}. \quad (37)$$

Each Σ_j is invariant modulo μ and the family is finite. If $p_j = \mu(\Sigma_j) > 0$, let $\mu_j = p_j^{-1}\mu|_{\Sigma_j}$. Then μ_j is still absolutely continuous with respect to Riemannian volume.

We record the elementary entropy decomposition across this finite invariant partition. For every finite measurable partition $\mathcal{A}$, joining with $\{\Sigma_j\}$ changes the entropy of

$\bigvee_{k=0}^{n-1} f^{-k} \mathcal{A}$ by at most the fixed number $H_\mu(\{\Sigma_j\})$. Hence

$$h_\mu(f, \mathcal{A}) = \sum_{j: p_j > 0} p_j h_{\mu_j}(f, \mathcal{A}). \quad (38)$$

Taking suprema gives “ $\leq$ ” in the corresponding system-entropy identity. For the reverse inequality, choose for each of the finitely many j a finite partition whose μ_j -entropy is within an arbitrary ε of $h_{\mu_j}(f)$ and take their common refinement. Therefore

$$h_\mu(f) = \sum_{j: p_j > 0} p_j h_{\mu_j}(f). \quad (39)$$

Consequently it suffices to prove

$$h_{\mu_j}(f) \geq \int \chi^+ d\mu_j \quad (40)$$

for every j . The case $j = 0$ is immediate. Fix henceforth a $j > 0$ and, to simplify notation, write $\eta = \mu_j$ and $\Sigma = \Sigma_j$.

Step 2: a uniform Oseledets block for an iterate. Fix $0 < \varepsilon < 10^{-2}$. Since every point of Σ has exactly j positive exponents, the sets on which the smallest positive exponent is at least $1/q$ exhaust Σ . Combining this observation with Oseledets, Egorov, and Lusin, we can choose a compact set $K \subset \Sigma$ with

$$\eta(K) > 1 - \varepsilon^2 \quad (41)$$

on which $E^0 \oplus E^u$ varies continuously and for which the Oseledets estimates are uniform after a sufficiently large time. More precisely, there are an integer $N \geq 1$ and numbers $\Lambda > B > 1$ such that, with $g = f^N$, for every $x \in K$ and every $n \geq 1$,

$$\begin{aligned} m(D_x g^n|_{E^u(x)}) &\geq \Lambda^n, \\ \|D_x g^n|_{E^0(x)}\| &\leq B^n, \\ \log J(D_x g^n|_{E^u(x)}) &\geq Nn(\chi^+(x) - \varepsilon). \end{aligned} \quad (42)$$

Here $m(L) = \inf_{\|v\|=1} \|Lv\|$, and $J(L|_E)$ denotes the j -dimensional volume Jacobian. The second line of (42) has the factor B^n ; omitting that exponent is precisely the published misprint in the 1981 article.

For completeness, the choice is elementary from Oseledets. First restrict to a set of measure $> 1 - \varepsilon^2/3$ on which the least positive exponent is bounded below by some $a_* > 0$. Choose $0 < b_* < a_*/3$. Uniformize, by Egorov, the asymptotic estimates

$$m(D_x f^m|_{E^u}) \geq e^{(a_* - b_*)m}, \quad \|D_x f^m|_{E^0}\| \leq e^{b_* m}, \quad \log J(D_x f^m|_{E^u}) \geq m(\chi^+(x) - \varepsilon)$$

for all sufficiently large m . Then choose N beyond that threshold and put $\Lambda = e^{(a_* - b_*)N}$ and $B = e^{b_* N}$; shrinking once more by Lusin gives the continuous splitting on K .

Because K is compact and the splitting is continuous there, the norms of its two projections are bounded by some Γ . Choose a graph slope $c > 0$ so small that later Jacobian comparisons can be made within ε . Then choose $\delta > 0$ satisfying the two

corrected smallness inequalities

$$\Gamma\delta(1+c) < \Lambda, \quad \Gamma\delta(1+c)^2 < c(\Lambda - B). \quad (43)$$

Step 3: uniform graph transport between returns to K . Since g is $C^{1+\alpha}$, fix a finite smooth atlas with Lebesgue radius $r_{\text{at}} > 0$ and let $A_g \geq 1$ be a global Lipschitz constant for g . There are constants $C_1 > 1$ and $0 < \theta \leq \alpha$ such that, whenever the paired orbit points $g^i x, g^i z$ remain within r_{at} for $0 \leq i \leq q$, the coordinate representatives satisfy

$$\|D_z g^q - D_x g^q\| \leq C_1^q d(x, z)^\theta. \quad (44)$$

Indeed, write the difference of the two derivative products as a telescoping sum, use the θ -Hölder bound on Dg , and use $d(g^i x, g^i z) \leq A_g^i d(x, z)$; increasing C_1 absorbs the geometric series and the finitely many chart/trivialization constants.

Choose $0 < \xi < 1$ so small that

$$C_1 \xi^\theta < \min\{1, \delta\}, \quad A_g \xi < \min\{1, r_{\text{at}}/2\}. \quad (45)$$

If $d(x, z) \leq \xi^q$, then for $0 \leq i \leq q$,

$$d(g^i x, g^i z) \leq A_g^i \xi^q = (A_g \xi)^i \xi^{q-i} < r_{\text{at}},$$

so (44) is applicable throughout the segment and gives $\|D_z g^q - D_x g^q\| \leq (C_1 \xi^\theta)^q < \delta$. Then the corrected graph-transform lemma applied to g^q and (42) gives: if $x, g^q x \in K$ and G is an $(E^0(x), E^u(x))$ graph of dispersion at most c contained in $B_{\xi^q}(x)$, then $g^q G$ is an $(E^0(g^q x), E^u(g^q x))$ graph of dispersion at most c . The same δ is valid for every q by the final observation in Auxiliary lemma 3.

Next choose $a_{\text{det}} > 0$ so small that whenever $x \in K$, $d(x, z) < a_{\text{det}}$, and $E \subset T_z M$ is a j -plane which, in the local trivialization, is a graph of dispersion at most c over $E^u(x)$, then

$$|\log J(D_z g|_E) - \log J(D_x g|_{E^u(x)})| < \varepsilon. \quad (46)$$

This follows from compactness and continuity of the logarithmic Jacobian on the Grassmann bundle; g is a diffeomorphism, so the Jacobian is uniformly bounded away from zero on j -planes.

For $x \in K$ and sufficiently small $r > 0$, use exponential coordinates and write the open product rectangle

$$D_r(x) := \{\exp_x(v_0 + v_u) : v_0 \in E^0(x), v_u \in E^u(x), \|v_0\| < r, \|v_u\| < r\}. \quad (47)$$

Uniform transversality on K and compactness give constants $0 < k_1 < k_2$ and $r_1 > 0$ such that

$$B_{k_1 r}(x) \subset D_r(x) \subset B_{k_2 r}(x) \quad (x \in K, 0 < r < r_1). \quad (48)$$

Now choose the final geometric radius a so that

$$0 < a < \min\{1, k_1 r_1, (k_1/k_2)a_{\text{det}}\}. \quad (49)$$

This separates the determinant-comparison radius from the Bowen radius and ensures

that every rectangle used below lies inside the range of (46).

Let $\tau(x)$ be the first positive return time of $x \in K$ to K . It is finite for a.e. $x \in K$ by recurrence. It is also integrable. Indeed, if $W_q = \{x \in K : \tau(x) = q\}$, then the sets $g^i W_q$, $0 \leq i < q$, are pairwise disjoint modulo null sets, so

$$\int_K \tau d\eta = \sum_{q \geq 1} q\eta(W_q) = \sum_{q \geq 1} \sum_{i=0}^{q-1} \eta(g^i W_q) \leq 1. \quad (50)$$

Define the measurable radius

$$\rho(x) := \begin{cases} \min\{a, (k_1/k_2)\xi^{\tau(x)}\}, & x \in K \text{ and } \tau(x) < \infty, \\ a, & x \notin K, \end{cases} \quad (51)$$

and change it arbitrarily on the null set of nonreturning points. Equation (50) implies $\log \rho \in L^1(\eta)$.

Step 4: most long orbit segments spend little time outside K . Apply Birkhoff, *Birkhoff pointwise ergodic theorem*, to $\mathbf{1}_{M \setminus K}$. Its limiting conditional expectation has integral $\eta(M \setminus K) < \varepsilon^2$. Hence the set on which that limit exceeds ε has measure at most ε . By Egorov, after deleting another set of measure at most ε , there are a measurable set K_1 and n_0 such that

$$\eta(K_1) > 1 - 2\varepsilon \quad (52)$$

and, for every $x \in K_1$ and $n \geq n_0$,

$$\#\{0 \leq i < n : g^i x \notin K\} \leq 2\varepsilon n. \quad (53)$$

Put $K' := K \cap K_1$. Then

$$\eta(M \setminus K') \leq 2\varepsilon + \varepsilon^2 \leq 3\varepsilon. \quad (54)$$

Step 5: unstable-slice volume estimate. Let m now denote normalized Riemannian volume on M . Fix $x \in K'$ which is Oseledets regular and recurrent to K . We estimate $m(B_n(g, \rho, x))$ along the infinite subsequence of return times n for which $g^n x \in K$.

Work in the exponential chart at x . For $y \in E^0(x)$ let $y + E^u(x)$ denote the corresponding affine unstable slice. For a return time n , define $\Lambda_n(y)$ to be the set of points z in that slice such that for every $0 \leq i \leq n$ with $g^i x \in K$,

$$g^i z \in D_{\rho(g^i x)/k_1}(g^i x). \quad (55)$$

Only the return times to K are imposed in (55). This slight strengthening of the book-keeping avoids any need for uniform angles away from the compact block K . By (48), every $z \in B_n(g, \rho, x)$ satisfies (55), and therefore

$$(y + E^u(x)) \cap B_n(g, \rho, x) \subset \Lambda_n(y). \quad (56)$$

We claim that at every return time $0 \leq r \leq n$ the set $g^r \Lambda_n(y)$, if nonempty, is an $(E^0(g^r x), E^u(g^r x))$ graph of dispersion at most c . At $r = 0$ this is immediate. Suppose

$r < s$ are consecutive return times. The constraint at time r and (48), (51) give

$$g^r \Lambda_n(y) \subset D_{\rho(g^r x)/k_1}(g^r x) \subset B_{k_2 \rho(g^r x)/k_1}(g^r x) \subset B_{\xi^{s-r}}(g^r x). \quad (57)$$

The graph-transport statement from Step 3 applies to g^{s-r} . Intersecting its image with the open product rectangle required by (55) at time s leaves an open subgraph. Induction proves the claim.

There is therefore a uniform constant $V_* > 0$ such that every terminal graph $g^n \Lambda_n(y)$ has j -dimensional volume at most V_* . This follows from (55) at time n , the fixed radius a/k_1 , the slope bound c , and compactness of K .

Put

$$C_* := \sup\{ |\log J(D_p g|_E)| : p \in M, E \in \text{Gr}_j(T_p M) \} < \infty. \quad (58)$$

Also put

$$C_f := \sup\{ |\log J(D_p f|_E)| : p \in M, E \in \text{Gr}_j(T_p M) \} < \infty. \quad (59)$$

Since $g = f^N$ and Jacobians multiply along transported j -planes,

$$C_* \leq N C_f. \quad (60)$$

For $z \in \Lambda_n(y)$, the change-of-variables formula on the j -graph gives

$$\text{vol}_j(g^n \Lambda_n(y)) = \int_{\Lambda_n(y)} J(D_z g^n|_{T_z \Lambda_n(y)}) d\text{vol}_j(z). \quad (61)$$

Let

$$S_n := \{0 \leq i < n : g^i x \in K\}. \quad (62)$$

At $i \in S_n$, Step 3 and (46) compare the tangent Jacobian of the transported graph with the reference unstable Jacobian to within ε in the logarithm. At $i \notin S_n$, both logarithmic Jacobians have absolute value at most C_* . Hence, using the multiplicativity of the Jacobian on the invariant bundle,

$$\begin{aligned} \log J(D_z g^n|_{T_z \Lambda_n(y)}) &\geq \sum_{i \in S_n} \log J(D_{g^i x} g|_{E^u(g^i x)}) - \varepsilon n - C_*(n - |S_n|) \\ &\geq \sum_{i=0}^{n-1} \log J(D_{g^i x} g|_{E^u(g^i x)}) - \varepsilon n - 2C_*(n - |S_n|) \\ &= \log J(D_x g^n|_{E^u(x)}) - \varepsilon n - 2C_*(n - |S_n|) \\ &\geq nN(\chi^+(x) - \varepsilon) - \varepsilon n - 4C_*\varepsilon n, \end{aligned} \quad (63)$$

where the final line uses (42) and (53). Together with (61) and the bound by V_* this yields, uniformly in y ,

$$\text{vol}_j(\Lambda_n(y)) \leq V_* \exp\{-n[N(\chi^+(x) - \varepsilon) - \varepsilon - 4C_*\varepsilon]\}. \quad (64)$$

Since the condition at time 0 confines the Bowen ball to a fixed small coordinate neighborhood of x , Fubini in the exponential chart and the uniform upper bound on

the Riemannian density give a constant C_F , independent of n , such that

$$m(B_n(g, \rho, x)) \leq C_F \sup_y \text{vol}_j((y + E^u(x)) \cap B_n(g, \rho, x)). \quad (65)$$

Equations (56), (64), and (65), along the return-time subsequence, imply

$$h_m(g, \rho, x) \geq N(\chi^+(x) - \varepsilon) - \varepsilon - 4C_*\varepsilon \quad \text{for a.e. } x \in K'. \quad (66)$$

The use of a subsequence is legitimate because h_m is defined by a limsup.

Step 6: entropy lower bound and removal of the block error. Auxiliary lemma 2 and (66) give

$$\begin{aligned} h_\eta(g) &\geq \int h_m(g, \rho, x) d\eta(x) \\ &\geq \int_{K'} [N(\chi^+(x) - \varepsilon) - \varepsilon - 4C_*\varepsilon] d\eta(x). \end{aligned} \quad (67)$$

Here the restriction from M to K' is legitimate because $m(B_n(g, \rho, x)) \leq 1$, hence $h_m(g, \rho, x) \geq 0$ everywhere it is defined. Let $L_\chi = \|\chi^+\|_\infty$. From (54),

$$\int_{K'} \chi^+ d\eta \geq \int_M \chi^+ d\eta - 3L_\chi \varepsilon. \quad (68)$$

Consequently

$$h_\eta(g) \geq N \int_M \chi^+ d\eta - \varepsilon(3NL_\chi + N + 1 + 4C_*). \quad (69)$$

The dependence of N on ε causes no loss here: (60) gives $C_*/N \leq C_f$, uniformly in the chosen iterate.

We also record, rather than hide, the standard iterate identity for entropy. For every finite partition $\mathcal{A}$,

$$h_\eta(g, \bigvee_{r=0}^{N-1} f^{-r} \mathcal{A}) = Nh_\eta(f, \mathcal{A}),$$

which gives $h_\eta(g) \geq Nh_\eta(f)$ after taking suprema. Conversely, for any finite $\mathcal{B}$, $h_\eta(g, \mathcal{B}) \leq Nh_\eta(f, \mathcal{B})$ by comparing the g -name with the corresponding f -name, so

$$h_\eta(g) = Nh_\eta(f). \quad (70)$$

Divide (69) by N and let $\varepsilon \downarrow 0$. We obtain

$$h_\eta(f) \geq \int_M \chi^+ d\eta. \quad (71)$$

This proves (40) for every positive unstable dimension. Summing with (39) gives

$$h_\mu(f) \geq \int_M \chi^+ d\mu. \quad (72)$$

Together with Ruelle's inequality (2),

$$h_\mu(f) = \int_M \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x) d\mu(x).$$

This is Pesin's entropy formula in the stated $C^{1+\alpha}$ absolutely continuous scope. $\square$

Theorem F11.7. Ledrappier-Young characterization of the SRB property

Statement. Let f be a C^2 diffeomorphism of a compact Riemannian manifold and let μ be an ergodic hyperbolic invariant probability with at least one positive Lyapunov exponent. Then the following are equivalent:

1. for a measurable partition ξ subordinate to Pesin unstable manifolds, the Rokhlin conditional measures μ_x^ξ are absolutely continuous with respect to the induced Riemannian measures on the unstable leaves (equivalently, this holds for every such subordinate partition); in the terminology of Definition F11.4, μ has the SRB unstable-conditional property;
2. μ satisfies Pesin's entropy formula

$$h_\mu(f) = \int_M \sum_{\lambda_i(x) > 0} m_i(x) \lambda_i(x) d\mu(x).$$

Scope note

The equivalence here concerns the unstable-conditional SRB property and Pesin entropy formula; physicality is a separate assertion.

Proof.

Proof strategy

Choose an increasing measurable partition subordinate to unstable leaves. The unstable Jacobian determines a canonical leafwise density through backward distortion. Conditional entropy is then compared with the integral of the unstable Jacobian; equality is characterized exactly by absolute continuity of the conditional measures. Combining this equality criterion with the entropy formula gives the equivalence between the SRB property and the entropy identity.

Write

$$E^u(x) = \bigoplus_{\lambda_i(x) > 0} E_i(x), \quad J^u(x) := |\det(D_x f|_{E^u(x)})|. \quad (1)$$

Because μ is ergodic and hyperbolic, the Lyapunov spectrum with multiplicity and $u := \dim E^u(x)$ are constant for almost every x . The hypothesis that there is at least one positive exponent says $u > 0$. Since f is a C^2 diffeomorphism of a compact manifold,

$$|\log J^u(x)| \leq u \max\{\log^+ \|D_x f\|, \log^+ \|D_x f^{-1}\|\} \quad (2)$$

is essentially bounded. Oseledets' theorem and Birkhoff's theorem give

$$\int \log J^u d\mu = \sum_{\lambda_i > 0} m_i \lambda_i. \quad (3)$$

Thus it remains to prove that the SRB unstable-conditional property is equivalent to

$$h_\mu(f) = \int \log J^u d\mu. \quad (4)$$

We shall do this without invoking *Pesin entropy formula*, whose hypothesis is ambient absolute continuity and is not available here.

Step 1: an increasing partition subordinate to W^u . We first record the exact measurable-partition package needed below. There is a full-measure invariant set S and a measurable partition η of S such that, for almost every $x \in S$,

$$f^{-1}\eta \text{ refines } \eta, \quad (5)$$

$$\eta(x) \subset W^u(x) \text{ and } \eta(x) \text{ contains an intrinsic neighborhood of } x \text{ in } W^u(x), \quad (6)$$

$$\bigcup_{n \geq 0} f^n \eta(f^{-n}x) = W^u(x), \quad (7)$$

$$\bigvee_{n \geq 0} f^{-n}\eta \text{ is the partition into points mod } 0. \quad (8)$$

Moreover the induced Riemannian leaf volumes m_x^u on the atoms can be chosen measurably, and whenever $y \in \eta(x)$ the product

$$\Delta(x, y) := \prod_{n=1}^{\infty} \frac{J^u(f^{-n}x)}{J^u(f^{-n}y)} \quad (9)$$

converges to a number in $(0, \infty)$. On each atom, $y \mapsto \Delta(x, y)$ is locally bounded above and below.

Here are the construction and the estimates, so that (5)–(9) are not hidden behind a “standard subordinate partition” citation. By *Pesin blocks and uniformization* choose a compact Pesin block Λ of positive measure, contained in the almost-everywhere constant hyperbolic index stratum, on which the chart radius, angle and stable/unstable contraction constants are uniform. Shrink to a small foliation box U meeting Λ in positive measure. *Pesin local stable/unstable manifold theorem* and the parameter-dependent graph transform used there give local unstable plaques of one fixed intrinsic size, depending continuously on the base point on this compact block.

Choose a transverse radius r for the box so that almost every orbit avoids its boundary at an exponentially fast prescribed scale. More explicitly, fix $0 < c < \min_i \lambda_i^+$ and let U_t , t in a compact interval, be the same box with transverse radius t . Fubini applied to

$$\sum_{n \geq 1} \mathbf{1}_{\{d(f^{-n}x, \partial U_t) < e^{-cn}\}} \quad (10)$$

shows that for almost every t the integral of this sum over x is finite: for each n ,

integration first in t bounds the bad set by a constant times e^{-cn} . Fix such a t . Borel-Cantelli then gives, for almost every x , a random $N(x)$ such that after every backward return $f^{-n}x \in U_t$, $n \geq N(x)$, the point $f^{-n}x$ is farther than e^{-cn} from the plaque-box boundary.

Let ζ be the measurable partition whose nontrivial atoms in U_t are the local unstable plaques of the box (and whose remaining part is the complement), and put

$$\eta := \bigvee_{n \geq 0} f^n \zeta \quad \text{on} \quad S := \bigcup_{k \in \mathbb{Z}} f^k(U_t \cap \Lambda). \quad (11)$$

The set S is invariant and has positive measure, hence ergodicity gives $\mu(S) = 1$. Formula (5) is immediate from (11). If x has a backward return at time n , then the corresponding factor in (11) restricts the atom at x to the forward image of a local unstable plaque at $f^{-n}x$. Backward contraction on the block is $O(e^{-\chi n})$ for some $\chi > c$. The boundary choice above therefore implies that, for all sufficiently large return times, this restriction contains one fixed intrinsic neighborhood of x in $W^u(x)$; the finitely many earlier restrictions can be absorbed by shrinking that neighborhood. This proves (6). Recurrence to $U_t \cap \Lambda$ and forward expansion of unstable plaques prove (7). The same recurrence, now applied to forward iterates, gives arbitrarily large n for which a uniform local plaque at $f^n x$ pulls back to an unstable disk of diameter $O(e^{-\chi n})$. Hence the atoms of $f^{-n}\eta$ shrink to x along the unstable leaf. Points not on the same unstable leaf are already separated by η . Thus (8) follows.

It remains to justify (9). On a compact Pesin block, if z, z' lie on one local unstable plaque, the graph transform gives

$$d(f^{-n}z, f^{-n}z') \leq C e^{-\chi n} d^u(z, z') \quad (12)$$

and exponential convergence of the tangent spaces. Since $f \in C^2$, the map

$$(p, V) \mapsto \log |\det(D_p f|_V)| \quad (13)$$

is Lipschitz on the compact set of u -planes in the unstable cone. Therefore

$$|\log J^u(f^{-n}z) - \log J^u(f^{-n}z')| \leq C' e^{-\chi' n} d^u(z, z')^\theta \quad (14)$$

for some $\chi' > 0$ and $0 < \theta \leq 1$ while the two backward points stay in the block chart. Between returns, tempered chart constants cost at most $e^{\varepsilon n}$; choose the Pesin error $\varepsilon < \chi'/2$. Summing by return blocks still gives a convergent geometric series. Thus

$$\sum_{n \geq 1} |\log J^u(f^{-n}z) - \log J^u(f^{-n}z')| < \infty, \quad (15)$$

which proves convergence and local two-sided bounds for (9). Measurability of the plaques, their induced volumes and the integral kernels follows from the measurable Lyapunov-chart construction and the compact Lusin-block exhaustion in *Pesin blocks and uniformization*.

Step 2: two elementary entropy lemmas. We use the refinement convention that

$\alpha \succeq \beta$ means that α refines β . For an increasing partition $f^{-1}\xi \succeq \xi$, put

$$h_\mu(f, \xi) := H_\mu(f^{-1}\xi \mid \xi). \quad (16)$$

We need two facts.

Lemma 2.1 (increasing-partition upper bound). If the quantity in (16) is finite, then

$$H_\mu(f^{-1}\xi \mid \xi) \leq h_\mu(f). \quad (17)$$

To prove it, let $\mathcal{A}$ be any finite partition measurable with respect to $\sigma(f^{-1}\xi)$. Choose a finite partition $\mathcal{B}$ measurable with respect to $\sigma(\xi)$ so fine that

$$H(\mathcal{A} \mid \mathcal{B}) \leq H(\mathcal{A} \mid \xi) + \varepsilon; \quad (18)$$

this is the usual martingale convergence of conditional probabilities, first for finite $\mathcal{A}$ and then summed over its atoms. Set $\mathcal{P} = \mathcal{A} \vee \mathcal{B}$. Since $\mathcal{P}$ is measurable with respect to $f^{-1}\xi$, every future partition $f^j\mathcal{P}$, $j \geq 1$, is measurable with respect to ξ (use $\xi \succeq f\xi \succeq f^2\xi \succeq \dots$). The finite-partition chain rule therefore gives

$$h_\mu(f, \mathcal{P}) = H\left(\mathcal{P} \mid \bigvee_{j \geq 1} f^j\mathcal{P}\right) \geq H(\mathcal{P} \mid \xi) = H(\mathcal{A} \mid \xi). \quad (19)$$

Taking the supremum over finite $\mathcal{A}$ in the relative sigma-algebra $\sigma(f^{-1}\xi)$ modulo $\sigma(\xi)$ yields (17). (The equality in (19) follows by telescoping the Shannon entropies of finite orbit joins and then using martingale convergence.)

Lemma 2.2 (Mañé finite-entropy diameter partition). Let $0 < \psi \leq 1$ be measurable and assume $\int -\log \psi \, d\mu < \infty$. There is a countable measurable partition $\mathcal{P}$ with $H_\mu(\mathcal{P}) < \infty$ and

$$\text{diam } \mathcal{P}(x) \leq \psi(x) \quad \text{for almost every } x. \quad (20)$$

Indeed put $A_k = \{e^{-(k+1)} < \psi \leq e^{-k}\}$. Compactness gives a constant C and finite partitions β_k of diameter at most $e^{-(k+1)}$ with $|\beta_k| \leq Ce^{d(k+1)}$. Refine each A_k by β_k . Writing $a_k = \mu(A_k)$, integrability gives $\sum k a_k < \infty$. The elementary estimate

$$\sum_k a_k \log(1/a_k) < \infty \quad (21)$$

follows by splitting into $\log(1/a_k) \leq k$ and its complement, where $a_k < e^{-k}$ and $a_k \log(1/a_k) = O(e^{-k/2})$. The entropy chain rule then gives

$$H(\mathcal{P}) \leq \sum_k a_k \log(1/a_k) + \sum_k a_k \log |\beta_k| < \infty, \quad (22)$$

and (20) is built into the construction.

Step 3: in the hyperbolic case the unstable partition realizes all entropy. We prove

$$H_\mu(f^{-1}\eta \mid \eta) = h_\mu(f). \quad (23)$$

The upper bound is Lemma 2.1. For the reverse inequality we construct a finite-entropy partition whose entire past name is contained in an η -atom.

Choose a compact Pesin block $R \subset U_t \cap \Lambda$ of positive measure, small enough that local product coordinates and the plaque separation used in Step 1 are uniform on R . For $x \in R$ let

$$r(x) = \min\{n \geq 1 : f^n x \in R\} \quad (24)$$

be the first return time. The tower levels $\{f^j x : x \in R, 0 \leq j < r(x)\}$ are disjoint modulo null sets and, by ergodicity, cover a full-measure set. Hence

$$\int_R r \, d\mu = 1. \quad (25)$$

(This is the one-line Kac identity; the equality follows by integrating the number of tower levels.)

Let $L \geq 1$ be a global Lipschitz constant for both f and f^{-1} . Fix the Pesin-block tempering allowance $\tau_R > 0$ used in the graph-transform charts (enlarging it, if necessary, so that all chart-to-Riemannian comparison factors accumulated over an excursion of length r are bounded by $C_R e^{\tau_R r}$). Choose

$$\lambda > \log L + \tau_R + 1$$

and a sufficiently small block scale $\delta > 0$. Define

$$\psi(x) = \begin{cases} \delta C_R^{-1} e^{-\lambda r(x)}, & x \in R, \\ \delta, & x \notin R, \end{cases} \quad (26)$$

where C_R dominates the fixed chart/metric comparison constants on R . By (25), $-\log \psi \in L^1(\mu)$. Lemma 2.2 gives a finite-entropy partition $\mathcal{P}_0$ with atom diameters bounded by ψ . Refine it by $\{R, M \setminus R\}$ and by the finitely many local plaque-component labels in the Step-1 box; call the result $\mathcal{P}$. Still $H(\mathcal{P}) < \infty$.

Put

$$\mathcal{P}^u := \bigvee_{n \geq 0} f^n \mathcal{P}. \quad (27)$$

Thus $y \in \mathcal{P}^u(x)$ means that $f^{-n}x$ and $f^{-n}y$ have the same $\mathcal{P}$ -name for every $n \geq 0$. We claim

$$\mathcal{P}^u(x) \subset \eta(x) \quad \text{for almost every } x. \quad (28)$$

Here is the hyperbolic argument. Let $-n_j$ be the successive backward return times of x to R . The refinement by R makes $f^{-n_j}y$ return to R at the same times. At a return $z = f^{-n_j}x$ with next forward return time $r = r(z)$, the diameter estimate gives

$$d(z, f^{-n_j}y) \leq \delta C_R^{-1} e^{-\lambda r}. \quad (29)$$

For $0 \leq k \leq r$, the global Lipschitz bound gives

$$d(f^k z, f^{k-n_j}y) \leq \delta C_R^{-1} e^{-\lambda r} L^k \leq \delta C_R^{-1} e^{-(\lambda - \log L)r}. \quad (30)$$

After conversion to the tempered Pesin charts this is at most $\delta e^{-(\lambda - \log L - \tau_R)r}$, hence smaller than the uniform graph-transform scale. Thus the exponentially small separation at each return compensates not only the global Lipschitz growth during the whole excursion but also every chart-tempering loss. At the return endpoints, decompose the displacement by the stable/unstable product chart. If its stable coordinate at the more recent return were nonzero, inverse iteration through the preceding controlled return blocks would expand that coordinate by at least $e^{(\chi_s - \varepsilon)n}$, whereas (29) keeps every return displacement bounded by δ . Letting the number of preceding returns tend to infinity forces the stable coordinate to vanish. Equivalently, the point at the return lies on the local unstable plaque of the corresponding return of x . This is just the uniqueness half of the graph transform in *Pesin local stable/unstable manifold theorem*, used with the uniform constants of R . The plaque-component label in $\mathcal{P}$ then puts both return points in the same local component. Propagating forward and using (11) proves (28).

Now let $\varepsilon > 0$. By the definition of metric entropy and the Ruelle bound (which also shows finiteness here), choose a finite partition $\mathcal{A}$ such that

$$h_\mu(f, \mathcal{A}) > h_\mu(f) - \varepsilon. \quad (31)$$

Set $\mathcal{Q} = \mathcal{P} \vee \mathcal{A}$; then $H(\mathcal{Q}) < \infty$ and $\mathcal{Q}^u \succeq \mathcal{P}^u \succeq \eta$. For

$$\mathcal{Q}_n := \bigvee_{j=1}^n f^{-j} \mathcal{Q}, \quad (32)$$

the conditional-entropy chain rule gives

$$H(f^{-n} \eta \mid \eta) \geq H(\mathcal{Q}_n \mid \eta) - H(\mathcal{Q}_n \mid f^{-n} \eta). \quad (33)$$

Since $\mathcal{Q}^u \succeq \eta$,

$$H(\mathcal{Q}_n \mid \eta) \geq H(\mathcal{Q}_n \mid \mathcal{Q}^u) = nh_\mu(f, \mathcal{Q}). \quad (34)$$

For the last equality, expand the left side one symbol at a time and shift by f^j ; at every step the conditioning partition becomes $\bigvee_{k \geq 1} f^k \mathcal{Q}$, giving the usual future formula for $h_\mu(f, \mathcal{Q})$.

By (8), the sigma-algebras of $f^{-m} \eta$ increase to the full Borel sigma-algebra. Hence martingale convergence gives an m_0 such that

$$H(\mathcal{Q} \mid f^{-m_0} \eta) < \varepsilon. \quad (35)$$

Expanding the second term of (33) and using invariance, for $j \leq n - m_0$ we obtain

$$H(f^{-j} \mathcal{Q} \mid f^{-n} \eta) = H(\mathcal{Q} \mid f^{-(n-j)} \eta) \leq \varepsilon, \quad (36)$$

while the remaining at most m_0 terms are bounded by $H(\mathcal{Q})$. Thus

$$H(\mathcal{Q}_n \mid f^{-n} \eta) \leq n\varepsilon + m_0 H(\mathcal{Q}). \quad (37)$$

Finally, increasingness and invariance give

$$H(f^{-n}\eta \mid \eta) = nH(f^{-1}\eta \mid \eta). \quad (38)$$

Divide (33) by n , use (34)–(38), let $n \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. We get

$$H(f^{-1}\eta \mid \eta) \geq h_\mu(f). \quad (39)$$

Together with Lemma 2.1 this proves (23).

Step 4: the canonical unstable leaf-volume family. For $x \in S$ define

$$L(x) := \int_{\eta(x)} \Delta(x, y) dm_x^u(y). \quad (40)$$

The local bounds from Step 1 and the local finiteness of the plaque volume show

$$0 < L(x) < \infty \quad \text{for almost every } x. \quad (41)$$

Define a probability on the atom $\eta(x)$ by

$$\nu_x(B) := \frac{1}{L(x)} \int_{B \cap \eta(x)} \Delta(x, y) dm_x^u(y). \quad (42)$$

This family is measurable and is equivalent, with a strictly positive density, to induced Riemannian volume on each atom.

Let

$$C_1(x) := (f^{-1}\eta)(x) = f^{-1}(\eta(fx)) \quad (43)$$

and put $q(x) = \nu_x(C_1(x))$. The cocycle relation

$$\Delta(fx, fy) = \frac{J^u(x)}{J^u(y)} \Delta(x, y) \quad (44)$$

and the leaf-volume change of variables $dm_{fx}^u(fy) = J^u(y)dm_x^u(y)$ give

$$q(x) = \frac{L(fx)}{J^u(x)L(x)}. \quad (45)$$

Consequently

$$-\log q(x) = \log J^u(x) + \log L(x) - \log L(fx). \quad (46)$$

We record the one-sided coboundary lemma needed here. If $d = g - g \circ f$ is finite almost everywhere and $d^- \in L^1(\mu)$, then also $d^+ \in L^1(\mu)$ and $\int d d\mu = 0$. Indeed, with the clipping $g_N = \max(-N, \min(g, N))$, monotonicity and the 1-Lipschitz property of the clipping map give

$$(g_N - g_N \circ f)^- \leq d^-.$$

Invariance gives equality of the positive and negative integrals of $g_N - g_N \circ f$. Fatou therefore makes d^+ integrable; once both parts of d are integrable, dominated convergence (using $|g_N - g_N \circ f| \leq |d|$) gives $\int d d\mu = 0$.

Apply this to $d = \log L - \log L \circ f = -\log q - \log J^u$. Since $0 < q \leq 1$ and $\log J^u$ is bounded on the compact manifold, $d^- \leq (\log J^u)^+$ is integrable. Consequently

$$\int -\log q \, d\mu = \int \log J^u \, d\mu. \quad (47)$$

Step 5: equality in unstable entropy forces the canonical conditionals. Let $\{\mu_x^\eta\}$ be the Rokhlin conditional probabilities of μ on η , and set

$$p(x) := \mu_x^\eta(C_1(x)). \quad (48)$$

Then

$$H_\mu(f^{-1}\eta \mid \eta) = \int -\log p(x) \, d\mu(x). \quad (49)$$

Inside each η -atom, $f^{-1}\eta$ is a countable partition (the atoms are open unstable pieces modulo leaf-volume-null boundaries). If its true and canonical child probabilities are (p_j) and (q_j) , respectively, then Gibbs' inequality gives

$$\sum_j p_j \log \frac{p_j}{q_j} \geq 0, \quad (50)$$

with equality if and only if $p_j = q_j$ for every j with positive mass. Integrating (50) over the quotient by η yields

$$\int \log \frac{p(x)}{q(x)} \, d\mu(x) \geq 0. \quad (51)$$

(The statement follows first for finitely many children from $\log t \leq t - 1$, and then for the countable partition by monotone truncation.)

Suppose now that

$$H_\mu(f^{-1}\eta \mid \eta) = \int \log J^u \, d\mu. \quad (52)$$

By (47), the integral in (51) is zero. Hence the true and canonical one-step child distributions agree almost everywhere. Apply the same argument to f^n . The partition $f^{-n}\eta$ refines η , its conditional entropy is n times (49), and its unstable Jacobian is

$$J_n^u(x) = \prod_{k=0}^{n-1} J^u(f^k x). \quad (53)$$

The canonical n -step probabilities obtained from (42) therefore agree with the true probabilities on every atom of $f^{-n}\eta$, for every $n \geq 1$. By (8), the union of these finite-level sigma-algebras generates the Borel sigma-algebra on almost every η -atom. The monotone-class theorem now gives

$$\mu_x^\eta = \nu_x \quad \text{for almost every } x. \quad (54)$$

By (42), the conditionals are absolutely continuous with respect to unstable Riemannian volume.

Step 6: absolutely continuous unstable conditionals give equality. Conversely

suppose the conditionals μ_x^η are absolutely continuous with respect to m_x^u . Write their normalized densities as G_x :

$$d\mu_x^\eta(y) = G_x(y) dm_x^u(y), \quad \int_{\eta(x)} G_x dm_x^u = 1. \quad (55)$$

For the child $C_1(x)$, the conditional probability is $p(x)$ from (48). The normalized restriction $\mu_x^\eta|_{C_1(x)}/p(x)$ pushes under f to μ_{fx}^η . Uniqueness of Rokhlin disintegration and the leaf-volume change of variables therefore imply, for almost every $y \in C_1(x)$,

$$G_{fx}(fy)J^u(y) = \frac{G_x(y)}{p(x)}. \quad (56)$$

Choose the Radon–Nikodym derivatives as a jointly measurable density on the measurable bundle of η -atoms and set $\Phi(y) = \log G_{\eta(y)}(y)$ on the full-measure set where the density is positive and finite. Equation (56) holds for μ_x^η -almost every y on almost every quotient atom; integrating this exceptional set over the quotient (Rokhlin disintegration) shows that it holds at $y = x$ for μ -almost every point after this measurable choice. Thus, without any pointwise evaluation of an arbitrarily chosen density representative, we get the information-function coboundary

$$-\log p(x) = \log J^u(x) + \Phi(fx) - \Phi(x) \quad \text{a.e.} \quad (57)$$

By (23) and Ruelle’s inequality, the left side has finite integral. Since $\log J^u$ is bounded, the coboundary in (57) is integrable; the truncation argument following (46) gives zero integral. Hence

$$H_\mu(f^{-1}\eta \mid \eta) = \int \log J^u d\mu. \quad (58)$$

Together with (23), this is Pesin’s formula (4).

Step 7: finish both implications and remove the choice of partition. If μ has the SRB unstable-conditional property, choose the partition η constructed above. Absolute continuity on any subordinate partition passes to a common measurable refinement by restricting and renormalizing the conditional densities; hence it holds for η . Step 6 and (23) give (4).

Conversely, assume Pesin’s formula. Equations (3), (4), and (23) give exactly (52). Step 5 therefore gives absolute continuity of the conditionals on η . If ξ is any other measurable partition subordinate to unstable manifolds, then $\xi \vee \eta$ is, almost everywhere, a countable local refinement on each unstable leaf after decomposing the atoms into connected plaque pieces. The conditional measures for $\xi \vee \eta$ are restrictions and normalizations of the absolutely continuous η -conditionals, hence are absolutely continuous. Disintegrating these conditionals once more over ξ expresses μ_x^ξ as a mixture of measures absolutely continuous with respect to the same induced leaf volume; therefore μ_x^ξ is absolutely continuous as well. Thus the property holds for every subordinate partition, proving the claimed equivalence.

No step asserts that the resulting SRB measure has a positive-Riemannian-volume basin. That physicality statement is deliberately outside the theorem’s scope. $\square$

Theorem F11.8. Hopf argument for hyperbolic ergodicity

Statement. Let $f \in C^{1+\alpha}(M)$, $\alpha > 0$, be a diffeomorphism of a compact Riemannian manifold preserving a probability μ equivalent to Riemannian volume, and assume that μ is hyperbolic. Use the Pesin local stable and unstable laminations on the full-measure regular set. Call the system *essentially su -accessible* if every measurable set E with the following property has $\mu(E) \in \{0, 1\}$: in every Pesin foliation box, for almost every local stable plaque and almost every local unstable plaque, E meets that plaque in either zero leaf-volume or full leaf-volume. Then f is ergodic. More precisely, for every $\varphi \in C(M)$ the forward and backward Birkhoff averages agree almost everywhere and are essentially constant along both stable and unstable plaques; essential su -accessibility forces this common average to be constant almost everywhere.

Proof. Let m denote Riemannian volume. Since μ is equivalent to m , the two measures have exactly the same null sets. This simple observation is what allows the absolute-continuity theorem, which is formulated using induced Riemannian volumes, to be used in a μ -ergodic argument.

We first isolate the leafwise Fubini consequence of *Absolute continuity of Pesin stable/unstable laminations*.

Claim 1 (strong leafwise null-set property). Let $R \subset M$ have full μ -measure. In every stable Pesin foliation box, for almost every local stable plaque (with respect to any smooth transverse parameterization), the complement of R has zero induced leaf-volume on that plaque. The analogous statement holds for unstable plaques.

Proof of Claim 1. Work in one stable foliation box from *Absolute continuity of Pesin stable/unstable laminations*. Choose a smooth one-parameter family of u -dimensional transversals $\{T_t\}$ filling a smaller box, where $u = \dim E^u$ and the parameter t ranges in a small s -disk. This smooth transverse family is an ordinary smooth foliation, so ordinary Fubini in its smooth coordinates disintegrates m into measures which are equivalent to induced volume on each T_t , with positive bounded smooth densities.

Fix a reference transversal T_0 and write $Z = T_0 \cap \mathcal{F}^s$ for the trace of the Pesin foliation block. Stable holonomy $h_t : Z \rightarrow T_t \cap \mathcal{F}^s$ has, by *Absolute continuity of Pesin stable/unstable laminations*, Jacobian bounded above and away from zero, uniformly after the box is shrunk. Write z for the coordinate on T_0 and define

$$\Psi(t, z) := h_t(z). \quad (18)$$

For fixed t , change of variables under h_t gives

$$dm_{T_t}(\Psi(t, z)) = Jh_t(z) dm_{T_0}(z), \quad C^{-1} \leq Jh_t(z) \leq C \quad (19)$$

for almost every $z \in Z$. On the other hand, because the family $\{T_t\}$ is a smooth transverse foliation, ordinary smooth Fubini gives

$$dm(q) = g(t, q) dm_{T_t}(q) dt, \quad 0 < c \leq g \leq C' \quad (20)$$

in the smaller box. Combining (19)–(20), for every nonnegative measurable F ,

$$\int F \, dm = \int \int F(\Psi(t, z)) g(t, \Psi(t, z)) Jh_t(z) \, dm_{T_0}(z) \, dt. \quad (21)$$

Thus ambient volume in the box is equivalent, with bounded positive density, to the restriction of the product measure $dt \, dm_{T_0}(z)$ to the measurable coordinates (t, z) .

For fixed $z \in Z$, the curve/s-disk $t \mapsto \Psi(t, z)$ parameterizes the stable plaque through z . The stable plaques are uniformly C^1 graphs in the Pesin box, so this parameterization has Jacobian bounded above and away from zero relative to stable leaf-volume. Therefore, if N has $m(N) = 0$, (21) and ordinary Fubini imply that for m_{T_0} -almost every $z \in Z$ the set of t with $\Psi(t, z) \in N$ has Lebesgue measure zero, and hence N has zero induced leaf-volume on the stable plaque through z . Taking $N = M \setminus R$ proves the stable assertion. The unstable assertion is identical, using unstable holonomy from *Absolute continuity of Pesin stable/unstable laminations*. $\square$

Fix now $\varphi \in C(M)$. By *Birkhoff pointwise ergodic theorem* applied to f and to f^{-1} , there is a full μ -measure set R_φ on which both limits

$$\varphi^+(x) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \varphi(f^n x), \quad \varphi^-(x) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \varphi(f^{-n} x) \quad (1)$$

exist. The invariant sigma-algebras of f and f^{-1} are identical. Hence both Birkhoff limits equal the same conditional expectation

$$g_\varphi := \mathbb{E}_\mu(\varphi \mid \mathcal{I}_f) \quad (2)$$

almost everywhere. Shrinking R_φ by a null set, we may assume

$$\varphi^+(x) = \varphi^-(x) = g_\varphi(x) \quad (x \in R_\varphi). \quad (3)$$

Step 1: stable and unstable constancy of Birkhoff averages. Suppose $x, y \in R_\varphi$ lie on the same local stable plaque. Then $d(f^n x, f^n y) \rightarrow 0$ exponentially by *Pesin local stable/unstable manifold theorem*. Since M is compact, φ is uniformly continuous. Thus

$$|\varphi(f^n x) - \varphi(f^n y)| \rightarrow 0. \quad (4)$$

A bounded sequence tending to zero has Cesàro mean tending to zero, so

$$|g_\varphi(x) - g_\varphi(y)| = \lim_{N \rightarrow \infty} \left| \frac{1}{N} \sum_{n=0}^{N-1} [\varphi(f^n x) - \varphi(f^n y)] \right| = 0. \quad (5)$$

Therefore g_φ is constant on the intersection of each stable plaque with R_φ .

If instead $x, y \in R_\varphi$ lie on the same local unstable plaque, then $d(f^{-n} x, f^{-n} y) \rightarrow 0$. Applying the same argument to the backward averages in (1) gives

$$g_\varphi(x) = g_\varphi(y). \quad (6)$$

Thus the common Birkhoff average is constant on regular points along both stable and unstable plaques.

Step 2: the level sets are essentially su -saturated. For $a \in \mathbb{R}$ set

$$E_a := \{x : g_\varphi(x) < a\}. \quad (7)$$

We claim that E_a is saturated modulo leaf-volume by both Pesin laminations in every foliation box.

Indeed Claim 1, applied to the full-measure set R_φ , says that on almost every stable plaque, R_φ has full leaf-volume. On such a plaque equation (5) says that g_φ takes one value at every point of R_φ on that plaque. Hence, modulo a leaf-volume-zero subset, either the entire plaque belongs to E_a or none of it does. This is exactly stable saturation modulo leaf-volume. Equation (6) and the unstable part of Claim 1 give the same conclusion for unstable plaques.

By essential su -accessibility,

$$\mu(E_a) \in \{0, 1\} \quad \text{for every } a \in \mathbb{R}. \quad (8)$$

Step 3: a measurable function whose sublevel sets have only zero or full measure is constant. Let

$$c := \inf\{a \in \mathbb{R} : \mu(E_a) = 1\}. \quad (9)$$

The set in (9) is nonempty because g_φ is bounded by $\|\varphi\|_\infty$, and it is bounded below for the same reason. For every $r < c$, (8) and the definition of c give $\mu(E_r) = 0$. Taking a countable union over rational $r < c$,

$$\mu\{g_\varphi < c\} = 0. \quad (10)$$

For every rational $r > c$, monotonicity of $a \mapsto E_a$ and the definition of the infimum give $\mu(E_r) = 1$; intersecting over rational $r > c$ yields

$$\mu\{g_\varphi \leq c\} = 1. \quad (11)$$

Together (10)–(11) show

$$g_\varphi = c \quad \mu\text{-almost everywhere.} \quad (12)$$

Since conditional expectation preserves integrals,

$$c = \int_M g_\varphi d\mu = \int_M \varphi d\mu. \quad (13)$$

Thus every continuous observable has almost-everywhere Birkhoff average equal to its space average.

Step 4: ergodicity. We now convert the conclusion for continuous functions into triviality of the invariant sigma-algebra. Let A be an invariant Borel set. Since M is a compact metric space and μ is a finite regular Borel measure, continuous functions

are dense in $L^1(\mu)$. Choose $\varphi_j \in C(M)$ with

$$\|\varphi_j - \mathbf{1}_A\|_{L^1(\mu)} \rightarrow 0. \quad (14)$$

Conditional expectation is an L^1 contraction. Because A is invariant, $\mathbb{E}(\mathbf{1}_A | \mathcal{I}_f) = \mathbf{1}_A$, while Step 3 gives

$$\mathbb{E}(\varphi_j | \mathcal{I}_f) = \int \varphi_j d\mu \quad \text{a.e.} \quad (15)$$

Therefore

$$\left\| \mathbf{1}_A - \int \varphi_j d\mu \right\|_{L^1} \leq \|\mathbf{1}_A - \varphi_j\|_{L^1} \rightarrow 0. \quad (16)$$

Also $\int \varphi_j d\mu \rightarrow \mu(A)$. Hence

$$\|\mathbf{1}_A - \mu(A)\|_{L^1} = 0. \quad (17)$$

An indicator can equal a constant almost everywhere only when that constant is 0 or 1. Thus $\mu(A) \in \{0, 1\}$ for every invariant Borel set A , which is precisely ergodicity.

This proves the theorem. $\square$

11.3 Worked examples

Worked example: Nonuniform Hyperbolicity and Pesin Theory: a model computation

The cat map on $\mathbb{T}^2$ is induced by

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det A = 1.$$

Its eigenvalues solve $\lambda^2 - 3\lambda + 1 = 0$:

$$\lambda_+ = \frac{3 + \sqrt{5}}{2} > 1, \quad \lambda_- = \frac{3 - \sqrt{5}}{2} = \lambda_+^{-1} < 1.$$

Because the derivative is the same matrix at every point, the Lyapunov exponents for Lebesgue measure are

$$\chi^+ = \log \lambda_+, \quad \chi^- = -\log \lambda_+.$$

The Oseledets splitting is constant and agrees with the eigendirections. Consequently the entire torus is one uniform Pesin block: the angles and contraction constants do not vary with x . Lebesgue measure is invariant because $\det A = 1$, and Pesin's entropy formula gives

$$h_{\text{Leb}}(A) = \log \lambda_+.$$

This uniformly hyperbolic example is the clean model to keep in mind when the same objects become only measurable in a genuinely nonuniform system.

Boundary case: what fails at the edge of the hypotheses

For the identity cocycle $Df_x = I$, every Lyapunov exponent is 0. No positive number χ can satisfy a uniform estimate like $\|Df^n v^s\| \leq Ce^{-\chi n} \|v^s\|$. The hyperbolic regular set required for Pesin stable/unstable manifolds is therefore empty. Oseledets' theorem still applies—it simply reports zero exponents—but the hyperbolic conclusions of Pesin theory do not follow.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

11.4 Chapter summary

What to remember

- Pesin theory is uniform hyperbolic geometry reconstructed on large-measure blocks.
- Lyapunov norms and tempered constants convert asymptotic exponents into finite-scale estimates.
- Absolute continuity is the bridge between invariant foliations and ambient measure.
- Entropy formulae quantify the information production associated with positive exponents.

Looking ahead

F12 turns from hyperbolic geometry to iterative solution methods. Newton, KAM, and Nash–Moser schemes repeatedly solve the kind of linearized equations developed in F08 while controlling the losses quantified in F07.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

11.5 Exercises and problems

Exercise 11.1

Difficulty. Core.

Related results. Lyapunov norm and Pesin regularity functions; Pesin local stable/unstable manifold theorem; Absolute continuity of Pesin stable/unstable laminations.

Problem. Work through the worked example ‘the cat map on T^2 with Lebesgue measure as a uniformly hyperbolic Pesin model’. For Lyapunov norm and Pesin regularity functions, Pesin local stable/unstable manifold theorem, and Absolute continuity of Pesin stable/unstable laminations, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 11.2

Difficulty. Core.

Related results. Pesin entropy formula; Ruelle inequality; Ledrappier–Young characterization of the SRB property.

Problem. Analyze the boundary example ‘a zero-exponent cocycle illustrating why Pesin hyperbolicity needs nonzero exponents’. For Pesin entropy formula, Ruelle inequality, and Ledrappier–Young characterization of the SRB property, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 11.3

Difficulty. Advanced.

Related results. Absolute continuity of Pesin stable/unstable laminations.

Problem. For Absolute continuity of Pesin stable/unstable laminations, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 11.4**Difficulty.** Advanced.**Related results.** Pesin entropy formula.**Problem.** Stress-test Pesin entropy formula. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.**Solution sketch.** Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.**Exercise 11.5****Difficulty.** Advanced.**Related results.** Lyapunov norm and Pesin regularity functions; Pesin local stable/unstable manifold theorem.**Problem.** Synthesis problem. Use Lyapunov norm and Pesin regularity functions and Pesin local stable/unstable manifold theorem to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.**Solution sketch.** State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.**Exercise 11.6****Difficulty.** Core.**Related results.** Hopf argument for hyperbolic ergodicity.**Problem.** Work directly with Hopf argument for hyperbolic ergodicity. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.**Exercise 11.7****Difficulty.** Core.**Related results.** Pesin blocks and uniformization.**Problem.** Work directly with Pesin blocks and uniformization. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.**Solution sketch.** First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include

the calculations rather than only a verbal discussion.

Normal Forms, Newton Schemes, Nash-Moser and Perturbative Reducibility

Why this chapter matters

Many hard existence and conjugacy theorems share the same engine: solve a linearized equation, use the solution to change coordinates, and exploit the fact that the new error is quadratic. Ordinary Newton iteration works when the inverse linearized operator is uniformly bounded. KAM adds small divisors. Nash-Moser adds smoothing when the inverse loses derivatives.

This chapter isolates those reusable mechanisms before they are specialized to Hamiltonian tori, cocycle spectral theory, or nonlinear PDE.

Prerequisites. F07–F09, differential calculus on Banach/Frechet scales, and basic normal-form algebra.

Learning goals

- Derive the quadratic residual identity behind Newton iteration.
- Understand homological equations in formal and analytic normal forms.
- See how smoothing compensates for derivative loss in Nash–Moser schemes.
- Interpret perturbative reducibility as a nonlinear conjugacy problem built from repeated linear solves.

Conceptual roadmap

Formal normal forms identify which terms are removable. Newton’s lemma explains why an accurate linear correction squares the error. Analytic KAM steps combine Newton with small-divisor estimates. Nash–Moser inserts smoothing when the inverse is not tame enough on one fixed space. Perturbative reducibility packages the same ideas for matrix cocycles.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

12.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F12.1. Normal form

A *normal form* is a representative obtained by an allowed class of coordinate or conjugacy changes in which all terms removable by the chosen homological operator, up to a stated order or scale, have been eliminated.

Definition F12.2. Newton scheme

A *Newton scheme* is an iteration $u_{n+1} = u_n + \Delta_n$ in which Δ_n solves or approximately solves $DF(u_n)\Delta_n = -F(u_n)$, with estimates arranged so that the residual decays superlinearly, typically quadratically.

Definition F12.3. Nash-Moser iteration

A *Nash-Moser iteration* is a Newton-type scheme on a scale of spaces in which smoothing operators are inserted to compensate for loss of derivatives in the inverse linearized operator.

Definition F12.4. Perturbative reducibility

A cocycle is *perturbatively reducible* in a specified regularity class if, under stated smallness and arithmetic assumptions, a conjugacy in that class transforms it to a constant cocycle or to the prescribed finite resonant normal form.

Definition F12.5. Graded Frechet space and model sequence scale

A *graded Frechet space* E is equipped with an increasing family of seminorms $\|\cdot\|_0 \leq \|\cdot\|_1 \leq \dots$ defining its topology. For a Banach space B , the model tame scale is $\Sigma(B) = \{x = (x_j)_{j \geq 0} : \|x\|_n := \sum_{j \geq 0} e^{nj} \|x_j\|_B < \infty \text{ for every } n\}$, up to the equivalent grading fixed in the formal theorem.

Definition F12.6. Tame map

Let E, F be graded Frechet spaces. A linear map $L : E \rightarrow F$ is *tame of degree r and base b* if, for every $n \geq b$, $\|Lx\|_n \leq C_n \|x\|_{n+r}$. A nonlinear map P is tame if locally $\|P(x)\|_n \leq C_n(1 + \|x\|_{n+r})$; it is smooth tame when all derivatives satisfy the corresponding local tame estimates.

Definition F12.7. Tame Frechet space

A graded Frechet space E is *tame* if it is a tame direct summand of a model sequence space: there are tame linear maps $i : E \rightarrow \Sigma(B)$, $p : \Sigma(B) \rightarrow E$ with $p \circ i = \text{id}_E$.

Definition F12.8. Smooth tame right-inverse family

If $P : U \subset E \rightarrow F$ is smooth tame, a family $\Psi(x) : F \rightarrow E$ is a *smooth tame family of right inverses* if $DP(x)\Psi(x)y = y$ and the joint map $(x, y) \mapsto \Psi(x)y$ is smooth tame.

Definition F12.9. Relative resonance for fibered rotation numbers

Let $\alpha \in \mathbb{R}/\mathbb{Z}$ be irrational and let $\rho \in \mathbb{R}/\mathbb{Z}$ be a fibered rotation number in the F09 normalization. We call ρ *resonant relative to* α if $\rho - k\alpha \in \mathbb{Z}$ for some $k \in \mathbb{Z}$. We call it *Diophantine relative to* α if there are $\kappa, \sigma > 0$ such that $|\sin \pi(\rho - k\alpha)| \geq \kappa|k|^{-\sigma}$ for every $k \neq 0$.

Definition F12.10. Analytic reducibility

An analytic $SL(2, \mathbb{R})$ cocycle (α, A) is *analytically reducible to a constant* if there are an analytic B (on the circle or, when explicitly allowed, its double cover) and a constant $A_* \in SL(2, \mathbb{R})$ such that $B(x + \alpha)A(x)B(x)^{-1} = A_*$.

12.2 Main results

Theorem F12.1. Formal Poincaré-Dulac normal form

Statement. Let

$$X(x) = Ax + \sum_{m \geq 2} X_m(x)$$

be a formal vector field at $0 \in \mathbb{C}^n$, where each X_m is a homogeneous polynomial vector field of degree m , and put A in Jordan normal form. Write $A = \Lambda + N$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ is the semisimple part and N is nilpotent. A monomial vector field $x^\alpha e_j$, $|\alpha| \geq 2$, is *resonant* when

$$\langle \alpha, \lambda \rangle = \lambda_j.$$

For every integer $M \geq 2$ there is a polynomial change of coordinates

$$\Phi_M(y) = y + \sum_{m=2}^M H_m(y), \quad H_m \text{ homogeneous of degree } m,$$

tangent to the identity and formally invertible, for which

$$(D\Phi_M(y))^{-1}X(\Phi_M(y)) = Ay + \sum_{m=2}^M R_m(y) + O(|y|^{M+1}),$$

and every R_m is a linear combination of resonant monomial vector fields. The choices can be made compatibly in M , producing a formal near-identity transformation to a formal Poincaré-Dulac normal form. In particular, if there are no resonances of degree at least 2, the vector field is formally linearizable.

Proof. For $m \geq 2$, let $\mathcal{H}_m$ be the finite-dimensional complex vector space of homogeneous polynomial vector fields of degree m on $\mathbb{C}^n$. For a linear vector field Ay , define the homological operator

$$\mathcal{L}_A : \mathcal{H}_m \longrightarrow \mathcal{H}_m, \quad \mathcal{L}_A H := DH(y) Ay - AH(y). \quad (1)$$

The sign in (1) is chosen so that a coordinate change $x = y + H(y)$ changes the degree- m term by subtracting $\mathcal{L}_A H$; we verify this explicitly below.

Step 1: the resonant/nonresonant splitting. Write the Jordan decomposition of the linear part as

$$A = \Lambda + N, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad [\Lambda, N] = 0,$$

with N nilpotent. On a monomial vector field $y^\alpha e_j \in \mathcal{H}_m$, $|\alpha| = m$, one computes

$$\mathcal{L}_\Lambda(y^\alpha e_j) = (\langle \alpha, \lambda \rangle - \lambda_j) y^\alpha e_j. \quad (2)$$

Hence

$$\mathcal{R}_m := \ker \mathcal{L}_\Lambda$$

is exactly the span of the resonant monomials of degree m .

Because $[\Lambda, N] = 0$, the induced operators $\mathcal{L}_\Lambda$ and $\mathcal{L}_N$ commute on $\mathcal{H}_m$. Moreover $\mathcal{L}_N$ is nilpotent. One way to see the latter is to choose a basis in which N is strictly upper triangular: substitution of Ny and multiplication by N both strictly move the corresponding monomial/basis indices, so repeated application of $\mathcal{L}_N$ is zero on the finite-dimensional space $\mathcal{H}_m$.

Decompose $\mathcal{H}_m$ into eigenspaces of the diagonal operator $\mathcal{L}_\Lambda$:

$$\mathcal{H}_m = \bigoplus_{\rho} E_{\rho}, \quad \mathcal{L}_\Lambda|_{E_{\rho}} = \rho I. \quad (3)$$

Commutation shows that every E_{ρ} is invariant under $\mathcal{L}_N$. If $\rho \neq 0$, then on E_{ρ}

$$\mathcal{L}_A = \rho I + \mathcal{L}_N = \rho (I + \rho^{-1} \mathcal{L}_N),$$

which is invertible because the inverse is the finite Neumann series

$$(\mathcal{L}_A|_{E_{\rho}})^{-1} = \rho^{-1} \sum_{q=0}^{Q-1} (-\rho^{-1} \mathcal{L}_N)^q \quad (4)$$

once Q is large enough that $\mathcal{L}_N^Q = 0$. Consequently every nonresonant eigenspace E_{ρ} , $\rho \neq 0$, is contained in $\text{im } \mathcal{L}_A$. Thus

$$\mathcal{H}_m = \mathcal{R}_m + \text{im } \mathcal{L}_A. \quad (5)$$

The sum need not be direct when $N \neq 0$, and no directness is needed.

It follows from (5) that for every $F_m \in \mathcal{H}_m$ we can choose $R_m \in \mathcal{R}_m$ and $H_m \in \mathcal{H}_m$

such that

$$F_m = R_m + \mathcal{L}_A H_m. \quad (6)$$

Step 2: one homogeneous normalization step. Suppose a formal vector field has already been normalized through degree $m - 1$:

$$Y(y) = Ay + R_2(y) + \cdots + R_{m-1}(y) + F_m(y) + O(|y|^{m+1}), \quad (7)$$

where each R_k is resonant. Choose H_m as in (6), and make the homogeneous change of variables

$$y = z + H_m(z). \quad (8)$$

Because $m \geq 2$, its derivative at the origin is the identity, so (8) has a formal inverse tangent to the identity.

Let $\tilde{Y}$ be the vector field in the z -coordinate:

$$\tilde{Y}(z) = (D(I + H_m)(z))^{-1} Y(z + H_m(z)). \quad (9)$$

Modulo terms of degree $m + 1$ and higher,

$$\begin{aligned} Y(z + H_m(z)) &= Az + AH_m(z) + R_2(z) + \cdots + R_{m-1}(z) + F_m(z) + O(|z|^{m+1}), \\ (D(I + H_m))^{-1} &= I - DH_m + O(|z|^{2m-2}). \end{aligned}$$

When the second line is multiplied into the first, the only contribution of degree exactly m coming from $-DH_m$ is $-DH_m(z)Az$. All products of DH_m with R_k , $k \geq 2$, have degree at least $m + 1$. Therefore the degree- m term of $\tilde{Y}$ is

$$F_m + AH_m - DH_m Az = F_m - \mathcal{L}_A H_m = R_m, \quad (10)$$

by (1) and (6). Since the change (8) starts in degree m , all terms of degrees $< m$ in (7) are unchanged. Thus one normalization step removes the nonresonant degree- m component while preserving every previously normalized degree.

Step 3: induction to arbitrary finite order. Starting with $m = 2$, apply Step 2 successively for $m = 2, 3, \dots, M$. The composition of these homogeneous near-identity changes is a polynomial map

$$\Phi_M(y) = y + \sum_{m=2}^M \hat{H}_m(y)$$

whose derivative at the origin is the identity. (The displayed $\hat{H}_m$ are the homogeneous jets of the composition; they need not equal the intermediate H_m literally.) The resulting transformed vector field is

$$(D\Phi_M(y))^{-1} X(\Phi_M(y)) = Ay + \sum_{m=2}^M R_m(y) + O(|y|^{M+1}), \quad (11)$$

with every $R_m \in \mathcal{R}_m$, hence every R_m a linear combination of resonant monomial vector fields by (2).

The construction at stage $M+1$ uses a coordinate change that is the identity through degree M . Therefore the M -jet of the normalizing transformation obtained at stage $M+1$ agrees with the M -jet already constructed. These compatible finite jets define a formal near-identity transformation

$$\Phi(y) = y + \sum_{m \geq 2} H_m(y)$$

with formal inverse, and (11) for every M means precisely that Φ conjugates X to a formal vector field containing only resonant nonlinear monomials.

Finally, if no resonance $\langle \alpha, \lambda \rangle = \lambda_j$ occurs for any $|\alpha| \geq 2$, then $\mathcal{R}_m = \{0\}$ for every $m \geq 2$. All R_m vanish in (11), so the formal normal form is simply Ay . Thus X is formally linearizable. $\square$

Proposition F12.2. Newton quadratic-error lemma

Statement. Let X, Y be Banach spaces, $U \subset X$ open, $F \in C^2(U, Y)$, $x \in U$, and $R: Y \rightarrow X$ bounded with $\|R\| \leq M$ and $\|I_Y - DF(x)R\| \leq \eta$. Put $e = F(x)$ and $\Delta = -Re$, assume $x + t\Delta \in U$ for $0 \leq t \leq 1$, and suppose $\sup_{0 \leq t \leq 1} \|D^2F(x + t\Delta)\| \leq K$. Then $\|F(x + \Delta)\| \leq \eta\|e\| + (K/2)M^2\|e\|^2$. In particular an exact right inverse ($\eta = 0$) gives a quadratic residual bound.

Proof. Set

$$e := F(x), \quad \Delta := -Re.$$

The assumed segment condition $x + t\Delta \in U$ for $0 \leq t \leq 1$ lets us apply the second-order Taylor formula along the one-dimensional path $t \mapsto x + t\Delta$. In Banach-space form,

$$F(x + \Delta) = F(x) + DF(x)\Delta + \int_0^1 (1-t)D^2F(x + t\Delta)[\Delta, \Delta] dt. \quad (1)$$

We estimate the linear part and the nonlinear remainder separately.

Because $\Delta = -Re$,

$$F(x) + DF(x)\Delta = e - DF(x)Re = (I_Y - DF(x)R)e. \quad (2)$$

The approximate right-inverse assumption therefore gives

$$\|F(x) + DF(x)\Delta\| \leq \eta\|e\|. \quad (3)$$

For the remainder in (1), the bound on the second derivative gives, for every $t \in [0, 1]$,

$$\|D^2F(x + t\Delta)[\Delta, \Delta]\| \leq K\|\Delta\|^2.$$

Hence

$$\begin{aligned} \left\| \int_0^1 (1-t)D^2F(x + t\Delta)[\Delta, \Delta] dt \right\| &\leq K\|\Delta\|^2 \int_0^1 (1-t) dt \\ &= \frac{K}{2}\|\Delta\|^2. \end{aligned} \quad (4)$$

Since $\|R\| \leq M$,

$$\|\Delta\| = \|Re\| \leq M\|e\|, \quad (5)$$

so (4) becomes

$$\|\text{remainder}\| \leq \frac{K}{2} M^2 \|e\|^2. \quad (6)$$

Combining (1), (3), and (6) yields the claimed estimate

$$\|F(x + \Delta)\| \leq \eta\|e\| + \frac{K}{2} M^2 \|e\|^2.$$

If R is an exact right inverse for $DF(x)$, then $DF(x)R = I_Y$, so $\eta = 0$ and

$$\|F(x + \Delta)\| \leq \frac{K}{2} M^2 \|F(x)\|^2.$$

Thus one exact Newton correction turns a residual of size ε into one of order ε^2 , provided the correction segment stays inside the region where the stated second-derivative bound holds. $\square$

Theorem F12.3. Abstract analytic Newton-KAM convergence lemma

Statement. Let $(X_h, \|\cdot\|_h)_{h_\infty \leq h \leq h_0}$ be a nested Banach scale whose restriction maps satisfy $\|u\|_{h'} \leq \|u\|_h$ for $h' < h$. Set $\sigma = h_0 - h_\infty > 0$,

$$h_n = h_\infty + \sigma 2^{-n}, \quad \delta_n = h_n - h_{n+1} = \sigma 2^{-n-1}.$$

Let $A, B > 0$, $p, q \geq 0$, let $\varepsilon_n \geq 0$, and suppose corrections $u_n \in X_{h_{n+1}}$ and iterates $x_{n+1} = x_n + u_n$ satisfy

$$\varepsilon_{n+1} \leq A \delta_n^{-q} \varepsilon_n^2, \quad \|u_n\|_{h_{n+1}} \leq B \delta_n^{-p} \varepsilon_n.$$

If

$$\varepsilon_0 \leq \frac{\sigma^q}{2^{2q+1}A},$$

then

$$\varepsilon_n \leq \frac{\sigma^q}{A} 2^{-2^n - q(n+2)},$$

and $\sum_n \|u_n\|_{h_\infty} < \infty$. Hence x_n converges in X_{h_∞} .

Scope note

The lemma is an abstract convergence engine. Existence of an invariant torus or reducibility requires a specialist Newton/KAM step that supplies the displayed recursive estimates.

Proof. By definition,

$$\delta_n = \sigma 2^{-n-1}, \quad \delta_{n+1} = \frac{1}{2} \delta_n. \quad (1)$$

Introduce the dimensionless error

$$b_n := 2^q A \delta_n^{-q} \varepsilon_n. \quad (2)$$

The quadratic recurrence gives

$$\begin{aligned} b_{n+1} &= 2^q A \delta_{n+1}^{-q} \varepsilon_{n+1} \\ &\leq 2^q A (2^q \delta_n^{-q}) (A \delta_n^{-q} \varepsilon_n^2) \\ &= (2^q A \delta_n^{-q} \varepsilon_n)^2 = b_n^2. \end{aligned} \quad (3)$$

Thus the shrinking analytic domain has been absorbed exactly into the normalization (2).

For $n = 0$, $\delta_0 = \sigma/2$, so the assumed smallness condition implies

$$\begin{aligned} b_0 &= 2^q A (\sigma/2)^{-q} \varepsilon_0 = 2^{2q} A \sigma^{-q} \varepsilon_0 \\ &\leq \frac{1}{2}. \end{aligned} \quad (4)$$

Iterating (3),

$$b_n \leq b_0^{2^n} \leq 2^{-2^n}. \quad (5)$$

Solving (2) for ε_n and using $\delta_n^q = \sigma^q 2^{-q(n+1)}$,

$$\begin{aligned} \varepsilon_n &= \frac{b_n \delta_n^q}{2^q A} \\ &\leq \frac{\sigma^q}{A} 2^{-2^n - q(n+2)}. \end{aligned} \quad (6)$$

This is the announced double-exponential error estimate.

We now estimate the total correction on the limiting space. Because the scale is nested and $h_\infty < h_{n+1}$,

$$\|u_n\|_{h_\infty} \leq \|u_n\|_{h_{n+1}} \leq B \delta_n^{-p} \varepsilon_n. \quad (7)$$

Substituting (6),

$$\begin{aligned} \|u_n\|_{h_\infty} &\leq B \sigma^{-p} 2^{p(n+1)} \frac{\sigma^q}{A} 2^{-2^n - q(n+2)} \\ &= \frac{B}{A} \sigma^{q-p} 2^{-2^n + (p-q)n + p - 2q}. \end{aligned} \quad (8)$$

The series

$$C_{p,q} := \sum_{n=0}^{\infty} 2^{-2^n + (p-q)n + p - 2q} \quad (9)$$

converges, because the negative double-exponential term -2^n dominates the linear term $(p-q)n$. Therefore

$$\sum_{n=0}^{\infty} \|u_n\|_{h_\infty} \leq \frac{B}{A} \sigma^{q-p} C_{p,q} < \infty. \quad (10)$$

For $m > n$, the iteration relation $x_{j+1} = x_j + u_j$, with all restrictions understood in

X_{h_∞} , gives

$$\|x_m - x_n\|_{h_\infty} \leq \sum_{j=n}^{m-1} \|u_j\|_{h_\infty}. \quad (11)$$

The right-hand side tends to zero as $m, n \rightarrow \infty$ by (10). Hence (x_n) is Cauchy in the Banach space X_{h_∞} , and there is $x_\infty \in X_{h_\infty}$ such that

$$x_n \longrightarrow x_\infty \quad \text{in } X_{h_\infty}. \quad (12)$$

Nothing in the argument constructs a Newton correction, solves a small-divisor equation, or identifies the limiting object with an invariant torus or a reduced cocycle. Those are specialist inputs. This lemma isolates only the reusable convergence mechanism once those inputs have produced the two hypotheses of the statement. $\square$

Theorem F12.4. Nash-Moser local-surjectivity theorem with a tame right inverse

Statement. Let E, F be tame Fréchet spaces, let $U \subset E$ be open, and let $P : U \rightarrow F$ be smooth tame. Fix $x_0 \in U$ and put $y_0 = P(x_0)$. Suppose there is an open neighborhood $U_0 \subset U$ of x_0 and a smooth tame family of continuous linear maps $\Psi(x) : F \rightarrow E$, $x \in U_0$, such that

$$DP(x)\Psi(x) = \text{id}_F \quad (x \in U_0).$$

Then there is an open neighborhood V of y_0 and a smooth tame map $G : V \rightarrow U_0$ satisfying

$$G(y_0) = x_0, \quad P(G(y)) = y \quad (y \in V).$$

In particular P is locally surjective at x_0 . The theorem asserts only this right-inverse/local-surjectivity conclusion. If one additionally has a smooth tame family of two-sided inverses of DP , the usual local inverse theorem follows by combining this result with the corresponding local-injectivity argument; the tame hypotheses for a concrete PDE must be verified separately in that application.

Proof.

Proof strategy

The argument follows Hamilton's one-sided surjectivity architecture, but every loss-of-derivatives mechanism is kept explicit. First reduce tame Fréchet spaces to model sequence scales and solve a smoothed horizontal-lift ODE. Weighted residual estimates then give a convergent selected local section. Finally differentiate the *same* algorithm: a one-sided tangent lift, arbitrary finite smoothing weights, and a one-high-factor tame induction give all parameter derivatives on one fixed low-norm neighborhood. The last step passes these uniform estimates to the limiting section.

Step A: reduction to model sequence spaces. Translate x_0 and y_0 to 0. By the definition of a tame Fréchet space, after tame embeddings and retractions we may identify E

and F with the ranges of tame projections $p : \Sigma(B) \rightarrow \Sigma(B)$ and $q : \Sigma(C) \rightarrow \Sigma(C)$. For the right-inverse problem extend P to

$$\tilde{P}(u, v) = P(pu) + (I - q)v, \quad (u, v) \in \Sigma(B) \times \Sigma(C). \quad (\text{NM.1})$$

If $\Psi(x)$ is a right inverse of $DP(x)$, then

$$\tilde{\Psi}(u, v)k = (\Psi(pu)qk, (I - q)k) \quad (\text{NM.2})$$

satisfies

$$D\tilde{P}(u, v)\tilde{\Psi}(u, v) = I_{\Sigma(C)}. \quad (\text{NM.3})$$

Both extended maps are smooth tame. A local right inverse of $\tilde{P}$ restricts to one of P : if $y \in \text{ran } q$ and $\tilde{P}(u, v) = y$, applying q gives $P(pu) = y$. It is therefore enough to work with model spaces. In the remainder we rename them $E = \Sigma(B)$, $F = \Sigma(C)$ and write $P(0) = 0$.

For a real index $s \geq 0$ temporarily extend the model norm by

$$\|x\|_s := \sum_{j \geq 0} e^{sj} \|x_j\|_B. \quad (\text{NM.4})$$

This does not change the Fréchet topology or the tame category. A finite shift of the grading is allowed by the definition of a tame Fréchet space. Since only finitely many maps are used in the existence argument, smooth tameness lets us choose an integer $r \geq 1$ and a low-norm neighborhood on which, for every $n \geq 0$,

$$\|P(x)\|_n \leq C_n \|x\|_{n+2r}, \quad (\text{NM.5})$$

$$\|DP(x)h\|_n \leq C_n (\|h\|_{n+r} + \|x\|_{n+2r} \|h\|_r), \quad (\text{NM.6})$$

$$\|D^2P(x)[h_0, h_1]\|_n \leq C_n (\|h_0\|_{n+r} \|h_1\|_r + \|h_0\|_r \|h_1\|_{n+r} + \|x\|_{n+2r} \|h_0\|_r \|h_1\|_r), \quad (\text{NM.7})$$

$$\|\Psi(x)k\|_n \leq C_n (\|k\|_n + \|x\|_{n+2r} \|k\|_0) \quad (\text{NM.8})$$

whenever $\|x\|_{2r} \leq 1$. Enlarging r absorbs all finite losses in these four estimates. Notice that higher derivatives are not needed until the parameter-smoothness stage.

Step B: the fixed smoothing family. Choose $\chi \in C^\infty(\mathbb{R}, [0, 1])$ with $\chi(s) = 0$ for $s \leq 0$ and $\chi(s) = 1$ for $s \geq 1$, and define

$$(S_t x)_j = \chi(t - j) x_j, \quad t \geq 0. \quad (\text{NM.9})$$

At each fixed t , S_t has only finitely many nonzero coordinate blocks, and $S_t x \rightarrow x$ in every seminorm. Directly from (NM.4), for $b \geq a$,

$$\|S_t x\|_b \leq C_{a,b} e^{(b-a)t} \|x\|_a, \quad (\text{NM.10})$$

$$\|(I - S_t)x\|_a \leq C_{a,b} e^{-(b-a)t} \|x\|_b. \quad (\text{NM.11})$$

The harmless constants account for the unit transition region of χ . The same estimates hold in F .

Fix once and for all

$$c > 2r + 1. \quad (\text{NM.12})$$

For a target y near 0, consider the smoothed horizontal-lift equation

$$\dot{x}(t) = c \Psi(S_t x(t)) S_t (y - P(x(t))), \quad x(0) = 0. \quad (\text{NM.13})$$

On every bounded time interval the right-hand side depends on x through only finitely many Banach coordinate blocks. Projecting to those blocks gives an ordinary Banach-space ODE; after solving it, the remaining coordinates are recovered by integration. Patching across the finitely many integer transition times gives a unique local solution. The same argument, applied with y as an additional Banach parameter, shows that the finite-time solution $x(t; y)$ is smooth in y .

Put

$$k(t) := y - P(x(t)). \quad (\text{NM.14})$$

Because $DP(S_t x)\Psi(S_t x) = I$, equation (NM.13) gives the exact residual identity

$$\dot{k} + c S_t k = g, \quad g := (DP(S_t x) - DP(x))\dot{x}. \quad (\text{NM.15})$$

This identity, rather than a formal Newton expansion, is the source of the time-integrability.

Step C: weighted residual lemma. If $u : [0, T] \rightarrow F$ solves

$$\dot{u} + c S_t u = v, \quad (\text{NM.16})$$

then for $p \geq 0$ and $0 < q < c$,

$$\int_0^T e^{qt} \|u(t)\|_p dt \leq C_{p,q} \|u(0)\|_{p+q} + C_{p,q} \int_0^T (e^{qt} \|v(t)\|_p + \|v(t)\|_{p+q}) dt. \quad (\text{NM.17})$$

Here is the proof. In coordinate j the propagator is

$$a_{j,s,t} := \exp\left(-c \int_s^t \chi(\tau - j) d\tau\right), \quad 0 \leq s \leq t, \quad (\text{NM.18})$$

and variation of constants gives

$$u_j(t) = a_{j,0,t} u_j(0) + \int_0^t a_{j,s,t} v_j(s) ds. \quad (\text{NM.19})$$

Before the j th coordinate has entered the smoothed region one only uses $a_{j,s,t} \leq 1$; after it has entered, the integral in (NM.18) grows linearly with slope 1, up to a bounded transition error. Consequently, for $q < c$,

$$\int_s^\infty e^{qt} a_{j,s,t} dt \leq C_{c,q} (e^{qj} + e^{qs}). \quad (\text{NM.20})$$

Multiply (NM.19) by e^{pj} , integrate in t , sum in j , and use Tonelli. The e^{qj} contribution is exactly the $(p+q)$ -norm, while e^{qs} produces the weighted p -norm. This proves (NM.17).

Step D: nonlinear error has one smoothing defect. Define

$$\mathcal{B}(x_0, x_1)[h_0, h_1] := \int_0^1 D^2P(x_0 + s(x_1 - x_0))[h_0, h_1] ds. \quad (\text{NM.21})$$

Then

$$g = -\mathcal{B}(x, S_t x)[(I - S_t)x, \dot{x}]. \quad (\text{NM.22})$$

The tame estimate (NM.7), the smoothing inequalities, and (NM.13)–(NM.8) imply, as long as $\|x(t)\|_{2r} \leq 1$,

$$\|\dot{x}(t)\|_{n+q} \leq C_{n,q} e^{qt} (\|k(t)\|_n + \|x(t)\|_{n+2r} \|k(t)\|_0), \quad (\text{NM.23})$$

$$\|g(t)\|_n \leq C_n \|x(t)\|_{n+2r} \|k(t)\|_0. \quad (\text{NM.24})$$

For completeness, the cancellation of exponential factors in (NM.24) is worth displaying. In the first tame term of (NM.7), use

$$\|(I - S_t)x\|_{n+r} \leq C e^{-rt} \|x\|_{n+2r}, \quad \|\dot{x}\|_r \leq C e^{rt} \|k\|_0 (1 + \|x\|_{2r}), \quad (\text{NM.25})$$

while in the second use

$$\|(I - S_t)x\|_r \leq C e^{-(n+r)t} \|x\|_{n+2r}, \quad \|\dot{x}\|_{n+r} \leq C e^{(n+r)t} \|k\|_0 (1 + \|x\|_{2r}). \quad (\text{NM.26})$$

The coefficient-high term is treated by $\|(I - S_t)x\|_r \leq C e^{-rt} \|x\|_{2r}$ and the first estimate in (NM.25). Thus every apparent growth factor is paired with the single smoothing defect in (NM.22).

Since $x(0) = 0$, equation (NM.23) yields

$$\|x(t)\|_q \leq C_q \int_0^t e^{qs} \|k(s)\|_0 ds \quad (\|x\|_{2r} \leq 1). \quad (\text{NM.27})$$

Substituting (NM.27) and (NM.24) into the weighted residual estimate gives the recursive inequality

$$\int_0^T e^{qt} \|k(t)\|_p dt \leq C \|y\|_{p+q} + C \left(\int_0^T e^{qt} \|k(t)\|_0 dt \right) \left(\int_0^T e^{(p+2r)s} \|k(s)\|_0 ds \right) \quad (\text{NM.28})$$

for the values of p, q used below. Taking $p = 0$, $q = 2r$, and writing

$$K_T := \int_0^T e^{2rt} \|k(t)\|_0 dt,$$

we obtain

$$K_T \leq C \|y\|_{2r} + C K_T^2. \quad (\text{NM.29})$$

Continuity in T and $K_0 = 0$ imply, after shrinking the target ball,

$$K_T \leq 2C \|y\|_{2r} \quad \text{for every } T \text{ in the interval of existence.} \quad (\text{NM.30})$$

Equation (NM.23) with $n = 2r$, $q = 0$ now gives

$$\|x(t)\|_{2r} \leq C \int_0^t e^{2rs} \|k(s)\|_0 ds \leq C' \|y\|_{2r}. \quad (\text{NM.31})$$

Taking the target ball smaller once more makes the right-hand side < 1 . Hence the bootstrap assumption $\|x(t)\|_{2r} \leq 1$ can never fail.

The only weighted exponents needed from now on are $2r$ and $2r + 1$; both are below c by (NM.12). From (NM.28), first with $p = 0$ and then with general p , one obtains

$$\int_0^T e^{qt} \|k(t)\|_0 dt \leq C_q \|y\|_q, \quad q \in \{2r, 2r + 1\}, \quad (\text{NM.32})$$

$$\int_0^T e^{qt} \|k(t)\|_p dt \leq C_{p,q} \|y\|_{p+q}, \quad q \geq p + 2r \text{ whenever that estimate is invoked.} \quad (\text{NM.33})$$

In the induction establishing the final all-seminorm estimate one never needs to choose q growing with the target seminorm. Indeed, using (NM.23), induction on $n \geq 2r$ gives, for every $q \geq 0$,

$$\int_0^T \|\dot{x}(t)\|_{n+q} dt \leq C_{n,q} e^{qT} \|y\|_n. \quad (\text{NM.34})$$

The induction step from n to $n + 1$ uses only (NM.32) with the fixed exponent $2r + 1$ and interpolation of the model norms to absorb the product $\|y\|_n \|y\|_{2r+1}$ into $C \|y\|_{n+1}$. Thus (NM.34) is a single-smoothing-schedule estimate, not a collection of proofs with a seminorm-dependent value of c . This is the point that makes the smoothing schedule independent of the target seminorm.

Taking $q = 0$ in (NM.34) yields

$$\int_0^\infty \|\dot{x}(t)\|_n dt \leq C_n \|y\|_n \quad (n \geq 2r). \quad (\text{NM.35})$$

Therefore the local solution cannot blow up at a finite time: if ω were maximal, the integral in (NM.35) would make $x(t)$ Cauchy in every seminorm as $t \uparrow \omega$, and the limiting point still lies in the low-norm ball where the finite-time ODE can be restarted. The solution is global and

$$G_0(y) := \lim_{t \rightarrow \infty} x(t; y) \quad (\text{NM.36})$$

exists in the full Fréchet topology, with

$$\|G_0(y)\|_n \leq C_n \|y\|_n. \quad (\text{NM.37})$$

Moreover $S_t x(t) \rightarrow G_0(y)$ and $S_t k(t) - k(t) \rightarrow 0$ in every fixed norm. Thus the right side of (NM.13) has the limit

$$c\Psi(G_0(y))(y - P(G_0(y))).$$

An integrable continuous function with a limit at infinity has limit zero; using injectivity of each right inverse $\Psi(G_0(y))$ gives

$$P(G_0(y)) = y. \quad (\text{NM.38})$$

Consequently the theorem's local-surjectivity conclusion is completely proved, and the selected right inverse already satisfies the zeroth-order tame estimate (NM.37).

Step E: exact parameter cancellation and the remaining top-order obstruction. The old rejected proof treated differentiation of (NM.13) as if the term $\Psi(S_t x)S_t v$ decayed in time. It does not. The correct variable is the differentiated residual. For a direction $v \in F$, let

$$z = D_y x(t; y)[v], \quad \ell = D_y k(t; y)[v] = v - DP(x)z. \quad (\text{NM.39})$$

Finite-time smooth dependence justifies these derivatives. Differentiating the already damped residual equation (NM.15) gives

$$\dot{\ell} + cS_t \ell = D_y g[v]. \quad (\text{NM.40})$$

Writing $A_t(x) := DP(S_t x) - DP(x)$, one has

$$D_y g[v] = DA_t(x)[z] \dot{x} + A_t(x) \dot{z}. \quad (\text{NM.41})$$

Both coefficients contain a smoothing defect. Explicitly,

$$DA_t(x)[z]h = D^2 P(S_t x)[S_t z, h] - D^2 P(x)[z, h] \quad (12.1)$$

$$= D^2 P(S_t x)[(S_t - I)z, h] + \int_0^1 D^3 P(x + s(S_t x - x))[S_t x - x, z, h] ds. \quad (\text{NM.42})$$

The first-variation flow equation is

$$\dot{z} = cD\Psi(S_t x)[S_t z]S_t k + c\Psi(S_t x)S_t \ell. \quad (\text{NM.43})$$

Thus the nondecaying forcing has been replaced by the damped residual ℓ , while every coefficient multiplying the unknown z or $\dot{z}$ in (NM.40) carries either $(I - S_t)x$ or $(I - S_t)z$. This is the exact cancellation required for a valid parameter argument.

The same structure persists at every derivative order. If I is a multi-index of target directions, set

$$z_I = D_y^{|I|} x[I], \quad \ell_I = D_y^{|I|} k[I]. \quad (\text{NM.44})$$

Then

$$\dot{\ell}_I + cS_t \ell_I = D_I(A_t(x)\dot{x}), \quad (\text{NM.45})$$

and every derivative of A_t admits a telescoping expansion in which each monomial contains at least one of

$$(I - S_t)x, \quad (I - S_t)z_J \quad (\emptyset \neq J \subseteq I). \quad (\text{NM.46})$$

Indeed, for $m \geq 0$,

$$D^m A_t(x)[h_1, \dots, h_m] = D^{m+1} P(S_t x)[S_t h_1, \dots, S_t h_m] - D^{m+1} P(x)[h_1, \dots, h_m], \quad (\text{NM.47})$$

and telescoping the arguments first, then the base point along the segment from x to $S_t x$, proves (NM.46). On the other hand, D_I of the flow (NM.13) is a finite Faà-di-Bruno sum in which every monomial contains either a residual derivative $S_t \ell_J$ or the

undifferentiated residual $S_t k$.

Equations (NM.45)–(NM.47) already reveal the smoothing-defect ideal, but a direct estimate still leaves an apparent top-order term $A_t(x)\dot{z}_I$ in (NM.45). Estimating that term directly at successively higher seminorms can create exactly the infinite upward loss chain. The corrected residual below removes that obstruction algebraically before any estimate is attempted.

Step F: corrected differentiated residual. For a first variation put

$$w := \ell - A_t(x)z. \quad (\text{NM.48})$$

Differentiate $A_t(x)z$ in time, use (NM.40), and cancel the term $A_t\dot{z}$. Since

$$\partial_t A_t(x)h = D^2P(S_t x)[\dot{S}_t x, h] \quad (\text{NM.49})$$

(where ∂_t keeps x fixed), one obtains the exact identity

$$\dot{w} + cS_t w = DA_t(x)[z]\dot{x} - DA_t(x)[\dot{x}]z - (\partial_t A_t)(x)z - cS_t(A_t(x)z). \quad (\text{NM.50})$$

The first two terms simplify further because D^2P is symmetric:

$$DA_t(x)[z]\dot{x} - DA_t(x)[\dot{x}]z = D^2P(S_t x)[(S_t - I)z, \dot{x}] - D^2P(S_t x)[(S_t - I)\dot{x}, z]. \quad (\text{NM.51})$$

Thus every coefficient multiplying the undifferentiated top variation z in (NM.50) contains one of the four integrable background objects

$$(I - S_t)x, \quad (I - S_t)\dot{x}, \quad \dot{S}_t x, \quad \dot{x}, \quad (\text{NM.52})$$

and the dangerous top derivative $\dot{z}$ has disappeared completely. The shell term $\dot{S}_t x$ enjoys the same one-step smoothing gain as $I - S_t$: in the model sequence norms,

$$\|\dot{S}_t x\|_a \leq C_{a,b} e^{-(b-a)t} \|x\|_b \quad (b > a), \quad (\text{NM.53})$$

because $\dot{x}(t - j)$ is supported where $|j - t| \leq 1$.

The same correction works at arbitrary parameter order. For $|I| = m \geq 1$ set

$$w_I := \ell_I - A_t(x)z_I. \quad (\text{NM.54})$$

Faà-di-Bruno expansion of $D_I(A_t(x)\dot{x})$ shows that the only terms containing the top derivative $\dot{z}_I$ are $A_t(x)\dot{z}_I$ itself. They cancel against the time derivative of $A_t(x)z_I$. Hence

$$\dot{w}_I + cS_t w_I = \mathcal{R}_I, \quad (\text{NM.55})$$

where every monomial of $\mathcal{R}_I$ is of one of the following types:

1. it contains only z_J, ℓ_J with $|J| < m$;
2. it contains one undifferentiated top factor z_I and at least one factor from the background defect ideal (NM.52);
3. it contains one factor $(I - S_t)z_I$ and at least one undifferentiated background factor $\dot{x}$.

No monomial contains $\dot{z}_I$. To verify the assertion, isolate the two top-order Faà-di-Bruno contributions $A_t \dot{z}_I + DA_t[z_I] \dot{x}$, differentiate $A_t z_I$, and use (NM.51); all remaining partitions of I are proper and hence belong to the first class. This is a finite algebraic argument at every fixed m .

This correction narrows the final analytic problem substantially. The base orbit estimates already proved above imply, for every fixed seminorm p ,

$$\sup_t \|x(t)\|_p < \infty, \quad \int_0^\infty \|\dot{x}(t)\|_p dt < \infty, \quad (\text{NM.56})$$

and the smoothing estimates give

$$\int_0^\infty (\|(I - S_t)x\|_p + \|(I - S_t)\dot{x}\|_p + \|\dot{S}_t x\|_p) dt < \infty \quad (\text{NM.57})$$

after paying one finite higher seminorm. Consequently the coefficient of z_I in (NM.55) is integrable in time. The remaining occurrence $(I - S_t)z_I$ is always multiplied by $\dot{x}$, so its one-step high-norm cost is accompanied by an already integrable low-norm factor. This is the precise triangular structure needed to close the all-orders estimate.

The preceding cancellation is enough to remove the differentiated top-time derivative, but it is not by itself a complete all-orders tame estimate. Two further points have to be handled before the all-orders tame conclusion follows: the tangent right inverse must agree with the derivative of the chosen one-sided algorithm, and the damping estimate must not have a fixed exponential-weight ceiling. Both points can be repaired exactly.

Step G: the exact tangent lift for a one-sided right inverse. Let $Q : X \supset U \rightarrow Y$ be smooth and let $R(x) : Y \rightarrow X$ be a smooth family satisfying

$$DQ(x)R(x) = I_Y. \quad (\text{NM.58})$$

Write

$$TQ(x, h) = (Q(x), DQ(x)h). \quad (\text{NM.59})$$

Differentiating (NM.58) in the direction h gives

$$D^2Q(x)[h, R(x)u] + DQ(x)DR(x)[h]u = 0. \quad (\text{NM.60})$$

Consequently

$$R^{[1]}(x, h)(u, v) := (R(x)u, DR(x)[h]u + R(x)v) \quad (\text{NM.61})$$

is an exact right inverse of $D(TQ)(x, h)$. Indeed,

$$\begin{aligned} D(TQ)(x, h)R^{[1]}(x, h)(u, v) &= \left(u, DQ(x)DR(x)[h]u + v + D^2Q(x)[R(x)u, h]\right) \\ &= (u, v) \end{aligned}$$

by (NM.60) and symmetry of D^2Q .

This formula matters specifically in the surjective, noninjective case. The algebraic right inverse for $D(TQ)$ printed in Hamilton, §III.1.7, is indeed a right inverse of $D(TQ)$,

but for a general chosen one-sided family R it need not generate the derivative of the chosen smoothing flow because it does not contain DR . Formula (NM.61) does. If

$$\dot{X} = a(t)R(S_t X)S_t(Y - Q(X)), \quad (\text{NM.62})$$

where $a(t)$ is independent of the target parameter, then differentiating in a target direction V gives

$$\begin{aligned} \dot{H} &= a(t) DR(S_t X)[S_t H]S_t(Y - Q(X)) \\ &\quad + a(t)R(S_t X)S_t(V - DQ(X)H). \end{aligned} \quad (\text{NM.63})$$

On the other hand, the smoothing algorithm for TQ with the diagonal smoothing (S_t, S_t) and the right inverse (NM.61) is exactly (NM.62)–(NM.63). Thus the tangent-functor identity is exact for arbitrary one-sided right-inverse families.

Step H: an all-weight damping lemma. The fixed constant c in (NM.13) is sufficient for the zeroth-order surjectivity proof, but a blind tangent iteration can encounter tame losses whose required exponential weights exceed c . To remove that ceiling, introduce a second, proof-local algorithm with the same smoothing family and with

$$\lambda(t) := \lambda_0 + t, \quad \lambda_0 \geq 1. \quad (\text{NM.64})$$

The basic linear estimate now holds at *every* finite weight. Suppose

$$\dot{u} + \lambda(t)S_t u = v. \quad (\text{NM.65})$$

Then, for every $p, q \geq 0$,

$$\begin{aligned} &\int_0^T e^{qt} \lambda(t) \|S_t u(t)\|_p dt \\ &\leq C_{p,q} \|u(0)\|_{p+q} + C_{p,q} \int_0^T (e^{qt} \|v(t)\|_p + \|v(t)\|_{p+q}) dt. \end{aligned} \quad (\text{NM.66})$$

Here is the complete coordinate proof. Put

$$a_{j,s,t} := \exp\left(-\int_s^t \lambda(\tau) \chi(\tau - j) d\tau\right). \quad (\text{NM.67})$$

For $q > 0$ define

$$K_{j,q}(s) := \int_s^\infty e^{qt} a_{j,s,t} dt.$$

With $b = \max\{s, j + 1\}$, the interval $[s, b]$ contributes at most $C_q(e^{qs} + e^{qj})$. On $[b, \infty)$ one has $\chi(t - j) = 1$ and hence, writing $t = b + r$,

$$a_{j,s,b+r} \leq \exp\left(-\lambda_0 r - br - \frac{r^2}{2}\right).$$

Therefore

$$K_{j,q}(s) \leq C_q(e^{qs} + e^{qj}). \quad (\text{NM.68})$$

Since $\partial_t a_{j,s,t} = -\lambda(t)\chi(t-j)a_{j,s,t}$, integration by parts gives, for $q > 0$,

$$D_{j,q}(s) := \int_s^\infty e^{qt} \lambda(t) \chi(t-j) a_{j,s,t} dt \quad (12.2)$$

$$= e^{qs} + qK_{j,q}(s) \leq C_q(e^{qs} + e^{qj}), \quad (\text{NM.69})$$

while $D_{j,0}(s) = 1$. Duhamel's formula for (NM.65), followed by Tonelli and multiplication by e^{pj} , now gives (NM.66). Unlike (NM.17), there is no restriction $q < c$.

Step I: the variable-damping surjectivity engine. Consider

$$\dot{X}(t) = \lambda(t)\Psi(S_t X(t))S_t K(t), \quad K(t) := y - P(X(t)), \quad X(0) = 0. \quad (\text{NM.70})$$

On every finite time interval λ is bounded. Hamilton's finite-time Banach-contraction argument for the smoothed vector field therefore applies verbatim: the factors S_t pay every fixed derivative loss on $[0, T]$, so the vector field is locally Lipschitz in one sufficiently high Banach seminorm and the solution depends smoothly on y for finite T . The estimates below are uniform in the terminal time and hence give continuation.

The residual satisfies

$$\dot{K} + \lambda(t)S_t K = G, \quad G := (DP(S_t X) - DP(X))\dot{X}. \quad (\text{NM.71})$$

The same one-defect calculation as in (NM.22)–(NM.26), with c replaced by $\lambda(t)$ and k by $S_t K$, gives, as long as $\|X(t)\|_{2r} \leq 1$,

$$\|\dot{X}(t)\|_a \leq C_a \lambda(t) e^{at} \|S_t K(t)\|_0, \quad (\text{NM.72})$$

$$\|G(t)\|_p \leq C_p \|X(t)\|_{p+2r} \lambda(t) \|S_t K(t)\|_0. \quad (\text{NM.73})$$

For example, the first tame term in the analogue of (NM.22) contains $\|(I - S_t)X\|_{p+r} \|\dot{X}\|_r$; the two factors contribute $e^{-rt} \|X\|_{p+2r}$ and $\lambda(t) e^{rt} \|S_t K\|_0$, respectively, so the exponentials cancel exactly. The remaining tame terms are identical.

Set

$$J_{p,q}(T) := \int_0^T e^{qt} \lambda(t) \|S_t K(t)\|_p dt. \quad (\text{NM.74})$$

Equation (NM.72) gives

$$\|X(t)\|_a \leq C_a J_{0,a}(t). \quad (\text{NM.75})$$

Applying (NM.66) to (NM.71), and then using (NM.73) and (NM.75), yields

$$J_{p,q}(T) \leq C_{p,q} \|y\|_{p+q} + C_{p,q} J_{0,q}(T) J_{0,p+2r}(T). \quad (\text{NM.76})$$

Indeed, the first nonlinear forcing integral is bounded by

$$\int_0^T e^{qt} \left(\int_0^t e^{(p+2r)s} \lambda(s) \|S_s K(s)\|_0 ds \right) \lambda(t) \|S_t K(t)\|_0 dt,$$

and the second is the same after using, for $s \leq t$, $e^{(p+q+2r)s} \leq e^{qt} e^{(p+2r)s}$. This proves (NM.76) without suppressing any time-dependent factor.

Take $p = 0, q = 2r$. After shrinking only the fixed low target ball $\|y\|_{2r} < \varepsilon$, continuity

in T gives

$$J_{0,2r}(T) \leq C\|y\|_{2r}, \quad \sup_t \|X(t)\|_{2r} \leq C\|y\|_{2r} < 1. \quad (\text{NM.77})$$

Thus the bootstrap never exits. With this low estimate fixed, (NM.76) at $p = 0$ can be absorbed for every $q \geq 0$:

$$J_{0,q}(T) \leq C_q\|y\|_q. \quad (\text{NM.78})$$

The model norms are log-convex in their real index. In particular, for $a, b \geq \rho$,

$$\|y\|_a\|y\|_b \leq \|y\|_\rho\|y\|_{a+b-\rho}. \quad (\text{NM.79})$$

This follows by applying convexity of $s \mapsto \log \|y\|_s$ to the two points a, b in the interval $[\rho, a + b - \rho]$; the zero vector is trivial. Hence, if $q \geq 2r$, (NM.76) and (NM.78) give

$$J_{p,q}(T) \leq C_{p,q}\|y\|_{p+q}, \quad (\text{NM.80})$$

where only $\|y\|_{2r}$ is required to be small. At $q = 0$ one obtains

$$J_{p,0}(T) \leq C_p\|y\|_{p+2r}. \quad (\text{NM.81})$$

Finally, the unsimplified form of the Ψ estimate gives

$$\|\dot{X}(t)\|_n \leq C_n\lambda(t)\left(\|S_t K(t)\|_n + e^{nt}\|X(t)\|_{2r}\|S_t K(t)\|_0\right). \quad (\text{NM.82})$$

Therefore

$$\int_0^\infty \|\dot{X}(t)\|_n dt \leq C_n\|y\|_{n+2r}, \quad (\text{NM.83})$$

$$\int_0^\infty e^{qt}\|\dot{X}(t)\|_n dt \leq C_{n,q}\|y\|_{n+q}, \quad q \geq 2r. \quad (\text{NM.84})$$

The second estimate uses (NM.80) and $J_{0,n+q}$; the interpolation (NM.79) absorbs every product of two high target norms against the single small norm $\|y\|_{2r}$.

It follows from (NM.83) that

$$\widehat{G}_0(y) := \lim_{t \rightarrow \infty} X(t; y) \quad (\text{NM.85})$$

exists in every seminorm and is tame with a finite loss. To verify the equation without assuming a pointwise limit of the now unbounded factor $\lambda(t)$, work coordinatewise. For fixed j and $t \geq j + 1$,

$$\dot{K}_j + \lambda(t)K_j = G_j.$$

By (NM.73), $G_j \in L^1(0, \infty)$, so variation of constants implies $K_j(t) \rightarrow 0$. For any p , the already proved high-seminorm bounds on X give $\sup_t \|K(t)\|_{p+1} < \infty$. Splitting the p -norm at a finite coordinate N and using

$$\sum_{j>N} e^{pj}\|K_j(t)\| \leq e^{-N}\|K(t)\|_{p+1}$$

then shows $K(t) \rightarrow 0$ in the full Fréchet topology. Continuity of P gives

$$P(\widehat{G}_0(y)) = y. \quad (\text{NM.86})$$

Thus (NM.70) supplies a second internally proved local-surjectivity selection, now with an all-weight tail estimate (NM.84).

Step J: fiberwise tangent induction on one fixed base neighborhood. Define recursively

$$Q_0 = P, \quad R_0 = \Psi, \quad Q_{m+1} = TQ_m, \quad (\text{NM.87})$$

and

$$R_{m+1}(X, H)(U, V) := (R_m(X)U, DR_m(X)[H]U + R_m(X)V). \quad (\text{NM.88})$$

Repeated use of (NM.60) gives

$$DQ_m(X)R_m(X) = I \quad (\text{NM.89})$$

for every m . Smooth tameness of P, Ψ and the tame chain rule imply that all Q_m, R_m are smooth tame. Equation (NM.63) also shows, inductively, that the finite-time map $T^m(y \mapsto X(t; y))$ is exactly the smoothing algorithm for (Q_m, R_m) with the same scalar schedule (NM.64). This is the tangent-algorithm identity used by Hamilton in §III.1.7; formula (NM.61) makes it exact for a genuinely one-sided chosen family.

The crucial point is that this iteration is used *fiberwise*, not as a new local-surjectivity theorem on the whole product. There is a canonical identification

$$T^m U \cong U \times E^{2^m - 1}. \quad (\text{NM.90})$$

Thus membership in the domain imposes a condition only on the base component $x \in U$; tangent coordinates are unrestricted vectors. The base component of the tangent algorithm is the already constructed orbit $X(t; y)$ and hence is controlled by (NM.77) on one fixed target neighborhood.

For distinct formal parameters $\theta_1, \dots, \theta_m$, let $j_m(y; v_1, \dots, v_m) \in T^m F$ be the tangent cube of the affine map $\theta \mapsto y + \sum_i \theta_i v_i$ at 0. In subset coordinates,

$$(j_m)_\emptyset = y, \quad (j_m)_{\{i\}} = v_i, \quad (j_m)_A = 0 \quad (|A| \geq 2). \quad (\text{NM.91})$$

Writing

$$T^m X_t(j_m(y; v_1, \dots, v_m)) = (z_A(t))_{A \subseteq \{1, \dots, m\}},$$

one has

$$z_\emptyset = X(t; y), \quad z_A = D_y^{|A|} X(t; y)[v_i : i \in A]. \quad (\text{NM.92})$$

This identity follows first for finite t from the Banach ODE parameter theorem quoted in Step I and then algebraically from the definition of the tangent functor.

We need two elementary tame-calculus observations. First, if Φ is smooth tame and $d \geq 1$, multilinearity of $D^d \Phi$ upgrades local tameness of the joint derivative map to a one-high-factor estimate. On a fixed low base neighborhood there are finite indices

b, r such that

$$\|D^d\Phi(x)[u_1, \dots, u_d]\|_n \leq C_{d,n} \left[(1 + \|x\|_{n+r}) \prod_{i=1}^d \|u_i\|_b + \sum_{i=1}^d \|u_i\|_{n+r} \prod_{j \neq i} \|u_j\|_b \right]. \quad (\text{NM.93})$$

To prove this rather than assume it, take a product neighborhood on which the joint derivative is tame. If none of the u_i vanishes, divide each u_i by a fixed multiple of $\|u_i\|_b$ so that all normalized fiber variables lie in that product neighborhood, apply the joint tame estimate, and multiply back by the product of the scaling factors. The case in which some $u_i = 0$ is trivial. This gives exactly (NM.93). The same argument applies to all finite compositions and contractions that occur below.

Second, repeated use of the log-convex estimate (NM.79) gives, for $a_1, \dots, a_N \geq \rho$,

$$\prod_{\nu=1}^N \|y\|_{a_\nu} \leq \|y\|_\rho^{N-1} \|y\|_{\rho + \sum_{\nu=1}^N (a_\nu - \rho)}. \quad (\text{NM.94})$$

Hence any finite product of high base norms can be reduced to one still higher base norm times powers of the *same fixed* low norm. On $\|y\|_\rho < \varepsilon$ those powers are absorbed into constants. This is why the induction must be performed on the individual tame monomials rather than by multiplying already-summed coarse derivative estimates.

We now prove the parameter estimate. Choose once and for all a finite $\rho \geq 2r$ large enough to dominate the low indices needed for the *first* linearization of the smoothed vector field and for the corrected residual in Step F. Shrinking ε if necessary, the base estimates of Step I hold whenever

$$\|y\|_\rho < \varepsilon. \quad (\text{NM.95})$$

This choice is independent of the tangent order m .

Fix $m \geq 1$ and a nonempty $I \subseteq \{1, \dots, m\}$. Differentiate (NM.71) in the directions indexed by I and make the same correction $w_I = \ell_I - A_t z_I$ as in Step F. There are only finitely many Faà-di-Bruno partitions at this order. After the correction every term is of one of the following three types:

1. a proper-partition monomial involving only $z_J, \dot{z}_J$ with $J \subsetneq I$;
2. an undifferentiated top variable z_I multiplied by a coefficient from the first linearization along the base orbit;
3. $(I - S_t)z_I$ multiplied by a base defect such as $\dot{X}$, with no occurrence of $\dot{z}_I$.

This is precisely the algebraic conclusion of (NM.54)–(NM.57). The crucial point is that the order-dependent higher derivatives of P and Ψ occur only in the first, proper-partition class. The coefficient of the top unknown comes from the same first linearization at every order. Consequently its low seminorm is controlled by the single base ball (NM.95).

For the variable-damping algorithm the same correction is exact with c replaced by $\lambda(t)$; differentiating $A_t z_I$ introduces no $\dot{\lambda}$ term. It is useful to record explicitly how the

corrected residual controls the top variation. Differentiating (NM.70) at order I and isolating the unique top blocks gives

$$\begin{aligned}\dot{z}_I &= \lambda \Psi(S_t X) S_t \ell_I + \lambda D\Psi(S_t X) [S_t z_I] S_t K + \mathcal{F}_I^{(0)} \\ &= \lambda \Psi(S_t X) S_t w_I + \lambda \Psi(S_t X) S_t (A_t z_I) + \lambda D\Psi(S_t X) [S_t z_I] S_t K + \mathcal{F}_I^{(0)},\end{aligned}\quad (\text{NM.95a})$$

where $\mathcal{F}_I^{(0)}$ contains only proper parameter partitions. The corrected residual satisfies

$$\dot{w}_I + \lambda(t) S_t w_I = \mathcal{R}_I, \quad (\text{NM.95b})$$

with the same three structural classes listed above, and for the affine jet (NM.91),

$$w_I(0) = \begin{cases} v_i, & I = \{i\}, \\ 0, & |I| \geq 2. \end{cases} \quad (\text{NM.95c})$$

Indeed $X(0; y) = 0$ is target-independent, while $K(0; y) = y - P(0)$ is affine in y , and $A_0(0) = 0$.

Put

$$W_{I,n,q}(T) := \int_0^T e^{qt} \lambda(t) \|S_t w_I(t)\|_n dt.$$

Applying the already proved all-weight estimate (NM.66) to (NM.95b) gives the literal bound

$$\begin{aligned}W_{I,n,q}(T) &\leq C_{n,q} \mathbf{1}_{|I|=1} \|v_i\|_{n+q} \\ &\quad + C_{n,q} \int_0^T (e^{qt} \|\mathcal{R}_I(t)\|_n + \|\mathcal{R}_I(t)\|_{n+q}) dt.\end{aligned}\quad (\text{NM.95d})$$

Thus no estimate of $\dot{z}_I$ at a still higher seminorm is ever fed back into the residual equation. Equation (NM.95a) then reconstructs z_I from $W_{I,n,q}$ plus: (a) proper-partition forcing, and (b) terms linear in z_I carrying either the derivative defect A_t or the residual K . The former has the smoothing defect $(I - S_t)X$ and the latter has the arbitrary weighted tails proved in Step I. This is the precise two-equation triangular system behind the Volterra majorant below.

For completeness, set

$$Z_{I,n}(T) := \sup_{0 \leq t \leq T} \|z_I(t)\|_n + \int_0^T \|\dot{z}_I(t)\|_n dt. \quad (\text{NM.96})$$

Assume first that, for a sufficiently large finite s_m to be chosen during the induction,

$$\|v_i\|_{s_m} \leq 1 \quad (1 \leq i \leq m). \quad (\text{NM.97})$$

We claim that s_m can be chosen so that, simultaneously for every $n \geq 0$ and every nonempty $I \subseteq \{1, \dots, m\}$,

$$Z_{I,n}(T) \leq C_{m,n} \left(1 + \|y\|_{n+s_m} + \sum_{i \in I} \|v_i\|_{n+s_m} \right) \quad (\text{NM.98})$$

uniformly in T .

The proof is an induction on $|I|$. For $|I| = 1$, differentiate the finite-time flow once. Equations (NM.95a)–(NM.95d) and (NM.66) give the same two-equation triangular estimate as the base engine, now with one inhomogeneous target direction. Choose an exponential weight larger than the finite loss in this first tangent equation. Because (NM.66) is available at every finite weight, the smoothing loss is paid without changing (NM.95). Applying (NM.93) to each coefficient produces one high factor only, and the term multiplying the top high norm uses only fixed low base quantities. Thus the resulting scalar majorant has the form

$$Z_{I,n}(T) \leq B_{I,n} + \int_0^T a_*(t) Z_{I,n}(t) dt, \quad (\text{NM.99})$$

where

$$\int_0^\infty a_*(t) dt \leq A_1 \quad (\text{NM.100})$$

with A_1 depending only on the fixed low base ball, while $B_{I,n} \leq C_n(1 + \|y\|_{n+s_1} + \|v_i\|_{n+s_1})$. The integrability in (NM.100) is exactly (NM.83)–(NM.84) together with the background terms listed in (NM.56)–(NM.57). Gronwall therefore proves (NM.98) for $|I| = 1$. Notice that the exponential factor is e^{A_1} ; no high seminorm of y occurs in it.

Suppose now that (NM.98) is known for all proper subsets of I . Insert those lower variations into the proper-partition terms of the corrected I -equation. At this fixed order there are only finitely many monomials and finitely many derivative losses. Choose s_m larger than all of them and choose in (NM.66) an exponential weight larger than their sum. Estimate each monomial from the outside inward using (NM.93): at each contraction only one branch is evaluated in the high seminorm and all other branches are evaluated at the chosen low index. If several high base norms are produced along that single branch, compress them by (NM.94). Because the primitive fiber leaves are the distinct directions v_i , this monomial-by-monomial procedure never creates more than one high fiber factor. Under the normalization (NM.97), all low fiber factors are bounded by 1. Hence the entire proper-partition forcing satisfies

$$B_{I,n} \leq C_{m,n} \left(1 + \|y\|_{n+s_m} + \sum_{i \in I} \|v_i\|_{n+s_m} \right). \quad (\text{NM.101})$$

The terms containing z_I itself are still exactly the second and third classes above. Use z_I as the unique high branch in (NM.93); the accompanying coefficient is then evaluated only in the fixed low seminorms already covered by (NM.95). The $(I - S_t)z_I$ terms acquire the required decay from (NM.11), and the all-weight estimate pays the resulting finite upward shift. Thus they again contribute the same Volterra coefficient a_* , now with

$$\int_0^\infty a_*(t) dt \leq A_m < \infty, \quad (\text{NM.102})$$

where A_m may depend on the order but only on the fixed low base bound and on normalized low fiber bounds. It does not depend on any high seminorm of y . Equations

(NM.99), (NM.101), and Gronwall prove (NM.98). This completes the finite induction on $|I|$. Since m was arbitrary, it closes every tangent order on the same base ball (NM.95).

Finally remove the normalization. If $v_i \neq 0$, put $a_i = \|v_i\|_{s_m}$ and $\hat{v}_i = v_i/a_i$; if some $v_i = 0$ the desired estimate is trivial. Multilinearity of z_I and (NM.98) give

$$\begin{aligned} & \sup_{t \geq 0} \|D_y^m X(t; y)[v_1, \dots, v_m]\|_n + \int_0^\infty \|D_y^m \dot{X}(t; y)[v_1, \dots, v_m]\|_n dt \\ & \leq C_{m,n} \left[(1 + \|y\|_{n+s_m}) \prod_{i=1}^m \|v_i\|_{s_m} + \sum_{i=1}^m \|v_i\|_{n+s_m} \prod_{j \neq i} \|v_j\|_{s_m} \right]. \end{aligned} \quad (\text{NM.103})$$

Only (NM.95) is a smallness condition. In particular no smallness of $\|y\|_{s_m}$ is used. This is the fixed-low-base estimate needed to close the parameter-derivative induction.

Step K: passage to the smooth tame limiting section. For finite t , the map $y \mapsto X(t; y)$ is smooth by the finite-time Banach ODE construction in Step I. Equation (NM.103) says, for every m, n , that $D_y^m X(t; y)$ is Cauchy as $t \rightarrow \infty$ in seminorm n , locally uniformly when the finitely many seminorms on the right-hand side are bounded. We now justify differentiation of the limit without invoking an external Fréchet inverse theorem.

For $m = 1$, apply the fundamental theorem of calculus to $X(t; y + h) - X(t; y)$ on the segment $y + sh$ and then pass to the locally uniform limit of $D_y X(t; \cdot)$. This proves that $\hat{G}_0$ from (NM.85) is C^1 and $D\hat{G}_0 = \lim_{t \rightarrow \infty} D_y X(t; \cdot)$. Repeating the same argument for $D_y^{m-1} X(t; \cdot)$ proves inductively that $\hat{G}_0$ is C^m and

$$D^m \hat{G}_0(y) = \lim_{t \rightarrow \infty} D_y^m X(t; y) \quad (\text{NM.104})$$

for every m . Hence $\hat{G}_0$ is smooth. Passing to the limit in (NM.103) gives precisely the one-high-factor local tame estimate for every derivative map, so the definition of a smooth tame map implies that $\hat{G}_0$ is smooth tame.

Equation (NM.86) already proves $P(\hat{G}_0(y)) = y$ on the fixed target neighborhood. Undo the translation and the tame retract/model-space reduction of Step A. The resulting map $G : V \rightarrow U_0$ satisfies

$$G(y_0) = x_0, \quad P(G(y)) = y,$$

and is smooth tame. This proves the theorem. $\square$

Lemma F12.5. Single analytic KAM conjugacy step near a diagonal constant

Statement. Let $\alpha \in \mathbb{R}^d$, let $D = \text{diag}(\lambda_1, \dots, \lambda_m) \in GL(m, \mathbb{C})$, and let $G : \mathbb{T}_h^d \rightarrow M_m(\mathbb{C})$ be bounded holomorphic. Fix $0 < \delta < h$, $N \geq 1$, and $\kappa > 0$, and form the low nonresonant block $G_{\mathcal{N}}$, low resonant block $G_{\mathcal{R}}$, high-frequency tail $G_{>N}$, and

correction X exactly as in Proposition F08.6. For the row-sum analytic norm

$$\|M\|_{r,*} := \sup_{z \in \mathbb{T}_r^d} \max_{1 \leq i \leq m} \sum_{j=1}^m |M_{ij}(z)|,$$

put

$$\varepsilon := \|G\|_{h,*}, \quad L := m\kappa^{-1}A_d(\delta), \quad Q := me^{-\pi\delta N}A_d(\delta/2).$$

If $L\varepsilon \leq \frac{1}{2}$ and $B := I - X$, then B is holomorphic and invertible on $\mathbb{T}_{h-\delta}^d$, with

$$\|B - I\|_{h-\delta,*} \leq L\varepsilon, \quad \|B^{-1} - I\|_{h-\delta,*} \leq 2L\varepsilon,$$

and the exact conjugated cocycle satisfies

$$B(x + \alpha)(D + G(x))B(x)^{-1} = D + E(x),$$

where

$$E = (G_{\mathcal{R}} + G_{>N} - X(\cdot + \alpha)G)B^{-1}$$

and

$$\|E\|_{h-\delta,*} \leq 2(\|G_{\mathcal{R}}\|_{h-\delta,*} + Q\varepsilon + L\varepsilon^2).$$

More generally, if $G_{\mathcal{R}} = C + R$ with C constant, then with $D_* := D + C$ one has

$$B(x + \alpha)(D + G(x))B(x)^{-1} = D_* + E_*(x)$$

and

$$\|E_*\|_{h-\delta,*} \leq 2(\|R\|_{h-\delta,*} + Q\varepsilon + L\varepsilon^2) + 2L\varepsilon\|C\|_*.$$

Thus one exact step removes the chosen low nonresonant modes, retains any genuinely resonant part, permits a constant resonant block to be absorbed into the next constant, and quantifies both strip loss and nonlinear error.

Proof. For a matrix-valued holomorphic function set

$$\|M\|_{r,*} := \sup_{z \in \mathbb{T}_r^d} \max_i \sum_j |M_{ij}(z)|. \quad (1)$$

The row-sum matrix norm is submultiplicative pointwise, hence

$$\|MN\|_{r,*} \leq \|M\|_{r,*} \|N\|_{r,*}. \quad (2)$$

It also satisfies

$$\|M\|_{r,*} \leq m\|M\|_{r,\max}. \quad (3)$$

Step 1: the homological correction in an algebra norm. By Matrix homological equation near a constant the trigonometric polynomial X satisfies

$$X(x + \alpha)D - DX(x) = G_{\mathcal{N}}(x) \quad (4)$$

and

$$\|X\|_{h-\delta, \max} \leq \kappa^{-1} A_d(\delta) \|G\|_{h, \max}. \quad (5)$$

Since $\|G\|_{h, \max} \leq \|G\|_{h, *} = \varepsilon$, (3)-(5) give

$$\boxed{\|X\|_{h-\delta, *} \leq L\varepsilon.} \quad (6)$$

The same proposition gives

$$\|G_{>N}\|_{h-\delta, \max} \leq e^{-\pi\delta N} A_d(\delta/2) \|G\|_{h, \max},$$

so

$$\boxed{\|G_{>N}\|_{h-\delta, *} \leq Q\varepsilon.} \quad (7)$$

Translation by the real vector α preserves every strip norm, and therefore

$$\|X(\cdot + \alpha)\|_{h-\delta, *} \leq L\varepsilon. \quad (8)$$

Step 2: exact invertibility of the near-identity change. Put $B = I - X$. The hypothesis $L\varepsilon \leq \frac{1}{2}$ and (6) imply $\|X\|_{h-\delta, *} \leq \frac{1}{2}$. The matrix-valued Neumann series

$$B^{-1} = (I - X)^{-1} = \sum_{n=0}^{\infty} X^n \quad (9)$$

converges normally on $\mathbb{T}_{h-\delta}^d$. Thus B^{-1} is holomorphic there and

$$\|B^{-1}\|_{h-\delta, *} \leq \frac{1}{1 - \|X\|_{h-\delta, *}} \leq 2, \quad (10)$$

while

$$\|B^{-1} - I\|_{h-\delta, *} = \|XB^{-1}\|_{h-\delta, *} \leq 2L\varepsilon. \quad (11)$$

Together with $B - I = -X$, this proves the two asserted bounds for the conjugacy.

Step 3: exact cancellation of the nonresonant block. Write $X_+(x) := X(x + \alpha)$. Define

$$A'(x) := B(x + \alpha)(D + G(x))B(x)^{-1}. \quad (12)$$

Instead of expanding B^{-1} only to first order, multiply (12) by B on the right:

$$\begin{aligned} A'B &= (I - X_+)(D + G) \\ &= D + G - X_+D - X_+G. \end{aligned} \quad (13)$$

The homological equation (4) says

$$X_+D - DX = G_N, \quad \text{hence} \quad -X_+D = -DX - G_N. \quad (14)$$

Substituting (14) into (13) and using $DB = D - DX$ gives

$$\begin{aligned} A'B &= DB + G - G_N - X_+G \\ &= DB + G_{\mathcal{R}} + G_{>N} - X_+G. \end{aligned} \quad (15)$$

Right multiplication by B^{-1} yields the exact identity

$$A' = D + (G_{\mathcal{R}} + G_{>N} - X_+G)B^{-1}. \quad (16)$$

Thus no Baker–Campbell–Hausdorff truncation and no hidden $O(\varepsilon^2)$ notation is needed.

By (2), (7), (8), (10), and $\|G\|_{h-\delta,*} \leq \varepsilon$,

$$\begin{aligned} \|E\|_{h-\delta,*} &\leq (\|G_{\mathcal{R}}\|_{h-\delta,*} + \|G_{>N}\|_{h-\delta,*} + \|X_+\|_{h-\delta,*}\|G\|_{h-\delta,*})\|B^{-1}\|_{h-\delta,*} \\ &\leq 2(\|G_{\mathcal{R}}\|_{h-\delta,*} + Q\varepsilon + L\varepsilon^2), \end{aligned} \quad (17)$$

which is the first remainder estimate.

Step 4: updating the constant without discarding genuine resonances. Suppose now that

$$G_{\mathcal{R}} = C + R, \quad (18)$$

where C is constant. From (16),

$$\begin{aligned} A' &= D + (C + R + G_{>N} - X_+G)B^{-1} \\ &= D + C + (R + G_{>N} - X_+G)B^{-1} + C(B^{-1} - I). \end{aligned} \quad (19)$$

Thus, with $D_* = D + C$,

$$E_* = (R + G_{>N} - X_+G)B^{-1} + C(B^{-1} - I). \quad (20)$$

Using (7)–(11) once more gives

$$\|E_*\|_{h-\delta,*} \leq 2(\|R\|_{h-\delta,*} + Q\varepsilon + L\varepsilon^2) + 2L\varepsilon\|C\|_*. \quad (21)$$

The estimate distinguishes the three mechanisms exactly: $Q\varepsilon$ is the analytically small high-frequency tail, $L\varepsilon^2$ is nonlinear, and R is a genuinely resonant low block that this step makes no attempt to invert. Only the constant part C has been absorbed into the next constant D_* . This is precisely the claimed one-step normal-form update. $\square$

Theorem F12.6. Perturbative Eliasson reducibility theorem for analytic $SL(2, \mathbb{R})$ cocycles

Statement. Fix $h > 0$, let $\alpha \in \mathbb{R}/\mathbb{Z}$ satisfy a Diophantine estimate $\|k\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma|k|^{-\tau}$ for every $k \neq 0$, and let $A_0 \in SL(2, \mathbb{R})$. There is

$$\varepsilon_0 = \varepsilon_0(h, \gamma, \tau, \|A_0\|) > 0$$

such that the following holds. If $A : \mathbb{T}_h \rightarrow SL(2, \mathbb{R})$ is real analytic, $\|A - A_0\|_h < \varepsilon_0$, and the fibered rotation number $\rho(\alpha, A)$ is either resonant or Diophantine with re-

spect to α in the sense of Definition F12.9, then (α, A) is analytically reducible to a constant. The reducing map is allowed to live on the double cover exactly in the resonant/topological cases where this is necessary, as specified in Definition F12.10.

Scope note

The theorem is a local dynamical reducibility result; spectral consequences and global almost reducibility require additional theorems.

Proof.

Proof strategy

Pass first to logarithmic coordinates near the constant cocycle. Each KAM step solves the nonresonant homological equation on a smaller strip, while a possible resonant mode is absorbed into the new constant part. Rotation-number arithmetic controls how often resonance can occur. Quadratic error estimates and summable strip losses yield convergence of the conjugacies; the resonant and relative-Diophantine alternatives then give the stated reducibility conclusion.

We write $\mathbb{T}_r = \{z \in \mathbb{C}/\mathbb{Z} : |\Im z| < r\}$ and $\|F\|_r = \sup_{\mathbb{T}_r} \|F(z)\|$ for a fixed submultiplicative matrix norm. All constants denoted by C may change from line to line; once the compact bound on the constant matrices has been fixed, they depend only on that bound and on harmless numerical choices. Dependence on the arithmetic constants is always displayed when it matters.

Step 1: logarithmic coordinates in the group. Put $H(x) = A(x)A_0^{-1}$. If $\|A - A_0\|_h$ is sufficiently small, then $\|H - I\|_h < 1/2$ and the uniformly convergent power series

$$\log H = \sum_{j \geq 1} \frac{(-1)^{j+1}}{j} (H - I)^j \quad (1)$$

defines a bounded holomorphic matrix F_0 on $\mathbb{T}_h$. Since $H(x) \in SL(2, \mathbb{R})$ for real x , the chosen logarithm is real on the real line and

$$\mathrm{tr} F_0 = \log \det H = 0.$$

Thus $F_0 : \mathbb{T}_h \rightarrow \mathfrak{sl}(2, \mathbb{C})$ is holomorphic, real on $\mathbb{T}$, and

$$A(x) = e^{F_0(x)} A_0, \quad \|F_0\|_h \leq C(A_0) \|A - A_0\|_h. \quad (2)$$

It is therefore enough to prove the theorem for a cocycle $(\alpha, e^F A_c)$ with $A_c \in SL(2, \mathbb{R})$ in a fixed compact neighborhood of A_0 and $\|F\|_r$ small.

Step 2: two elementary rotation-number lemmas. We need two facts that are not contained in the degree-zero formulation *Fibered rotation number*.

Lemma 2.1 (degree formula on the double cover). Let $B : \mathbb{R}/2\mathbb{Z} \rightarrow SL(2, \mathbb{R})$ be continuous and suppose its projective action has degree $d \in \mathbb{Z}$, meaning that a lift of the

projective coordinate change has the form

$$h_x(y) = y + dx + u(x, y), \quad u(x + 2, y) = u(x, y), \quad u(x, y + 1) = u(x, y), \quad (3)$$

with u bounded. If

$$A'(x) = B(x + \alpha)A(x)B(x)^{-1} \quad (4)$$

is 1-periodic, then, in the projective-line normalization of F09,

$$\rho(\alpha, A') = \rho(\alpha, A) + d\alpha \pmod{\mathbb{Z}}. \quad (5)$$

(The sign is reversed if the opposite convention for d is chosen.)

Indeed, use the same lifted-conjugacy computation as in Step 6 of *Fibered rotation number*. If $(x_n, y_n) = \tilde{F}_A^n(x, y)$, then the two fiber displacements differ by

$$h_{x+n\alpha}(y_n) - h_x(y) - (y_n - y) = dn\alpha + u(x + n\alpha, y_n) - u(x, y). \quad (6)$$

After division by n , the bounded u -term disappears and (5) follows. For

$$D_m(x) := R_{-\pi mx}, \quad (7)$$

the projective coordinate is $y \mapsto y - mx$, so $d = -m$ and

$$\rho(\alpha, D_m(\cdot + \alpha)A(\cdot)D_m(\cdot)^{-1}) = \rho(\alpha, A) - m\alpha. \quad (8)$$

If m is odd, D_m is 2-periodic as an $SL(2, \mathbb{R})$ map but its projective action is 1-periodic. This is exactly the double-cover phenomenon allowed in *the definition of analytic reducibility on the circle or double cover*.

Lemma 2.2 (uniform square-root comparison with a constant). For every $M \geq 1$ there are $c_M, C_M > 0$ such that, whenever $A_c \in SL(2, \mathbb{R})$, $\|A_c\| \leq M$, and $G : \mathbb{T} \rightarrow SL(2, \mathbb{R})$ satisfies $\|G - A_c\|_0 < t < c_M$, compatible real lifts can be chosen with

$$\text{dist}_{\mathbb{R}/\mathbb{Z}}(\rho(\alpha, G), \rho(\alpha, A_c)) \leq C_M \sqrt{t}. \quad (9)$$

To prove this, first note that the least singular value of A_c is at least M^{-1} . Hence the angle between $A_c v$ and $G(x)v$ is at most $C_M t$, uniformly in x and nonzero v . In lifted projective coordinates this gives, after enlarging C_M once to absorb the fixed angle-to-projective-coordinate normalization, for $s = C_M t$ the pointwise order

$$f_{R_{-s}A_c}(y) \leq f_{G,x}(y) \leq f_{R_s A_c}(y) \quad (10)$$

after choosing the three lifts in the same local lift branch. Fiber maps are increasing, so induction preserves this order under iteration; dividing the ordered displacements by n and using *Fibered rotation number* gives

$$\rho(\alpha, R_{-s}A_c) \leq \rho(\alpha, G) \leq \rho(\alpha, R_s A_c) \quad (11)$$

for the corresponding real representatives.

It remains to compare the two constant rotation numbers. For a constant matrix C ,

its projective rotation number is zero when C is nonelliptic (with the lift chosen near that of A_c), while in the elliptic case

$$\operatorname{tr} C = 2 \cos(\pi \rho(C)) \quad (12)$$

up to the locally fixed sign/lift convention. Since $|\operatorname{tr}(R_s A_c) - \operatorname{tr} A_c| \leq C_M |s|$ and the inverse of $u \mapsto \cos u$ is uniformly $1/2$ -Hölder on $[-1, 1]$,

$$\operatorname{dist}(\rho(R_s A_c), \rho(A_c)) \leq C_M \sqrt{|s|} \leq C_M \sqrt{t}. \quad (13)$$

This also covers a crossing through $|\operatorname{tr} C| = 2$ and is the reason for the square root. Combining (11)–(13) proves (9).

For later use, (2) and (9) imply, uniformly for $\|A_c\| \leq M$,

$$\operatorname{dist}(\rho(\alpha, e^F A_c), \rho(\alpha, A_c)) \leq C_M \sqrt{\|F\|_0}. \quad (14)$$

Step 3: the finite-scale analytic Lie-algebra KAM step. We now prove the local engine used in the iteration. For a constant A_c write $T_c = \operatorname{Ad}_{A_c}$ on $\mathfrak{sl}(2, \mathbb{C})$. For $z = e^{2\pi i k \alpha}$ the Fourier homological operator is

$$L_{k,c} := zI - T_c. \quad (15)$$

If A_c is elliptic, let

$$\beta(A_c) := \rho(\alpha, A_c) \in \mathbb{R}/\mathbb{Z}; \quad (16)$$

then the eigenvalues of T_c are $1, e^{2\pi i \beta(A_c)}, e^{-2\pi i \beta(A_c)}$. If A_c is nonelliptic, the other two eigenvalues of T_c are positive reciprocal real numbers (with the parabolic case obtained by coalescence at 1).

The following determinant estimate avoids any unjustified uniform diagonalization near a parabolic constant. Since $\dim \mathfrak{sl}_2 = 3$,

$$\det(zI - T_c) = (z - 1)(z - \lambda^2)(z - \lambda^{-2}), \quad (17)$$

where λ, λ^{-1} are the eigenvalues of A_c , counted with algebraic multiplicity. If $\lambda^2 = a > 0$ and $|z| = 1$, then

$$|z - a| |z - a^{-1}| = a^{-1} |z - a|^2 = a^{-1} (a - 1)^2 + |z - 1|^2 \geq |z - 1|^2. \quad (18)$$

Thus for every nonelliptic A_c ,

$$|\det L_{k,c}| \geq |z - 1|^3. \quad (19)$$

For $\|A_c\| \leq M$, the entries of $\operatorname{adj}(L_{k,c})$ are bounded by C_M . Hence the Diophantine condition on the base gives, for $0 < |k| \leq N$,

$$\|L_{k,c}^{-1}\| \leq C_M \gamma^{-3} N^{3\tau} \quad (A_c \text{ nonelliptic}). \quad (20)$$

If A_c is elliptic and

$$\|\beta(A_c) - k\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \eta \quad (0 < |k| \leq N), \quad (21)$$

then the same adjugate argument, now using the three factors in (17), gives

$$\|L_{k,c}^{-1}\| \leq C_M \gamma^{-1} N^\tau \eta^{-2} \quad (0 < |k| \leq N). \quad (22)$$

We shall use the harmless common majorant

$$H(M, \gamma, \tau; N, \eta) := C_M \gamma^{-3} N^{3\tau+2} \eta^{-2}. \quad (23)$$

The two surplus powers of N absorb Fourier summation and all fixed norm-equivalence constants.

Let now $F : \mathbb{T}_r \rightarrow \mathfrak{sl}(2, \mathbb{C})$ be holomorphic, real on $\mathbb{T}$, with $\varepsilon = \|F\|_r$. In a nonresonant situation satisfying (21), define

$$\widehat{Y}(k) = \begin{cases} -L_{k,c}^{-1} \widehat{F}(k), & 0 < |k| \leq N, \\ 0, & k = 0 \text{ or } |k| > N. \end{cases} \quad (24)$$

Reality of F and of T_c gives reality of Y . Fourier decay and (23) imply for $0 < \delta < r$

$$\|Y\|_{r-\delta} \leq C \delta^{-1} H(M, \gamma, \tau; N, \eta) \varepsilon. \quad (25)$$

The homological equation is exactly

$$Y(x + \alpha) - \text{Ad}_{A_c} Y(x) = -(T_N F(x) - \widehat{F}(0)). \quad (26)$$

Set

$$A'_c = e^{\widehat{F}(0)} A_c \quad (27)$$

and define F' by the exact identity

$$e^{Y(x+\alpha)} e^{F(x)} A_c e^{-Y(x)} = e^{F'(x)} A'_c. \quad (28)$$

For $\|Y\|_{r-\delta} + \varepsilon$ small, the logarithm defining F' is well defined. Taylor's formula for the analytic multiplication/logarithm maps on the finite-dimensional Lie algebra, together with (26), gives

$$\|F'\|_{r-\delta} \leq C_M (\varepsilon + \|Y\|_{r-\delta})^2 + C \delta^{-1} e^{-2\pi\delta N} \varepsilon. \quad (29)$$

This is the group-valued counterpart of *Single analytic KAM conjugacy step near a diagonal constant*: no determinant error is introduced because every correction is $e^Y \in SL(2, \mathbb{R})$. The degree is zero, so (5) reduces to the invariance already proved in *Fibered rotation number*.

We next treat one low resonance. Fix $0 < a < 10^{-2}$ and put

$$\eta = \varepsilon^a. \quad (30)$$

For sufficiently small ε , the base Diophantine estimate implies that there is *at most one* nonzero m , $|m| \leq N$, with

$$\|\beta(A_c) - m\alpha\| < \eta. \quad (31)$$

Indeed, two such integers $m \neq m'$ would give

$$\|(m - m')\alpha\| < 2\eta,$$

whereas $\|(m - m')\alpha\| \geq \gamma(2N)^{-\tau}$; our later choice of N makes $2\eta < \gamma(2N)^{-\tau}$. A nonelliptic constant has no such fiber resonance and is always handled by (20).

Assume (31). Then A_c is elliptic. A real constant conjugacy P puts it into the rotation form

$$PA_cP^{-1} = R_{\pi\beta(A_c)}. \quad (32)$$

Although P can become large near the parabolic boundary, (31) and the base Diophantine condition give

$$\|\beta(A_c)\|_{\mathbb{R}/\mathbb{Z}} \geq \|m\alpha\| - \eta \geq \frac{1}{2}\gamma N^{-\tau} \quad (33)$$

for small ε . To make the size of this normal form explicit, let $e^{i\phi}$ be an eigenvalue of A_c , so $\phi = \pi\beta(A_c)$ modulo π . The spectral projection

$$\Pi_+ = \frac{A_c - e^{-i\phi}I}{e^{i\phi} - e^{-i\phi}}$$

has norm at most $C_M|\sin\phi|^{-1}$. Taking the real and imaginary parts of a normalized vector in $\text{im}\Pi_+$ gives a real basis in which A_c is R_ϕ , with condition number bounded by a fixed multiple of $1 + \|\Pi_+\|$. By (33), therefore,

$$\|P\| + \|P^{-1}\| \leq C_M\gamma^{-2}N^{2\tau}. \quad (34)$$

(The exponent 2τ is deliberately wasteful; only polynomial control is used below.) Now conjugate by $D_m(x) = R_{-\pi m x}$ on the double cover. Because all rotations commute,

$$D_m(x + \alpha)R_{\pi\beta(A_c)}D_m(x)^{-1} = R_{\pi(\beta(A_c) - m\alpha)}. \quad (35)$$

The transformed perturbation is still an exponential of a traceless real matrix; in the complex eigenbasis of Ad_{R_θ} the three Fourier families are merely shifted by $0, +m, -m$. Consequently, if

$$r_1 = \frac{r}{8}, \quad r_+ = \frac{r}{16}, \quad N = \left\lceil \frac{1}{\pi r} \log \frac{1}{\varepsilon} \right\rceil, \quad (36)$$

then the transformed perturbation on $\mathbb{T}_{r_1}$ has norm at most

$$\bar{\varepsilon} \leq C_M\gamma^{-C}N^C e^{2\pi N r_1 \varepsilon} \leq C_M\gamma^{-C}N^C \varepsilon^{3/4}. \quad (37)$$

Here the only exponential factor comes from shifting the two off-diagonal Fourier families by $\pm m$; the constant phase $m\alpha$ has modulus one. The new constant angle $\tilde{\beta} = \beta(A_c) - m\alpha$ has $\|\tilde{\beta}\| < \eta$. Since $\gamma(8N)^{-\tau} \gg \eta$ for small ε , every nonzero Fourier divisor up to $4N$ is now bounded below solely by the base Diophantine estimate. Apply (24)–(29), from $r_1 = r/8$ to $r_+ = r/16$ and with cutoff $4N$, to the transformed cocycle, leaving only the zero Fourier mode to be absorbed into the next constant. The generic Fourier-tail factor is

$$e^{-2\pi(r_1 - r_+)4N} \bar{\varepsilon} \leq C_M\gamma^{-C}N^C \varepsilon^{5/4}, \quad (38)$$

while the quadratic term is $C_M \gamma^{-C} N^C \varepsilon^{3/2}$. With a fixed sufficiently small, these estimates show that the resulting perturbation F_+ on $\mathbb{T}_{r/16}$ satisfies

$$\|F_+\|_{r/16} \leq \varepsilon^{6/5} \quad (39)$$

provided ε is below a threshold depending only on r, γ, τ, M . The resulting constant A_+ remains in a fixed compact set. We also record explicitly why its angle stays on the resonant scale. Before the final nonresonant solve the constant angle is $\tilde{\beta}$, hence has distance $< \eta$ from zero. The only subsequent constant change is absorption of the transformed zero Fourier mode, whose norm is at most $\bar{\varepsilon}$. Apply the square-root comparison of Step 2 to the two constant cocycles before and after this absorption. By (37),

$$\text{dist}(\rho(\alpha, A_+), \tilde{\beta}) \leq C_M \sqrt{\bar{\varepsilon}} \leq C_M \gamma^{-C} N^C \varepsilon^{3/8}.$$

Since $a < 10^{-2}$ and N is only logarithmic in ε^{-1} , the last quantity is $o(\varepsilon^a) = o(\eta)$. Shrinking the universal smallness threshold once therefore gives

$$\text{dist}_{\mathbb{R}/\mathbb{Z}}(\rho(\alpha, A_+), 0) \leq 2\eta. \quad (40)$$

By (8), the complete resonant step has degree $-m$ and changes the fibered rotation number by $-m\alpha$. This proves the finite-scale resonant KAM step, including the double-cover and degree bookkeeping that were missing from the previous draft.

Step 4: an almost-reduction sequence with controlled degree growth. Fix

$$r_n = h16^{-n}, \quad E_n = E_0^{(6/5)^n}, \quad \eta_n = E_n^a, \quad N_n = \left\lceil \frac{1}{\pi r_n} \log \frac{1}{E_n} \right\rceil. \quad (41)$$

Choose E_0 so small that all estimates in Step 3 hold uniformly for constants in a compact set

$$\mathcal{K} := \{C \in SL(2, \mathbb{R}) : \|C\| + \|C^{-1}\| \leq M\} \quad (42)$$

containing a neighborhood of A_0 and a fixed neighborhood of the rotation subgroup. This compactness assertion does not presuppose finite resonance. At a nonresonant stage the constant changes only by absorption of a zero mode of size $O(E_n)$. At a resonant stage, after the possibly large constant normalizer P and the degree map D_m , the new constant itself is the small-angle rotation in (35), followed by a zero-mode update of size $O(\bar{\varepsilon})$; hence it lies in the fixed rotation neighborhood, independently of the size of P . Since the sums of all subsequent zero-mode updates converge, increasing M once makes the induction self-consistent.

Starting from (A_0, F_0) , apply at stage n the nonresonant step if the constant is nonelliptic or if

$$\|\beta(A_n) - k\alpha\| \geq \eta_n \quad (0 < |k| \leq N_n), \quad (43)$$

and otherwise apply the unique resonant step. Let us make explicit that the same target exponent $6/5$ is available in the nonresonant branch, so no fiber arithmetic is being hidden here. Take $\delta = 15r_n/16$ in (25)–(29). With $N = N_n$ and $\eta = \eta_n = E_n^a$, the common inverse bound (23) is E_n^{-2a} times a polynomial in r_n^{-1} and $\log E_n^{-1}$. Thus the

quadratic contribution in (29) is bounded by

$$P_n E_n^{2-4a},$$

where P_n grows at most exponentially in n . On the other hand, (41) gives

$$e^{-2\pi(15r_n/16)N_n} E_n \leq E_n^{1+30/16} = E_n^{23/8}$$

(up to an inessential constant coming from the ceiling in N_n). Because $a < 10^{-2}$, both exponents are strictly larger than $6/5$; the superexponential decay of E_n absorbs P_n . Hence, after making E_0 smaller once, the nonresonant output is at most $E_n^{6/5} = E_{n+1}$. The resonant branch has the same output bound by (37)–(39). Restricting further to r_{n+1} when necessary, Step 3 therefore gives

$$W_n(x + \alpha) e^{F_n(x)} A_n W_n(x)^{-1} = e^{F_{n+1}(x)} A_{n+1}, \quad \|F_{n+1}\|_{r_{n+1}} \leq E_{n+1}. \quad (44)$$

Here $W_n = e^{Y_n}$ has degree zero in a nonresonant step, while in a resonant step it is a product of a constant conjugacy, D_{m_n} , and a near-identity correction, with

$$0 < |m_n| \leq N_n. \quad (45)$$

Let $m_n = 0$ in the nonresonant case and put

$$d_n := m_0 + \cdots + m_{n-1}. \quad (46)$$

The sign convention in (8) and Lemma 2.1 gives the exact identity

$$\rho_0 := \rho(\alpha, e^{F_0} A_0) = \rho(\alpha, e^{F_n} A_n) + d_n \alpha \pmod{\mathbb{Z}}. \quad (47)$$

Moreover, whenever stage n is resonant, (40), (47), and the rotation comparison (14) give

$$\text{dist}_{\mathbb{R}/\mathbb{Z}}(\rho_0 - d_{n+1}\alpha, 0) \leq 2\eta_n + C\sqrt{E_{n+1}} \leq 3E_n^a \quad (48)$$

for small E_0 .

The cutoffs grow only exponentially in the stage number. Indeed, from (41),

$$N_n \leq C(h, E_0) \left(16 \cdot \frac{6}{5}\right)^n, \quad |d_n| \leq \sum_{j < n} N_j \leq C_0 D^n \quad (49)$$

for fixed $C_0, D > 1$. In contrast,

$$E_n^a = \exp\{-a|\log E_0|(6/5)^n\} \quad (50)$$

decays faster than every exponential in n .

Step 5: the arithmetic hypothesis forces only finitely many resonant steps. First suppose that ρ_0 is Diophantine with respect to α in the repaired *the definition of relative resonant and Diophantine fibered rotation numbers* and is not resonant. Thus for some $\kappa, \sigma > 0$,

$$|\sin \pi(\rho_0 - k\alpha)| \geq \kappa |k|^{-\sigma} \quad (k \neq 0). \quad (51)$$

If there were infinitely many resonant stages, apply (48) along them. When $d_{n+1} \neq 0$, (51), $|\sin \pi t| \leq \pi \operatorname{dist}(t, \mathbb{Z})$, and (49) give

$$\operatorname{dist}(\rho_0 - d_{n+1}\alpha, \mathbb{Z}) \geq \frac{\kappa}{\pi} |d_{n+1}|^{-\sigma} \geq cD^{-\sigma n}, \quad (52)$$

which contradicts the superexponential upper bound (48)–(50) for large n . If $d_{n+1} = 0$ along a subsequence, the fixed positive number $\operatorname{dist}(\rho_0, \mathbb{Z})$ contradicts (48) directly. Hence there are only finitely many resonant stages.

Now suppose ρ_0 is resonant:

$$\rho_0 - k_0\alpha \in \mathbb{Z} \quad (53)$$

for some $k_0 \in \mathbb{Z}$. At a resonant stage, (48) becomes

$$\|(k_0 - d_{n+1})\alpha\|_{\mathbb{R}/\mathbb{Z}} \leq 3E_n^a. \quad (54)$$

If $k_0 - d_{n+1} \neq 0$, the base Diophantine condition and (49) give an exponential lower bound, again contradicting (50) at every sufficiently late resonant stage. Thus an infinite sequence of resonant stages would force $d_{n+1} = k_0$ at some late one. After this happens the transformed fibered rotation number is zero by (47). At any still later proposed resonance $m \neq 0$, (14) and (43) would imply

$$\|m\alpha\| \leq \eta_n + C\sqrt{E_n}, \quad (55)$$

whereas $\|m\alpha\| \geq \gamma N_n^{-\tau}$; (41) and (50) contradict (55) for large n . Hence the resonant case also has only finitely many nonzero-degree steps.

Let q be larger than the last resonant stage. By taking q still larger through the subsequent degree-zero stages, $\|F_q\|_{r_q}$ can be made as small as desired while $r_q > 0$ remains a positive number.

There are now two terminal arithmetic alternatives. In the nonresonant Diophantine case, put

$$\rho_* := \rho_0 - d_q\alpha. \quad (56)$$

For $k + d_q \neq 0$, (51) gives a polynomial lower bound for $\rho_* - k\alpha$; for the single exceptional index $k = -d_q$, the lower bound is the fixed positive number $\operatorname{dist}(\rho_0, \mathbb{Z})$ because we are in the nonresonant branch. After decreasing the constant, there are therefore $\kappa_*, \sigma_* > 0$ such that

$$|\sin \pi(\rho_* - k\alpha)| \geq \kappa_*(1 + |k|)^{-\sigma_*} \quad (k \in \mathbb{Z}). \quad (57)$$

In the resonant case, in fact

$$\rho_* = 0. \quad (58)$$

For if $\rho_* = (k_0 - d_q)\alpha$ were nonzero, the cutoffs $N_n \rightarrow \infty$ and the comparison (14) would eventually make the fixed nonzero index $k_0 - d_q$ satisfy the resonance test (43), contradicting the definition of q as being after the last resonant stage.

Step 6: terminal positive-strip Newton-KAM convergence. We restart the iteration from (A_q, F_q) , now on the fixed positive strip r_q . Set

$$s_j = \frac{r_q}{2} + \frac{r_q}{2^{j+1}}, \quad \delta_j = s_j - s_{j+1} = \frac{r_q}{2^{j+2}}. \quad (59)$$

Thus $s_j \downarrow r_q/2 > 0$ and the total future strip loss is finite. At stage j choose a Fourier cutoff

$$M_j = \left\lceil C\delta_j^{-1} \log \frac{1}{\varepsilon_j} \right\rceil, \quad \varepsilon_j = \|G_j\|_{s_j}. \quad (60)$$

In the Diophantine terminal case (57), degree-zero conjugacy keeps the fibered rotation number equal to ρ_* . By (14), the constant part C_j satisfies

$$\text{dist}(\rho(\alpha, C_j), \rho_*) \leq C\sqrt{\varepsilon_j}. \quad (61)$$

For sufficiently small terminal error (obtained by choosing q farther along the degree-zero preparatory tail), (57), (60), and (61) show that every divisor with $0 < |k| \leq M_j$ has the polynomial lower bound required in (22). Thus every nonconstant low mode is solved by (24); the entire zero Fourier coefficient is deliberately left unsolved and is absorbed into C_{j+1} .

In the terminal resonant case (58), (61) says simply $\text{dist}(\rho(\alpha, C_j), 0) \leq C\sqrt{\varepsilon_j}$. For $0 < |k| \leq M_j$, the base estimate $\|k\alpha\| \geq \gamma M_j^{-\tau}$ dominates this square-root error once ε_j is small. Hence all *nonzero* Fourier modes are again safely invertible. The only persistent resonance is the zero mode, which is precisely the mode absorbed into the next constant. No further nonzero-degree conjugacy is required.

Consequently (25) and (29), with the choices (59)–(60), give constants $A, B, p, q > 0$ such that

$$\varepsilon_{j+1} \leq A\delta_j^{-q}\varepsilon_j^2, \quad \|Y_j\|_{s_{j+1}} \leq B\delta_j^{-p}\varepsilon_j. \quad (62)$$

Taking q far enough along the preparatory sequence makes the initial terminal error satisfy the explicit smallness hypothesis of *Abstract analytic Newton-KAM convergence lemma*. That theorem yields

$$\sum_{j \geq 0} \|Y_j\|_{r_q/2} < \infty, \quad \varepsilon_j \longrightarrow 0. \quad (63)$$

Since $e^{Y_j} = I + O(Y_j)$ on the common strip, the ordered product of the near-identity corrections converges uniformly on $\mathbb{T}_{r_q/2}$ to an invertible holomorphic $SL(2, \mathbb{C})$ map B_∞ . Reality on the real axis is preserved at every step, so $B_\infty(x) \in SL(2, \mathbb{R})$ for real x . The constants converge as well because

$$\|C_{j+1} - C_j\| \leq C\varepsilon_j \quad (64)$$

and $\sum_j \varepsilon_j < \infty$. Passing to the limit in the exact conjugacy identities gives

$$B_\infty(x + \alpha)e^{F_q(x)}A_qB_\infty(x)^{-1} = A_* \quad (65)$$

for a constant $A_* \in SL(2, \mathbb{R})$.

Finally compose B_∞ with the finitely many analytic conjugacies from Steps 3–5 and with the initial logarithmic normalization. The resulting map is analytic on a positive strip. It is 1-periodic if every degree step was even and otherwise is 2-periodic; in either case it is exactly a circle or double-cover conjugacy in the sense of *the definition of analytic reducibility on the circle or double cover*. Equation (65) therefore proves analytic reducibility to a constant. No spectral conclusion and no global almost-

reducibility assertion has been used or proved. $\square$

Theorem F12.7. Herman-Yoccoz smooth circle linearization theorem

Statement. Let $f \in \text{Diff}_+^\infty(\mathbb{T})$ have irrational rotation number α . If α is Diophantine, i.e. there are $\gamma > 0$ and $\tau > 0$ such that

$$\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-\tau} \quad (q \geq 1),$$

then there exists $h \in \text{Diff}_+^\infty(\mathbb{T})$ satisfying

$$h \circ f \circ h^{-1} = R_\alpha.$$

This is the global smooth theorem; no smallness assumption on $f - R_\alpha$ is imposed.

Proof.

Proof strategy

The proof has two logically distinct stages. Continued-fraction return towers and distortion estimates first upgrade the topological conjugacy to a quantitative C^1 conjugacy. One then solves successive cohomological equations along the convergents and bootstraps finite differentiability, using the Diophantine growth law to pay every small divisor. Since the differentiability order is arbitrary for a smooth map, the conjugacy is C^∞ .

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and choose a lift $F : \mathbb{R} \rightarrow \mathbb{R}$ of f with $F(x+1) = F(x)+1$. We divide what follows into the global C^1 part and the finite-order bootstrap. Every constant denoted by C may increase from line to line and may depend on the fixed map and on the finite differentiability order currently being used, but never on a return denominator or an iterate index.

Step 1: arithmetic normalization. Let p_n/q_n be the continued-fraction convergents of α , and set

$$\alpha_n := (-1)^n (q_n \alpha - p_n) > 0. \quad (\text{HY.1})$$

The hypothesis

$$\|q\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \gamma q^{-\tau} \quad (\text{HY.2})$$

forces $\tau \geq 1$ by Dirichlet approximation. Put

$$\beta := \tau - 1 \geq 0. \quad (\text{HY.3})$$

Then

$$\left| \alpha - \frac{p}{q} \right| \geq \gamma q^{-2-\beta}, \quad (\text{HY.4})$$

and F05 gives

$$\frac{1}{q_{n+1} + q_n} < \alpha_n < \frac{1}{q_{n+1}}, \quad \alpha_{n+2} < \frac{1}{2} \alpha_n, \quad (\text{HY.5})$$

together with the equivalent Diophantine growth estimates

$$q_{n+1} \leq Cq_n^{1+\beta}, \quad \alpha_{n+1} \geq c\alpha_n^{1+\beta}. \quad (\text{HY.6})$$

Since f is C^∞ , *Denjoy theorem* supplies a topological conjugacy to R_α . In particular, the cyclic ordering of all closest-return intervals is exactly the cyclic ordering for the rigid rotation. *Unique ergodicity of irrational circle diffeomorphisms under Denjoy hypotheses* also supplies the unique invariant probability μ .

For the rest of the C^1 argument fix an integer

$$k > 2\beta + 1, \quad (\text{HY.7})$$

and use only the C^k norm of f .

Step 2: closest-return towers and the elementary distortion package. Take two consecutive convergents p/q and P/Q , with the orientation chosen so that the two corresponding closest returns approach a point from opposite sides. For $x \in \mathbb{T}$, let I_x be the shorter oriented arc from x to $f^q x$, let

$$d(x) := |I_x|, \quad M := \max_x d(x), \quad m := \min_x d(x), \quad (\text{HY.8})$$

and let J_x be the arc from $f^{-q}x$ to $f^q x$ that contains x .

The continued-fraction best-approximation property implies the following standard tower facts: the interiors of

$$f^i(I_x), \quad 0 \leq i < Q, \quad (\text{HY.9})$$

are pairwise disjoint, whereas the arcs $f^i(J_x)$, $0 \leq i < Q$, cover the circle. Indeed, conjugating by the Denjoy homeomorphism sends their endpoints to the corresponding endpoints for the rigid rotation; for the rotation, disjointness and covering are precisely the closest-return ordering of two consecutive convergents. Because a circle homeomorphism preserves cyclic order, the same statements hold for f .

Put $V = \text{Var}(\log Df)$. If $y \in I_x$ and $0 \leq i < Q$, then the intervals $f^j(I_x)$, $0 \leq j < i$, have disjoint interiors. Therefore

$$\begin{aligned} |\log Df^i(y) - \log Df^i(x)| &\leq \sum_{j=0}^{i-1} \text{Var}_{f^j(I_x)}(\log Df) \\ &\leq V. \end{aligned} \quad (\text{HY.10})$$

This is the local tower form of Denjoy distortion. By the mean-value theorem, for some $\xi_i \in I_x$,

$$|f^i(I_x)| = Df^i(\xi_i)d(x),$$

so (HY.10) yields

$$Df^i(x) \leq e^V \frac{|f^i(I_x)|}{d(x)}. \quad (\text{HY.11})$$

Consequently, for every integer $\ell \geq 1$,

$$\sum_{i=0}^{Q-1} (Df^i(x))^\ell \leq C \frac{M^{\ell-1}}{d(x)^\ell}. \quad (\text{HY.12})$$

Indeed, $|f^i(I_x)| \leq M$ after changing C by the bounded-distortion factor and, by (HY.9), the sum of the lengths is at most one.

We now record the higher-derivative consequence used twice below. For a C^k orientation-preserving diffeomorphism g , set

$$u_g := \log Dg, \quad Sg := D^2u_g - \frac{1}{2}(Du_g)^2. \quad (\text{HY.13})$$

Repeated use of the chain and product rules gives universal polynomials $A_r, C_r, E_{r,j}$ with the following weighted property: if the variable $D^j u$ is assigned weight j , every monomial in the formula of total derivative order r has total weight r . More concretely,

$$S(g \circ h) = (Sg \circ h)(Dh)^2 + Sh, \quad (\text{HY.14})$$

$$D^r u_{g \circ h} = (D^r u_g \circ h)(Dh)^r + D^r u_h + C_r((D^j u_g) \circ h, D^j u_h : 1 \leq j < r), \quad (\text{HY.15})$$

and differentiating (HY.14) $r - 1$ times gives a sum in which every term contains one derivative of $Sg \circ h$, a power of Dh , and lower derivatives of u_h , with total weight $r + 1$. The assertions follow by induction on r : differentiating a monomial either raises one derivative index by one or differentiates Dh , which introduces $Du_h Dh$; in both cases the total weight rises by exactly one. Thus no term of the wrong scaling can appear.

From (HY.14) and (HY.12), for $0 \leq n \leq Q$,

$$|Sf^n(x)| \leq C \frac{M}{d(x)^2}. \quad (\text{HY.16})$$

The same bound with $d(x)$ replaced by m holds in sup norm. If $u = u_{f^n}$, then Du has a zero on the circle. At a point where $|Du|$ is maximal, $D^2u = 0$; hence (HY.13) gives

$$\|Du\|_0^2 \leq 2\|Sf^n\|_0 \leq C \frac{M}{m^2}. \quad (\text{HY.17})$$

The covering by $f^i(J_z)$, where $d(z) = m$, and (HY.10) transport this global estimate back to a point x and give the sharper pointwise version

$$|D \log Df^n(x)| \leq C \frac{M^{1/2}}{d(x)}. \quad (\text{HY.18})$$

Here is the transport calculation. Choose $t \in J_z$ and $0 \leq i < Q$ with $x = f^i(t)$. Since iterates commute,

$$Du_{f^{n+i}}(t) = Du_{f^n}(x) Df^i(t) + Du_{f^i}(t).$$

The two terms other than $Du_{f^n}(x)$ satisfy (HY.17); the mean-value theorem on I_t , together with (HY.10), gives $Df^i(t)^{-1} \leq Cm/d(x)$. This is (HY.18).

The weighted-polynomial induction following (HY.15) now gives, for $1 \leq r \leq k-1$,

$$\boxed{|D^r \log Df^n(x)| \leq C \left(\frac{M^{1/2}}{d(x)} \right)^r, \quad 0 \leq n \leq Q.} \quad (\text{HY.19})$$

For completeness, suppose the estimate is known through order r . In the $(r-1)$ -fold derivative of (HY.14), every proper monomial has weighted degree $r+1$; substituting the induction bounds therefore contributes at most $C(M^{1/2}/d(x))^{r+1}$. The unique term containing the new derivative $D^{r+1}u_{f^n}$ is recovered from $D^{r-1}Sf^n = D^{r+1}u_{f^n} + C_{r+1}(Du_{f^n}, \dots, D^r u_{f^n})$. The sum over the orbit that occurs after differentiating (HY.14) is controlled by (HY.12); its power of M is exactly canceled by the weighted degree. This yields the same bound at order $r+1$. The initial orders are (HY.18) and $D^2u = Sf^n + \frac{1}{2}(Du)^2$. Hence the induction closes.

Step 3: improved Denjoy inequality. Choose z with $d(z) = m$. The oriented closest-return displacement has constant sign; hence at its minimum z its derivative vanishes, so $Df^q(z) - 1 = 0$. Moreover *Denjoy distortion inequality* gives a uniform two-sided bound on $Df^{\pm q}$. Therefore the fixed enlargement from $f^{-q}z$ to $f^{2q}z$ has length $O(m)$. For arbitrary x , choose $t \in J_z$ and $0 \leq i < Q$ with $x = f^i(t)$. Commutation of f^q and f^i gives

$$\log Df^q(x) = \log Df^q(t) + \log Df^i(f^q t) - \log Df^i(t). \quad (\text{HY.20})$$

Both I_t and the arc joining t to z inside J_z have length $O(m)$, by the tower distortion just proved. Applying (HY.17) to the first term (using its zero at z) and (HY.18) to the difference in the second term gives

$$\boxed{\|\log Df^q\|_0 \leq CM^{1/2}, \quad \|Df^q - 1\|_0 \leq CM^{1/2}.} \quad (\text{HY.21})$$

The second inequality follows from the first because $M \rightarrow 0$ along closest returns, and finitely many initial indices are absorbed into C .

Step 4: the next-return comparison. Let

$$\tilde{d}(x) := |f^Q x - x|, \quad a := \frac{|Q\alpha - P|}{|q\alpha - p|}. \quad (\text{HY.22})$$

We prove

$$\boxed{|\tilde{d}(x) - ad(x)| \leq C(M^{(k-1)/2}d(x) + M^{1/2}\tilde{d}(x)).} \quad (\text{HY.23})$$

Only the proof of this estimate is delicate, so we spell out the two alternatives.

First note a mean-value identity. For a lift displacement $\phi_q(x) = F^q(x) - x - p$, invariance of μ gives

$$\int \phi_q d\mu = q\alpha - p. \quad (\text{HY.24})$$

Indeed, if $\phi_1 = F - \text{id}$, then $F^q(x) - x = \sum_{j=0}^{q-1} \phi_1(f^j x)$; invariance gives $\int (F^q - x)d\mu = q \int \phi_1 d\mu$, and the uniform translation-number limit from *Existence of rotation number* forces $\int \phi_1 d\mu = \alpha$. Applying the same formula to the two oppositely oriented closest returns, and then the one-dimensional mean-value theorem on the two adjacent return

arcs, yields points

$$y \in [x, f^q x], \quad z \in [f^Q x, x]$$

for which

$$\tilde{d}(y) = a d(z). \quad (\text{HY.25})$$

Here is the integral calculation. Let Φ be the real-valued lift of the identity coordinate on the oriented arc from $f^Q x$ to $f^q x$. Invariance of μ , applied first to the Q -translate of the q -arc and then to the q -translate of the Q -arc, gives

$$\begin{aligned} \int_x^{f^q x} (f^Q - \text{id}) d\mu &= \int_{f^Q x}^{f^{Q+q} x} \Phi d\mu - \int_x^{f^q x} \Phi d\mu \\ &= \int_{f^q x}^{f^{q+Q} x} \Phi d\mu - \int_x^{f^Q x} \Phi d\mu \\ &= \int_x^{f^Q x} (f^q - \text{id}) d\mu. \end{aligned} \quad (\text{HY.26a})$$

The two oriented arc masses are

$$\mu([x, f^q x]) = |q\alpha - p|, \quad \mu([f^Q x, x]) = |Q\alpha - P|,$$

because the Denjoy conjugacy sends μ to Haar measure and sends the endpoints to rotation endpoints. Applying the mean-value theorem for integrals to the two sides of (HY.26a) therefore gives $y \in [x, f^q x]$ and $z \in [f^Q x, x]$ satisfying (HY.25).

Now consider $\tilde{d}$ on the q -return intervals. If it is monotone on one such interval, then (HY.21) and the local distortion estimate show that its relative oscillation is small:

$$\left| \frac{\tilde{d}(y)}{\tilde{d}(x)} - 1 \right| \leq CM^{1/2} \quad (y \in I_x). \quad (\text{HY.26})$$

Indeed, translate the two points into a fixed bounded enlargement of a minimal return interval by the covering tower, write the two lengths by the mean-value theorem, and integrate (HY.18); the integration length is $O(d(x))$, so the factor $d(x)^{-1}$ disappears.

If $\tilde{d}$ is monotone on no q -return interval, take an integer $N > k/2$ and a bounded enlargement K_x consisting of $2N + 1$ consecutive q -return arcs. Nonmonotonicity supplies at least $2N \geq k$ distinct zeros of $Df^Q - 1$, equivalently of $\log Df^Q$, in K_x . Repeated Rolle gives a zero in K_x for each derivative $D^j \log Df^Q$, $0 \leq j \leq k - 1$. The enlargement has length $O(d(x))$, and (HY.19) gives on it

$$|D^{k-1} \log Df^Q| \leq C \left(\frac{M^{1/2}}{d(x)} \right)^{k-1}.$$

Integrating successively from the successive zeros yields

$$\|\log Df^Q\|_{C^0(K_x)} \leq CM^{(k-1)/2}. \quad (\text{HY.27})$$

Since $D\tilde{d} = Df^Q - 1$, integration over an interval of length $O(d(x))$ gives

$$|\tilde{d}(y) - \tilde{d}(x)| \leq CM^{(k-1)/2}d(x), \quad y \in I_x. \quad (\text{HY.28})$$

Apply (HY.26) or (HY.28) to the point y in (HY.25); for d , the improved Denjoy inequality gives

$$|d(z) - d(x)| \leq CM^{1/2}d(x). \quad (\text{HY.29})$$

Substitution into (HY.25) proves (HY.23).

Taking maxima and minima in (HY.23) gives, whenever $CM^{1/2} < 1$,

$$M_+ \leq M \frac{a + CM^{(k-1)/2}}{1 - CM^{1/2}}, \quad m_+ \geq m \frac{a - CM^{(k-1)/2}}{1 + CM^{1/2}}. \quad (\text{HY.30})$$

Notice especially that the nonlinear contribution in the first numerator becomes $CM^{(k+1)/2}$ after multiplication by M . This is the exponent used below.

Step 5: the Diophantine dichotomy and two-sided return comparability. Return to q_n and write

$$d_n(x) = |F^{q_n}(x) - x - p_n|, \quad M_n = \max d_n, \quad m_n = \min d_n. \quad (\text{HY.31})$$

By topological conjugacy, $M_n \rightarrow 0$. Also the invariant-measure identity (HY.24) implies

$$m_n \leq \alpha_n \leq M_n. \quad (\text{HY.32})$$

From (HY.30) and $\alpha_{n+2} \leq \alpha_n/2$, for all large n ,

$$M_{n+2} \leq \frac{2}{3}M_n. \quad (\text{HY.33})$$

Thus every positive power of M_n is summable (after separating even and odd indices).

Choose $\theta > 0$ so small that

$$\frac{k+1}{2} - \theta > (1 + \beta + \theta)(1 + \theta). \quad (\text{HY.34})$$

This is possible by (HY.7). For large n , arrange also

$$CM_n^{1/2} \leq \frac{1}{2}, \quad \alpha_{n+1} \geq \alpha_n^{1+\beta+\theta}. \quad (\text{HY.35})$$

The last inequality follows from (HY.6) because the fixed constant can be absorbed by increasing n .

We now reproduce the finite alternative rather than referring to it. Pick $A > 1$ so large that the convergent product

$$\prod_{n \geq n_0} \frac{1 + M_n^\theta}{1 - CM_n^{1/2}} \quad (\text{HY.36})$$

is at most A , and increase A further so that

$$4CA^{-((k-1)/2-\theta)} \leq 1. \quad (\text{HY.36a})$$

Choose $n_1 \geq n_0$ with $M_{n_1} < A^{-2}$, set $a_{n_1} = A^{-2}$, and choose $0 < \rho_{n_1} < 1$ so that

$$M_{n_1} = a_{n_1} \alpha_{n_1}^{\rho_{n_1}}. \quad (\text{HY.37})$$

Inductively maintain

$$M_n \leq a_n \alpha_n^{\rho_n}, \quad A^{-2} \leq a_n \leq A^{-1}, \quad 0 < \rho_n < 1. \quad (\text{HY.38})$$

There are two cases.

If

$$CM_n^{(k-1)/2-\theta} \leq \frac{\alpha_{n+1}}{\alpha_n}, \quad (\text{HY.39})$$

then (HY.30) gives

$$M_{n+1} \leq M_n \frac{(\alpha_{n+1}/\alpha_n)(1 + M_n^\theta)}{1 - CM_n^{1/2}}.$$

Set $\rho_{n+1} = \rho_n$ and

$$a_{n+1} = a_n \frac{1 + M_n^\theta}{1 - CM_n^{1/2}}. \quad (\text{HY.40})$$

The product choice keeps $a_{n+1} \leq A^{-1}$, and because $\alpha_{n+1} \leq \alpha_n$, (HY.38) propagates.

If (HY.39) fails, then, using $CM_n^{1/2} \leq 1/2$,

$$M_{n+1} \leq 4CM_n^{(k+1)/2-\theta}. \quad (\text{HY.41})$$

Set $a_{n+1} = a_n$ and $\rho_{n+1} = (1 + \theta)\rho_n$. From (HY.35), (HY.38), and (HY.34),

$$\begin{aligned} M_{n+1} &\leq 4Ca_n^{(k+1)/2-\theta} \alpha_n^{\rho_n((k+1)/2-\theta)} \\ &\leq a_n \alpha_n^{(1+\theta)\rho_n} \\ &\leq a_n \alpha_{n+1}^{(1+\theta)\rho_n}, \end{aligned} \quad (\text{HY.42})$$

where (HY.36a) pays the coefficient. Thus (HY.38) again propagates. Since every occurrence of the second case multiplies ρ_n by $1 + \theta$, while $\rho_n < 1$, the second case can occur only finitely many times. Eventually only (HY.39) occurs, and the telescope in (HY.40) gives

$$\boxed{\sup_n \frac{M_n}{\alpha_n} < \infty.} \quad (\text{HY.43})$$

The lower recurrence in (HY.30), together with (HY.43), gives for all large n

$$\frac{\alpha_{n+1}}{m_{n+1}} \leq \frac{\alpha_n}{m_n} \frac{1 + CM_n^{1/2}}{1 - CM_n^{(k-1)/2}(\alpha_n/\alpha_{n+1})}. \quad (\text{HY.44})$$

By (HY.6) and $M_n \leq C\alpha_n$,

$$M_n^{(k-1)/2} \frac{\alpha_n}{\alpha_{n+1}} \leq C\alpha_n^{(k-1)/2-\beta}. \quad (\text{HY.45})$$

The exponent is positive because $k > 2\beta + 1$, and the right-hand side is summable by (HY.5). Hence the product in (HY.44) converges and

$$\sup_n \frac{\alpha_n}{m_n} < \infty. \quad (\text{HY.46})$$

Combining (HY.43), (HY.46), and (HY.32),

$$c\alpha_n \leq d_n(x) \leq C\alpha_n \quad (n \geq 0, x \in \mathbb{T}). \quad (\text{HY.47})$$

Step 6: every iterate has a uniform derivative bound. Fix $i \in \mathbb{Z}$ and x . For even n large, the mean-value theorem on the interval from x to $F^{q_n}x - p_n$ gives a point ξ_n such that

$$d_n(F^i x) = Df^i(\xi_n)d_n(x). \quad (\text{HY.48})$$

Thus (HY.47) yields

$$c \leq Df^i(\xi_n) \leq C.$$

The closest-return intervals shrink uniformly to x , so $\xi_n \rightarrow x$. Continuity of Df^i gives

$$c \leq Df^i(x) \leq C \quad \text{for all } i \in \mathbb{Z}, x \in \mathbb{T}. \quad (\text{HY.49})$$

The constants do not depend on i .

Step 7: construction of the C^1 conjugacy from bounded iterates. We include this step so that no external bounded-coboundary theorem is needed. Let

$$\varphi(x) := \log Df(x).$$

Equation (HY.49) says that all two-sided Birkhoff sums

$$S_n \varphi(x) = \log Df^n(x) \quad (\text{HY.50})$$

are uniformly bounded. Consider the skew product

$$\mathcal{F}(x, t) = (fx, t + \varphi(x)) \quad \text{on } \mathbb{T} \times \mathbb{R}. \quad (\text{HY.51})$$

The orbit of any $(x_0, 0)$ has bounded vertical coordinate, so its closure contains a compact minimal $\mathcal{F}$ -invariant set K . Its projection to $\mathbb{T}$ is nonempty, compact, and f -invariant; the Denjoy conjugacy makes f minimal, hence the projection is the whole circle.

If both (x, a) and (x, b) belonged to K , with $a \neq b$, then the vertical translation $V_{b-a}K$ would be another compact minimal $\mathcal{F}$ -invariant set meeting K . Two minimal invariant compact sets that meet are equal, so $V_{b-a}K = K$. Iterating this nonzero translation would make K vertically unbounded, a contradiction. Thus every fiber of the projection

contains exactly one point. The compact graph is therefore

$$K = \{(x, u(x)) : x \in \mathbb{T}\}$$

for a continuous function u , and invariance gives

$$u(fx) - u(x) = \varphi(x) = \log Df(x). \quad (\text{HY.52})$$

Set $\rho = e^{-u}$, normalize it so that $\int_0^1 \rho = 1$, and define a degree-one lift

$$H(x) := \int_0^x \rho(t) dt. \quad (\text{HY.53})$$

Then $H \in C^1$, $DH = \rho > 0$, and (HY.52) gives

$$DH(Fx)DF(x) = DH(x).$$

Hence $H(Fx) - H(x)$ is constant. Its translation number is the rotation number of F , so the constant is α :

$$H \circ F = H + \alpha. \quad (\text{HY.54})$$

Thus the normalized circle conjugacy is a C^1 diffeomorphism. This establishes the HY-A estimate.

Step 8: Hölder norms, interpolation, and composition. We now prove enough of Yoccoz's higher-order engine for the frozen smooth theorem. For a noninteger $s = r + \sigma$, with $r \in \mathbb{N}$ and $0 < \sigma < 1$, use

$$\|v\|_s := \max_{0 \leq j \leq r} \|D^j v\|_0 + [D^r v]_\sigma, \quad [w]_\sigma := \sup_{x \neq y} \frac{|w(x) - w(y)|}{|x - y|^\sigma}. \quad (\text{HY.55})$$

At integer orders use the usual C^r norm. Product estimates follow directly from $[uv]_\sigma \leq \|u\|_0[v]_\sigma + \|v\|_0[u]_\sigma$.

We need the following interpolation inequality on the circle. If $0 \leq s_0 < s < s_1$, then

$$\|v\|_s \leq C \|v\|_{s_0}^{1-\vartheta} \|v\|_{s_1}^\vartheta, \quad \vartheta = \frac{s - s_0}{s_1 - s_0}, \quad (\text{HY.56})$$

provided one replaces either endpoint by an arbitrarily close noninteger order when it is an integer. A proof is obtained by applying the elementary finite-difference estimate

$$\|D^j v\|_0 \leq C(h^{s_1-j} \|v\|_{s_1} + h^{s_0-j} \|v\|_{s_0})$$

to difference quotients at scale h , choosing $h = (\|v\|_{s_0}/\|v\|_{s_1})^{1/(s_1-s_0)}$, and applying the same argument to $D^r v(x) - D^r v(y)$ for the terminal Hölder seminorm. This also proves the limiting cases by continuity of the exponents.

The Faà-di-Bruno formula and induction on r give the composition bound

$$\|v \circ \psi\|_s \leq C_s P_s(\|D\psi\|_{s-1}, \|D\psi^{-1}\|_0) \|v\|_s \quad (\text{HY.57})$$

whenever ψ is a $C^{\max(1,s)}$ diffeomorphism and the displayed norms are finite; P_s is a

polynomial with nonnegative coefficients. The proof is literal: every derivative of $v \circ \psi$ is a sum over partitions of the derivative order, and the Hölder seminorm of each product is bounded by putting the seminorm on one factor and sup norms on the rest.

A second Faà-di-Bruno induction gives universal weighted polynomials P_r, Q_r such that

$$\frac{D^{r+1}g}{Dg} = P_r(Du_g, \dots, D^r u_g), \quad D^r u_g = Q_r\left(\frac{D^2g}{Dg}, \dots, \frac{D^{r+1}g}{Dg}\right). \quad (\text{HY.58})$$

Consequently, on any family for which Dg and $(Dg)^{-1}$ are uniformly bounded,

$$C_s^{-1} \|Dg - 1\|_s \leq 1 + \|u_g\|_s \leq C_s(1 + \|Dg - 1\|_s), \quad (\text{HY.59})$$

with the harmless constants absorbed when the zero-order term is already bounded. All iterates of f satisfy the required derivative bounds by (HY.49).

Step 9: a closest-return estimate at arbitrary finite order. Let $\gamma_0 \geq 0$ be such that the normalized conjugacy H is $C^{1+\gamma_0}$. In this proof we shall ultimately use only $\gamma_0 = 0$, which is already established. From $F^n = H^{-1} \circ R_{n\alpha} \circ H$, the mean-value theorem, (HY.57), and the periodicity of derivatives of H give for every $0 \leq s \leq \gamma_0$

$$\|F^n - \text{id} - n\alpha\|_s \leq C_s \|n\alpha\|_{\mathbb{R}/\mathbb{Z}}. \quad (\text{HY.60})$$

Here and below the lift is adjusted by the nearest integer before taking a periodic norm.

Fix a differentiability order $k \geq 3$, and define

$$\Delta_j := \|D^{k-1} \log Df^{q_j}\|_0 + \alpha_j. \quad (\text{HY.61})$$

Applying (HY.19) to the preceding closest-return tower and using (HY.47), $\alpha_{j-1} \asymp q_j^{-1}$, gives the crude endpoint estimate

$$\Delta_j \leq C q_j^{(k-1)/2}. \quad (\text{HY.62})$$

Interpolating between the low estimate (HY.60) and the endpoint (HY.62), and using (HY.59), yields for $0 \leq s \leq k-1$

$$\boxed{\|\log Df^{q_j}\|_s \leq C q_{j+1}^{-1} (q_{j+1} \Delta_j)^{\rho(s)}, \quad \rho(s) := \max\left(0, \frac{s+1-\gamma_0}{k-\gamma_0}\right).} \quad (\text{HY.63})$$

To verify the exponent, when $s \leq \gamma_0 - 1$ use (HY.60) directly. Otherwise first interpolate the periodic displacement $f^{q_j} - \text{id} - p_j$ between orders γ_0 and k , then differentiate once and use (HY.59); the interpolation parameter is exactly $(s+1-\gamma_0)/(k-\gamma_0)$. Since $\alpha_j \asymp q_{j+1}^{-1}$, the low endpoint contributes the factor q_{j+1}^{-1} , while the quotient of high to low endpoints is $q_{j+1} \Delta_j$. This proves (HY.63).

For $\gamma_0 = 0$, (HY.62) and $q_{j+1} \leq C q_j^{1+\beta}$ imply

$$(q_{j+1} \Delta_j)^{1/k} q_j^{-1} \leq C q_j^{-\varepsilon_0}, \quad \varepsilon_0 := \frac{1}{2} - \frac{1+2\beta}{2k} > 0, \quad (\text{HY.64})$$

provided $k > 1 + 2\beta$. Thus the block induction below starts from the already proved C^1 conjugacy.

Step 10: block induction. For $0 \leq n \leq q_{j+1}/q_j$, write

$$\log Df^{nq_j} = \sum_{a=0}^{n-1} (\log Df^{q_j}) \circ f^{aq_j}. \quad (\text{HY.65})$$

Differentiate r times. Faà-di-Bruno expresses the derivative as a sum over a and over partitions of r . The term in which all r derivatives fall on $\log Df^{q_j}$ is controlled by (HY.63); every proper-partition term contains a lower derivative of $\log Df^{q_j}$ times a universal weighted polynomial in lower derivatives of $\log Df^{aq_j}$. Inducting on r , using the product estimate and (HY.64), gives

$$\|\log Df^{nq_j}\|_s \leq Cq_j^{-1}(q_{j+1}\Delta_j)^{\rho(s)}, \quad 0 \leq s \leq k-1, \quad 0 \leq n \leq q_{j+1}/q_j. \quad (\text{HY.66})$$

We record the exponent check because it is the load-bearing part. If a proper partition splits an order r into two positive weighted pieces, the two interpolation exponents ρ_1, ρ_2 satisfy

$$\max(\rho_1, \rho_2) \leq \rho(s) + \frac{1}{k - \gamma_0}. \quad (\text{HY.67})$$

The additional $1/(k - \gamma_0)$ is paid by the small factor in (HY.64); each sum over a contributes at most q_{j+1}/q_j , which is exactly canceled by the q_{j+1}^{-1} in (HY.63) and the remaining q_j^{-1} . Thus every proper-partition term is bounded by the right side of (HY.66). This proves the induction for integer orders; the terminal Hölder seminorm is handled by putting the seminorm on one factor, exactly as in (HY.57).

Step 11: the shifted top-derivative recurrence. The crude estimate (HY.62) is not enough for the global Ostrowski product. We now improve it to

$$\Delta_j \leq Cq_j. \quad (\text{HY.68})$$

This is where the shift by α_j in (HY.61) is essential.

Write the continued-fraction relation

$$q_{j+1} = aq_j + q_{j-1}, \quad a \geq 1. \quad (\text{HY.69})$$

Differentiate the composition identity $f^{q_{j+1}} = f^{q_{j-1}} \circ f^{aq_j}$ at order $k-1$ for $\log D$. The universal formula (HY.15) splits the result into three groups:

$$X = (D^{k-1} \log Df^{q_{j-1}} \circ f^{aq_j})(Df^{aq_j})^{k-1},$$

$$Y = D^{k-1} \log Df^{aq_j},$$

and the sum Z of all proper-partition terms. To avoid losing the location of the high derivative, introduce

$$\Delta'_j := \sup_{0 \leq t \leq j, m \geq 0} \|(D^{k-1} \log Df^{q_t} \circ f^m)(Df^m)^{k-1}\|_0 + \alpha_t. \quad (\text{HY.70})$$

By (HY.49), $\Delta_j \asymp \Delta'_j$ up to a fixed constant. The X term is at most Δ'_{j-1} . Expanding Y as a sum of a translates of the q_j block gives at most $a\Delta'_j$. Every term of Z contains only derivatives of order at most $k-2$; applying (HY.66), the interpolation exponent bound (HY.67), and (HY.64) gives

$$\|Z\|_0 \leq C a q_j^{1-\varepsilon_1} \quad (\text{HY.71})$$

for some $\varepsilon_1 > 0$. The same estimate survives precomposition by an arbitrary iterate because of (HY.49). Hence, with

$$\Theta_j := \max_{0 \leq t \leq j} \frac{\Delta'_t}{q_t},$$

(HY.69) implies

$$\Theta_{j+1} \leq \Theta_j (1 + C q_j^{-\varepsilon_1}). \quad (\text{HY.72})$$

The product converges because q_j grows at least geometrically every two indices. Thus Θ_j is bounded and (HY.68) follows.

For clarity about (HY.71): a typical proper term has the form

$$(D^r \log D f^{q_{j-1}} \circ f^{a q_j}) (D f^{a q_j})^r P(D \log D f^{a q_j}, \dots, D^{k-2} \log D f^{a q_j}),$$

with total weight $k-1$ and $r \leq k-2$. Put the largest derivative in C^0 , interpolate it between the controlled low endpoint and Δ_j , and bound all other factors by lower interpolation powers. Weighted homogeneity makes the total interpolation exponent no larger than the one in (HY.67); (HY.64) then contributes the factor $q_j^{-\varepsilon_1}$. Summing the a translates gives (HY.71). This is the entire top-order bookkeeping; there is no hidden second high derivative.

Step 12: one-step gain from C^1 to any prescribed finite order. Insert (HY.68) into (HY.66). Using $q_{j+1} \leq C q_j^{1+\beta}$, for $\gamma_0 = 0$ we obtain

$$\|\log D f^{n q_j}\|_s \leq C q_j^{\varrho_k(s)}, \quad \varrho_k(s) := \frac{(2+\beta)(s+1)}{k} - 1, \quad (\text{HY.73})$$

for $0 \leq n \leq q_{j+1}/q_j$. Thus if

$$s < g_k(0) := \frac{k - (2+\beta)}{2+\beta}, \quad (\text{HY.74})$$

there is $\delta > 0$ with

$$\|\log D f^{n q_j}\|_s \leq C q_j^{-\delta}. \quad (\text{HY.75})$$

Every integer $N \geq 0$ has an Ostrowski expansion

$$N = \sum_{j=0}^J b_j q_j, \quad 0 \leq b_j \leq q_{j+1}/q_j. \quad (\text{HY.76})$$

This follows by the greedy algorithm, starting with the largest $q_J \leq N$: the recurrence $q_{j+1} = a_{j+1} q_j + q_{j-1}$ forces the displayed digit bound at each subtraction. Put $N_j =$

$\sum_{t \leq j} b_t q_t$. From

$$\log Df^{N_j} = (\log Df^{b_j q_j}) \circ f^{N_{j-1}} + \log Df^{N_{j-1}} \quad (\text{HY.77})$$

and the Faà-di-Bruno composition estimate, the low uniform derivative bound (HY.49), and (HY.75), one obtains

$$1 + \|\log Df^{N_j}\|_s \leq (1 + \|\log Df^{N_{j-1}}\|_s)(1 + Cq_j^{-\delta}). \quad (\text{HY.78})$$

The product $\prod_j (1 + Cq_j^{-\delta})$ converges, hence

$$\boxed{\sup_{N \in \mathbb{Z}} \|\log Df^N\|_s < \infty \quad \text{for every } s < g_k(0).} \quad (\text{HY.79})$$

Negative iterates follow from $\log Df^{-N} = -(\log Df^N) \circ f^{-N}$ and the same composition estimate.

This establishes the HY-B estimate at the scope needed here. We do not need to iterate the sharp finite-regularity exponent map: because the original map is C^∞ , k is at our disposal.

Step 13: recover the same conjugacy in arbitrary finite regularity. Fix an integer $R \geq 1$. Choose k so large that there is a noninteger $s > R - 1$ satisfying $s < g_k(0)$. Equation (HY.79), together with (HY.59), gives uniform C^{1+s} bounds on all iterates after subtracting their lift translations. For each $N \geq 1$ define the degree-one averaged lift

$$H_N(x) := \frac{1}{N} \sum_{j=0}^{N-1} (F^j(x) - j\alpha) - c_N, \quad (\text{HY.80})$$

where c_N is chosen so that $H_N(0) = 0$. Since $DF^j \geq c > 0$ by (HY.49),

$$DH_N = \frac{1}{N} \sum_{j=0}^{N-1} DF^j \geq c, \quad (\text{HY.81})$$

so every H_N is a circle diffeomorphism. The uniform iterate bounds give a uniform C^{1+s} bound on H_N .

A direct telescoping computation gives

$$H_N(Fx) - H_N(x) = \frac{F^N(x) - x}{N} = \alpha + \frac{F^N(x) - x - N\alpha}{N}. \quad (\text{HY.82})$$

For a circle homeomorphism lift, the standard rotation-number estimate $|F^N(x) - x - N\alpha| \leq 1$ (proved in *Existence of rotation number*) makes the last term converge uniformly to zero. By Ascoli and the compact embedding $C^{1+s} \hookrightarrow C^R$, a subsequence converges in C^R to a normalized degree-one map $\hat{H}$. Equations (HY.81) and (HY.82) pass to the limit:

$$D\hat{H} \geq c, \quad \hat{H} \circ F = \hat{H} + \alpha. \quad (\text{HY.83})$$

Thus $\hat{H}$ is a C^R conjugacy.

Two conjugacies from the same irrational circle diffeomorphism to the same rotation differ by a rotation: if H_1, H_2 are normalized lifts, then $H_2 \circ H_1^{-1}$ commutes with R_α ; its

displacement is a continuous function invariant under the dense orbit of R_α , hence constant. With the normalization at zero the constant vanishes. Therefore $\widehat{H}$ is the C^1 conjugacy constructed in Step 7. Since R was arbitrary, the same H belongs to C^R for every R , so

$$H \in C^\infty. \quad (\text{HY.84})$$

Its derivative is everywhere positive, hence its inverse is also C^∞ . Descending to the circle and writing h for the induced diffeomorphism, (HY.54) becomes

$$h \circ f \circ h^{-1} = R_\alpha.$$

This is the asserted global smooth Herman-Yoccoz theorem. $\square$

12.3 Worked examples

Worked example: Normal Forms, Newton Schemes, Nash-Moser and Perturbative Reducibility: a model computation

Consider the scalar equation $F(u) = u^2 - 2 = 0$. Starting from $u_0 = 3/2$, Newton's correction is

$$\Delta_0 = -\frac{F(u_0)}{F'(u_0)} = -\frac{\frac{9}{4} - 2}{3} = -\frac{1}{12},$$

so

$$u_1 = \frac{3}{2} - \frac{1}{12} = \frac{17}{12}.$$

The old residual is $F(u_0) = 1/4$, while

$$F(u_1) = \frac{289}{144} - 2 = \frac{1}{144}.$$

The residual has dropped from $1/4$ to $1/144$, reflecting the quadratic cancellation. Now compare this with the linear homological equation

$$\omega u'(x) = \varepsilon \cos(2\pi x), \quad \omega \neq 0.$$

A zero-average solution is

$$u(x) = \frac{\varepsilon}{2\pi\omega} \sin(2\pi x).$$

Here the factor $1/\omega$ is the inverse linearized cost. In a quasiperiodic problem, ω is replaced mode by mode by divisors $\langle k, \omega \rangle$; controlling all of them is exactly where F08 enters a Newton/KAM step.

Boundary case: what fails at the edge of the hypotheses

Suppose the homological operator has a resonant mode on which its divisor is zero. For example, over a rational rotation $x \mapsto x + p/q$, the mode $e^{2\pi i q x}$ satisfies

$$e^{2\pi i q(x+p/q)} - e^{2\pi i q x} = 0.$$

The linearized operator is singular on that mode. Newton's formula cannot be applied there without first changing the normal form, adding a parameter, or retaining the resonant term. This is why "invert the derivative" is not a complete proof strategy in KAM-type problems.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

12.4 Chapter summary

What to remember

- Newton iteration succeeds because the linear term in the new residual is canceled.
- Normal forms separate removable and resonant terms.
- KAM controls small divisors; Nash-Moser controls derivative loss by smoothing.
- Reducibility theorems are structured nonlinear conjugacy problems, not isolated tricks.

Looking ahead

F13 supplies a different common toolkit: subharmonic functions, logarithmic potentials, capacity, Green functions, and Herglotz representations. These tools later connect dynamical growth to spectral and complex-analytic geometry.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

12.5 Exercises and problems

Exercise 12.1

Difficulty. Core.

Related results. Formal Poincaré–Dulac normal form; Newton quadratic-error lemma; Abstract analytic Newton–KAM convergence lemma.

Problem. Work through the worked example 'one Newton/KAM correction for a

scalar nonlinear equation and a single Fourier mode'. For Formal Poincaré–Dulac normal form, Newton quadratic-error lemma, and Abstract analytic Newton–KAM convergence lemma, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 12.2

Difficulty. Core.

Related results. Nash–Moser local-surjectivity theorem with a tame right inverse; Single KAM conjugacy step for near-constant cocycles; Perturbative Eliasson reducibility theorem for analytic $SL(2, \mathbb{R})$ cocycles.

Problem. Analyze the boundary example 'a resonant homological equation whose linearized inverse is singular'. For Nash–Moser local-surjectivity theorem with a tame right inverse, Single KAM conjugacy step for near-constant cocycles, and Perturbative Eliasson reducibility theorem for analytic $SL(2, \mathbb{R})$ cocycles, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 12.3

Difficulty. Advanced.

Related results. Abstract analytic Newton–KAM convergence lemma.

Problem. For Abstract analytic Newton–KAM convergence lemma, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 12.4

Difficulty. Advanced.

Related results. Nash–Moser local-surjectivity theorem with a tame right inverse.

Problem. Stress-test Nash–Moser local-surjectivity theorem with a tame right inverse. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 12.5

Difficulty. Advanced.

Related results. Formal Poincaré–Dulac normal form; Newton quadratic-error lemma.

Problem. Synthesis problem. Use Formal Poincaré–Dulac normal form and Newton quadratic-error lemma to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 12.6

Difficulty. Core.

Related results. Herman–Yoccoz smooth circle linearization theorem.

Problem. Work directly with Herman–Yoccoz smooth circle linearization theorem. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Potential, Subharmonic and Herglotz Theory

Why this chapter matters

Subharmonic functions are the natural home for logarithms of analytic quantities. Zeros, poles, growth rates, equilibrium measures, and spectral measures can all be encoded through Laplacians and potentials. This makes potential theory a shared language between complex dynamics and one-dimensional spectral theory.

The chapter emphasizes normalization. Factors of 2π , the sign of logarithmic energy, and the choice of Green function matter; worked examples make these conventions concrete.

Prerequisites. Complex analysis, distributions, harmonic functions, weak convergence of measures, and basic Hilbert-space spectral intuition.

Learning goals

- Compute Riesz measures and use Riesz decomposition.
- Apply Jensen and Poisson–Jensen formulae to zeros and boundary values.
- Understand logarithmic capacity and equilibrium measures through energy minimization.
- Recognize Herglotz functions as analytic encodings of positive measures.

Conceptual roadmap

Subharmonicity turns Δu into a positive measure. Riesz decomposition reverses that passage locally. Jensen and Poisson–Jensen compare interior mass with boundary averages. Logarithmic energy selects equilibrium measures and capacity, while Green functions and harmonic measure encode the same geometry. Herglotz representation is the upper-half-plane counterpart in which positivity of the imaginary part encodes a measure.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

13.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F13.1. Subharmonic function

A function u on a planar domain is *subharmonic* if it is upper semicontinuous and satisfies the sub-mean inequality on every compactly contained disk. When $u \in L^1_{\text{loc}}$, this is equivalent to Δu being a positive distribution.

Definition F13.2. Riesz measure

With the normalization used here, the *Riesz measure* of a subharmonic function is $\mu_u = (2\pi)^{-1} \Delta u$.

Definition F13.3. Logarithmic capacity

For compact $K \subset \mathbb{C}$, set $V_K = \inf_{\mu \in \mathcal{P}(K)} \iint \log |z - w|^{-1} d\mu(z) d\mu(w)$. The *logarithmic capacity* is $\text{cap}(K) = e^{-V_K}$, with capacity zero when $V_K = +\infty$.

Definition F13.4. Herglotz function

A *Herglotz (Nevanlinna) function* is a holomorphic map $m : \mathbb{C}_+ \rightarrow \mathbb{C}_+$. Its canonical representation has the form $m(z) = a + bz + \int_{\mathbb{R}} ((t - z)^{-1} - t/(1 + t^2)) d\mu(t)$, where $a \in \mathbb{R}$, $b \geq 0$, and $\int (1 + t^2)^{-1} d\mu(t) < \infty$.

13.2 Main results

Theorem F13.1. Riesz measure for subharmonic functions

Statement. Let $\Omega \subset \mathbb{C}$ be open and let $u : \Omega \rightarrow [-\infty, \infty)$ be subharmonic and not identically $-\infty$ on any connected component. Then $u \in L^1_{\text{loc}}(\Omega)$ and its distributional Laplacian $\langle \Delta u, \varphi \rangle := \int_{\Omega} u \Delta \varphi dA$ is a positive distribution. Hence there is a unique positive Radon measure ν_u such that $\langle \Delta u, \varphi \rangle = \int \varphi d\nu_u$ for all $\varphi \in C_c^\infty(\Omega)$. The normalized Riesz measure is $\mu_u = (2\pi)^{-1} \nu_u$. If $u \in C^2(\Omega)$, then $d\mu_u = (2\pi)^{-1} (\Delta u) dA$.

Proof. We prove the theorem in four steps.

Step 1: local integrability. It is enough to work on one connected component of Ω , so assume Ω is connected and $u \not\equiv -\infty$. Because u is upper semicontinuous, it is bounded above on every compact subset of Ω .

Let Ω_0 be the set of points having a neighborhood on which $|u|$ is integrable. This set is open by definition. It is nonempty: choose x_0 with $u(x_0) > -\infty$ and a disk $\overline{B}(x_0, r) \subset \Omega$. The disk-average sub-mean inequality (obtained by integrating the circle sub-mean

inequalities in the radius) gives

$$u(x_0) \leq \frac{1}{\pi r^2} \int_{B(x_0, r)} u \, dA. \quad (1)$$

The positive part of u is integrable on this disk because u is bounded above there. Since the right-hand side of (1) is not $-\infty$, the negative part is integrable as well. Hence $x_0 \in \Omega_0$.

We claim that $\Omega \setminus \Omega_0$ is also open. Let $x \notin \Omega_0$ and choose $r > 0$ with $\overline{B(x, 2r)} \subset \Omega$. If $y \in B(x, r)$, set

$$\rho := r - |x - y| > 0.$$

Then $B(x, \rho) \subset B(y, r)$. Since $x \notin \Omega_0$, $|u|$ is not integrable on $B(x, \rho)$, and therefore not on $B(y, r)$. On the other hand, u is bounded above on $\overline{B(y, r)}$. Thus

$$\int_{B(y, r)} u \, dA = -\infty.$$

The disk sub-mean inequality forces

$$u(y) = -\infty.$$

So $u \equiv -\infty$ on $B(x, r)$, and in particular no point of this ball belongs to Ω_0 . Hence the complement of Ω_0 is open.

The connected set Ω is therefore the disjoint union of the two open sets Ω_0 and $\Omega \setminus \Omega_0$. Since $\Omega_0 \neq \emptyset$, connectedness gives $\Omega_0 = \Omega$. Thus

$$u \in L^1_{\text{loc}}(\Omega). \quad (2)$$

The same argument applies componentwise.

Step 2: positivity of the distributional Laplacian. Let $\varphi \in C_c^\infty(\Omega)$ satisfy $\varphi \geq 0$. Choose $U \Subset \Omega$ containing $\text{supp } \varphi$. By (2), we may convolve u on a slightly larger relatively compact set with a standard nonnegative mollifier. For sufficiently small ε , write

$$u_\varepsilon := u * \chi_\varepsilon \quad \text{on } U.$$

Convolution with a nonnegative mollifier preserves the sub-mean inequality, so u_ε is smooth and subharmonic. Moreover,

$$u_\varepsilon \longrightarrow u \quad \text{in } L^1(U). \quad (3)$$

For a C^2 subharmonic function, the classical Laplacian is nonnegative. Hence, by integration by parts,

$$\int_U u_\varepsilon \Delta \varphi \, dA = \int_U \varphi \Delta u_\varepsilon \, dA \geq 0. \quad (4)$$

Because $\Delta \varphi$ is bounded and compactly supported, (3) allows us to pass to the limit in (4):

$$\langle \Delta u, \varphi \rangle := \int_\Omega u \Delta \varphi \, dA \geq 0. \quad (5)$$

Thus Δu is a positive distribution.

Step 3: a positive distribution is a Radon measure. We include the order-zero argument because it is the point where a measure, rather than a general distribution, enters. Let $K \Subset \Omega$. Choose $\psi \in C_c^\infty(\Omega)$ with $\psi \geq 1$ on K and $\psi \geq 0$. If $\varphi \in C_c^\infty(\Omega)$ is real-valued and supported in K , then pointwise

$$-\|\varphi\|_\infty \psi \leq \varphi \leq \|\varphi\|_\infty \psi.$$

Positivity of Δu therefore gives

$$|\langle \Delta u, \varphi \rangle| \leq \|\varphi\|_\infty \langle \Delta u, \psi \rangle. \quad (6)$$

Thus on every compact set the distribution is bounded in the uniform norm: it is locally of order zero. By uniform approximation of compactly supported continuous functions by smooth ones, (6) extends Δu uniquely to a positive linear functional on $C_c(\Omega)$. The Riesz–Markov representation theorem then gives a unique positive Radon measure ν_u such that

$$\langle \Delta u, \varphi \rangle = \int_\Omega \varphi d\nu_u \quad (\varphi \in C_c^\infty(\Omega)). \quad (7)$$

Uniqueness follows because smooth compactly supported functions are dense in C_c on each fixed compact support.

By the normalization fixed in *the definition of Riesz measure*, the Riesz measure is

$$\mu_u := \frac{1}{2\pi} \nu_u. \quad (8)$$

Step 4: the smooth case. If $u \in C^2(\Omega)$, ordinary integration by parts gives

$$\langle \Delta u, \varphi \rangle = \int_\Omega (\Delta u) \varphi dA.$$

Comparing with (7),

$$d\nu_u = (\Delta u) dA, \quad d\mu_u = \frac{1}{2\pi} (\Delta u) dA.$$

This proves all clauses of the theorem. □

Theorem F13.2. Riesz decomposition theorem

Statement. Let $\Omega \subset \mathbb{C}$ be open, let u be subharmonic on Ω and not identically $-\infty$ on any connected component, and let $G \Subset \Omega$ be open. With the normalized Riesz measure $\mu_u = (2\pi)^{-1} \Delta u$ from Theorem F13.1, define

$$p_G(z) := \int_G \log |z - w| d\mu_u(w), \quad z \in G,$$

where the integral is understood in the extended-valued potential-theoretic sense. Then $p_G \in L^1_{\text{loc}}(G)$ is subharmonic, $\Delta p_G = 2\pi \mu_u|_G$ distributionally, and there is a

harmonic function h_G on G such that

$$u(z) = p_G(z) + h_G(z) \quad (z \in G).$$

The harmonic term is uniquely determined once the potential p_G is fixed.

Proof. Let

$$\mu := \mu_u|_G.$$

Because $G \Subset \Omega$ and μ_u is a Radon measure, $\mu_u(\overline{G}) < \infty$; in particular, μ is finite. Define

$$p_G(z) = \int_G \log |z - w| d\mu(w). \quad (1)$$

We first justify the analytic meaning of this expression.

Step 1: local integrability of the logarithmic potential. Fix a compact set $K \Subset G$. Since G is bounded, there is a constant M such that

$$\log^+ |z - w| \leq M \quad (z \in K, w \in G). \quad (2)$$

For the negative part, choose $R_0 > 0$ with $G - K \subset B(0, R_0)$. Translation invariance of planar Lebesgue measure gives, uniformly in w ,

$$\int_K \log^- |z - w| dA(z) \leq \int_{B(0, R_0)} \log^- |\zeta| dA(\zeta) < \infty, \quad (3)$$

because in polar coordinates the possible singularity is controlled by $\int_0^1 r |\log r| dr < \infty$. Tonelli's theorem applied to the nonnegative function $\log^- |z - w|$ and finiteness of μ therefore show that

$$\int_K \int_G \log^- |z - w| d\mu(w) dA(z) < \infty. \quad (4)$$

Together with (2), this proves that p_G is finite almost everywhere and belongs to $L^1_{\text{loc}}(G)$. It may equal $-\infty$ at exceptional points (for example at atoms of μ), which is allowed for a subharmonic function.

Step 2: compute the distributional Laplacian. For fixed $w \in \mathbb{C}$, the planar fundamental-solution identity is

$$\Delta_z \log |z - w| = 2\pi \delta_w \quad (5)$$

in the sense of distributions. Let $\varphi \in C_c^\infty(G)$. Using (4) to justify Fubini on the compact support of φ and then (5), we obtain

$$\begin{aligned} \langle \Delta p_G, \varphi \rangle &= \int_G p_G(z) \Delta \varphi(z) dA(z) \\ &= \int_G \left(\int_G \log |z - w| \Delta \varphi(z) dA(z) \right) d\mu(w) \\ &= 2\pi \int_G \varphi(w) d\mu(w). \end{aligned} \quad (6)$$

Thus

$$\Delta p_G = 2\pi\mu = 2\pi\mu_u|_G = \Delta u|_G. \quad (7)$$

We also verify directly that the representative in (1) is subharmonic, rather than appealing to a converse distribution theorem. On every compact set of z -values the kernels $\log|z - w|$ are bounded above by one constant independent of $w \in G$. Hence the reverse-Fatou argument applied to that upper bound shows that p_G is upper semi-continuous. Moreover, for every closed disk $\overline{D(z_0, r)} \subset G$, the submean inequality for $z \mapsto \log|z - w|$ and Tonelli's theorem give

$$p_G(z_0) \leq \frac{1}{2\pi} \int_0^{2\pi} p_G(z_0 + re^{i\theta}) d\theta. \quad (7a)$$

Thus p_G is subharmonic. In particular, the potential used below is the pointwise potential-theoretic representative, not merely an L^1_{loc} equivalence class.

Step 3: a distribution with zero Laplacian has a harmonic representative. Set

$$v := u - p_G \in L^1_{\text{loc}}(G).$$

By (7), $\Delta v = 0$ distributionally. We recall the short Weyl-lemma argument needed here rather than hiding it as an additional black box. Let $B \Subset G$ be a disk and choose a standard mollifier ρ_ϵ with ϵ smaller than the distance from B to ∂G . Then

$$v_\epsilon := v * \rho_\epsilon$$

is smooth near B and

$$\Delta v_\epsilon = (\Delta v) * \rho_\epsilon = 0. \quad (8)$$

Hence every v_ϵ is harmonic on B . Moreover, $v_\epsilon \rightarrow v$ in $L^1(B)$ as $\epsilon \downarrow 0$.

Choose a smaller disk $B' \Subset B$. The mean-value property applied to $|h|$, which is subharmonic whenever h is harmonic, gives the standard interior estimate

$$\sup_{B'} |h| \leq C_{B', B} \|h\|_{L^1(B)} \quad (9)$$

for harmonic h on B . Applying (9) to $h = v_\epsilon - v_\delta$ shows that the family v_ϵ is uniformly Cauchy on B' because it is Cauchy in $L^1(B)$. Its uniform limit on B' is harmonic and agrees almost everywhere there with v . By covering G by such disks, the local harmonic limits agree on overlaps (they agree almost everywhere there), so they glue to a harmonic function h_G on G satisfying

$$u = p_G + h_G \quad \text{almost everywhere on } G. \quad (10)$$

Step 4: upgrade the identity to every point. Both u and $p_G + h_G$ are subharmonic on G , and they are equal almost everywhere by (10). A subharmonic function is the canonical upper-semicontinuous representative of its L^1_{loc} class: for each z its value is recovered as the limit of its averages over shrinking circles (or disks) at radii for which the averages are defined. Hence two subharmonic functions that agree almost

everywhere agree everywhere. Therefore

$$u(z) = p_G(z) + h_G(z) \quad \text{for every } z \in G. \quad (11)$$

Finally, once p_G is fixed by (1), the harmonic term is necessarily $h_G = u - p_G$, so it is unique. This completes the local Riesz decomposition with the normalization from *Riesz measure for subharmonic functions*. $\square$

Theorem F13.3. Subharmonic compactness / Hartogs-type lemma

Statement. Let $\Omega \subset \mathbb{C}$ be a domain and let (u_n) be subharmonic on Ω , locally uniformly bounded above: for every $K \Subset \Omega$, $\sup_n \sup_K u_n < \infty$. Then either the full sequence converges to $-\infty$ locally uniformly, or there is a subsequence u_{n_j} that converges in $L^1_{\text{loc}}(\Omega)$ to a subharmonic function u . Moreover, whenever subharmonic $u_n \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$, with u taken in its upper-semicontinuous subharmonic representative,

$$\limsup_{n \rightarrow \infty} \sup_K (u_n - \psi) \leq \sup_K (u - \psi)$$

for every compact $K \Subset \Omega$ and every continuous ψ on K . In particular $\limsup_n u_n(z) \leq u(z)$ for every z , equality holds almost everywhere, and if $\psi \geq u$ on K , then for every $\varepsilon > 0$ one has $u_n \leq \psi + \varepsilon$ on K for all sufficiently large n .

Proof. Let $\Omega \subset \mathbb{C}$ be a domain and let (u_n) be subharmonic on Ω . Assume throughout the compactness part that the family is locally uniformly bounded above:

$$\forall K \Subset \Omega \quad \sup_n \sup_K u_n < \infty. \quad (1)$$

We first prove the local estimate that prevents a normalized subharmonic sequence from developing uncontrolled negative mass.

Step 1: a disk estimate. Choose disks

$$K_0 \Subset D_0 \Subset D_1 \Subset \Omega, \quad (2)$$

where K_0 is a nonempty compact set. We claim that for every $A > 0$ there is a constant $C = C(K_0, D_0, D_1, A)$ with the following property. If v is subharmonic on a neighborhood of $\overline{D_1}$,

$$v \leq 0 \quad \text{on } D_1, \quad v(a) \geq -A \quad \text{for some } a \in K_0, \quad (3)$$

then

$$\int_{D_0} |v| dA \leq C, \quad \mu_v(D_0) \leq C, \quad (4)$$

where $\mu_v = (2\pi)^{-1} \Delta v$ is the Riesz measure from *Riesz measure for subharmonic functions*.

It is enough first to prove this when v is smooth. Let B be a disk satisfying

$$K_0 \cup \overline{D_0} \Subset B \Subset D_1 \quad (5)$$

and let $G_B(z, w)$ be the positive Green kernel of B , normalized by $\Delta_z G_B(z, w) = -2\pi\delta_w$. Green's formula gives

$$v(z) = H(z) - \int_B G_B(z, w) d\mu_v(w), \quad (6)$$

where H is harmonic on B and has the boundary values of v . Since $v \leq 0$, the maximum principle gives $H \leq 0$. Evaluating (6) at the point a in (3),

$$0 \leq \int_B G_B(a, w) d\mu_v(w) = H(a) - v(a) \leq A. \quad (7)$$

Because K_0 and $\overline{D_0}$ lie a positive distance from ∂B , there is $c > 0$, independent of $a \in K_0$, such that

$$G_B(a, w) \geq c \quad (w \in D_0). \quad (8)$$

Indeed, on the compact set $K_0 \times \overline{D_0}$ the Green function is positive and continuous away from the diagonal, while it tends to $+\infty$ on the diagonal; therefore its infimum is positive. Equations (7)–(8) give

$$\mu_v(D_0) \leq A/c. \quad (9)$$

We also need an L^1 bound. First, $-H$ is a nonnegative harmonic function and $-H(a) \leq A$ by (7) and $v(a) \geq -A$. Harnack's inequality, uniformly for $a \in K_0$, yields

$$\sup_{D_0}(-H) \leq C_1 A. \quad (10)$$

Next there is C_2 such that

$$\int_{D_0} G_B(z, w) dA(z) \leq C_2 G_B(a, w) \quad (a \in K_0, w \in B). \quad (11)$$

For completeness, split B into a compact inner disk $B' \ni K_0 \cup \overline{D_0}$ and its outer annulus. For $w \in B'$, the left side of (11) is uniformly bounded because the logarithmic singularity is integrable, whereas $G_B(a, w)$ has a positive uniform lower bound for $a \in K_0$ (and is infinite at $a = w$). For w in the outer annulus, $z \mapsto G_B(z, w)$ is positive harmonic on a fixed neighborhood of $K_0 \cup \overline{D_0}$; Harnack's inequality compares its values on D_0 with its value at any $a \in K_0$, uniformly. This proves (11). Integrating (6), then using (7), (10), and (11), gives

$$\int_{D_0} (-v) dA \leq C_1 A \text{ area}(D_0) + C_2 \int_B G_B(a, w) d\mu_v(w) \leq C_3 A. \quad (12)$$

Since $v \leq 0$, (12) is the desired L^1 estimate.

For a general subharmonic v , convolve on a slightly smaller disk with a standard nonnegative radial mollifier. The regularizations v_ε are smooth and subharmonic, satisfy $v_\varepsilon \leq 0$ where the convolution is taken inside D_1 , and by the submean inequality

$$v_\varepsilon(a) \geq v(a) \geq -A. \quad (13)$$

The smooth estimates are uniform in ε . Since subharmonic functions are locally integrable and convolution converges in L^1_{loc} , the first estimate in (4) passes to v . The

distributional Laplacians converge weakly, so the positive measure bound passes to the Riesz measure as well. This proves the disk estimate.

Step 2: local L^1 bounds unless the sequence collapses. Suppose (u_n) does not converge to $-\infty$ locally uniformly. Then there are a compact set $K_0 \Subset \Omega$, a number A , and a subsequence, still denoted (u_n) , such that

$$\sup_{K_0} u_n \geq -A. \quad (14)$$

Choose $a_n \in K_0$ with $u_n(a_n) \geq -A - 1$. After passing to a subsequence, $a_n \rightarrow a \in K_0$.

Take a small disk D about a with $\overline{D} \Subset \Omega$. The local upper bound (1) gives a number M with $u_n \leq M$ on a slightly larger disk. Apply Step 1 to $v_n = u_n - M$ and the compact set containing all large a_n . We obtain a uniform L^1 bound for (u_n) on D .

This bound propagates through overlapping disks. Indeed, if D' overlaps D in a set E of positive area, a uniform $L^1(E)$ bound and a uniform upper bound imply that for every n there is $b_n \in E$ with

$$u_n(b_n) \geq -C_E; \quad (15)$$

otherwise the negative part would have unbounded integral on E . Step 1, applied with the points b_n in a compact subset of the overlap, gives an L^1 bound on a smaller concentric part of D' . A finite chain of overlapping disks joins a to any prescribed compact subset of the connected domain Ω . Consequently,

$$\sup_n \|u_n\|_{L^1(K)} < \infty \quad (K \Subset \Omega). \quad (16)$$

The same argument gives locally uniform bounds for the Riesz masses

$$\sup_n \mu_{u_n}(K) < \infty \quad (K \Subset \Omega). \quad (17)$$

Step 3: L^1_{loc} precompactness. Fix disks

$$K \Subset G \Subset \Omega. \quad (18)$$

By (17) the restricted measures $\mu_{u_n}|_G$ have uniformly bounded mass on every slightly smaller disk. After passing to a subsequence, they converge weakly there to a positive Radon measure. Use *Riesz decomposition theorem* on a disk G_1 with $K \Subset G_1 \Subset G$:

$$u_n(z) = p_n(z) + h_n(z), \quad p_n(z) = \int_{G_1} \log |z - w| d\mu_{u_n}(w), \quad (19)$$

where h_n is harmonic on G_1 .

The logarithmic potentials are precompact in $L^1(K)$. To see this without hiding the logarithmic singularity, truncate the kernel:

$$k_\delta(z, w) := \max\{\log |z - w|, \log \delta\}. \quad (20)$$

On compact subsets the truncated kernel is bounded and continuous, so weak convergence of the measures gives convergence of the corresponding truncated potentials.

On the other hand,

$$\sup_w \int_K |\log |z - w| - k_\delta(z, w)| dA(z) \leq 2\pi \int_0^\delta r |\log r| dr \longrightarrow 0. \quad (21)$$

Multiplying by the uniform local mass bound in (17) shows that the truncation error is uniform in n . Hence a subsequence of (p_n) converges in $L^1(K)$.

By (16), the harmonic remainders $h_n = u_n - p_n$ are uniformly bounded in L^1 on a slightly larger compact disk. The mean-value formula and its differentiated versions therefore give uniform bounds for every derivative of h_n on K . Arzelà–Ascoli yields a locally uniformly convergent subsequence of (h_n) . Thus (u_n) has an $L^1(K)$ -convergent subsequence. A diagonal argument over a countable exhaustion of Ω gives

$$u_{n_j} \longrightarrow u \quad \text{in } L^1_{\text{loc}}(\Omega). \quad (22)$$

The limit has a subharmonic representative. Indeed, distributional convergence in (22) gives

$$\langle \Delta u, \varphi \rangle = \lim_j \langle \Delta u_{n_j}, \varphi \rangle \geq 0 \quad (0 \leq \varphi \in C_c^\infty(\Omega)). \quad (23)$$

Thus Δu is a positive distribution. Mollifying u gives smooth functions with nonnegative Laplacian. Their decreasing-radius disk means converge almost everywhere to u and define the usual upper-semicontinuous representative u^* ; the submean inequality passes to the limit, so u^* is subharmonic and $u^* = u$ almost everywhere. We henceforth write u for this representative.

We have proved the compactness dichotomy: if the original sequence does not converge to $-\infty$ locally uniformly, then it has an L^1_{loc} -convergent subsequence to a subharmonic function. This proves the first part of the theorem.

Step 4: Hartogs' compact-limsup inequality. Now suppose directly that

$$u_n \longrightarrow u \quad \text{in } L^1_{\text{loc}}(\Omega), \quad (24)$$

where all functions are subharmonic and u is represented by its upper-semicontinuous subharmonic version. We first prove the pointwise upper estimate

$$\limsup_{n \rightarrow \infty, z' \rightarrow z} u_n(z') \leq u(z). \quad (25)$$

If $z_n \rightarrow z$, choose $r > 0$ with $\overline{D(z, 2r)} \Subset \Omega$. For all large n , the disk $D(z_n, r)$ lies in $D(z, 2r)$. The submean inequality gives

$$u_n(z_n) \leq \frac{1}{\pi r^2} \int_{D(z_n, r)} u_n(\zeta) dA(\zeta). \quad (26)$$

Because of L^1 convergence and because the symmetric difference of $D(z_n, r)$ and $D(z, r)$ has area tending to zero while (u_n) is uniformly L^1 -integrable on $D(z, 2r)$, the right side has limsup at most

$$\frac{1}{\pi r^2} \int_{D(z, r)} u(\zeta) dA(\zeta). \quad (27)$$

Letting $r \downarrow 0$ gives (25), because disk means of a subharmonic function converge downward to its value.

Let $K \Subset \Omega$ be compact and let ψ be continuous on K . Put

$$S := \sup_K (u - \psi).$$

If the desired inequality failed, there would be $\varepsilon > 0$, a subsequence, and points $z_n \in K$ such that

$$u_n(z_n) - \psi(z_n) > S + \varepsilon. \quad (28)$$

After another subsequence $z_n \rightarrow z \in K$. Equation (25) and continuity of ψ give

$$\limsup_n (u_n(z_n) - \psi(z_n)) \leq u(z) - \psi(z) \leq S,$$

contradicting (28). Hence

$$\boxed{\limsup_{n \rightarrow \infty} \sup_K (u_n - \psi) \leq \sup_K (u - \psi).} \quad (29)$$

Taking $K = \{z\}$ gives $\limsup_n u_n(z) \leq u(z)$. If $\psi \geq u$ on K , then (29) says that for every $\varepsilon > 0$,

$$u_n \leq \psi + \varepsilon \quad \text{on } K \quad (30)$$

for all sufficiently large n .

Finally, equality $\limsup_n u_n = u$ holds almost everywhere. If on a set of positive area one had $\limsup_n u_n \leq u - \delta$ for some $\delta > 0$, then for some positive-measure subset and some N one would have $u_n \leq u - \delta/2$ for every $n \geq N$. That contradicts L^1 convergence in (24). Taking the union over rational $\delta > 0$ proves the almost-everywhere equality. The compactness and Hartogs claims are complete. $\square$

Theorem F13.4. Jensen formula

Statement. Let f be nonzero and holomorphic on a neighborhood of $\overline{D(0, r)}$, and assume f has no zero on $|z| = r$. Let $m = \text{ord}_0(f)$, write $f(z) = z^m g(z)$ with $g(0) \neq 0$, and list the nonzero zeros $a_1, \dots, a_N$ in $|z| < r$ with multiplicity. Then $\frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta = \log |g(0)| + m \log r + \sum_{j=1}^N \log(r/|a_j|)$. For $m = 0$ this is the standard Jensen formula.

Proof. Because f is holomorphic on a neighborhood of the compact disk $\overline{D(0, r)}$ and is not identically zero, it has only finitely many zeros in that closed disk. By hypothesis none lies on the boundary. Let $m = \text{ord}_0(f)$ and list the remaining zeros in $0 < |z| < r$, with multiplicity, as

$$a_1, \dots, a_N.$$

Factoring these zeros gives

$$f(z) = z^m \prod_{j=1}^N (z - a_j) h(z), \quad (1)$$

where h is holomorphic and has no zero on $\overline{D(0, r)}$. Consequently $\log |h|$ is harmonic on a neighborhood of the closed disk, and the harmonic mean-value property gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log |h(re^{i\theta})| d\theta = \log |h(0)|. \quad (2)$$

We next compute the contribution of one nonzero zero a with $|a| < r$. Write

$$re^{i\theta} - a = re^{i\theta} \left(1 - \frac{a}{r}e^{-i\theta}\right).$$

Thus

$$\log |re^{i\theta} - a| = \log r + \log \left|1 - \frac{a}{r}e^{-i\theta}\right|. \quad (3)$$

Since $|a|/r < 1$, the function

$$z \mapsto \log(1 - z) = - \sum_{n=1}^{\infty} \frac{z^n}{n}$$

can be taken with its analytic branch on the disk $|z| < 1$. Taking real parts at $z = (a/r)e^{-i\theta}$ and integrating term by term (the series converges uniformly on the circle $|z| = |a|/r < 1$), every nonconstant Fourier mode has mean zero. Hence

$$\frac{1}{2\pi} \int_0^{2\pi} \log \left|1 - \frac{a}{r}e^{-i\theta}\right| d\theta = 0.$$

Together with (3),

$$\frac{1}{2\pi} \int_0^{2\pi} \log |re^{i\theta} - a| d\theta = \log r. \quad (4)$$

The zero at the origin contributes even more directly:

$$\frac{1}{2\pi} \int_0^{2\pi} \log |(re^{i\theta})^m| d\theta = m \log r. \quad (5)$$

Taking logarithms of absolute values in (1), integrating around the boundary, and using (2), (4), and (5), we obtain

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta = m \log r + N \log r + \log |h(0)|. \quad (6)$$

Now write $f(z) = z^m g(z)$. From (1),

$$g(0) = \prod_{j=1}^N (-a_j) h(0),$$

so

$$\log |g(0)| = \sum_{j=1}^N \log |a_j| + \log |h(0)|. \quad (7)$$

Substituting (7) into (6) and regrouping terms gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta = \log |g(0)| + m \log r + \sum_{j=1}^N \log \frac{r}{|a_j|}.$$

If $m = 0$, then $g = f$ and $g(0) = f(0)$, which is the standard form of Jensen's formula. $\square$

Theorem F13.5. Poisson-Jensen formula

Statement. Let $R > 0$, and let $f \neq 0$ be meromorphic on a neighborhood of $\overline{D(0, R)}$, with no zero or pole on $|z| = R$. List the zeros $a_1, \dots, a_M$ and poles $b_1, \dots, b_N$ in $D(0, R)$ with multiplicity. For $z \in D(0, R)$ which is neither a zero nor a pole of f , set

$$P_R(z, \theta) = \frac{R^2 - |z|^2}{|Re^{i\theta} - z|^2}.$$

Then

$$\begin{aligned} \log |f(z)| &= \frac{1}{2\pi} \int_0^{2\pi} P_R(z, \theta) \log |f(Re^{i\theta})| d\theta \\ &\quad + \sum_{j=1}^M \log \left| \frac{R(z - a_j)}{R^2 - \bar{a}_j z} \right| - \sum_{\ell=1}^N \log \left| \frac{R(z - b_\ell)}{R^2 - \bar{b}_\ell z} \right|. \end{aligned}$$

At $z = 0$, whenever $f(0) \neq 0, \infty$, this reduces to the meromorphic Jensen formula.

Proof. For $a \in D(0, R)$ define the radius- R disk factor

$$B_a(\zeta) := \frac{R(\zeta - a)}{R^2 - \bar{a}\zeta}. \quad (1)$$

The denominator does not vanish in $\overline{D(0, R)}$, because its possible zero has modulus $R^2/|a| > R$. Thus B_a is holomorphic near the closed disk and has one simple zero at a . If $|\zeta| = R$, then

$$|R^2 - \bar{a}\zeta| = |\bar{\zeta}\zeta - \bar{a}\zeta| = |\zeta| |\zeta - a| = R|\zeta - a|,$$

so

$$|B_a(\zeta)| = 1 \quad (|\zeta| = R). \quad (2)$$

This elementary boundary identity is the reason for choosing the factor (1).

List zeros and poles with repetition according to multiplicity and set

$$g(\zeta) := f(\zeta) \prod_{j=1}^M B_{a_j}(\zeta)^{-1} \prod_{\ell=1}^N B_{b_\ell}(\zeta). \quad (3)$$

At each zero a_j of f , the corresponding reciprocal factors in (3) cancel exactly its zero order. At each pole b_ℓ , the corresponding factors B_{b_ℓ} cancel exactly its pole order. Hence all apparent singularities in (3) are removable, and g extends holomorphically to a neighborhood of $\overline{D(0, R)}$. By construction it has no zero in the closed disk.

Therefore

$$u(\zeta) := \log |g(\zeta)|$$

is harmonic on a neighborhood of the closed disk. The Poisson formula for a harmonic function on $D(0, R)$ gives, for every $z \in D(0, R)$,

$$\log |g(z)| = \frac{1}{2\pi} \int_0^{2\pi} P_R(z, \theta) \log |g(Re^{i\theta})| d\theta, \quad (4)$$

where

$$P_R(z, \theta) = \frac{R^2 - |z|^2}{|Re^{i\theta} - z|^2}.$$

On the boundary circle, (2) and (3) imply

$$|g(Re^{i\theta})| = |f(Re^{i\theta})|. \quad (5)$$

Since the hypothesis excludes zeros and poles of f on the boundary, the logarithm in (5) is finite and continuous there.

Now take logarithms of absolute values in (3). At a point z which is neither a zero nor a pole of f ,

$$\log |g(z)| = \log |f(z)| - \sum_{j=1}^M \log |B_{a_j}(z)| + \sum_{\ell=1}^N \log |B_{b_\ell}(z)|. \quad (6)$$

Insert (5) into (4), then solve (6) for $\log |f(z)|$. Using the explicit expression (1) for each factor gives

$$\begin{aligned} \log |f(z)| &= \frac{1}{2\pi} \int_0^{2\pi} P_R(z, \theta) \log |f(Re^{i\theta})| d\theta \\ &\quad + \sum_{j=1}^M \log \left| \frac{R(z - a_j)}{R^2 - \overline{a_j}z} \right| - \sum_{\ell=1}^N \log \left| \frac{R(z - b_\ell)}{R^2 - \overline{b_\ell}z} \right|, \end{aligned}$$

which is the asserted Poisson-Jensen formula.

Finally suppose $f(0) \neq 0, \infty$. Then 0 is absent from both divisor lists, $P_R(0, \theta) = 1$, and

$$|B_a(0)| = \frac{|a|}{R}.$$

Hence the formula at $z = 0$ becomes

$$\log |f(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta + \sum_{j=1}^M \log \frac{|a_j|}{R} - \sum_{\ell=1}^N \log \frac{|b_\ell|}{R},$$

the meromorphic Jensen formula with zeros and poles counted with multiplicity. $\square$

Theorem F13.6. Herglotz representation theorem

Statement. Let m be holomorphic on $\mathbb{C}_+$ and satisfy $\operatorname{Im} m(z) \geq 0$ for every $z \in \mathbb{C}_+$. Then there are unique $a \in \mathbb{R}$, $b \geq 0$, and a positive Borel measure σ on $\mathbb{R}$ satisfying

$$\int_{\mathbb{R}} \frac{d\sigma(t)}{1+t^2} < \infty$$

such that

$$m(z) = a + bz + \int_{\mathbb{R}} \left(\frac{1}{t-z} - \frac{t}{1+t^2} \right) d\sigma(t).$$

Conversely every such triple defines a holomorphic function with nonnegative imaginary part. Moreover, its imaginary part is strictly positive throughout $\mathbb{C}_+$ exactly when $b > 0$ or $\sigma \neq 0$; the remaining case $b = 0$, $\sigma = 0$ is the real constant $m \equiv a$.

Proof. We divide the argument into the disk representation and the conformal transfer to the upper half-plane.

Step 1: the Herglotz representation on the unit disk. Let F be holomorphic on the unit disk $\mathbb{D}$ and suppose

$$u := \operatorname{Re} F \geq 0.$$

For $0 < r < 1$, define a finite positive measure on the unit circle $\mathbb{T}$ by

$$d\mu_r(e^{i\theta}) := u(re^{i\theta}) \frac{d\theta}{2\pi}. \quad (1)$$

The mean-value property gives

$$\mu_r(\mathbb{T}) = u(0) \quad (2)$$

for every r . Thus the family (μ_r) has uniformly bounded mass. Choose a sequence $r_j \uparrow 1$. By weak-* compactness of finite measures on the compact space $\mathbb{T}$, after passing to a subsequence we may assume

$$\mu_{r_j} \xrightarrow{*} \mu \quad (3)$$

for a finite positive Borel measure μ on $\mathbb{T}$.

Fix $z \in \mathbb{D}$. For j sufficiently large, $|z| < r_j$; applying the Poisson formula to u on the disk of radius r_j yields

$$u(z) = \int_{\mathbb{T}} P_{z/r_j}(\xi) d\mu_{r_j}(\xi), \quad (4)$$

where

$$P_w(\xi) = \operatorname{Re} \frac{\xi + w}{\xi - w} = \frac{1 - |w|^2}{|\xi - w|^2}.$$

As $j \rightarrow \infty$, the functions P_{z/r_j} converge uniformly on $\mathbb{T}$ to P_z . Using (2) and weak-* convergence in (3), we may pass to the limit in (4) and obtain

$$u(z) = \int_{\mathbb{T}} P_z(\xi) d\mu(\xi). \quad (5)$$

Now define

$$H(z) := \int_{\mathbb{T}} \frac{\xi + z}{\xi - z} d\mu(\xi). \quad (6)$$

For z in a compact subset of $\mathbb{D}$, the integrand and all of its z -derivatives are uniformly bounded in ξ , so (6) defines a holomorphic function. By (5),

$$\operatorname{Re} H(z) = u(z) = \operatorname{Re} F(z).$$

Hence $F - H$ is holomorphic with identically zero real part. By the open mapping theorem (or the Cauchy-Riemann equations), it is a purely imaginary constant. Thus

$$F(z) = i\beta + \int_{\mathbb{T}} \frac{\xi + z}{\xi - z} d\mu(\xi), \quad \beta \in \mathbb{R}. \quad (7)$$

The pair (β, μ) is unique. Indeed, the real part determines all Poisson integrals of μ . Expanding for $|z| < 1$,

$$\frac{\xi + z}{\xi - z} = 1 + 2 \sum_{n \geq 1} z^n \bar{\xi}^n, \quad (8)$$

shows that these Poisson integrals determine all Fourier coefficients of μ , hence the finite measure itself. Evaluating the imaginary part of (7) at $z = 0$ then determines β .

Step 2: transfer the representation to $\mathbb{C}_+$. Let m be holomorphic on $\mathbb{C}_+$ with $\operatorname{Im} m \geq 0$. Use the Cayley maps

$$\zeta = \frac{z - i}{z + i}, \quad z = i \frac{1 + \zeta}{1 - \zeta}, \quad (9)$$

and define

$$F(\zeta) := -i m \left(i \frac{1 + \zeta}{1 - \zeta} \right). \quad (10)$$

Then F is holomorphic on $\mathbb{D}$ and $\operatorname{Re} F = \operatorname{Im} m \geq 0$. Apply (7) to obtain a unique $\beta \in \mathbb{R}$ and finite positive measure μ on $\mathbb{T}$.

Separate the possible atom at 1:

$$\mu = b \delta_1 + \mu_0, \quad b := \mu(\{1\}) \geq 0, \quad \mu_0(\{1\}) = 0. \quad (11)$$

The point 1 is exactly the boundary point corresponding to infinity under (9). Since

$$\frac{1 + \zeta}{1 - \zeta} = -iz,$$

the contribution of $b\delta_1$ to $m = iF$ is precisely

$$ib \frac{1 + \zeta}{1 - \zeta} = bz. \quad (12)$$

Every $\xi \in \mathbb{T} \setminus \{1\}$ can be written uniquely as

$$\xi = \frac{t - i}{t + i} \quad (t \in \mathbb{R}). \quad (13)$$

Push μ_0 forward under this correspondence to a finite positive measure ν on $\mathbb{R}$. A direct substitution of (9) and (13) gives

$$i \frac{\xi + \zeta}{\xi - \zeta} = \frac{1 + tz}{t - z} = (1 + t^2) \left(\frac{1}{t - z} - \frac{t}{1 + t^2} \right). \quad (14)$$

Define the positive Borel measure σ by

$$d\sigma(t) := (1 + t^2) d\nu(t). \quad (15)$$

Since ν is finite,

$$\int_{\mathbb{R}} \frac{d\sigma(t)}{1 + t^2} = \nu(\mathbb{R}) < \infty. \quad (16)$$

Multiplying (7) by i and using (11)–(15) now yields

$$m(z) = a + bz + \int_{\mathbb{R}} \left(\frac{1}{t - z} - \frac{t}{1 + t^2} \right) d\sigma(t), \quad (17)$$

where $a := -\beta \in \mathbb{R}$. Uniqueness of a, b, σ follows from uniqueness of β, μ : the mass at 1 gives b , and the Cayley correspondence (13) together with (15) is one-to-one for the remaining measure.

Step 3: converse and strict positivity. Conversely, suppose $a \in \mathbb{R}$, $b \geq 0$, and σ is a positive Borel measure satisfying (16). On compact subsets of $\mathbb{C}_+$,

$$\frac{1}{t - z} - \frac{t}{1 + t^2} = \frac{1 + tz}{(t - z)(1 + t^2)} \quad (18)$$

is bounded in absolute value by a compact-dependent constant times $(1 + t^2)^{-1}$; the same is true after one or more z -derivatives. Thus the integral in (17) converges locally uniformly and defines a holomorphic function.

Writing $z = x + iy$ with $y > 0$, the subtraction term in (17) is real and

$$\operatorname{Im} \frac{1}{t - z} = \frac{y}{(t - x)^2 + y^2}.$$

Therefore

$$\operatorname{Im} m(z) = by + \int_{\mathbb{R}} \frac{y}{(t - x)^2 + y^2} d\sigma(t) \geq 0. \quad (19)$$

If $b > 0$ or $\sigma \neq 0$, the right-hand side is strictly positive for every $z \in \mathbb{C}_+$, because its integrand is strictly positive. If $b = 0$ and $\sigma = 0$, then (17) reduces to the real constant $m \equiv a$. This proves both the converse and the exact strictness criterion. $\square$

Theorem F13.7. Logarithmic equilibrium measure and capacity

Statement. Let $K \subset \mathbb{C}$ be compact and nonpolar, equivalently $V_K < +\infty$ in Definition F13.3. Then there is a unique Borel probability measure $\mu_K \in \mathcal{P}(K)$ such that

$$I(\mu_K) := \iint \log \frac{1}{|z - w|} d\mu_K(z) d\mu_K(w) = V_K.$$

This μ_K is the logarithmic equilibrium measure of K , and

$$\text{cap}(K) = e^{-V_K} = e^{-I(\mu_K)} > 0.$$

Proof. Put

$$k(z, w) := \log \frac{1}{|z - w|} \in (-\infty, +\infty], \quad k(z, z) := +\infty.$$

Because K is compact and nonpolar, it contains at least two points; in particular its diameter $D := \text{diam } K$ is positive. On $K \times K$,

$$k(z, w) \geq -\log D, \quad (1)$$

so the logarithmic energy of a probability measure can never be $-\infty$.

Step 1: lower semicontinuity of the energy. For an integer $M > -\log D$, set

$$k_M(z, w) := \min \left\{ \log \frac{1}{|z - w|}, M \right\}. \quad (2)$$

Equivalently, $k_M(z, w) = -\log \max\{|z - w|, e^{-M}\}$. Thus k_M is a bounded continuous function on $K \times K$, and $k_M \uparrow k$ pointwise as $M \rightarrow \infty$. Hence, for every $\mu \in \mathcal{P}(K)$, monotone convergence applied after adding the harmless constant $\log D$ gives

$$I(\mu) = \iint k \, d\mu \, d\mu = \sup_M \iint k_M \, d\mu \, d\mu. \quad (3)$$

If $\mu_j \rightarrow \mu$ weak-* in $\mathcal{P}(K)$, then $\mu_j \otimes \mu_j \rightarrow \mu \otimes \mu$ weak-* on $K \times K$. Therefore, for every fixed M ,

$$\iint k_M \, d\mu_j \, d\mu_j \longrightarrow \iint k_M \, d\mu \, d\mu.$$

Taking first $\liminf_j$ and then the supremum over M in (3) yields

$$I(\mu) \leq \liminf_{j \rightarrow \infty} I(\mu_j). \quad (4)$$

Thus I is weak-* lower semicontinuous on $\mathcal{P}(K)$.

Step 2: existence of a minimizer. The space $\mathcal{P}(K)$ is weak-* compact. Since K is nonpolar, $V_K < +\infty$, so there is a sequence $(\mu_j) \subset \mathcal{P}(K)$ with $I(\mu_j) \downarrow V_K$. Passing to a weak-* convergent subsequence, say $\mu_j \rightarrow \mu_K$, and using (4),

$$I(\mu_K) \leq \liminf_j I(\mu_j) = V_K.$$

The reverse inequality $I(\mu_K) \geq V_K$ follows from the definition of V_K . Consequently

$$I(\mu_K) = V_K < +\infty. \quad (5)$$

So an equilibrium measure exists.

Step 3: conditional strict positivity of logarithmic energy. We record the precise

fact needed for uniqueness. If σ is a compactly supported finite signed measure on $\mathbb{C} \cong \mathbb{R}^2$ with total mass $\sigma(\mathbb{C}) = 0$, and its logarithmic energy is finite in the polarization sense, then

$$I(\sigma) := \iint \log \frac{1}{|z-w|} d\sigma(z)d\sigma(w) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \frac{|\widehat{\sigma}(\xi)|^2}{|\xi|^2} d\xi \geq 0. \quad (6)$$

Moreover equality holds only for $\sigma = 0$.

For completeness, here is the standard justification of (6). We use the Fourier convention $\widehat{f}(\xi) = \int e^{-ix \cdot \xi} f(x) dx$. In distributions on $\mathbb{R}^2$,

$$\Delta \left(\log \frac{1}{|x|} \right) = -2\pi \delta_0. \quad (7)$$

Consequently the Fourier transform of $\log(1/|x|)$, modulo a distribution supported at $\xi = 0$ (which represents the arbitrary additive constant of the logarithmic kernel), is $2\pi|\xi|^{-2}$. When paired against $\sigma \otimes \sigma$, that zero-frequency term disappears because $\widehat{\sigma}(0) = \sigma(\mathbb{C}) = 0$. Regularizing σ by a smooth compactly supported approximate identity and truncating $|\xi|^{-2}$ first to $\varepsilon \leq |\xi| \leq R$ makes every operation an ordinary Fubini/Plancherel calculation. Letting $\varepsilon \downarrow 0$ and $R \uparrow \infty$ gives (6) by monotone convergence; near $\xi = 0$ there is no singularity because compact support and zero total mass imply $\widehat{\sigma}(\xi) = O(|\xi|)$. This same regularization is the usual polarization definition when the two positive parts have finite logarithmic energy.

The right-hand side of (6) is nonnegative. If it is zero, then $\widehat{\sigma}(\xi) = 0$ for almost every $\xi \neq 0$. The Fourier transform of a finite measure is continuous and also vanishes at $\xi = 0$, hence it is identically zero. Uniqueness of the Fourier transform of finite measures gives $\sigma = 0$. This proves the asserted strictness.

Step 4: uniqueness of the minimizer. Suppose μ and ν are two equilibrium measures. Both have the finite energy V_K by (5). The usual truncated-kernel polarization (which is legitimate here because the two self-energies are finite and the kernel is bounded below on $K \times K$) gives, for $\sigma := \mu - \nu$,

$$I\left(\frac{\mu + \nu}{2}\right) = \frac{I(\mu) + I(\nu)}{2} - \frac{1}{4}I(\sigma) = V_K - \frac{1}{4}I(\sigma). \quad (8)$$

The left-hand side is the energy of a probability measure on K , so it is at least V_K . Equation (8) therefore implies $I(\sigma) \leq 0$. By (6), $I(\sigma) \geq 0$. Thus $I(\sigma) = 0$, and the strictness statement following (6) yields $\sigma = 0$, i.e. $\mu = \nu$. The equilibrium measure is unique.

Step 5: capacity. Finally, the definition of logarithmic capacity defines $\text{cap}(K) = e^{-V_K}$. Combining this definition with (5) gives

$$\text{cap}(K) = e^{-V_K} = e^{-I(\mu_K)}.$$

Since $V_K < +\infty$ for a nonpolar compact set, this number is strictly positive, as claimed. $\square$

Theorem F13.8. Green function with pole at infinity

Statement. Let $K \subset \mathbb{C}$ be compact and nonpolar, let μ_K be its equilibrium measure, let $V_K = -\log \text{cap}(K)$, and let Ω_K be the unbounded component of $\widehat{\mathbb{C}} \setminus K$. Put

$$U_K(z) := \int_K \log \frac{1}{|z-w|} d\mu_K(w).$$

Assume the explicit regularity condition that U_K is finite and continuous on $\overline{\Omega_K} \cap \mathbb{C}$ and satisfies $U_K = V_K$ on $\partial\Omega_K$ (the standard potential-theoretic regularity condition equivalent to boundary regularity for the Green function at infinity). Then

$$g_K(z) := V_K - U_K(z), \quad z \in \Omega_K,$$

is positive and harmonic on Ω_K , extends continuously to $\overline{\Omega_K} \cap \mathbb{C}$ with boundary value 0, and satisfies

$$g_K(z) = \log |z| - \log \text{cap}(K) + o(1) \quad (z \rightarrow \infty).$$

These properties characterize g_K uniquely.

Scope note

The positivity assertion is made on the unbounded component Ω_K ; bounded complementary components are outside this statement.

Proof. Let

$$U_K(z) = \int_K \log \frac{1}{|z-w|} d\mu_K(w), \quad g_K(z) = V_K - U_K(z), \quad z \in \Omega_K. \quad (1)$$

By *Logarithmic equilibrium measure and capacity*, μ_K is a probability measure supported on the compact set K , and

$$V_K = -\log \text{cap}(K) < \infty. \quad (2)$$

We verify the asserted properties one at a time.

Step 1: harmonicity on the unbounded component. Fix a compact set $L \Subset \Omega_K$. Its distance from K is positive:

$$\rho := \text{dist}(L, K) > 0.$$

For each fixed $w \in K$, the function $z \mapsto \log(1/|z-w|)$ is harmonic on a neighborhood of L . All of its derivatives of any fixed order are uniformly bounded on $L \times K$ by constants depending on ρ . Differentiation may therefore be passed under the integral in (1). In particular,

$$\Delta U_K(z) = \int_K \Delta_z \log \frac{1}{|z-w|} d\mu_K(w) = 0, \quad z \in L.$$

Since $L \Subset \Omega_K$ was arbitrary, U_K , and hence g_K , is harmonic on Ω_K .

Step 2: boundary values. The regularity hypothesis says exactly that U_K is finite and

continuous on $\overline{\Omega_K} \cap \mathbb{C}$ and

$$U_K(\zeta) = V_K \quad (\zeta \in \partial\Omega_K). \quad (3)$$

Consequently (1) extends continuously to every finite boundary point and

$$g_K(\zeta) = 0 \quad (\zeta \in \partial\Omega_K). \quad (4)$$

No assertion about a bounded component of $\mathbb{C} \setminus K$ is needed or used.

Step 3: normalization at infinity. Let $R_K := \sup_{w \in K} |w|$. For $|z| > 2R_K$, uniformly in $w \in K$,

$$\log \frac{1}{|z - w|} = -\log |z| - \log \left| 1 - \frac{w}{z} \right|. \quad (5)$$

The second term tends uniformly to 0 as $|z| \rightarrow \infty$. Since μ_K has total mass 1, integrating (5) yields

$$U_K(z) = -\log |z| + o(1). \quad (6)$$

Using (2),

$$\begin{aligned} g_K(z) &= V_K - U_K(z) \\ &= \log |z| + V_K + o(1) \\ &= \log |z| - \log \text{cap}(K) + o(1). \end{aligned} \quad (7)$$

This is the required pole-at-infinity normalization.

Step 4: positivity. Equation (7) implies that there is R_0 so large that

$$g_K(z) > 0 \quad \text{whenever } z \in \Omega_K, \quad |z| = R \geq R_0. \quad (8)$$

Fix $z_0 \in \Omega_K$, and choose $R > \max\{R_0, |z_0|\}$. Let D_R be the component of $\Omega_K \cap \{|z| < R\}$ containing z_0 . The function g_K is harmonic on D_R and continuous on its boundary. On the portion of ∂D_R contained in $\partial\Omega_K$, it is 0 by (4); on the portion contained in $|z| = R$, it is positive by (8). The minimum principle therefore gives

$$g_K \geq 0 \quad \text{on } D_R. \quad (9)$$

Since g_K is not identically zero on Ω_K by the asymptotic (7), the strong minimum principle gives

$$g_K(z) > 0 \quad (z \in \Omega_K). \quad (10)$$

Step 5: uniqueness. Suppose h is another function on Ω_K which is harmonic and positive there, extends continuously to the finite boundary with value 0, and satisfies the same normalization

$$h(z) = \log |z| - \log \text{cap}(K) + o(1) \quad (z \rightarrow \infty). \quad (11)$$

Set

$$q = h - g_K.$$

Then q is harmonic on Ω_K , continuous with zero finite boundary values, and

$$q(z) = o(1) \quad (z \rightarrow \infty). \quad (12)$$

Given $\varepsilon > 0$, choose R so large that $|q(z)| < \varepsilon$ whenever $z \in \Omega_K$ and $|z| = R$. Applying the maximum principle to q and $-q$ on each bounded component of $\Omega_K \cap \{|z| < R\}$ that meets a fixed compact subset of Ω_K , and using the zero boundary values on $\partial\Omega_K$, gives

$$|q| \leq \varepsilon$$

on that compact subset. Since ε is arbitrary, $q \equiv 0$. Thus $h = g_K$, proving uniqueness. $\square$

Proposition F13.9. Harmonic measure and conformal invariance

Statement. Let $D, D' \subset \mathbb{C}$ be bounded domains regular for the continuous Dirichlet problem, and let $\phi : D \rightarrow D'$ be conformal and extend to a homeomorphism $\overline{D} \rightarrow \overline{D}'$. For $z \in D$, let ω_D^z be harmonic measure, characterized by $H_g(z) = \int_{\partial D} g d\omega_D^z$ for every $g \in C(\partial D)$, where H_g is the harmonic extension of g . Then $\phi_*\omega_D^z = \omega_{D'}^{\phi(z)}$; equivalently, $\omega_D^z(E) = \omega_{D'}^{\phi(z)}(\phi(E))$ for every Borel $E \subset \partial D$.

Proof. We first recall why the representing measure exists in the stated setting. Fix $z \in D$. For $g \in C(\partial D)$ let H_g denote the unique harmonic function on D , continuous on $\overline{D}$, with boundary values g . Define

$$L_z(g) := H_g(z).$$

The maximum principle shows that L_z is a positive linear functional on $C(\partial D)$ and

$$|L_z(g)| \leq \|g\|_\infty.$$

Moreover $L_z(1) = 1$. By the Riesz–Markov representation theorem there is therefore a unique Borel probability measure ω_D^z on ∂D such that

$$H_g(z) = \int_{\partial D} g d\omega_D^z \quad (g \in C(\partial D)). \quad (1)$$

This is harmonic measure at z .

Now let $\phi : D \rightarrow D'$ be as in the statement and let $g \in C(\partial D')$. Let H'_g be the harmonic extension of g to D' . Because conformal maps preserve harmonicity in the plane,

$$H'_g \circ \phi$$

is harmonic on D . Since ϕ extends homeomorphically to the boundaries, its boundary values are $g \circ \phi$ on ∂D . Uniqueness of the continuous Dirichlet problem on D therefore gives

$$H_{g \circ \phi} = H'_g \circ \phi. \quad (2)$$

Evaluate (2) at z and use (1) in the two domains:

$$\begin{aligned}
 \int_{\partial D} g(\phi(\xi)) d\omega_D^z(\xi) &= H_{g \circ \phi}(z) \\
 &= H'_g(\phi(z)) \\
 &= \int_{\partial D'} g(\eta) d\omega_{D'}^{\phi(z)}(\eta).
 \end{aligned} \tag{3}$$

The left-hand side of (3) is, by definition, the integral of g against the pushforward measure $\phi_*\omega_D^z$. Since (3) holds for every continuous boundary function g , uniqueness in Riesz–Markov implies

$$\boxed{\phi_*\omega_D^z = \omega_{D'}^{\phi(z)}}. \tag{4}$$

Applying equality of the two measures in (4) to Borel subsets of $\partial D'$ gives the equivalent set formula stated in the theorem. $\square$

13.3 Worked examples

Worked example: Potential, Subharmonic and Herglotz Theory: a model computation

Take $u(z) = \log |z|$. It is harmonic on $\mathbb{C} \setminus \{0\}$ and, in the distributional sense,

$$\Delta \log |z| = 2\pi\delta_0.$$

With the normalization $\mu_u = (2\pi)^{-1}\Delta u$, the Riesz measure is therefore δ_0 .

Now let m be normalized arclength on the unit circle. Its logarithmic potential is

$$U^m(z) = \int_{|w|=1} \log \frac{1}{|z-w|} dm(w).$$

Jensen's formula gives

$$U^m(z) = \begin{cases} 0, & |z| \leq 1, \\ -\log |z|, & |z| \geq 1. \end{cases}$$

Thus the equilibrium energy of the unit circle is 0, so

$$\text{cap}(\mathbb{T}) = e^0 = 1.$$

The Green function of the exterior domain with pole at infinity is correspondingly $g(z) = \log |z|$ for $|z| > 1$, vanishing continuously at the circle.

Boundary case: what fails at the edge of the hypotheses

For the singleton $K = \{0\}$, the only probability measure supported on K is δ_0 . Its

logarithmic self-energy contains

$$\log \frac{1}{|0-0|} = +\infty,$$

so $V_K = +\infty$ and

$$\text{cap}(\{0\}) = e^{-\infty} = 0.$$

A point is therefore polar in planar logarithmic potential theory. This shows why capacity, rather than ordinary length or area, is the right smallness notion for several later exceptional sets.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

13.4 Chapter summary

What to remember

- The Laplacian of a subharmonic function is a positive Riesz measure.
- Jensen-type formulae convert analytic zeros into integral identities.
- Capacity is defined through minimum logarithmic energy and detects polar sets.
- Herglotz functions encode positive measures through analyticity and positivity of the imaginary part.

Looking ahead

F14 turns from potential theory to entropy and transfer operators. Both chapters share a variational viewpoint: one optimizes an energy or pressure functional and studies the measure that realizes the optimum.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

13.5 Exercises and problems

Exercise 13.1

Difficulty. Core.

Related results. Riesz measure for subharmonic functions; Riesz decomposition theorem; Jensen formula.

Problem. Work through the worked example 'u(z)=log|z| and the equilibrium measure of the unit circle'. For Riesz measure for subharmonic functions, Riesz decomposition theorem, and Jensen formula, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 13.2

Difficulty. Core.

Related results. Poisson–Jensen formula; Herglotz representation theorem; Logarithmic equilibrium measure and capacity.

Problem. Analyze the boundary example 'a polar set/capacity-zero example'. For Poisson–Jensen formula, Herglotz representation theorem, and Logarithmic equilibrium measure and capacity, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 13.3

Difficulty. Advanced.

Related results. Jensen formula.

Problem. For Jensen formula, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 13.4

Difficulty. Advanced.

Related results. Poisson–Jensen formula.

Problem. Stress-test Poisson–Jensen formula. Choose one genuinely load-bearing

hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 13.5

Difficulty. Advanced.

Related results. Riesz measure for subharmonic functions; Riesz decomposition theorem.

Problem. Synthesis problem. Use Riesz measure for subharmonic functions and Riesz decomposition theorem to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 13.6

Difficulty. Core.

Related results. Green function with pole at infinity.

Problem. Work directly with Green function with pole at infinity. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 13.7

Difficulty. Core.

Related results. Harmonic measure and conformal invariance.

Problem. Work directly with Harmonic measure and conformal invariance. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 13.8

Difficulty. Core.

Related results. Subharmonic compactness / Hartogs-type lemma.

Problem. Work directly with Subharmonic compactness / Hartogs-type lemma. Restate its hypotheses in the notation of this volume, construct the simplest non-trivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Symbolic Dynamics and Thermodynamic Formalism

Why this chapter matters

Symbolic dynamics replaces geometry by sequences of symbols, often without losing the essential orbit structure. Thermodynamic formalism then assigns weights to orbit segments and studies the exponential growth of their partition sums. Pressure unifies entropy and energy; transfer operators turn the variational theory into spectral analysis.

This chapter is useful because complicated hyperbolic systems can often be coded by subshifts of finite type. A theorem proved symbolically can then be transferred back to geometry.

Prerequisites. F01–F02, basic functional analysis, finite-state Markov chains, and elementary spectral theory.

Learning goals

- Compute pressure and equilibrium states in finite symbolic models.
- Use the Ruelle transfer operator and its spectral gap.
- Understand the variational principle and Gibbs property.
- Derive statistical consequences such as decay of correlations and the central limit theorem from spectral perturbation.

Conceptual roadmap

Subshifts of finite type provide the model. Pressure is defined dynamically and characterized variationally. The Ruelle–Perron–Frobenius theorem gives a leading eigenvalue/eigenfunction/eigenmeasure for Hölder potentials. This produces the unique Gibbs equilibrium state in the mixing case. Markov coding transports the conclusions to basic hyperbolic sets.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

14.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F14.1. Subshift of finite type

Given a finite 0-1 matrix A , the associated *subshift of finite type* consists of bi-infinite symbol sequences (x_n) with $A_{x_n x_{n+1}} = 1$ for every n , together with the left shift σ .

Definition F14.2. Topological pressure

For continuous $T : X \rightarrow X$ on compact metric X and $\phi \in C(X)$, write $S_n \phi(x) = \sum_{j=0}^{n-1} \phi(T^j x)$ and $d_n(x, y) = \max_{0 \leq j < n} d(T^j x, T^j y)$. The topological pressure $P_T(\phi)$ is the exponential growth rate of the weighted sums $\sum_{x \in E} e^{S_n \phi(x)}$ over maximal (n, ε) -separated sets, followed by $\varepsilon \downarrow 0$.

Definition F14.3. Transfer operator

For a finite-to-one system and potential ϕ , the *Ruelle transfer operator* is $(\mathcal{L}_\phi g)(x) = \sum_{Ty=x} e^{\phi(y)} g(y)$ on the function space specified by the theorem.

Definition F14.4. Equilibrium and Gibbs states

An invariant probability μ is an *equilibrium state* for ϕ if it attains $h_\mu(T) + \int \phi d\mu = P_T(\phi)$. A Gibbs state additionally satisfies the stated exponential cylinder or Bowen-ball weight estimates.

Definition F14.5. Basic hyperbolic set

A nonempty compact invariant hyperbolic set Λ is *basic* if it is locally maximal and $f|_\Lambda$ is topologically transitive.

Definition F14.6. Rectangle and Markov partition

At a local-product scale, a *rectangle* $R \subset \Lambda$ is closed under the bracket operation $[x, y] = W_{\text{loc}}^u(x) \cap W_{\text{loc}}^s(y)$. A finite family of proper rectangles is a *Markov partition* when stable and unstable plaques map across rectangles according to the Markov inclusion rules stated in the coding theorem.

14.2 Main results

Theorem F14.1. Entropy/generating-partition interface

Statement. Let $(X, \mathcal{B}, \mu, T)$ be a probability-preserving system and let $\mathcal{P}$ be a finite or countable measurable partition with $H_\mu(\mathcal{P}) < \infty$. If T is invertible modulo μ and

$$\sigma\left(\bigvee_{n \in \mathbb{Z}} T^{-n}\mathcal{P}\right) = \mathcal{B} \quad \text{modulo } \mu\text{-null sets,}$$

then $h_\mu(T) = h_\mu(T, \mathcal{P})$. Without invertibility, the same conclusion holds under the forward-generator hypothesis

$$\sigma\left(\bigvee_{n \geq 0} T^{-n}\mathcal{P}\right) = \mathcal{B} \quad \text{modulo } \mu\text{-null sets.}$$

Here $h_\mu(T)$ is the supremum of $h_\mu(T, \mathcal{Q})$ over finite measurable partitions $\mathcal{Q}$.

Proof. We use the standard notation

$$\mathcal{A}_m^n := \bigvee_{j=m}^n T^{-j}\mathcal{A} \quad (m \leq n), \quad (1)$$

whenever the indicated iterates are defined. For a finite-entropy partition $\mathcal{A}$,

$$h_\mu(T, \mathcal{A}) := \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu(\mathcal{A}_0^{n-1}). \quad (2)$$

The limit exists by subadditivity. The system entropy is

$$h_\mu(T) := \sup_{\mathcal{Q} \text{ finite}} h_\mu(T, \mathcal{Q}). \quad (3)$$

We first prove the comparison mechanism used in both versions of the theorem.

Step 1: comparison with an approximating partition. Let $\mathcal{Q}$ and $\mathcal{R}$ have finite Shannon entropy. The chain rule and conditioning-reduces-entropy give

$$\begin{aligned} H_\mu(\mathcal{Q}_0^{n-1}) &\leq H_\mu(\mathcal{R}_0^{n-1}) + H_\mu(\mathcal{Q}_0^{n-1} \mid \mathcal{R}_0^{n-1}) \\ &\leq H_\mu(\mathcal{R}_0^{n-1}) + \sum_{j=0}^{n-1} H_\mu(T^{-j}\mathcal{Q} \mid T^{-j}\mathcal{R}) \\ &= H_\mu(\mathcal{R}_0^{n-1}) + nH_\mu(\mathcal{Q} \mid \mathcal{R}), \end{aligned} \quad (4)$$

where the last equality uses measure preservation. Dividing by n and letting $n \rightarrow \infty$ yields

$$\boxed{h_\mu(T, \mathcal{Q}) \leq h_\mu(T, \mathcal{R}) + H_\mu(\mathcal{Q} \mid \mathcal{R}).} \quad (5)$$

The same inequality is valid when $\mathcal{Q}$ is countable of finite entropy, by applying the

chain rule first to finite coarsenings and passing to the limit.

Step 2: finite windows of $\mathcal{P}$ have the same entropy rate. First suppose T is invertible. Put

$$\mathcal{R}_N := \mathcal{P}_{-N}^N. \quad (6)$$

Since $\mathcal{P} \preceq \mathcal{R}_N$,

$$h_\mu(T, \mathcal{P}) \leq h_\mu(T, \mathcal{R}_N). \quad (7)$$

On the other hand,

$$(\mathcal{R}_N)_0^{n-1} = \mathcal{P}_{-N}^{n-1+N}. \quad (8)$$

By invariance and the elementary bound $H_\mu(\mathcal{A} \vee \mathcal{B}) \leq H_\mu(\mathcal{A}) + H_\mu(\mathcal{B})$,

$$H_\mu(\mathcal{P}_{-N}^{n-1+N}) \leq H_\mu(\mathcal{P}_0^{n-1}) + 2NH_\mu(\mathcal{P}). \quad (9)$$

After division by n , the boundary term disappears. Hence

$$\boxed{h_\mu(T, \mathcal{R}_N) = h_\mu(T, \mathcal{P}).} \quad (10)$$

If T is not assumed invertible, take instead

$$\mathcal{R}_N := \mathcal{P}_0^N. \quad (11)$$

Then $(\mathcal{R}_N)_0^{n-1} = \mathcal{P}_0^{n-1+N}$, and exactly the same argument, now with only N terminal boundary iterates, proves (10).

Step 3: the generator makes finite windows approximate every finite partition. Let $\mathcal{Q} = \{Q_1, \dots, Q_s\}$ be a finite measurable partition. In the invertible case, the generator hypothesis says

$$\sigma \left(\bigvee_{j \in \mathbb{Z}} T^{-j} \mathcal{P} \right) = \mathcal{B} \quad \text{mod } \mu, \quad (12)$$

so the increasing sigma-algebras $\mathcal{G}_N := \sigma(\mathcal{R}_N)$ generate $\mathcal{B}$ modulo null sets. In the noninvertible case the same conclusion follows from the forward-generator hypothesis with (11).

For each atom Q_i , martingale convergence gives

$$p_{i,N} := \mathbb{E}_\mu(\mathbf{1}_{Q_i} \mid \mathcal{G}_N) \longrightarrow \mathbf{1}_{Q_i} \quad \text{a.e. and in } L^1. \quad (13)$$

Using $\eta(t) = -t \log t$ with $\eta(0) = 0$, the conditional entropy can be written

$$H_\mu(\mathcal{Q} \mid \mathcal{R}_N) = \int_X \sum_{i=1}^s \eta(p_{i,N}) d\mu. \quad (14)$$

The integrand is bounded by s/e , while (13) implies $\eta(p_{i,N}) \rightarrow \eta(\mathbf{1}_{Q_i}) = 0$ almost everywhere. Dominated convergence therefore gives

$$\boxed{H_\mu(\mathcal{Q} \mid \mathcal{R}_N) \longrightarrow 0.} \quad (15)$$

Combining (5), (10), and (15),

$$h_\mu(T, \mathcal{Q}) \leq h_\mu(T, \mathcal{P}) \quad \text{for every finite } \mathcal{Q}. \quad (16)$$

Taking the supremum over finite $\mathcal{Q}$ yields

$$h_\mu(T) \leq h_\mu(T, \mathcal{P}). \quad (17)$$

Step 4: reverse inequality for a countable finite-entropy generator. If $\mathcal{P}$ is finite, (3) immediately gives $h_\mu(T, \mathcal{P}) \leq h_\mu(T)$. Suppose now

$$\mathcal{P} = \{P_1, P_2, \dots\}, \quad H_\mu(\mathcal{P}) < \infty.$$

Define the finite coarsening

$$\mathcal{P}^{(m)} := \{P_1, \dots, P_m, \bigcup_{j>m} P_j\}. \quad (18)$$

Then

$$H_\mu(\mathcal{P} \mid \mathcal{P}^{(m)}) = \sum_{j>m} -\mu(P_j) \log \frac{\mu(P_j)}{\sum_{k>m} \mu(P_k)} \rightarrow 0. \quad (19)$$

For completeness, (19) follows directly from $H_\mu(\mathcal{P}) < \infty$: if $r_m = \sum_{j>m} \mu(P_j)$, then

$$H_\mu(\mathcal{P} \mid \mathcal{P}^{(m)}) = \sum_{j>m} -\mu(P_j) \log \mu(P_j) + r_m \log r_m, \quad (20)$$

and both terms tend to 0.

Apply (5) with $\mathcal{Q} = \mathcal{P}$ and $\mathcal{R} = \mathcal{P}^{(m)}$:

$$h_\mu(T, \mathcal{P}) \leq h_\mu(T, \mathcal{P}^{(m)}) + H_\mu(\mathcal{P} \mid \mathcal{P}^{(m)}) \leq h_\mu(T) + H_\mu(\mathcal{P} \mid \mathcal{P}^{(m)}). \quad (21)$$

Letting $m \rightarrow \infty$ gives

$$h_\mu(T, \mathcal{P}) \leq h_\mu(T). \quad (22)$$

Together with (17), this proves

$$\boxed{h_\mu(T) = h_\mu(T, \mathcal{P})}. \quad (23)$$

The proof used negative iterates only in the explicitly invertible case; (11)–(15) show that the forward-generator formulation is fully self-contained for a noninvertible transformation. $\square$

Theorem F14.2. Variational principle for topological pressure

Statement. Let (X, d) be a nonempty compact metric space, let $T : X \rightarrow X$ be continuous, and let $\phi \in C(X, \mathbb{R})$. With topological pressure defined by the separated-set

formula of Definition F14.2 (equivalently by weighted spanning sets),

$$P(T, \phi) = \sup_{\mu \in \mathcal{M}_T(X)} \left(h_\mu(T) + \int_X \phi d\mu \right).$$

The equality is understood in $\mathbb{R} \cup \{+\infty\}$; in particular it also covers systems of infinite topological entropy.

Proof. Write

$$\mathcal{M}_T(X) := \{\mu \in \mathcal{P}(X) : T_*\mu = \mu\}.$$

By *Invariant measures form a compact convex set* this is a nonempty compact convex subset of the weak-* compact space of Borel probability measures. For a finite measurable partition $\mathcal{P}$, write

$$\mathcal{P}^n := \bigvee_{j=0}^{n-1} T^{-j}\mathcal{P}, \quad h_\mu(T, \mathcal{P}) := \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu(\mathcal{P}^n), \quad h_\mu(T) := \sup_{\mathcal{P}} h_\mu(T, \mathcal{P}), \quad (1)$$

where the supremum is over finite measurable partitions. The limit in (1) exists by the usual subadditivity of partition entropy. We prove the theorem in five steps. The two substantial inequalities are proved independently, so that no existence theorem for equilibrium states is being used.

Step 1: separated and spanning definitions give the same pressure. For $n \geq 1$ and $\varepsilon > 0$, let $Z_n(\phi, \varepsilon)$ be the separated partition sum from *the definition of topological pressure*. Call a finite set $F \subset X$ (n, ε) -spanning if for every $x \in X$ there is $y \in F$ with $d_n(x, y) < \varepsilon$, and put

$$Q_n(\phi, \varepsilon) := \inf_F \sum_{y \in F} e^{S_n \phi(y)}. \quad (2)$$

Compactness of (X, d_n) makes both finite: a maximal separated set is finite and spanning, while ϕ is bounded.

Let

$$\omega_\phi(r) := \sup\{|\phi(x) - \phi(y)| : d(x, y) < r\}; \quad (3)$$

then $\omega_\phi(r) \downarrow 0$ as $r \downarrow 0$. A maximal (n, ε) -separated set is (n, ε) -spanning, so

$$Q_n(\phi, \varepsilon) \leq Z_n(\phi, \varepsilon). \quad (4)$$

Conversely, let E be (n, ε) -separated and let F be $(n, \varepsilon/2)$ -spanning. Assign each $x \in E$ a point $y(x) \in F$ with $d_n(x, y(x)) < \varepsilon/2$. No two distinct points of E can have the same assigned point, because then their d_n -distance would be $< \varepsilon$. Moreover

$$S_n \phi(x) \leq S_n \phi(y(x)) + n\omega_\phi(\varepsilon/2).$$

Hence

$$Z_n(\phi, \varepsilon) \leq e^{n\omega_\phi(\varepsilon/2)} Q_n(\phi, \varepsilon/2). \quad (5)$$

Taking $n^{-1} \log$, then limsups, and finally $\varepsilon \downarrow 0$, equations (4)–(5) prove equivalence of

the two pressure definitions. We henceforth use the separated version.

Step 2: a block-entropy estimate. We isolate an elementary estimate used in the first variational inequality. Let σ be any probability measure, let $\mathcal{P}$ have m atoms, and put

$$\mu_n := \frac{1}{n} \sum_{j=0}^{n-1} T_*^j \sigma. \quad (6)$$

For every fixed integer $q \geq 1$,

$$\frac{1}{n} H_\sigma(\mathcal{P}^n) \leq \frac{1}{q} H_{\mu_n}(\mathcal{P}^q) + \frac{2q \log m}{n}. \quad (7)$$

(If $m = 1$, the right-hand error is zero.)

To prove (7), fix a residue $r \in \{0, \dots, q-1\}$. Decompose the time interval $\{0, \dots, n-1\}$ into consecutive q -blocks whose initial indices are congruent to r modulo q , together with at most $2q$ leftover single times at the two ends. Subadditivity of entropy under joins and the identity $H_\sigma(T^{-j}\mathcal{Q}) = H_{T_*^j\sigma}(\mathcal{Q})$ give

$$H_\sigma(\mathcal{P}^n) \leq \sum_{\substack{0 \leq j \leq n-q \\ j \equiv r \pmod{q}}} H_{T_*^j\sigma}(\mathcal{P}^q) + 2q \log m. \quad (8)$$

Sum (8) over the q residues. Every admissible block initial time occurs once; adding the remaining nonnegative entropy terms if necessary yields

$$q H_\sigma(\mathcal{P}^n) \leq \sum_{j=0}^{n-1} H_{T_*^j\sigma}(\mathcal{P}^q) + 2q^2 \log m. \quad (9)$$

Entropy of a fixed finite partition is concave in the probability measure, so

$$\frac{1}{n} \sum_{j=0}^{n-1} H_{T_*^j\sigma}(\mathcal{P}^q) \leq H_{\mu_n}(\mathcal{P}^q). \quad (10)$$

Equations (9)–(10) prove (7).

Step 3: topological pressure is at most the supremum of measure free energies. Fix $\varepsilon > 0$. Choose integers $n_k \rightarrow \infty$ for which

$$\frac{1}{n_k} \log Z_{n_k}(\phi, \varepsilon) \longrightarrow P(T, \phi, \varepsilon), \quad (11)$$

and choose finite (n_k, ε) -separated sets E_k satisfying

$$Z'_k := \sum_{x \in E_k} e^{S_{n_k} \phi(x)} \geq e^{-1} Z_{n_k}(\phi, \varepsilon). \quad (12)$$

Give E_k the probability weights

$$p_{k,x} := \frac{e^{S_{n_k}\phi(x)}}{Z'_k}, \quad \sigma_k := \sum_{x \in E_k} p_{k,x} \delta_x, \quad (13)$$

and average them along the orbit:

$$\mu_k := \frac{1}{n_k} \sum_{j=0}^{n_k-1} T_*^j \sigma_k. \quad (14)$$

By weak-* compactness of $\mathcal{P}(X)$, pass to a subsequence such that $\mu_k \rightarrow \mu$ weak-*. For $f \in C(X)$,

$$\int f \circ T d\mu_k - \int f d\mu_k = \frac{1}{n_k} \left(\int f \circ T^{n_k} d\sigma_k - \int f d\sigma_k \right), \quad (15)$$

whose absolute value is at most $2\|f\|_\infty/n_k$. Passing to the limit in (15) gives $T_*\mu = \mu$.

We now choose a finite Borel partition $\mathcal{P} = \{P_1, \dots, P_m\}$ such that

$$\text{diam } P_i < \varepsilon, \quad \mu(\partial P_i) = 0. \quad (16)$$

For completeness, take a finite cover by sufficiently small balls and choose each radius outside the countable set of radii whose sphere has positive μ -measure; disjointifying the balls produces (16), since every new boundary is contained in a finite union of the chosen sphere boundaries.

Every atom of $\mathcal{P}^{n_k}$ contains at most one point of E_k : two points in the same atom would have d_{n_k} -distance $< \varepsilon$. The elementary log-sum identity for the Gibbs weights (13) therefore gives exactly

$$\log Z'_k = H_{\sigma_k}(\mathcal{P}^{n_k}) + \int S_{n_k}\phi d\sigma_k. \quad (17)$$

The potential term is

$$\int S_{n_k}\phi d\sigma_k = n_k \int \phi d\mu_k. \quad (18)$$

Apply the block estimate (7) to σ_k . For every fixed q ,

$$\frac{1}{n_k} \log Z'_k \leq \frac{1}{q} H_{\mu_k}(\mathcal{P}^q) + \int \phi d\mu_k + \frac{2q \log m}{n_k}. \quad (19)$$

Because μ is invariant and the boundaries in (16) are μ -null, every atom of $\mathcal{P}^q$ is a μ -continuity set: its boundary is contained in a finite union of $T^{-j}(\partial P_i)$, each of μ -measure zero. Weak-* convergence therefore gives convergence of the finite vector of atom masses and hence

$$H_{\mu_k}(\mathcal{P}^q) \longrightarrow H_\mu(\mathcal{P}^q). \quad (20)$$

Using (11)–(12) and letting $k \rightarrow \infty$ in (19),

$$P(T, \phi, \varepsilon) \leq \frac{1}{q} H_\mu(\mathcal{P}^q) + \int \phi d\mu. \quad (21)$$

Let $q \rightarrow \infty$ and use (1):

$$P(T, \phi, \varepsilon) \leq h_\mu(T, \mathcal{P}) + \int \phi d\mu \leq h_\mu(T) + \int \phi d\mu. \quad (22)$$

Since $\mu \in \mathcal{M}_T(X)$, and then $\varepsilon \downarrow 0$,

$$P(T, \phi) \leq \sup_{\eta \in \mathcal{M}_T(X)} \left(h_\eta(T) + \int \phi d\eta \right). \quad (23)$$

Notice that no equilibrium state was assumed to exist: the measure μ in this argument may depend on ε .

Step 4: every invariant measure free energy is at most topological pressure.

Fix $\mu \in \mathcal{M}_T(X)$ and a finite measurable partition $\mathcal{P} = \{P_1, \dots, P_m\}$. We prove

$$P(T, \phi) \geq h_\mu(T, \mathcal{P}) + \int \phi d\mu. \quad (24)$$

Since a finite measurable partition may be changed on a μ -null set without changing its entropy, take Borel representatives for its atoms. Fix $0 < \eta < 1/8$. By regularity of the Borel probability μ , choose pairwise disjoint compact sets $K_i \subset P_i$ such that

$$B := X \setminus \bigcup_{i=1}^m K_i, \quad \mu(B) < \eta^2. \quad (25)$$

If at least two K_i are nonempty, put

$$\rho := \min_{i \neq j, K_i, K_j \neq \emptyset} d(K_i, K_j) > 0; \quad (26)$$

if fewer than two are nonempty, choose any $\rho > 0$. Choose $0 < \varepsilon < \rho$ so small that

$$\omega_\phi(\varepsilon) < \eta. \quad (27)$$

Let

$$N_n(x) := \sum_{j=0}^{n-1} \mathbf{1}_B(T^j x), \quad G_n := \{x : N_n(x) \leq \eta n\}. \quad (28)$$

Invariance of μ and (25) give

$$\int N_n d\mu = n\mu(B) < \eta^2 n,$$

so Markov's inequality gives

$$a_n := \mu(G_n) \geq 1 - \eta. \quad (29)$$

Let $\nu_n := a_n^{-1} \mu|_{G_n}$.

For every atom A of $\mathcal{P}^n$ with $\nu_n(A) > 0$, choose $x_A \in A \cap G_n$ so that

$$S_n \phi(x_A) \geq \frac{1}{\nu_n(A)} \int_A S_n \phi d\nu_n - 1. \quad (30)$$

(The harmless constant 1 could be replaced by any positive error.) The finite log-sum inequality $\log \sum_A e^{b_A} \geq H(p) + \sum_A p_A b_A$ applied to $p_A = \nu_n(A)$ and $b_A = S_n \phi(x_A)$ gives

$$H_{\nu_n}(\mathcal{P}^n) + \int S_n \phi d\nu_n \leq 1 + \log \sum_A e^{S_n \phi(x_A)}. \quad (31)$$

Choose a maximal (n, ε) -separated subset $E_n \subset G_n$. Total boundedness of (X, d_n) makes E_n finite, and maximality means that for every $x \in G_n$ there is $y \in E_n$ with

$$d_n(x, y) < \varepsilon. \quad (32)$$

Assign each selected x_A such a $y = y(A)$.

We compare their $\mathcal{P}$ -names. Both x_A and y belong to G_n , so each orbit segment visits B at at most ηn times. At a time j when neither $T^j x_A$ nor $T^j y$ lies in B , both points lie in compact cores $K_i, K_{i'}$. If their partition symbols were different, $i \neq i'$, then (26) would give $d(T^j x_A, T^j y) \geq \rho > \varepsilon$, contradicting (32). Hence the two names can differ in at most $2\eta n$ positions.

For $m \geq 2$ put

$$B_{m,\eta}(n) := \sum_{0 \leq j \leq \lfloor 2\eta n \rfloor} \binom{n}{j} (m-1)^j, \quad (33)$$

and put $B_{1,\eta}(n) := 1$. Thus at most $B_{m,\eta}(n)$ different $\mathcal{P}^n$ -names can be assigned to any fixed $y \in E_n$. The standard binomial estimate gives, for $m \geq 2$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log B_{m,\eta}(n) \leq b_m(\eta) := H_2(2\eta) + 2\eta \log(m-1), \quad (34)$$

where $H_2(t) = -t \log t - (1-t) \log(1-t)$; set $b_1(\eta) = 0$. In particular $b_m(\eta) \rightarrow 0$ as $\eta \downarrow 0$.

Equation (32) and uniform continuity give

$$S_n \phi(x_A) \leq S_n \phi(y(A)) + n\omega_\phi(\varepsilon). \quad (35)$$

Group the atoms A by their assigned point $y(A)$ and use (33)–(35):

$$\sum_A e^{S_n \phi(x_A)} \leq B_{m,\eta}(n) e^{n\omega_\phi(\varepsilon)} \sum_{y \in E_n} e^{S_n \phi(y)} \leq B_{m,\eta}(n) e^{n\omega_\phi(\varepsilon)} Z_n(\phi, \varepsilon). \quad (36)$$

Combining (31) and (36),

$$\frac{1}{n} \log Z_n(\phi, \varepsilon) \geq \frac{1}{n} H_{\nu_n}(\mathcal{P}^n) + \frac{1}{n} \int S_n \phi d\nu_n - \frac{1}{n} \log B_{m,\eta}(n) - \omega_\phi(\varepsilon) - \frac{1}{n}. \quad (37)$$

It remains only to compare the conditional measure ν_n with μ ; this is where the loss $O(\eta)$ enters. Put $M := \|\phi\|_\infty$ and $L := \log m + M$ (with $\log 1 = 0$). If $a_n < 1$, let $\xi_n = (1 - a_n)^{-1} \mu|_{X \setminus G_n}$; if $a_n = 1$, omit all ξ_n terms. Then

$$\mu = a_n \nu_n + (1 - a_n) \xi_n. \quad (38)$$

The entropy of a mixture is at most the mixture of the entropies plus the binary entropy,

hence

$$H_\mu(\mathcal{P}^n) \leq \log 2 + a_n H_{\nu_n}(\mathcal{P}^n) + (1 - a_n)n \log m. \quad (39)$$

Also, by invariance of μ ,

$$n \int \phi d\mu = a_n \int S_n \phi d\nu_n + (1 - a_n) \int S_n \phi d\xi_n, \quad (40)$$

and the last integral is at most nM . Add (39)-(40):

$$a_n \left(H_{\nu_n}(\mathcal{P}^n) + \int S_n \phi d\nu_n \right) \geq H_\mu(\mathcal{P}^n) + n \int \phi d\mu - \log 2 - (1 - a_n)nL. \quad (41)$$

Since $a_n \geq 1 - \eta > 1/2$ and $|n^{-1} H_\mu(\mathcal{P}^n) + \int \phi d\mu| \leq L$, a direct division of (41) gives the convenient uniform bound

$$\frac{1}{n} H_{\nu_n}(\mathcal{P}^n) + \frac{1}{n} \int S_n \phi d\nu_n \geq \frac{1}{n} H_\mu(\mathcal{P}^n) + \int \phi d\mu - 4\eta L - \frac{2 \log 2}{n}. \quad (42)$$

Indeed, writing $\delta_n = 1 - a_n \leq \eta$, the difference between the right side of (41) divided by a_n and the first two terms on the right of (42) is bounded below by $-2\delta_n L/a_n - (\log 2)/(na_n)$, which is at least the displayed error.

Insert (42) into (37), take the limsup in n , and use (34) and the entropy limit in (1):

$$P(T, \phi, \varepsilon) \geq h_\mu(T, \mathcal{P}) + \int \phi d\mu - 4\eta L - b_m(\eta) - \omega_\phi(\varepsilon). \quad (43)$$

Our choice (27) makes the last term $< \eta$. Since $P(T, \phi) = \lim_{r \downarrow 0} P(T, \phi, r) \geq P(T, \phi, \varepsilon)$, letting $\eta \downarrow 0$ in (43) proves (24).

Now take the supremum of (24) over all finite measurable partitions. By (1),

$$P(T, \phi) \geq h_\mu(T) + \int \phi d\mu. \quad (44)$$

Finally take the supremum over $\mu \in \mathcal{M}_T(X)$:

$$P(T, \phi) \geq \sup_{\mu \in \mathcal{M}_T(X)} \left(h_\mu(T) + \int \phi d\mu \right). \quad (45)$$

Step 5: combine the inequalities and check the extended-real case. Equations (23) and (45) give the desired equality. Because ϕ is continuous on compact X , it is bounded, so the potential integral is always finite. If some invariant measure has $h_\mu(T) = +\infty$, then its finite-partition entropies are arbitrarily large; the proof of (24), applied one partition at a time, forces $P(T, \phi) = +\infty$. Thus no subtraction of infinities occurs, and the equality is valid in $\mathbb{R} \cup \{+\infty\}$ exactly as stated. $\square$

Theorem F14.3. Ruelle-Perron-Frobenius theorem for Hölder potentials on mixing SFTs

Statement. Let (Σ_A^+, σ) be a topologically mixing one-sided subshift of finite type over a finite alphabet, let $0 < \alpha \leq 1$, and let $\phi \in C^\alpha(\Sigma_A^+, \mathbb{R})$. On C^α , the Ruelle operator $\mathcal{L}_\phi$ has a positive simple isolated spectral-radius eigenvalue $\lambda > 0$, a strictly positive Hölder eigenfunction h , and an eigenprobability ν for $\mathcal{L}_\phi^*$, normalized by $\nu(h) = 1$. There are $C > 0$ and $0 < \tau < 1$ such that $\|\lambda^{-n} \mathcal{L}_\phi^n f - h\nu(f)\|_\alpha \leq C\tau^n \|f\|_\alpha$ for every $f \in C^\alpha$.

Proof. Let A be the primitive $q \times q$ zero-one matrix defining the one-sided mixing subshift Σ_A^+ , and write σ for the left shift. Fix $0 < \theta < 1$ and use the standard metric

$$d_\theta(x, y) = \theta^{N(x, y)},$$

where $N(x, y)$ is the first coordinate at which x and y differ. Put $C^\alpha = C^\alpha(\Sigma_A^+)$ with norm

$$\|f\|_\alpha = \|f\|_\infty + [f]_\alpha,$$

and for real $\phi \in C^\alpha$ define

$$(\mathcal{L}_\phi f)(x) = \sum_{\sigma y = x} e^{\phi(y)} f(y). \quad (1)$$

Because the alphabet is finite, the sum in (1) is finite and $\mathcal{L}_\phi$ is bounded on C^α .

Step 1: bounded distortion. If an admissible word $w = w_0 \dots w_{n-1}$ may be prepended to both x and x' , then for $0 \leq j < n$ the points $\sigma^j(wx)$ and $\sigma^j(wx')$ agree for their first $n - j$ coordinates. Hence

$$\begin{aligned} |S_n \phi(wx) - S_n \phi(wx')| &\leq [\phi]_\alpha \sum_{j=0}^{n-1} d_\theta(\sigma^j(wx), \sigma^j(wx'))^\alpha \\ &\leq [\phi]_\alpha \sum_{j=0}^{n-1} \theta^{\alpha(n-j)} d_\theta(x, x')^\alpha \\ &\leq D_\phi d_\theta(x, x')^\alpha, \end{aligned} \quad (2)$$

where

$$D_\phi := \frac{[\phi]_\alpha \theta^\alpha}{1 - \theta^\alpha}. \quad (3)$$

The important point is that D_ϕ is independent of n and of the admissible inverse branch.

Choose $m \geq 1$ with $A^m > 0$. Thus every symbol can be connected to every other symbol by an admissible word of length m . For $a > 0$ let $\mathcal{C}_a$ be the cone of strictly positive Hölder functions satisfying

$$|\log f(x) - \log f(x')| \leq a d_\theta(x, x')^\alpha \quad \text{whenever } x_0 = x'_0. \quad (4)$$

If $x_0 = x'_0$, the inverse branches of σ^m ending at x and x' are canonically paired by the same length- m word. Estimate (2), together with the fact that prepending m symbols

multiplies distance by θ^m , gives

$$\mathcal{L}_\phi^m(\mathcal{C}_a) \subset \mathcal{C}_{a'}, \quad a' \leq D_\phi + \theta^{\alpha m} a. \quad (5)$$

Choose

$$a > \frac{D_\phi}{1 - \theta^{\alpha m}}. \quad (6)$$

Then $a' < a$ in (5).

Primitivity now supplies the second ingredient. There are only finitely many symbols and finitely many length- m connecting words. On the normalized section

$$\mathcal{S}_a := \{f \in \mathcal{C}_a : \|f\|_\infty = 1\},$$

condition (4), compactness of the shift space, (2), and the existence of a connecting word between every pair of symbols give constants $0 < c_a < C_a < \infty$ such that every normalized image of $\mathcal{L}_\phi^m(\mathcal{S}_a)$ satisfies

$$c_a \leq f(x) \leq C_a \quad \text{and} \quad |\log f(x) - \log f(y)| \leq C_a d_\theta(x, y)^\alpha \quad (7)$$

for all x, y . In particular the image has finite diameter in the Hilbert projective metric of the positive cone. The strict inclusion $a' < a$ is what keeps that diameter away from infinity at the boundary of the cone.

We use the standard Birkhoff cone-contraction theorem in its Banach-cone form: a positive linear map whose image has finite projective diameter is a strict contraction in Hilbert metric; on a normalized section of a normal Hölder cone such as (7), projective convergence is equivalent, with uniform constants, to convergence in $\|\cdot\|_\alpha$. Consequently there is $0 < \tau_0 < 1$ such that the projective action of $\mathcal{L}_\phi^m$ contracts by at most τ_0 on $\mathcal{C}_a$. There is therefore a unique invariant ray in this cone. Choose $h > 0$ on that ray. Since $\mathcal{L}_\phi$ commutes with $\mathcal{L}_\phi^m$, it sends the unique invariant ray to itself, so

$$\mathcal{L}_\phi h = \lambda h \quad (8)$$

for some $\lambda > 0$. Compactness gives

$$0 < \min h \leq \max h < \infty, \quad (9)$$

and $h \in C^\alpha$.

Step 2: the dual eigenprobability. Let $\mathcal{P}(\Sigma_A^+)$ be the weak-* compact convex space of Borel probabilities. For $\eta \in \mathcal{P}(\Sigma_A^+)$ define

$$\Phi(\eta) := \frac{\mathcal{L}_\phi^* \eta}{\eta(\mathcal{L}_\phi 1)}. \quad (10)$$

The denominator is strictly positive, and Φ is a continuous self-map of $\mathcal{P}(\Sigma_A^+)$. Schauder's fixed-point theorem gives a probability ν and a number $\lambda_\nu > 0$ such that

$$\mathcal{L}_\phi^* \nu = \lambda_\nu \nu. \quad (11)$$

Pair (11) with the positive eigenfunction h . Using (8),

$$\lambda_\nu \nu(h) = \nu(\mathcal{L}_\phi h) = \lambda \nu(h).$$

Since $\nu(h) > 0$, $\lambda_\nu = \lambda$. Rescale h by a positive constant so that

$$\nu(h) = 1. \quad (12)$$

Step 3: normalize to a Markov operator. Define

$$Pf := \frac{1}{\lambda h} \mathcal{L}_\phi(hf), \quad d\mu := h d\nu. \quad (13)$$

Then $P1 = 1$, and for continuous f ,

$$\int Pf d\mu = \lambda^{-1} \nu(\mathcal{L}_\phi(hf)) = \nu(hf) = \int f d\mu,$$

so $P^*\mu = \mu$.

Conjugation by the strictly positive Hölder function h preserves the normal Hölder-cone structure used in Step 1. Hence the same projective-contraction argument, now on the affine section normalized by $\mu(f) = 1$, gives constants $C_0 > 0$ and $0 < \tau < 1$ such that

$$\left\| P^n f - \left(\int f d\mu \right) 1 \right\|_\alpha \leq C_0 \tau^n \|f\|_\alpha \quad (14)$$

for every positive f lying in a fixed enlarged cone.

We explain the extension from the cone to all of C^α , since this is needed for the spectral statement. There is a constant K_a depending only on the chosen cone such that, for every real $f \in C^\alpha$,

$$g_\pm := K_a \|f\|_\alpha \pm f$$

are strictly positive and belong to that enlarged cone. Apply (14) to g_+ and g_- and subtract. The constant parts cancel, giving (14) for every real Hölder f , after enlarging C_0 . Applying the real result to real and imaginary parts gives (14) for every complex $f \in C^\alpha$.

Step 4: return to the Ruelle operator and identify the spectrum. Iterating (13) gives

$$\lambda^{-n} \mathcal{L}_\phi^n f = h P^n(f/h). \quad (15)$$

Multiplication by h and by h^{-1} are bounded operators on C^α , and

$$\int (f/h) d\mu = \nu(f).$$

Therefore (14) and (15) yield constants $C > 0$, $0 < \tau < 1$ such that

$$\boxed{\|\lambda^{-n} \mathcal{L}_\phi^n f - h \nu(f)\|_\alpha \leq C \tau^n \|f\|_\alpha} \quad (n \geq 0). \quad (16)$$

This is the claimed quantitative rank-one convergence.

It remains only to spell out the spectral consequences. If $\mathcal{L}_\phi f = \lambda f$, then (16) gives $f = h\nu(f)$, so the λ -eigenspace is one-dimensional. The same argument for the dual shows that the dual eigenspace is one-dimensional. If there were a nontrivial Jordan vector g with $(\mathcal{L}_\phi - \lambda I)g = h$, then $\lambda^{-n}\mathcal{L}_\phi^n g$ would contain a term growing linearly in n , contradicting the bounded convergence in (16). Thus the eigenvalue is algebraically simple.

Finally the codimension-one closed space $\ker \nu$ is invariant under $\mathcal{L}_\phi$, and (16) gives

$$\|\mathcal{L}_\phi^n|_{\ker \nu}\| \leq C(\tau\lambda)^n.$$

The spectral-radius formula therefore places the spectrum on $\ker \nu$ inside $|z| \leq \tau\lambda$. Since λ itself is an eigenvalue, it is the spectral radius; it is positive, simple, and isolated, and all remaining spectrum lies in a strictly smaller disk. This proves every assertion of the theorem. $\square$

Theorem F14.4. Existence and uniqueness of Gibbs/equilibrium state on mixing SFT

Statement. Under the one-sided mixing-SFT and Hölder-potential hypotheses of Theorem F14.3, let (λ, h, ν) be its normalized Ruelle-Perron-Frobenius triple and put $d\mu_\phi = h d\nu$. Then μ_ϕ is σ -invariant, $P(\phi) = \log \lambda$, and there is $C_G \geq 1$ such that for every nonempty n -cylinder $[w]$ and every $x \in [w]$,

$$C_G^{-1} \leq \frac{\mu_\phi([w])}{\exp(S_n \phi(x) - nP(\phi))} \leq C_G.$$

Moreover μ_ϕ is the unique invariant probability satisfying $h_\eta(\sigma) + \int \phi d\eta = P(\phi)$. Every invariant Gibbs state for the same potential and pressure equals μ_ϕ .

Proof. Let (λ, h, ν) be the normalized Ruelle-Perron-Frobenius triple from *Ruelle-Perron-Frobenius theorem for Hölder potentials on mixing SFTs*:

$$\mathcal{L}_\phi h = \lambda h, \quad \mathcal{L}_\phi^* \nu = \lambda \nu, \quad \nu(h) = 1, \quad (1)$$

with $h > 0$ Hölder. Put

$$d\mu = h d\nu, \quad (Pf)(x) := \frac{1}{\lambda h(x)} \mathcal{L}_\phi(hf)(x). \quad (2)$$

Ruelle-Perron-Frobenius theorem for Hölder potentials on mixing SFTs gives

$$P1 = 1, \quad P^* \mu = \mu, \quad (3)$$

and constants $C_0 > 0$, $0 < \tau < 1$ such that

$$\left\| P^n f - \left(\int f d\mu \right) 1 \right\|_{C^\alpha} \leq C_0 \tau^n \|f\|_{C^\alpha} \quad (4)$$

for every Hölder f .

Step 1: μ is shift invariant. For every continuous g ,

$$P(g \circ \sigma) = gP1 = g. \quad (5)$$

Indeed, in the sum defining P , all one-step preimages y of x satisfy $g(\sigma y) = g(x)$. Using (3) and (5),

$$\int g \circ \sigma d\mu = \int P(g \circ \sigma) d\mu = \int g d\mu. \quad (6)$$

Hence $\sigma_*\mu = \mu$.

Step 2: bounded distortion and the Gibbs cylinder estimate. Let $[w] = [w_0 \dots w_{n-1}]$ be a nonempty n -cylinder and put

$$S_n\phi(x) := \sum_{j=0}^{n-1} \phi(\sigma^j x). \quad (7)$$

Hölder continuity gives a uniform distortion constant $D_\phi < \infty$ such that

$$|S_n\phi(x) - S_n\phi(x')| \leq D_\phi \quad (x, x' \in [w], n \geq 1). \quad (8)$$

Also compactness and strict positivity give constants

$$0 < h_- \leq h \leq h_+ < \infty. \quad (9)$$

Write

$$F_i := \{z \in \Sigma_A^+ : A_{i,z_0} = 1\}. \quad (10)$$

Every F_i which can occur as a follower set is a nonempty clopen set. The eigenprobability ν has full support. To see this, if U is a nonempty cylinder, choose a nonzero continuous $f \geq 0$ supported in U . Primitivity supplies N for which $\mathcal{L}_\phi^N f > 0$ everywhere; then

$$\lambda^N \nu(f) = \nu(\mathcal{L}_\phi^N f) > 0.$$

Thus $\nu(U) > 0$. Since only finitely many F_i occur,

$$c_F := \min_i \nu(F_i) > 0, \quad C_F := \max_i \nu(F_i) \leq 1. \quad (11)$$

For $z \in F_{w_{n-1}}$ there is exactly one n -step preimage $y_w(z) \in [w]$. Using (1),

$$\lambda^n \mu([w]) = \int \mathcal{L}_\phi^n(h1_{[w]})(z) d\nu(z) = \int_{F_{w_{n-1}}} e^{S_n\phi(y_w(z))} h(y_w(z)) d\nu(z). \quad (12)$$

Fix $x \in [w]$. Equations (8)–(12) imply

$$c_F h_- e^{-D_\phi} \leq \frac{\mu([w])}{\exp(S_n\phi(x) - n \log \lambda)} \leq C_F h_+ e^{D_\phi}. \quad (13)$$

After enlarging one constant $C_G \geq 1$ we obtain the uniform Gibbs bound

$$\boxed{C_G^{-1} \leq \frac{\mu([w])}{\exp(S_n \phi(x) - n \log \lambda)} \leq C_G.} \quad (14)$$

Step 3: identify topological pressure. For each admissible word w of length n , choose $x_w \in [w]$ and set

$$Z_n := \sum_{|w|=n} \exp(S_n \phi(x_w)). \quad (15)$$

Summing (14) over all nonempty n -cylinders gives

$$C_G^{-1} \leq e^{-n \log \lambda} Z_n \leq C_G. \quad (16)$$

Hence

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n = \log \lambda. \quad (17)$$

We briefly check that the left side is the separated-set pressure in *the definition of topological pressure* rather than a new definition. Use the standard symbolic metric d_θ . For any $\theta < \varepsilon < 1$, one point from each nonempty n -cylinder is (n, ε) -separated, while an (n, ε) -separated set contains at most one point from each n -cylinder. The distortion estimate (8) changes the corresponding weights by at most the factor e^{D_ϕ} . Therefore the exponential rate in (17) is exactly $P(\phi)$. Thus

$$\boxed{P(\phi) = \log \lambda.} \quad (18)$$

Step 4: the normalized predecessor kernel. Define

$$\gamma(x) := \frac{e^{\phi(x)} h(x)}{\lambda h(\sigma x)}. \quad (19)$$

For every $y \in \Sigma_A^+$,

$$\sum_{\sigma x=y} \gamma(x) = 1, \quad (20)$$

because $\mathcal{L}_\phi h = \lambda h$. Thus γ is a strictly positive normalized predecessor kernel on every admissible inverse branch. Moreover, if η is any σ -invariant probability, then

$$\int \log \gamma d\eta = \int \phi d\eta - \log \lambda, \quad (21)$$

since the two integrals of $\log h$ and $\log h \circ \sigma$ cancel by invariance.

Let $\mathcal{P}$ be the finite time-zero partition and let

$$\mathcal{F} := \bigvee_{j \geq 1} \sigma^{-j} \mathcal{P} \quad (22)$$

be the future sigma-algebra. Since $\mathcal{P}$ is a forward generator, *Entropy/generating-*

partition interface gives

$$h_\eta(\sigma) = h_\eta(\sigma, \mathcal{P}). \quad (23)$$

The elementary block-entropy identity gives

$$h_\eta(\sigma, \mathcal{P}) = \lim_{n \rightarrow \infty} H_\eta(\mathcal{P} \mid \mathcal{P}_1^n) = H_\eta(\mathcal{P} \mid \mathcal{F}), \quad (24)$$

where the last passage follows from martingale convergence of conditional probabilities. Thus no general pressure variational principle is being used.

For each symbol a , let

$$p_a := \mathbb{E}_\eta(1_{[a]} \mid \mathcal{F}). \quad (25)$$

At a future tail $y = (x_1, x_2, \dots)$ define

$$q_a(y) := \begin{cases} \gamma(ay), & ay \in \Sigma_A^+, \\ 0, & \text{otherwise.} \end{cases} \quad (26)$$

Then $\sum_a p_a = \sum_a q_a = 1$ almost everywhere. The pointwise log-sum (relative-entropy) inequality says

$$-\sum_a p_a \log p_a + \sum_a p_a \log q_a \leq 0, \quad (27)$$

with the usual convention $0 \log 0 = 0$; if $p_a > 0$ then admissibility forces the corresponding $q_a > 0$ on the relevant conditional fiber. Integrating (27), using (24), and using conditional expectation to rewrite the second term as $\int \log \gamma d\eta$, we obtain from (21)

$$h_\eta(\sigma) + \int \phi d\eta - \log \lambda \leq 0. \quad (28)$$

Combining with (18),

$$\boxed{h_\eta(\sigma) + \int \phi d\eta \leq P(\phi)} \quad (29)$$

for every invariant probability η .

Step 5: μ attains equality. The transfer identity

$$P((g \circ \sigma)f) = gPf \quad (30)$$

implies, for bounded future-measurable test functions, that the conditional probability under μ of the predecessor a is exactly q_a . More explicitly, for bounded ψ depending only on the future,

$$\int 1_{[a]} \psi \circ \sigma d\mu = \int P(1_{[a]} \psi \circ \sigma) d\mu = \int \psi q_a d\mu. \quad (31)$$

Hence $p_a = q_a$ for μ -almost every future. Equality holds pointwise in (27), and (28) becomes

$$h_\mu(\sigma) + \int \phi d\mu = \log \lambda = P(\phi). \quad (32)$$

Thus μ is an equilibrium state.

Step 6: equality forces the Ruelle kernel. Suppose an invariant probability η attains equality in (29). The integrand in (27) is nonpositive and its integral is now zero. Therefore its relative entropy vanishes almost everywhere, so

$$p_a = q_a \quad \text{for every symbol } a, \text{ for } \eta\text{-a.e. future.} \quad (33)$$

For every bounded Hölder f , conditional expectation with respect to the future therefore gives

$$\mathbb{E}_\eta(f \mid \mathcal{F})(x) = (Pf)(\sigma x). \quad (34)$$

Integrating and using σ -invariance of η yields

$$\int f d\eta = \int Pf d\eta. \quad (35)$$

Iterating,

$$\int f d\eta = \int P^n f d\eta. \quad (36)$$

By the rank-one convergence (4), $P^n f \rightarrow \mu(f)$ uniformly. Hence

$$\int f d\eta = \int f d\mu \quad (37)$$

for every Hölder f . Locally constant functions are Hölder and are uniformly dense in $C(\Sigma_A^+)$, so Riesz-Markov gives

$$\eta = \mu. \quad (38)$$

Thus μ is the unique equilibrium state.

Step 7: every invariant Gibbs state is the same measure. Let η be an invariant probability satisfying a Gibbs bound for the same potential and pressure:

$$C^{-1}e^{S_n\phi(x)-nP(\phi)} \leq \eta([x_0 \dots x_{n-1}]) \leq Ce^{S_n\phi(x)-nP(\phi)}. \quad (39)$$

For the time-zero partition $\mathcal{P}$,

$$H_\eta(\mathcal{P}_0^{n-1}) = - \int \log \eta([x_0 \dots x_{n-1}]) d\eta(x). \quad (40)$$

Taking logarithms in (39), integrating, and using invariance gives

$$\left| \frac{1}{n} H_\eta(\mathcal{P}_0^{n-1}) - P(\phi) + \int \phi d\eta \right| \leq \frac{\log C}{n}. \quad (41)$$

Let $n \rightarrow \infty$ and invoke *Entropy/generating-partition interface* for the forward generating partition $\mathcal{P}$:

$$h_\eta(\sigma) + \int \phi d\eta = P(\phi). \quad (42)$$

Thus every invariant Gibbs state is an equilibrium state, and Step 6 forces it to equal μ . Together with (14), (18), and (32), this proves all parts of the theorem. $\square$

Theorem F14.5. Bowen coding for hyperbolic basic sets

Statement. Let Λ be a basic hyperbolic set of a C^1 diffeomorphism f . For every $\delta > 0$ there is a finite Markov partition $\{R_1, \dots, R_m\}$ of Λ with $\text{diam } R_i < \delta$. If $A_{ij} = 1$ when $\text{int}_\Lambda R_i \cap f^{-1}(\text{int}_\Lambda R_j) \neq \emptyset$, the associated subshift of finite type (Σ_A, σ) admits a continuous surjective semiconjugacy $\pi : \Sigma_A \rightarrow \Lambda$ satisfying $\pi \circ \sigma = f \circ \pi$. The map π is finite-to-one and is one-to-one on points whose orbit avoids all partition boundaries.

Proof.

Proof strategy

Construct small rectangles using local product structure, refine them until the Markov property holds, and record allowed transitions in a finite directed graph. A bi-infinite admissible itinerary determines nested stable/unstable strips whose intersection is a single orbit, producing the coding map. The remaining work verifies continuity, surjectivity, semiconjugacy, and the precise finite-to-one ambiguity on rectangle boundaries.

Throughout the proof the bracket convention is

$$[x, y] = W_{\text{loc}}^u(x) \cap W_{\text{loc}}^s(y).$$

Since a basic set is locally maximal, *Stable manifold theorem for hyperbolic sets*–004 give uniform local stable and unstable manifolds, a continuous bracket taking values in Λ , and shadowing inside Λ . We first construct the Markov partition and then construct its symbolic coding.

Step 1: one common hyperbolic scale. Choose $\rho > 0$ so that 4ρ is smaller than the local-product bracket radius and so small that uniqueness in *Shadowing lemma* applies to every orbit staying within 4ρ of a given orbit. In particular, after shrinking ρ if necessary,

$$d(f^n x, f^n y) < 4\rho \quad (n \in \mathbb{Z}), \quad x, y \in \Lambda, \quad \implies \quad x = y. \quad (1)$$

Thus 4ρ is an expansivity scale for $f|_\Lambda$.

Let $\delta > 0$ be prescribed. Apply shadowing with an accuracy $0 < \beta < \min\{\rho, \delta/8\}$. Choose the corresponding pseudo-orbit tolerance $\alpha > 0$ so small that every α -pseudo-orbit in Λ is uniquely β -shadowed by a point of Λ . By uniform continuity of f , choose

$$0 < r < \alpha/2, \quad d(x, y) < r \implies d(fx, fy) < \alpha/2. \quad (2)$$

Let $P = \{p_1, \dots, p_q\} \subset \Lambda$ be a finite r -net and set

$$\Sigma(P) := \{a = (a_n)_{n \in \mathbb{Z}} \in \{1, \dots, q\}^{\mathbb{Z}} : d(fp_{a_n}, p_{a_{n+1}}) < \alpha \ \forall n\}. \quad (3)$$

This is a two-sided subshift of finite type. Denote by $\theta(a)$ the unique point whose orbit β -shadows (p_{a_n}) . Uniqueness immediately gives

$$\theta \circ \sigma = f \circ \theta. \quad (4)$$

The usual compactness argument also gives continuity: if $a^{(k)}$ and $b^{(k)}$ agree on longer and longer central blocks while their images stay a fixed positive distance apart, take subsequential limits; both limiting points β -shadow the same limiting pseudo-orbit, contradicting uniqueness. Finally θ is onto. Given $x \in \Lambda$, choose $d(f^n x, p_{a_n}) < r$. Then (2) gives $d(fp_{a_n}, p_{a_{n+1}}) < \alpha$, and x itself r -shadows that pseudo-orbit; uniqueness yields $x = \theta(a)$.

For each symbol i put

$$T_i := \theta\{a \in \Sigma(P) : a_0 = i\}. \quad (5)$$

The T_i are compact and cover Λ , and $\text{diam } T_i \leq 2\beta < \delta/4$. They are rectangles. Indeed, if $x = \theta(a)$ and $y = \theta(b)$ belong to T_i , the sequence obtained by using the past a_n for $n \leq 0$ and the future b_n for $n \geq 0$ is still admissible because $a_0 = b_0 = i$. Its shadowing point lies simultaneously on $W_{\text{loc}}^u(x)$ and $W_{\text{loc}}^s(y)$, hence equals $[x, y]$ and belongs to T_i .

We shall use the following precise, code-relative pre-Markov fact. If $a \in \Sigma(P)$, $x = \theta(a)$, $a_0 = i$, and $a_1 = j$, then

$$f(W^s(x, T_i)) \subset W^s(fx, T_j), \quad f(W^u(x, T_i)) \supset W^u(fx, T_j). \quad (6)$$

For the stable inclusion, let $y \in W^s(x, T_i)$ and choose a name b of y with $b_0 = i$. Because $y = [y, x]$, the symbolic sequence using the past of b and the future of a has shadowing point y ; after shifting, its zeroth symbol is $a_1 = j$. Hence $fy \in T_j$, and of course $fy \in W^s(fx)$. The unstable inclusion is the same argument for f^{-1} . Notice that (6) is asserted only when the transition $i \rightarrow j$ comes from a single symbolic name of x ; this is exactly the form used below and avoids the false claim that arbitrary overlapping coarse rectangles are already Markov.

Step 2: stable and unstable sides of the coarse rectangles. For a small rectangle T define

$$\partial^s T := \{x \in T : x \notin \text{int}_{W_{\text{loc}}^u(x) \cap \Lambda} W^u(x, T)\}, \quad \partial^u T := \{x \in T : x \notin \text{int}_{W_{\text{loc}}^s(x) \cap \Lambda} W^s(x, T)\}. \quad (7)$$

Then

$$\partial_\Lambda T = \partial^s T \cup \partial^u T. \quad (8)$$

Indeed, an interior point of T is interior in both plaque directions. Conversely, if x is interior in both plaque directions, choose small relative plaque neighborhoods $U^u \subset W^u(x, T)$ and $U^s \subset W^s(x, T)$. Continuity of the bracket and local product coordinates show that $\{[u, s] : u \in U^u, s \in U^s\}$ is a relative neighborhood of x contained in T , so $x \in \text{int}_\Lambda T$. Thus (8) follows. The set $\partial^s T$ is saturated by the local stable plaques that it meets, and $\partial^u T$ is saturated by the local unstable plaques that it meets; this is immediate in the product coordinates of *Local rectangles for a hyperbolic set*.

For $x \in \Lambda$ let

$$\mathcal{T}(x) := \{T_i : x \in T_i\}, \quad \mathcal{T}^*(x) := \{T_k : T_k \cap T_i \neq \emptyset \text{ for some } T_i \in \mathcal{T}(x)\}. \quad (9)$$

Define the good set

$$Z^* := \{x : W_{\text{loc}}^s(x) \cap \partial^s T_k = \emptyset, W_{\text{loc}}^u(x) \cap \partial^u T_k = \emptyset \text{ for every } T_k \in \mathcal{T}^*(x)\}. \quad (10)$$

It is open and dense. Here is the local argument, included because this fact is load-bearing. In a product box, $\partial^s T_k$ is a closed union of stable plaques and $\partial^u T_k$ a closed union of unstable plaques. Their stable and unstable saturations therefore remain closed plaque walls. None contains a product-open set: if, for example, a stable saturation contained $U^u \times U^s$, then every unstable coordinate in U^u would be a boundary coordinate of T_k , contradicting the definition of boundary in the unstable slice. Hence each such wall is nowhere dense. The finite cover has only finitely many relevant walls in each product box, so their complement is open dense. On a sufficiently small product box the finite incidence set $\mathcal{T}^*(x)$ can only decrease away from rectangle boundaries; intersecting these finitely many complements proves (10) is open dense.

Step 3: Bowen's four-way refinement. Whenever $T_j \cap T_k \neq \emptyset$, partition T_j according to the two incidence tests

$$W^u(x, T_j) \cap T_k \neq \emptyset, \quad W^s(x, T_j) \cap T_k \neq \emptyset.$$

Thus

$$\begin{aligned} T_{jk}^1 &:= T_j \cap T_k, \\ T_{jk}^2 &:= \{x \in T_j : W^u(x, T_j) \cap T_k \neq \emptyset, W^s(x, T_j) \cap T_k = \emptyset\}, \\ T_{jk}^3 &:= \{x \in T_j : W^u(x, T_j) \cap T_k = \emptyset, W^s(x, T_j) \cap T_k \neq \emptyset\}, \\ T_{jk}^4 &:= \{x \in T_j : W^u(x, T_j) \cap T_k = \emptyset, W^s(x, T_j) \cap T_k = \emptyset\}. \end{aligned} \tag{11}$$

If both plaques meet T_k , say at u and s , then $x = [u, s] \in T_k$ because T_k is a rectangle; hence the first line really is the “both meet” case. Each T_{jk}^ν is a rectangle. In local product coordinates the two incidence tests depend on separate plaque coordinates, and

$$W^s([x, y], T_j) = W^s(y, T_j), \quad W^u([x, y], T_j) = W^u(x, T_j), \tag{12}$$

with our bracket convention, so each of the four truth-value combinations is bracket closed.

For $x \in Z^*$ every incidence test in (11) is locally constant at x . Indeed, a change of the unstable-plaque test can occur only when the local unstable plaque crosses an unstable boundary of a relevant T_k , and a change of the stable-plaque test only when the stable plaque crosses a stable boundary; both events were excluded in (10). Therefore x lies in the relative interior of the unique T_{jk}^ν encoding each relevant test. Put

$$Q(x) := \bigcap \{ \text{int}_\Lambda T_{jk}^\nu : T_j \in \mathcal{T}(x), T_k \in \mathcal{T}^*(x), x \in T_{jk}^\nu \}. \tag{13}$$

This is a nonempty open rectangle containing x . Only finitely many distinct $Q(x)$ occur. Moreover, if $y \in Q(x) \cap Z^*$, then $\mathcal{T}(y) = \mathcal{T}(x)$ and every four-way incidence value is the same, so $Q(y) = Q(x)$. Hence the distinct sets

$$Q_1, \dots, Q_m \tag{14}$$

form a partition of the dense set Z^* into pairwise disjoint open rectangles.

Set

$$R_i := \overline{Q_i}^\Lambda. \tag{15}$$

Continuity of the bracket makes each R_i a closed rectangle. We claim

$$Q_i = \text{int}_\Lambda R_i. \quad (16)$$

The inclusion $Q_i \subset \text{int}_\Lambda R_i$ is clear. For the reverse, let $z \in R_i \setminus Q_i$. Then $z \notin Z^*$, since a point of $Z^* \cap R_i$ has the same locally constant incidence data as nearby points of Q_i and hence belongs to Q_i . Thus one of the finitely many plaque walls defining (10) passes through z . By the definition of a boundary wall, every product neighborhood of z contains points on which the corresponding incidence test has the opposite value. Since all other walls are nowhere dense, such points may be chosen in Z^* ; they then belong to a cell $Q_j \neq Q_i$. Consequently no relative neighborhood of z is contained in R_i , proving $z \notin \text{int}_\Lambda R_i$. This proves (16). Therefore each R_i is proper. Their interiors are pairwise disjoint, and because $Z^* \subset \bigcup_i R_i$ is dense while the finite union on the right is closed,

$$\Lambda = \bigcup_{i=1}^m R_i. \quad (17)$$

Every Q_i is contained in a coarse T_j , hence so is R_i ; therefore

$$\text{diam } R_i \leq 2\beta < \delta. \quad (18)$$

Step 4: the refined rectangles satisfy the Markov property. We first prove the good-point propagation lemma. Suppose

$$x, y \in Z^* \cap f^{-1}(Z^*), \quad Q(x) = Q(y), \quad y \in W_{\text{loc}}^s(x). \quad (19)$$

Then

$$Q(fx) = Q(fy). \quad (20)$$

First we show $\mathcal{T}(fx) = \mathcal{T}(fy)$. Let T_j contain fx . Choose a symbolic name c of fx with $c_0 = j$ and put $a = \sigma^{-1}c$. Then $x = \theta(a)$, say $a_0 = s$, $a_1 = j$, so $x \in T_s$. Equality $Q(x) = Q(y)$ implies $\mathcal{T}(x) = \mathcal{T}(y)$, hence $y \in T_s$. The code-relative inclusion (6) gives $fy \in T_j$. Interchanging x, y gives the converse implication.

Now fix $T_j \in \mathcal{T}(fx) = \mathcal{T}(fy)$ and a coarse rectangle T_k meeting T_j . Because fx and fy are on the same local stable plaque,

$$W^s(fx, T_j) = W^s(fy, T_j), \quad (21)$$

so their stable incidence with T_k agrees. Suppose, for a contradiction, that their unstable incidence differs. Choose

$$fz \in W^u(fx, T_j) \cap T_k, \quad W^u(fy, T_j) \cap T_k = \emptyset. \quad (22)$$

Use the name of x selected above to obtain from the unstable half of (6) that $z \in W^u(x, T_s)$. Choose a name of fz whose zeroth symbol is the symbol of T_k and shift it back; if its time-zero coarse rectangle is T_t , then $z \in T_t$ and

$$f(W^s(z, T_t)) \subset W^s(fz, T_k). \quad (23)$$

Since $z \in T_s \cap T_t$, the rectangle T_t belongs to $\mathcal{T}^*(x) = \mathcal{T}^*(y)$. The points x and y have the same four-way incidence data relative to (T_s, T_t) ; because $z \in W^u(x, T_s) \cap T_t$, there is $w \in W^u(y, T_s) \cap T_t$. The bracket

$$v := [z, w] = [z, y] \quad (24)$$

lies in $W^s(z, T_t) \cap W^u(y, T_s)$. Applying f and using (23) and the unstable part of (6) gives

$$fv = [fz, fy] \in W^s(fz, T_k) \cap W^u(fy, T_j) \subset T_k, \quad (25)$$

contradicting (22). Thus the unstable incidence also agrees, proving (20).

We pass from good points to whole plaques. Let $\partial^s R_i$ and $\partial^u R_i$ be defined as in (7). For sufficiently small plaque radius, the stable saturation of $\bigcup_i \partial^s R_i$ and the unstable saturation of $\bigcup_i \partial^u R_i$ are closed nowhere-dense sets, by the same product box argument used after (10). Hence there is an open dense set $G \subset Z^* \cap f^{-1}(Z^*)$ such that, for every $x \in G$, the set $G \cap W^s(x, Q(x))$ is dense in $W^s(x, Q(x))$. For each such y , (20) gives $Q(fy) = Q(fx)$, so continuity yields

$$f(W^s(x, R(x))) \subset W^s(fx, R(fx)), \quad (26)$$

where $R(x) = \overline{Q(x)}$.

Now suppose $\text{int } R_i \cap f^{-1}(\text{int } R_j) \neq \emptyset$. Choose x in this open transition set and in G ; then $R_i = R(x)$ and $R_j = R(fx)$. For an arbitrary $x' \in R_i \cap f^{-1}R_j$, local product coordinates give

$$W^s(x', R_i) = \{[z, x'] : z \in W^s(x, R_i)\}. \quad (27)$$

Using bracket equivariance and (26),

$$\begin{aligned} f(W^s(x', R_i)) &= \{[fz, fx'] : z \in W^s(x, R_i)\} \\ &\subset W^s(fx', R_j). \end{aligned} \quad (28)$$

Applying the same argument to f^{-1} gives

$$f(W^u(x', R_i)) \supset W^u(fx', R_j). \quad (29)$$

In particular (28)–(29) hold when $x' \in \text{int } R_i$ and $fx' \in \text{int } R_j$. Thus $\mathcal{R} = \{R_1, \dots, R_m\}$ is a Markov partition in the precise sense of *the definition of rectangle and Markov partition*, and (18) gives the requested diameter bound.

Step 5: finite admissible itineraries are realized. Define

$$A_{ij} = 1 \iff \text{int}_\Lambda R_i \cap f^{-1}(\text{int}_\Lambda R_j) \neq \emptyset. \quad (30)$$

We prove the standard strip lemma explicitly. Call a subrectangle $S \subset R_i$ *stable-full* if $W^s(x, R_i) \subset S$ for every $x \in S$, and *unstable-full* if $W^u(x, R_i) \subset S$ for every $x \in S$. For $A_{ij} = 1$ put

$$D_{ij} := \overline{\text{int } R_i \cap f^{-1}(\text{int } R_j)}^{R_i}. \quad (31)$$

The Markov property implies that D_{ij} is stable-full in R_i and $f(D_{ij})$ is unstable-full in R_j .

To verify the first assertion, take transition points $x_k \rightarrow x \in D_{ij}$ and a point $y \in W^s(x, R_i)$. In local product coordinates choose $y_k \in W^s(x_k, R_i)$ with $y_k \rightarrow y$. Equation (28) gives $fy_k \in R_j$, hence $y \in D_{ij}$ after taking limits. For the second assertion, let $w \in W^u(fx, R_j)$ with $x \in D_{ij}$. Approximate x by transition points $x_k \in \text{int } R_i$ with $fx_k \in \text{int } R_j$ and approximate w inside the target unstable plaques by $w_k \in W^u(fx_k, R_j)$. The unstable Markov inclusion (29) gives $v_k \in W^u(x_k, R_i)$ with $fv_k = w_k$; after taking limits, $w \in f(D_{ij})$. Thus $f(D_{ij})$ is unstable-full.

If $U \subset R_i$ is a nonempty unstable-full rectangle with nonempty interior, then $U \cap D_{ij}$ has nonempty interior: choose $u \in \text{int } U$ and $d \in \text{int } D_{ij}$; the bracket $[u, d]$ lies on the unstable plaque of u and the stable plaque of d , so it lies in both sets, and choosing u, d in small product neighborhoods gives a product-open intersection. Moreover

$$U' := \overline{f(\text{int}(U \cap D_{ij}))}^{R_j} \quad (32)$$

is a nonempty unstable-full subrectangle of R_j : stable-fullness of D_{ij} supplies the whole transition strip in the source. If a point of $f(\text{int}(U \cap D_{ij}))$ is moved along its target unstable plaque inside R_j , the unstable Markov inclusion (29), together with the unstable-fullness of U , realizes nearby points as images of points of $\text{int}(U \cap D_{ij})$; closure then supplies the plaque endpoints.

Starting with $U_0 = R_{i_0}$ and applying (31)–(32) inductively therefore shows that every admissible word $(i_0, \dots, i_n)$ has a point x with

$$f^k x \in \text{int } R_{i_k}, \quad 0 \leq k \leq n. \quad (33)$$

Applying the same construction after a time shift realizes every admissible word on an arbitrary finite interval $[-N, N]$.

Step 6: the coding map. For $a = (a_n)_{n \in \mathbb{Z}} \in \Sigma_A$ define

$$K_N(a) := \bigcap_{n=-N}^N f^{-n} R_{a_n}. \quad (34)$$

By Step 5, $K_N(a)$ is nonempty and compact, and the sets decrease with N . Their total intersection is a singleton. Indeed, if x, y are both in it, then $f^n x$ and $f^n y$ lie in the same rectangle for every n , so by (18)

$$d(f^n x, f^n y) \leq 2\beta < 4\rho \quad (n \in \mathbb{Z}). \quad (35)$$

Expansivity (1) gives $x = y$. Set

$$\pi(a) := \bigcap_{N \geq 0} K_N(a). \quad (36)$$

Then directly

$$\pi \circ \sigma = f \circ \pi. \quad (37)$$

The map π is continuous. If $a^{(k)}$ and $b^{(k)}$ agree on $[-k, k]$, any subsequential limits x of $\pi(a^{(k)})$ and y of $\pi(b^{(k)})$ satisfy $d(f^n x, f^n y) \leq \max_i \text{diam } R_i < 4\rho$ for every fixed n ; (1)

gives $x = y$.

For surjectivity put

$$Y := \Lambda \setminus \bigcup_{n \in \mathbb{Z}} f^n \left(\bigcup_{i=1}^m \partial_\Lambda R_i \right). \quad (38)$$

The R_i are proper, so their boundaries are nowhere dense; Baire's theorem makes Y dense. Every $y \in Y$ lies in a unique rectangle at every time, say $f^n y \in \text{int } R_{a_n}$. Then $A_{a_n a_{n+1}} = 1$ and $\pi(a) = y$. Given arbitrary $x \in \Lambda$, choose $y_k \in Y$ tending to x , take their names $a^{(k)}$, pass to a convergent subsequence in compact Σ_A , and use continuity of π . Hence π is onto. The same argument shows that every point whose full orbit avoids all partition boundaries has exactly one symbolic name.

Step 7: the m^2 finite-fiber bound. It remains to prove that boundary ambiguity is finite. We first establish a finite-word no-diamond statement. Suppose $a, b \in \pi^{-1}(x)$ and, for some $N \geq 1$,

$$a_{-N} = b_{-N}, \quad a_N = b_N. \quad (39)$$

By Step 5 choose $p, q \in \Lambda$ such that

$$f^k p \in \text{int } R_{a_k}, \quad f^k q \in \text{int } R_{b_k}, \quad -N \leq k \leq N. \quad (40)$$

Because x belongs to both R_{a_k} and R_{b_k} , these two rectangles intersect. By (18),

$$d(f^k p, f^k q) \leq \text{diam } R_{a_k} + \text{diam } R_{b_k} \leq 4\beta < 4\rho,$$

and Step 1 placed 4ρ below the bracket radius. Therefore

$$z_k := [f^k p, f^k q] \quad (41)$$

is defined for $-N \leq k \leq N$, and bracket equivariance gives $z_k = f^{k+N} z_{-N} = f^{k-N} z_N$.

At time $-N$ the two symbols agree, so z_{-N} is an interior point of the common rectangle and lies on the stable plaque of $f^{-N} q$. The stable Markov inclusion, iterated forward along the b -itinerary, shows

$$z_k \in \text{int } R_{b_k} \quad (-N \leq k \leq N). \quad (42)$$

The word “interior” is preserved here because $q_k = f^k q$ is interior and f maps a relative neighborhood of z_k in the stable plaque homeomorphically into the target stable plaque; the transverse plaque coordinate is the interior coordinate of q_k . Similarly, at time N the common endpoint symbol makes z_N an interior point on the unstable plaque of $f^N p$. Iterating the unstable Markov inclusion backward along the a -itinerary gives

$$z_k \in \text{int } R_{a_k} \quad (-N \leq k \leq N). \quad (43)$$

Since distinct Markov rectangles have disjoint interiors, (42)–(43) imply

$$a_k = b_k \quad (-N \leq k \leq N). \quad (44)$$

This is exactly the finite-word no-diamond statement needed below; no claim about equality of the tails is made.

Now fix $x \in \Lambda$. If $\#\pi^{-1}(x) > m^2$, choose $m^2 + 1$ distinct names. There exists N large enough that their restrictions to $[-N, N]$ are pairwise distinct. Among the endpoint pairs (a_{-N}, a_N) there are at most m^2 possibilities, so two of the chosen names have the same endpoint pair. Equation (44) then says their central blocks are identical, a contradiction. Hence

$$\#\pi^{-1}(x) \leq m^2 \quad (x \in \Lambda). \quad (45)$$

Thus π is finite-to-one. Together, (17)–(18), (28)–(30), and (36)–(45) give every conclusion in the statement. $\square$

Theorem F14.6. Equilibrium states for Axiom A/basic hyperbolic systems

Statement. Let $f : M \rightarrow M$ be a C^1 diffeomorphism, let Λ be a basic hyperbolic set, and let $\phi : \Lambda \rightarrow \mathbb{R}$ be Hölder continuous. Then there is a unique f -invariant Borel probability μ_ϕ on Λ such that

$$h_{\mu_\phi}(f) + \int_{\Lambda} \phi d\mu_\phi = P(f|_{\Lambda}, \phi) = \sup_{\eta \in \mathcal{M}_f(\Lambda)} \left(h_\eta(f) + \int_{\Lambda} \phi d\eta \right).$$

Equivalently, every Hölder potential on a basic hyperbolic set has a unique equilibrium state. No mixing hypothesis on $f|_{\Lambda}$ is required.

Proof. Write $T = f|_{\Lambda}$. Fix a Hölder exponent $0 < \alpha \leq 1$ and a constant H_ϕ such that

$$|\phi(x) - \phi(y)| \leq H_\phi d(x, y)^\alpha \quad (x, y \in \Lambda). \quad (1)$$

We divide the proof into three parts. The first obtains the exact symbolic model needed by thermodynamic formalism, the second proves the symbolic equilibrium theorem at the transitive (not necessarily mixing) level, and the third proves that the finite-to-one coding preserves the variational problem.

Step 1: an irreducible Markov coding with Hölder coding map. Choose from *Bowen coding for hyperbolic basic sets* a Markov partition $\mathcal{R} = \{R_1, \dots, R_m\}$ whose diameter is below a fixed local hyperbolic scale, and let

$$\pi : (\Sigma_A, \sigma) \longrightarrow (\Lambda, T) \quad (2)$$

be its two-sided symbolic coding. Thus π is continuous, onto, finite-to-one, and

$$\pi \circ \sigma = T \circ \pi. \quad (3)$$

The proof of *Bowen coding for hyperbolic basic sets* gives the uniform fiber bound

$$\#\pi^{-1}(x) \leq M := m^2 \quad (x \in \Lambda). \quad (4)$$

The transition matrix A is irreducible. Indeed, fix i, j . Since $\text{int}_\Lambda R_i$ and $\text{int}_\Lambda R_j$ are nonempty relatively open sets and $T|_\Lambda$ is transitive, for some $n \geq 1$

$$U := \text{int}_\Lambda R_i \cap T^{-n}(\text{int}_\Lambda R_j) \neq \emptyset. \quad (5)$$

Put $B = \bigcup_{r=1}^m \partial_\Lambda R_r$. Every R_r is proper, so $\partial_\Lambda R_r$ is nowhere dense; hence the finite union $\bigcup_{k=1}^{n-1} T^{-k}B$ is nowhere dense. Choose $x \in U$ outside this union. Then for $0 \leq k \leq n$ the point $T^k x$ lies in the interior of a unique rectangle, say R_{i_k} , with $i_0 = i$ and $i_n = j$. Consecutive symbols satisfy $A_{i_k i_{k+1}} = 1$. Thus there is a path from every i to every j .

We also need a quantitative regularity fact not stated in *Bowen coding for hyperbolic basic sets*: π is Hölder for a standard symbolic metric. Fix $0 < \vartheta < 1$ and write

$$d_\vartheta(a, b) = \vartheta^{N(a, b)}, \quad N(a, b) := \max\{N \geq 0 : a_k = b_k \text{ for } |k| \leq N\}, \quad (6)$$

with the usual convention when $a = b$. Uniform hyperbolicity gives constants $C_h > 0$ and $0 < \lambda_h < 1$ for contraction on local stable manifolds and backward contraction on local unstable manifolds.

Suppose $a_k = b_k$ for $|k| \leq N$, put $x = \pi(a)$ and $y = \pi(b)$, and splice

$$c_k = \begin{cases} b_k, & k < 0, \\ a_k, & k \geq 0. \end{cases} \quad (7)$$

The splice is admissible because $a_0 = b_0$; put $z = \pi(c)$. Since a and c have the same future, their coded orbits remain in the same small rectangles for all nonnegative times. The tail characterization in *Stable manifold theorem for hyperbolic sets* therefore gives $z \in W_{\text{loc}}^s(x)$. Since they also have the same symbols from time $-N$ through 0, forward stable contraction from time $-N$ to time 0 gives

$$d(x, z) \leq C_1 \lambda_h^N. \quad (8)$$

Similarly b and c have the same past, hence $z \in W_{\text{loc}}^u(y)$, and their symbols agree from time 0 through N ; backward unstable contraction from time N to time 0 gives

$$d(y, z) \leq C_1 \lambda_h^N. \quad (9)$$

Consequently

$$d(\pi a, \pi b) \leq 2C_1 \lambda_h^N. \quad (10)$$

Choosing $0 < \gamma \leq 1$ with $\lambda_h \leq \vartheta^\gamma$ yields

$$d(\pi a, \pi b) \leq C_\pi d_\vartheta(a, b)^\gamma. \quad (11)$$

Thus

$$\Psi := \phi \circ \pi \quad (12)$$

is Hölder on Σ_A .

Step 2: the equilibrium theorem for an irreducible two-sided SFT. We prove the precise symbolic statement needed below rather than importing a two-sided theorem. Let p be the period of the irreducible matrix A . The alphabet has the canonical cyclic decomposition

$$\mathcal{A} = C_0 \sqcup \cdots \sqcup C_{p-1}, \quad A(C_r, C_s) \neq 0 \implies s \equiv r + 1 \pmod{p}. \quad (13)$$

Put

$$\Sigma_r := \{a \in \Sigma_A : a_0 \in C_r\}. \quad (14)$$

Then $\sigma(\Sigma_r) = \Sigma_{r+1}$, and after relabelling admissible p -blocks the system

$$S := \sigma^p|_{\Sigma_0} \quad (15)$$

is a topologically mixing two-sided finite SFT. Put

$$F := S_p \Psi = \sum_{r=0}^{p-1} \Psi \circ \sigma^r \quad \text{on } \Sigma_0. \quad (16)$$

It is Hölder.

We first reduce a Hölder potential on a mixing two-sided SFT to a cohomologous future-dependent Hölder potential. We record the argument because *Existence and uniqueness of Gibbs/equilibrium state on mixing SFT* is deliberately stated for one-sided shifts. Work in any mixing two-sided finite SFT (Σ, S) with Hölder potential G . For each symbol choose once and for all an admissible left-infinite past ending at that symbol. For $x \in \Sigma$, let x^* have this fixed past and the same coordinates as x for $k \geq 0$. Then

$$u(x) := \sum_{n=0}^{\infty} (G(S^n x) - G(S^n x^*)) \quad (17)$$

converges uniformly: the two arguments in the n th summand agree on a central block of length at least n , so the summand is $O(\vartheta^{cn})$ for some $c > 0$. The same estimate, splitting the sum at half the first coordinate where two points differ, proves that u is Hölder (possibly with a smaller exponent).

Define

$$G^+ := G - u + u \circ S. \quad (18)$$

A cancellation of the two series gives the useful explicit formula

$$G^+(x) = G(x^*) + \sum_{n=1}^{\infty} (G(S^n x^*) - G(S^{n-1}(Sx)^*)). \quad (19)$$

The two arguments in the n th difference agree on a central block of length at least $n - 1$, so (19) converges uniformly and is Hölder. More importantly, every term in (19) depends only on $(x_0, x_1, \dots)$. Therefore G^+ descends to a Hölder potential on the corresponding mixing one-sided SFT. For every invariant probability η ,

$$\int G \, d\eta = \int G^+ \, d\eta \quad (20)$$

because the coboundary integrates to zero.

Apply this construction to (Σ_0, S, F) . *Existence and uniqueness of Gibbs/equilibrium state on mixing SFT* gives a unique equilibrium state for the resulting one-sided Hölder potential. Its natural extension is the unique equilibrium state ν_0 for (Σ_0, S, F) . For completeness, the natural-extension assertion uses no extra theorem: a one-sided

invariant probability determines a unique two-sided invariant probability by

$$\hat{\eta}([a_{-r} \dots a_s]) := \eta([a_{-r} \dots a_s]), \quad (21)$$

where on the right the word is viewed, by invariance, as a cylinder beginning at time 0. The consistency relations are exactly invariance. Conversely, projection to nonnegative coordinates recovers the one-sided measure. The time-zero partition has identical finite block distributions on the two systems, so *Entropy/generating-partition interface* (or directly the block-entropy definition) gives identical metric entropies. Since G^+ depends only on the future, the potential integrals also agree. Hence equilibrium and uniqueness pass between the one-sided system and its natural extension, and (20) transfers them back from G^+ to G .

Return to the period- p system. Define

$$\nu_\Psi := \frac{1}{p} \sum_{r=0}^{p-1} \sigma_*^r \nu_0. \quad (22)$$

It is σ -invariant. If η is any σ -invariant probability on Σ_A , invariance and the cyclic permutation imply $\eta(\Sigma_r) = 1/p$. Thus

$$\eta_0 := p \eta|_{\Sigma_0} \quad (23)$$

is S -invariant and

$$\eta = \frac{1}{p} \sum_{r=0}^{p-1} \sigma_*^r \eta_0. \quad (24)$$

The time-zero generator and the block-entropy chain rule give the standard cyclic identities

$$h_\eta(\sigma) = \frac{1}{p} h_{\eta_0}(S), \quad \int \Psi d\eta = \frac{1}{p} \int F d\eta_0. \quad (25)$$

Indeed, $h_\eta(\sigma^p) = p h_\eta(\sigma)$, while the finite σ^p -invariant partition $\{\Sigma_0, \dots, \Sigma_{p-1}\}$ has zero entropy rate and decomposes $h_\eta(\sigma^p)$ as the average of the entropies of the normalized restrictions; these restrictions are conjugate by powers of σ . The integral identity follows immediately from (24). Therefore

$$h_\eta(\sigma) + \int \Psi d\eta = \frac{1}{p} \left(h_{\eta_0}(S) + \int F d\eta_0 \right). \quad (26)$$

Uniqueness of ν_0 now shows that ν_Ψ is the unique equilibrium state for the Hölder potential Ψ on the irreducible two-sided SFT (Σ_A, σ) .

Step 3: a finite-to-one factor lemma at exact scope. We need two elementary facts for the particular factor (2).

(a) *Every invariant measure downstairs has an invariant lift.* Let $\eta \in \mathcal{M}_T(\Lambda)$. We first choose a Borel section of π . Enumerate $\mathbb{Z}$ as $e_1, e_2, \dots$. In the nonempty finite fiber $\pi^{-1}(x)$ choose successively the smallest possible symbol at coordinate e_1 , then at e_2 , and so on, always retaining at least one point of the fiber. This selects a unique sequence $s(x) \in \pi^{-1}(x)$. The selector is Borel: for every finite cylinder $C \subset \Sigma_A$ the set $\pi(C)$ is

compact, and at each finite stage the event that a candidate with a prescribed next symbol exists is therefore Borel. Hence all coordinates of s are Borel.

Put $\tilde{\eta}_0 = s_*\eta$ and form the Cesàro averages

$$\tilde{\eta}_N := \frac{1}{N} \sum_{k=0}^{N-1} \sigma_*^k \tilde{\eta}_0. \quad (27)$$

Any weak-* limit $\tilde{\eta}$ is σ -invariant. By (3) and $T_*\eta = \eta$,

$$\pi_*\tilde{\eta}_N = \frac{1}{N} \sum_{k=0}^{N-1} T_*^k \eta = \eta, \quad (28)$$

so $\pi_*\tilde{\eta} = \eta$.

(b) *The finite-to-one factor preserves metric entropy.* Let ν be any σ -invariant probability and put $\mu = \pi_*\nu$. We prove

$$h_\nu(\sigma) = h_\mu(T). \quad (29)$$

The inequality $h_\mu(T) \leq h_\nu(\sigma)$ is the elementary monotonicity of entropy under factors (pull back any finite partition of Λ). For the reverse inequality we spell out the relative-entropy calculation.

Let $\mathcal{P}$ be the time-zero symbol partition of Σ_A . It is a finite two-sided generator. Hyperbolicity gives an expansivity constant $\varepsilon_0 > 0$ for $T|_\Lambda$; choose a finite Borel partition $\mathcal{Q}$ of Λ with all atoms of diameter $< \varepsilon_0$. Then $\mathcal{Q}$ is a two-sided generator: two points with the same $\mathcal{Q}$ -name for all $n \in \mathbb{Z}$ remain ε_0 -close for all iterates and hence are equal. Since Λ is compact metric, the resulting Borel itinerary map is injective between standard Borel spaces, so its inverse on its image is Borel; hence the two-sided joins of $\mathcal{Q}$ generate $\mathcal{B}(\Lambda)$. Put

$$\mathcal{C} := \pi^{-1}\mathcal{B}(\Lambda). \quad (30)$$

This is a σ -invariant sigma-algebra and, because $\mathcal{Q}$ is a generator,

$$\mathcal{C} = \sigma \left(\bigvee_{k \in \mathbb{Z}} \sigma^{-k} \pi^{-1} \mathcal{Q} \right) \mod \nu. \quad (31)$$

For a finite partition $\mathcal{A}$ write $\mathcal{A}_0^{n-1} = \bigvee_{k=0}^{n-1} \sigma^{-k} \mathcal{A}$. The conditional chain rule and martingale convergence give

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{1}{n} H_\nu(\mathcal{P}_0^{n-1} \vee \pi^{-1} \mathcal{Q}_0^{n-1}) \\ &= h_\mu(T, \mathcal{Q}) + H_\nu \left(\mathcal{P} \left| \mathcal{C} \vee \bigvee_{j=1}^{\infty} \sigma^j \mathcal{P} \right. \right). \end{aligned} \quad (32)$$

Here is the finite-window verification of the only nonformal step. After expanding the conditional entropy of $\mathcal{P}_0^{n-1}$ by the chain rule, the k th summand, shifted back to time 0, is conditioned on an increasing piece of the past $\bigvee_{j=1}^k \sigma^j \mathcal{P}$ and on the factor window

$\bigvee_{r=k-n+1}^k \sigma^r \pi^{-1} \mathcal{Q}$. For $L \leq k \leq n-1-L$ this conditioning contains the fixed window

$$\bigvee_{j=1}^L \sigma^j \mathcal{P} \vee \bigvee_{|j| \leq L} \sigma^{-j} \pi^{-1} \mathcal{Q}.$$

Let first $n \rightarrow \infty$ (the at most $2L$ end terms contribute $O(L/n)$), and then $L \rightarrow \infty$. Conditional entropies decrease to the right-hand conditional entropy in (32) by (31). The opposite inequality follows because the full conditioning sigma-algebra is stronger than every finite window. This proves (32). Since both $\mathcal{P}$ and $\mathcal{Q}$ are generators, *Entropy/generating-partition interface* identifies the left side with $h_\nu(\sigma)$ and the first term on the right with $h_\mu(T)$.

It remains to show that the relative term in (32) is zero. Equivalently, by the same conditional chain-rule limit,

$$h_\nu(\sigma, \mathcal{P} \mid \mathcal{C}) = \lim_{n \rightarrow \infty} \frac{1}{n} H_\nu(\mathcal{P}_0^{n-1} \mid \mathcal{C}). \quad (33)$$

Disintegrate ν over μ along the fibers of π . Conditioned on $x \in \Lambda$, the partition $\mathcal{P}_0^{n-1}$ is supported on at most $\#\pi^{-1}(x) \leq M$ atoms, because the entire conditional measure lives on the finite fiber $\pi^{-1}(x)$. Hence

$$H_\nu(\mathcal{P}_0^{n-1} \mid \mathcal{C}) \leq \int \log \#\pi^{-1}(x) d\mu(x) \leq \log M. \quad (34)$$

Dividing by n gives zero in (33), and (29) follows.

Step 4: equality of pressures and uniqueness downstairs. Let

$$P_\Sigma := P(\sigma, \Psi), \quad P_\Lambda := P(T, \phi). \quad (35)$$

Variational principle for topological pressure identifies both pressures with the corresponding suprema of free energy. If $\nu \in \mathcal{M}_\sigma(\Sigma_A)$ and $\mu = \pi_* \nu$, then (12) and (29) give

$$h_\nu(\sigma) + \int \Psi d\nu = h_\mu(T) + \int \phi d\mu \leq P_\Lambda. \quad (36)$$

Taking the supremum gives $P_\Sigma \leq P_\Lambda$.

Conversely, if $\eta \in \mathcal{M}_T(\Lambda)$, part (a) gives an invariant lift $\tilde{\eta}$ and part (b) gives

$$h_\eta(T) + \int \phi d\eta = h_{\tilde{\eta}}(\sigma) + \int \Psi d\tilde{\eta} \leq P_\Sigma. \quad (37)$$

Thus $P_\Lambda \leq P_\Sigma$, and hence

$$\boxed{P_\Lambda = P_\Sigma}. \quad (38)$$

Let ν_Ψ be the unique symbolic equilibrium state constructed in Step 2 and set

$$\mu_\phi := \pi_* \nu_\Psi. \quad (39)$$

Equations (29), (36), and (38) show

$$h_{\mu_\phi}(T) + \int \phi d\mu_\phi = P_\Lambda, \quad (40)$$

so μ_ϕ is an equilibrium state.

Finally let η be any equilibrium state downstairs and choose an invariant lift $\tilde{\eta}$. Equations (37)–(38) imply

$$h_{\tilde{\eta}}(\sigma) + \int \Psi d\tilde{\eta} = P_\Sigma.$$

Symbolic uniqueness from Step 2 gives $\tilde{\eta} = \nu_\Psi$, hence $\eta = \pi_*\tilde{\eta} = \mu_\phi$. This proves existence and uniqueness for every Hölder potential on every basic hyperbolic set, including the nonmixing finite-period case. $\square$

Theorem F14.7. Exponential decay from transfer-operator spectral gap

Statement. Under the hypotheses of Theorem F14.3, let $\mu_\phi = h\nu$ be the Gibbs/equilibrium probability from Theorem F14.4. Then there are $C > 0$ and $0 < \tau < 1$ such that for every $f \in C^\alpha(\Sigma_A^+)$, every $g \in L^\infty(\mu_\phi)$, and every $n \geq 0$,

$$\left| \int f(g \circ \sigma^n) d\mu_\phi - \int f d\mu_\phi \int g d\mu_\phi \right| \leq C\tau^n \|f\|_{C^\alpha} \|g\|_{L^\infty(\mu_\phi)}.$$

In particular μ_ϕ is exponentially mixing for Hölder observables.

Proof. Let $\mu = \mu_\phi = h\nu$ be the Gibbs/equilibrium probability from *Existence and uniqueness of Gibbs/equilibrium state on mixing SFT*, and let

$$(Pf)(x) = \frac{1}{\lambda h(x)} \mathcal{L}_\phi(hf)(x) \quad (1)$$

be the normalized transfer operator from *Ruelle-Perron-Frobenius theorem for Hölder potentials on mixing SFTs*. Then

$$P1 = 1, \quad P^*\mu = \mu, \quad (2)$$

and there are $C_0 > 0$, $0 < \tau < 1$ such that

$$\left\| P^n f - \left(\int f d\mu \right) 1 \right\|_{C^\alpha} \leq C_0 \tau^n \|f\|_{C^\alpha} \quad (3)$$

for every $f \in C^\alpha$ and every $n \geq 0$.

The key algebraic identity is

$$P((g \circ \sigma)f) = gPf. \quad (4)$$

It follows directly from the definition: every one-step preimage y of x satisfies $g(\sigma y) = g(x)$. Iterating (4),

$$P^n((g \circ \sigma^n)f) = gP^n f. \quad (5)$$

Because each fiber of σ is finite, the same formula defines P on bounded Borel functions, so (4)–(5) remain pointwise identities there. The invariance identity $\int Pu \, d\mu = \int u \, d\mu$, proved first for continuous u , extends to bounded Borel u by the monotone-class theorem. Thus for every $g \in L^\infty(\mu)$, using a bounded Borel representative,

$$\int f(g \circ \sigma^n) \, d\mu = \int P^n((g \circ \sigma^n)f) \, d\mu = \int g P^n f \, d\mu. \quad (6)$$

Subtract the product of the means:

$$\begin{aligned} \left| \int f(g \circ \sigma^n) \, d\mu - \int f \, d\mu \int g \, d\mu \right| &\leq \|g\|_{L^\infty(\mu)} \left\| P^n f - \int f \, d\mu \right\|_\infty \\ &\leq C_0 \tau^n \|f\|_{C^\alpha} \|g\|_{L^\infty(\mu)}. \end{aligned} \quad (7)$$

This is the claimed exponential correlation estimate. Taking g Hölder as well gives the customary Hölder–Hölder exponential mixing statement. $\square$

Theorem F14.8. Central limit theorem from spectral perturbation

Statement. Under the Theorem F14.3 and the Theorem F14.4, let μ_ϕ be the Gibbs/equilibrium state and let $\psi \in C^\alpha(\Sigma_A^+, \mathbb{R})$ satisfy $\int \psi \, d\mu_\phi = 0$. Put $S_n \psi = \sum_{j=0}^{n-1} \psi \circ \sigma^j$. Then

$$\sigma_\psi^2 := \lim_{n \rightarrow \infty} \frac{1}{n} \int (S_n \psi)^2 \, d\mu_\phi = \int \psi^2 \, d\mu_\phi + 2 \sum_{k=1}^{\infty} \int \psi(\psi \circ \sigma^k) \, d\mu_\phi$$

exists, with an absolutely convergent series. If $\sigma_\psi^2 > 0$, then $S_n \psi / (\sigma_\psi \sqrt{n}) \Rightarrow \mathcal{N}(0, 1)$. If $\sigma_\psi^2 = 0$, then $S_n \psi / \sqrt{n} \rightarrow 0$ in $L^2(\mu_\phi)$. In particular every bounded measurable coboundary $\psi = u - u \circ \sigma$ has $\sigma_\psi^2 = 0$.

Scope note

Vanishing asymptotic variance is asserted for bounded measurable coboundaries; no converse coboundary criterion is included here.

Proof. Let $\mu = \mu_\phi$ be the Gibbs/equilibrium probability from *Existence and uniqueness of Gibbs/equilibrium state on mixing SFT*, and let

$$(Pf)(x) = \frac{1}{\lambda_\phi h(x)} \mathcal{L}_\phi(hf)(x) \quad (1)$$

be the normalized transfer operator. Here λ_ϕ is the leading Ruelle eigenvalue from *Ruelle–Perron–Frobenius theorem for Hölder potentials on mixing SFTs*. Thus

$$P1 = 1, \quad P^* \mu = \mu, \quad (2)$$

and 1 is an algebraically simple eigenvalue of P while the rest of the spectrum on C^α lies in a disk of radius strictly smaller than 1.

Fix a real observable $\psi \in C^\alpha(\Sigma_A^+)$ satisfying $\mu(\psi) = 0$, and write

$$S_n \psi = \sum_{j=0}^{n-1} \psi \circ \sigma^j. \quad (3)$$

Step 1: existence and Green-Kubo formula for the asymptotic variance. Set

$$C_k := \int \psi (\psi \circ \sigma^k) d\mu, \quad k \geq 0. \quad (4)$$

Since ψ is bounded and Hölder, *Exponential decay from transfer-operator spectral gap* gives constants $C > 0$ and $0 < \tau < 1$ such that

$$|C_k| \leq C\tau^k \|\psi\|_{C^\alpha} \|\psi\|_\infty \quad (k \geq 1). \quad (5)$$

Consequently $\sum_{k \geq 1} |C_k| < \infty$.

By invariance of μ and centering,

$$\begin{aligned} \int (S_n \psi)^2 d\mu &= \sum_{j,\ell=0}^{n-1} \int (\psi \circ \sigma^j)(\psi \circ \sigma^\ell) d\mu \\ &= nC_0 + 2 \sum_{k=1}^{n-1} (n-k)C_k. \end{aligned} \quad (6)$$

Dividing by n and using absolute summability,

$$\sigma_\psi^2 := \lim_{n \rightarrow \infty} \frac{1}{n} \int (S_n \psi)^2 d\mu = C_0 + 2 \sum_{k=1}^{\infty} C_k. \quad (7)$$

In particular $\sigma_\psi^2 \geq 0$, because it is a limit of nonnegative numbers. Formula (7) is the Green-Kubo formula.

Step 2: analytic twisted transfer operators. For complex z near 0 define

$$P_z f := P(e^{z\psi} f). \quad (8)$$

The Hölder space C^α is a Banach algebra, and $z \mapsto e^{z\psi}$ is entire with values in C^α . Hence $z \mapsto P_z$ is an analytic family of bounded operators on C^α .

We use the frozen analytic perturbation theorem for an isolated algebraically simple eigenvalue. Because 1 is a simple isolated eigenvalue of $P = P_0$, there exist $r > 0$, analytic functions

$$z \mapsto \lambda(z) \in \mathbb{C}, \quad z \mapsto \Pi_z \in \mathcal{B}(C^\alpha), \quad (9)$$

for $|z| < r$, and constants $C_1 > 0$, $0 < \rho < 1$ such that, after shrinking r if necessary,

$$P_z^n = \lambda(z)^n \Pi_z + N_z^n, \quad \|N_z^n\|_{C^\alpha \rightarrow C^\alpha} \leq C_1 \rho^n, \quad (10)$$

uniformly for $|z| \leq r/2$. Here

$$\lambda(0) = 1, \quad \Pi_0 f = \mu(f) 1. \quad (11)$$

We may also assume $\lambda(z) \neq 0$ for $|z| < r$.

Define

$$a(z) := \int \Pi_z 1 d\mu, \quad b_n(z) := \int N_z^n 1 d\mu. \quad (12)$$

Then a is analytic and $a(0) = 1$. Cauchy's integral formula applied to the analytic remainder gives, for each fixed $j \geq 0$,

$$|b_n^{(j)}(0)| \leq C_j \rho^n. \quad (13)$$

Step 3: the twisted operator generates the Birkhoff moment-generating function. The normalized transfer operator satisfies

$$\int f(g \circ \sigma) d\mu = \int (Pf)g d\mu. \quad (14)$$

From (8) and (14), induction gives

$$M_n(z) := \int e^{zS_n\psi} d\mu = \int P_z^n 1 d\mu. \quad (15)$$

Indeed, (14) transfers one shifted factor at a time. Combining (10), (12), and (15),

$$M_n(z) = \lambda(z)^n a(z) + b_n(z). \quad (16)$$

Step 4: first and second derivatives of the dominant eigenvalue. Choose an analytic eigenvector v_z for $\lambda(z)$, normalized by $\mu(v_z) = 1$. Thus

$$P_z v_z = \lambda(z) v_z, \quad v_0 = 1. \quad (17)$$

Differentiate at $z = 0$ and apply μ . Since $P^* \mu = \mu$ and $P'_0 f = P(\psi f)$,

$$\lambda'(0) = \mu(P'_0 1) = \mu(P\psi) = \mu(\psi) = 0. \quad (18)$$

On the other hand,

$$M_n''(0) = \int (S_n \psi)^2 d\mu. \quad (19)$$

Differentiate (16) twice at 0. Using $\lambda(0) = a(0) = 1$, $\lambda'(0) = 0$, and (13),

$$M_n''(0) = n\lambda''(0) + a''(0) + b_n''(0). \quad (20)$$

Divide by n and let $n \rightarrow \infty$. Equations (7), (13), and (19)–(20) give

$$\lambda''(0) = \sigma_\psi^2. \quad (21)$$

Equivalently, because $\lambda(0) = 1$ and $\lambda'(0) = 0$,

$$(\log \lambda)''(0) = \sigma_\psi^2. \quad (22)$$

Step 5: the nondegenerate central limit theorem. Assume $\sigma_\psi^2 > 0$, and write $\sigma_\psi > 0$ for its positive square root. For fixed $t \in \mathbb{R}$ set

$$z_n := \frac{it}{\sigma_\psi \sqrt{n}}. \quad (23)$$

By (18) and (21), Taylor expansion at 0 gives

$$\lambda(z) = 1 + \frac{\sigma_\psi^2}{2} z^2 + O(|z|^3). \quad (24)$$

Hence

$$\lambda(z_n) = 1 - \frac{t^2}{2n} + O(n^{-3/2}), \quad (25)$$

and therefore

$$\lambda(z_n)^n \longrightarrow e^{-t^2/2}. \quad (26)$$

Also $a(z_n) \rightarrow 1$, while (10)–(12) give $b_n(z_n) \rightarrow 0$. Evaluating (16) at z_n yields

$$\int \exp\left(it \frac{S_n \psi}{\sigma_\psi \sqrt{n}}\right) d\mu \longrightarrow e^{-t^2/2}. \quad (27)$$

Lévy's continuity theorem gives

$$\boxed{\frac{S_n \psi}{\sigma_\psi \sqrt{n}} \Longrightarrow \mathcal{N}(0, 1).} \quad (28)$$

Step 6: the zero-variance and coboundary cases. If $\sigma_\psi^2 = 0$, then (7) directly gives

$$\left\| \frac{S_n \psi}{\sqrt{n}} \right\|_{L^2(\mu)}^2 = \frac{1}{n} \int (S_n \psi)^2 d\mu \longrightarrow 0. \quad (29)$$

Thus the $\sqrt{n}$ -normalized sums converge to 0 in L^2 , hence in probability.

Finally, if

$$\psi = u - u \circ \sigma \quad (30)$$

for some bounded measurable u , then

$$S_n \psi = u - u \circ \sigma^n. \quad (31)$$

Therefore $\|S_n \psi\|_{L^2} \leq 2\|u\|_\infty$, and (7) implies $\sigma_\psi^2 = 0$. No converse Livšic criterion is asserted here. $\square$

Proposition F14.9. Pressure derivative / linear response at symbolic level

Statement. Let $t \mapsto \phi_t \in C^\alpha(\Sigma_A^+, \mathbb{R})$ be a C^1 curve of Hölder potentials on a topologically mixing finite one-sided SFT, and let μ_t be the normalized RPF/Gibbs state. Then the simple RPF eigenvalue, its spectral projection, and $\mu_t(f)$ for each fixed Hölder f vary C^1 locally in t , and

$$\frac{d}{dt}P(\phi_t) = \int \dot{\phi}_t d\mu_t.$$

A C^k or real-analytic potential curve gives the corresponding local regularity of the symbolic RPF data.

Scope note

This is a symbolic linear-response statement; no geometric smooth-system linear-response theorem is included.

Proof. Let $I \subset \mathbb{R}$ be an interval and let

$$t \mapsto \phi_t \in C^\alpha(\Sigma_A^+, \mathbb{R})$$

be a C^1 curve. Write

$$\mathcal{L}_t := \mathcal{L}_{\phi_t}. \tag{1}$$

We separate the operator-perturbation step from the pressure calculation.

Step 1: the transfer operator is a C^1 operator curve. For $f \in C^\alpha(\Sigma_A^+)$,

$$(\mathcal{L}_t f)(x) = \sum_{\sigma y=x} e^{\phi_t(y)} f(y).$$

Because C^α is a Banach algebra and the exponential map is smooth on it, the map $t \mapsto e^{\phi_t}$ is C^1 in C^α . The finite-preimage formula above therefore gives a C^1 map

$$t \mapsto \mathcal{L}_t \quad \text{in } \mathcal{B}(C^\alpha, C^\alpha),$$

with derivative

$$\dot{\mathcal{L}}_t f = \mathcal{L}_t(\dot{\phi}_t f). \tag{2}$$

Indeed, multiplication by a Hölder function is bounded on C^α , and the finite sum over inverse branches preserves the operator-norm differentiation.

Step 2: the simple RPF spectral data vary C^1 locally. Fix $t_0 \in I$. By *Ruelle-Perron-Frobenius theorem for Hölder potentials on mixing SFTs* the leading RPF eigenvalue $\lambda_{t_0} > 0$ is simple and isolated on C^α . Choose a positively oriented circle Γ around λ_{t_0} which contains no other point of the spectrum. Operator-norm continuity of $\mathcal{L}_t$ implies that, after shrinking to a neighborhood J of t_0 , the same circle lies in the resolvent set

of $\mathcal{L}_t$ and isolates exactly one spectral point. Define the Riesz projection

$$\Pi_t = \frac{1}{2\pi i} \int_{\Gamma} (zI - \mathcal{L}_t)^{-1} dz. \quad (3)$$

The resolvent identity and Step 1 show that $t \mapsto \Pi_t$ is C^1 on J . Its rank is locally constant and equals one.

The same construction for the dual operator gives a one-dimensional dual spectral subspace. Hence one may choose local C^1 right and left eigendata h_t and ν_t so that

$$\mathcal{L}_t h_t = \lambda_t h_t, \quad \mathcal{L}_t^* \nu_t = \lambda_t \nu_t, \quad \nu_t(h_t) = 1, \quad (4)$$

with the positive RPF branches selected at t_0 and continued on J . The normalization is legitimate because positivity gives $\nu_{t_0}(h_{t_0}) > 0$, so the relevant pairing remains nonzero after shrinking J if necessary. Put

$$d\mu_t = h_t d\nu_t.$$

Then μ_t is exactly the normalized RPF/Gibbs probability from *Existence and uniqueness of Gibbs/equilibrium state on mixing SFT*. For every fixed $f \in C^\alpha$,

$$t \longmapsto \mu_t(f) = \nu_t(h_t f) \quad (5)$$

is therefore C^1 locally.

Step 3: differentiate the leading eigenvalue. Differentiate the first equation in (4):

$$\dot{\mathcal{L}}_t h_t + \mathcal{L}_t \dot{h}_t = \dot{\lambda}_t h_t + \lambda_t \dot{h}_t. \quad (6)$$

Apply ν_t . Since $\nu_t \mathcal{L}_t = \lambda_t \nu_t$ and $\nu_t(h_t) = 1$, the two terms involving $\dot{h}_t$ cancel, and we obtain

$$\dot{\lambda}_t = \nu_t(\dot{\mathcal{L}}_t h_t). \quad (7)$$

Using (2) and the dual eigenrelation,

$$\begin{aligned} \dot{\lambda}_t &= \nu_t(\mathcal{L}_t(\dot{\phi}_t h_t)) \\ &= \lambda_t \nu_t(\dot{\phi}_t h_t) = \lambda_t \int_{\Sigma_A^+} \dot{\phi}_t d\mu_t. \end{aligned} \quad (8)$$

Consequently

$$\frac{d}{dt} \log \lambda_t = \int_{\Sigma_A^+} \dot{\phi}_t d\mu_t. \quad (9)$$

Existence and uniqueness of Gibbs/equilibrium state on mixing SFT identifies the symbolic pressure with the leading RPF eigenvalue,

$$P(\phi_t) = \log \lambda_t. \quad (10)$$

Combining (9)–(10) gives the claimed pressure derivative

$$\boxed{\frac{d}{dt}P(\phi_t) = \int_{\Sigma_A^+} \dot{\phi}_t d\mu_t.} \quad (11)$$

Step 4: higher regularity and the scope boundary. If $t \mapsto \phi_t$ is C^k , then $t \mapsto \mathcal{L}_t$ is C^k in operator norm. Repeated differentiation of the resolvent in (3) shows that Π_t , λ_t , and normalized right/left eigendata may be chosen C^k locally; hence so may $\mu_t(f)$ for every fixed Hölder f . If the potential curve is real analytic, complexifying the Hölder space and applying analytic perturbation to the isolated simple eigenvalue gives the corresponding local analyticity.

All of these statements concern the fixed symbolic system Σ_A^+ while its potential varies. We have not differentiated a Markov coding, a smooth map, or an SRB state of a varying geometric dynamical system, so no geometric linear-response theorem is asserted. $\square$

14.3 Worked examples

Worked example: Symbolic Dynamics and Thermodynamic Formalism: a model computation

Consider the full two-shift $\{0, 1\}^{\mathbb{Z}}$ and the locally constant potential

$$\phi(x) = a \mathbf{1}_{\{x_0=1\}}.$$

For a word of length n containing k ones, $S_n \phi = ak$. Summing over all 2^n words gives the partition sum

$$Z_n = \sum_{k=0}^n \binom{n}{k} e^{ak} = (1 + e^a)^n.$$

Therefore

$$P(\phi) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n = \log(1 + e^a).$$

The equilibrium measure is Bernoulli with

$$p := \mu(x_0 = 1) = \frac{e^a}{1 + e^a}.$$

Indeed

$$P'(a) = \frac{e^a}{1 + e^a} = p,$$

so the pressure derivative recovers the mean of the observable $\mathbf{1}_{\{x_0=1\}}$. A second derivative gives

$$P''(a) = p(1 - p),$$

the one-step variance, foreshadowing spectral-perturbation formulas for the central limit theorem.

Boundary case: what fails at the edge of the hypotheses

Let the transition matrix be

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The subshift consists of exactly two fixed sequences, $\dots 000\dots$ and $\dots 111\dots$. For the zero potential, both Dirac measures are invariant and both attain pressure 0. The system is not mixing, and the equilibrium state is not unique. This is why mixing/irreducibility hypotheses are load-bearing in the standard uniqueness theorem.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

14.4 Chapter summary

What to remember

- Pressure combines entropy with the average of a potential.
- The transfer operator converts dynamical weighting into a spectral problem.
- Mixing subshifts with Hölder potentials have a unique Gibbs/equilibrium state.
- A spectral gap yields quantitative statistical laws, not merely existence of an invariant measure.

Looking ahead

F15 studies parameters rather than phase points. The measure estimates used to construct good parameter sets are the parameter-space analogue of controlling exceptional orbit sets.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

14.5 Exercises and problems

Exercise 14.1

Difficulty. Core.

Related results. Variational principle for topological pressure; Ruelle–Perron–Frobenius theorem for Hölder potentials on mixing SFTs; Existence and uniqueness of Gibbs/equilibrium state on mixing SFT.

Problem. Work through the worked example ‘the full two-shift with a locally constant potential’. For Variational principle for topological pressure, Ruelle–Perron–Frobenius theorem for Hölder potentials on mixing SFTs, and Existence and uniqueness of Gibbs/equilibrium state on mixing SFT, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 14.2

Difficulty. Core.

Related results. Bowen coding for hyperbolic basic sets; Equilibrium states for Axiom A/basic hyperbolic systems; Exponential decay from transfer-operator spectral gap.

Problem. Analyze the boundary example ‘a reducible subshift showing failure of uniqueness without mixing assumptions’. For Bowen coding for hyperbolic basic sets, Equilibrium states for Axiom A/basic hyperbolic systems, and Exponential decay from transfer-operator spectral gap, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 14.3

Difficulty. Advanced.

Related results. Existence and uniqueness of Gibbs/equilibrium state on mixing SFT.

Problem. For Existence and uniqueness of Gibbs/equilibrium state on mixing SFT, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how

its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 14.4

Difficulty. Advanced.

Related results. Bowen coding for hyperbolic basic sets.

Problem. Stress-test Bowen coding for hyperbolic basic sets. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 14.5

Difficulty. Advanced.

Related results. Variational principle for topological pressure; Ruelle–Perron–Frobenius theorem for Hölder potentials on mixing SFTs.

Problem. Synthesis problem. Use Variational principle for topological pressure and Ruelle–Perron–Frobenius theorem for Hölder potentials on mixing SFTs to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 14.6

Difficulty. Lab.

Related results. None.

Problem. Computational laboratory: carry out the reproducible experiment ‘the full two-shift with a locally constant potential’. Use the computation only as numerical evidence or visualization, not as proof. Include a refinement or convergence test and identify one numerical artifact that could mislead a reader.

Solution sketch. State parameters and truncation/time horizon, compare with an analytically controlled identity or exactly solvable case, repeat after refinement, and explain what the numerical experiment can and cannot establish.

Exercise 14.7

Difficulty. Core.

Related results. Central limit theorem from spectral perturbation.

Problem. Work directly with Central limit theorem from spectral perturbation. Re-state its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 14.8

Difficulty. Core.

Related results. Pressure derivative / linear response at symbolic level.

Problem. Work directly with Pressure derivative / linear response at symbolic level. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Exercise 14.9

Difficulty. Core.

Related results. Entropy/generating-partition interface.

Problem. Work directly with Entropy/generating-partition interface. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Parameter Families, Transversality and Exclusion

Why this chapter matters

Many dynamical theorems are true not for every parameter but for a large Cantor-like set of parameters. The proof pattern is remarkably stable: identify a bad event at each scale, use transversality to estimate the size of the corresponding parameter set, remove it, and prove that the total loss is controlled.

This chapter isolates that logic. It also keeps distinct three notions that are frequently confused: topological genericity, measure-theoretic prevalence, and positive/full Lebesgue measure.

Prerequisites. Lebesgue measure, Borel-Cantelli, elementary differential topology, and F05 resonance estimates.

Learning goals

- Use transversality to convert geometric conditions into quantitative parameter estimates.
- Build nested good-parameter sets by removing resonant intervals or slabs.
- Apply summability and Borel-Cantelli correctly.
- Distinguish Baire residuality from measure-theoretic largeness.

Conceptual roadmap

A parameter derivative bound supplies transversality. Each scale produces a union of bad intervals/slabs. A union bound estimates the loss at that scale. If the losses are summable, the nested intersection retains positive measure and often asymptotically full relative measure. Phase-parameter transfer principles then transport estimates between orbit geometry and parameter geometry.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is designed to control it?* Keep these answers in view while reading the proofs.

15.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F15.1. Transversality

Submanifolds $A, B \subset M$ meet *transversely* at $x \in A \cap B$ if $T_x A + T_x B = T_x M$. For parameter families, transversality is expressed by the corresponding surjectivity condition for the evaluation map.

Definition F15.2. Parameter exclusion

Parameter exclusion is an iterative construction that removes parameters violating scale-dependent nonresonance, expansion, or distortion conditions. The removed sets and their quantitative measure bounds are part of the construction.

Definition F15.3. Cantor parameter set

A *Cantor parameter set* here is a compact perfect totally disconnected set obtained as the intersection of nested closed good-parameter sets after specified resonant intervals or slabs are removed.

Definition F15.4. Baire genericity

A property is *Baire generic* if it holds on a residual set, i.e. a countable intersection of open dense sets. Residuality is not identified with full Lebesgue measure or prevalence without a separate theorem.

15.2 Main results

Theorem F15.1. Transversality and persistence of intersections

Statement. Let $F : B_r(x_0) \times B_s(\lambda_0) \rightarrow \mathbb{R}^m$ be C^1 , with $F(x_0, \lambda_0) = 0$, and put $A = D_x F(x_0, \lambda_0)$. Assume A is invertible, $M = \|A^{-1}\|$, and $\sup \|D_x F(x, \lambda) - A\| \leq (2M)^{-1}$ on the product neighborhood. For every λ with $M\|F(x_0, \lambda)\| \leq r/2$, there is a unique zero $x(\lambda) \in B_r(x_0)$ of $F(\cdot, \lambda)$, and $\|x(\lambda) - x_0\| \leq 2M\|F(x_0, \lambda)\|$. The zero depends C^1 on λ and satisfies $D_\lambda x = -(D_x F)^{-1} D_\lambda F$. In local coordinates, a transverse intersection of complementary-dimensional C^1 submanifolds is exactly such an invertible zero and therefore persists uniquely under sufficiently small C^1 parameter perturbations.

Proof. Fix a parameter λ satisfying

$$M\|F(x_0, \lambda)\| \leq \frac{r}{2}. \quad (1)$$

Define

$$\Psi_\lambda(x) := x - A^{-1}F(x, \lambda).$$

For every (x, λ) in the stated product neighborhood,

$$D_x \Psi_\lambda(x) = I - A^{-1}D_x F(x, \lambda) = A^{-1}(A - D_x F(x, \lambda)).$$

Hence

$$\|D_x \Psi_\lambda(x)\| \leq M \cdot \frac{1}{2M} = \frac{1}{2}. \quad (2)$$

Because the Euclidean ball is convex, the mean-value estimate from (2) gives

$$\|\Psi_\lambda(x) - \Psi_\lambda(y)\| \leq \frac{1}{2}\|x - y\| \quad (x, y \in B_r(x_0)). \quad (3)$$

Thus Ψ_λ is a contraction once we know it maps the ball into itself.

At the center,

$$\|\Psi_\lambda(x_0) - x_0\| = \|A^{-1}F(x_0, \lambda)\| \leq M\|F(x_0, \lambda)\| \leq \frac{r}{2}. \quad (4)$$

For $x \in B_r(x_0)$, (3) and (4) yield

$$\begin{aligned} \|\Psi_\lambda(x) - x_0\| &\leq \|\Psi_\lambda(x) - \Psi_\lambda(x_0)\| + \|\Psi_\lambda(x_0) - x_0\| \\ &\leq \frac{1}{2}\|x - x_0\| + \frac{r}{2} \leq r. \end{aligned} \quad (5)$$

So Ψ_λ maps the closed ball $\overline{B_r(x_0)}$ into itself. The contraction mapping theorem gives a unique fixed point $x(\lambda)$ in that ball. The fixed-point equation is

$$x(\lambda) = x(\lambda) - A^{-1}F(x(\lambda), \lambda),$$

and since A^{-1} is injective,

$$F(x(\lambda), \lambda) = 0. \quad (6)$$

Uniqueness of the fixed point is therefore uniqueness of the zero in the controlled ball.

The contraction estimate also gives the announced displacement bound. Indeed,

$$\begin{aligned} \|x(\lambda) - x_0\| &\leq \|\Psi_\lambda(x(\lambda)) - \Psi_\lambda(x_0)\| + \|\Psi_\lambda(x_0) - x_0\| \\ &\leq \frac{1}{2}\|x(\lambda) - x_0\| + M\|F(x_0, \lambda)\|. \end{aligned}$$

Moving the first term to the left gives

$$\boxed{\|x(\lambda) - x_0\| \leq 2M\|F(x_0, \lambda)\|}. \quad (7)$$

At every such zero, the derivative $D_x F$ remains invertible. Indeed,

$$D_x F = A[I + A^{-1}(D_x F - A)],$$

and the second factor is invertible by a Neumann series because its perturbation from I has norm at most $1/2$. The classical implicit function theorem therefore shows that

the uniquely selected zero $x(\lambda)$ depends C^1 on λ . Differentiating $F(x(\lambda), \lambda) = 0$ gives

$$D_x F(x(\lambda), \lambda) D_\lambda x(\lambda) + D_\lambda F(x(\lambda), \lambda) = 0,$$

so

$$D_\lambda x(\lambda) = -D_x F(x(\lambda), \lambda)^{-1} D_\lambda F(x(\lambda), \lambda). \quad (8)$$

Finally, consider two complementary-dimensional submanifolds written in local coordinates as parameter-dependent embeddings

$$a(s, \lambda), \quad b(t, \lambda)$$

into $\mathbb{R}^m$, with $\dim s + \dim t = m$. Their intersections solve

$$\mathcal{F}(s, t, \lambda) := a(s, \lambda) - b(t, \lambda) = 0. \quad (9)$$

The derivative of $\mathcal{F}$ in (s, t) has image

$$T_x A_\lambda + T_x B_\lambda.$$

Because the source and target dimensions are both m , this derivative is invertible exactly when the two tangent spaces span $\mathbb{R}^m$, that is, exactly when the intersection is transverse. Therefore a transverse intersection satisfies the zero theorem above and persists, uniquely in the chosen local chart, under sufficiently small C^1 parameter perturbations. $\square$

Lemma F15.2. Parameter exclusion union bound

Statement. Let $\Lambda \subset [-R, R]^d$ be measurable and suppose a bad parameter set E is contained in a finite union of resonant slabs $S_\alpha = \{\lambda \in \Lambda : |\langle k_\alpha, \lambda \rangle - m_\alpha| < \varepsilon_\alpha\}$, with $k_\alpha \in \mathbb{Z}^d \setminus \{0\}$. Then $\text{Leb}(E) \leq 2(2R)^{d-1} \sum_\alpha \varepsilon_\alpha / |k_\alpha|_\infty$. In particular, any scale-by-scale family with summable right-hand sides has summable excluded measure.

Proof. By hypothesis,

$$E \subset \bigcup_{\alpha \in A} S_\alpha,$$

where A is finite and

$$S_\alpha = \{\lambda \in \Lambda : |\langle k_\alpha, \lambda \rangle - m_\alpha| < \varepsilon_\alpha\}.$$

Therefore monotonicity and finite subadditivity of Lebesgue measure give

$$\text{Leb}(E) \leq \text{Leb}\left(\bigcup_{\alpha \in A} S_\alpha\right) \leq \sum_{\alpha \in A} \text{Leb}(S_\alpha). \quad (1)$$

For each α , *Resonant slab measure estimate* applies inside the same cube $[-R, R]^d$ and yields

$$\text{Leb}(S_\alpha) \leq 2(2R)^{d-1} \frac{\varepsilon_\alpha}{|k_\alpha|_\infty}. \quad (2)$$

Substituting (2) into (1),

$$\text{Leb}(E) \leq 2(2R)^{d-1} \sum_{\alpha \in A} \frac{\varepsilon_\alpha}{|k_\alpha|_\infty}. \quad (3)$$

For the scale-by-scale consequence, let E_n be the excluded set at stage n and let M_n denote the right-hand side of (3) for the resonances used at that stage. If

$$\sum_{n=1}^{\infty} M_n < \infty,$$

then (3) gives

$$\sum_{n=1}^{\infty} \text{Leb}(E_n) \leq \sum_{n=1}^{\infty} M_n < \infty.$$

Thus the excluded measures are summable, which is precisely the input needed by the subsequent Borel-Cantelli parameter-exclusion theorem. $\square$

Theorem F15.3. Cantor parameter construction with positive/full asymptotic measure

Statement. Let $G_0 \subset \mathbb{R}^d$ be compact with $0 < \text{Leb}(G_0) < \infty$. For $n \geq 1$, let $B_n \subset G_{n-1}$ be relatively open and measurable, set $G_n = G_{n-1} \setminus B_n$, and suppose $\text{Leb}(B_n) \leq M_n$. Then the nested limit $G_\infty = \bigcap_{n \geq 0} G_n$ is compact and

$$\text{Leb}(G_\infty) \geq \text{Leb}(G_0) - \sum_{n=1}^{\infty} M_n.$$

In particular it has positive measure whenever the total bound is smaller than $\text{Leb}(G_0)$. For a family of such constructions indexed by $\varepsilon \rightarrow 0$, if $\sum_n M_n(\varepsilon) = o(\text{Leb}(G_0(\varepsilon)))$, then $\text{Leb}(G_\infty(\varepsilon))/\text{Leb}(G_0(\varepsilon)) \rightarrow 1$. In dimension one, if every G_n is a finite union of pairwise disjoint nondegenerate closed intervals, each component of G_n contains at least two components of G_{n+1} , and the maximal component length tends to zero, then G_∞ is a Cantor set (compact, perfect, and totally disconnected).

Proof. Because B_n is relatively open in the compact set G_{n-1} ,

$$G_n = G_{n-1} \setminus B_n$$

is compact. Inductively, all G_n are compact and

$$G_0 \supset G_1 \supset G_2 \supset \cdots. \quad (1)$$

Hence

$$G_\infty := \bigcap_{n=0}^{\infty} G_n \quad (2)$$

is compact as an intersection of closed subsets of the compact set G_0 .

The sets removed at different stages are pairwise disjoint. Indeed, if $m > n$, then

$B_m \subset G_{m-1} \subset G_n$, whereas $B_n \cap G_n = \emptyset$. Therefore

$$G_0 \setminus G_N = \bigsqcup_{n=1}^N B_n \quad (3)$$

and

$$\text{Leb}(G_N) = \text{Leb}(G_0) - \sum_{n=1}^N \text{Leb}(B_n) \geq \text{Leb}(G_0) - \sum_{n=1}^N M_n. \quad (4)$$

Since G_0 has finite measure, continuity of Lebesgue measure from above for the decreasing sequence (1) gives

$$\text{Leb}(G_\infty) = \lim_{N \rightarrow \infty} \text{Leb}(G_N). \quad (5)$$

Letting $N \rightarrow \infty$ in (4),

$$\boxed{\text{Leb}(G_\infty) \geq \text{Leb}(G_0) - \sum_{n=1}^{\infty} M_n.} \quad (6)$$

Thus if the total exclusion bound is strictly smaller than $\text{Leb}(G_0)$, the surviving set has positive measure. In applications *Parameter exclusion union bound* supplies the numbers M_n from the resonant slabs at stage n .

For an ε -dependent family, divide (6) by $\text{Leb}(G_0(\varepsilon)) > 0$:

$$1 \geq \frac{\text{Leb}(G_\infty(\varepsilon))}{\text{Leb}(G_0(\varepsilon))} \geq 1 - \frac{\sum_n M_n(\varepsilon)}{\text{Leb}(G_0(\varepsilon))}. \quad (7)$$

If the last quotient tends to zero, the squeeze theorem yields

$$\frac{\text{Leb}(G_\infty(\varepsilon))}{\text{Leb}(G_0(\varepsilon))} \rightarrow 1. \quad (8)$$

This is the precise meaning of full asymptotic measure here.

It remains to justify the optional one-dimensional Cantor conclusion. Assume now $d = 1$ and the interval hypotheses in the statement. Write

$$\text{mesh}(G_n) := \max\{|I| : I \text{ is an interval component of } G_n\} \rightarrow 0. \quad (9)$$

Compactness of G_∞ was already proved.

To prove total disconnectedness, take distinct $x, y \in G_\infty$. Choose n so large that

$$\text{mesh}(G_n) < |x - y|.$$

Then x and y cannot lie in the same interval component of G_n . Since the components are pairwise disjoint closed intervals, they are separated by an open gap. Thus no connected subset of G_∞ can contain both x and y . Every connected subset of G_∞ is therefore a singleton.

Finally we prove that G_∞ has no isolated point. Fix $x \in G_\infty$ and let I_n be the interval component of G_n containing x . By hypothesis, I_n contains at least two interval

components of G_{n+1} . One contains x ; choose another one and call it J_{n+1} . The same splitting hypothesis applies to J_{n+1} and to every descendant component. Selecting one descendant at every later stage gives a nested sequence of nonempty compact intervals. Their intersection is nonempty by compactness; choose

$$y_n \in G_\infty \cap J_{n+1}.$$

Then $y_n \neq x$ and, because both points lie in I_n ,

$$|y_n - x| \leq |I_n| \leq \text{mesh}(G_n) \longrightarrow 0. \quad (10)$$

Hence every point of G_∞ is an accumulation point. The limit set is compact, perfect, and totally disconnected, so it is a Cantor set. $\square$

Proposition F15.4. Borel-Cantelli lemma and dynamical parameter use

Statement. Let Λ be a finite-measure parameter space and B_n subset Λ measurable. If $\sum_n \text{Leb}(B_n) < \infty$, then almost every parameter belongs to only finitely many B_n . In a staged exclusion construction this means almost every parameter is excluded only finitely often.

Scope note

The divergent-series direction requires an additional independence or quasi-independence hypothesis.

Proof. Let $(\Lambda, \mathcal{A}, \text{Leb})$ be the finite-measure parameter space and let $B_n \in \mathcal{A}$. Define the limsup set

$$\limsup_{n \rightarrow \infty} B_n := \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} B_n.$$

A parameter λ belongs to this set exactly when it belongs to infinitely many of the bad sets B_n .

By hypothesis,

$$\sum_{n=1}^{\infty} \text{Leb}(B_n) < \infty.$$

Borel-Cantelli lemmas, the first Borel-Cantelli lemma, therefore gives

$$\text{Leb}\left(\limsup_{n \rightarrow \infty} B_n\right) = 0. \quad (1)$$

Consequently, for almost every $\lambda \in \Lambda$ there exists an index $N(\lambda)$ such that

$$\lambda \notin B_n \quad \text{for every } n \geq N(\lambda). \quad (2)$$

This is the precise meaning of “the parameter is excluded only finitely often”: after some parameter-dependent finite stage, that parameter no longer lies in any of the stage- n exclusion sets.

Nothing in the argument uses independence, mixing, or any relation between distinct B_n . Conversely, if the series $\sum_n \text{Leb}(B_n)$ diverges, (1) no longer follows from Borel-Cantelli alone. A conclusion asserting infinitely many visits in the divergent case requires a separate independence or quasi-independence estimate, exactly as stated. $\square$

Lemma F15.5. Parameter derivative distortion control

Statement. Let f be a C^2 diffeomorphism on an open interval containing $J, f(J), \dots, f^n(J)$, and suppose

$$|(\log |f'|)'(x)| \leq L$$

on their union. Then for all $x, y \in J$,

$$\left| \log \frac{|(f^n)'(x)|}{|(f^n)'(y)|} \right| \leq L \sum_{k=0}^{n-1} |f^k(J)|.$$

Hence if the sum on the right is at most S , the derivative distortion of f^n on J is at most e^{LS} . More generally, if $\Phi : I \rightarrow \Phi(I)$ is any C^1 interval diffeomorphism with $\text{osc}_I \log |\Phi'| \leq D$, then every Lebesgue-measurable $E \subset I$ satisfies

$$e^{-D \frac{|\Phi(I)|}{|I|}} |E| \leq |\Phi(E)| \leq e^{D \frac{|\Phi(I)|}{|I|}} |E|.$$

Thus the same conclusion applies to any phase-parameter or parameter coordinate map once its logarithmic derivative distortion is bounded by D .

Proof. Because f is a diffeomorphism on every interval appearing in the orbit of J , the chain rule gives, for $x \in J$,

$$(f^n)'(x) = \prod_{k=0}^{n-1} f'(f^k(x)). \quad (1)$$

Taking absolute values and logarithms,

$$\log |(f^n)'(x)| = \sum_{k=0}^{n-1} \log |f'(f^k(x))|. \quad (2)$$

Thus for $x, y \in J$,

$$\begin{aligned} \left| \log \frac{|(f^n)'(x)|}{|(f^n)'(y)|} \right| &\leq \sum_{k=0}^{n-1} \left| \log |f'(f^k(x))| - \log |f'(f^k(y))| \right| \\ &\leq L \sum_{k=0}^{n-1} |f^k(x) - f^k(y)|. \end{aligned} \quad (3)$$

Both $f^k(x)$ and $f^k(y)$ lie in the interval $f^k(J)$, so

$$|f^k(x) - f^k(y)| \leq |f^k(J)|. \quad (4)$$

Substitution of (4) into (3) yields

$$\left| \log \frac{|(f^n)'(x)|}{|(f^n)'(y)|} \right| \leq L \sum_{k=0}^{n-1} |f^k(J)|. \quad (5)$$

If the interval-length sum is at most S , exponentiating (5) gives

$$e^{-LS} \leq \frac{|(f^n)'(x)|}{|(f^n)'(y)|} \leq e^{LS}, \quad (6)$$

which is the asserted derivative-distortion bound.

We now prove the general coordinate-map conclusion. Let $\Phi : I \rightarrow \Phi(I)$ be a C^1 interval diffeomorphism and suppose

$$\text{osc}_I \log |\Phi'| := \sup_{x,y \in I} |\log |\Phi'(x)| - \log |\Phi'(y)|| \leq D. \quad (7)$$

Put

$$m := \inf_I |\Phi'|, \quad M := \sup_I |\Phi'|.$$

Then (7) implies $M/m \leq e^D$. By the mean-value theorem (equivalently, by integrating $|\Phi'|$ over I),

$$\frac{|\Phi(I)|}{|I|}$$

lies between m and M . Denote this average derivative by a . Since $m \leq a \leq M$ and $M/m \leq e^D$,

$$e^{-D}a \leq |\Phi'(x)| \leq e^D a \quad (x \in I). \quad (8)$$

For every Lebesgue-measurable $E \subset I$, the one-dimensional change-of-variables formula therefore gives

$$|\Phi(E)| = \int_E |\Phi'(x)| dx. \quad (9)$$

Using (8) in (9),

$$e^{-D}a|E| \leq |\Phi(E)| \leq e^D a|E|.$$

Replacing a by $|\Phi(I)|/|I|$ proves

$$\boxed{e^{-D} \frac{|\Phi(I)|}{|I|} |E| \leq |\Phi(E)| \leq e^D \frac{|\Phi(I)|}{|I|} |E|}. \quad (10)$$

The proof of (10) depends only on the distortion bound (7), so it applies without change when Φ is a phase-parameter holonomy or a parameter coordinate map whose logarithmic derivative has been controlled by some other argument. $\square$

Theorem F15.6. Abstract phase-parameter transfer principle

Statement. Let $(I_j)_{j \in J}$ be a finite or countable family of pairwise disjoint bounded intervals and, for each j , let $\Phi_j : I_j \rightarrow J_j$ be a C^1 interval diffeomorphism. Assume

uniform constants $D \geq 0$ and $0 < a \leq b < \infty$ such that

$$\operatorname{osc}_{I_j} \log |\Phi'_j| \leq D, \quad a \leq \frac{|J_j|}{|I_j|} \leq b \quad (j \in J),$$

and assume the phase intervals have overlap multiplicity at most $M < \infty$: every point belongs to at most M of the J_j . For a measurable phase set E , put

$$E_{\text{par}} := \bigcup_{j \in J} \Phi_j^{-1}(E \cap J_j).$$

Then, branchwise,

$$e^{-D} \frac{|E \cap J_j|}{|J_j|} \leq \frac{|\Phi_j^{-1}(E \cap J_j)|}{|I_j|} \leq e^D \frac{|E \cap J_j|}{|J_j|},$$

and globally

$$\frac{e^{-D}}{b} |E \cap \bigcup_j J_j| \leq |E_{\text{par}}| \leq \frac{Me^D}{a} |E \cap \bigcup_j J_j|.$$

Thus any phase-set measure estimate transfers quantitatively to parameter space once distortion, transversality scale, and branch multiplicity are controlled.

Scope note

This proposition is an abstract transfer principle; model-specific phase-parameter theorems require additional dynamical input.

Proof. Fix a branch j , and put

$$F_j := \Phi_j^{-1}(E \cap J_j) \subset I_j.$$

Apply the second part of *Parameter derivative distortion control* to $\Phi_j : I_j \rightarrow J_j$ and the measurable set F_j . Since $\Phi_j(F_j) = E \cap J_j$, it gives

$$e^{-D} \frac{|J_j|}{|I_j|} |F_j| \leq |E \cap J_j| \leq e^D \frac{|J_j|}{|I_j|} |F_j|. \quad (1)$$

Solving (1) for $|F_j|/|I_j|$ yields the branchwise estimate

$$e^{-D} \frac{|E \cap J_j|}{|J_j|} \leq \frac{|F_j|}{|I_j|} \leq e^D \frac{|E \cap J_j|}{|J_j|}. \quad (2)$$

This proves the first displayed conclusion.

The scale bounds

$$a \leq \frac{|J_j|}{|I_j|} \leq b$$

are equivalent to

$$\frac{1}{b} \leq \frac{|I_j|}{|J_j|} \leq \frac{1}{a}. \quad (3)$$

Multiplying (2) by $|I_j|$ and using (3),

$$\frac{e^{-D}}{b}|E \cap J_j| \leq |F_j| \leq \frac{e^D}{a}|E \cap J_j|. \quad (4)$$

Because the parameter intervals I_j are pairwise disjoint, the sets $F_j \subset I_j$ are pairwise disjoint. Hence countable additivity gives

$$|E_{\text{par}}| = \sum_j |F_j|. \quad (5)$$

Summing the lower estimate in (4) and using

$$\sum_j |E \cap J_j| \geq \left| E \cap \bigcup_j J_j \right| \quad (6)$$

gives

$$|E_{\text{par}}| \geq \frac{e^{-D}}{b} \left| E \cap \bigcup_j J_j \right|. \quad (7)$$

For the upper estimate, the overlap-multiplicity hypothesis means

$$\sum_j \mathbf{1}_{J_j}(y) \leq M \quad \text{for a.e. } y.$$

Therefore, by Tonelli,

$$\sum_j |E \cap J_j| = \int_E \sum_j \mathbf{1}_{J_j}(y) dy \leq M \left| E \cap \bigcup_j J_j \right|. \quad (8)$$

Summing the upper estimate in (4) and using (8) yields

$$|E_{\text{par}}| \leq \frac{Me^D}{a} \left| E \cap \bigcup_j J_j \right|. \quad (9)$$

Equations (7) and (9) are exactly the asserted global transfer bounds. All constants arise from the three advertised sources: logarithmic Jacobian distortion D , branch-scale transversality a, b , and phase multiplicity M . No dynamical information beyond those hypotheses has been used. $\square$

Proposition F15.7. Genericity notions and Baire residuality

Statement. In a Baire space, every countable intersection of open dense sets is dense. A set is residual when it contains such an intersection, and a property is Baire-generic when it holds on a residual set. Measure-generic, prevalent, and full-measure statements are recorded separately and are not inferred from residuality without an additional theorem.

Proof. Let X be a Baire space. By the defining Baire property, whenever $U_1, U_2, \dots$ are open dense subsets of X , their countable intersection

$$G := \bigcap_{n=1}^{\infty} U_n$$

is dense in X .

Now let $R \subseteq X$ be residual in the convention fixed in this volume. Thus there exist open dense sets U_n such that

$$G = \bigcap_{n=1}^{\infty} U_n \subseteq R.$$

Since G is dense, every nonempty open set $V \subseteq X$ meets G . Because $G \subseteq R$, the same V meets R . Hence R is dense. Accordingly, a property that holds on a residual set holds on a dense set of Baire-generic points, which is precisely the convention used in this project.

The final clause is a logical boundary. The argument above uses only topology: open sets, density, and countable intersections. It does not introduce a measure, a probability class, or the translation structure used in definitions of prevalence. Therefore this proposition alone cannot imply that a residual set has full measure, is measure-generic, or is prevalent. Any such implication must be supplied by a separate theorem with the extra structure and hypotheses stated explicitly. This proves both the Baire-residuality assertion and the advertised separation of genericity notions. $\square$

15.3 Worked examples

Worked example: Parameter Families, Transversality and Exclusion: a model computation

On $[0, 1]$, fix $\tau > 1$ and $\gamma > 0$. For each integer $q \geq 1$, remove

$$B_q = \bigcup_{p=0}^q \left\{ \omega : \left| \omega - \frac{p}{q} \right| < \gamma q^{-(\tau+1)} \right\}.$$

There are at most $q + 1$ intervals, each of length $2\gamma q^{-(\tau+1)}$, so

$$|B_q| \leq 2\gamma(q + 1)q^{-(\tau+1)} \leq 4\gamma q^{-\tau}.$$

Hence

$$\left| \bigcup_{q \geq 1} B_q \right| \leq 4\gamma \sum_{q \geq 1} q^{-\tau}.$$

Because $\tau > 1$, the series converges. For sufficiently small γ , the complement

has positive measure. A point in that complement satisfies

$$\left| \omega - \frac{p}{q} \right| \geq \gamma q^{-(\tau+1)}$$

for every p, q . This elementary construction is the one-dimensional prototype of a KAM Cantor set.

Boundary case: what fails at the edge of the hypotheses

The classical middle-thirds Cantor construction removes one third of every surviving interval at each stage. After n steps the remaining length is

$$\left(\frac{2}{3} \right)^n \rightarrow 0.$$

The limiting set is compact, perfect, and totally disconnected—a genuine Cantor set—but it has zero Lebesgue measure. Thus the word “Cantor” says nothing by itself about positive measure. A parameter-exclusion proof must include quantitative summability if it wants a measure conclusion.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

15.4 Chapter summary

What to remember

- Transversality turns dynamical degeneracy into a codimension/measure estimate in parameter space.
- Positive-measure Cantor constructions depend on summable losses, not on topology alone.
- Borel-Cantelli controls infinitely repeated bad events when their measures are summable.
- Residual, prevalent, and full-measure sets are different notions of largeness.

Looking ahead

F16 applies hyperbolic ideas at the level of operators on spaces of dynamical systems. Transversality to stable laminations then becomes the mechanism behind universal scaling laws.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

15.5 Exercises and problems**Exercise 15.1**

Difficulty. Core.

Related results. Transversality and persistence of intersections; Parameter exclusion union bound; Cantor parameter construction with positive/full asymptotic measure.

Problem. Work through the worked example 'remove intervals around rationals to build a Diophantine Cantor set'. For Transversality and persistence of intersections, Parameter exclusion union bound, and Cantor parameter construction with positive/full asymptotic measure, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

Exercise 15.2

Difficulty. Core.

Related results. Borel-Cantelli parameter-exclusion interface; Parameter derivative distortion control; Abstract phase-parameter transfer principle.

Problem. Analyze the boundary example 'a nonsummable exclusion scheme whose good set can have zero measure'. For Borel-Cantelli parameter-exclusion interface, Parameter derivative distortion control, and Abstract phase-parameter transfer principle, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.

Solution sketch. Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.

Exercise 15.3

Difficulty. Advanced.

Related results. Cantor parameter construction with positive/full asymptotic measure.

Problem. For Cantor parameter construction with positive/full asymptotic measure, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.

Solution sketch. Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.

Exercise 15.4

Difficulty. Advanced.

Related results. Borel-Cantelli parameter-exclusion interface.

Problem. Stress-test Borel-Cantelli parameter-exclusion interface. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.

Solution sketch. Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.

Exercise 15.5

Difficulty. Advanced.

Related results. Transversality and persistence of intersections; Parameter exclusion union bound.

Problem. Synthesis problem. Use Transversality and persistence of intersections and Parameter exclusion union bound to derive a nontrivial corollary in a model from this volume. Write the logical chain explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

Exercise 15.6

Difficulty. Core.

Related results. Genericity notions and Baire residuality.

Problem. Work directly with Genericity notions and Baire residuality. Restate its hypotheses in the notation of this volume, construct the simplest nontrivial example satisfying them, and derive one immediate corollary from the theorem. Then exhibit one modification of the example for which a hypothesis fails.

Solution sketch. First make a complete hypothesis checklist. Verify the example, apply the theorem once, derive the corollary explicitly, and finally identify the failed hypothesis in the modified example. The later full solution should include the calculations rather than only a verbal discussion.

Abstract Renormalization and Hyperbolicity

Why this chapter matters

Renormalization studies dynamics by repeatedly zooming in on a distinguished return structure and rescaling it. The resulting renormalization operator acts not on phase space but on a space of dynamical systems. If that operator has a hyperbolic fixed point or hyperbolic invariant set, the stable-manifold machinery of F10 becomes a theory of universality.

This chapter deliberately stays abstract. It develops the functional-analytic engine that later specialist volumes can apply to concrete Feigenbaum, critical-circle, cocycle, or near-parabolic renormalizations.

Prerequisites. F10, Banach-space differential calculus, compact/quasicompact operators, and basic spectral projections.

Learning goals

- Apply stable-manifold theory to maps on Banach spaces.
- Understand hyperbolic persistence and stable laminations for renormalization operators.
- Read universal scaling as transversality to a stable lamination.
- Use compact/quasicompact spectral theory to isolate finite-dimensional unstable directions.

Conceptual roadmap

A hyperbolic fixed point of the renormalization operator has stable and unstable spectral directions. Stable-manifold theory produces the codimension- k set of systems converging to the fixed point. A one-parameter family crossing that stable set transversely is expanded in the unstable coordinate under renormalization, which forces geometric parameter scaling. Quasicompactness is what makes the unstable spectrum finite-dimensional in many infinite-dimensional settings.

Check your understanding

Before reading the formal statements, try to answer two questions in words: *What is the principal obstruction in this chapter?* and *Which hypothesis is de-*

signed to control it? Keep these answers in view while reading the proofs.

16.1 Core definitions

Intuition

The definitions below are the vocabulary used by the formal results. Read each together with the first theorem in which it appears; the point is not to memorize notation in isolation.

Definition F16.1. Renormalization operator

A *renormalization operator* $\mathcal{R}$ acts on a specified function, moduli, or Banach space by taking a dynamically defined return or restriction and rescaling it to a fixed normalization.

Definition F16.2. Hyperbolic fixed point

A fixed point x_* of a C^1 map $\mathcal{R}$ on a Banach space or Banach manifold is *hyperbolic* if $\sigma(D\mathcal{R}(x_*))$ is disjoint from the unit circle and the tangent space splits into invariant stable and unstable spectral subspaces.

Definition F16.3. Stable lamination

A *stable lamination* is a locally varying family of immersed leaves $W^s(x)$ over a hyperbolic invariant set such that points on the same leaf have forward iterates converging to the same asymptotic orbit at the stated rate.

Definition F16.4. Universality

Universality means that suitably normalized geometric, dynamical, or parameter scaling data converge to quantities determined by a renormalization fixed point or hyperbolic class rather than by the initial system. The scaling observable must always be named explicitly.

16.2 Main results

Theorem F16.1. Banach-space hyperbolic fixed-point stable manifold theorem

Statement. Let F be a C^r map, $r \geq 1$, on a Banach space or Banach manifold with fixed point p . Suppose $DF(p)$ has a topological invariant splitting $E^s \oplus E^u$, the restriction to E^u is invertible, and

$$r(DF(p)|_{E^s}) < 1, \quad r((DF(p)|_{E^u})^{-1}) < 1.$$

Then F has a C^r local stable manifold tangent to E^s , locally characterized by forward orbits converging exponentially to p . If F is a local C^r diffeomorphism, it also has a C^r local unstable manifold tangent to E^u .

Proof. It is enough to prove the assertion in a Banach chart. Translate the fixed point to the origin and write

$$F(x) = Lx + R(x), \quad L = DF(0), \quad R(0) = 0, \quad DR(0) = 0. \quad (1)$$

By hypothesis there is a topological L -invariant splitting

$$X = E^s \oplus E^u, \quad (2)$$

$L_u := L|_{E^u}$ is invertible, and

$$r(L_s) < 1, \quad r(L_u^{-1}) < 1, \quad (3)$$

where $L_s = L|_{E^s}$. Let π_s, π_u be the bounded projections associated with (2).

Step 1: adapted norms. Choose numbers

$$\max\{r(L_s), r(L_u^{-1})\} < a < b < 1. \quad (4)$$

The spectral-radius formula gives equivalent norms on E^s and E^u for which

$$\|L_s\| \leq a, \quad \|L_u^{-1}\| \leq a. \quad (5)$$

Indeed, for a bounded operator A with $r(A) < a$, the norm $\|v\|_A := \sup_{n \geq 0} a^{-n} \|A^n v\|_0$ is finite and equivalent to the original norm after multiplying the first finitely many iterates by a common constant. Apply this to L_s and L_u^{-1} . Equip (2) with the maximum product norm. Then $\|\pi_s\|, \|\pi_u\| \leq 1$ and (5) remains true.

Since $DR(0) = 0$, choose $\rho > 0$ so small that F is defined on $B_\rho(0)$ and

$$\|DR(x)\| \leq \eta \quad (\|x\| \leq \rho), \quad (6)$$

where $\eta > 0$ will be chosen below. Thus

$$\|R(x) - R(y)\| \leq \eta \|x - y\| \quad (7)$$

whenever the segment joining x and y stays in $B_\rho(0)$. No cutoff extension is introduced; all fixed points below are constructed inside this ball.

Step 2: the weighted sequence space and the Lyapunov-Perron map. Let

$$\mathcal{Y}_b := \left\{ z = (z_n)_{n \geq 0} : \|z\|_b := \sup_{n \geq 0} b^{-n} \|z_n\| < \infty \right\}. \quad (8)$$

This is a Banach space. For a small stable coordinate $\xi \in E^s$, define $\mathcal{T}_\xi$ on sequences

by

$$\begin{aligned} (\mathcal{T}_\xi z)_n^s &= L_s^n \xi + \sum_{j=0}^{n-1} L_s^{n-1-j} \pi_s R(z_j), \\ (\mathcal{T}_\xi z)_n^u &= - \sum_{j=n}^{\infty} L_u^{n-1-j} \pi_u R(z_j). \end{aligned} \tag{9}$$

For $j \geq n$ the notation in the second line means $L_u^{n-1-j} = (L_u^{-1})^{j-n+1}$.

Suppose $\|z\|_b \leq \rho$. Then $\|z_j\| \leq b^j \rho \leq \rho$, so (6)-(7) apply. For two such sequences z, w , (5), (7), and geometric summation give

$$\begin{aligned} b^{-n} \|(\mathcal{T}_\xi z - \mathcal{T}_\xi w)_n^s\| &\leq \eta \|z - w\|_b \sum_{j=0}^{n-1} a^{n-1-j} b^{j-n} \\ &\leq \frac{\eta}{b-a} \|z - w\|_b, \end{aligned} \tag{10}$$

and

$$\begin{aligned} b^{-n} \|(\mathcal{T}_\xi z - \mathcal{T}_\xi w)_n^u\| &\leq \eta \|z - w\|_b \sum_{j=n}^{\infty} a^{j-n+1} b^{j-n} \\ &\leq \frac{a\eta}{1-ab} \|z - w\|_b. \end{aligned} \tag{11}$$

Hence, with

$$q := \max \left\{ \frac{\eta}{b-a}, \frac{a\eta}{1-ab} \right\}, \tag{12}$$

we have

$$\|\mathcal{T}_\xi z - \mathcal{T}_\xi w\|_b \leq q \|z - w\|_b. \tag{13}$$

Choose η so small that $q < 1/2$.

Because $R(0) = 0$, the same estimates with $w = 0$ show

$$\|\mathcal{T}_\xi z\|_b \leq \|\xi\| + q \|z\|_b. \tag{14}$$

Thus if

$$\|\xi\| \leq (1-q)\rho_0, \quad 0 < \rho_0 < \rho, \tag{15}$$

then $\mathcal{T}_\xi$ maps the closed ρ_0 -ball of $\mathcal{Y}_b$ into itself and is a contraction there. Banach's fixed-point theorem therefore gives a unique fixed sequence

$$z(\xi) = (z_n(\xi))_{n \geq 0} \in \mathcal{Y}_b, \quad \|z(\xi)\|_b \leq \rho_0. \tag{16}$$

Step 3: the fixed sequence is exactly a forward orbit. Subtract the two formulas (9) at times $n+1$ and n . The stable part gives

$$z_{n+1}^s = L_s z_n^s + \pi_s R(z_n), \tag{17}$$

while cancellation of the first term in the infinite unstable sum gives

$$z_{n+1}^u = L_u z_n^u + \pi_u R(z_n). \tag{18}$$

Hence

$$z_{n+1} = Lz_n + R(z_n) = F(z_n). \quad (19)$$

The stable coordinate at time zero is

$$\pi_s z_0 = \xi. \quad (20)$$

Define

$$h^s(\xi) := \pi_u z_0(\xi) = - \sum_{j=0}^{\infty} L_u^{-j-1} \pi_u R(z_j(\xi)). \quad (21)$$

Then the local stable set produced by the construction is the graph

$$W_{\text{loc}}^s(0) := \{\xi + h^s(\xi) : \|\xi\| < (1-q)\rho_0\}. \quad (22)$$

By (16), every point on this graph satisfies

$$\|F^n(\xi + h^s(\xi))\| = \|z_n(\xi)\| \leq \rho_0 b^n, \quad (23)$$

so its forward orbit converges exponentially to 0.

Step 4: converse characterization of the local stable set. Suppose y has a forward orbit $y_n = F^n y$ contained in $B_{\rho_0}(0)$ and satisfying

$$\sup_{n \geq 0} b^{-n} \|y_n\| < \infty. \quad (24)$$

Iterating the stable equation forward gives the first formula in (9) with $\xi = \pi_s y$. For the unstable component, rewrite

$$y_n^u = L_u^{-1} y_{n+1}^u - L_u^{-1} \pi_u R(y_n) \quad (25)$$

and iterate it to time $N > n$:

$$y_n^u = L_u^{n-N} y_N^u - \sum_{j=n}^{N-1} L_u^{n-1-j} \pi_u R(y_j). \quad (26)$$

By (5) and (24),

$$\|L_u^{n-N} y_N^u\| \leq a^{N-n} C b^N = C b^n (ab)^{N-n} \rightarrow 0. \quad (27)$$

Letting $N \rightarrow \infty$ gives the second formula in (9). Thus (y_n) is a fixed point of $\mathcal{T}_{\pi_s y}$, and uniqueness implies $y = z_0(\pi_s y)$. This proves the stated local characterization by exponentially convergent forward orbits.

Step 5: C^r regularity and tangency. Consider the fixed-point equation

$$\Phi(\xi, z) := z - \mathcal{T}_\xi z = 0 \quad (28)$$

on the open part of $E^s \times \mathcal{Y}_b$ for which all coordinates lie in $B_\rho(0)$. Because F is C^r , the substitution operator $z \mapsto (R(z_n))_{n \geq 0}$ is C^r on a smaller sequence ball. To see the only point specific to the weighted space, note that continuity of $D^j R$ at 0 gives uniform bounds on $D^j R$ in a sufficiently small ball, and products of j weighted variations carry

the factor $b^{jn} \leq b^n$ for $j \geq 1$. The geometric Green sums in (9) therefore converge uniformly also after differentiating up to order r .

Moreover

$$D_z \Phi = I - D_z \mathcal{T}_\xi \quad (29)$$

is invertible throughout the chosen neighborhood because $\|D_z \mathcal{T}_\xi\| \leq q < 1$. The Banach implicit-function theorem yields $\xi \mapsto z(\xi)$ of class C^r , hence h^s and the graph (22) are C^r .

At $\xi = 0$, $z(0) = 0$. Since $DR(0) = 0$, differentiating (9) there gives

$$D_\xi z_n(0)\eta = L_s^n \eta, \quad Dh^s(0)\eta = 0. \quad (30)$$

Consequently

$$T_0 W_{\text{loc}}^s(0) = E^s. \quad (31)$$

Equation (19) also gives local forward invariance after the graph radius is shrunk once more.

Step 6: unstable manifold when a local inverse exists. If F is a local C^r diffeomorphism, let $G = F^{-1}$ near 0. Its derivative is L^{-1} . Relative to the splitting $E^u \oplus E^s$, the stable restriction of $DG(0)$ is L_u^{-1} , while the inverse of its unstable restriction is L_s . Both have spectral radius < 1 by (3). Applying Steps 1–5 to G produces a C^r local stable manifold tangent to E^u ; viewed for F , this is the local unstable manifold $W_{\text{loc}}^u(0)$.

Finally, on a Banach manifold choose a chart centered at p . The preceding construction is invariant under the chart change because it is characterized by the orbit property and tangency. Mapping the two local graphs back to the manifold proves the theorem. $\square$

Theorem F16.2. Hyperbolic set persistence for Banach dynamical systems

Statement. Let F be a C^1 diffeomorphism of a Banach space or Banach manifold and let Λ be a compact uniformly hyperbolic F -invariant set. There are a fixed orbit-tube radius and a C^1 neighborhood of F such that every sufficiently small C^1 perturbation G has a unique continuous continuation map $h_G : \Lambda \rightarrow M$, uniformly close to the inclusion, satisfying

$$G \circ h_G = h_G \circ F.$$

The map h_G is a homeomorphism onto a compact G -invariant set Λ_G , and Λ_G is uniformly hyperbolic; its stable and unstable bundles are graph-close to those of Λ and converge uniformly to them as $G \rightarrow F$ in C^1 . Uniqueness is among continuations whose full G -orbits stay in the fixed tube around the corresponding F -orbits.

Scope note

Identifying Λ_G with a maximal invariant set in a neighborhood additionally requires a local-maximality/isolating hypothesis.

Proof. We prove the Banach-space statement. The final manifold clause is explained

at the end. All constants below are uniform in the base point $x \in \Lambda$.

Step 1: the Green operator for the linearized orbit equation. For $x \in \Lambda$ write

$$x_n := F^n x, \quad A_n := DF(x_n), \quad n \in \mathbb{Z}, \quad (1)$$

and let P_n^s, P_n^u be the projections onto $E_{x_n}^s, E_{x_n}^u$. For integers n, m , denote by $A(n, m)$ the evolution from time m to time n : for $n > m$,

$$A(n, m) = A_{n-1} \cdots A_m,$$

for $n = m$ it is the identity, and for $n < m$ it is the corresponding inverse product. Hyperbolicity gives

$$\|A(n, m)P_m^s\| \leq C\lambda^{n-m} \quad (n \geq m), \quad (2)$$

and

$$\|A(n, m)P_m^u\| \leq C\lambda^{m-n} \quad (n \leq m). \quad (3)$$

On $\ell^\infty(\mathbb{Z}, X)$ consider

$$(\mathcal{L}_x v)_n := v_{n+1} - A_n v_n. \quad (4)$$

For $e = (e_n) \in \ell^\infty$, define

$$(\mathcal{G}_x e)_n := \sum_{j=-\infty}^{n-1} A(n, j+1)P_{j+1}^s e_j - \sum_{j=n}^{\infty} A(n, j+1)P_{j+1}^u e_j. \quad (5)$$

The two series converge absolutely by (2)-(3), and

$$\|\mathcal{G}_x e\|_\infty \leq K \|e\|_\infty, \quad K := \frac{2C}{1-\lambda}. \quad (6)$$

A direct subtraction of successive terms in (5) gives $\mathcal{L}_x \mathcal{G}_x e = e$. Conversely, if $\mathcal{L}_x v = 0$ and v is bounded, then its stable component at time 0 would grow without bound under backward iteration unless it vanishes, and its unstable component would grow without bound under forward iteration unless it vanishes. Hence $v = 0$. Thus

$$\mathcal{L}_x^{-1} = \mathcal{G}_x, \quad \|\mathcal{L}_x^{-1}\| \leq K. \quad (7)$$

This is the bounded-solution form of the exponential dichotomy, proved here with its uniform constant.

Step 2: a uniform local neighborhood. Choose $r > 0$ so small that the closed $2r$ -neighborhood of Λ lies in U and, by uniform continuity of DF on a sufficiently small neighborhood of the compact set Λ ,

$$\|DF(y) - DF(x)\| \leq \delta \quad \text{whenever } x \in \Lambda, \quad |y - x| \leq 2r. \quad (8)$$

The number $\delta > 0$ will be fixed below. Shrink the eventual C^1 perturbation size so that on this neighborhood

$$\|G - F\|_{C^0} \leq \delta, \quad \|DG - DF\| \leq \delta. \quad (9)$$

Then, whenever $\|v\| \leq r$,

$$\|DG(x_n + v) - A_n\| \leq 2\delta. \quad (10)$$

Step 3: every F -orbit in Λ has a unique nearby G -orbit. For fixed $x \in \Lambda$ and $v = (v_n) \in \ell^\infty(\mathbb{Z}, X)$ with $\|v\|_\infty \leq r$, put

$$\mathcal{N}_{x,G}(v)_n := G(x_n + v_n) - F(x_n) - A_n v_n. \quad (11)$$

At $v = 0$,

$$\|\mathcal{N}_{x,G}(0)\|_\infty \leq \delta. \quad (12)$$

By (10) and the mean-value theorem,

$$\|\mathcal{N}_{x,G}(v) - \mathcal{N}_{x,G}(w)\|_\infty \leq 2\delta\|v - w\|_\infty. \quad (13)$$

Define

$$\mathcal{T}_{x,G}(v) := \mathcal{G}_x \mathcal{N}_{x,G}(v). \quad (14)$$

Choose δ so small that

$$q := 2K\delta < \frac{1}{2}, \quad K\delta \leq \frac{r}{2}. \quad (15)$$

Then $\mathcal{T}_{x,G}$ is a contraction with constant at most q , and for $\|v\|_\infty \leq r$,

$$\|\mathcal{T}_{x,G}v\|_\infty \leq K\delta + qr \leq r. \quad (16)$$

Thus it has a unique fixed point $v^x = (v_n^x)_{n \in \mathbb{Z}}$ in the closed r -ball, and in fact

$$\|v^x\|_\infty \leq \frac{K\delta}{1 - q}. \quad (17)$$

The fixed-point equation and (4) imply

$$v_{n+1}^x - A_n v_n^x = G(x_n + v_n^x) - F(x_n) - A_n v_n^x,$$

so

$$z_n^x := x_n + v_n^x \quad \text{satisfies} \quad z_{n+1}^x = G(z_n^x) \quad (18)$$

for every $n \in \mathbb{Z}$. Define

$$h_G(x) := z_0^x = x + v_0^x. \quad (19)$$

By (17), choosing δ still smaller makes $\sup_{x \in \Lambda} |h_G(x) - x| < \eta$ and also places the image inside the prescribed neighborhood V .

Uniqueness is important. If (z_n) is any G -orbit with $\sup_n |z_n - x_n| \leq r$, then $v_n = z_n - x_n$ is a fixed point of $\mathcal{T}_{x,G}$, hence $v = v^x$. Thus there is exactly one G -orbit in this fixed tube around the F -orbit of x .

Step 4: conjugacy and injectivity. The orbit of Fx is the shift $(x_{n+1})_{n \in \mathbb{Z}}$. The shifted sequence (v_{n+1}^x) produces a G -orbit staying within the same tube of that shifted F -orbit. By the uniqueness just proved, it is the fixed sequence associated with Fx . At time zero,

$$h_G(Fx) = G(h_G(x)). \quad (20)$$

We next prove the local expansivity needed for injectivity. Shrink r if necessary so that

$$\|DF(x+w) - DF(x)\| \leq \delta \quad (x \in \Lambda, |w| \leq 2r), \quad (21)$$

with $K\delta < 1$. Suppose $x, y \in \Lambda$ satisfy

$$|F^n x - F^n y| \leq 2r \quad (n \in \mathbb{Z}). \quad (22)$$

Put $d_n = F^n y - F^n x$. Then

$$d_{n+1} - A_n d_n = R_n(d_n), \quad (23)$$

where

$$R_n(w) := F(x_n + w) - F(x_n) - A_n w$$

satisfies $|R_n(w)| \leq \delta|w|$ on the $2r$ -ball. Equation (7) therefore gives

$$\|d\|_\infty \leq K\delta\|d\|_\infty. \quad (24)$$

Hence $d = 0$ and $x = y$. Thus $F|_\Lambda$ is expansive at scale $2r$.

If $h_G(x) = h_G(y)$, then (20) gives $h_G(F^n x) = h_G(F^n y)$ for all $n \in \mathbb{Z}$. By (17),

$$|F^n x - F^n y| \leq |F^n x - h_G(F^n x)| + |h_G(F^n y) - F^n y| < 2r,$$

so expansivity gives $x = y$. Therefore h_G is injective.

Step 5: continuity of the continuation map. The only subtlety here is that two nearby points of a hyperbolic set need not have uniformly close full orbits. The exponential Green kernel resolves this.

Fix the time-zero coordinate of (5). For every $N \geq 1$, split both sums into $|j| \leq N$ and tails. By (2)–(3), the operator norm of the two tails is at most

$$C_1 \lambda^N \quad (25)$$

with C_1 independent of x . The same estimate holds at every iterate of the contraction (14), because all nonlinear terms stay uniformly bounded in the fixed r -ball.

Start the fixed-point iteration from $v^{(0)} = 0$ and set $v^{(m+1)} = \mathcal{T}_{x,G} v^{(m)}$. Uniform contraction gives

$$\|v^{(m)} - v^x\|_\infty \leq r q^m. \quad (26)$$

For fixed m and N , replace every Green sum occurring in the first m iterations by its truncation to $|j| \leq N$. Its time-zero value depends only on the finite orbit segment

$$F^{-N'} x, \dots, F^{N'} x \quad (27)$$

for some finite $N' = N'(m, N)$ and depends continuously on those points, because F, G, DF and the splitting projections are continuous on the compact set under consideration. The total error made by the truncations is bounded by $C_2 \lambda^N$, uniformly in x . Given $\epsilon_0 > 0$, first choose m so that $r q^m < \epsilon_0/3$, next choose N so that $C_2 \lambda^N < \epsilon_0/3$,

and finally use continuity of the finite orbit segment (27). This proves

$$x_j \rightarrow x \implies v_0^{x_j} \rightarrow v_0^x. \quad (28)$$

Hence h_G is continuous. A continuous injection from compact Λ into the Hausdorff space X is a homeomorphism onto its image. Set

$$\Lambda_G := h_G(\Lambda). \quad (29)$$

Then Λ_G is compact and, by (20), G -invariant.

Step 6: roughness of the hyperbolic splitting. Fix $x \in \Lambda$ and its shadowed orbit $z_n = h_G(x_n)$. Put

$$B_n := DG(z_n). \quad (30)$$

By (9)–(10) and (17), after shrinking the perturbation,

$$\sup_n \|B_n - A_n\| \leq \varepsilon_1, \quad (31)$$

where ε_1 can be made arbitrarily small.

We give the stable graph construction; the unstable one is the same argument backwards. Choose a rate a with $\lambda < a < 1$. For a prescribed $\xi \in E_{x_0}^s$, seek a forward B -orbit $w = (w_n)_{n \geq 0}$ with $P_0^s w_0 = \xi$ in the weighted space

$$\|w\|_a := \sup_{n \geq 0} a^{-n} |w_n| < \infty. \quad (32)$$

Writing $D_n = B_n - A_n$, the equation $w_{n+1} = B_n w_n$ is equivalent to

$$\begin{aligned} w_n^s &= A(n, 0)\xi + \sum_{j=0}^{n-1} A(n, j+1)P_{j+1}^s D_j w_j, \\ w_n^u &= - \sum_{j=n}^{\infty} A(n, j+1)P_{j+1}^u D_j w_j. \end{aligned} \quad (33)$$

Exactly the geometric-series estimates used in (5) show that the right-hand side is a contraction on the weighted sequence space when ε_1 is small, with contraction constant

$$q_1 \leq C_3 \varepsilon_1 \left(\frac{1}{a - \lambda} + \frac{1}{1 - a\lambda} \right) < 1. \quad (34)$$

Thus (33) has a unique solution depending linearly on ξ . Its initial unstable component defines a bounded linear map

$$\Gamma_x^s : E_x^s \longrightarrow E_x^u, \quad E_{G, h_G(x)}^s := \{\xi + \Gamma_x^s \xi : \xi \in E_x^s\}. \quad (35)$$

The contraction estimate gives

$$\|\Gamma_x^s\| \leq C_4 \varepsilon_1, \quad (36)$$

uniformly in x . Shifting a solution of (33) by one time step and using uniqueness shows

$$B_0 E_{G, h_G(x)}^s = E_{G, h_G(Fx)}^s. \quad (37)$$

Moreover the weighted bound gives

$$|B(n, 0)v^s| \leq C_5 a^n |v^s| \quad (v^s \in E_{G, h_G(x)}^s). \quad (38)$$

Applying the same construction to the backward cocycle produces $E_{G, h_G(x)}^u$ as a graph over E_x^u , with

$$|B(-n, 0)v^u| \leq C_5 a^n |v^u|. \quad (39)$$

For small ε_1 , the two graph norms are so small that the graph subspaces remain complementary. Hence

$$X = E_{G, h_G(x)}^s \oplus E_{G, h_G(x)}^u \quad (40)$$

is a uniformly hyperbolic splitting for G on Λ_G .

The graph fixed points depend continuously on x . The proof is the same finite-tail argument as in Step 5, now with the contraction (33): truncate the exponentially decaying sums, use continuity of the finite orbit segment, and then let the tail length tend to infinity. Consequently the two bundles in (40) are continuous. Estimate (36) and its unstable analogue also prove that their graph norms tend uniformly to zero as $\|G - F\|_{C^1} \rightarrow 0$.

Step 7: uniqueness and the Banach-manifold version. The uniqueness assertion in the theorem is precisely the tube uniqueness in Step 3: any conjugacy $\tilde{h}$ whose orbit $G^n \tilde{h}(x) = \tilde{h}(F^n x)$ stays in the fixed r -tube of $F^n x$ must use the unique shadowing orbit and hence equals h_G .

If the phase space is a C^1 Banach manifold, compactness of Λ provides finitely many charts and tangent trivializations with uniformly bounded transition derivatives. Carry out Steps 1–6 in these coordinates; shrinking the common radius ensures that each successive orbit equation is written inside one of the chosen charts. Equivalent norms and finite chart changes only alter the uniform constants. The constructions agree on chart overlaps by their uniqueness, so they patch to the asserted manifold continuation. $\square$

Theorem F16.3. Stable lamination near a codimension- k hyperbolic set

Statement. Let F be a C^r diffeomorphism of a Banach manifold, $r \geq 1$, and let Λ be a compact uniformly hyperbolic invariant set with invariant splitting $E^s \oplus E^u$ and $\dim E^u = k < \infty$. There exist $\rho > 0$, $C \geq 1$, and $0 < \theta < 1$ such that every $x \in \Lambda$ has a C^r embedded local stable disk $W_\rho^s(x)$ tangent to E_x^s and of codimension k , with local invariance and

$$d(F^n y, F^n x) \leq C \theta^n d(y, x) \quad (y \in W_\rho^s(x), n \geq 0).$$

Conversely, after fixing the same local tube and rate, every point whose forward orbit stays in that tube and converges to the orbit of x at the prescribed exponential rate lies on the corresponding local stable disk. In local trivializations the stable

graph maps depend continuously on x in the C^1 topology, and the coherent plaque family forms the local stable lamination.

Scope note

Transverse differentiability stronger than the stated continuous C^1 dependence requires an additional bunching/regular-holonomy theorem.

Proof. Let

$$T_x M = E_x^s \oplus E_x^u, \quad x \in \Lambda, \quad (1)$$

be the continuous invariant hyperbolic splitting. By compactness there are constants $C \geq 1$ and $0 < \lambda < 1$ such that for every $x \in \Lambda$ and $n \geq 0$,

$$\|DF_x^n v^s\| \leq C\lambda^n \|v^s\|, \quad \|DF_x^{-n} v^u\| \leq C\lambda^n \|v^u\|. \quad (2)$$

The norms in (2) may be taken from any fixed finite family of equivalent chart norms covering the compact set.

Step 1: uniform local orbit equations. Fix $x \in \Lambda$ and write $x_n = F^n x$. Choose Banach charts centered at x_n and tangent trivializations over a finite atlas covering Λ . In those charts the map from a neighborhood of x_n to a neighborhood of x_{n+1} has the form

$$f_n(v) = A_n v + r_n(v), \quad A_n = DF(x_n), \quad r_n(0) = 0, \quad Dr_n(0) = 0. \quad (3)$$

Here the domain and target Banach spaces are the corresponding tangent models; the invariant splitting (1) is carried along by A_n .

Because the atlas is finite, F is C^r , and Λ is compact, there is a common radius $\rho > 0$ such that all relevant coordinate maps are defined on the ρ -balls and

$$\|Dr_n(v)\| \leq \eta \quad (\|v\| \leq \rho, \text{ extall } n, x), \quad (4)$$

where $\eta > 0$ can be made arbitrarily small by decreasing ρ . The same finite-atlas argument gives uniform bounds for $D^j r_n$, $2 \leq j \leq r$, on a slightly smaller ball.

Let P_n^s, P_n^u be the projections associated with (1), and let $A(n, m)$ denote the linear evolution from time m to time n . Then (2) is equivalent, after increasing C if necessary, to

$$\|A(n, m)P_m^s\| \leq C\lambda^{n-m} \quad (n \geq m), \quad (5)$$

and

$$\|A(n, m)P_m^u\| \leq C\lambda^{m-n} \quad (n \leq m). \quad (6)$$

Step 2: nonautonomous Lyapunov-Perron contraction. Choose

$$\lambda < b < 1 \quad (7)$$

and let $\mathcal{Y}_{x,b}$ be the Banach space of forward tangent sequences $v = (v_n)_{n \geq 0}$ with

$$\|v\|_b := \sup_{n \geq 0} b^{-n} \|v_n\| < \infty. \quad (8)$$

For $\xi \in E_x^s$, define

$$\begin{aligned} (\mathcal{T}_{x,\xi} v)_n^s &= A(n, 0)\xi + \sum_{j=0}^{n-1} A(n, j+1)P_{j+1}^s r_j(v_j), \\ (\mathcal{T}_{x,\xi} v)_n^u &= - \sum_{j=n}^{\infty} A(n, j+1)P_{j+1}^u r_j(v_j). \end{aligned} \quad (9)$$

The second series uses the inverse evolution on the unstable bundle. For two sequences in the closed ρ_0 -ball of $\mathcal{Y}_{x,b}$, with $0 < \rho_0 < \rho$, (4)-(6) give

$$\begin{aligned} b^{-n} \|\Delta(\mathcal{T}^s)_n\| &\leq C\eta \|v - w\|_b \sum_{j=0}^{n-1} \lambda^{n-1-j} b^{j-n} \\ &\leq \frac{C\eta}{b-\lambda} \|v - w\|_b, \end{aligned} \quad (10)$$

and

$$\begin{aligned} b^{-n} \|\Delta(\mathcal{T}^u)_n\| &\leq C\eta \|v - w\|_b \sum_{j=n}^{\infty} \lambda^{j+1-n} b^{j-n} \\ &\leq \frac{C\eta\lambda}{1-\lambda b} \|v - w\|_b. \end{aligned} \quad (11)$$

Choose ρ so small that

$$q := \max \left\{ \frac{C\eta}{b-\lambda}, \frac{C\eta\lambda}{1-\lambda b} \right\} < \frac{1}{2}. \quad (12)$$

Also

$$\|\mathcal{T}_{x,\xi} v\|_b \leq C\|\xi\| + q\|v\|_b. \quad (13)$$

Thus, for

$$\|\xi\| \leq \varepsilon_s := \frac{(1-q)\rho_0}{C}, \quad (14)$$

$\mathcal{T}_{x,\xi}$ is a self-map and contraction of the closed ρ_0 -ball. It has a unique fixed point $v(x, \xi)$. All constants are independent of $x \in \Lambda$.

Step 3: stable disks, tangency, and contraction. Exactly as in the fixed-point calculation, subtracting successive formulas in (9) shows that the fixed sequence satisfies

$$v_{n+1} = f_n(v_n). \quad (15)$$

Its stable time-zero coordinate is $P_0^s v_0 = \xi$. Define

$$h_x^s(\xi) := P_0^u v_0(x, \xi), \quad W_{\text{loc}}^s(x) := \kappa_x^{-1} \{ \xi + h_x^s(\xi) : \|\xi\| < \varepsilon_s \}, \quad (16)$$

where κ_x is the chosen chart centered at x . The contraction fixed-point theorem with parameters, or equivalently the Banach implicit-function theorem applied to $v - \mathcal{T}_{x,\xi} v = 0$, gives C^r dependence on ξ . Uniform bounds on $D^j r_n$ justify differentiating the convergent Green sums. At $\xi = 0$ we have $v = 0$, and because $Dr_n(0) = 0$,

$$D_\xi h_x^s(0) = 0. \quad (17)$$

Therefore

$$T_x W_{\text{loc}}^s(x) = E_x^s. \quad (18)$$

Since $\dim E_x^u = k$, every stable disk has codimension k .

The weighted bound and (13) give

$$\|v_n(x, \xi)\| \leq C_0 b^n \|\xi\| \quad (19)$$

for a uniform C_0 . On a smaller graph ball the map $\xi \mapsto \xi + h_x^s(\xi)$ is bi-Lipschitz with constants independent of x , because $Dh_x^s(0) = 0$ and the graph slopes can be made uniformly small. Changing back to the finite atlas therefore gives constants $C_1 \geq 1$ and $0 < \theta < 1$ (take any $b < \theta < 1$ after shrinking charts if desired) such that

$$d(F^n y, F^n x) \leq C_1 \theta^n d(y, x), \quad y \in W_{\text{loc}}^s(x), \quad n \geq 0, \quad (20)$$

as long as the orbit is read in the chosen local plaques.

Shifting the fixed sequence one step gives the Lyapunov-Perron fixed sequence based at Fx . By uniqueness,

$$F(W_{\text{loc}}^s(x)) \cap B_\rho(Fx) \subset W_{\text{loc}}^s(Fx). \quad (21)$$

After choosing the standard nested plaque radii, a smaller plaque at x is mapped entirely into the plaque at Fx ; this is the usual local invariance.

Step 4: converse characterization. Suppose y lies in the common chart tube of the forward orbit of x and its coordinate differences w_n satisfy

$$\sup_{n \geq 0} b^{-n} \|w_n\| < \infty. \quad (22)$$

Variation of constants gives the stable formula in (9). On the unstable component, iterate the equation backward from a terminal time N :

$$w_n^u = A(n, N) w_N^u - \sum_{j=n}^{N-1} A(n, j+1) P_{j+1}^u r_j(w_j). \quad (23)$$

By (6) and (22),

$$\|A(n, N) w_N^u\| \leq C \lambda^{N-n} C' b^N = C C' b^n (\lambda b)^{N-n} \rightarrow 0. \quad (24)$$

Letting $N \rightarrow \infty$ gives the unstable formula in (9). Hence the entire sequence is the unique fixed point corresponding to $\xi = P_0^s w_0$, and $y \in W_{\text{loc}}^s(x)$. Thus the plaques are exactly the local forward orbits converging at the prescribed exponential rate.

Step 5: continuous C^1 variation with the base point. Fix $x_0 \in \Lambda$. For x near x_0 , use one fixed member of the finite atlas for the time-zero neighborhood and compatible members for any prescribed finite orbit segment. The coefficients A_n , projections $P_n^{s,u}$, and remainders r_n then vary continuously with x on that finite segment.

The fixed point in (9) may be approximated uniformly in x in two stages. First perform m contraction iterates starting from the zero sequence; the error is at most $\rho_0 q^m$. Second truncate every infinite unstable Green sum at time N ; by (6), (8), and the

uniform bound on the nonlinear terms, the discarded tail is at most $C_2(\lambda b)^N$. For fixed m, N , only a finite orbit segment and finitely many continuously varying chart data are used. Hence the time-zero graph h_x^s varies continuously in the uniform norm.

Differentiate the fixed-point equation in ξ . The derivative $D_\xi v$ satisfies a linear fixed-point equation whose contraction part is bounded by the same $q < 1$; its coefficients are $Dr_n(v_n)$ and the hyperbolic evolution. Repeat the same contraction-iteration and tail truncation argument. It follows that

$$x_j \rightarrow x \implies h_{x_j}^s \rightarrow h_x^s \text{ in } C^1 \quad (25)$$

after the natural local trivializations. Notice that this proves precisely continuous transverse dependence in the C^1 topology. No higher transverse differentiability follows from the hypotheses used here.

Step 6: lamination structure. Choose $x_0 \in \Lambda$ and a small k -dimensional disk Σ through x_0 tangent to $E_{x_0}^u$. By (18), (25), and transversality, after shrinking the neighborhood every stable plaque meeting it intersects Σ in at most one point, and every plaque sufficiently close to that of x_0 intersects Σ once. If two of the plaques intersect, their forward orbits converge exponentially to each other; applying the converse characterization in a smaller common tube shows that their overlapping pieces belong to the same stable leaf. Thus the plaque family is coherent.

Let $T \subset \Sigma$ be the set of these intersection points. In a fixed splitting chart, the plaques are graphs over one stable ball, indexed by $t \in T$, and (25) says that the graph depends continuously on t in the C^1 topology. Write $P_t : B^s \rightarrow M$ for the resulting C^r plaque parametrization. Then

$$\Phi : B^s \times T \longrightarrow M, \quad \Phi(\xi, t) = P_t(\xi), \quad (26)$$

is a homeomorphism onto the local union of plaques, with C^r embeddings in the B^s variable. These are exactly local lamination charts. Hence the stable disks form the asserted local stable lamination.

All constructions are independent of the auxiliary chart choices by the uniqueness in Step 4, so they patch over the compact hyperbolic set. This completes the proof. $\square$

Proposition F16.4. Renormalization horseshoe abstract model

Statement. Let R be a C^1 diffeomorphism of a Banach manifold and let Λ be a compact uniformly hyperbolic invariant Cantor set with $\dim E^u < \infty$. Assume a finite small Markov coding has been verified for this model: a finite 0-1 matrix A , the two-sided subshift (Σ_A, σ) , Markov rectangles of diameter below a stable/unstable plaque scale, and a continuous surjective finite-to-one coding $\pi : \Sigma_A \rightarrow \Lambda$ with $R \circ \pi = \pi \circ \sigma$. Then identical future symbols imply membership in the corresponding local stable plaque, identical past symbols imply membership in the local unstable plaque, and symbolic splicing agrees with the local-product bracket whenever the splice is admissible. If π is injective, it is a topological conjugacy and identifies the full asymptotic stable/unstable equivalence relations. Under sufficiently small C^1 perturbations, the hyperbolic continuation transports the same symbolic factor

(and conjugacy when injective) to the continued set.

Scope note

The proposition starts from a verified finite Markov coding; existence of such a coding for an arbitrary Banach hyperbolic Cantor set is not part of this result.

Proof. We prove the repaired abstract interface. Let $R : M \rightarrow M$ be a C^1 diffeomorphism of a Banach manifold, and let $\Lambda \subset M$ be a compact uniformly hyperbolic invariant set with invariant splitting $E^s \oplus E^u$ and $\dim E^u < \infty$. Fix a plaque scale from *Stable lamination near a hyperbolic invariant set*: there is $\rho > 0$ such that, for $x, y \in \Lambda$,

$$d(R^n x, R^n y) < \rho \quad (n \geq 0) \implies y \in W_{\text{loc}}^s(x), \quad (1)$$

and, by applying the same theorem to R^{-1} ,

$$d(R^{-n} x, R^{-n} y) < \rho \quad (n \geq 0) \implies y \in W_{\text{loc}}^u(x). \quad (2)$$

After decreasing the plaque scale if necessary, the corresponding local stable and unstable plaques have the usual unique local-product intersection.

Assume now that a finite Markov coding has been verified for this model. Precisely, let $Q_1, \dots, Q_m$ be Markov rectangles covering Λ with

$$\text{diam } Q_i < \rho \quad (1 \leq i \leq m), \quad (3)$$

let $A = (A_{ij})$ be the associated transition matrix, and let

$$\Sigma_A := \{\xi = (\xi_n)_{n \in \mathbb{Z}} : A_{\xi_n \xi_{n+1}} = 1 \text{ for all } n \in \mathbb{Z}\} \quad (4)$$

with left shift σ . Let

$$\pi : \Sigma_A \longrightarrow \Lambda \quad (5)$$

be the continuous surjective finite-to-one coding map furnished by the verified Markov construction, so that

$$R \circ \pi = \pi \circ \sigma, \quad R^n \pi(\xi) \in Q_{\xi_n} \quad (n \in \mathbb{Z}). \quad (6)$$

The proof identifies exactly what hyperbolicity adds to this symbolic coding and what persists under perturbation.

Step 1: future symbolic agreement gives geometric stability. Let $\xi, \eta \in \Sigma_A$ satisfy

$$\xi_n = \eta_n \quad (n \geq 0). \quad (7)$$

Write $x = \pi(\xi)$, $y = \pi(\eta)$. By (6), for each $n \geq 0$, both $R^n x$ and $R^n y$ lie in the same rectangle Q_{ξ_n} . Hence by (3),

$$d(R^n x, R^n y) < \rho \quad (n \geq 0). \quad (8)$$

The converse tube characterization (1) therefore gives

$$y \in W_{\text{loc}}^s(x). \quad (9)$$

Stable lamination near a hyperbolic invariant set then gives constants $C \geq 1$ and $0 < \theta < 1$, uniform on the compact hyperbolic set, such that

$$d(R^n x, R^n y) \leq C\theta^n d(x, y) \quad (n \geq 0). \quad (10)$$

Thus agreement of the nonnegative itinerary is precisely a symbolic sufficient condition for local stable equivalence, with quantitative exponential contraction.

Step 2: past symbolic agreement gives geometric instability. If instead

$$\xi_n = \eta_n \quad (n \leq 0), \quad (11)$$

then, for every $n \geq 0$, the points $R^{-n}x$ and $R^{-n}y$ lie in the same rectangle $Q_{\xi_{-n}}$. Equations (2)–(3) give

$$y \in W_{\text{loc}}^u(x), \quad (12)$$

and the unstable version of *Stable lamination near a hyperbolic invariant set* yields exponential contraction under backward iteration. Hence past combinatorics encode the local unstable direction exactly as future combinatorics encode the local stable direction.

Step 3: symbolic splicing is the local-product bracket. Suppose $\xi, \eta \in \Sigma_A$ are close enough that the local product bracket $[x, y]$ is defined, where

$$x = \pi(\xi), \quad y = \pi(\eta), \quad [x, y] := W_{\text{loc}}^u(x) \cap W_{\text{loc}}^s(y). \quad (13)$$

Assume that the past-future splice $\zeta \in \Sigma_A$ is admissible and is defined by

$$\zeta_n = \xi_n \quad (n \leq 0), \quad \zeta_n = \eta_n \quad (n \geq 0), \quad (14)$$

with the common central symbol required by admissibility. Set $z = \pi(\zeta)$. Step 2 applied to ζ and ξ gives $z \in W_{\text{loc}}^u(x)$, while Step 1 applied to ζ and η gives $z \in W_{\text{loc}}^s(y)$. By uniqueness of local product intersections,

$$\pi(\zeta) = z = [x, y]. \quad (15)$$

Thus the geometric local product is exactly the symbolic operation of joining a past to a future whenever that splice is admissible.

Step 4: when the coding is one-to-one, it is a conjugacy and identifies the asymptotic stable/unstable relations. The previous steps require only semiconjugacy. Suppose now that the model-specific theorem gives injectivity of π , as happens when the renormalization invariant set has already been proved to be a genuine shift horseshoe rather than merely a Markov factor. Since Σ_A is compact and Λ is Hausdorff, the continuous bijection π is a homeomorphism. Equation (6) then says that

$$\pi \circ \sigma = R \circ \pi \quad (16)$$

is a topological conjugacy.

Define the asymptotic stable and unstable relations by

$$\begin{aligned}\eta \sim_s \xi &\iff d_\Sigma(\sigma^n \eta, \sigma^n \xi) \longrightarrow 0 \quad (n \rightarrow +\infty), \\ \eta \sim_u \xi &\iff d_\Sigma(\sigma^{-n} \eta, \sigma^{-n} \xi) \longrightarrow 0 \quad (n \rightarrow +\infty).\end{aligned}\tag{17}$$

For any compatible metric on the compact shift space, uniform continuity of π and π^{-1} , together with (16), gives

$$\begin{aligned}\eta \sim_s \xi &\iff d(R^n \pi \eta, R^n \pi \xi) \longrightarrow 0, \\ \eta \sim_u \xi &\iff d(R^{-n} \pi \eta, R^{-n} \pi \xi) \longrightarrow 0.\end{aligned}\tag{18}$$

Thus a genuine horseshoe conjugacy identifies the full topological stable/unstable equivalence relations. In particular, identical future coordinates imply the local stable-plaque conclusion of Step 1 and identical past coordinates imply the local unstable-plaque conclusion of Step 2. We deliberately do not claim the converse with “identical from time zero” at Markov boundaries without an additional boundary-free coding convention.

Step 5: the symbolic model persists with the hyperbolic set. Let G be a sufficiently small C^1 perturbation of R . *Hyperbolic set persistence for Banach dynamical systems* supplies the hyperbolic continuation

$$h_G : \Lambda \longrightarrow \Lambda_G, \quad G \circ h_G = h_G \circ R,\tag{19}$$

where h_G is a homeomorphism onto the compact uniformly hyperbolic set Λ_G . Define

$$\pi_G := h_G \circ \pi : \Sigma_A \longrightarrow \Lambda_G.\tag{20}$$

Then

$$\begin{aligned}G \circ \pi_G &= G \circ h_G \circ \pi = h_G \circ R \circ \pi \\ &= h_G \circ \pi \circ \sigma = \pi_G \circ \sigma.\end{aligned}\tag{21}$$

Thus the same subshift remains a continuous surjective finite-to-one symbolic model for $G|_{\Lambda_G}$. If π was injective, then so is π_G , because h_G is injective; therefore the conjugacy case also persists.

Finally, the stable/unstable interpretation transports to Λ_G . For example, if ξ and η have the same future, Step 1 gives $d(R^n \pi \xi, R^n \pi \eta) \rightarrow 0$. Since $G^n \circ h_G = h_G \circ R^n$ on Λ and h_G is uniformly continuous on the compact set Λ ,

$$d(G^n \pi_G \xi, G^n \pi_G \eta) \longrightarrow 0.\tag{22}$$

After choosing the perturbation and plaque boxes uniformly small, the local stable-plaque conclusion follows from the hyperbolicity of Λ_G ; the past/unstable statement is identical for negative time. This proves persistence of both the symbolic factor and its stable/unstable meaning.

The proposition deliberately does not claim that every Banach-manifold hyperbolic Cantor set automatically possesses a finite Markov partition. In finite-dimensional

basic-set settings such a coding is supplied by the Bowen theorem proved above as *Bowen coding for hyperbolic basic sets*. In concrete renormalization settings, the existence and possible injectivity of the coding are model-specific inputs (for example the renormalization horseshoe theorems of Lyubich and their later extensions). The present Foundation result proves the common hyperbolic interface once that coding has been verified. $\square$

Theorem F16.5. Transversality to stable lamination implies universal scaling

Statement. Let R be a C^1 local diffeomorphism near a hyperbolic fixed point p with one-dimensional unstable space and real unstable multiplier λ_u , $|\lambda_u| > 1$. Assume there is a C^1 transverse linearizing coordinate Ψ on a neighborhood U such that $\Psi^{-1}(0) = W_{\text{loc}}^s(p) \cap U$, $D\Psi_p$ is nonzero on E_p^u , and

$$\Psi(Rz) = \lambda_u \Psi(z)$$

whenever $z, Rz \in U$. Let γ be a C^1 parameter curve with $\gamma(0) = p$ and $(\Psi \circ \gamma)'(0) = c \neq 0$. For every sufficiently small fixed $a \neq 0$ and all large n , there is a unique small parameter t_n with

$$\Psi(\gamma(t_n)) = a\lambda_u^{-n}.$$

Whenever the orbit segment stays in U , this is equivalent to $\Psi(R^n \gamma(t_n)) = a$, and

$$t_n = \frac{a}{c} \lambda_u^{-n} + o(|\lambda_u|^{-n}), \quad \frac{t_n}{t_{n+1}} \rightarrow \lambda_u, \quad \frac{|t_n|}{|t_{n+1}|} \rightarrow |\lambda_u|.$$

Bare topological transversality without the stated C^1 linearizing/holonomy coordinate is not sufficient for this conclusion.

Proof. Put

$$\psi(t) := (\Psi \circ \gamma)(t). \tag{1}$$

Because $\gamma(0) = p$ and $p \in W_{\text{loc}}^s(p) = \Psi^{-1}(0)$,

$$\psi(0) = 0. \tag{2}$$

Transversality is exactly

$$c := \psi'(0) \neq 0. \tag{3}$$

Indeed, $T_p W_{\text{loc}}^s(p) = \ker D\Psi_p$, because Ψ is a submersion transverse to the one-dimensional unstable direction. Hence $\gamma'(0) \notin T_p W_{\text{loc}}^s(p)$ if and only if $D\Psi_p \gamma'(0) \neq 0$, which is (3).

Step 1: solve the renormalization levels uniquely. Apply *Transversality and persistence of intersections* in the one-dimensional case to

$$F(t, s) = \psi(t) - s \tag{4}$$

at $(t, s) = (0, 0)$. Since $D_t F(0, 0) = c \neq 0$, after shrinking the intervals there is a unique

C^1 inverse branch

$$\tau : (-\eta, \eta) \longrightarrow (-\varepsilon_0, \varepsilon_0), \quad \psi(\tau(s)) = s, \quad \tau(0) = 0. \quad (5)$$

Moreover,

$$\tau'(0) = \frac{1}{c}. \quad (6)$$

Fix $a \neq 0$ so small that $a\lambda_u^{-n} \in (-\eta, \eta)$ for all sufficiently large n , and define

$$t_n := \tau(a\lambda_u^{-n}). \quad (7)$$

Then $t_n \rightarrow 0$, and (5) gives

$$\Psi(\gamma(t_n)) = \psi(t_n) = a\lambda_u^{-n}. \quad (8)$$

The local uniqueness in (5) proves uniqueness of t_n among parameters sufficiently close to zero.

If the orbit segment $\gamma(t_n), R\gamma(t_n), \dots, R^n\gamma(t_n)$ lies in U , repeated use of the transverse linearization identity $\Psi(Rz) = \lambda_u\Psi(z)$ gives

$$\Psi(R^n\gamma(t_n)) = \lambda_u^n\Psi(\gamma(t_n)) = a. \quad (9)$$

Conversely, under the same orbit-in- U qualification, the equality $\Psi(R^n\gamma(t)) = a$ is equivalent to $\Psi(\gamma(t)) = a\lambda_u^{-n}$. Thus the scalar level formulation and the renormalized target formulation are exactly equivalent whenever the iterates remain in the linearizing neighborhood.

Step 2: first-order parameter asymptotics. Differentiability of τ at zero means

$$\tau(s) = \tau(0) + \tau'(0)s + o(|s|) = \frac{s}{c} + o(|s|) \quad (s \rightarrow 0). \quad (10)$$

Substitute $s = a\lambda_u^{-n}$. By (7),

$$t_n = \frac{a}{c}\lambda_u^{-n} + o(|\lambda_u|^{-n}). \quad (11)$$

This separates the nonuniversal prefactor a/c , which depends on the chosen target level and parameter family, from the exponential rate, which is the unstable multiplier of the renormalization fixed point.

Step 3: the universal ratio. Equation (11) may be written

$$t_n = \frac{a}{c}\lambda_u^{-n}(1 + \varepsilon_n), \quad \varepsilon_n \rightarrow 0. \quad (12)$$

Since $a/c \neq 0$, we have $t_n \neq 0$ for all sufficiently large n . Therefore

$$\frac{t_n}{t_{n+1}} = \lambda_u \frac{1 + \varepsilon_n}{1 + \varepsilon_{n+1}} \longrightarrow \lambda_u. \quad (13)$$

Taking absolute values gives

$$\frac{|t_n|}{|t_{n+1}|} \longrightarrow |\lambda_u|. \quad (14)$$

If $\lambda_u < 0$, the signed parameters alternate sides for all large n ; (13) records the orien-

tation reversal and (14) records the geometric scaling factor.

Nothing in this argument promotes mere topological transversality to a differentiable transverse coordinate. The C^1 linearizing/holonomy coordinate Ψ is therefore a genuine hypothesis of the abstract Foundation interface. In a concrete renormalization theory it must be proved by the relevant transversal-conformality or holonomy theorem rather than inferred from *Stable lamination near a hyperbolic invariant set* alone. $\square$

Proposition F16.6. Compact/quascompact derivative spectral splitting

Statement. Let L be a bounded operator on a complex Banach space and let $r_{\text{ess}}(L)$ be its essential spectral radius. If $\rho > r_{\text{ess}}(L)$ and $|z| = \rho$ misses $\sigma(L)$, then the spectrum with $|z| > \rho$ is a finite set of isolated eigenvalues of finite algebraic multiplicity. The sum P_ρ of their Riesz projections has finite-dimensional range E_ρ and

$$X = E_\rho \oplus F_\rho, \quad F_\rho = \ker P_\rho,$$

with both summands L -invariant, $\sigma(L|_{E_\rho}) = \sigma(L) \cap \{|z| > \rho\}$ and $\sigma(L|_{F_\rho}) \subset \{|z| < \rho\}$. In particular compact operators have discrete nonzero spectrum of finite algebraic multiplicity. If $r_{\text{ess}}(L) < 1$ and the unit circle misses $\sigma(L)$, then there is an invariant hyperbolic splitting $X = E^s \oplus E^u$ with E^u finite-dimensional, $r(L|_{E^s}) < 1$, and $r((L|_{E^u})^{-1}) < 1$.

Proof. Let X be a complex Banach space, let $L \in \mathcal{B}(X)$, and let

$$\pi : \mathcal{B}(X) \longrightarrow \mathcal{B}(X)/\mathcal{K}(X) \tag{1}$$

be the quotient onto the Calkin algebra. We use the Fredholm essential spectrum

$$\sigma_{\text{ess}}(L) := \sigma(\pi(L)) \tag{2}$$

and hence

$$r_{\text{ess}}(L) = r(\pi(L)) = \max\{|z| : z \in \sigma_{\text{ess}}(L)\}. \tag{3}$$

Step 1: spectrum outside the essential spectral radius is discrete Fredholm spectrum. Fix z with

$$|z| > r_{\text{ess}}(L). \tag{4}$$

Then $z \notin \sigma(\pi(L))$, so $\pi(L - zI)$ is invertible in the Calkin algebra. By Atkinson's theorem, $L - zI$ is Fredholm. On the connected exterior region

$$\Omega := \{z \in \mathbb{C} : |z| > r_{\text{ess}}(L)\}, \tag{5}$$

the Fredholm index is locally constant. For $|z| > \|L\|$ the Neumann series shows that $L - zI$ is invertible, hence has index zero. Therefore

$$\text{ind}(L - zI) = 0 \quad (z \in \Omega). \tag{6}$$

The analytic Fredholm theorem applied to the analytic family $z \mapsto L - zI$ now says

that, because the family is invertible for large $|z|$, its noninvertible points in Ω are isolated and the resolvent is meromorphic there with finite-rank principal parts. Consequently every point of

$$\sigma(L) \cap \Omega \quad (7)$$

is an isolated eigenvalue of finite algebraic multiplicity. Equivalently, the associated Riesz projection has finite rank. This is the only functional-analytic discreteness input needed below.

Step 2: only finitely many eigenvalues lie outside the separating circle. Let

$$\rho > r_{\text{ess}}(L), \quad \sigma(L) \cap \{|z| = \rho\} = \emptyset. \quad (8)$$

The spectrum is compact. Since the compact circle in (8) is disjoint from it, there is $\varepsilon > 0$ such that

$$\sigma(L) \cap \{z : \rho - \varepsilon \leq |z| \leq \rho + \varepsilon\} = \emptyset. \quad (9)$$

Every spectral point with $|z| > \rho$ therefore lies in the compact annulus

$$\rho + \varepsilon \leq |z| \leq \|L\| + 1 \quad (10)$$

and, by Step 1, is isolated in the spectrum. An infinite subset of the compact set (10) would have an accumulation point; that point would belong to $\sigma(L)$ by closedness and would still lie in Ω , contradicting isolation. Hence

$$\sigma_{>}(L) := \sigma(L) \cap \{|z| > \rho\} = \{\lambda_1, \dots, \lambda_m\} \quad (11)$$

is finite (possibly empty), and each λ_j has finite algebraic multiplicity.

Step 3: Riesz projections and the invariant direct sum. Choose disjoint positively oriented small circles Γ_j around the λ_j , containing no other spectral points, and define

$$P_j := \frac{1}{2\pi i} \int_{\Gamma_j} (zI - L)^{-1} dz. \quad (12)$$

Each P_j is a bounded projection commuting with L , its range is the finite-dimensional generalized eigenspace associated with λ_j , and $P_i P_j = 0$ for $i \neq j$. Put

$$P_\rho := \sum_{j=1}^m P_j, \quad E_\rho := \text{Ran } P_\rho, \quad F_\rho := \ker P_\rho. \quad (13)$$

Then $P_\rho^2 = P_\rho$, so

$$X = E_\rho \oplus F_\rho, \quad (14)$$

and both summands are L -invariant because $P_\rho L = L P_\rho$. Furthermore E_ρ is finite-dimensional.

The Riesz decomposition theorem gives the exact spectral separation

$$\sigma(L|_{E_\rho}) = \sigma_{>}(L), \quad (15)$$

and

$$\sigma(L|_{F_\rho}) = \sigma(L) \setminus \sigma_{>}(L). \quad (16)$$

By (8), the remaining spectrum contains no point on $|z| = \rho$; hence

$$\sigma(L|_{F_\rho}) \subset \{|z| < \rho\}. \quad (17)$$

Equations (11)–(17) prove the first assertion of the proposition.

Step 4: the compact-operator specialization. If L is compact, then $\pi(L) = 0$, so

$$r_{\text{ess}}(L) = 0. \quad (18)$$

Apply Steps 1–3 with a circle of radius ρ chosen strictly between 0 and any specified nonzero spectral modulus. Every nonzero spectral point is therefore an isolated eigenvalue of finite algebraic multiplicity. Since $\sigma(L)$ is compact, 0 is the only possible accumulation point of nonzero eigenvalues. This is the usual discrete-spectrum theorem for a compact operator.

Step 5: hyperbolicity when the unit circle is avoided. Assume now

$$r_{\text{ess}}(L) < 1, \quad \sigma(L) \cap \{|z| = 1\} = \emptyset. \quad (19)$$

Use the Riesz separation above with $\rho = 1$ and write

$$E^u := E_1, \quad E^s := F_1. \quad (20)$$

Then E^u is finite-dimensional and

$$\sigma(L|_{E^s}) \subset \{|z| < 1\}. \quad (21)$$

Compactness of the spectrum in (21) gives

$$r(L|_{E^s}) < 1. \quad (22)$$

On E^u , every spectral value has modulus > 1 . Thus 0 is not in $\sigma(L|_{E^u})$, so $L|_{E^u}$ is invertible, and spectral mapping gives

$$r((L|_{E^u})^{-1}) = \max_{\lambda \in \sigma(L|_{E^u})} |\lambda|^{-1} < 1. \quad (23)$$

If $E^u = \{0\}$ the same statement holds with the usual zero-dimensional convention. Combining (14), (20), (22), and (23) yields the asserted hyperbolic splitting

$$\boxed{X = E^s \oplus E^u, \quad r(L|_{E^s}) < 1, \quad r((L|_{E^u})^{-1}) < 1, \quad \dim E^u < \infty.} \quad (24)$$

This is exactly the spectral input required by the abstract renormalization hyperbolicity interface. $\square$

16.3 Worked examples

Worked example: Abstract Renormalization and Hyperbolicity: a model computation

Consider the linear renormalization model

$$\mathcal{R}(x, y) = (2x, y/3)$$

on $\mathbb{R}^2$. The origin is a hyperbolic fixed point. The unstable space is $E^u = \mathbb{R}(1, 0)$ and the stable space is $E^s = \mathbb{R}(0, 1)$. Hence

$$W^s(0) = \{(0, y) : y \in \mathbb{R}\}.$$

Take a parameter curve $\gamma(t) = (t, b)$, which crosses $W^s(0)$ transversely at $t = 0$ when $b = 0$ (or after translating the stable leaf in an affine version). The unstable coordinate after n renormalizations is $2^n t$. If a fixed reference value $a \neq 0$ is reached at scale n , then

$$2^n t_n = a, \quad t_n = a 2^{-n}.$$

Therefore

$$\frac{t_n}{t_{n+1}} = 2.$$

The universal scaling factor is exactly the unstable eigenvalue. Nonlinear renormalization theorems replace this exact identity by an asymptotic one, but the mechanism is the same.

Boundary case: what fails at the edge of the hypotheses

Replace the model by

$$\mathcal{R}(x, y) = (x, y/3).$$

The derivative has eigenvalue 1 in the x -direction, so the fixed point is not hyperbolic. The unstable/stable spectral splitting required by the theorem is absent, and the equation $x_n = x_0$ produces no geometric scaling sequence $t_n \sim \lambda^{-n}$. A unit-modulus eigenvalue is therefore not a harmless technicality: it destroys the hyperbolic mechanism behind universality.

Common pitfall

Do not use the worked model as a substitute for checking hypotheses in a new system. The point of the example is to expose the mechanism; the theorem statements that precede it specify the actual scope.

16.4 Chapter summary

What to remember

- Renormalization converts self-similarity questions into dynamics on a function or moduli space.
- Hyperbolicity of the renormalization operator creates stable laminations and finite-codimension universality classes.
- Transverse parameter families inherit scaling from unstable eigenvalues.
- Quasiconpact spectral theory is the infinite-dimensional device that isolates finitely many unstable modes.

Looking ahead

This closes the common Foundation sequence. The specialist books can now use these tools without reintroducing them: KAM uses F05–F12 and F15, spectral theory relies heavily on F04, F08–F09, F13–F15, and complex dynamics draws on F10–F16.

Check your understanding

Without looking back, state one major theorem from this chapter, list its load-bearing hypotheses, and explain the worked example in a way that makes clear where those hypotheses appear. Then describe the boundary example and identify the first theorem conclusion that can no longer be justified.

16.5 Exercises and problems

Exercise 16.1

Difficulty. Core.

Related results. Banach-space hyperbolic fixed-point stable manifold theorem; Hyperbolic set persistence for Banach dynamical systems; Stable lamination near a codimension- k hyperbolic set.

Problem. Work through the worked example ‘a two-dimensional hyperbolic fixed point model for an abstract renormalization operator’. For Banach-space hyperbolic fixed-point stable manifold theorem, Hyperbolic set persistence for Banach dynamical systems, and Stable lamination near a codimension- k hyperbolic set, list every hypothesis in mathematical form, verify it in the model, compute the objects appearing in the conclusions, and then state the resulting conclusions. Do not invoke a result until its hypotheses and normalizations have been checked.

Solution sketch. Begin with the definitions used in the problem. Build a hypothesis/conclusion table for each result, carry out the model calculations, and end by matching each computed quantity with the corresponding conclusion. A complete solution should show all intermediate calculations.

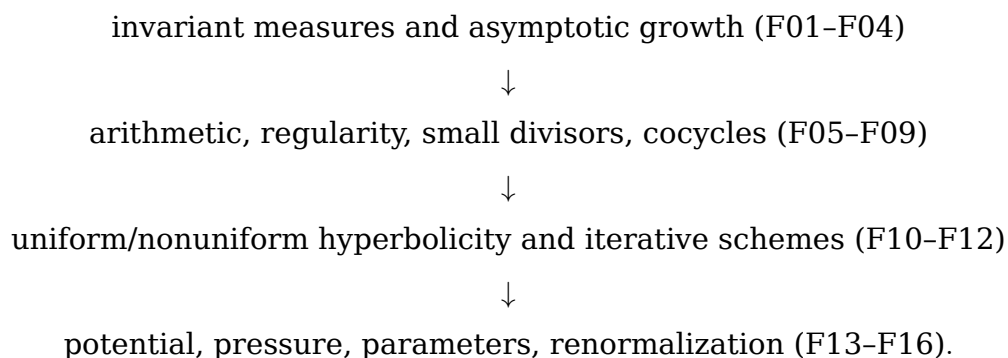
Exercise 16.2**Difficulty.** Core.**Related results.** Renormalization horseshoe abstract model; Transversality to stable lamination implies universal scaling; Compact/quasiconvex derivative spectral splitting.**Problem.** Analyze the boundary example 'a unit-modulus derivative eigenvalue destroying hyperbolicity'. For Renormalization horseshoe abstract model, Transversality to stable lamination implies universal scaling, and Compact/quasiconvex derivative spectral splitting, identify the first hypothesis that fails, prove that it fails, and determine exactly which conclusion is no longer justified. Where possible, replace the lost conclusion by the strongest statement that remains true.**Solution sketch.** Make a hypothesis/conclusion table, give the failed estimate or counterexample explicitly, and finish with a mathematically precise replacement statement.**Exercise 16.3****Difficulty.** Advanced.**Related results.** Stable lamination near a codimension- k hyperbolic set.**Problem.** For Stable lamination near a codimension- k hyperbolic set, isolate one load-bearing intermediate lemma in a proof. State that lemma with complete hypotheses and prove it in the simplest nontrivial setting available in this volume. Do not use the parent theorem as a black box.**Solution sketch.** Separate the elementary, analytic, algebraic, or geometric inputs from the chosen step. Prove only the intermediate lemma, then explain how its conclusion is used in the next step of an eventual proof of the theorem.**Exercise 16.4****Difficulty.** Advanced.**Related results.** Renormalization horseshoe abstract model.**Problem.** Stress-test Renormalization horseshoe abstract model. Choose one genuinely load-bearing hypothesis, weaken or remove it, and either construct a counterexample or prove that a weaker conclusion survives. State the modified claim before analyzing it.**Solution sketch.** Identify where the chosen hypothesis enters the argument. If giving a counterexample, compute the failure mechanism; if proving a weaker result, state the replacement conclusion with all surviving hypotheses and justify each inference.**Exercise 16.5****Difficulty.** Advanced.**Related results.** Banach-space hyperbolic fixed-point stable manifold theorem; Hyperbolic set persistence for Banach dynamical systems.**Problem.** Synthesis problem. Use Banach-space hyperbolic fixed-point stable manifold theorem and Hyperbolic set persistence for Banach dynamical systems to derive a nontrivial corollary in a model from this volume. Write the logical chain

explicitly and show why no single result in the list gives the final conclusion by itself.

Solution sketch. State the target corollary first. Check that the output of each result satisfies the hypotheses of the next one, and isolate any elementary lemma needed between successive steps.

A final map of the Foundation series

The common Foundation can be remembered as four linked layers:



The specialist series should now be read as applications and extensions of these shared tools rather than as independent reintroductions of the same foundations.